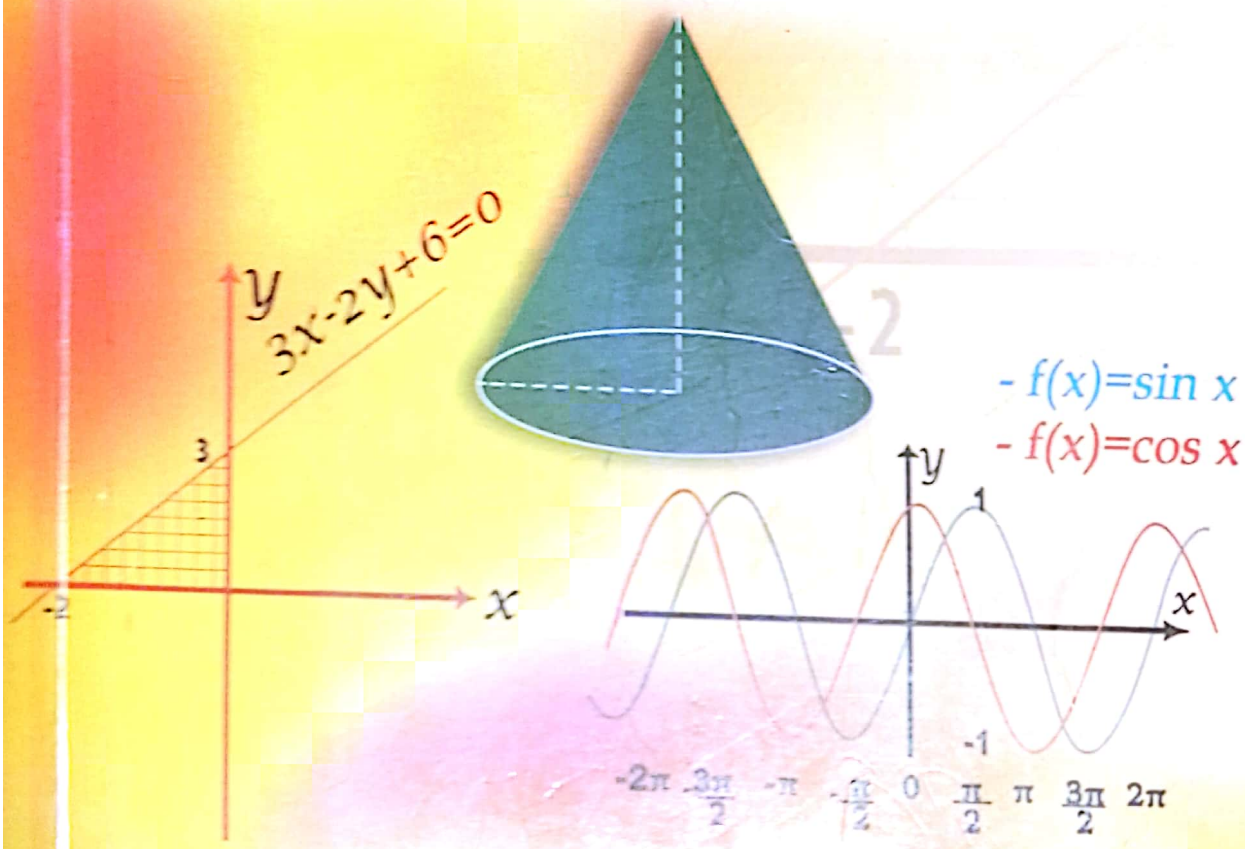
Extreme Series

MATHEMATICS

GRADE 9 - 10



Based on Commonly Used
National and International Curriculum

Third Edition

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UNIT ONE

The Number System

Factors and the least common multiple

Factor: means to write as a product

Illustrative example

1. List the factor of 6

Solution: First we write the different factorization of 6.

$6 = 6.1$

$6 = 2.3$

$\therefore$ The factor of 6 are 1,2,3 and 6.

2. List the factor of 20

Solution: $20 = 1.20$

$20 = 2.10$

$20 = 4.5$

$\therefore$ The factor of 20 are 1,2,4,5,10 and 20.

Definition: **Prime Number and Composite Numbers**

A prime number. *Is a natural number greater than 1 that has exactly two distinct factors only, 1 and itself. The first few prime numbers are 2,3,5,7,11,13,17,. . . etc.*

A composite number. A natural number that has more than two factor

The first few composite numbers are 4,6,8,9,10,12,14,. . . etc

Note, 0 and 1 are neither prime nor composite.

Prime factorization: A natural number written as the product of its **prime factors.**

For example, $30 = 2.3.5$

- When checking to see if one number is factor of another, it is generally faster to use the following divisibility tests than to divide.

Divisibility test

The number is divisible by

- 2 if the unit's digit is 0,2,4,6 or 8.

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- 3 if the sum of the digits is divisible by 3.
- 4 if the number named by the last two digits is divisible by 4.
- 5 If its unit is either 0 or 5.
- 6 If it is divisible by 2 and 3.
- 8 If the number formed by its last three digit is divisible by 8.
- 9 if the sum of the digit is divisible by 9.
- 10 if its unit's digit is 0.

Illustrative Example

3. Determine the prime factorization of each of the following
   a) 45 b) 276 c) 294 d) 4092

Solution:

a) Using divisibility test.

$45 \div 3 = 15$

$15 \div 3 = 5 \longleftarrow$ prime number

$\therefore 45 = 3^2 \times 5$

b) $276 \div 2 = 138$

$138 \div 2 = 69$

$69 \div 3 = 23 \longleftarrow$ prime number

$\therefore 276 = 2^2 \times 3 \times 23.$

c) $294 \div 2 = 147 \leftarrow$ it is divisible by 3

$147 \div 3 = 49$

$49 \div 7 = 7 \longleftarrow$ prime number

$\therefore 294 = 2 \times 3 \times 7^2$

d) $4092 \div 2 = 2046$

$2046 \div 2 = 1023 \longleftarrow$ it is divisible by 3

$1023 \div 3 = 341 \longleftarrow$ prime number

$\therefore 4092 = 2^2 \times 3 \times 341$

4. Determine whether each of the following number divisible by 2,3,4,5,6,9 and 10
   a) 341 c) 3162
   b) 657 d) 444 e) 5124

Solution:

a) 341 is not divisible by any one of 2,3,4,5,6,9 and 10

b) by the rule of divisibility, 6+5+7 = 18 ← divisible by 3

$\therefore$ 657 is divisible by 3 and 9

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c) By the rule of divisibility test:

3+1+6+2 = 12 ← divisible by 3

And the unit digit in 3162 is 2 divisible by 2

∴ 3162 is divisible by 3 and 2.

d) 444 is divisible by 2,3, and 4

e) since the last two digits 24 divisible by 4.

And 5+1+2+4 = 12 ← divisible by 3

∴ 5124 is divisible by 2,3 and 4

The Greatest common factor (GCF)

Consider the common factor of 24 and 80

$F_{24} = \{1,2,3,4,6,8,12,24\}$

$F_{80} = \{1,2,4,5,8,10, 16, 20, 40,80\}$

$F_{24} \cap F_{80} = \{1,2,4,8\}$ ← common factor.

The set $F_{24} n F_{80}$ contains all the possible common factor of 24 and 80 among the factor, 8 is the greatest

∴ GCF (24,80) = 8 ← highest common factor

Definition: *The greatest common factor (GCF) is the greatest integer which is a factor of each of the given integers.*

The two method of finding GCF

1. prime factorization
2. the divisible algorithm

1. Prime factorization method

Step 1: determine the prime factorization of each of the given Integer

Step 2: Form the **GCF** of the given numbers as the product of every factor that appears in each of the factorization but taken the least number of times it occurs in any one of them.

Illustrative Example

5. Find the GCF of each of the following.

a) 60, 72 b) 84,150,198 c) 60,72,180

Solution:

a) $60 = 2\times2\times3\times5 = 2^2 \times 3 \times 5$ ⟵ Prime factorization

$72 = 2\times2\times2\times3\times3 = 2^3 \times 3^3$ ⟵ Prime factorization

Now, collect different factors with smallest exponent, take the product of these factor

∴ GCF (60,72) = $2^2 \times 3 = 12$

b) $84 = 2\times2\times3\times7 = 2^2 \times 3 \times 7$ ⟵ Prime factorization

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$150 = 2\times3\times5\times5 = 2\times3\times5^2$ ⟵ Prime factorization

$198 = 2\times3\times3\times11 = 2\times3^2\times11$ ⟵ Prime factorization

Hence, collect different factors with smallest exponent, then take the product of these factor.

$\therefore$ GCF (84,150,198) $= 2\times3 = 6$

c) $60 = 2^2\times3\times5$

$72 = 2^3\times3^2$

$180 = 2^2\times3^2\times5$

$\therefore$ GCF (60, 72,180) $= 2^2\times3 = 12$

2. The division algorithm

(Euclid's division Algorithm)

Given positive integers a and b, then there exist unique non-negative integers q and r such that

$a = qb + r$, with $0 < r < b$

$b = q_1r + r_1$ with $0 < r_1 < r$

$\therefore$ GCF (a,b) = r if $r_1 = 0$ (the last non zero-reminder)

And GCF (a,b) = GCF(b,r)

Illustrative Example

6. Find the GCF of the following

a) GCF(224,84)
b) GCF (963,657)
c) GCF(4500,1350)
d) GCF(3997,2947)
e) GCF (7286,1684)
f) GCF(27,24)
g) GCF(143,39)

Solution: **Using division algorithm**

a) $224 = 2.84 + 56$
$84 = 1.56 + 28$
$56 = 2.(28) + 0$
$\therefore$ GCF(224,84) = 28

Explanation:
Divide the bigger number by smaller number until you get zero remainder.

Note GCF (224, 84) = GCF (84,56) = GCF(56.28) = 28

b) GCF (963, 657)
$963 = 1.657 + 306$
$567 = 2.(306) + 45$
$306 = 6.(45) + 36$
$45 = 1.36 + 9$
$36 = 4.9 + 0$

Explanation
Divide 963 by 657 keep r = 306
Divide 657 by 306 keep, r = 45
Divide 306 by 45 remainder 36
Divide 45 by 36 remainder 9
36 divide by 9 remainder 0
$\therefore$ 9 ⟵ GCF

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$\therefore$ GCF (963, 657) = 9

Note GCF (963,657) = GCF (306, 45) = . . . GCF (36,9) = 9

c) GCF (4500,1350)

4500 = 3(1350) + 450

1350 = 3(450) + 0 $\therefore$ GCF (4500,1350) = 450

d) GCF (3997,2947)

3997 = 1(2947) + 1050

2947 = 2(1050)+ 847

1050 = (1)(847) + 203

847 = (4)(203) + 35

203 = 5(35) +28

35 = 1(28) + 7

28 = 4(7) + 0 $\therefore$ GCF (3997,2947) = 7

Note

GCF (3997,2947) = GCF (2947,1054) = GCF (1050,847)
= GCF (847,203) = GCF (203,35) = GCF (35, 28)
= GCF (28,7) = 7

e) GCF (7286,1684)

$\Rightarrow$7286 = 4(1684) + 550

$\Rightarrow$1684 = 3(550) + 34

$\Rightarrow$550 = 16(34) + 6

$\Rightarrow$ 34 = 5(6) + 4

$\Rightarrow$ 6 = (1)(4) + 2

$\Rightarrow$4 = 2(2) + 0

$\therefore$ GCF (7286, 1684) = 2

Note: GCF (7286,1684) = GCF(1684,550) = GCF(550, 34) = GCF(4.2)=2

f) GCF(72,24)

27 = 1(24) + 3

24 = 8(3) + 0 $\therefore$ GCF (27,24) = 3

g) GCF(143,39)

143 = 3(39) + 26

39 = 1× 26 + 13

36 = 2(13) + 0 $\therefore$ GCF (143, 39) = 13

- If GCF (a,b) = 1, then they are called relatively prime.

Finding least common Multiples

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Definition. *For any two natural **a** and **b** the least common multiple (LCM) of a and b denoted by LCM (a,b) is the smallest multiple of both **a** and **b**.*

To find the least common Multiple (LCM)

- find the prime factorization of each number.
- Identify the prime factors that appear in each factorization, and
- Multiply these prime factors, using each factor the greatest number of times that it occur in any of the factorization.

Illustrative example

7. Find a) LCM (8,12) c. LCM(18,20,24)
b) LCM (12,18,45) d. LCM(15,18)

Solution: Lets evaluate each of the prime factor,

a) $8 = 2 \times 2 \times 2 = 2^3 \longleftarrow$ prime factor
$12 = 2 \times 2 \times 3 = 2^2 \times 3 \longleftarrow$ prime factor
$\therefore$ LCM $(8, 12) = 2^3 \times 3 = 24$

b) LCM (12, 18, 45)
Now, to find the LCM, we use prime factor method.
$12 = 2^2 \times 3$
$18 = 2 \times 3 \times 3 = 2 \times 3^2$
$45 = 3 \times 3 \times 5 = 3^2 \times 5$
$\therefore$ LCM $(12,18,45) = 3^2 \times 2^2 \times 5 = 180$

c) LCM (18, 20, 24)
$18 = 2 \times 3^2$
$20 = 2^2 \times 5$
$45 = 3^2 \times 5$ $\therefore$ LCM $(18,20, 45) = 3^2 \times 2^2 \times 5 = 180$

d) LCM (15,18)
$15 = 3 \times 5$
$18 = 2 \times 3^2$ $\therefore$ LCM $(15,18) = 3^2 \times 2 \times 5 = 90$

Relationship between the number the LCM and the GCF

For any natural number **a** and **b** GCF (a,b) × LCM (a,b) = a×b

Illustrative example

8. If the GCF (x,36) is 4 and LCM (x,36) is 288 find the number x.

Solution: We have: GCF(x,36) = 4 and LCM (X,36) = 288
Now to find x, use GCF(a,b) LCM (a,b) = a.b. Hence,

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$(X)\times(36)=4\times288$

$\Rightarrow x = \frac{4\times288}{36} = 32 \quad \therefore x = 32$

i.e. GCF (32,36) = 4 and LCM (32, 36) = 288

9. Given GCF (18,x) = 6 and LCM (18,x) = 72

Find X.

Solution:

$(18)(X) = (6)(72)$

$\uparrow \quad \uparrow$

GCF LCM

$\Rightarrow X = \frac{(6)(72)}{18} = 24 \quad \therefore x = 24$

10. The GCF of two number is 12 and the LCM is 144. If one of the number is 36 find the other number.

Solution:

Let a be the other number hence GCF (a, 36) = 12 and LCM (a, 36) = 144

$\Rightarrow (a)(36) = (12)(144)$

$\Rightarrow a = \frac{(12)(144)}{36} = 48 \quad \therefore$ the other number, $a = 48$

11. If x is an odd natural number such that LCM(40,x) = 1400, then x is equal to

A. 35 B. 175 C. 25 D. 875

Solution: Here

$40 = 2\times2\times2\times5 = 2^3\times5$

$X = \frac{}{}$

And $1400 = 2\times2\times2\times5\times5\times7 = 2^3\times5^2\times7$

$\therefore x = 5^2\times7 = 175$ (It should not take multiple of 2, because x is an odd number.

$\therefore$ Answer B

12. There are between 50 and 60 eggs in a basket. If one counts them out 3 at a time, he has 2 left over, but if he counts them out 5 at a time, he has 4 left over. How many eggs are there in the basket .. . . ESLCE

A) 56 B) 52 C) 59 D) 57

Solution:

A) $56 = 3(18) + 2$ when he counts as by 3 remainder 2

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$56 = 5(11) + 1$ when he counts by 5 the remainder is 1

$\therefore$ 56 is not

B) $52 = 3(17) + 1$, when he counts by 3 left 1

$52 = 5(10) + 2$ when he counts by 5 left 2

$\therefore$ 52 is not

C) $59 = 3(19) + 2$ he counts by 3 left 2

$59 = 5(11) + 4$ he counts by 5 left 4

Thus, there are 59 eggs in the basket **$\therefore$ Answer C**

13) If x is the GCF and y is the LCM of the integers 360,105 and 98, then which of the following statement is true

A) $x = 2.3$ and $y = 2.3^2.5.7^2$ C) $x = 1$ and $y = 2^3.3^2.5.7^2$

B) $x = 3.5$ and $y = 2^2.3^2.5.7^2$ D) $x = 1$ and $y = 2^3.3.5.7^2$

Solution: Given GCF (360,105,98) = x

LCM(360,105,98) = y $360 = 2\times2\times2\times3^2\times5\times1$

$105 = 3\times5\times7\times1$

$98 = 2\times7\times7\times1$

Since, no common factor except 1

$\therefore$ GCF (360, 105,98) = x = 1 ⟵ relative prime.

LCM (360,105,98) = $2^3.3^2.5.7^2 = y$ **$\therefore$ Answer C**

14) Find, the value of q and r, using division algorithm, for each of the following

a) $230 = 13q+r$ b) $369 = 21q+r$ c) $235 = 11q+r$

Solution: Note, **q = quotient, r = remainder**

	Computation:	Division algorithm	Application of Euclid
a)	17 13 | 230 130 100 −(91) 9←r	230=(13×17)+9 $\therefore$ q = 17, r = 9	GCF(230,13)=GCF(13,9) Note. GCF(a,b) = GCF(b,r))
b)	17 21 | 369 −21 159 −(147) 12←r	369 = (21×17)+12 $\therefore$ q = 17 r = 12	GCF (369,21)= GCF (21,12)

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c) **Exercise left for you**

15) According to the division algorithm, what should be the value of q and r respectively so that 215 = 11q + r . . . EGSECE.

A) 18 and 7 B) 20 and (– 9) C) 19 and 6 D) 6 and 19

Solution: To find q and r, divide 215 by 11

$\Rightarrow$

$$\begin{array}{r|l} & 19 \leftarrow q \\ \hline 11 & 215 \\ & -(11) \\ \hline & 105 \\ & -(99) \\ \hline & 6 \leftarrow r \end{array}$$

$\Rightarrow 215 = 11\times19+6$

$\therefore$ **Answer: C**

Real Number system

- *Together, the **rational** numbers and irrational numbers form the set of real numbers.*

A. Rational numbers

- A rational number is a number that can be expressed as a quotient of two integer. The set of rational number is denoted by

$Q=\left\{\frac{a}{b}: a \text{ and } b \text{ are integers and } b \neq 0\right\}$

Theorem: *Between any to distinct rational numbers* **a** and **b**, *there exist another rational number.*

To proof . Let a<b $\Rightarrow$ a+a<a+b$\Rightarrow$ 2a<a+b $\Rightarrow$a< $\frac{a+b}{2}$

Again a<b $\Rightarrow$ a+b<b+b $\Rightarrow$a+b<2b $\Rightarrow$ $\frac{a+b}{2}$ <b

Therefore, $a < \frac{a+b}{2} < b$

- It follows that there are infinitely many rational numbers between two rational number.
  - The set of **n** rational numbers between **a** and **b** is obtained by

$a+\frac{b-a}{n-1}, a+\frac{2(b-a)}{n+1} \ldots . a+\frac{n(b-a)}{n+1}$

16) Find two rational number between:

a) $\frac{1}{3}$ *and* $\frac{1}{2}$ b) $\frac{1}{3}$ *and* $\frac{3}{4}$ c) $\frac{1}{4}$ *and* $\frac{1}{3}$

Solution: Here, n = 2

a) $a=\frac{1}{3}, \ b=\frac{1}{2}$, then $\frac{b-a}{n+1}=\frac{\frac{1}{2}-\frac{1}{3}}{2+1}=\frac{3-2}{18}=\frac{1}{18}$

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$\therefore$ Two rational number between $\frac{1}{3}$ and $\frac{1}{2}$ are:

$\frac{1}{3}+\frac{1}{18}, \frac{1}{3}+\frac{2}{18} \Rightarrow \frac{7}{18}, \frac{8}{18}$, i.e. $\frac{1}{3}<\frac{7}{18}<\frac{4}{9}<\frac{1}{2}$

b) $a=\frac{1}{3}, b=\frac{3}{4}$, n = 2, then $\frac{b-a}{n+1}=\frac{\frac{3}{4}-\frac{1}{3}}{2+1}=\frac{9-4}{36}=\frac{5}{36}$

$\therefore$ Two rational number between $\frac{1}{3}$ and $\frac{3}{4}$ are:

$\frac{1}{3}+\frac{5}{36}, \frac{1}{3}+\frac{10}{36} \Rightarrow \frac{17}{36}, \frac{22}{36}$, that is $\frac{1}{3}<\frac{17}{36}<\frac{11}{18}<\frac{3}{4}$

c) **Exercise left for you.**

17) Arrange each of the following rational number in increasing order

$$\frac{2}{3}, \frac{11}{12}, \frac{16}{27}, \frac{11}{18},$$

Solution: $\frac{16}{27}<\frac{11}{18}<\frac{2}{3}<\frac{11}{12}$

Representation of rational numbers by Decimal

- *Any fraction of integers can be expressed as a decimal by dividing the numerator by the denominator.*
- To change a common fraction in to a decimal form, divide the numerator by denominator.

Example :

a) $\frac{3}{4}$ =

$$\begin{array}{r|l} & 0.75 \\ \hline 4 & 3.00 \\ & -28 \\ \hline & 20 \\ & -20 \\ \hline & 0 \end{array}$$

$\therefore \frac{3}{4}=0.75 \leftarrow$ decimal form

b) $\frac{7}{8}$ =

$$\begin{array}{r|l} & 0.875 \\ \hline 8 & 7.00 \\ & -64 \\ \hline & 60 \\ & (-56) \\ \hline & 40 \\ & -40 \\ \hline & 0 \end{array}$$

$\therefore \frac{7}{8}=0.875 \leftarrow$ Terminating decimal

Note: If the division process is stop with remainder 0, then it is called **terminating decimals**.

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If the division process is continue (Non – stoping), then it is called **Repeating (Non – terminating decimal)**.

Illustrative Examples

18. Give the decimal form of the following tractions

a) $\frac{7}{11}$ b) $\frac{11}{6}$ c) $\frac{10}{7}$

Solution: Here

a) $11\overline{)7.00}\ 0.6363\ldots$

-66

40

-33

70

-66

40

-33

7

$\therefore \frac{7}{11} = 0.6363\ldots = 0.\overline{63}$

b) $6\overline{)11}\ 1.833$

-6

50

-48

20

-18

20

-18

2

$\therefore \frac{11}{6} = 1.833\ldots = 1.8\overline{3}$

Expressing Repeating decimals as fractions (Quotient of two integer)

Method (1) put the given number equal to . . . . (1)

(2) Multiplying both side of equation (1) by 10^{k+p}, where K is non-repeating and p is repeating digit after the decimal point

(3) Multiplying both side of equation (1) by 10^{k}, where k is non-repeating digit after decimal point.

(4) subtract equation (3) from (2).

- *The point of a decimal that repeats* is called **the repetend**. A bar over the repetend indicates **the repeating digit.**

Illustrative Example

19) Express each of the following decimal as fraction.

a) $0.\overline{5}$ b) $0.\overline{47}$ c) $0.2\overline{36}$ d) $-0.4\overline{2}$

Solution: Here p = repeating digit, k = non – repeating

a) let $d = 0.\overline{5} = 0.555\ldots$

$10d = 5.\overline{5}$ ⟵ Multiplying d by 10^1, since P = 1

$10d - d = 5.\overline{5} - 0.\overline{5} \Rightarrow 9d = 5 \Rightarrow d = \frac{5}{9}$

b) let $d = 0.\overline{47}$, p = 2, k = 0 (No- none- repeating)

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$100d - d = 47.\overline{47} - 0.\overline{47}$

$100d - d = 47 \Rightarrow 99d = 47 \Rightarrow d = \frac{47}{99}$

c) let $d = 0.2\overline{36}$, $p = 2$, $k = 1 \Rightarrow 10^{p+k} = 10^3$

$1000d = 236.\overline{36}$

$10d = 2.\overline{36}$ ⟵ (since $k = 1 \Rightarrow 10^K = 10^1$)

$1000d - 10d = 234 \Rightarrow 990d = 234 \Rightarrow d = \frac{234}{990}$

d) let $d = -0.4\bar{2}$, $k = 1$ $p = 1 \Rightarrow 10^{p+k} = 10^2 = 100$

$100d = -42.\bar{2}$ ⟵ Multiplying d by 100

$10d = -4.\bar{2}$ ⟵ Multiplying d by 10

$100d - 10d = -42.\dot{2} - (-4.\bar{2}) = 90d = -38 \Rightarrow d = \frac{-38}{90} = \frac{-19}{45}$

In general, to short, we use the formula $d = \frac{d(10^{k+p} - 10^k)}{10^{k+p} - 10^k}$ is used to convert the decimal to fractional form

Where, **k**- non repeating and **p** is repeating digit after decimal point

Illustrative Example

20) Express each of the following decimal as fraction

a) $4.23\overline{575}$ b) $-2.1\overline{174}$ b) $-1.00\overline{791}$ d) $0.00\overline{79}$

Solution: Here, Let k = non – repeating, P = repeating

a) $k = 2$, $p = 3$, now, $d = \frac{(4.23\overline{575})10^{2+3} - (4.23\overline{575})10^2}{10^5 - 10^2}$

$\Rightarrow d = \frac{423575.\overline{575} - 423.\overline{575}}{100{,}000 - 100} = \frac{423152}{99900}$

b) let $k = 1$, $p = 2 \Rightarrow d = \frac{(-2.1\overline{74})(10^{2+1}) - (-2.1\overline{74})10^2}{10^3 - 10}$

$\Rightarrow d = \frac{-2174.\overline{74} - (-21.\overline{74})}{1000 - 10} = \frac{-2153}{990}$

c) Let $k = 2$, $p = 3 \Rightarrow d = \frac{(-1.00\overline{791})(10^{2+3}) - (-1.00\overline{791})(10^2)}{100{,}000 - 100}$

$\therefore d = \frac{-100791.\overline{791} + 100.\overline{791}}{99900} = \frac{-100691}{99900}$

d) Exercise left for you.

21) Find the value of $\frac{x}{y}$ if

a) $x = 0.2\bar{2}$ and $y = -0.\bar{3}$ b) $x = 0.\dot{4}$ and $y = 3.\dot{5}$

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Solution: Here

a) $x = 0.2\dot{2} \Rightarrow x = \frac{22.\dot{2} - 2.\dot{2}}{100-10} = \frac{20}{90} = \frac{2}{9}$

$y = 0.\dot{3} \Rightarrow y = -\frac{(3.\overline{3} - 0.\overline{3})}{10^1 - 10^0} = -\frac{3}{9}$

$\therefore \frac{x}{y} = \frac{2}{9} \cdot \left(\frac{-9}{3}\right) = -\frac{2}{3}$

- $x = 0.\overline{4} \Rightarrow y = 3 + 0.\overline{5} = 3 + \frac{(5.5-0.5)}{10-1} = 3 + \frac{5}{9} = \frac{32}{9}$

  $\therefore \frac{x}{y} = \left(\frac{4}{9}\right)\left(\frac{9}{32}\right) = \frac{4}{32} = \frac{1}{8}$

22) Let $x = 0.\overline{12}$ and $y = 0.\overline{345}$, then find

a) x + y b) y – x c) 2x + y

Solution: Here $x = 0.\overline{12} = 0.121212\ldots$ $y = 0.\overline{345} = 0.345345\ldots$

a) $x+y = 0.\overline{12} + 0.\overline{345} = 0.\overline{466557}$

b) $y - x = 0.\overline{345} - 0.\overline{12} = 0.\overline{224133}$

c) $2x+y = 0.\overline{24} + 0.\overline{345} = (0.242424\ldots) + (0.345345\ldots) = 0.\overline{587769}$

23) Let $x = 0.\overline{4}$, and $y = 2.3\overline{5}$, then find

a) x+y b) x – y

Solution: $x = 0.\overline{4} = 0.4444\ldots$ $y = 2.3\overline{5} = 2.3555\ldots$

a) $x+y = 0.\overline{4} + 2.3\overline{5} = 2.7999\ldots = \frac{279.9 - 27.9}{100-10} = \frac{252}{90}$

b) $x - y = 0.\overline{4} - 2.3\overline{5} = -1.9\overline{1} = -\frac{(191.\overline{1} - 19.\overline{1})}{100-10} = \frac{-172}{90}$

24) If $x = 0.15017017017\ldots$ and $y = 0.40252250052\ldots$ then $3x+2y$:

a) $\frac{113}{90}$ b) $\frac{135}{100}$ c) $\frac{216}{148}$ d) $\frac{1355}{10^3 - 10^2}$

Solution: $3x = 0.45051051051\ldots$ $2y = 0.805045045\ldots$

$\Rightarrow 3x+2y = 1.25555555\ldots = 1.25\overline{5} = \frac{125.\overline{5} - 12.\overline{5}}{100-10} = \frac{113}{90}$, **∴ Answer. A**

25) If $x = 2.3515515551\ldots$ and $y = 0.021373373337\ldots$ Then $10x + 10^3y$ is equal to:

a) 44.9 b) 44.89 c) $44\frac{8}{9}$ d) $\frac{405}{9}$

Solution: $10x = 23.515515551\ldots$ $10^3y = 21.373373337\ldots$

$\Rightarrow 10x+10^3y = 44.888\ldots = \frac{448.8 - 44.8}{10-10^0} = 44\frac{8}{9}$

∴ Answer. C

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26) If a = 1.323323332. . ., b = 0.101101110. . .and $\frac{1}{ac-bc} = 27$,then wh

is the value of c

a) $\frac{1}{33}$ b) $\frac{1}{30}$ c) $\frac{33}{100}$ d) 0.3

Solution: we have $\frac{1}{ac-bc} = \frac{1}{c(a-b)} = 27$

Explanation:

a = 1.323323332 b = 0.101101110

$a - b = 1.222\ldots = 1.\dot{2} = \frac{12.\bar{2}-1.\bar{2}}{10-1} = \frac{11}{9}$

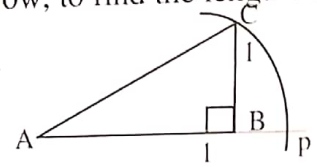
$\Rightarrow c(a-b) = \frac{1}{27} \Leftrightarrow c\left(\frac{11}{9}\right) = \frac{1}{27} \Rightarrow c = \frac{1}{27} \times \frac{9}{11} = \frac{1}{33}$ ∴ **Answer. A**

B. Irrational number

To measure length or distance, we use rational numbers. For example 4 meter $8\frac{3}{4}$ killogram. However, there are lengths which cannot be measure in terms of rational number.

To show this, let us consider a right angled triangle, right angle at B such that $\overline{AB} = \overline{BC} = 1\ unit$.

Now, to find the length AC



- $AB^2+BC^2=AC^2 \Rightarrow AC^2 = 1+1=2$

$AC=\sqrt{2} \Rightarrow \overline{AC} = \overline{AP} = \sqrt{2} \leftarrow$ radius

Thus, the length of the hypotenuse is $\sqrt{2}$ unit. Instead of using the term "non-rational" we say that $\sqrt{2}$ is **an irrational number**

Definition:- *A number which is not rational number is called* **an irrationa number.**

For example: $\sqrt{2}, \sqrt{3}, \sqrt{5}, \ldots$ are irrational numbers

A number which is not perfect square or not perfect cube are irrational.

Example:

a) $\sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{6}, \sqrt{7}, \sqrt{8}, \sqrt{10}$ . . . . etc are irrational (they are no perfect square)

b) $\sqrt[3]{2}, \sqrt[3]{3}, \sqrt[3]{5}, \sqrt[3]{4}, \sqrt[3]{6}, \sqrt[3]{7}, \sqrt[3]{9}, \ldots$ etc are irrational number

- Neither terminating nor repeating decimals are irrational number.

Example. a) 0.01011011101. . .Irrational number

c) 0.231331333133331. . . . Irrational number.

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Facts: i) The sum of two irrational numbers is **not always** an irrational numbers. Example: $\sqrt{2} + (-\sqrt{2}) = 0 \leftarrow$ rationl.

ii) The product of two irrational numbers is **not always** an irrational numbers.

Example: a) $\sqrt{2} \times (\sqrt{2}) = \sqrt{4} = 2 \leftarrow$ rational.

b) $\sqrt{2} \times \sqrt{8} = \sqrt{16} = 4 \leftarrow$ rational.

iii) The set of irrational number is **not closed** under the operation, **addition, subtraction, multiplication and division.**

Illustrative Example

27) If **a** and **b** are two irrational number given by a=2.16210211021110…and b=1.530120112011112…then a+b is

A) an irrational number B. $\frac{3323}{900}$ c) 3.692 D) 3.69

Solution: a = 2.16210211021110… Irr b =1.530120112011112..Irr

However, a+b = 3.69222… rational

$\therefore$ a+b = $\frac{3692.2-369.2}{1000-100} = \frac{3323}{900}$, **$\therefore$ Answer: B**

C. Real Numbers

A number is called a real number, if and only if it is either a rational number or an irrational number.

- *The set of real numbers is closed under the operation addition, subtraction, multiplication and division (except by 0)*

Note: Every point of the number line corresponds to a real number. This is not true of rational numbers.

Classifying the Numbers in a set

Illustrative Example

28) List the numbers in set: $\left\{\sqrt[5]{-3}, \sqrt[6]{-3}\ -3, \frac{4}{3}, 0.15, \sqrt{3}, \pi, 2.131313\ldots\right\}$

that are: a) Natural, b) integers c) rational

d) Irrational e) Real number

Solution: a) Not natural b) – 3 is integer

c) $-3, \frac{4}{3}$, 0.15 and 2.131313…. are **rational**

d) $-\sqrt{3}, \sqrt[5]{-3}$, and π are **irrational**

e) $-3, \frac{4}{3}$, 0.15, π,2.131313. . . $\sqrt[5]{-3}$, and $\sqrt{3}$ are **Real number**

Note $\sqrt[6]{-3}$ is neither, however, it is called complex number

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29. Given $M = \left\{-\sqrt{7}, \frac{-1}{4}, \frac{22}{7}, \sqrt[5]{-2}, \sqrt[4]{-3}\right\}$, Which one of the following statements is true about the elements of set **M**? EGSECE 2005

a) $\sqrt[4]{-3}$ is a real number but $\sqrt[5]{-2}$ **is not.**

b) All the elements of **M** are real numbers,

c) All elements of **M** excepts $\sqrt[4]{-3}$ are real number

d) All elements of $M/\{-\sqrt{7}, \pi\}$ are rational number.

Solution: we know that: • $\sqrt{7}, \pi, \sqrt[5]{-2}$ ⟵ Irrational

• $-\frac{1}{4}, \frac{22}{7}$ ⟵ rational

• $-\sqrt{7}, \pi, \sqrt[5]{-2}, -\frac{1}{4}$, and $\frac{22}{7}$ **are** Real number However, $\sqrt[4]{-3}$ is not real number which is complex number (you will learn in grade 11)

∴ Answer: C

30. Which of the following sets does not represent the set of all real numbers.

A. The set of all decimal numbers.

B. The set of all non-terminating, repeating decimals.

C. The set of rational or irrational numbers D. A and C

Three important properties of order

i) Trichotomy Property:

For all real numbers a and b, one and only one of these statements is true.

$a < b$, $a = b$ or $a > b$

ii) Transitive Property of order

For all real numbers a,b and c

i) If $a < b$ and $b < c$, then $a < c$ ii) If $a < b$, then $a + c < b + c$

if $a > b$, then $a + c > b + c$

31) Let x = rational number, and y = = Irrational then which one of the following is necessarily true?

A) $\frac{y}{x}$ is real number

B) xy is an irrational number

C) x+y is an irrational

D. $\frac{x}{y}$ is an irrational

Solution: Let x = 0, so that

A. $\frac{y}{x} = \frac{y}{0}$ is not real number

B. $xy = 0. y = 0$ ← rational

C) x+y=0+y=y← irrational

D) $\frac{x}{y} = \frac{0}{y} = 0$← rational

∴ **Answer: C**

32) Find the number x, on the number line located between such that

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a) $x=\sqrt{15}$, b) $x = 4-\sqrt{25}$ c) $x = \sqrt{47} - 6$

Solution: a) $\sqrt{9} < \sqrt{15} < \sqrt{16} \Rightarrow 3 < \sqrt{15} < 4$

b) $-\sqrt{25} < -\sqrt{24} < -\sqrt{16} \Rightarrow -5 < -\sqrt{24} < -4$

$\Rightarrow -5+4 < 4 - \sqrt{24} < 4+4 \longleftarrow$ adding (4)

$\Rightarrow -1 < 4 - \sqrt{24} < 0$, Located between – 1 and 0

c) $\sqrt{36} < \sqrt{47} < \sqrt{49} \Rightarrow 6 < \sqrt{47} < 7$.

$\Rightarrow 6 - 6 < \sqrt{47} - 6 < 7 - 6 \longleftarrow$ subtracting 6 on both side

$\Rightarrow 0 < \sqrt{47} - 6 < 1 \longleftarrow$ located between 0 and 1

33) On the number line, $\sqrt{35} - 6$ is located between… EGSECE.

A. -2 and – 1 B. – 1 and 0 C. 1 and 2 D. 0 and 1

Solution: Recall that $\sqrt{25} < \sqrt{35} < \sqrt{36} \Rightarrow 5 < \sqrt{35} < 6$

$\Rightarrow 5 - 6 < \sqrt{35} - 6 < 6 - 6 \longleftarrow$ subtracting (6)

$\Rightarrow -1 < \sqrt{35} - 6 < 0$, $\therefore$ **Answer: B**

The n^{th} Root of real number

The principal nth root of a real number **a**, denoted by $\sqrt[n]{a}$ is defined as: $\sqrt[n]{a}$ **=b** means $\mathbf{a = b^n}$, where

- $a \geq 0$ and $b \geq 0$ if **n is even**.
- a and b are any real numbers if **n is odd**.

Note: If a is negative and n is even then $\sqrt[n]{a}$ is not defined **(no real nth root)**

Example: $\sqrt[4]{-2}, \sqrt[6]{-3}, \sqrt[8]{-5}$ . . . not real number

- The symbol $\sqrt[n]{\ }$ is called a radical sign, the expression $\sqrt[n]{a}$ is called a radical, n is called index, and b is radicand.

Properties of Radical

For all real number a,b, and $\sqrt[n]{a}$, $\sqrt[m]{b}$ is defined and n is positive integer n, then:

i) $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$

ii) $\frac{\sqrt[n]{a}}{\sqrt[n]{b}} = \sqrt[n]{\frac{a}{b}}, b \neq 0$

iii) $\sqrt[n]{a^m} = a^{m/n} = \left(\sqrt[n]{a}\right)^m$

iv) $\sqrt[m]{\sqrt[n]{a}} = \sqrt[mn]{a}$

v) $a^m \cdot a^n = a^{m+n}$

vi) $a^n \cdot b^n = (ab)^n$

- For any three positive rela numbers a,b, and c and all integer n. if $a^n < b < c^n$, then $a < \sqrt[n]{b} < c$

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Illustrative Example

34) Simplify:

a) $(16\times81)^{1/4}$ b) $\sqrt[3]{81}\times\sqrt[3]{9}$ c) $\frac{\sqrt[3]{32}}{\sqrt[3]{216}}$ d) $\sqrt[3]{\frac{27}{16}}$

e) $(-0.008)^{2/3}$ f) $\sqrt[5]{0.00032}$ g) $\sqrt[3]{4}\cdot\sqrt[3]{22}$

h) $\left(5\sqrt[3]{4}\right)\div\left(3\sqrt{2}\right)$ i) $\sqrt{2}\cdot\sqrt[3]{3}\cdot\sqrt[4]{4}$

Solution: Here

a) $(16\times81)^{1/4}=(16)^{1/4}\cdot(81)^{1/4}=(2^4)^{1/4}\cdot(3^4)^{1/4}=(2)(3)=6$

b) $\sqrt[3]{81}\times\sqrt[3]{9}=\sqrt[3]{81.9}=\sqrt[3]{3^4.3^2}=(3^6)^{1/3}=3^2=9$

c) $\frac{\sqrt[3]{32}}{\sqrt[3]{216}}=\sqrt[3]{\frac{32}{216}}=\sqrt[3]{\frac{4}{27}}=\frac{1}{3}\sqrt[3]{4}$

d) $\sqrt[3]{\frac{27}{16}}=\sqrt[3]{\frac{3^3}{2^3\cdot2}}=\frac{3}{2}\sqrt[3]{\frac{1}{2}}=\frac{3}{2\sqrt[3]{2}}$

e) $(-0.008)^{2/3}=(-8\times10^{-3})^{2/3}=(-2^3)^{2/3}\cdot(10^{-3})^{2/3}$
$=(4)(0.01)=0.04$

f) $\sqrt[5]{0.00032}=(32\times10^{-5})^{1/5}=(2^5)^{1/5}(10^{-5})=2(0.1)=0.2$

g) $\sqrt[3]{4}\cdot\sqrt[3]{22}=\sqrt[3]{4\times22}=\sqrt[3]{2^3.11}=2\sqrt[3]{11}$ **Explanation:**

$\sqrt[3]{4}=4^{1/3\cdot2/2}=\sqrt[6]{16}$

h) $\frac{5\sqrt[3]{4}}{3\sqrt{2}}=\frac{5}{3}\cdot\frac{\sqrt[6]{4^2}}{\sqrt[6]{2^3}}=\frac{5}{3}\sqrt[6]{\frac{16}{8}}=\frac{5}{3}\sqrt[6]{2}$

i) LCM (2,3,4) = 12, then

- $\sqrt{2}=2^{1/2}=2^{\frac{1\cdot6}{2\cdot6}}=2^{\frac{6}{12}}=\sqrt[12]{2^6}=\sqrt[12]{64}$
- $\sqrt[3]{3}=3^{1/3}=3^{\frac{1\cdot4}{3\cdot4}}=3^{\frac{4}{12}}=\sqrt[12]{3^4}=\sqrt[12]{81}$
- $\sqrt[4]{4}=4^{1/4}=4^{\frac{1\cdot3}{4\cdot3}}=4^{\frac{3}{12}}=\sqrt[12]{4^3}=\sqrt[12]{64}$ now

$\Rightarrow\sqrt[12]{64}\cdot\sqrt[12]{81}\cdot\sqrt[12]{64}=\sqrt[12]{64\times81\times64}=2^{12}\cdot3^4$

$=(2^{12}\cdot3^4)^{1/2}=2\cdot3^{\frac{1}{3}}=2\sqrt[3]{3}$

$\therefore\sqrt{2}\cdot\sqrt[3]{3}\cdot\sqrt[4]{4}=2\sqrt[3]{3}$

- State the two consecutive integers the number between

a) $\sqrt[3]{34}$ b) $\sqrt[3]{83}$ c) $\sqrt[4]{15}$ d) $\sqrt[3]{20}$

Solution:

a) $\sqrt[3]{27}<\sqrt[3]{34}<\sqrt[3]{64}\Rightarrow3<\sqrt[3]{34}<4$

b) $\sqrt[3]{64}<\sqrt[3]{83}<\sqrt[3]{125}\Rightarrow4<\sqrt[3]{83}<5$

c) $\sqrt[4]{1}<\sqrt[4]{15}<\sqrt[4]{16}=1<\sqrt[4]{15}<2$

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d) Exercise left for you

35) $(0.0081)^{-3/4}$ is equal to _____

A. $\frac{1}{27}$ B. $\frac{10}{27}$ C. $\frac{100}{27}$ D. $\frac{1000}{27}$

Solution: Here

$(0.0081)^{-3/4} = (81\times10^{-4})^{-3/4} = (3^4)^{-3/4}\cdot(10^{-4})^{-3/4}$

$= (3^{-3})(10^3) = \frac{1000}{27}$, ∴ **Answer: D**

36) $\left(\frac{8^{1/4}\times4^{-\frac{1}{6}}}{2^{\frac{1}{6}}}\right)^{-8}$ is equal to

A. 2 B. $\frac{1}{2}$ C. 4 D. $\frac{1}{4}$

Solution: Here, the expression is equivalent to:

$$\left(\frac{2^{3/4}\cdot2^{-\frac{2}{6}}}{2^{\frac{1}{6}}}\right)^{-8} = \left(2^{\frac{3}{4}-\frac{2}{6}-\frac{1}{6}}\right)^{-8} = \left(2^{\frac{3}{4}-\frac{1}{2}}\right)^{-8} = \left(2^{2/8}\right)^{-8}$$

$$= 2^{-2} = \frac{1}{4} \quad \therefore \textbf{Answer D}$$

37) Simplify:

a) $\sqrt[3]{\sqrt{125}}\times\sqrt{45}$ b) $\sqrt[3]{\sqrt{343}}\times\sqrt{28}$ c) $\sqrt[4]{\sqrt[3]{16}}\times\sqrt[3]{3}$

d) $\sqrt[6]{\sqrt[4]{27}}\times\sqrt{28}$ e) $\sqrt[3]{0.2\sqrt{0.00032}}$ f) $\sqrt{0.00016\sqrt[4]{0.016}}$

Solution: Recall $\sqrt[m]{\sqrt[n]{a}} = \sqrt[mn]{a} = (a)^{1/mn}$

a) $\sqrt[3]{\sqrt{125}}\times\sqrt{45} = (125)^{1/6}\cdot(45)^{1/2} = (5^3)^{1/6}\cdot9^{1/2}\cdot5^{1/2}$

$= 5^{1/2}\cdot5^{1/2} = (3^2)^{1/2}\cdot\sqrt{5}\cdot\sqrt{5} = 3.5 = 15$

b) $\sqrt[3]{\sqrt{343}}\times\sqrt{28} = (343)^{1/6}\cdot(28)^{1/2} = (7^3)^{1/6}\cdot(4.7)^{1/2}$

$= 7^{1/2}\cdot7^{1/2} = 4^{1/2}\cdot\sqrt{7}\cdot\sqrt{7}\cdot\sqrt{4} = 2\times7 = 14$

c) $\sqrt[4]{\sqrt[3]{16}}\times\sqrt[3]{4} = (16)^{1/12}\cdot4^{1/3} = (4^2)^{1/12}\cdot4^{1/3}.$

$= 4^{1/6}\cdot4^{1/3} = 4^{1/6+\frac{1}{3}} = 4^{\frac{1}{2}} = \sqrt{4} = 2$

d) $\sqrt[6]{\sqrt[4]{27}}\cdot\sqrt[4]{\sqrt{27}} = (27)^{\frac{1}{24}}\cdot27^{\frac{1}{8}} = 27^{1/24+\frac{1}{8}} = 27^{\frac{4}{24}} = (3^3)^{\frac{1}{24}} = \sqrt{3}$

e) $\sqrt[3]{0.2\sqrt{0.00032}} = (0.2\times0.00032)^{\frac{1}{6}} = (2\times32\times10^{-6})^{\frac{1}{6}}$

$= (2^6\cdot10^{-6})^{\frac{1}{6}} = 2\times10^{-1} = 0.2$

f) $\sqrt{0.00016\sqrt[4]{0.016}} = (0.00016\times0.016)^{\frac{1}{8}} = (16\times16\times10^{-5}\times10^{-3})^{\frac{1}{8}}$

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$= (2^8 \cdot 10^{-8})^{\frac{1}{8}} = 2 \times 10^{-1} = 0.2$

38) Let '**a**' be real number, then which of the following statement is not true.

A. If a > o, then $a > \sqrt{a}$ B. If a > 1, then $a > \sqrt{a}$

C. If a > 1, then $\sqrt{a} > \sqrt[3]{a}$ D. If $0 < a < 1$ then $\sqrt{a} > \sqrt[3]{a}$

Solution:

A. F, for example $\frac{1}{4} > \sqrt{\frac{1}{4}} = \frac{1}{4} > \frac{1}{2}$ B. T C. T D. T ∴ **Answer:** A

If n ≥ 2 is an integer and a is a real number we have

$$\sqrt[n]{a^n} = \begin{cases} a \text{ if } n \text{ is odd} \\ |a| \text{ if } n \text{ is even} \end{cases}$$

Examples • $\sqrt[5]{-32} = \sqrt[5]{(-2)^5} = -2$

- $\sqrt[3]{x^3} = x$ *and* $\sqrt[3]{-64} = \sqrt[3]{(-4)^3} = -4$
- $\sqrt[4]{(-2^4)} = |-2| = 2$ *and* $\sqrt[6]{(-3)^6} = |-3| = 3$
- $\sqrt[4]{x^4} = |x|$ • $\sqrt{x^2} = |x|$

Illustrative Example

39. Simplify: state restriction where necessary.

a) $\sqrt{x^2y^3}$ b) $\sqrt[6]{x^6y^{12}}$ c) $\sqrt[4]{3x^2y^3} \cdot \sqrt[4]{2x^2 \cdot y}$ d) $\sqrt[5]{x^7}$

e) $\sqrt{x^9y^2}$ f) $\sqrt[4]{x^6y^3}$ g) $\sqrt[4]{32x^3y^4}$ h) $\sqrt{18x^2y^5}$ i) $\sqrt{24x^3y^2}$

Solution: The explanation here

a) $\sqrt{x^2y^3} = \sqrt{x^2 \cdot y^2 \cdot y} = \sqrt{y}\,|x|y,\ y \geq 0$

b) $\sqrt[6]{x^6y^{12}} = \sqrt[6]{x^6 \cdot x^6 \cdot y^6} = x^6|y|$

c) $\sqrt[4]{3x^2y^3} \cdot \sqrt[4]{2x^2 \cdot y} = \sqrt[6]{6x^4y^4} = \sqrt{6}\,|xy|$

d) $\sqrt[5]{x^7} = \sqrt[5]{x^5 \cdot x^2} = x\sqrt[5]{x^2} \longleftarrow \sqrt[5]{x^5} = x$

e) $\sqrt{x^9y^2} = \sqrt{x^8 \cdot x} \cdot \sqrt{y^2} = |x|^4|y|\sqrt{x},\ x \geq 0 = x^4|y|\sqrt{x},\ x \geq 0$

f) $\sqrt[4]{x^6 \cdot y^3} = \sqrt[4]{x^4 \cdot x^2} \cdot \sqrt[4]{y^3} = |x|\sqrt[4]{x^2y^3},\ y \geq 0$

g) $\sqrt[4]{32x^3y^4} = \sqrt[4]{2^5x^3 \cdot y^4} = 2|y|\sqrt[4]{2x^3},\ x \geq 0$

h) $\sqrt{18x^2y^5} = \sqrt{9.2.x^2.y^4.y} = 3|x|y^2\sqrt{2y},\ y \geq 0$

i) $\sqrt{24x^3y^2} = 2\sqrt{6x}|y|x,\ x \geq 0$

40) $\sqrt{16\sqrt[3]{a^6b^{12}}}$ is equal to

A. $4a^2b^4$ B. $4|a|b^2$ C. $4ab^2$. D. $16|a|b^2$

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Solution: $4\sqrt[6]{a^6b^{12}} = 4|a|b^2$, **∴ Answer: B**

41) $(0.008)^{-2/3} \times \sqrt[3]{-64-y^3}$ is equal to:

A. $-200y^2$ B. $0.002y$ C. $-8y^3$ D. $-100y$

Solution: $\Rightarrow (8\times10^{-3})^{-2/3}\cdot(-4y) = (2^3\times10^{-3})^{-2/3}(-4y)$

$= (2^{-2})(10^2)(-4y) = \frac{-400y}{4} = -100y$, **∴ Answer: D**

42) $(0.01)^{-\frac{1}{2}}\sqrt[3]{\frac{-343y6}{1000}}$ is equal to

A. $\frac{7y^2}{20}$ B. $-7y^2$ C. $\frac{-7y^2}{100}$ D) $\frac{-7\sqrt{10}y^2}{100}$

Solution: $(0.01)^{-\frac{1}{2}}\sqrt[3]{\frac{-343y6}{1000}} = \frac{(10^{-2})^{-1/2}\cdot(-7^3)^{1/3}\cdot(y^6)^{1/3}}{10}$

$= \frac{(10)(-7)(y^2)}{10} = -7y^2$, **∴ Answer: B**

43) $\left(\sqrt[5]{64\sqrt[3]{\frac{-a^{15}}{8}}}\right)$ is equal to. A. $-\frac{1}{2a}$ B. 2a C. -2a D. $\frac{1}{2a}$

Solution: $\sqrt[5]{\frac{64}{2}(-a^5)} = \sqrt[5]{-32a^5} = -2a$, **∴ Answer: B**

44) If x is any real number, which of the following is necessarily true?

A. $\sqrt{x^3} = x\sqrt{x}$ B. $\sqrt[3]{x^3\sqrt{x^6}} = x^2$ C. $\sqrt[3]{x^2\sqrt[3]{x^3}} = x$ D. $\sqrt{\sqrt[3]{x^6}} = x$

Solution: Here

A. $\sqrt{x^3} = \sqrt{x^2\cdot x} = |x|\sqrt{x}, x \geq 0.$

B. $\sqrt[3]{x^3\sqrt{x^6}} = x\sqrt[6]{x^6} = x|x| \longleftarrow \sqrt[3]{x^3} = x$

C. $\sqrt[3]{x^2\sqrt[3]{x^3}} = \sqrt[3]{x^2\cdot x} = \sqrt[3]{x^3} = x$

D. $\sqrt{\sqrt[3]{x^6}} = \sqrt[6]{x^6} = |x|$, **∴ Answer: C**

45) The simplification of $\left(\left(\left(\frac{32}{243}\right)^4\right)^{1/5} - \frac{5}{9^2}\right)\times\frac{81}{22}$

A. $\frac{1}{2}$ B. $\frac{3341}{891}$ C. $\frac{49}{22}$ D. $\frac{17}{22}$

Solution: $\left(\left(\left(\frac{2^5}{3^5}\right)^{4/5} - \frac{5}{81}\right)\frac{81}{22} = \left(\frac{16}{81} - \frac{5}{81}\right)\frac{81}{22} = \frac{11}{81}\cdot\frac{81}{22} = \frac{1}{2}\right)$ ∴ **Answer: A**

46) a∈R, then the simplest form of $\sqrt{20\sqrt[3]{125a^2}}$

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A. $10\sqrt{5}|a|$ B. $10\sqrt{5}a$ C. $10|a|$ D. $10a$

Solution: $\sqrt{20(5)a^2} = \sqrt{100a^2} = 10|a|$, ∴ **Answer: C**

47) The we simplify: $\sqrt[3]{\frac{1}{3}} \cdot \sqrt[3]{\frac{3}{4}} \cdot \sqrt[3]{\frac{1}{2}}$ we obtain

A. $\frac{1}{8}$ B. $\frac{1}{2}$ C. $\frac{1}{4}$ D. $\frac{\sqrt[3]{3}}{2}$

Solution: $\sqrt[3]{\frac{1}{3}} \cdot \sqrt[3]{\frac{3}{4}} \cdot \sqrt[3]{\frac{1}{2}} = \left(\frac{1}{3} \cdot \frac{3}{4} \cdot \frac{1}{2}\right)^{\frac{1}{3}}$

$= \left(\frac{1}{8}\right)^{1/3} = \sqrt[3]{\frac{1}{8}}, = \frac{1}{2}$ ∴ **Answer: B**

48) Simplify

a. $\frac{\sqrt[4]{3}}{\sqrt[4]{27}}$ b. $\sqrt[3]{\frac{3}{4}}$ c) $\sqrt[3]{\frac{2}{3}}$ d) $\sqrt[4]{\frac{4}{9}}$ e) $\sqrt[3]{\frac{2}{7}}$

Solution:

a) $\frac{\sqrt[4]{3}}{\sqrt[4]{27}} = \sqrt[4]{\frac{3}{27} \cdot \frac{3}{3}}$ ⟵ Multiplying by $\frac{3}{3}$

$= \frac{\sqrt[4]{9}}{3}$ ⟵ since $\sqrt[4]{81} = 3$

b) $\sqrt[3]{\frac{3}{4}} = \sqrt[3]{\frac{3}{4} \cdot \frac{2}{2}} = \sqrt[3]{\frac{6}{8}} = \frac{\sqrt[3]{6}}{\sqrt[3]{8}} = \frac{\sqrt[3]{6}}{2}$ c) $\sqrt[3]{\frac{2}{3}} = \sqrt[3]{\frac{2 \cdot 9}{3 \cdot 9}} = \frac{\sqrt[3]{18}}{\sqrt[3]{27}} = \frac{\sqrt[3]{18}}{3}$

d) $\sqrt[4]{\frac{4}{9}} = \sqrt[4]{\frac{4}{3^2} \cdot \frac{3^2}{3^2}} = \frac{\sqrt[4]{36}}{\sqrt[4]{7^3}} = \frac{\sqrt[4]{36}}{3}$ e) $\sqrt[3]{\frac{2}{7}} = \sqrt[3]{\frac{2 \cdot 7^2}{7 \cdot 7^3}} = \frac{\sqrt[3]{98}}{\sqrt[3]{7^3}} = \frac{\sqrt[3]{98}}{7}$

49) Express in a single radical

a) $3\sqrt{2}$ b) $3\sqrt[3]{4}$ c) $2\sqrt[4]{5}$ d) $a\sqrt[3]{b}$

Solution: Explain as:

a) $3\sqrt{2} = \sqrt{3^2 \cdot 2} = \sqrt{18}$ c) $2\sqrt[4]{5} = \sqrt[4]{2^4.5} = \sqrt[4]{80}$

b) $3\sqrt[3]{4} = \sqrt{3^3 \cdot 4} = \sqrt[3]{108}$ d) $a\sqrt[3]{b} = \sqrt[3]{a^3 b}$

Comparing nth root (Radicals)

If two root of the same index can be easily compared by just comparing their radicals. Hence $\sqrt[n]{a} > \sqrt[n]{b}$ if a > b

- In order to compare two roots of different order, we first all reduce them to the root of the same index.

Illustrative Example

49) Which is greater (compare)

a) $\sqrt[3]{3}$ *or* $\sqrt[4]{5}$ b) $\sqrt{3}$ *or* $\sqrt[3]{5}$ c) $\sqrt[3]{2}$ *or* $\sqrt[5]{3}$

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Solution:

a) LCM (3,4) = 12. Now, $\sqrt[3]{3} = 3^{1/3 \cdot 1/4} = 3^{\frac{4}{12}} = \sqrt[12]{3^4} = \sqrt[12]{81}$

- $\sqrt[4]{5} = 5^{1/4 \cdot 3/3} = 5^{3/12} = \sqrt[12]{5^3} = \sqrt[12]{125}$

Therefore $\sqrt[12]{125} > \sqrt[12]{81} \Rightarrow \sqrt[4]{5} > \sqrt[3]{3}$

b) LCM (2,3) = 6. Then. $\sqrt{3} = 3^{1/2 \cdot 3/3} = 3^{3/6} = \sqrt[6]{3^3} = \sqrt[6]{27}$

- $\sqrt[3]{5} = 5^{1/3 \cdot 2/2} = 5^{2/6} = \sqrt[6]{5^2} = \sqrt[6]{25}$

$\Rightarrow \sqrt[6]{27} > \sqrt[6]{25} \Leftrightarrow \sqrt{3} > \sqrt[3]{5}$.

c) LCM (3,5) = 15. Then $\sqrt[3]{2} = 2^{1/3 \cdot 5/5} = 2^{5/15} = \sqrt[15]{2^5} = \sqrt[15]{32}$

- $\sqrt[5]{3} = 3^{1/5 \cdot 3/3} = 3^{3/15} = \sqrt[15]{3^3} = \sqrt[15]{27}$

Hence. $\sqrt[15]{32} > \sqrt[15]{27}$ $\therefore \sqrt[3]{2} > \sqrt[5]{3}$

Addition and subtraction of Radical

- Similar radical can be added and subtracted.
- Radicals that have the same **index** and the same **radicand** are said to be like radicals.

Example a) $5\sqrt{3} + 6\sqrt{3}$ are like radical b) $\sqrt[3]{2} + 2\sqrt[3]{2}$ are like radical

So that a) $5\sqrt{3} + 6\sqrt{3} = (5+6)\sqrt{3} = 11\sqrt{3}$

b) $\sqrt[3]{2} + 2\sqrt[3]{2} = (1+2)\sqrt[3]{2} = 3\sqrt[3]{2} = \sqrt[3]{3^3 \cdot 2} = \sqrt[3]{54}$

Illustrative Example

50) Simplify

a) $\sqrt{162} - \sqrt{32} + \sqrt{200}$ b) $4\sqrt{32} + \sqrt{72} + \sqrt{50}$

c) $2\sqrt{63} + \sqrt{112} + \sqrt{28}$ d) $3\sqrt{75} - 2\sqrt{48} + \sqrt{12}$

Solution:

a) $\sqrt{162} - \sqrt{32} + \sqrt{200} = \sqrt{2.9.9} - \sqrt{4.4.2} + \sqrt{2.10.10}$

$= 9\sqrt{2} - 4\sqrt{2} + 10\sqrt{2}$

$= (9-4+10)\sqrt{2} = 15\sqrt{2}$

b) $4\sqrt{32} + \sqrt{72} + \sqrt{50} = 4\sqrt{16.2} + \sqrt{6.6.2} + \sqrt{5.5.2}$

$= 16\sqrt{2} + 6\sqrt{2} + 5\sqrt{2} = 27\sqrt{2}$

c) $2\sqrt{63} + \sqrt{112} + \sqrt{28} = 2\sqrt{9 \cdot 7} + \sqrt{4 \cdot 4 \cdot 7} + \sqrt{4 \cdot 7}$

$= 6\sqrt{7} + 4\sqrt{7} + 2\sqrt{7} = 12\sqrt{7}$

d) $3\sqrt{75} - 2\sqrt{48} + \sqrt{12} = 3\sqrt{5 \cdot 5 \cdot 3} - 2\sqrt{4 \cdot 4 \cdot 3} + \sqrt{2 \cdot 2 \cdot 3}$

$= 15\sqrt{3} - 8\sqrt{3} + 2\sqrt{3} = 9\sqrt{3}$

51) Simplify by combining similar terms

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a) $\sqrt{18}+\sqrt[4]{64}+\sqrt[4]{8}$ b) $\sqrt[3]{320}+3\sqrt[3]{625}-8\sqrt[3]{40}$

c) $\sqrt[3]{24}+\sqrt[3]{81}-\sqrt[3]{192}$ d) $2\sqrt[3]{4}+5\sqrt[3]{32}-\sqrt[3]{108}$

Solution: Here:

a) $\sqrt{18}+\sqrt[4]{64}+\sqrt[4]{8}=\sqrt{9\cdot 2}+\sqrt[4]{16}\cdot\sqrt[4]{4}+\sqrt[6]{2^3}$

$=3\sqrt{2}+2\cdot\sqrt{2}+\sqrt{2}=6\sqrt{2}$

b) $\sqrt[3]{320}+3\sqrt[3]{625}-8\sqrt[3]{40}=\sqrt[3]{2^6\cdot 5}+3\sqrt[3]{5^3\cdot 5}-8\sqrt[3]{2^3\cdot 5}$

$=4\sqrt[3]{5}+15\sqrt[3]{5}-16\sqrt[3]{5}=3\sqrt[3]{5}$

c) $\sqrt[3]{24}+\sqrt[3]{81}-\sqrt[3]{192}=\sqrt[3]{2^3}+3\sqrt[3]{3^3\cdot 3}-\sqrt[3]{2^3\cdot 2^3\cdot 3}$

$=2\sqrt[3]{3}+3\sqrt[3]{3}-4\sqrt[3]{3}=\sqrt[3]{3}$

d) Exercise left for you!

52) Simplify each of the following

a) $\sqrt[3]{0.008}+\sqrt[4]{0.0016}$ b) $\frac{\sqrt{0.09}}{\sqrt[3]{0.008}}\left(\frac{\frac{1}{3}-\frac{23}{9}}{\frac{8}{3}\left(\frac{1}{2}+\frac{1}{3}\right)}\right)$

b) $\frac{\sqrt{0.09}}{\sqrt[3]{0.008}}\left(\frac{\frac{1}{3}-\frac{23}{9}}{\frac{8}{3}\left(\frac{1}{2}+\frac{1}{3}\right)}\right)=\frac{\sqrt{\frac{9}{100}}}{\sqrt[3]{\frac{8}{1000}}}\left(\frac{\frac{2}{9}-\frac{23}{9}}{\frac{8}{3}\cdot\frac{5}{6}}\right)=\frac{3}{10}\left(\frac{-20}{9}\right)\left(\frac{10}{2}\cdot\frac{18}{40}\right)=-\frac{3}{2}$

Rationalization of Radical

- The process of transforming radical expression to the rational number is called **Rationalization**
  - The rationalizing factor of $\sqrt[n]{a^k}$ *is* $\sqrt[n]{a^{n-k}}$ where k < n
  - The rationalize, multiply the numerator and denominator by $\sqrt[n]{a^{n-k}}$ for k < n

Rationalizing denominator

Factor in the denominator	Rationalizing factor	Resulting factor
$\sqrt{a}$	$\sqrt{a^{2-1}}=\sqrt{a}$	$\sqrt{a}\cdot\sqrt{a}=a$
$\sqrt[3]{a}$	$\sqrt[3]{a^{3-1}}=\sqrt[3]{a^2}$	$\sqrt[3]{a}\cdot\sqrt[3]{a^2}=\sqrt[3]{a^3}=a$
$\sqrt[7]{a^3}$	$\sqrt[7]{a^{7-3}}=\sqrt[7]{a^4}$	$\sqrt[7]{a^3}\cdot\sqrt[7]{a^4}=\sqrt[7]{a^7}=a$

Illustrative Example

53) Rationalize the denominator in each of the following

a) $\frac{2\sqrt{3}}{\sqrt{7}}$ b) $\sqrt{\frac{3}{5}}$ c) $\frac{3\sqrt{7}}{\sqrt{11}}$ d) $\sqrt{\frac{63}{5}}$

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Solution:

a) $\frac{2\sqrt{3}}{\sqrt{7}} = \frac{2\sqrt{3}}{\sqrt{7}}\cdot\frac{\sqrt{7}}{\sqrt{7}} = \frac{2\sqrt{21}}{7}$ ← Rationalizing factor $\frac{\sqrt{7}}{\sqrt{7}}$

b) $\sqrt{\frac{3}{5}} = \sqrt{\frac{3}{5}\cdot\frac{5}{5}} = \frac{\sqrt{15}}{5}$ ← Rationalizing factor $\frac{\sqrt{5}}{\sqrt{5}}$

c) $\frac{3\sqrt{7}}{\sqrt{11}} = \frac{3\sqrt{7}}{\sqrt{11}}\cdot\frac{\sqrt{11}}{\sqrt{11}} = \frac{3\sqrt{77}}{\sqrt{11}}$ ← Rationalizing factor $\frac{\sqrt{11}}{\sqrt{11}}$

d) $\sqrt{\frac{63}{5}} = \sqrt{\frac{63}{5}}\frac{\sqrt{5}}{\sqrt{5}} = \frac{\sqrt{63.5}}{5} = \frac{3\sqrt{35}}{5}$ ← Rationalizing factor $\frac{\sqrt{5}}{\sqrt{5}}$

54) Rationalize the denominator of the following

a) $\frac{1}{\sqrt[3]{2}}$ b) $\sqrt[3]{\frac{3}{5}}$ c) $\sqrt[4]{\frac{4}{9}}$ d) $\frac{5}{\sqrt[3]{7}}$

Solution: Here Use **R.F** $\sqrt[n]{a^k} = \sqrt[n]{a^{n-k}}$, (k < n)

a) $\frac{1}{\sqrt[3]{2}} = \frac{1\cdot\sqrt[3]{2^2}}{\sqrt[3]{2}.\sqrt[3]{2^2}} = \frac{\sqrt[3]{4}}{\sqrt[3]{2^3}} = \frac{\sqrt[3]{4}}{2}$ ← R.F $= \frac{\sqrt[3]{4}}{\sqrt[3]{4}}$

b) $\sqrt[3]{\frac{3}{5}} = \sqrt[3]{\frac{3}{5}\cdot\frac{5^2}{5^2}} = \sqrt[3]{\frac{75}{5^3}} = \frac{\sqrt[3]{75}}{5}$ ← R.F $= \frac{\sqrt[3]{5^2}}{\sqrt[3]{5^2}}$

c) $\sqrt[4]{\frac{4}{9}} = \sqrt[4]{\frac{4}{9}\cdot\frac{9}{9}} = \sqrt[4]{\frac{36}{3^4}} = \frac{\sqrt[3]{36}}{3}$ ← R.F $= \frac{\sqrt[4]{9}}{\sqrt[4]{9}}$

d) $\frac{5}{\sqrt[3]{7}} = \frac{5}{\sqrt[3]{7}}\frac{\sqrt[3]{7^2}}{\sqrt[3]{7^2}} = \frac{5\sqrt[3]{49}}{7}$ ← R.F $= \frac{\sqrt[3]{49}}{\sqrt[3]{49}}$

Rationalizing binomial $(\sqrt{a} + \sqrt{b})$

- The rationalizing factor of $\sqrt{a} + \sqrt{b}$ is $\sqrt{a} - \sqrt{b}$ and $\sqrt{a} - \sqrt{b}$ is $\sqrt{a} + \sqrt{b}$.
- $(\sqrt{a} + \sqrt{b})(\sqrt{a} - \sqrt{b}) = (\sqrt{a}(\sqrt{a} - \sqrt{b}) + \sqrt{b}(\sqrt{a} - \sqrt{b})$
  $= a - \sqrt{ab} + \sqrt{ab} - b = a - b$
- $(\sqrt{a} + \sqrt{b})\cdot(\sqrt{a} - \sqrt{b}) = \sqrt{a}^2 - \sqrt{b}^2 = a - b$

Illustrative Example

55) Find the value of x if

a) $x = (\sqrt{3} + \sqrt{2})(\sqrt{3} - \sqrt{2})$ c) $x = (3 + \sqrt{5})(3 + \sqrt{5})$

b) $x = (\sqrt{5} - \sqrt{2})(\sqrt{5} + \sqrt{2})$ d) $x = (\sqrt{a} - \sqrt{b})(\sqrt{a} - \sqrt{b})$

Solution: Here

a) $x = (\sqrt{3} + \sqrt{2})(\sqrt{3} - \sqrt{2}) = \sqrt{3}(\sqrt{3} - \sqrt{2}) + \sqrt{2}(\sqrt{3} - \sqrt{2}) = 3 - \sqrt{6} + \sqrt{6} - 2 = 1$

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b) $x = (\sqrt{5}-\sqrt{2})(\sqrt{5}+\sqrt{2}) = \sqrt{5}(\sqrt{5}+\sqrt{2}) - \sqrt{2}(\sqrt{5}+\sqrt{2})$

$\Rightarrow x = 5 + \sqrt{10} - \sqrt{10} - \sqrt{4} = 1 \longleftarrow$ rational

d) $x = (3+\sqrt{5})(3+\sqrt{5}) = (3+\sqrt{5})^2 = 9 + 6\sqrt{5} + 5 =$

$14 + 6\sqrt{5} \longleftarrow$ (irrational)

d) $x = (\sqrt{a}-\sqrt{b})(\sqrt{a}-\sqrt{b}) = (\sqrt{a}-\sqrt{b})^2 = a - 2\sqrt{ab} + b$

Note. $\frac{1}{\sqrt{a}+\sqrt{b}} = \frac{1}{\sqrt{a}+\sqrt{b}} \cdot \frac{\sqrt{a}-\sqrt{b}}{\sqrt{a}-\sqrt{b}} = \frac{\sqrt{a}-\sqrt{b}}{a-b}$

$$\frac{1}{\sqrt{a}-\sqrt{b}} = \frac{1}{\sqrt{a}-\sqrt{b}} \cdot \frac{\sqrt{a}+\sqrt{b}}{\sqrt{a}+\sqrt{b}} = \frac{\sqrt{a}+\sqrt{b}}{a-b}$$

Simplify and identify whether, it is rational or irrational

a) $\sqrt[8]{2+\sqrt{3}} \cdot \sqrt[8]{2-\sqrt{3}}$

b) $\sqrt[5]{8-4\sqrt{2}} \cdot \sqrt[5]{8+4\sqrt{2}}$

e) $\sqrt[5]{5+2\sqrt{3}} \cdot \sqrt[5]{5-2\sqrt{3}}$

c) $\sqrt[4]{6-2\sqrt{5}} \cdot \sqrt[4]{6+2\sqrt{5}}$

d) $\sqrt[3]{5-2\sqrt{2}}\sqrt[3]{5+3\sqrt{2}}$

f) $\sqrt[3]{6+2\sqrt{2}}\sqrt[3]{6-3\sqrt{2}}$

Solution: Remember that $\sqrt[n]{a}\sqrt[n]{b} = \sqrt[n]{ab}$

a) $\sqrt[8]{(2+\sqrt{3})(2-\sqrt{3})} = \sqrt[8]{4-3} = \sqrt[8]{1} = 1 \longleftarrow$ rational

b) $\sqrt[5]{(8-4\sqrt{2})(8+4\sqrt{2})} = \sqrt[5]{64-32} = \sqrt[5]{32} = 2 \longleftarrow$ rational

c) $\sqrt[4]{(6-2\sqrt{5})(6+2\sqrt{5})} = \sqrt[4]{16} = 2 \longleftarrow$ rational

d) Exercise left for you

e) $\sqrt[5]{(5+2\sqrt{3})(5-2\sqrt{3})} = \sqrt[5]{25-12} = \sqrt[5]{13} \longleftarrow$ irrational

f) $\sqrt[3]{(6+2\sqrt{2})(6-2\sqrt{2})} = \sqrt[3]{36-8} = \sqrt[5]{28} \longleftarrow$ irrational

55. c) $\dfrac{\left(\sqrt{2+\sqrt{2+\sqrt{2}}}\right)^4 - \left[2-\sqrt{2+\sqrt{2}}\right]^2}{\left(\left(2+\sqrt[4]{2+\sqrt{2}}\right)\right)^2}$ is equal to

A) 1 B) 2 C) 3 D) 4

Solution: $\dfrac{\left(\left(2+\sqrt{2+\sqrt{2}}\right)^{1/2}\right)^4 - \left[4+2+\sqrt{2}+4\sqrt{2+\sqrt{2}}\right]}{4\sqrt{2+\sqrt{2}}}$

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$$= \frac{\left(2+\sqrt{2+\sqrt{2}}\right)^2-\left(6+\sqrt{2}-4\sqrt{2+\sqrt{2}}\right)}{4\sqrt{2+\sqrt{2}}} = \frac{4+2+\sqrt{2}+4\sqrt{2+\sqrt{2}}-6-\sqrt{2}+4\sqrt{2+\sqrt{2}}}{4\sqrt{2+\sqrt{2}}}$$

$$= \frac{8\sqrt{2+\sqrt{2}}}{4\sqrt{2+\sqrt{2}}} = 2, \qquad \therefore \textbf{Answer: B}$$

56) Express each of the following with rational

a) $\frac{1}{2-\sqrt{5}}$ b) $\frac{3}{\sqrt{3}+5}$ c) $\frac{\sqrt{2}+3}{\sqrt{2}-1}$ d) $\frac{1+\sqrt{3}}{2\sqrt{2}-3\sqrt{3}}$ e) $\frac{1}{2\sqrt{5}-\sqrt{3}}$ f) $\frac{1+\sqrt{2}}{1-\sqrt{2}}$

Solution: Here

a) $\frac{1}{2-\sqrt{5}} = \frac{1}{2-\sqrt{5}} \cdot \frac{2+\sqrt{5}}{2+\sqrt{5}} = \frac{2+\sqrt{5}}{4-5} = -2-\sqrt{5} \longleftarrow \text{R.F} = \frac{2+\sqrt{5}}{2+\sqrt{5}}$

b) $\frac{3}{\sqrt{3}+5} = \frac{3}{\sqrt{3}+5} \cdot \frac{\sqrt{3}-5}{\sqrt{3}-5} = \frac{\sqrt{3}-5}{3-25} = \frac{\sqrt{3}-5}{-22} \longleftarrow \text{R.F} = \frac{\sqrt{3}-5}{\sqrt{3}-5}$

c) $\frac{\sqrt{2}+3}{\sqrt{2}-1} = \frac{\sqrt{2}+3}{\sqrt{2}-1} \cdot \frac{\sqrt{2}+1}{\sqrt{2}+1} = \frac{\sqrt{2}\,(\sqrt{2}+1)+(\sqrt{2}+1)}{2-1} \longleftarrow \text{R.F} = \frac{\sqrt{2}+1}{\sqrt{2}-1}$

$= \sqrt{4} = \sqrt{2}+3\sqrt{2}+3 = 5+4\sqrt{2}$

d) $\frac{1+\sqrt{3}}{2\sqrt{2}-3\sqrt{3}} + \frac{1+\sqrt{3}}{2\sqrt{2}-3\sqrt{3}} \cdot \frac{2\sqrt{2}+3\sqrt{2}}{2\sqrt{2}+3\sqrt{2}} = \frac{1(2\sqrt{2}+3\sqrt{2})+\sqrt{3}(2\sqrt{2}+3\sqrt{2})}{(2\sqrt{2})^2-(3\sqrt{2})^2}$

$= \frac{2\sqrt{2}+3\sqrt{2}+2\sqrt{6}+3\sqrt{6}}{8-18} = \frac{5(\sqrt{2}+\sqrt{6})}{10} = \frac{\sqrt{2}+\sqrt{6}}{-2}$

e) $\frac{1}{2\sqrt{5}-\sqrt{3}} = \frac{1}{2\sqrt{5}-\sqrt{3}} \cdot \frac{2\sqrt{3}+\sqrt{3}}{2\sqrt{5}+\sqrt{3}} = \frac{2\sqrt{5}+\sqrt{3}}{(2\sqrt{5})^2-(\sqrt{3})^2} = \frac{2\sqrt{5}+\sqrt{3}}{17}$

f) Exercise left for you!

56 b. $\sqrt{5\sqrt{k}}$ is equal to: A $\sqrt{5k}$ B $\sqrt[4]{5k}$ C) $\sqrt[4]{25k}$

Solution: $\sqrt{5\sqrt{k}} = \sqrt{\sqrt{25k}} = \left((25k)^{1/2}\right)^{1/2} = (25k)^{1/4} = \sqrt[4]{25k}$

56 c. $\sqrt{2\sqrt[3]{a}}$ is equal to A. $\sqrt{6a}$ B. $\sqrt[3]{2a}$ C. $\sqrt[6]{2a}$ D. $\sqrt[6]{8a}$

Solution: $\sqrt{2\sqrt[3]{a}} = \sqrt{\sqrt[3]{8a}} = \sqrt[6]{8a}$ ∴ Answer: D

57) If both x and y are rational numbers find the values of x and y if

a) $\frac{\sqrt{3}-1}{\sqrt{3}+1} = x + y\sqrt{3}$ b) $\frac{5+2\sqrt{3}}{7+4\sqrt{3}} = x + y\sqrt{3}$

c) $\frac{(\sqrt{2}+1)^2}{3-\sqrt{2}} = x + y\sqrt{2}$ d) $\frac{\sqrt{2}+\sqrt{3}}{3\sqrt{2}-2\sqrt{3}} = x - y\sqrt{6}$

Solution: Here

a) $\frac{\sqrt{3}-1}{\sqrt{3}+1} = x + y\sqrt{3} \Rightarrow \frac{\sqrt{3}-1}{\sqrt{3}+1} \cdot \frac{\sqrt{3}-1}{\sqrt{3}-1} = x + y\sqrt{3}$

$\Rightarrow x + y\sqrt{3} = \frac{(\sqrt{3}-1)^2}{3-1} = \frac{3-2\sqrt{3}+1}{2} = 2-\sqrt{3} \Rightarrow x = 2,\ y = -1$

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b) $x + y\sqrt{3} = \frac{5+2\sqrt{3}}{7+4\sqrt{3}} \cdot \frac{7-4\sqrt{3}}{7-4\sqrt{3}} = \frac{5(7-4\sqrt{3})+2\sqrt{3}(7-4\sqrt{3})}{49-48}$

$\Rightarrow x + y\sqrt{3} = 35 - 20\sqrt{3} + 14\sqrt{3} - 24 = 11 - 6\sqrt{3}$

$\Rightarrow x = 11, y = -6$

c) $x + y\sqrt{2} = \frac{(\sqrt{2}+1)^2}{3-\sqrt{2}} \cdot \frac{3+\sqrt{2}}{3+\sqrt{2}} = \frac{(2+2\sqrt{2}+1)(3+\sqrt{2})}{49-2} = \frac{(3+2\sqrt{2})(3+\sqrt{2})}{7}$

$\Rightarrow x+y\sqrt{2} = \frac{3(3+\sqrt{2})+2\sqrt{2}(3+\sqrt{2})}{7} = \frac{9+3\sqrt{2}+6\sqrt{2}+4}{7} = \frac{13}{7} + \frac{9\sqrt{2}}{7}$

$\Rightarrow x = \frac{13}{7}, \ y = \frac{9}{7}$

d) Exercise left for you.

58) If $x = \frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}}$ and $y = \frac{\sqrt{5}-\sqrt{3}}{\sqrt{5}+\sqrt{3}}$, then find a) x+y b) x − y c) $x + \frac{1}{x}$

Solution: Here

a) $x + y = \frac{\sqrt{5}+\sqrt{3}}{\sqrt{5}-\sqrt{3}} + \frac{\sqrt{5}-\sqrt{3}}{\sqrt{5}+\sqrt{3}} = \frac{(\sqrt{5}+\sqrt{3})^2+(\sqrt{5}-\sqrt{3})^2}{(\sqrt{5})^2-(\sqrt{3})^2}$

$\Rightarrow (x+y) = \frac{5+3+2\sqrt{15}+5+3-2\sqrt{15}}{5-3} = \frac{16}{2} = 8$

b) Exercise left for you

c) Exercise left for you

59) If $x = 9 + 4\sqrt{5}$, find the value of $\sqrt{x} - \frac{1}{\sqrt{x}}$

Solution: let $\sqrt{x} - \frac{1}{\sqrt{x}} = y$ squaring both side

$\left(\sqrt{x} - \frac{1}{\sqrt{x}}\right)^2 = y^2 \Rightarrow x + \frac{1}{x} - 2 = y^2$

$\Rightarrow x = 9 + 4\sqrt{5} \Rightarrow \frac{1}{x} = \frac{1}{9+4\sqrt{5}} \cdot \frac{9-4\sqrt{5}}{9-4\sqrt{5}} = \frac{9-4\sqrt{5}}{81-80} = 9 - 4\sqrt{5}$

$\therefore y^2 = 9+4\sqrt{5} + 9 - 4\sqrt{5} - 2 = 16 \Rightarrow y\sqrt{16} = 4 \qquad \therefore \sqrt{x} - \frac{1}{\sqrt{x}} = 4$

60) Simplify:

a) $\sqrt{\frac{3}{2}} - \sqrt{\frac{2}{3}}$ b) $\sqrt{\frac{3}{5}} + \sqrt{\frac{5}{3}}$ c) $\frac{1}{\sqrt{3}+1} + \frac{1}{\sqrt{3}-1}$ d) $\left(\frac{\sqrt{2}+\sqrt{3}}{\sqrt{2}-\sqrt{3}}\right) - \left(\frac{\sqrt{2}-\sqrt{3}}{\sqrt{2}+\sqrt{3}}\right)$

Solution: Here

a) $\sqrt{\frac{3}{2}} - \sqrt{\frac{2}{3}} = \frac{\sqrt{3}}{\sqrt{2}} - \frac{\sqrt{2}}{\sqrt{3}} = \frac{\sqrt{3}^2-\sqrt{2}^2}{\sqrt{6}} = \frac{3-1}{\sqrt{6}} = \frac{2}{\sqrt{6}} \cdot \frac{\sqrt{6}}{\sqrt{6}} = \frac{\sqrt{6}}{3}$

b) Exercise left for you.

c) $\frac{1}{\sqrt{3}+1} + \frac{1}{\sqrt{3}-1} = \frac{\sqrt{3}-1+\sqrt{3}+1}{(\sqrt{3})^2-1} = \frac{2\sqrt{3}}{2} = \sqrt{3}$

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d) $\left(\frac{\sqrt{2}+\sqrt{3}}{\sqrt{2}-\sqrt{3}}\right)-\left(\frac{\sqrt{2}-\sqrt{3}}{\sqrt{2}+\sqrt{3}}\right)=\frac{(\sqrt{2}+\sqrt{3})^2-(\sqrt{2}-\sqrt{3})^2}{(\sqrt{2}-\sqrt{3})(\sqrt{2}+\sqrt{3})}$

$=\frac{2+3+2\sqrt{6}-(2+3-2\sqrt{6})}{\sqrt{2}^2-\sqrt{3}^2}=\frac{4\sqrt{6}}{2-3}=-4\sqrt{6}$

61) $\frac{2\sqrt{3}}{\sqrt{3}+\sqrt{2}}-\frac{\sqrt{3}}{\sqrt{3}-\sqrt{2}}$ is equal to **EGSECE**

A. $\sqrt{3}$ B. $3+\sqrt{6}$ C. $\frac{\sqrt{3}}{\sqrt{3}+2}$ D. $3-3\sqrt{6}$

Solution: $\frac{2\sqrt{3}}{\sqrt{3}+\sqrt{2}}-\frac{\sqrt{3}}{\sqrt{3}-\sqrt{2}}=\frac{2\sqrt{3}(\sqrt{3}-\sqrt{2})-\sqrt{3}(\sqrt{3}+\sqrt{2})}{(\sqrt{3}+\sqrt{2})(\sqrt{3}-\sqrt{2})}$

$\frac{6-2\sqrt{6}-3-\sqrt{6}}{3-2}=3-3\sqrt{6}$, **∴ Answer: D**

62) $\frac{2\sqrt{3}+3\sqrt{2}}{\sqrt{2}+\sqrt{3}}$ is equal to **EGSECE**

A. $\sqrt{6}$ B. $5\sqrt{6}$ C. 5 D. $-\sqrt{6}$

Solution: $\frac{2\sqrt{3}+3\sqrt{2}}{\sqrt{2}+\sqrt{3}}=\frac{2\sqrt{3}+3\sqrt{2}}{\sqrt{2}+\sqrt{3}}\cdot\frac{\sqrt{2}-\sqrt{3}}{\sqrt{2}-\sqrt{3}}$

$=\frac{2\sqrt{3}(\sqrt{2}-\sqrt{3})+3\sqrt{2}(\sqrt{2}-\sqrt{3})}{(\sqrt{2}+\sqrt{3})(\sqrt{2}-\sqrt{3})}=\frac{2\sqrt{6}-6+6-3\sqrt{6}}{2-3}=\sqrt{6}$

∴ Answer: A

63) $\left(\frac{\sqrt{2}-1}{\sqrt{2}+1}-\frac{\sqrt{2}+1}{\sqrt{2}-1}\right)\div 4\sqrt{2}$ is . . . **EGSECE**

A. 0 B. 1 C. – 1 D. $\frac{-\sqrt{7}}{14}$

Solution: $\frac{(\sqrt{2}-1)^2-(\sqrt{2}+1)^2}{(\sqrt{2}+1)(\sqrt{2}-1)}\div 4\sqrt{2}=\frac{2+1-2\sqrt{2}-(2+1+2\sqrt{2})}{2-1}\cdot\frac{1}{4\sqrt{2}}=\frac{4\sqrt{2}}{4\sqrt{2}}=-1,$

∴ Answer: C

Finding the square root of the form $\left(\sqrt{a+2\sqrt{ab}+b}\right)$

Observe that $(\sqrt{a}+\sqrt{b})^2=\sqrt{a}^2+2\sqrt{ab}+\sqrt{b}^2$

$\Rightarrow(\sqrt{a}+\sqrt{b})^2=a+2\sqrt{ab}+b$

$\therefore \sqrt{a}+\sqrt{b}=\sqrt{a+2\sqrt{ab}+b}=\sqrt{(\sqrt{a}+\sqrt{b})^2}$

$|\sqrt{a}-\sqrt{b}|=\sqrt{a-2\sqrt{ab}+b}=\sqrt{(\sqrt{a}-\sqrt{b})^2}$

Illustrative Example

64) **Simplify**: a) $\frac{\sqrt{a-2\sqrt{ab}+b}}{\sqrt{a}-\sqrt{b}}$ b) $(\sqrt{a}+\sqrt{b})\left(\frac{1}{\sqrt{a}}-\frac{1}{\sqrt{b}}\right)$

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Solution: a) $\left(\frac{\sqrt{a-2\sqrt{ab}+b}}{\sqrt{a}-\sqrt{b}}\right) = \frac{\sqrt{(a-\sqrt{b})^2}}{(\sqrt{a}-\sqrt{b})} = \frac{|\sqrt{a}-\sqrt{b}|}{\sqrt{a}-\sqrt{b}} = \pm 1$

We conclude that $|\sqrt{a}-\sqrt{b}| = \sqrt{a-2\sqrt{ab}+b}$

b) $(\sqrt{a}+\sqrt{b})\left(\frac{1}{\sqrt{a}}-\frac{1}{\sqrt{b}}\right) = (\sqrt{a}+\sqrt{b})\frac{(\sqrt{b}-\sqrt{a})}{\sqrt{ab}} = \frac{\sqrt{b}^2-\sqrt{a}^2}{\sqrt{ab}} = \frac{b-a}{\sqrt{ab}}$

65) **Simplify**: a) $\sqrt{5+2\sqrt{6}}$ b) $\sqrt{8-2\sqrt{15}}$

c) $\sqrt{9+2\sqrt{14}}$ d) $\sqrt{11+2\sqrt{30}}$

Solution: Here

a) Let $\sqrt{x}+\sqrt{y} = \sqrt{5+2\sqrt{6}}$ ⟵ squaring both side

$\Rightarrow (\sqrt{x}+\sqrt{y})^2 = \left(\sqrt{5+2\sqrt{6}}\right)^2$

$\Rightarrow x + y + 2\sqrt{xy} = 5 + 2\sqrt{6}$ ⟵ compare

$x + y = 5$ and $xy = 6 \Leftrightarrow x = 3, y = 2$ Or $x = 2, y = 3$

$\therefore \sqrt{5+2\sqrt{6}} = \sqrt{3}+\sqrt{2}$

b) Let $(\sqrt{x}-\sqrt{y})^2 = \sqrt{8-2\sqrt{15}}^2$ ⟵ squaring both sides

$\Rightarrow (\sqrt{x}-\sqrt{y})^2 = \left(\sqrt{8-2\sqrt{15}}\right)^2$

$\Rightarrow x + y - 2\sqrt{xy} = 8 - 2\sqrt{15}$ ⟵ compare

$\Rightarrow x + y = 8$ and $xy = 15$ Hence, $x = 5$, and $y = 3$

$\therefore \sqrt{8-2\sqrt{15}} = \sqrt{15} - \sqrt{3}$

65) Express each of the following with rational denominator

a) $\frac{1}{\sqrt{2-\sqrt{3}}}$ b) $\frac{1}{\sqrt{5+2\sqrt{6}}}$

Solution: Here

a) $\frac{1}{\sqrt{2-\sqrt{3}}} = \frac{1}{\sqrt{2-\sqrt{3}}} \cdot \frac{\sqrt{2+\sqrt{3}}}{\sqrt{2+\sqrt{3}}}$ ⟵ Rationalizing factor $\left(\frac{\sqrt{2+\sqrt{3}}}{\sqrt{2+\sqrt{3}}}\right)$

$= \frac{\sqrt{2+\sqrt{3}}}{\sqrt{(2-\sqrt{3})(2+\sqrt{3})}} = \frac{\sqrt{2+\sqrt{3}}}{\sqrt{4-3}} = \frac{\sqrt{2+\sqrt{3}}}{1}$

b) $\frac{1}{\sqrt{5+2\sqrt{6}}} = \frac{1}{\sqrt{5+2\sqrt{6}}} = \frac{\sqrt{5-2\sqrt{6}}}{\sqrt{25-24}} = \frac{\sqrt{5-2\sqrt{6}}}{1}$

c) Exercise left for you (Tip!). Ans. $(\sqrt{7}-\sqrt{2})$

d) Exercise left for you (Tip!). Ans. $(\sqrt{6}-\sqrt{5})$

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65) The simplified form of $\sqrt{11-2\sqrt{30}}$ is equal to

A. $\sqrt{6}+\sqrt{5}$ B. $\sqrt{5}-\sqrt{6}$ C. $\sqrt{6}-\sqrt{5}$ D. $\sqrt{5}-\sqrt{3}$

Solution: Let $\sqrt{x}-\sqrt{y}=\sqrt{11-2\sqrt{30}}$ ⟵ squaring both side

$\Rightarrow (\sqrt{x}-\sqrt{y})^2=\left(\sqrt{11-2\sqrt{30}}\right)^2 \Rightarrow x+y-2\sqrt{xy}=11-2\sqrt{30}$

- Now, compare, x+y = 11 and xy = 30
- The sum of two number x and y is 11 and the product of two number x and y is 30.

$x=\sqrt{6}$, and $y=\sqrt{5}$

$\therefore \sqrt{11-2\sqrt{30}}=\sqrt{6}-\sqrt{5}$

66) $\frac{\sqrt{5-2\sqrt{6}}\cdot\sqrt{5+2\sqrt{6}}}{\sqrt[3]{-125}}$ is equal to

A. $\frac{-\sqrt{13}}{5}$ B. $\frac{\sqrt{13}}{5}$ C. $\frac{1}{5}$ D. $\frac{-1}{5}$

Solution: $\frac{\sqrt{5-2\sqrt{6}}\cdot\sqrt{5+2\sqrt{6}}}{\sqrt[3]{-125}}=\frac{\sqrt{(5-2\sqrt{6})(5+2\sqrt{6}}}{-5}$

$=\frac{\sqrt{25-24}}{-5}=\frac{-1}{5}$, **∴ Answer: D**

67) If $x=\frac{5-2\sqrt{3}}{5+2\sqrt{3}}$, then $x+\frac{1}{x}$ is equal to

A. $\frac{13}{17+20\sqrt{3}}$ B. $\frac{6}{17}$ C. $\frac{5\sqrt{3}}{2+6\sqrt{3}}$ D. $\frac{74}{13}$

Solution: $x=\frac{5-2\sqrt{3}}{5+2\sqrt{3}} \Rightarrow \frac{1}{x}=\frac{5+2\sqrt{3}}{5-2\sqrt{3}}$

$\Rightarrow x+\frac{1}{x}=\frac{5-2\sqrt{3}}{5+2\sqrt{3}}+\frac{5+2\sqrt{3}}{5-2\sqrt{3}}=\frac{(5-2\sqrt{3})^2+(5+2\sqrt{3})^2}{(5+2\sqrt{3})(5-2\sqrt{3})}$

$=\frac{25+12-20\sqrt{3}+25+12+20\sqrt{3}}{25-12}=\frac{74}{13}$, **∴ Answer: D**

68) The simplification of $\sqrt{\sqrt{11}-\sqrt{10}} \div \sqrt{\sqrt{11}+\sqrt{10}}$ gives

A. $21-2\sqrt{110}$ B. $\sqrt{11}-\sqrt{10}$ C. $\frac{21-2\sqrt{110}}{21}$ D. $11-2\sqrt{110}$

Solution: $\frac{\sqrt{\sqrt{11}-\sqrt{10}}}{\sqrt{\sqrt{11}+\sqrt{10}}}=\sqrt{\frac{\sqrt{11}-\sqrt{10}}{\sqrt{11}+\sqrt{10}}\cdot\frac{\sqrt{11}-\sqrt{10}}{\sqrt{11}-\sqrt{10}}}$

$=\sqrt{\frac{(\sqrt{11}-\sqrt{10})^2}{11-10}}=\sqrt{11}-\sqrt{10}$, **∴ Answer: B**

69) Suppose $\sqrt{2}=1.41$, $\sqrt{3}=1.73$, $\sqrt{5}=2.23$ and $\sqrt{6}=2.44$

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Find a) $\frac{1+\sqrt{2}}{\sqrt{5}+\sqrt{3}}+\frac{1-\sqrt{2}}{\sqrt{5}-\sqrt{3}}$ b) $\frac{2+\sqrt{3}}{2-\sqrt{3}}+\frac{2-\sqrt{3}}{2+\sqrt{3}}+\frac{\sqrt{3}-1}{\sqrt{3}+1}$

Solution: The expression is the same as

a) $\frac{(1+\sqrt{2})(\sqrt{5}-\sqrt{3})+(\sqrt{5}+\sqrt{3})(1-\sqrt{2})}{(\sqrt{5}+\sqrt{3})(\sqrt{5}-\sqrt{3})}+\frac{\sqrt{5}-\sqrt{3}+\sqrt{10}-\sqrt{6}+\sqrt{5}+\sqrt{3}-\sqrt{10}-\sqrt{6}}{5-3}$

$= \frac{2\sqrt{5}-2\sqrt{6}}{2} = \sqrt{5}-\sqrt{6} = 2.23 - 2.44 = -0.21$

b) $\left(\frac{2+\sqrt{3}}{2-\sqrt{3}}+\frac{2-\sqrt{3}}{2+\sqrt{3}}\right)+\frac{\sqrt{3}-1}{\sqrt{3}+1} = \frac{(2+\sqrt{3})^2+(2-\sqrt{3})^2}{(2-\sqrt{3})(2+\sqrt{3})}+\frac{\sqrt{3}-1}{\sqrt{3}+1}\cdot\frac{\sqrt{3}-1}{\sqrt{3}-1}$

$= \frac{4+4\sqrt{3}+3+4+3-4\sqrt{3}}{4-3} = \frac{3+1-2\sqrt{3}}{3-1}$

$= 14+\frac{4-2\sqrt{3}}{2} = 14+2-2\sqrt{3} = 16-\sqrt{3} = 16 - 1.73 = 14.27$

69 b) If $a = \frac{1}{3+\sqrt{8}}$ and $b = \frac{1}{3-\sqrt{8}}$, Then find

a) $(a+b)^2$ b) $(a-b)^2$ c) (ab)

Solution: a) $(a+b)^2 = \left(\frac{1}{3+\sqrt{8}}+\frac{1}{3-\sqrt{8}}\right)^2 = \left(\frac{3-\sqrt{8}+3+\sqrt{8}}{9-8}\right)^2 = 36$

b) $(a-b)^2 = \left(\frac{1}{3+\sqrt{8}}+\frac{1}{3-\sqrt{8}}\right)^2 = \left(\frac{3-\sqrt{8}-3-\sqrt{8}}{9-8}\right)^2 = (-2\sqrt{8})^2 = 32$

70) Simplify each of the following:

a) $\sqrt{9+2\sqrt{14}}-\sqrt{9-2\sqrt{14}}$ b) $\frac{\sqrt{7+\sqrt{13}}}{2}+\frac{\sqrt{7-\sqrt{13}}}{2}$

c) $\sqrt{9-2\sqrt{14}}-\sqrt{9+2\sqrt{14}}$ d) $\frac{\sqrt{7+\sqrt{13}}}{2}-\frac{\sqrt{7-\sqrt{13}}}{2}$

Solution: Here

a) Let $x = \sqrt{9+2\sqrt{14}}-\sqrt{9-2\sqrt{14}}$ ⟵ squaring both side

$\Rightarrow x^2 = \left(\sqrt{9+2\sqrt{14}}-\sqrt{9-2\sqrt{14}}\right)^2$

$\Rightarrow x^2 = 9+2\sqrt{14}+9-2\sqrt{14}-2\sqrt{(9+2\sqrt{14})(9-2\sqrt{14})}$

$\Rightarrow x^2 = 18-2\sqrt{81-56} = 18-2(5)$

$\Rightarrow x^2 = 18-10 = 8 \Rightarrow x = \sqrt{8} = 2\sqrt{2}$

b) Let $x = \frac{\sqrt{7+\sqrt{13}}}{2}+\frac{\sqrt{7-\sqrt{13}}}{2}$ ⟵ squaring both side

$\Rightarrow x^2 = \left(\frac{\sqrt{7+\sqrt{13}}}{2}+\frac{\sqrt{7-\sqrt{13}}}{2}\right)^2 = \frac{7+\sqrt{13}}{4}+\frac{7-\sqrt{13}}{4}+\frac{2}{4}\sqrt{(7+\sqrt{13})(7+\sqrt{}$

$\Rightarrow x^2 = \frac{14}{4} = \frac{\sqrt{49-13}}{2} = \frac{7}{3}+\frac{6}{2} = \frac{13}{2}$

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$\therefore x = \sqrt{\frac{2}{13}}$

c) Exercise left for you. $\left(\sqrt{9-2\sqrt{14}}\right) - \left(\sqrt{9+2\sqrt{14}}\right) = -\sqrt{8}$

d) Exercise left for you

70 b) Identify, the given expression is, irrational rational.

a) $\sqrt{2+\sqrt{3}} - \sqrt{2-\sqrt{3}}$ b) $\sqrt{3+2\sqrt{2}} - \sqrt{3-2\sqrt{2}}$

c) $\sqrt{2+\sqrt{3}} + \sqrt{2-\sqrt{3}}$

Solution: a) Let $\sqrt{2+\sqrt{3}} - \sqrt{2-\sqrt{3}} = x \longleftarrow$ squaring both side

$\Rightarrow \left(\sqrt{2+\sqrt{3}} - \sqrt{2-\sqrt{3}}\right)^2 = x^2 \longrightarrow$ use $(a-b)^2 = a^2+b^2 - 2ab$

$\Rightarrow x^2 = 2+\sqrt{3} + 2 - \sqrt{3} - 2\sqrt{(2+\sqrt{3})(2-\sqrt{3})}$

$\Rightarrow x^2 = 4 - 2\sqrt{4-3} = 4 - 2(1) = 2 \Rightarrow x = \sqrt{2}$

$\therefore \sqrt{2+\sqrt{3}} - \sqrt{2-\sqrt{3}} = \sqrt{2} \longleftarrow$ Irrational

b) Let $\sqrt{3+2\sqrt{2}} - \sqrt{3-2\sqrt{2}} = x, \longleftarrow$ squaring both side

$\Rightarrow \left(\sqrt{3+2\sqrt{2}} - \sqrt{3-2\sqrt{2}}\right)^2 = x^2 \rightarrow$ use $(a-b)^2 = a^2+b^2-2ab$

$\Rightarrow 3+2\sqrt{2} + (3-2\sqrt{2}) - 2\sqrt{(3+2\sqrt{2})(3-2\sqrt{2})}$

$= x^2 \rightarrow$ use $\sqrt{k}^2 = k,\ k \geq 0$

$\Rightarrow x^2 = 6 - 2\sqrt{9-8} = 6 - 2 = 4 \Rightarrow x = \sqrt{4} = 2$

$\therefore \sqrt{3+2\sqrt{2}} - \sqrt{3-2\sqrt{2}} = 2 \longleftarrow$ rational

c) We have done in (a),

$x^2 = 4 + 2\sqrt{4-3} = 4+2 = 6 \Rightarrow x = \sqrt{6}$

$\therefore \sqrt{2+\sqrt{3}} + \sqrt{2-\sqrt{3}} = \sqrt{6} \longleftarrow$ Irrational

71) Simplifying: a) $\frac{1}{\sqrt{3}-\sqrt{2}-1}$ b) $\frac{1}{\sqrt{3}+\sqrt{2}+1}$

Solution: a) $\frac{1}{\sqrt{3}-\sqrt{2}-1} = \frac{1}{(\sqrt{3}-1)-\sqrt{2}} \cdot \left(\frac{(\sqrt{3}-1)+\sqrt{2}}{(\sqrt{3}-1)+\sqrt{2}}\right)$

$= \frac{(\sqrt{3}-1)+\sqrt{2}}{(\sqrt{3}-1)^2-\sqrt{2}^2} = \frac{(\sqrt{3}-1)+\sqrt{2}}{3+1-2\sqrt{3}-2} = \frac{\sqrt{3}+\sqrt{2}-1}{2-2\sqrt{3}}$, again

$= \frac{(\sqrt{3}-1)+\sqrt{2}}{2(1-\sqrt{3})} \cdot \frac{(\sqrt{3}+1)}{(\sqrt{3}+1)} = \frac{\sqrt{3}^2-1+\sqrt{2}(\sqrt{3}+1)}{2(1-3)} = \frac{3-1+\sqrt{6}+\sqrt{2}}{2(1-\sqrt{3})} = \frac{2+\sqrt{6}+\sqrt{2}}{-4}$

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b) Exercise left for you Ans. $\left(\frac{\sqrt{2}+2-\sqrt{6}}{4}\right.$

73) Simplify: a) $(9^{-1})^4 \times (3^5) \times \left(\frac{1}{8}\right)^3 \times (27^{-2})^3 \times 27^6$

b) $(27)^{1/3} \times 16 \times (27^{-2} \times 16)^{1/2}$

Solution: First make the base equal

a) $(3^2)^4 \times 3^{15} \times (3^{-3})^3 \times (3^{-6})^3 \times 3^{18} \longleftarrow a^x \cdot a^y + a^x + y$

$= 3^{-8+15-9-18+18} = 3^{-2} = \frac{1}{9}$

b) $(3^3)^{2/3} \times 2^4 \times (3^{-6} \times 2^4)^{1/2} = 3^2 \times 3^{-3} \times 2^4 \times 2^2$

$= 3^{2-3} \times 2^{4+2} = 3^{-1} \cdot 2^6 = \frac{64}{3}$

74) When the expression $\frac{(0.4)(5.02)}{\frac{251}{5}(\sqrt{0.0016}}$ is simplified it gives.

A. 1 B. $\frac{1}{2}$ C. 2.5 D. $\frac{1}{0.04}$

Solution: $\Rightarrow (4 \times 10^{-1})\,(502 \times 10^{-1}) \left(\frac{5}{251}\right) \times \left(\frac{1}{0.04}\right) =$

$\Rightarrow 4 \times 502 \times \frac{5}{251} \times \frac{5}{4 \times 10^{-2}} \times 10^{-1} \times 10^{-2} = (10)(10^{-1}) = 1,$

∴ Answer A

75) Which of the following gives an irrational number?.. . (NEAEA)

A. $\sqrt{2}\left(\sqrt{8}+\sqrt{2}\right)$ C. $(12)^{3/2} \cdot 3^{3/2}$

B. $\sqrt[3]{3}\left(\sqrt[3]{9}-\frac{1}{\sqrt[3]{3}}\right)$ D. $\sqrt{18}$

Solution:

A. $\sqrt{2}\left(\sqrt{8}+\sqrt{2}\right) = \sqrt{16}+\sqrt{4} = 4+2 = 6 \longleftarrow$ rational

B. $\sqrt[3]{3}\left(\sqrt[3]{9}-\frac{1}{\sqrt[3]{3}}\right) = \sqrt[3]{27} - 1 = 3 - 1 = 2 \longleftarrow$ rational

C. $(12)^{3/2} \cdot 3^{3/2} = (36)^{3/2} = (6^2)^{3/2} = 6^3 \longleftarrow$ rational

D. $\sqrt{18} = \sqrt{9} \cdot \sqrt{2} = 3\sqrt{2} \longleftarrow$ irrational **∴ Answer D**

76) The simplified form of the expression $\frac{5^2 \times 3^7 \times 10^3}{100 \times 81 \times 125}$ is equal to:

A. $\frac{15}{2}$ B. 54 C. 81 D. $\frac{27}{5}$

Solution: $\frac{5^2 \times 3^7 \times 10^3}{100 \times 81 \times 125} = \frac{5^2}{5^3} \cdot \frac{10^3}{10^2} \cdot \frac{3^7}{3^4} = \frac{10 \times 3^3}{5} = 2 \times 27 = 54$ **∴Answer: B**

Limits of Accuracy

Estimation

- Rounding, decimal places and significant figure (digit) are common typ of approximation

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A. Rounding

- Rounding to approximate the exact answer by a number that ends in a given number of zeros

To round whole number

- Under line the place to which you are rounding.
- Look at the next digit to the under lined digit called the critical digit. If the next digit is 5 or more add 1 to the under lined digit as it is (unchanged). Then replace all the next digit with 0.

Illustrative Example

77) Round the number 86,472 to the nearest

a) 10,000 b) 1000 c) 100 d) 10

Solution: a) 10,000 place occupies 8, so that under line 8

- The critical digit (the digit next to 9 is 6, a, 5 6 greater than 5, we add 1 to 8 $\Rightarrow 8 + 1 = 9$

$\therefore$ 8 6,472 $\times$ 90,000 ⟵ Round up.

b) 1000 place is 6, then 86472, critical digit is 4 since $4 < 5$, so that 86472, $\approx$ 86,000 ⟵

c) 86,472 $\approx$ 86,500 ⟶ since $7 > 5 \Rightarrow 4 + 1 = 5$

d) 86,472 $\approx$ 86,470 ⟶ since $2 < 5$ so that no, adding.

78) Rounding the following to the nearest 10

a) 375 b) 593 c) 74 d) 13 f) 16 g) 4 h) 7

Solution:

	275	593	74	13	16	04	07
Nearest to 10	380	590	70	10	20	1	10

79) If the population of certain city is 3,647,281 If this figure reports to varies group find the this figure to the nearest

a) 100,000 b) 10,000 c) 1000 d) 100 e) 10

Solution: Here, rounding 3,647,281

Nearest to	100,000	10,000	..1000	100	10
Under lined digit	6	4	7	2	8
Critical digit	4<5	7>5	2<5	8>5	1<5
3,647,281≈	3,600,000	3,650,000	3,647,000	3,647,300	3,647,280

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80) A certain publishing enterprise printed 83,462 copies of mathematics student Text books for Grade 10. If you are asked on a radio to report the number of these text books to the nearest 100, then what would be the correct figure that you will report? EGSECE 2005

A. 83,000 B. 83,500 C. 84,000 D. 83,400

Solution: The 100 place digit is 4, the next digit is $6 > 5$ so that the 100 place digit will be $4 + 1 = 5$, $83,\underline{4}62 \approx 83,500$, ∴ Answer: B

Rounding Decimals

- As rounding whole number, we can round a decimal to a given place.
- The relationship between the place to which we are rounding and the number of decimal places

Rounding to the nearest	**Means that are rounded decimal has**
• Tenth $\left(\frac{1}{10}\right)$	One decimal place (1d.p)
• Hundredth $\left(\frac{1}{100}\right)$	Two decimal place (2d.p)
• Thousandth $\left(\frac{1}{1000}\right)$	Three decimal place (3d.p)

Illustrative Example

81) Round 94.735 to the nearest

a) tenth (1d.p) b) hundredth (2d.p) c) thousandth (3d.p)
d) whole number e) ten

Solution: Here

a) $94.\underline{7}35 \approx 94.70$ c) $94.7\underline{3}5 \approx 94.735$ d) $9\underline{4}.735 \approx 95$
b) $94.73\underline{5} \approx 94.74$ e) $94.735 \approx 90$

82) Round 4.038 to: a) 1d.p b) 2d.p

Solution: a) $4.\underline{0}38 \approx 4.0$ b) $4.0\underline{3}8 \approx 4.04$

C. Significant figure

- Significant digit (figure) is any digit that denote an actual physical amount.

Rules to determine whether a zero is significant or not

i) Zeros are significant when all zeros between non-zero digits

Example: 34.06 – has four significant figure 3,4,0 and 6
500 – has one significant figure, 5
40 – has one significant figure, 4

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ii) Zeros as significant figure when it comes after non-zero digits of decimal point

Example 0.340 – has three significant figures 3,4,0
0.34 – has two significant figure, 3,4
0.3400 – has four significant figure,
0.034 – has two significant figure 3,4
0.006 – has one significant figure 6

iii) All non zero digit are significant figure

83) How many significant figures are there in 0.001050. . . **NAEAE**
A. 5 B. 4 C. 6, D. 7

Solution: since there is no non – zero digit before zero
∴ 0.001050 has 4 significant figure, **∴ Answer: B**

84) Write 618.27 in a) 4 S.f b) 3 S.f c) 2S.f

Solution:
a) 618.27 ≈ 618.3 b) 618.27≈ 618 c) 618.27≈620

85) Write 0.0863 in: a) 2S.f b) 1S.f

Solution: a) 0.0863≈ 0.086 b) 0.0863 ≈ 0.09

Scientific notation (Exponential notation)

Scientific notation is a way of expressing very large or very small numbers as factor of the power of 10

That is $N \times 10^m$, where $1 \leq N < 10$ and m is any integer.

Example: Express the following numbers in scientific notation each with 3 significant figure

a) 0.00000025 b) 8,670,000 c) 300300

Solution: Here

a) 2.50×10^{-7} b) 8.67×10^6 c) 3.030×10^5

Changing from Scientific Notation to Decimal.

Illustrative Example

86) Write each number as a decimal

a) 2.1×10^4 b) 2.1×10^4 c) 4.36×10^{-5} d) 3×10^{-6}

Solution: Here

a) $2.1\times 10^4 = 2.1\times 10000 = 21000 \leftarrow$ move 4 unit to right from the decimal point

b) $2.1\times 10^{-4} = 2.1\times \frac{1}{10,000} = 0.00021 \leftarrow$ move 4 unit to the left from the decimal point

c) $4.36\times 10^{-5} = 0.0000436 \leftarrow$ move 5 unit to left from the dc.p

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d) $3\times10^{-6} = 0.000003 \leftarrow$ move 6 unit to left

87) The number 2.15×10^{-3} is equal to . . . **EGSEC 2005**

A. 0.00215 B. 0.0215 C. 0.215 D. 0.000215

Solution: Here

A. 2.15×10^{-3} B. 2.15×10^{-2} C. 2.15×10^{-1} D. 2.15×10^{-4},

∴ Answer A

88) The scientific notation for 0.00023. . . .

A. 2.3×10^{-3} B. 2.3×10^{-3} C. 2.3×10^{-4} D. 2.3×10^{4}

∴ Answer C

Solution: $0.00023 = 2.3\times 10^{-4}$,

89) Find the scientific notation

a) $(0.00012)^2$ b) $(0.002)^5$ c) $(0.03)^4$

Solution: Here, recall that $(ab)^n = a^n b^n$

a) $(0.000012)^2 = (12\times 10^{-6})^2 = 12^2\times(10^{-6})^2$

$= 144\times 10^{-12} = 1.44\times 10^{2}\times 10^{-12} = 1.44\times 10^{-10}$

b) $(0.002)^5 = (2\times10^{-3})^5 = 2^5\times10^{-15} = 32\times10^{-15} = 3.2\times10^{-14}$

c) $(0.03)^4=(3\times10^{-2})^4=3^4\times10^{-8} = 81\times10^{-8} = 8.1\times10\times10^{-8} = 8.1\times10^{-7}$

$(0.03)^4 = 0.00000081 \longleftarrow$ decimal notation

Accuracy

- Here, we will see how to give **upper bounds and lower bounds** for data to a specified accuracy (for example number rounded off or number expressed to a given number of significant
  - The lower bounds of x is x – 0.05 if x is 1 d.p.
    The upper bounds of x is x + 0.05 if x is 1d.p.
  - The lower bounds of x is x – 0.005 if x is 2 d.p.
    The upper bounds of x is x + 0.005 if x is 2d.p.

Example: Find the lower and upper bounds of the following number correct to two significant figure

a) 4.7 b) 0.83 c) 0.035 d) 340

Solution: a) lower bound of 4.7 is 4.7 – 0.05 = 4.65

Upper bounds of 4.7 is 4.7 + 0.05 = 4.75

$\Rightarrow 4.65 \leq 4.7 < 4.74$

b) $0.83 - 0.005 \leq 0.83 < 0.83 + 0.005 = 0.825 < 0.83 < 0.835$

c) $0.035-0.0005\leq0.035<0.035+0.0005=0.0345\leq0.035<0.0355$

d) $340 - 5 \leq 340 < 340 + 5 = 335 \leq 340 < 345$

Effect on, addition, subtraction and multiplication

Example – Find lower and upper bounds of calculation:

a) 4.3. + 2.8 b) 3.2×2.6

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Solution: $4.3 - 0.05 \leq 4.3 < 4.3 + 0.05$

Lower boundary $\longrightarrow 4.25 \leq 4.3 < 4.35 \longleftarrow$ upper boundary of 3.4

$2.8 - 0.05 \leq 2.8 < 2.8 + 0.05 = 2.75 \leq 2.8 < 2.85$

Then $4.25 \leq 4.3 < 4.3 < 4.35$ and $2.75 \leq 2.8 < 2.85$

$\Rightarrow 4.25 + 2.75 \leq 4.3 + 2.8 < 4.25 + 2.85$

$7.00 \leq 4.3 + 2.8 < 7.20$

$\therefore$ The minimum sum = 7.00 ⟵ lower bound sum.

$\therefore$ The maximum sum = 7.20 ⟵ upper bound sum.

b) Let x = 3.2 and y = 2.6

$\Rightarrow 3.2 - 0.05 \leq 3.2 < 3.2 + 0.05 = 3.15 \leq 3.2 < 3.25$

$\Rightarrow 2.6 - 0.05 \leq 2.6 < 2.6 + 0.05 = 2.55 \leq 2.6 < 2.65$

Therefore, $3.15 \times 2.55 \leq 3.2 \times 2.6 < 3.25 \times 3.65$

$8.0325 \leq 3.2 \times 3.2 \times 2.6 < 8.6125$

Solved Problem

91) Which of the following pairs are **not twins prime**

A) 5 and 7 B) 17 and 19 C) 3 and 5 D) 12 and 17

Solution: *Prime numbers that differ by 2 are called twins prime:* A,B and C are twins, However 13 and 17 are not twin prime **Answer: D**

92) The GCF (97,54) is equal to _____

A) 11 B) 10 C) 1 D) 43

Solution: using division algorithm

$97 = 54(1) + 43$ ⟵ Divide 97 by 54, rem = 43

$54 = 43(1) + 11$ ⟵ Divide 54 by 43, rem = 11

$43 = 11(1) + 10$ ⟵ Divide 43 by 11, rem = 10

$11 = 10(1) + 1$ ⟵ Divide 11 by 10, rem = 1

$10 = 10(1) + 0$ ⟵ Divide 10 by 10, rem = 0

$\therefore$ GCF (97.54) = 1 ⟵ Relative prime

Explanation: GCF (a,b) = 1, then a and b are relative prim

93) If $0.3 < x < 0.9$ and $0.01 < y\ 0.03$, then find A) x + y B) xy C) $\frac{x}{y}$

Solution: a) $0.3 + 0.01 < x + y < 0.9 + 0.03$

$\Rightarrow 0.31 < x + y < 0.93$

b) $(0.3)(0.01) < xy < (0.9)(0.03)$

$\Rightarrow 0.003 < xy < 0.027$

c) max. of x = 0.9, min. of y = 0.01 $\Rightarrow$ max $\frac{x}{y} = \frac{0.9}{0.01} = 90$

min of x = 0.3 and max of y = 0.03

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$\Rightarrow$ min of $\frac{x}{y} = \frac{0.3}{0.03} = 10$ $\quad \therefore\ 10 < \frac{x}{y} < 90$

94) $\sqrt[3]{(-512)^{-1}}$ is equal to

A. $\frac{1}{8}$ B. $-\frac{1}{8}$ C. 8 D.– 8

Solution: $\left(\sqrt[3]{(-512)}\right)^{-1} = \left(\sqrt[3]{(-2)^9}\right)^{-1} = \left((-2^9)^{\frac{1}{3}}\right)^{-1}$

$= (-2^3)^{-1} = (-8)^{-1} = -\frac{1}{8}$, **∴ Answer: B**

95) Simplify

a) $\sqrt{\sqrt{3}-\sqrt{2}} \div \sqrt{\sqrt{3}+\sqrt{2}}$ b) $\sqrt{\sqrt{5}+2\sqrt{2}} \cdot \sqrt{2\sqrt{2}-\sqrt{5}}$

Solution: use $\sqrt{a} \cdot \sqrt{b} = \sqrt{ab}$

a) $\frac{\sqrt{\sqrt{3}-\sqrt{2}}}{\sqrt{\sqrt{3}+\sqrt{2}}} = \sqrt{\frac{(\sqrt{3}-\sqrt{2})}{\sqrt{3}+\sqrt{2}} \cdot \frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}-\sqrt{2}}} = \sqrt{\frac{(\sqrt{3}-\sqrt{2})^2}{3-2}} = \sqrt{3}-\sqrt{2}$

b) $\sqrt{\sqrt{5}+2\sqrt{2}} \cdot \sqrt{2\sqrt{2}-\sqrt{5})} = \sqrt{(2\sqrt{2}+\sqrt{5})(2\sqrt{2}-\sqrt{5})} = \sqrt{8-5}$

96) Simplify a) $\sqrt[4]{3x^4} - \sqrt[4]{48x^4}$ b) $\sqrt{5x^2} - \sqrt{245x^2}$

Solution: a) $\sqrt[4]{3x^4} - \sqrt[4]{48x^4} = |x|\sqrt[4]{3} - 2|x|\sqrt[4]{3} = -|x|\sqrt[4]{3}$

b) $\sqrt{5x^2} - \sqrt{(49)(5)x^2} = |x|\sqrt{5} - 7|x|\sqrt{5} = -6|x|\sqrt{5}$

97) Which one of the following is a prime number

A. 161 B. 277 C. 711 D. 687

Solution: A. $161 = 23 \times 7 \longleftarrow$ composite

C. $711 = 237 \times 3 \times 1 \longleftarrow$ composite

D. $687 = 229 \times 3 \times 1 \longleftarrow$ composite **∴ Answer B**

98) $\sqrt[4]{a^2 \cdot \sqrt[3]{a^2}}$, for $a \geq 0$ is equal to

A. $\sqrt[3]{a^2}$ B. $\sqrt[6]{a^5}$ C. a D. $\sqrt[3]{a^4}$

Solution: $\sqrt[4]{a^2 \cdot \sqrt[3]{a^2}} = \left(a^2 \cdot a^{2/3}\right)^{1/4}$

$= \left(a^{2+\frac{2}{3}}\right)^{1/4} = a^{\frac{8}{3}\cdot\frac{1}{4}} = \sqrt[3]{a^2}$, **∴ Answer A**

99) Find the value of r if

A) GCF (375,225) = GCF (225, r)

B) GCF (224,188) = GCF (188, r)

C) GCF (882,402) = GCF (402, r)

Solution: Using the division algorithm: $a = (qb) + r$ with $0 \leq r < b$, a

GCF (a,b) = GCF (b,r)

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A) $375 = 1\times225 + 150 \Rightarrow a = 375, q = 1, b = 225, r = 150$
That is GCF (375, 225) = GCF (225, 150), then r = 150

B) $224 = 1\times188 + 36 \Rightarrow a = 224, q = 1\ b = 188, r = 36$
That is GCF (224,188) = GCF (188, 36) = 4, $\therefore$ r = 36

C) $882 = 2\times402 + 78 \Rightarrow a = 882, q = 2, b = 402, r = 78$
$\therefore$ GCF (882,402) = GCF (402, 78) = 6, $\therefore$ r = 78

94) Three bells toll at intervals of 4 sec, 6dec, 9sec respectively. They start together; after what times will they next toll together.
A) 18sec B) 24 sec C) 27 sec D) 36 sec

Solution: Use LCM(4,6,9)
$4 = 2^2 \times 2 = 2^2$, $6 = 2 \times 3 = 2 \times 3$
$9 = 3 \times 3 = 3^2 \therefore$ LCM $(4,6,9) = 2^2 \cdot 3^2 = 36 \therefore$ **Answer D**

101) If 90,146, and 230 are each divided by the same number, the remainder is 6. What is the greatest possible value of the divisor?
A) 14 B) 56 C) 12 D) 28

Solution: we know that $90 = 84 + 6$
$146 + 140 + 6$
$230 = 224 + 6$
Thus, GCF(84, 140,224) = 28 ⟵ greatest common divisor of 90,146,230, **$\therefore$ Answer D**

Supplementary Exercise on Unit One

1. Classify each number into one of the following three categories:
A: divisible by 3 and not divisible by 9.
B: divisible by 9 C: not divisible by 3
a) 8,883, b) 504 c) 10,011 d) 986 e) 31,986 f) 1737
g) 989 h) 3078 i) 80,838 j) 176,367

2. State whether each statement is true or false. (If the answer is false, give counter example.
a) If a number is divisible by 10, then it is also divisible by 5
b) If a number is divisible by 5, then it is also divisible 10
c) If a number is divisible by 15, it is also divisible by 5
d) If a number is divisible by 4 and 6, then it is also divisible by 24

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UNIT TWO

Solution of Equation

- **Equations involving Radicals and exponents.**

When the variable in an equation occurs is a *square root, cube root* and so on the equation is called **a radical equation**

- **Solving radical equation**

Illustrative Example

1. Find the real solution set of the equation:

a) $\sqrt[3]{4x+5}=2$ b) $\sqrt{3x+4}=3$ c) $(3x-5)^{\frac{1}{4}}=2$

Solution: Remember that $\sqrt[n]{a}=b \Leftrightarrow b^n=a$

a) $\sqrt[3]{4x+5}=2 \Rightarrow 4x+5=2^3=8 \Leftrightarrow 4x+5-5=8-5$

$\Rightarrow 4x=3 \Rightarrow x=\frac{3}{4}$

b) $\sqrt{3x+4}=3 \Leftrightarrow 3x+4=3^2 \Rightarrow 3x=9-4=5 \quad \therefore x=\frac{5}{3}$

c) $(3x-5)^{1/4}=2 \Leftrightarrow 3x-5=2^4=16 \Rightarrow 3x=21 \Rightarrow x=\frac{21}{3}=7$

Exponential equation. When the variable in an equation occurs in a exponent is called **exponential equation.**

To solve the exponential equation use the following rule. For a>0 i) $a^x=a^y$ if and only if x = y ii) $a^x \cdot a^y=a^{x+y}$ iii) $(a^m)^n=a^{mn}$ iv) $a^x \cdot b^x=(ab)^x$

Illustrative Example

2. Solve each of the following equations

a) $16^x=2^{8x+5}$ b) $(\sqrt{15})^x 5^{2x-1}=0.2$ c) $3^{2x}=27\left(\frac{1}{3}\right)^{5x+9}$

d) $\left(\frac{9}{25}\right)^{x-1}\left(\frac{125}{27}\right)^x=\left(\frac{3}{5}\right)^{x+2}$ e) $(5^x)^3 \cdot 8^x=10\sqrt{10}$

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f) $4^{2x}\cdot 8^{1-x}=16^{2x+1}$ g) $\sqrt[3]{64^{2x-1}}=4^x$ h) $\left(\sqrt[3]{6}\right)^x=\sqrt[5]{36}$

Solution: Express each sides with same base

a) $16^x=2^{8x+5}\Leftrightarrow 2^{4x}=2^{x+5}\Rightarrow 4x=8x+5$

$\Rightarrow 4x-8x=5\Rightarrow -4x=5$, therefore, $x=\frac{-5}{4}$

b) $\sqrt{5}^x\cdot 5^{2x-1}=0.2\Leftrightarrow 5^{\frac{1}{2}x}\cdot 5^{2x-1}=\frac{2}{10}=\frac{1}{5}$

$\Rightarrow 5^{\frac{1}{2}x+2x-1}=5^{-1}$ ⟵ use $a^x\cdot a^y=a^{x+y}$

$\Rightarrow \frac{5}{2}x-1=-1\Rightarrow \frac{5}{2}x=-1+1=0$ therefore, $x=0$

c) $3^{2x}=27\left(\frac{1}{3}\right)^{5x+9}\Leftrightarrow 3^{2x}=3^3\cdot 3^{-5x-9}\Rightarrow 3^{2x}=3^{-5x-9+3}$

$\Leftrightarrow 2x=-5x-6\Rightarrow 2x+5x=-6$

$\Rightarrow 7x=-6$, therefore, $x=\frac{-6}{7}$

d) $\left(\frac{9}{25}\right)^{x-1}\left(\frac{125}{27}\right)^{x}=\left(\frac{3}{5}\right)^{x+2}\Rightarrow\left(\frac{3}{5}\right)^{2x+2}\left(\frac{5}{3}\right)^{3x}=\left(\frac{3}{5}\right)^{x+2}$

$\Rightarrow\left(\frac{3}{5}\right)^{2x+2}\left(\frac{5}{3}\right)^{3x}=\left(\frac{3}{5}\right)^{x+2}$

$\Rightarrow\left(\frac{3}{5}\right)^{2x-2-3x}=\left(\frac{3}{5}\right)^{x+2}\Leftrightarrow -x-2=x+2$

$\Rightarrow -2x=4$, therefore, $x=-2$

e) $(5^x)^3\cdot 8^x=10\sqrt{10}\Leftrightarrow 5^{3x}\cdot 2^{3x}=10^1\cdot 10^{\frac{1}{2}}$

$\Leftrightarrow (5.2)^{3x}=10^{1+\frac{1}{2}}\Rightarrow 10^{3x}=10^{\frac{3}{2}}$

$\Leftrightarrow 3x=\frac{3}{2}$, therefore $x=\frac{1}{2}$

f) $4^{2x}\cdot 8^{1-x}=16^{2x+1}\Leftrightarrow 2^{4x}\cdot 2^{3-3x}=2^{8x+4}\Rightarrow 2^{4x+3-3x}=2^{8x+4}$

$\Leftrightarrow 4x-3x+3=8x+4\Rightarrow x+3=8x+4$

$\Rightarrow x-8x=4-3\Rightarrow -7x=1$, therefore, $x=-\frac{1}{7}$

g) $\sqrt[3]{64^{2x-1}}=4^x\Leftrightarrow 64^{\frac{2x-1}{3}}=4^x\Rightarrow (4^3)^{\frac{2x-1}{3}}=4^x$

$\Rightarrow 4^{2x-1}=4^x\Leftrightarrow 2x-1=x$

$\Rightarrow 2x-x=-1\Rightarrow x=-1$

h) $\left(\sqrt[3]{6}\right)^x=\sqrt[5]{36}\Leftrightarrow 6^{\frac{1}{3}x}=36^{\frac{1}{5}}\Rightarrow 6^{\frac{1}{3}x}=6^{\frac{2}{5}}\Rightarrow\frac{1}{3}x=\frac{2}{5}$ $x=\frac{6}{5}$

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3. The solution set of $25^{1-x} = \sqrt{125}$ . . . EGSECE 2005

Solution: $25^{1-x} = \sqrt{125} \Leftrightarrow 5^{2-2x} = 5^{\frac{2}{3}}$

$\Leftrightarrow 2 - 2x = \frac{3}{2} \Rightarrow 2 - \frac{3}{2} = 2x$, therefore $x = \frac{1}{4}$

Linear Equations in Two Variables

Remember that: $ax+b = c$ for $a \neq 0$ is linear equation in **one variable**,
Example: $2x - 5 = 7$, $3x+6 = 11$, ... etc are linear equation in one variabl
However, the equation of the form $ax+by = c$ for $a \neq 0$, $b \neq c$ and $c \in IR$ is called **a linear equations in two variable s**

Note, • The two variables are x and y, a and b are coefficient of variable.
• The pair of values of x and y which satisfies the given equation is called **solutions** of the equation.

Example: $2x+3y = 12$ is linear equation in two variables.

- $2x+3y = 12$ is reduced to $3y = 12 - 2x$

  $\Rightarrow y = \frac{-2}{3}x + \frac{12}{3} = \frac{-2}{3}x + 4$

  When $x = 3 \Rightarrow y = \frac{-2}{3}(-3) + 4 = 2 \Rightarrow (3,2)$

  When $x = -3 \Rightarrow y = \frac{-2}{3}(-3) + 4 = 6 \Rightarrow (-3,2)$

  When $x = 0 \Rightarrow y = \frac{-2}{3}(0) + 4 = 4 \Rightarrow (0,4)$

  When $x = 9 \Rightarrow y = \frac{-2}{3}(9) + 4 = -2 \Rightarrow (9,-2)$

  ⋮ ⋮ ⋮

This shows that unlimited pair of x and y which will satisfies the equatio
Therefore, the set $\{(3,2), (-3,6), (0,4), (9,-2)\}$ is some of the solutions the equation $2x+3y = 12$.

Illustrative Example

4. Find the value of k if the indicated ordered pairs are solution of the equation
   a) $(2,k)$: $4x+5y = -7$
   b) $(2,k)$: $4x+5y = k$
   c) $(k, k-1)$: $3x+4y = 10$
   d) $(3,-2)$: $2x+5y = 3k$

Solution:

a) $(2,k)$ is solution of $4x+5y = -7$, so that
   $4(2) + 5k = -7 \Rightarrow 5k = -7 - 8 = -15 \Rightarrow k = -3$
b) $(2,k)$ is solution of $4x+5y = k$, then $4(2)+5k = k$

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$\Rightarrow 8+5k = k \Rightarrow k - 5k = 8$, therefore, $k = \frac{-8}{4} = -2$

c) $(k,k-1)$ is solution of $3x+4y = 10$

$\Rightarrow 3x+4(k-1) = 10 \Leftrightarrow 3k+4k - 4 = 10$

$\Rightarrow 7k = 10 + 4 = 14$, therefore, $k = \frac{14}{7} = 2$

d) $(3, -2)$ is solution of $2x+5y = 3k - 2$

$\Rightarrow 2(3) + 5(-2) = 3k - 2 \Leftrightarrow 3k-2 = 6 - 10$

$\Rightarrow 3k - 2 = -4 \Rightarrow 3k = -4+2 = -2 \quad \Rightarrow k = \frac{-2}{3}$

System of linear equation in two variables

System means group or set of two or more things

- A group or set of two or more linear equation is called **system of linear equation.**

Example: $\begin{cases} 2x + 3y = 4 \\ 3x - 2y = -7 \end{cases}$ is system of two linear equation in two variables

- *Solution to a system of two linear equation in two variables the ordered pair (x,y) that satisfies both equation of the system is called solution set.*

Example: $\begin{cases} 4x - y = 3 \\ 3x - y = 1 \end{cases}$, the ordered pair (2,5) is the solution of the system, since satisfies both equation: $4(2) - 5 = 8-5=3$ and $3(2)-5 = 1$

The methods of solving systems of linear equation

i) substitution method ii) elimination method iii) Graphic method

i) substitution method – is one way to solve a linear system of equation.

- In the substitution method, start by solving one of the equation for y
- Then substitute that expression in to the other equation and simplify.

Illustrative Example

5. Solve each of the following system of equation by substitution

a) $\begin{cases} 2x + y = 4 \\ 3x + 5y = -1 \end{cases}$ b) $\begin{cases} 2x - 3y = 8 \\ 4x + 5y = -6 \end{cases}$ c) $\begin{cases} 3x - 4y = 7 \\ 4x + 3y = 1 \end{cases}$

d) $\begin{cases} 3x - 2y = 5 \\ 6x - 4y = -2 \end{cases}$ e) $\begin{cases} 8x - 4y = 16 \\ 4x - 2y = 8 \end{cases}$ f) $\begin{cases} 2x + 7y = -6 \\ x + \frac{1}{2}y = -3 \end{cases}$

Solution: First solve for y, then put in 2nd equation:

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a) $\begin{cases} 2x + y = 4 \\ 3x + 5y = -1 \end{cases}$, Now, from $2x+y = 4 \Rightarrow y = 4 - 2x$

put $y = 4 - 2x$ in $3x + 5y = -1 \Leftrightarrow 3x+5(4 - 2x) = -1$

$\Rightarrow 3x+20 - 10x = -1 \Rightarrow -7x = -1 - 20 = -21$

$\Rightarrow -7x = -21 \Rightarrow x = \frac{21}{7} = 3$ and $y = 4 - 2(3) = -2$

Therefore, the solution is $(3, -2)$

b) $\begin{cases} 2x - 3y = 8 \\ 4x + 5y = -6 \end{cases}$, first solve for y, from $2x - 3y = 8$

$\Rightarrow -3y = 8 - 2x \Leftrightarrow y = \frac{-8}{3} + \frac{2}{3}x.$

Put $y = \frac{-8}{3} + \frac{2}{3}x$ in $4x+5y = -6$

$\Rightarrow 4x+5\left(\frac{-8}{3} + \frac{2}{3}x\right) = -6 \Leftrightarrow 4x+\frac{-40}{3} + \frac{10x}{3} = -6$

$\Leftrightarrow 4x+\frac{10}{3}x = -6 + \frac{40}{3} \Rightarrow \frac{22x}{3} = \frac{-18+40}{3}$

$\Rightarrow \frac{22x}{3} = \frac{-18+40}{3} \Rightarrow x = 1$

To find for y, $y = \frac{-8}{3} + \frac{2}{3}(1) = \frac{-6}{3} = -2$ Therefore, the solution is $(1,-2)$

c) $\begin{cases} 3x - 4y = 7 \\ 4x + 3y = 1 \end{cases}$, solving for y from $3x - 4y = 2$

$\Rightarrow -4y = 7 - 3x \Leftrightarrow \frac{3}{4}x - \frac{7}{4}$

To find for x, put $\frac{3}{4}x - \frac{7}{4}$ *in* $4x + 3y = 1$

$\Rightarrow 4x+3\left(\frac{3}{4}x - \frac{7}{4}\right) = 1 \Leftrightarrow 4x+\frac{9x}{4} - \frac{21}{4} = 1$

$\Rightarrow 4x + \frac{9x}{4} = 1+\frac{21}{4} \Leftrightarrow \frac{25x}{4} = \frac{25}{4}$

$\Rightarrow x = \left(\frac{25}{4}\right)\left(\frac{4}{25}\right) = 1$

- **To find for y**, $y = \frac{3}{4}(1) - \frac{7}{4} = -\frac{4}{4} = -1$ Thereofre, the solution is $(1,-1)$

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d) $\begin{cases} 3x - 2y = 5 \\ 6x - 4y = -2 \end{cases}$, solving for y from $3x - 2y = 5$

$\Rightarrow -2y = -3x+5 \Leftrightarrow y = \frac{3}{2}x - \frac{5}{2}$

To find x, put $y = \frac{3}{2}x - \frac{5}{2}$ in $6x - 4y = -2$

$\Rightarrow 6x - 4\left(\frac{3}{2}x - \frac{5}{2}\right) = -2 \Leftrightarrow 6x - 6x + 10 = -2$

$\Rightarrow 0 + 10 = -2 \Rightarrow 10 = -2$ which is always False

Therefore, the system has no solution.

e) $\begin{cases} 8x - 4y = 16 \\ 4x - 2y = 8 \end{cases}$, solving for y, from $8x - 4y = 16$

$\Rightarrow -4y = -8x + 16 \Leftrightarrow y = 2x - 4$

To find for x, put $y = 2x - 4$ in $4x - 2y = 8$

$\Rightarrow 4x - 2(2x - 4) = 8 \Leftrightarrow 4x - 4x + 8 = 8$

$\Rightarrow 8 = 8$ which is always True

Therefore, the system **has infinite solution**:

The solution of the system is $\{(x,y): 4x - 2y = 8\}$ or $\{(x,y): 2x-y=4\}$ or $\{(x.y): y = 2x - 4\}$

f) $\begin{cases} 2x + 7y = -6 \\ x + \frac{1}{2}y = -3 \end{cases}$, solving for y, from $2x + 7y = -6$

$\Rightarrow 7y = -2x - 6 \Leftrightarrow y = \frac{-2}{7}x - \frac{6}{7}$

To find x, put $y = \frac{-2}{7}x - \frac{6}{7}$ in $x + \frac{1}{2}y = -3$

$\Rightarrow x + \frac{1}{2}\left(\frac{-2}{7}x - \frac{6}{7}\right) = -3 \Leftrightarrow x - \frac{x}{7} - \frac{3}{7} = -3$

$\Leftrightarrow \frac{7x-x}{7} = -3 + \frac{3}{7} \Rightarrow \frac{6x}{7} = \frac{-18}{7} \Rightarrow x = \left(\frac{-18}{7}\right)\left(\frac{7}{6}\right) = -3$

To find for y, put $x = -3$, $y = \frac{-2}{7}(-3) - \frac{6}{7} = \frac{6}{7} - \frac{6}{7} = 0$

Therefore, the solution $(-3, 0)$

ii) Elimination method

- The elimination method is another method of solving a linear system of equations.
- In the elimination method, start by making the coefficient of the **x – term** or **y – terms** equal but opposite sign
- Then subtract the two equations to get an equation in only one variable.

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Example a) $\begin{cases} 3x+5y=-4 \\ x-4y=-7 \end{cases}$, now the coefficient of x to make equal and opposite, multiply 2nd row equation by (–3), and add together

$$-3\begin{cases} 3x+5y=-4 \\ x-4y=-7 \end{cases} \Rightarrow \frac{\begin{cases} 3x+5y=-4 \\ -3x+12y=21 \end{cases}}{0+17y=17} \Rightarrow y=1$$

To find for x, put y = 1 in 3x+5y = –4

$\Rightarrow 3x+5(1)=-4 \Leftrightarrow 3x=-4-5=-9$

$\Rightarrow 3x=-9 \Rightarrow x=-3$

Therefore, the solution (–3,1)

Illustrative Example

6. Solve each system of equations by the elimination method.

a) $\begin{cases} 3x-y=10 \\ 4x+3y=-4 \end{cases}$ b) $\begin{cases} 3x+6y=11 \\ 2x+4y=9 \end{cases}$

c) $\begin{cases} 4x+5y=2 \\ 8x-15y=9 \end{cases}$ d) $\begin{cases} 4x-2y=9 \\ 2x-y=3 \end{cases}$

Solution: Here, first make the coefficient

a) $\begin{cases} 3x-y=10 \\ 4x+3y=-4 \end{cases}$

To find for x, put y = –4 in 3x – y = 10

$\Rightarrow 3x-(-4)=10 \Leftrightarrow 3x=10-4=6 \Rightarrow x=2$

$\therefore$ The solution (2,-4)

b) $\begin{cases} 3x+6y=11 \\ 2x+4y=9 \end{cases} \Rightarrow \begin{matrix} 2 \\ -3 \end{matrix}\begin{cases} 3x+5y=11 \\ 2x+4y=9 \end{cases} \Rightarrow \frac{\begin{cases} 6x+12y=22 \\ -6x-12y=-27 \end{cases}}{0+0=-5 \Rightarrow 0=5}$ which i

always false

Therefore, the system has no solution

c) $\begin{cases} 4x+5y=2 \\ 8x-15y=9 \end{cases} \Rightarrow -22\begin{cases} 4x+5y=2 \\ 8x+15y=9 \end{cases} \Rightarrow \frac{+\begin{cases} -8x-10y=-4 \\ 8x-15y=9 \end{cases}}{0-25y=5}$

$$\therefore y=\frac{-5}{25}=\frac{-1}{5}$$

To find x, put $y=\frac{-1}{5}$ in 4x + 5y = 2

$$\Rightarrow 4x+5\left(\frac{-1}{5}\right)=2 \Leftrightarrow 4x-1=2$$

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$\Rightarrow 4x = 3 \Rightarrow x = \frac{3}{4}$

$\therefore$ The solution $\left(\frac{3}{4}, \frac{-1}{5}\right)$

d) Exercise left for you.

Solving system of non – linear system of equation

Illustrative Example

7. a) $\begin{cases} \frac{2}{x} + \frac{3}{y} = -2 \\ \frac{4}{x} - \frac{5}{y} = 1 \end{cases}$ b) $\begin{cases} \frac{3}{x} + \frac{4}{y} = \frac{5}{2} \\ \frac{5}{x} - \frac{3}{y} = \frac{7}{4} \end{cases}$

Solution: Let $u = \frac{1}{x}$ and $v = \frac{1}{y}$, Then

$$\Rightarrow \begin{cases} 2u + 3v = -2 \\ 4u - 5v = 1 \end{cases} \Rightarrow -2 \begin{cases} 2u + 3v = -2 \\ 4u - 5v = 1 \end{cases}$$

$$\Rightarrow \begin{array}{l} \begin{cases} -4u - 6v = 4 \\ 4u - 5v = 1 \end{cases} \\ \hline 0 - 11v = 5 \end{array} \quad + \quad \Rightarrow v = \frac{-5}{11} \Rightarrow y = \frac{-11}{5}$$

To find for u, put $v = \frac{-5}{11}$ in $2u + 3v = -2$

$\Rightarrow 2u + 3\left(\frac{-5}{11}\right) = -2 \Rightarrow 2u = -2 + \frac{15}{11} = \frac{-7}{11}$

$\Rightarrow u = \left(\frac{-7}{11}\right)\left(\frac{1}{2}\right) = \frac{-7}{22} = \frac{1}{x} \Rightarrow x = \frac{-22}{7}$

$\therefore$ The solution is $(x,y) = \left(\frac{-22}{7}, \frac{-11}{5}\right)$

b) $\begin{cases} \frac{3}{x} + \frac{4}{y} = \frac{5}{2} \\ \frac{5}{x} - \frac{3}{y} = \frac{7}{4} \end{cases} \Rightarrow \begin{array}{l} 5 \begin{cases} \frac{3}{x} + \frac{4}{y} = \frac{5}{2} \\ \frac{5}{x} - \frac{3}{y} = \frac{7}{4} \end{cases} \Rightarrow \begin{cases} \frac{15}{x} + \frac{20}{y} = \frac{25}{2} \\ \frac{-15}{x} + \frac{9}{y} = \frac{-21}{4} \end{cases} \\ -3 \end{array}$

$$0 + \frac{20+9}{y} = \frac{25}{2} - \frac{21}{4}$$

$\Rightarrow \frac{29}{y} = \frac{50}{4} - \frac{21}{4} = \frac{29}{4} \Rightarrow = \frac{1}{y} = \frac{1}{4} \Rightarrow y = 4$

To find for x, put $y = 4$ in $\frac{3}{x} + \frac{4}{y} = \frac{5}{2}$

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b) $\frac{15}{-10} = \frac{3}{k} \Rightarrow 15k = -30 \Rightarrow k = \frac{-30}{15} = -2$

c) $\frac{k}{9} = \frac{4}{k} \Rightarrow k^2 = 36 \Rightarrow k = \pm\sqrt{36} = \pm 6$

15. Which one of the following systems of linear equations has infinitely many solution . . . EGSECE 2004.

a) $\begin{cases} 3x + y = -1 \\ -x - \frac{1}{3}y = \frac{1}{3} \end{cases}$ b) $\begin{cases} x + y = 6 \\ 5x - 5y = 0 \end{cases}$

c) $\begin{cases} x + 3y = 2 \\ 2x - 4y = 1 \end{cases}$ d) $\begin{cases} x - 2y = 5 \\ -3x + 6y = 1 \end{cases}$

Solution: Use ratio test, if the system has infinite many solution, all the ratio equal

a) $\frac{3}{-1} = \frac{\frac{1}{-1}}{3} = \frac{\frac{1}{-1}}{3} \Rightarrow -3 = -3 = -3,$ **∴ Answer A**

Explanation: $\begin{cases} x + 3y = 2 \\ 2x - 4y = 1 \end{cases}$ and $\begin{cases} x + y = 6 \\ 5x - 5y = 0 \end{cases}$ have one solution

However, $\begin{cases} x - 2y = 5 \\ -3x + 6y = 1 \end{cases}$ has no solution.

16. If $\begin{cases} x + ky = 15 \\ 2x + y = 9 \end{cases}$ the system has no solution.

Then the value of k is . . . EGCESE

a) 2 b) –2 c) $\frac{1}{2}$ d) $\frac{-1}{2}$

Solution: The system has no solution if $\frac{1}{2} = \frac{k}{1} \Rightarrow k = \frac{1}{2},$ **∴ Answer C**

17. If the system $\begin{cases} 3x + ay = 7 \\ 2x + 5y = b \end{cases}$ has one solution then the value of a and b is equal to . . . EGSECE

a) $a \neq \frac{15}{2}$, $b \in R$ c) $a = \frac{15}{2}$ and $b = 7$ only

b) $a = \frac{15}{2}$ and $b \in R$ d) $a \neq \frac{15}{2}$ and $b = 7$ only

Solution: Here, ratio: $\frac{3}{2} = \frac{a}{5}$, $b \in R$

$\Leftrightarrow 2a \neq 15 \Rightarrow a \neq \frac{15}{2}$ and $b \in R,$ **∴ Answer A**

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18. The equation $3x + 5y = 2$ has no solution to which one of the following equation?

a) $x + \frac{5}{3}y = \frac{2}{3}$ b) $6x + 10y = 6$ c) $3x - 5y = -2$ d) $1.5x + 2.5y = 1$

Solution: $3x + 5y = 2$ to equation:

a) $x + \frac{5}{3}y = \frac{2}{3}$ has ratio: $3 = 3 = 3 \Rightarrow$ infinite solution.

b) $6x + 10y = 6$, ratio: $\frac{3}{6} = \frac{5}{10} \neq \frac{2}{6} \Rightarrow \frac{1}{2} = \frac{1}{2} \neq \frac{1}{3}$

$\therefore$ The system has no solution.

c) $3x - 5y = -2$, ratio: $\frac{3}{3} \neq \frac{-5}{5} \Rightarrow 1 \neq -1$

$\therefore$ The system has one solution

d) $1.5x + 2.5y = 1$, ratio $\frac{3}{1.5} = \frac{5}{2.5} = 2 \Rightarrow 2 = 2 = 2$

$\therefore$ The system has infinite many solution

18a) The value of k for which, the equation $\begin{cases} \frac{x-1}{2} = \frac{2y+4}{5} \\ \frac{x+6}{4} = \frac{y-2}{k} \end{cases}$ are no solution is: a) 10 b) 5 c) 2 d) –5

We have. $\frac{x-1}{2} = \frac{2y+4}{5} \Leftrightarrow 5x - 5 = 4y + 8 \Rightarrow 5x - 4y = 13$

and $\frac{x+6}{4} = \frac{y-2}{k} \Leftrightarrow kx + 6k = 4y - 8 \Rightarrow kx - 4y = -8 - 6k$

$\therefore \frac{5}{k} = \frac{-4}{-4} \Rightarrow \frac{5}{k} = 1 \Rightarrow k = 5$ **$\therefore$ Answer B**

19. Find the value of b and c, if

a) The graph of $y=2x^2+bx+c$ passing through (–2,–3) and (1,15)

b) The graph of $y=x^2+bx+$ passing through (–1, –9) and (–4,–5)

Solution: a) since (–2, –3) and (1,15) on the graph so that:

$\Rightarrow$ (–2, –3): $-3 = 2(-2)^2 + b(-2) + c \Leftrightarrow 8 - 2b + c = -3 \ldots$ (1)

$\Rightarrow$ (1,15): $15 = 2(1)^2 + b(1) + c \Leftrightarrow b + c + 2 = 15 \ldots\ldots$ (2)

From (1) and (2) we get

$$\begin{cases} -2b + c = -11 \\ b + c = 13 \end{cases} \Rightarrow \begin{array}{l} \begin{cases} -2b+c=-11 \\ +\ \begin{cases} -b-c=-13 \end{cases} \end{cases} \\ \hline -3b+0=-24 \Rightarrow b=8 \end{array}$$

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$\Rightarrow yx = 10y + x = 10y + 2y = 12y$

$\Rightarrow xy - yx = 21y - 12y = 27 \Rightarrow y = \frac{27}{9} = 3$

$\therefore x = 6$, thus, the original number will be 63

26. The sum of the digits of a certain two digit number is 13. Twice the tens digit exceeds the unit digit by 2. Find the number

Solution: $\begin{cases} x+y=13 \\ 2x-y=2 \end{cases}$

$3x=15 \Rightarrow x = 5 \text{ and } y = 8$

$\therefore$ The number 58

27. A rectangle is 5cm longer than it is wide. If the perimeter is 42cm, what are the length and width

Solution: Let W = the width, and ℓ = the length

$\Rightarrow \ell = W + 5$ and perimeter: $P = 2\ell + 2w$

$\Rightarrow 2(w + 5) + 2w = 42 \Leftrightarrow 2x + 2w = 42 - 10$

$\Rightarrow 4w = 32 \Rightarrow W = 8, \ell = 13$

28. Hana had invested 30,000 Birr in two bonds which pay 7% and 9% simple interest. The annual interest is 2,460 Birr
Find the amount invested at each rate.

Solution: Let x = amount invested at 7% $\Rightarrow 7\%x = 0.07x$

Let y = amount invested at 9%. $\Rightarrow 9\%y = 0.09y$

$\Rightarrow \begin{cases} x + y = 30{,}000 \\ 0.07x + 0.09y = 2{,}460 \end{cases}$

From $x + y = 30{,}000 \Rightarrow x = 30{,}000 - y$

$\Rightarrow 2100 - 0.07y + 0.09y = 2{,}460$

$\Rightarrow 0.02y - 2{,}460 - 2100 = 360$

$\Rightarrow y = \frac{360}{0.02} = 18{,}000, x = 12{,}000$

29. One evening, 1500 movie tickets were sold at a theater. Tickets cost Birr for adult and 5 Birr for students. Total receipts were 9500 Birr How many adult tickets were sold.

Solution: Let x = number of adult tickets.

Let y = number of student tickets.

$\Rightarrow \begin{cases} x + y = 1500 \\ 7x + 5y = 9500 \end{cases} \Rightarrow \begin{cases} 5x+5y=7500 \\ 7x+5y=9500 \end{cases}$

$-2x+0=-2000$

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a) $|x+4|+|x|+5x$ if $x<-4$ b) $|2x|+5x+3$ if $x<0$

Solution: a) if $|x+4|=-x-4$ if $x<-4$ and $|x|=-x$ when $x<0$

$\therefore |x+4|+|x|+5x=-x-4-x+5x=3x-4$ if $x<-4$

b) Exercise left for you.

Illustrative Example

36. Solve each of the follow equation:

a) $|x|=\pi$ b) $|x+5|=2-\sqrt{3}$ c) $|x|=-3$ d) $|x-2|=5$

e) $|3-5x|=6$ f) $|\sqrt[3]{x}-1|=9$ g) $\sqrt[5]{|x+1|}=2$ h) $\sqrt[4]{|x|}=\frac{1}{2}$

i) $|x-1|=\sqrt{2}-1$ j) $|x^3-1|=7$ k) $|x^2-4|=5$ l) $|2x^3|=54$

Solution: Recall that $|x|=k \Leftrightarrow x=k$ or $x=-k$

a) $|x|=\pi \Leftrightarrow x=\pi$ or $x=-\pi$, $\therefore$ S.S $=\{\pi,-\pi\}$

b) $|x+5|=2-\sqrt{3} \Leftrightarrow x+5=2-\sqrt{3}$ or $x+5=\sqrt{3}-2$

$\Rightarrow x=2-\sqrt{3}-5$ or $x=\sqrt{3}-2-5$ $\therefore$ S.S $=\{-3,\sqrt{3},\sqrt{3},-7\}$

c) $|x|=-3$ has no solution, since $|x|\geq 0$ ($|x|$ is always non-negative)

d) $|x-2|=5$ means $x-2=5$ or $x-2=-5$

$\Rightarrow x=7$ or $x=-5+2=-3$ $\therefore$ S.S $=\{-3,7\}$

e) $|3-5x|=6$ means $3-5x=6$ or $3-5x=-6$

$\Rightarrow -5x=6-3$ or $-5x=-6-3$

$\Rightarrow x=\frac{-3}{5}$ or $x=\frac{9}{5}$ $\therefore$ S.S $=\left\{\frac{-3}{5},\frac{9}{5}\right\}$

f) $|\sqrt[3]{x}-1|=9$ means $\sqrt[3]{x}-1=9$ or $\sqrt[3]{x}-1=-9$

$\Rightarrow \sqrt[3]{x}=10$ or $\sqrt[3]{x}=-8$

$\Rightarrow x^{\frac{1}{3}}=10$ or $x^{\frac{1}{3}}=-8$

$\Rightarrow x=10^3=1000$ or $x=(-8)^3=-512$ $\therefore$ S.S $=\{-512,1,000\}$

g) $|\sqrt[5]{x+1}|=2$ means $\sqrt[5]{x+1}=2$ or $\sqrt[5]{x+1}=-2$

$\Rightarrow (x+1)^{1/5}=2$ or $(x+1)^{1/5}=-2$

$\Leftrightarrow x+1=2^5$ or $x+1=(-2)^5=-32$

$\Rightarrow x=32-1=31$ or $x=-32-1=-33$ $\therefore$ S.S $=\{-33,11\}$

h) $|\sqrt[4]{x}|=\frac{1}{2}$ means $\sqrt[4]{x}=\frac{1}{2}$ or $\sqrt[4]{x}=-\frac{1}{2}$

$\Rightarrow x=\left(\frac{1}{2}\right)^4$ or $x=\left(-\frac{1}{2}\right)^4$ $\therefore x=\frac{1}{16}$ or $x=\frac{1}{16}$ $\therefore$ S.S $=\left\{\frac{1}{16}\right\}$

i) $|x-1|=\sqrt{2}-1$ means $x-1=\sqrt{2}-1$ or $x-1=-(\sqrt{2}-1)$

$\Rightarrow x=\sqrt{2}-1+1$ or $x=-\sqrt{2}-1+1=2-\sqrt{2}$

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Case II, if $x \in \left(-\infty, \frac{5}{2}\right)$, then $|2x-5| = -2x+5$

$\therefore |2x-5| = x+8 \Rightarrow -2x+5 = x+8$

$\Rightarrow -2x - x = 8-5$

$-3x = 3$

$x = -1 \in \left(-\infty, \frac{5}{2}\right)$,

$\therefore S.S = \{13, -1\}$. **Answer D**

42. The solution set of the equation $|x-1| = 1-x$

A) $\{x:x<1\}$ B) $\{x:x \geq 1\}$ C) $\{x:x>1\}$ D) $\{x:x \leq 1\}$

Solution: Recall that $|x-1| = \begin{cases} x-1 \text{ if } x-1 \geq 0, & x \geq 1 \\ -x+1 \text{ if } x-1 < 0, & x < 1 \end{cases}$

Case I, if $x \in [1, \infty)$, then $|x-1| = x-1$

$\therefore |x-1| = 1-x \Rightarrow x-1-x$

$\Rightarrow 2x = 2 \Rightarrow x = 1$

Case II, if $x \in (-\infty, 1)$, then $|x-1| = -x+1$

$\therefore |x-1| = 1-x \Rightarrow -x+1 = 1-x$

$\Rightarrow -x+x = 1-1 \Rightarrow 0 = 0$ (True)

$\therefore S.S = (-\infty, 1)$

Thus, the solution set is **case I ∪ case II**

$= \{1\} \cup (-\infty, 1) = (-\infty, 1]$, **∴ Answer D**

43. The solution set of $|x-\sqrt{2}| = \sqrt{2}-1$ is

A) $\{2\sqrt{2}-1, 1\}$ B) $\{2\sqrt{2}+1, -1\}$ C) $\{1-2\sqrt{2}, 1\}$ D) $\{2\sqrt{2}, 1\}$

Solution: $|x-\sqrt{2}| = \sqrt{2}-1$ means $x-\sqrt{2}+1$

$\Rightarrow x-\sqrt{2} = \sqrt{2}-1$ or $x = -\sqrt{2} = -\sqrt{2}+1$

$x = \sqrt{2}+\sqrt{2}-1$ or $x = -\sqrt{2}+\sqrt{2}+1 = 1$

$\therefore x = 2\sqrt{2}-1$, or $x = 1$, **∴ Answer A**

44. Solve each of the following

A) $|x-2|+|x-2| = 8$ B) $|x+3|+|x+3| = 4$

Solution: use, the notion $|k|+|k| = 2|k|$

a) $|x-2|+|x-2| = 8 \Rightarrow 2|x-2| = 8$

$\Rightarrow |x-2| = \frac{8}{2} = 4$

$|x-2| = 4$ means $x-2 = 4$ or $x-2 = -4$

$\Rightarrow x = 2+4 = 6$ or $x = -4+2 = -2$

$\therefore S.S = \{6, -2\}$

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b) $|x+3| + |x+3| = 4$ is the same as $2|x+3| = 4$

$\Rightarrow x = 2 - 3$ or $x = -2 - 3 = -5$

$\therefore$ S.S $= \{-1, -5\}$

45. Solve each of the following

a) $|5 - x| + |x - 3| = 3$ b) $|3x + 2| + |x - 3| = 5$

Solution: Here

a. $|5 - x| = \begin{cases} 5 - x, if\ 5 - x \geq 0, \text{ i.e. } x \leq 5 \\ -5 + x\ if\ 5 - x < 0, i.e.\ x > 5 \end{cases}$

$|x - 3| = \begin{cases} x - 3, if x \geq 3 \\ -x + 3\ if\ x < 3 \end{cases}$

	$-\infty$		3		5		∞
$\|5 - x\|$	$5 - x$		$5 - x$		$-5+ x$		
$\|x - 3\|$	$-x + 3$		$x - 3$		$x - 3$		

- To solve, we consider three cases.

Case I. assume $x \in (-\infty, 3)$

Then $|5 - x| + |x - 3| = 3 \Rightarrow 5 - x + 3 - x = 3$

$\Rightarrow -2x + 8 = 3 \Rightarrow -2x = -5$

$\therefore x = \frac{5}{2} \in (-\infty, 3)$

$\therefore$ S.S$_1$ is $\left\{x: x = \frac{5}{2}\right\}$

Case II. assume $x \in (3, 5)$

$\therefore |5 - x| + |x - 3| = 3$ implies $5 - x + x - 3 = 3$

$\Rightarrow 2 = 3$ is always false $\therefore$ S.S$_2$ is $\emptyset$

Case III. assume $x \in [5, \infty)$

Then $|5 - x| + |x - 3| = 3$ implies $x - 5 + x - 3 = 3$

$\Rightarrow 2x - 8 = 3 \Rightarrow 2x = 11$

$\therefore x = \frac{11}{2} \in [5, \infty)$ $\therefore$ S.S$_3$ is $\left\{x: x = \frac{11}{2}\right\}$

Thus, all the possible solution set is $\left\{\frac{5}{2}, \frac{11}{2}\right\}$

b. $|3x+2| = \begin{cases} 3x + 2\ if x \geq \frac{-2}{3} \\ -3x - 2\ if\ x < \frac{-2}{3} \end{cases}$ $|x - 3| = \begin{cases} x - 3\ if\ x \geq 3 \\ -x + 3\ if\ x < 3 \end{cases}$

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	$-\infty$	$\frac{-2}{3}$	3	∞
$\|3x+2\|$	$-3x-2$	$3x+2$	$3x+2$	
$\|x-3\|$	$-x+3$	$-x+3$	$x-3$	
$\|3x+2\|+\|x-3\|$	$-4x+1$	$2x+5$	$4x-1$	

- We need to consider three cases

Case I, When $x \in \left(-\infty, \frac{-2}{3}\right)$

$|3x+2| + |x-3| = -3x-2+-x+3 = 5$

$\Rightarrow -4x+1 = 5 \Rightarrow -4x = 4$

$\therefore x = -1 \in \left(-\infty, \frac{-2}{3}\right)$ $\therefore S.S_1$ is $\{x : x = -1\}$

Case II, When $x \in \left[\frac{-2}{3}, 3\right)$

$|3x+2| + |x-3| = 2x + 5 = 5$

$\Rightarrow 2x = 5 - 5 = 0$

$x = 0 \in \left[\frac{-2}{3}, 3\right)$ $\therefore S.S_2$ is $\{x : x = 0\}$

Case III, When $x \in [3, \infty)$

$|3x+2| + |x-3| = 3x+2 + x - 3 = 5$

$4x - 1 = 5 \Rightarrow 4x = 6$

$\therefore x = \frac{6}{4} = \frac{3}{2} \notin [3,\infty)$ $\therefore S.S_3 = \emptyset$

Therefore, all the possible solution set is $\{-1,0\}$

46. Solve each of the following

a) $|x+4| + |x-3| = 0$ b) $|x+2| + |x-8| = 0$

Solution: Remember that absolute value is always positive

a) Thus there is no real number x, that makes $|x+4| + |x-3| = 0$, therefore, $S.S = \emptyset$

b) No real number x, that makes $|x+2| + |x-8| = 0$ $\therefore S.S = \emptyset$

47. Solve each of the following

a) $|x-5| + |x-5| = 0$ b) $|x-3| - |x+4| = 0$

c) $|x+6| + |x+6| = 0$ d) $|2x+4| - |x-7| = 0$

Solution: a) $|x-5| + |x-5| = 0$ implies $2|x-5| = 0$

$|x-5| = 0$ if $x = 5$ $\therefore S.S_1$ is $\{x : x = 5\}$

b) $|x-3| - |x+4| = 0 \Rightarrow |x-3| = |x+4|$

implies, $x - 3 = \pm (x+4)$

$\Rightarrow x - 3 = -x - 4$ or $x - 3 = x + 4$

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In the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, the expression $b^2 - 4ac$ determines the existence of the roots (solution).

- Any quadratic equation $ax^2 + bx + c = 0$ for $a \neq 0$
  - i) If $b^2 - 4ac > 0$, then the equation *has two solution (roots)*
  - ii) If $b^2 - 4ac = 0$, then the equation *has only one solution.*
  - iii) If $b^2 - 4ac < 0$ then the equation *has no real solution.*
- The expression $b^2 - 4ac$ is called the **discriminator.**

58. Solve each of the following equations, by using quadratic formula, if there is a solution.

a) $x^2 - 3x + 1 = 0$ b) $x^2 + 4x - 2 = 0$
c) $6x^2 - 12x + 7 = 0$ d) $2x^2 + 3x - 5 = 0$
e) $9x^2 - 12x + 4 = 0$ f) $4x^2 - 12x + 10 = 0$

Solution: Determine the value of $b^2 - 4ac$

a) $x^2 - 3x + 1 = 0$, $a = 1$, $b = -3$ and $c = 1$

$\Rightarrow b^2 - 4ac = (-3)^2 - 4(1)(1) = 9 - 4 = 5 > 0$

Since $b^2 - 4ac > 0$,

Therefore, the equation $x^2 - 3x+1 = 0$ has **two solution.**

$$\Rightarrow x = \frac{-b \pm \sqrt{b^2-4ac}}{2a} = \frac{-(-3) \pm \sqrt{(-3)^2-4(1)(-2)}}{2(1)}$$

$$\therefore x = \frac{3+\sqrt{5}}{2} \text{ or } x = \frac{3-\sqrt{5}}{2}$$

b) $x^2 + 4x - 2 = 0$, $a = 1$, $b = 4$, $c = -2$

$\Rightarrow b^2 - 4ac = (4)^2 - 4(1)(2) = 16 = 8 = 24 > 0$

Therefore, the equation $x^2 + 4x - 2 = 0$ **has two solution.**

$$\Rightarrow \frac{-b \pm \sqrt{b^2-4ac}}{2a} = \frac{-4 \pm \sqrt{16-4(1)(-2)}}{2(1)}$$

$$\therefore x = \frac{-4 \pm \sqrt{24}}{2a} = -2 \pm \sqrt{6} \qquad \therefore x = -2 - \sqrt{6} \text{ or } x = -2 + \sqrt{6}$$

c) $6x^2 - 12x + 7 = 0$, $a = 6$ $b = -12$, $c = 7$

$\Rightarrow b^2 - 4ac = (-12)^2 - 4(6)(7) = 144 - 168 = -24 < 0$

$\therefore$ The equation $6x^2 - 12x + 7 = 0$ **has no real solution** because $b^2 - 4ac < 0$

d) $2x^2 + 3x - 5 = 0$, $a = 2$, $b = 3$ $c = -5$

$\Rightarrow b^2 - 4ac = (3)^2 = - 4(2)(-5) = 9 + 40 = 49 > 0$

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$$\therefore x = \frac{-\sqrt{2} \pm \sqrt{(\sqrt{26})^2 - 4(1)(-6)}}{2(1)} = \frac{-\sqrt{2} \pm \sqrt{26}}{2}, \quad \therefore \text{Answer A}$$

61. The solution set of $2x^2 + 4x + 1 = 0$ . . . .EGSECE 2004

A. $\left\{\frac{-\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right\}$ B. $\left\{-1\frac{\sqrt{2}}{2}, -1\frac{\sqrt{2}}{2}\right\}$ C. $\emptyset$ D. $\{-1+\sqrt{2}, -1-\sqrt{2}\}$

Solution: a = 2, b = 4, c = 1

$$\Rightarrow x = \frac{-4 \pm \sqrt{16 - 4(2)(1)}}{2(2)} = \frac{-4\sqrt{8}}{4} = \frac{-4 \pm 2\sqrt{2}}{4}$$

$$\therefore S.S = \left\{-1 - \frac{\sqrt{2}}{2}, -1 + \frac{\sqrt{2}}{2}\right\}, \qquad \therefore \textbf{Answer: B}$$

62. The solution set of $\frac{1}{x} - 3x = 2$ is

A. $\left\{-1, \frac{1}{3}\right\}$ B. $\left\{1, \frac{1}{2}\right\}$ C. $\{2, -3\}$ D. $\{-1, 3\}$

Solution: $\frac{1}{x} - 3x = 2 \Rightarrow x\left(\frac{1}{x} - 3x\right) = 2x,$

$\Rightarrow 1 - 3x^2 = 2x \longleftarrow$ multiplying both side by x

$\Rightarrow 3x^2 + 2x - 1 = 0$ a = 3, b = 2, c = −1

$$\Rightarrow x = \frac{-2 \pm \sqrt{4 - 4(3)(-1)}}{2(3)} = \frac{-2 \pm \sqrt{16}}{6} = \frac{-2 \pm 4}{6}$$

$$\therefore x = -1 \text{ or } x = \frac{-2+4}{6} = \frac{2}{6} = \frac{1}{3} \therefore S.S = \left\{-1, \frac{1}{3}\right\}, \quad \therefore \text{Answer A}$$

63. In each of the following, what expression k must added to in order to make perfect square

a) $x^2 + 5ax$ b) $2x^2 + 3x$ c) $x^2 - 4ax$

Solution: In order to make perfect square means making the expression to have exactly one solution, that is we set $b^2 - 4ac = 0$

a) Let $x^2 + 5ax + k = 0$ is perfect expression if

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$b^2 - 4ac = 0 \Rightarrow (5a)^2 - 4(1)k = 0$, a = 1, b = 5 a, c = k

$\Rightarrow 25a^2 - 4k = 0$ ⟵ solving for k

$\Rightarrow 4k = 25a^2 \Rightarrow k = \frac{25}{4}a^2$

Therefore the expression become $x^2 + 5ax + \frac{25}{4}a^2$

b) Let $2x^2 + 3x + k = 0$ is perfect expression if

$b^2 - 4ac = 0 \Rightarrow 3^2 - 4(2)k = 0 \Rightarrow 9 - 8k = 0$

Therefore, the expression become $2x^2 + 3x + \frac{9}{8}$

c) $a = 1, b = -4a, c = k \Rightarrow (-4a)^2 - 4(1)k = 0$

$\Rightarrow 4k = 16a^2 \Rightarrow k = 4a^2$

$\therefore x^2 - 4a + 4a2$ is perfect expression

Sum and products of solution (roots) of quadratic equation $ax^2 + bx + c = 0$

Note: The solution to a quadratic equation are called **Roots.**

Let r_1 and r_2 be roots of $ax^2 + bx + c = 0$

Let $r_1 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$ and $r_2 = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$, then

i) $r_1 + r_2 = \frac{-b}{a}$ ⟵ sum of the root (solution)

ii) $r_1 \cdot r_2 = \frac{c}{a}$ ⟵ product of the root

This relationship is called **Viet's theorem**

Illustrative Example

64. Without solving, find

i) sum of the root $(r_1 + r_2)$,

ii) product of the root. $(r_1 \cdot r_2)$ for each quadratic equation given below

a) $2x^2 + 6x - 3 = 0$

b) $x^2 - 7x + 6 = 0$

c) $x^2 - 7x + 6 = 0$

d) $-5x + 4x + 1 = 0$

Solution: Recall that: for $ax^2 + bx + c = 0$ has sum of root,

$r_1 + r_2 - \frac{b}{a}$, and product of root, $r_1 \cdot r_2 = \frac{c}{a}$

a) $2x^2 + 6x - 3 = 0$, a = 2, b = 6, c = –3, then

i) sum of its root: $r_1 + r_2 = \frac{-b}{a} = \frac{-6}{2} = -3$

ii) product of its root: $r_1 \cdot r_2 = \frac{c}{a} = \frac{-3}{2}$

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b) $x^2 - 7x + 6 = 0$, $a = 1$, $b = -7$, $c = 6$, then

i) sum of its root: $r_1 + r_2 = \frac{-b}{a} = \frac{-(-7)}{1} = 7$

ii) product of its root: $r_1 \cdot r_2 = \frac{c}{a} = \frac{6}{1} = 6$

c) $3x^2 + x + 5 = 0$, $a = 3$, $b = 1$, $c = 5$, then

i) sum of its root: $r_1 + r_2 = \frac{-b}{a} = \frac{-1}{3}$

ii) product of its root: $r_1 \cdot r_2 = \frac{c}{a} = \frac{5}{3}$

d) Exercise left for you.

65. In each of the following quadratic equation find the value of k if sum the root

a) $2x^2 - kx + 4 = 0$ is $\frac{-11}{4}$ b) $(k + 2)x^2 + 5x + 2k = 0$ is 3

Solution: Recall that sum of the root: $r_1 + r_2 = \frac{-b}{a}$

a) $a = 2$, $b = -k$, $c = 4$, then $r_1 + r_2 = \frac{-b}{a} = \frac{-(-k)}{2} = \frac{-11}{4}$

$\Rightarrow \frac{k}{2} = \frac{-11}{4} \Rightarrow k = \frac{-22}{4} = \frac{-11}{2}$

b) $a = k + 2$, $b = 5$, $c = 2k$, then sum of root $r_1 + r_2 = \frac{-b}{a} = \frac{-5}{k+2} = 3$

$\Rightarrow \frac{-5}{k+2} = 3 \Leftrightarrow 3k + 6 = -5 \Rightarrow 3k = -11$

$\therefore k = \frac{-11}{3}$

66. Find the value of k, if the product of the roots of the equation:

a) $(k + 2)x^2 + 5x + 2k = 0$ is $\frac{2}{3}$

b) $(k - 2)x^2 - 4x + k = 0$ is $\frac{-4}{5}$

Solution: Remember that, product of the root: $r_1 \cdot r_2 = \frac{c}{a}$

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a) $a = k + 2, b = 5, c = 2k$, then $r_1 \cdot r_2 = \frac{c}{a} = \frac{2k}{k+2} = \frac{2}{3}$

$\Rightarrow \frac{2k}{k+2} = \frac{2}{3} \Rightarrow 6k = 2k + 4 \Rightarrow 4k = 4$

$\therefore k = 1$

b) $a = k - 2, b = 4, c = k$ then $r_1 \cdot r_2 = \frac{c}{a} = \frac{k}{k-2} = \frac{-4}{5}$

$\Rightarrow \frac{k}{k-2} = \frac{-4}{5} \Rightarrow 5k = -4k + 8 \Rightarrow 5k + 4k = 8, \quad \therefore k = \frac{8}{9}$

67. Find the roots of the quadratic equation if

a) $(2k+2)x^2 + (4 - 4k)x + k - 2$ has roots which are reciprocal of each other

b) $Kx^2 - (2x + 2)x + 3k - 5 = 0$ has the sum of its root equal to twice the product of its root

c) $4x^2 - 8x + 2k - 1 = 0$ has one root equal to three times the other

Solution:

a) Let r and $\frac{1}{r}$ be roots, then

i) Product of root: (r) $\left(\frac{1}{r}\right) = \frac{c}{a} = \frac{k-2}{2k+2} = 1$

$\Rightarrow \frac{k-2}{2k+2} = 1 \longleftarrow$ since $r \cdot \frac{1}{r} = 1$

$\Rightarrow k - 2 = 2k + 2 \Rightarrow k - 2k = 2 + 2$

$\Rightarrow -k = 4, \qquad \therefore k = -4$

Therefore, the equation will be: $-6x^2 + 20x - 6 = 0$

i.e. $-3x^2 + 10x - 3 = 0$, and root $\frac{1}{3}$ and 3

b) $a = k, b = -(2k + 2), x = 3k - 5$ and $r_1 + r_2 = 2r_1 \cdot r_2$

$\Rightarrow \frac{-b}{a} = 2\left(\frac{a}{c}\right) \Leftrightarrow \frac{2k+2}{k} = 2\left(\frac{3k-5}{k}\right)$

$\Rightarrow \frac{k+1}{k} = \frac{3k-5}{k} \Rightarrow 3k - 5 = k + 1$

$\Rightarrow 3k - k = 1 + 5 = 2k = 6, k = 3$

$\therefore$ The equation will be: $3x^2 - 8x + 4 = 0$ without 2 and $\frac{2}{3}$

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c) $a = 4, b = -8, c = 2k - 1$ and $r_1 = 3r_2$, then $r_1 + r_2 =$

$$3r_2 + r_2 = \frac{-b}{a} = \frac{8}{4} = 2$$

$$\Rightarrow 4r_2 = 2 \Rightarrow r_2 = \frac{2}{4} = \frac{1}{2}, \quad S.O.r = 2 \quad 2\left(\frac{1}{2}\right) = \frac{3}{2}$$

Therefore, $r_1 \cdot r_2 = \frac{c}{a} = \frac{2k-1}{4}$

$$\Rightarrow \left(\frac{3}{2}\right)\left(\frac{1}{2}\right) = \frac{2k-1}{4} \Leftrightarrow 2x - 1 = 0 \quad \Rightarrow 2k = 4 \Rightarrow k = 2$$

$\therefore$ The equation will be $4x^2 - 8x + 3 = 0$ with root $\frac{1}{2}$ and $\frac{3}{2}$

68. If one of the roots of the equation $x^2 - 8x + k = 0$ exceeds the other by 4 then k is equal to . . . .

A. 12 B. 16 C. 6 D. 15

Solution: $a = 1, b = -8, c = k$ and $r_1 = r_2 + 4$

- sum of its root $r_1 + r_2 = \frac{-b}{a} = \frac{-(-8)}{1} = 8$

$$\Rightarrow r_2 + r_2 + 4 = 8 \Rightarrow 2\sqrt{2} = 8 - 4 = 4$$

$$\therefore r_2 = \frac{4}{2} = 2, \text{ thus } r_1 = r_2 + 4 = 2 + 4 = 6$$

Therefore, product of the root, $r_1 \cdot r_2 = \frac{c}{a} = k = (2)(6) = 12$ $\therefore$ Answer A

69. One of the roots of the equation $x^2 - 2kx + 12$ is 3.
What is the other root? . . . EGSECE

A. -4 B. 4 C. -3 D. 3

Solution: It is given that $r_1 = 3$ now we find r_2

$$\Rightarrow r_1 \cdot r_2 = \frac{c}{a} = \frac{12}{1} \longleftarrow \text{product rule}$$

$$3r_2 = 12 \Rightarrow r_2 = 4$$

$\therefore$ Answer B

Determining the equation from its root

For any quadratic equation $ax^2 + bx + c = 0$, $a \neq 0$ can be expressed in terms of its root r_1 and r_2 is given by:

$x_2 - (r_1 + r_2)x + r_1 \cdot r_2 = 0$ that is

$x^2 - (\text{sum of root})x + (\text{product of the root}) = 0$

Illustrative Example

70. Determine the quadratic equation, if two roots of the equation are

a) -2 and 3 b) $\frac{4}{3}$ and $\frac{-3}{5}$ c) $2-\sqrt{2}$ and $2+\sqrt{2}$

Solution: Remember that any quadratic equation is expressed by:

- $x^2 - (\text{sum of root})x + (\text{product of root}) = 0$

a) $x^2 - (-2+3)x + (-2)(3) = 0 \Rightarrow x^2 - x - 6 = 0$

b) when roots $\frac{4}{3}$ *and* $\frac{-3}{5}$

$$\Rightarrow x^2 - \left(\frac{4}{3} + \frac{-3}{5}\right)x + \left(\frac{4}{3}\cdot\left(\frac{-3}{5}\right)\right) = 0 \Rightarrow x^2 - \frac{11}{15}x - \frac{4}{5} = 0$$

$\Rightarrow 15x^2 - 11x - 12 = 0 \leftarrow$ multiplying by (15)

c) when roots $2-\sqrt{2}$ and $2+\sqrt{2}$

$\Rightarrow x^2 - (2-\sqrt{2}+2+\sqrt{2})x + (2-\sqrt{2})(2+\sqrt{2}) = 0$

$\Rightarrow x^2 - 4x + 2 = 0$

71. The roots of the equation $x^2 + mx - n = 0$ are -2 and 3, what are the values of m and n respectively. . . EGSECE

A. 6 and -1 B. -1 and 6 C. 1 and 6 D. -6 and 1

Solution: $x^2 + mx - n = x^2 - (-2+3)x + (-2)(3) = 0$

$= x^2 - x - 6 \quad \therefore m = -1$, and $n = 6$ **∴ Answer B**

72. If the solutions of the equation $2x^2 + (k-3)x - 7 = 0$ are $\frac{7}{2}$ and -1, then what is the value of k . . . EGSECE 2005/13

A. -2 B. $\frac{-34}{7}$ C. 8 D. 2

Solution: sum of the roots, $r_1 + r_2 = \frac{-b}{a} \quad \frac{-(k-3)}{2}$

$$\Rightarrow \frac{7}{2} + (-1) = \frac{-k+3}{2} \Rightarrow \frac{-k+3}{2} = \frac{5}{2}$$

$\therefore -k + 2 = 5 \Rightarrow k = -2$ **∴ Answer A**

73. Express the way how the solution of the equation $ax^2+bx + c = 0$ is related to $cx^2 + bx + a = 0$

Solution: The solution of $ax^2 + bx + c = 0$are the reciprocals of the solution of the equation $cx^2 + bx + a = 0$

Example: $2x^2 + 8x + 6 = 0$, and $6x^2 + 8x + x = 0$

The solution of $2x^2 + 8x + 6 = 0$are $2x^2 + 6x + 2x + 6 = 0$

$\Rightarrow 2x(x+3) + 2(x+3) = 0 \Rightarrow x = -1$, or $x = -3$

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But, the solution of $6x^2+8x+2=0$ are $6x^2+2x+6x+2=0 \Rightarrow 2x(3x+1)+2(3x+1)$

$\Rightarrow (2x+2)(3x+1)=0 \Rightarrow x=-1$ or $x=\dfrac{-1}{3}$

Application of Viet's theorem

Illustrative Example

74. If A and B are roots of $2x^2+4x-1=0$, then find:

a) $A+B$ b) AB c) A^2+B^2 d) $A-B$ e) A^2-B^2

f) $\dfrac{1}{A+1}+\dfrac{1}{B+1}$ g) $\dfrac{1}{A+3}-\dfrac{1}{B+3}$ h) $\dfrac{1}{A}+\dfrac{1}{B}$ i) A^3-B^3 j) A^3+B^3

Solution: a) $A+B=\dfrac{-4}{2}=-2 \longleftarrow$ Sum of the root

b) $AB=\dfrac{-1}{2} \longleftarrow$ product of the root

c) Recall that $(A+B)^2=A^2+2AB+B^2$, thus,

$\Rightarrow A^2+B^2=(A+B)^2-2AB \longleftarrow A+B=\dfrac{-4}{2}=-2$

$=(-2)^2-2\left(\dfrac{1}{2}\right) \longleftarrow AB=-\dfrac{1}{2}$

$\therefore A^2+B^2=4+1=5$

d) Remember that $(A-B)^2=A^2+B^2-2AB$

$\Rightarrow A-B=\pm\sqrt{A^2+B^2-2AB} \longleftarrow A^2+B^2=5$

$\therefore A-B=\pm\sqrt{5-2\left(-\frac{1}{2}\right)}=\pm\sqrt{6}$

e) $A^2-B^2=(A-B)(A+B)=(\pm\sqrt{6})(-2)=(\pm\sqrt{6})(-2)=\pm2\sqrt{6}$

f) $\dfrac{1}{A+1}+\dfrac{1}{B+1}=\dfrac{B+1+A+1}{(A+1)(B+1)}=\dfrac{A+b+2}{AB+A+B+1}=\dfrac{-2+2}{-\frac{1}{2}+(-2)+1}=0$

g) $\dfrac{1}{A+3}-\dfrac{1}{B+3}=\dfrac{B+3-(A+3)}{(A+3)(B+3)}=\dfrac{B-A}{AB+3A+3B+9}$

$=\dfrac{B-A}{AB+3(A+B)+9}=\dfrac{\pm\sqrt{6}}{-\frac{1}{2}+3(-2)+9}=\dfrac{\pm2\sqrt{6}}{5}$

h) $\dfrac{1}{A}+\dfrac{1}{B}=\dfrac{A+B}{AB}=\dfrac{-2}{-\frac{1}{2}}=4$

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i) $A^3 - B^3 = (A - B)(A^2 + B^2 + AB) = (\sqrt{6})(5 + \frac{1}{2}) = \frac{9\sqrt{6}}{2}$

75. If α and β are the roots of $3x^2 - 2x - 6 = 0$ and $\alpha > \beta$, which of the following is not true

A. $\alpha^2 + \beta^2 = \frac{40}{9}$ C. $\alpha^2 - \beta^2 = \frac{2\sqrt{19}}{9}$

B. $\alpha - \beta = \frac{2\sqrt{19}}{3}$ D. $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ are roots of $3x^2 - 2x + 3 = 0$

Solution: we have, $\alpha + \beta = \frac{-(-2}{3} = \frac{2}{3}$ $\alpha\beta = \frac{-6}{3} = -2$

A. $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = \left(\frac{2}{3}\right)^2 - \left(\frac{-6}{3}\right) = \frac{40}{9}$

B. $\alpha - \beta = \sqrt{\alpha^2 + \beta^2 - 2\alpha\beta} = \sqrt{\frac{40}{9} - 2(-2)} = \frac{2\sqrt{19}}{3}$

C. $\alpha^2 - \beta^2 = (\alpha - \beta)(\alpha + \beta) = \left(\frac{2\sqrt{19}}{9}\right)\left(\frac{2}{3}\right) = \frac{4\sqrt{19}}{9}$

D. If $\frac{1}{\alpha}$ and $\frac{1}{\beta}$ are the roots, then

$$x^2 - \left(\frac{1}{\alpha} + \frac{1}{\beta}\right)x + \frac{1}{\alpha} \cdot \frac{1}{\beta} = 0 \Rightarrow x^2 - \left(\frac{\beta + \alpha}{\alpha\beta}\right)x = \frac{1}{\alpha\beta} = 0$$

$$\Rightarrow x^2 - \left(-\frac{2}{3} \cdot \left(\frac{1}{2}\right)\right)x = \frac{1}{-2} = 0 = x^2 + \frac{1}{3}x - \frac{1}{2} = 0$$

$\Rightarrow 6x^2 + 2x - 3 = 0 \longleftarrow$ multiplying by (6) ∴ **Answer D**

76. If a and b are the roots of $x^2 + 3x + 2 = 0$, then find

a) $\frac{1}{a^2} + \frac{1}{b^2}$ b) $\frac{1}{a} - \frac{1}{b}$ c) $\frac{1}{a^2} - \frac{1}{b^2}$

Solution: We have $a + b = -3$ and $ab = 2$

a) $\frac{1}{a^2} + \frac{1}{b^2} = \frac{a^2 + b^2}{(ab)^2} = \frac{(ab)^2 - 2ab}{(ab)^2} = \frac{(-3)^2 - 2(2)}{(2)^2} = \frac{9-4}{4} = \frac{5}{4}$

b) $\frac{1}{a} - \frac{1}{b} \longleftarrow$ exercise left for you

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c) $\frac{1}{a^2}-\frac{1}{b^2}$ = Exercise left for you.

77. a) If r_1 and r_2 be the roots of $7x^2+6x+c=0$ If $r_1=2r_2$ then find
a) r_2 and r_1, b) c c) $r_1^2+r_2^2$, d) $r_1^2-r_2^2$,

Solution: $r_1+r_2=\frac{-6}{7}$. $r_1\cdot r_2=\frac{c}{7}$, thus,

a) $r_1=2r_2 \Rightarrow r_1+r_2=2r_2+r_2=\frac{-6}{7}\Rightarrow 3r_2=\frac{-6}{7}, r_2=\frac{-2}{7}$

$\therefore r_1=2r_2=2\left(\frac{-2}{7}\right)=\frac{-4}{7}$, Therefore , $r_1=\frac{-4}{7}$ are $r_2=\frac{-2}{7}$

b) $r_1\cdot r_2=\frac{c}{7}\Rightarrow c=7\ r_1\cdot r_2=7\left(\frac{-4}{7}\cdot\frac{-2}{7}\right)=\frac{8}{7}$

c) Exercise left for you d) Exercise left for you

77b) Find the value of k, if the equation have the indicated solution.

a) $2x^2+(k^2-8)x+7=0$, has root $\frac{7}{2}$ and -1

b) $(k+4)x^2+5x+2=0$, has root $\frac{-3}{2}$ and -1

c) $4x^2+11x+k^2+2k=0$, has root $\frac{-3}{2}$ and -2

Solution: a) sum of roots: $r_1+r_2=\frac{-b}{a}=\frac{-(k^2-8)}{2}$

$\Rightarrow\frac{7}{2}+(-1)=\frac{8-k^2}{2}\Rightarrow\frac{8-k^2}{2}=\frac{5}{2}\Rightarrow 8-k^2=5\Rightarrow -k^2=5-8$

$\Rightarrow k^2=3\Rightarrow k=\pm\sqrt{3}$

b) $r_1+r_2=\frac{-b}{a}=\frac{-5}{k+4}$ ⟵ sum of roots

$\Rightarrow\frac{-5}{k+4}=\frac{-3}{2}+(-1)=\frac{-5}{2}$ ⟵ $r_1+r_2=\frac{-3}{2}+(-1)$

$\Rightarrow k+4=2\Rightarrow k=-2$

c) $r_1\cdot r_2=\frac{c}{a}=\frac{k^2+2k}{4}$ ⟵ product of roots

$\Rightarrow\frac{(-2)(-3)}{4}=\frac{k^2+2k}{4}$ ⟵ $r_1\cdot r_2=\frac{-3}{4}\cdot(-2)=\frac{6}{4}$

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$$\Rightarrow \frac{k^2+2k}{4} = \frac{6}{4} \Rightarrow k^2 + 2k - 6 = 0$$

$$\Rightarrow k^2 + 2k - 6 = 0 \Leftrightarrow k = \frac{-2\pm\sqrt{4-4(-6)}}{2} \therefore k = \frac{-2\pm\sqrt{28}}{2} = -1\pm\sqrt{7}$$

Word problems

Here we are going to see word problems which is related to quadratic equation.

Illustrative Example

78. **Geometry**: Find the dimension of a rectangle whose perimeter is 26m and whose area is $40m^2$.

Solution: Let ℓ = length and w = width perimeter = $2\ell + 2w = 26$

$\Rightarrow \ell + w = 13$ Area = $\ell \cdot w = 40m^2$

$A(\ell) = \ell\cdot(13 - \ell) = 40 \longleftarrow w = 12 - \ell$

$\Rightarrow 13\ell - \ell^2 - 40 = 0 \Rightarrow \ell^2 - 13\ell + 40 = 0$

$\Rightarrow (\ell - 8)(\ell + 5) = 0 \Rightarrow \ell = 8$ or $\ell = -5$

Width (W) = 13 – 8 = 5

∴ The dimension of rectangle are 8 and 5

79. **Geometry**: If the area of the rectangle is $6m^2$ and its length 4m longer than its width find the dimensions.

Solution: Let x = be length, y = width

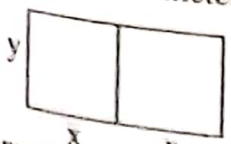
$\Rightarrow x = y + 4$ and area = $xy = 6m^2 \Rightarrow xy = y(y+4)=6 \Rightarrow y^2+4y-6=0$

$$\Rightarrow y = \frac{4\pm\sqrt{16-4(-6)}}{2} = -2 + \sqrt{10}m \longleftarrow \text{width}$$

$x = y + 4 = -2+\sqrt{10} + 4 = (2+\sqrt{10})m \longleftarrow$ length.

80. **Geometry**: A gardener wants to fence in her plot of land in two equal rectangular section. If she has 48 meters of fence and the area of entire plot of land is $96m^2$. Find the dimension of the garden.

Solution: Perimeter (the total fence) = 4x + 3y = 48m



$\Rightarrow 3y = 48 - 4x$

$\Rightarrow y = 16 - \frac{4}{3}x$

Area of rectangle (A) = (x+ x)y = 2xy = 96

$\Rightarrow xy = 48 \longleftarrow$ dividing by (2)

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$$\Rightarrow A = x\,x\left(16 - \frac{4}{3}x\right) = 48 = 16x - \frac{4}{3}x^2 = 48 \Rightarrow \frac{4}{3}x^2 - 16x + 48$$

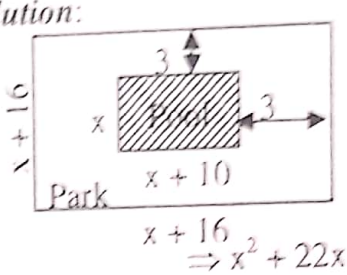
- $x^2 - 12x + 36 = 0$ ⟵ Multiplying by $\left(\frac{3}{4}\right)$

$$\Rightarrow (x-6)(x-6) = 0 \Rightarrow x = 6$$

Hence $y = 16 - \frac{4}{3}x = 16 - \frac{4}{3}(4) = 8m$ ∴ The dimension is 12m by

81. **Swimming pool**: A rectangular swimming pool is 10m longer than width 3m and it has a surrounding park of width 3m on each side. If area of the pool and park is $200m^2$. Find the dimensions of the pool the outside edge of the park.

Solution:



Let x = width of the pool.

x + 10 = length of the pool.

- Then the dimension of the outside p will be x+6 and x+16

Area: $A = (x+6)(x+16) = 200m^2$

$$\Rightarrow x^2 + 22x + 96 = 200$$

$$\Rightarrow x^2 + 22x + 96 - 200 = 0 \Rightarrow x^2 + 22x - 104 = 0$$

$$\Rightarrow (x + 26)(x - 4) = 0 \Rightarrow x = -26, \text{ or } x = 4$$

x cannot be negative:

∴ The dimension of the pool is 4 by 14.

∴ The dimension of the park is 10 by 20.

82. The sum of two number is 13 and their product is 42. Find t numbers.

Solution: Let x and y be the two number

$x + y = 13 \Rightarrow y = 13 - x$

Product: $xy = 42 \Rightarrow x(13 - x) = 42$

$\Rightarrow 13x - x^2 = 42 \Leftrightarrow x^2 - 13x + 42 = 0$

$(x - 7)(x - 6) = 0$; $x = 6$ or $x = 7$

∴ The two numbers are 6 and 7

83. One leg of a right triangle exceeds the other leg by 4, and the hypote is 8 more than the smaller leg. Find the dimension of the leg.

Solution: let x = length of smaller leg, so that x + 4, the other leg. T hypotenuse will be x + 8 $\Rightarrow x^2 + (x+4)^2 = (x+$

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$\Rightarrow x^2 + x^2 + 8x + 16 = x^2 + 16x + 64$

$\Rightarrow x^2 - 8x - 48 = 0 \Leftrightarrow (x - 12) + 16 = x^2 + 16x + 64$

$\therefore x = 12$ or $x = 4$, but $x = 4$ cannot be length

$\therefore$ The length will be 12,16,20

84. **Flight of a projectile** : If a ball is projected vertically upward from height of h_0 feet with an initial of V_0 ft/sec, then its height after t second is $h(t) = -16t^2 + V_0t + h_0$.
If the initial velocity $V_0 = 160$ft/sec on which the ball is projected vertically from the ground
a) At what time the height of the ball is **400** ft.
b) How long will it take for the ball to reach height of 336 feet.

Solution: a) $V_0 = 160$ft/sec

$h(t) = -16t^2 + 160t \leftarrow$ $h_0 = 0$, from the ground

$\Rightarrow -16t^2 + 160t = 400 \Rightarrow 16t^2 - 160t + 400 = 0$

$\Rightarrow t^2 - 10t + 25 = 0 \Leftrightarrow (t - 5)^2 = 0$

$\therefore t = 5$ sec.

b) $16t^2 - 160t + 336 = 0 \Rightarrow t^2 = 10t + 21 = 0$

$\Rightarrow (t - 3)(t - 7) = 0 = t = 3$sec or $t = 7$ sec

85. The length of rectangle is 3 unit longer than twice its width. If the area of the rectangle is 44 sq. unit, then what is the perimeter of the rectangle.

A. 30 unit B. 48 unit C. 15 unit D. 28 unit.

Solution: Let x = length of rectangle

Let x = width of rectangle

$\Rightarrow x = 3 + 2y \leftarrow$ length of rectangle

Area: $xy = 44 \Rightarrow (3x+2y)(y) = 44$

$\Rightarrow 2y^2 + 3y - 44 = 0$

$$\therefore y = \frac{-3 \pm \sqrt{9 - 4(2)(-44)}}{2(2)} = \frac{-3 \pm \sqrt{361}}{4} = \frac{-3 \pm 19}{4} = 4$$

$\therefore x = 3 + 2y = 3 + 2(4) = 11$

$\therefore$ Perimeter: $2x + 2y = 2(11) + 2(4) = 30$

86. What is the solution set of the equation $4^{2x+2} = 4^{2x-1} + 126$

A. $\left\{\frac{3}{4}\right\}$ B. {6} C. { } D. {3}

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Solution: $4^{2x+2} = 4^{2x-1} + 126 \Rightarrow 4^2 \cdot 4^{2x} = 4^{2x} \cdot 4^{-1} + 126$

$\Rightarrow 16.\ 4^{2x} = \frac{1}{4} \cdot 4^{2x} + 126, \leftarrow$ let $4^{2x} = y$

$\Rightarrow 16y = \frac{1}{4} y + 126 \Rightarrow 16y - \frac{1}{4} y = 126$

$\Rightarrow \frac{63y}{4} = 126 \Rightarrow 126 \times \frac{4}{63} = 8 \Rightarrow y = 8 \Leftrightarrow 4^{2x} = 2^3 \Rightarrow 2^{4x} = 2^3$

$\Leftrightarrow 4x = 3 \Rightarrow x = \frac{3}{4}$ **∴ Answer A**

87. The solution set of the equation: $(x + 8)(x - 3) = 3x$

A. $\{-4, 6\}$ B. $\{-6,4\}$ C. $\{-8,3\}$ D. $\{-3,2\}$

Solution: $(x + 8)(x - 3) = 3x \Rightarrow x^2 + 8x - 3x - 24 = 3x$

$\Rightarrow x^2 + 5x - 24 - 3x = 0 \Rightarrow x^2 + 2x - 24 = 0$

$\Rightarrow (x - 4)(x + 6) = 0 \Rightarrow x = 4$ or $x = -6$ **∴ Answer B**

88. Suppose the quadratic equation $ax^2 + x - c = 0$ has two real roots r_1 and r_2. If $r_1 + r_2 = r_1 \cdot r_2 = \frac{-1}{2}$, then c is equal to

A. -2 B. 2 C. 1 D. -1

Solution: $r_1 + r_2 = \frac{-1}{a} = \frac{-1}{2} \Rightarrow a = 2$

$r_1 \cdot r_2 = \frac{c}{a} = \frac{-1}{2} \Rightarrow \frac{c}{2} = \frac{-1}{2} \Rightarrow c = -1$ **∴ Answer D**

89. The base of a triangle is 2cm longer than its altitude and area of $5cm^2$. Find the base of triangle.

Solution: Let x = be altitude, then base $x + 2$

Area $A = \frac{1}{2} x(x + 2) = 5 \Rightarrow x^2 + 2x = 10$

$\Rightarrow x^2 + 2x - 10 = 0 \Rightarrow a = 1, b = 2, c = -10$

$\therefore x = \frac{-2 \pm \sqrt{4 - 4(1)(-10)}}{2} = -1 + \sqrt{11} \leftarrow$ altitude

The base: $x + 2 = -1 + \sqrt{11} + 2 = 1 + \sqrt{11}$

90. If $x^2 + px - 8 = 0$ has roots r and s such that $r = -2s$, then s is equal to

A. 2 or 4 B. 2 C. 4 D. 2 or – 2

Solution: Recall that, $r.s = \frac{-8}{1} \Rightarrow (-2s)\,s = -8$

$\Rightarrow s^2 = 4 \Rightarrow s = 2$ or -2, ∴ **Answer D**

91. Find two consecutive integers such that the sum of their squares is 613.

Solution: Let x and x + 1 be two consecutive integer so that $x^2+(x+1)^2 = 613$

$\Rightarrow x^2 + x^2 + 2x + 1 = 613 \Rightarrow 2x^2 + 2x + 1 - 613 = 0$

$\Rightarrow 2x^2 + 2x - 612 = 0 \Leftrightarrow x^2 + x - 306 = 0$

$\Leftrightarrow (x - 17)(x + 18) = 0 \Rightarrow x = 17$ or $x = -18$

92. What is the solution set of $3x^2 + \sqrt{15}\,x + 1 = 0$?

A. $\left\{\frac{-15 \pm \sqrt{3}}{6}\right\}$ B. $\left\{\frac{-3 \pm \sqrt{15}}{6}\right\}$

C. $\left\{\frac{-\sqrt{15} \pm \sqrt{3}}{6}\right\}$ D. $\left\{\frac{\sqrt{15} \pm 3\sqrt{3}}{6}\right\}$

Solution: a = 3, b = $\sqrt{15}$, C = 1, using $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$\therefore x = \frac{-\sqrt{15} \pm \sqrt{15 - 4(3)(1)}}{2(3)} = \frac{-\sqrt{15} \pm \sqrt{3}}{6}$, ∴ **Answer C**

93. Solve: a) $(x^2 + 7x + 12)^{\frac{1}{5}} = (x^2 + 3)^{\frac{1}{5}}$ b) $(2x)^{\frac{2}{3}} = (6 - 10x)^{\frac{1}{3}}$

Solution: a) $x^2 + 7x + 12 = x^2 + 3 \longleftarrow a^n = b^n \Leftrightarrow a = b$

$\Rightarrow 7x + 12 = 3 \Rightarrow 7x = -9 \Rightarrow x = x = \frac{-9}{7}$

b) $\left((2x)^2\right)^{\frac{1}{3}} = (6 - 10x)^{\frac{1}{3}} \Leftrightarrow (2x)^2 = 6 - 10x$

$\Rightarrow 4x^2 + 10x - 6 = 0 \Leftrightarrow x = \frac{-10 \pm \sqrt{100 - 4(4)(-6)}}{8} = \frac{-10 \pm 14}{8}$

$\therefore x = -3$ or $x = \frac{1}{2}$

94. Solve: a) $(4x - \sqrt{3})^2 = 24$ b) $(\sqrt{2}x + \sqrt{2})^2 - 27 = 0$

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UNIT THREE

Further on sets

Sets and Elements

Set: Set is collection of any object, things, or number.

- The individual objects inset are called elements of set.
- A set is known by its element.
- A set is well defined if and only if it can be decided that a given object is an element of the set.

Description of sets

A set may be described by three method

i. Verbal method

For example:

A = The set of even number between 1 and 9. This method may be used when there is large number of elements or when all the elements cannot be named.

ii. The listing method.

(The roster or tabulation method)

This method used when listing each element of the set with in the brace.

For example: the set of all even number between 1 and 9 by listing method is A = {2, 4, 6, 8}

iii. Set builder method

Set builder method consists in enclosing within the brace a general element and describing it.

Thus: A = {x|x is an even integer, $1 < x < 9$}

Illustrative Example

1. Describe each of the following in verbal and set builder method
   a) A = {3, 5, 7, 9} b) A = {-6, -7, -8, -9, -10} c) A = {1, 3, 5, ...99}

Solution: Verbal method:

a) A consists of set of all odd integer between 2 and 10

Set builder method

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A = {x|x is an odd number and $2 < x < 10$}

b) **verbal method** A = The set of integer between -5 and -11.

Set builder method A={x|x ∈ z and $-11 < x < -5$} or A={x|x∈z and $-10 \leq x$

c) **verbal method:**

A = The set of odd number greater than or equal to 1 and less than or equal to 99

Set builder method: A = {x|x is odd and $1 \leq x \leq 99$}

2. Describe each of the following sets by

i. set builder method ii. listing method

a) The set of letters in the word "Extreme-mathematics"

b) The set of subcity in Addis Ababa

c) The set of integer greater than 0 but less than 19

Solution:

a) i) set builder method. A = {x|x is letter in the word " Extreme mathe

ii) listing method.

A = {E,x,t,r,e,m,a,h,i,c,s}

Note : Repeating elements in a set does not add new elements to the set

{m,m} = {m} , {e,e} = {e}

b) i) Set builder method A = {x|x is sub city in Addis Ababa}

ii) Listing method

A = {Gullele, Arada, Addisketema, Bole, Ledta, Kolfe, Akaki, Lafto, Kir

c) i) Set builder method A = {x|x is integer and $0 < x < 19$}

ii) listing method A = {1, 2, 3, 4, …18}

The notation of set

Empty set: The set which contain no element denoted by ∅ or {}

Example: A = {x|x ∈N and $1 < x < 2$} ∴ A = ∅

B = {x|x ∈ R and $x^2 = -1$}

No number whose square is negative ∴ B = ∅

Finite set: A set is finite, if it has a finite number of elements.

Infinite set: If a set has an infinite number of elements it is an infinite

Illustrative Example

3. Identify, whether

i) Empty ii) Finite iii) infinite set

a) A = {x|x ∈ R and $0 < x < 1$}

b) A = {x|x is multiple of 2}

c) A = {x|x ∈ z and $1 < x < 2$}

d) A = {x|x is a grain of sand on the sea shore}

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Solution: a) Infinite b) infinite c) $\varnothing$ and finite d) finite

Note: Empty set is finite set.

Subset: set A is subset of B. denoted by $A \subseteq B$ if every element of A is an element of B.

Example. Let $A = \{1, 2\}$ $B = \{1, 2, 3, 4\}$

$\Rightarrow A \subseteq B$

Note: For any set A

i) $\varnothing \subseteq A$ ii) $A \subseteq A$

iii) If $A \subseteq B$ and $A \neq B$ then A is called Proper subset of B, denoted by $A \subset B$

Power set: *The set of all subsets of set A is called the power set of A. It is denoted by P(A).*

Example: Let $A = \{0, 1\}$ then $\varnothing, \{0\}, \{1\}, \{0, 1\}$ are the subset of A

Thus, the power set of A $P(A) = \{\varnothing, \{0\}, \{1\}, \{0, 1\}\}$

Illustrative Example

4. Let $A = \{a, \{b, c\}\}$

a) list elements of A. b) list all subset of A. c) list power set of A.

Solution:

a) $a \in A$, $\{b, c\} \in A$ ∴ Elements of A are a, and $\{b, c\}$

b) subset of A are $\varnothing, \{a\}, \{a, \{b, c\}\}, \{\{b,c\}\}$

c) Power set $P(A) = \{\varnothing, \{a\}, \{a, \{b, c\}\}, \{\{b,c\}\}\}$

5. Which of the following is true or false justify

a) $\{1, 2, 3\} \subseteq \{2, 3, 1\}$

b) $\{3\} \subseteq \{\{3\}\}$

c) $\varnothing \subseteq \{\{3\}\}$

d) $\{a, b, a, b, c\} \subseteq \{a, b, c\}$

e) $\{2\} \subseteq \{\{2, 3\}\}$

f) $\{1\} \in \{\{1, 2\}\}$

Solution: a) True

b) False, since $\{3\} \in \{\{3\}\}$ but it is true if $\{\{3\}\} \subseteq \{\{3\}\}$

c) True, $\varnothing$ is subset of any set.

d) True, explanation: $\{a, a\} = \{a\}$

$\{b, b\} = \{b\}$

e) False

f) False, explanation: $\{1, 2\} \in \{\{1, 2\}\}$

Universal Set

The universal set is the totality of element under consideration as elements of any set.

Universal set including all possible elements.

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Equal set: *If every element of one set is also an element of another set.*

- Equal set have exactly the same element.
- If sets A, B are equal we write,

$A = B$

If $A \subseteq B$ and $B \subseteq A$ then $A = B$

Example: a) If $A = \{2, 4, 6, 8\}$ and $B = \{6, 4, 8, 2\}$ then $A = B$

b) $A = \{2, 4, 6, 8\}$ and $B = \{2, 2, 4, 4, 6, 8, 8\}$

$\Rightarrow A = B \quad \Rightarrow n(A) = n(B)$

Equivalent set: *Two sets are said to be equivalent if there is a one-to-one correspondence between the elements of thee two sets.*

Example: Let $A = \{1, 2, 3\}$, $B = \{a, b, c\}$

$A \leftrightarrow B$, $A = \{1, 2, 3\}$ $B = \{a, b, c\}$ $n(A) = n(B)$

Note: Equal sets are equivalent set.

Illustrative Example

6. Which of the following sets are equivalent to $A = \{\{1, 2\}\}$

A. $S = \emptyset$ B. $E = \{\emptyset\}$ C. $F = \{1, 2\}$ D. $G = \{x | x < x\}$

Solution: $A = \{\{1, 2\}\}$, $\therefore n(A) = 1$

A. $n(S) = 0$ B. $n(E) = 1$ C. $n(F) = 2$ D. $n(G) = 0$

$\therefore A \leftrightarrow E$ since $n(A) = n(E)$ **$\therefore$ Answer B**

Operation on set

Union of set $(A \cup B)$: The union of two sets A and B the set of all elemer belonging or B or to both A and B

That is $A \cup B = \{x | x \in A$ or $x \in B$ or $x \in$ both A, B$\}$

- Note: i) $A \cup B = B \cup A$, and $A \cup A = A$

Intersection $(A \cap B)$: The intersection of two sets A and B is the set of elements which belong to both A and B.

That is: $A \cap B = \{x | x \in A$ and $x \in B\}$

Note $A \cap B = B \cap A$ and $A \cap A = A$

Note: For any set A and B:

a) $n(A) + n(B) = n(A \cap B) + n(A \cup B)$

b) if $(A \cap B) = \emptyset$, then $n(A \cup B) = n(A) + n(B)$

$A \cap B \subseteq A \cup B$

c) $A \cap B \subseteq A$ and $A \cap B \subseteq B$

d) $A \subseteq A \cup B$ and $B \subseteq A \cup B$

e) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

f) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

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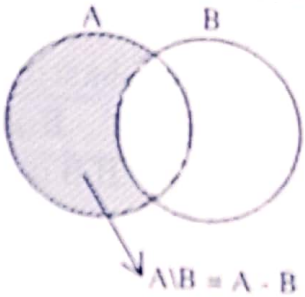
Difference and symmetric difference set.

The relative complement (difference) of two sets

Difference (A – B): The difference of two set A and B is the set of all elements belonging to A but not to B.

- $A - B = \{x | x \in A \text{ and } x \notin B\} = A \backslash B$
- $B - A = \{x | x \in B \text{ and } x \notin A\} = B \backslash A$



Note: i) A – B, read A less B

ii) $A - B = A \backslash B$

iii) $A - B = A \backslash B = A - (A \cap B)$

iv) $B - A = B \backslash A = B - (A \cap B)$

Illustrative example

7. Let $A = \{a, \{b, c\}\}$, and $B = \{a, b, c\}$

Then find

a) The relative complement of A with respect to B.

b) The complement of B with respect to A

c) $A \backslash B$ d) $B \backslash A$ e) $A \cap B$ f) $A \cup B$

Solution: Note: Relative complement of A w.r.t B = $B \backslash A$

Then a) $B \backslash A = \{b, c\}$

b) $A \backslash B = \{\{b, c\}\}$ ⟵Relative complement of B with

c) $A \backslash B = \{\{b, c\}\}$ respect to A

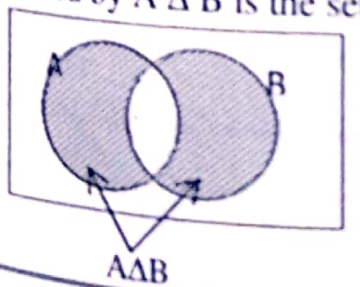
d) $B \backslash A = \{b, c\}$ e) $A \cap B = \{a\}$ f) $A \cup B = \{a, b, c, \{b, c\}\}$

Complement of a set. If A is subset of universal set U then elements belongs to U but not belongs to A is called the complement of A denoted by A'

That is $A' = \{x | x \in \cup \text{ and } x \notin A\}$ $\therefore A' = U - A = U \backslash A$

Symmetric difference between two sets

Let A and B be any two sets. The symmetric difference between A and B, de noted by $A \Delta B$ is the set of all elements in $A \cup B$ that are not in $A \cap B$.



That is

$A \Delta B = \{x | x \in A \cup B \text{ and } x \notin A \cap B\}$

$A \Delta B = A \cup B \backslash A \cap B$

Note: $A \Delta B = A \cup B$ if $A \cap B = \varnothing$

Illustrative example

8. Let $A = \{2, \{3, 4\}\}$ and $B = \{2, \{4\}, 3\}$ then find

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a. $A \Delta B$ b. $A \backslash B$ c. $B \backslash A$ d. $A \cap B$ e. $A \cup B$

Solution: a) $A \Delta B = A \cup B \backslash A \cap B = \{3\{4\}\{3, 4\}\}$

b) $A \backslash B = \{\{3, 4\}\}$ c) $B \backslash A = \{3, \{4\}\}$ d) $A \cap B = \{2\}$

e) $A \cup B = \{2, 3, \{4\}, \{3, 4\}\}$

9. If $B \subseteq A$, then which of the following is false

A. $A' \Delta B' = B' \backslash A'$ B. $A' \subseteq B'$ C. $A' \backslash B' = \varnothing$ D. $A \cap B = B$

Solution: Note that: If $B \subseteq A$ then $A' \subseteq B'$

A. $\therefore A' \Delta B' = (A' \backslash B') \cup (B' \backslash A') = \varnothing \cup (B' \backslash A') = B' \backslash A'$

B. $B \subseteq A \Rightarrow A' \subseteq B'$ C. Since $A' \subseteq B' \Rightarrow A' \backslash B' = \varnothing$

D. $A \cap B \neq B$ **$\therefore$ Answer D**

The laws of operations on sets

1. $A \cup A = A$ and $A \cap A = A$
2. $A \cup B = B \cup A$ and $A \cap B = B \cap A$
3. Distributive law
   i. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ ii. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
4. Identity law
   i) $A \cup \varnothing = A$ ii) $A \cap U = A$
5. Complement law
   i) $A \cup A' = U$ ii) $A \cap A' = \varnothing$ iii) $U' = \varnothing$ iv) $\varnothing' = U$
6. Demorgan's law
   i) $(A \cup B)' = A' \cap B'$ ii) $(A \cap B)' = A' \cup B'$

Ordered pairs

The element of a set {a, b} need not be listed in any particular order because {a, b} = {b, a}. In the case of ordered pair the order in which the elements are given has a special meaning. This is because, often, the way of arrangement or order of elements is very important.

Let A, B be two sets and let $a \in A$, $b \in B$

Then (a, b) is an ordered pair of

- first component a. - second component b.

In general $(a, b) \neq (b, a)$ unless $a = b$

Illustrative Example

10. If $(4x - 3, 5) = (9, 2y + 7)$ Find the value of x, y

Solution: $4x - 3 = 9$ and $2y + 7 = 5$

$\Rightarrow 4x = 12$ and $2y = 5 - 7 = -2 \Rightarrow x = 3$ and $y = -1$ $\therefore x = 3$ and $y = -1$.

Cartesian product

The set of all possible pairs (a, b) where $a \in A$ and $b \in B$ is called the

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Cartesian product of A and B and is written:

$A \times B = \{(a, b) / a \in A, b \in B\}$

Note: i) $n(A \times B) = n(A) \times n(B)$

ii) $A \times (B \cap C) = (A \times B) \cap (A \times C)$

iii) $A \times (B \cup C) = (A \times B) \cup (A \times C)$

Illustrative Example

11. Let $A = \{1, 2\}$, $B = \{a, b\}$ then find a) $A \times B$ b) $A \times A$ c) $B \times B$ d) $B \times A$

Solution: a) $A \times B = \{(1, a), (1, b), (2, a), (2, b)\}$

b) $A \times A = \{(1, 1), (1, 2), (2, 1), (2, 2)\}$

c) $B \times B = \{(a, b), (a, a), (b, a), (b, b)\}$

12. If $n(A) = 4$ and $n(B) = 5$ then, Find $n(A \times B)$

Solution: $n(A \times B) = n(A) \times n(B) = (4)(5) = 20$

13. Let $A = \{x | x = 0, 1, -1\}$ and $B = \{y | y = 2x - 1\}$ then Find $A \times B$

Solution: First, $A = \{0, 1, -1\}$ and

$B = \{2(0)-1, 2(1) - 1, 2(-1)-1\} = \{-1, 1, -3\}$

$\therefore A \times B = \{(0,-1), (0,1), (0,-3), (1,-1), (1,1), (1,-3), (-1,-1), (-1,1), (-1,-3)\}$

14. If $A \cap B = \{-1, 0, 1\}$ and $B \cap C = \{2, -1, 0\}$, then $Bn(A \cup C)$ is equal to ..... EGSECE 2001.

A. $\{0, -1\}$ B. $\{-1, 0, 1\}$ C) $\{0, -1, 2\}$ D) $\{-1, 0, 1, 2\}$

Solution: Recall that (Distributive property)

$Bn(A \cup C) = (B \cap A) \cup (B \cap C) = (A \cap B) \cup (B \cap C)$

$\therefore Bn(A \cup C) = \{-1,0,1\} \cup \{2, -1, 0\} = \{-1, 0, 1, 2\}$ **$\therefore$ Answer D.**

15. If A is the subset of B, then which of the following is always true?

A. $A \cup B = A$ C. $A \backslash B = \varnothing$

B. $A' \cap B' = B'$ D. $A \cap B = B$

Solution: If $A \subseteq B$, then $A \backslash B = \varnothing$ **$\therefore$ Answer C.**

16. If $A = \{\varnothing, \{\{\varnothing\}\}\}$ and $B = \{\{\varnothing\}, \{\varnothing, \{\varnothing\}\}\}$ then which of the following is not true..... EGSECE

A. $B \backslash A = B$ C. $A \cap B = \varnothing$

B. $A \backslash B = A$ D. $A \cap B = \{\varnothing\}$

Solution: Note: If $A \cap B = \varnothing$ then i) $A \backslash B = A$

ii) $B \backslash A = B$ **$\therefore$ Answer D.**

17. In a medical examination of 100 persons, it was found that 18 persons

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are short sighted 15 are diabetic, and 7 are both short sighted and diabetic of the 100 person examined, the number of those who are neither diabetic nor short sighted is:

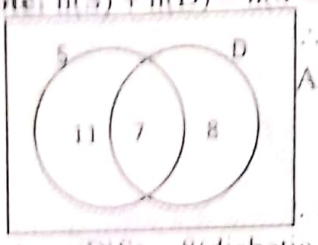
A. 74 B. 67 C. 60 D. 78

Solution: Let U = universal set = 100

Let S be short sighted = 18

Let D be diabetic = 15

Note: $n(S) + n(D) = n(S \cup D) + n(S \cap D)$ $\Rightarrow 18 + 15 = n(S \cup D) + 7$



$\therefore n(S \cup D) = 33 - 7 = 26$

And $n(U \backslash S \cup D) = n(U) - n(S \cup D)$

$= 100 - 26$

$= 74$

$\therefore$ neither diabetic nor short sighted is 74.

Note: $n(D \backslash S) = 8$(diabetic only) and $n(S \backslash D) = 11$

18. If for sets A, B and C, $A \cup C = \{1,2,3,6\}$ and $B \cup C = \{1,8,7,6,3\}$, then $(A \cap B) \cup C$ is equal to: ------ ESLCE.

A. {1, 8, 7, 6, 3} C. {7, 8}

B. {1, 3, 6} D. {1, 2, 3, 6}

Solution: we use distributive property

$(A \cap B) \cup C = C \cup (A \cap B) = (C \cup A) \cap (C \cup B)$

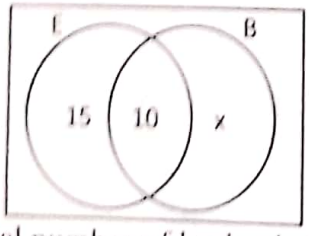
$= (A \cup C) \cap (B \cup C) = \{1, 2, 3, 6\} \cap \{1, 8, 7, 6, 3\}$

$= \{1, 3, 6\}$ **$\therefore$ Answer B**

19. In a class of 50 students, 15 play foot ball only, 10 play both foot ball and basket ball and 5 play neither foot ball nor basket ball. How many students play basket ball? ..... EGSECE

A. 25 B. 30 C. 20 D. 10

Solution:



- Let F be football player
- Let B be basket ball player

$(F \cup B)' = 5$

$\Rightarrow (F \cup B)' = U - F \cup B$

$\Rightarrow 5 = 50 - (15 + 10 + x)$

$\Rightarrow 5 = 25 - x$ $\therefore x = 20$

$\therefore$ Total number of basket ball player $10 + 20 = 30$ **$\therefore$ Answer B**

Note: 20 only basket ball player i.e. $B \backslash F = 20$

20. For any sets A and B, which of the following is not true?

A. $(A \backslash B) \cup (A \cap B) = A$ C. $(A \backslash B) \cup (B \backslash A) = (A \cup B) \backslash A \cap B$.

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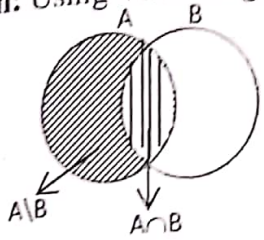
B. $A\backslash(A \cap B) = A\backslash B$ D. $A \cup (B\backslash A) = A$

Solution: Using Venn diagram

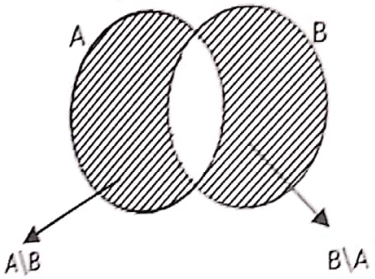
A.



$\therefore (A\backslash B) \cup (A \cap B) = A$

B.

- $A\backslash B = A - (A \cap B) = A\backslash(A \cap B)$
- $B\backslash A = B - (A \cap B) = B\backslash(A \cap B)$



C. $A \Delta B = (A\backslash B) \cup (B\backslash A, - A \cup B\backslash A \cap B$

D. $A \cup (B\backslash A) = A \cup B$ **∴ Answer D**

21. Let $A = \{d, \{e\}, \{ef\}\}$. Which of the following statement is true about A?

A. $\{\{f\}\} \subset A$ B. $\{ef\} \subseteq A$ C. $\{e\} \subseteq A$ D. $\{\{ef\}\} \subset A$

Solution: $\{ef\} \in A$ but $\{ef\} \not\subseteq A$ $\{e\} \in A$ but $\{e\} \not\subseteq A$ $\{\{ef\}\} \subset A$

∴ Answer D

22. Let A, B and C be sets such that $A \cap C = \varnothing$ and $B \subseteq C$.

Then $A \Delta (B\backslash C)$ is equal to:

A. B B. A C. $A \cup B$ D. $A \cup C$

Solution: since $B \subseteq C \Rightarrow B\backslash C = \varnothing$

$\therefore A \Delta (B\backslash C) = A \Delta \varnothing = A$ **∴ Answer B**

23. For any two sets A and B. $(A' \cup B)'$ is equal to:

A. A B. $A\backslash B$ C. $B\backslash A$ D. B

Solution: By Demorgan's law

$(A' \cup B)' = (A')' \cap B' = A \cap B' = A\backslash B$ **∴ Answer B**

Note: $A \cap B' = A\backslash B$ $A' \cap B = B\backslash A$.

24. Let $A = \{1, 2\}$, $B = \{1, 3, 4\}$, and $C = \{2, 3\}$

Then $A \times (B \cap C)$ is equal to: ESLCE

A. $\{(3, 1), (3, 2)\}$ C. $\{(1, 3), (3, 2)\}$

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B. {(3, 1), (2, 3)} D. {(1, 3), (2, 3)}

Solution: $A \times (B \cap C) =$

First $B \cap C = \{3\} \therefore \{1, 2\} \times \{3\} = \{(1, 3), (2, 3)\} \therefore$ **Answer D**

25. If $A = \{-1, 0, \{-1, 1\}\}$ and $B = \{-1, 0, 1\}$, then which of the following is true.

A. $A \cap B = B$ B. $A \cup B = A$

C. $A \cap B \backslash \{-1, 0\} = \varnothing$ D. $B \backslash A = \varnothing$

Solution: A. $A \cap B = \{-1, 0\}$ B. $A \cup B = \{-1, 0, 1, \{-1, 1\}\}$

C. $A \cap B \backslash \{-1, 0\} = \varnothing$ D. $B \backslash A = \{1\}$ **∴ Answer C**

26. The number of proper subsets of the set $\{x, \{y,z\}\}$ is

A. 8 B. 6 C. 7 D. 3

Solution: Let $A = \{x, \{y, z\}\} \therefore n(A) = 2$

The number of proper subset of n element is $2^n - 1$

$\therefore 2^2 - 1 = 4 - 1 = 3$ **∴ Answer D**

27. Which of the following is not a subset of the set $\{0, \varnothing\}$

A. $\varnothing$ B. $\{0\}$ C. 0 D. $\{\varnothing\}$

Solution: Let $A = \{0, \varnothing\}$

$\Rightarrow$ subset of A are, $\varnothing$, $\{0\}$, $\{\varnothing\}$ and $\{0, \varnothing\}$

Hence 0 is elements of A but not subset **∴ Answer C**

28. Given $A = \{1, \{2\}, \{1, 2\}\}$ and $B = \}1, \{1, 2,3\}\}$ which one of the following is not true?

A. $B \backslash A = \{1, 2, 3\}$

B. $A \cup B = \{1, \{2\}, \{1, 2\}, \{1, 2, 3\}\}$

C. $A \Delta B = \{\{2\}, \{1, 2\}, \{1, 2, 3\}\}$ D. $A \backslash B = \{\{2\}, \{1, 2\}\}$

Solution: A. $B \backslash A = \{\{1, 2, 3\}\}$

B. $A \cup B = \{1, \{2\}, \{1,2\}, \{1, 2, 3\}\}$

C. $A \Delta B = A \cup B \backslash A \cap B = \{\{2\}, \{1, 2\}, \{1, 2, 3\}\}$

D. $A \backslash B = \{\{2\},\{1,2\}\}$ **∴ Answer A**

29. If A, B, and C are any three sets in the same universal set then which of the following is not equal to $A \backslash (B' \cup C)$ …. EGSECE

A. $A \cap (B \backslash C)$ B. $A \cap (C \backslash B)$ C. $A \cap B \cap C'$ D. $B \cap (A \backslash C)$

Solution: Recall that $A \backslash B = A \cap B'$

$\Rightarrow A \backslash (B' \cup C) = A \cap (B' \cup C)'$ Explanation:

$= A \cap (B \cap C')$ ← By Demorgan's law

$= A \cap (B \backslash C)$

And $A \cap (B \cap C') = B \cap (A \cap C')$

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$= B \cap (A \backslash C)$ ∴ **Answer B**

30. If A and B are two sets such that $n(A) = 12$, $n(B) = 15$ and $n(A \cap B) = 4$, then $n(A \Delta B)$ is equal to:

A. 23 B. 19 C. 8 D. 11

Solution: First, we find that $n(A \cup B)$, so that,

$n(A) + n(B) = n(A \cap B) + n(A \cup B)$

$\Rightarrow 12 + 15 = 4 + n(A \cup B)$ ∴ $n(A \cup B) = 27 - 4 = 23$

Now, $n(A \Delta B) = n(A \cup B) - n(A \cap B) = 23 - 4 = 19$ ∴ **Answer B**

Note, we can find $n(A \backslash B)$, $n(B \backslash A)$

$\Rightarrow n(A \backslash B) = n(A) - n(A \cap B) = 12 - 4 = 8$

and $n(B \backslash A) = n(B) - n(A \cap B)$

$= 15 - 4 = 11$

Method II

$n(A \Delta B) = n(A \backslash B) + n(B \backslash A) = 8 + 11 = 19$

31. If $(A \backslash B) \cup B = A$ which of the following is always true ...EGSECE

a. $B = \{\}$ B. $A = B$ C. $B \subseteq A$ D. $A \subseteq B$

Solution: Let $B = \{\} = \varnothing$

A. $(A \backslash B) \cup B \Rightarrow (A \backslash \varnothing) \cup \varnothing = A$

B. If $A = B$, then $(A \backslash A) \cup B = \varnothing \cup B = B$ ∴ **Answer A**

32. Given $A = \{1, \{1, 2\}\}$ and $B = \{2, \{1, 2\}\}$ then what is $A \Delta B$.

A. $\{1, 2\}$ B. $\{\{1, 2\}\}$ C. $\varnothing$ D. $\{1,2,\{1,2\}\}$

Solution: $A \Delta B = A \cup B \backslash A \cap B = (A \cup B) - (A \cap B)$

$= \{1, 2\}$ ∴ **Answer A**

33. If $A = \{x | x$ is non-negative integer and $x^3 = x\}$ then the number of proper subsets of A is ..... EGSECE 2001 E.C

A. 7 B. 3 C. 4 D. 1

Solution: We have, $x^3 = x$

$\Rightarrow x^3 - x = 0$

$\Rightarrow x(x^2 - 1) = 0$ ∴ $x = 0, x = 1$ ∴ $A = \{x | x = 0, 1\} = \{0, 1\}$

∴ $n(A) = 2$, number of proper subset of $A = 2^n - 1$

$\Rightarrow 2^2 - 1 = 4 - 1 = 3$ ∴ **Answer B**

34. Out of 85 football players 20 play for neither the national team nor for the major clubs. If 40 of the players play for the national team and 15 play for both three national team a major clubs. What is the number of players playing for the major clubs..

A. 40 B. 25 C. 65 D. 45

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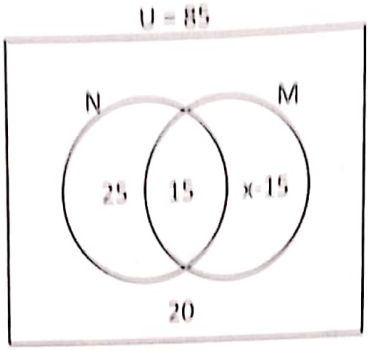
Solution: Let U = 85 ⟵ Universal



let N = 40 ⟵ National team

let M = x ⟵ major club,

now, we have $n(N \cup M)' = 20$,

$(N \cup M)' = U \setminus (N \cup M)$

$n(N \cup M)' = n(U) - n(N \cup M)$

$20 = 85 - (25 + 15 + x - 15)$

$\Rightarrow 85 - 25 - x = 20$

$\Rightarrow 60 - x = 20$

$\therefore x = 40$ ⟵ major clubs player

∴ **Answer A**

Note, x – 15 ⟵ major clubs player only

$\Rightarrow 40 - 15 \Rightarrow 25$

$n(M \setminus N) = 25$ and $N \setminus M = 25$

35. Let $A \cap B = \{1, 2\}$, $A \cup B = \{1, 2, 3, 4, 5\}$, then the $A \Delta B$ is equal to

A. {3, 4, 5} B. {1, 2, 3, 4, 5}

C. {1, 3,5} D. {1, 2}

Solution: $\therefore A \Delta B = A \cup B \setminus A \cap B = \{3, 4, 5\}$ ∴ Answer A

36. The number of subsets of the set

$A = \{x | x \text{ is even integer and } -1 \leq x \leq 3\}$ is

A. 4 B. 2 C. 32 D. 8

Solution: $A = \{0, 2\}$

∴ number subset of A is $2^n \Rightarrow 2^2 = 4$ ∴ Answer A

37. In a given universal set U, let A and B be sets. The simplified form of

$(A \cup B) \cap (A' \cup B')'$

A. ∅ B. U C. $A \cup B$ D. $A \cap B$

Solution: Here,

$(A \cup B) \cap (A' \cup B')' = (A \cup B) \cap (A \cap B) = A \cap B$ ∴ Answer D

38. In Elon Academy, there 500 students. The numbers of members of mathematics clubs are 100 and 180 are member of physics club. If 45 students are members of both clubs, what is the number of students that are neither members of mathematics club nor members of physics club?

A. 304 B. 395 C. 220 D. 262

Solution: Let U = 500 ⟵ Elon Academy student

Let P = 180 ⟵ member of physics club

Let M = 100 ⟵ member of physics club

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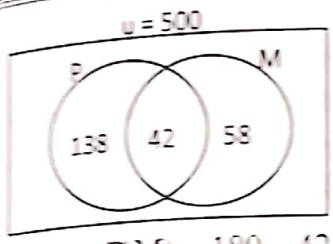
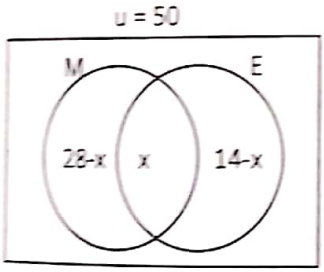


Let P∩M = 42 ⟵ member of both club

Now, ⇒ n(P\M) = 180 – 42 = 138 ⟵ only physics club
⇒ n(M\P) = 100 – 42 = 58 ⟵ only maths club
⇒ n(P ∪ M)′ = 500 – (138 + 42 + 58)
= 500 – 238
= 262 ⟵ members of neither club. ∴ Answer D

39. Out of 50 student in a class 28 students have taken maths test and 14 student have taken English test: if 12 are have taken neither of the test. then how many students are have taken of
a) only maths test b) only English test c) both test.

Solution: Let U = 50 ⟵ student in class.
Let M = 28 ⟵ student take maths test.
Let E = 14 ⟵ student take English test.
Let (M ∪ E)′ = 12 ⟵ student take neither test.
n(M ∪ E)′ = n(U) – n(M ∪ E)



⇒ 50 – (28 – x + x + 14 – x) = 12
⇒ 50 – (42 – x) = 12
⇒ 8 + x = 12
∴ x = 4
∴ only math test, M\E = 28 – 4 = 24
∴ only English test E\M = 14 – 4 = 10
∴ both test 4.

40. Let n(A\B) = 6, n(B\A) = 8 and n(A) = 11
Find a) n(B) b) n(A ∩ B) c) n(A ∪ B)

Solution: n(A\B) = n(A) – n(A ∩ B)
6 = 11 – n(A ∩ B) ⇒ n(A ∩ B) = 5
To find B, n(B\A) = n(B) – n(A ∩ B)
8 = n(B) – 5 ⇒ n(β) = 8 + 5 = 13
To find n(A ∪ B)
We use n(A) + n(B) = n(A ∩ B) + n(A ∪ B)
11 + 13 = 5 + n(A ∪ B)

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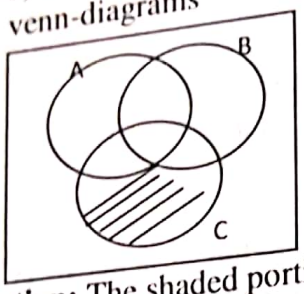
$24 - 5 = n(A \cup B)$

$19 = n(A \cup B)$ $\therefore n(A \cup B) = 19$

41. a) Which of the following set operation represent the shaded portion venn-diagrams



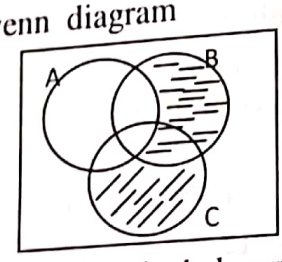
A) $C \cap (A \cap B)'$ or $C \backslash A \cap B$
B. $C \cap (A \cup B)'$ or $C \backslash A \cup B$
C. $C' \cap (A \cup B)$ or $A \cup B \backslash C$
D. $A \cup B \backslash C$

Solution: The shaded portion indicate the intersection of the complement of $A \cup B$ and C

i.e. $(A \cup B)' \cap C = C \cap (A \cup B)'$

$\therefore$ Answer B. i.e. $C \backslash A \cup B$.

42. Which of the following set operation represent the shaded portion of venn diagram



A. $C \cap (B \cup A')$
B. $A' \cap (B \cup C)$ or $B \cup C \backslash A$
C. $C \cup (A' \cap B)$
D. $(A \cup B) \backslash C$

Solution: The shaded portion is $B \cup C \backslash A = A' \cap (B \cup C)$

$\therefore$ Answer B

43. If $n(A) = 10$, $n(B) = 12$ and $n(A \cap B) = 2$ find

a) $n(A \backslash B)$ b) $n(B \backslash A)$ c) $n(A \cup B)$ d) $n(A \Delta B)$

Solution: a) $n(A \backslash B) = n(A) - n(A \cap B) = 10 - (2) = 8$

b) $n(B \backslash A) = n(B) - n(A \cap B) = 12 - 2 = 10$

c) $n(A) + n(B) = n(A \cup B) + n(A \cap B)$

$10 + 12 = n(A \cup B) + 2 \Rightarrow n(A \cup B) = 22 - 2 = 20$

d) $n(A \Delta B) = n(A \backslash B) + n(B \backslash A) = 8 + 10 = 18$

44. If A and B are sets such that $A \cap B = \varnothing$ $n(A) = 8$ and $n(B) = 9$, Fin

a) $n(A \cup B)$ b) $n(A \backslash B)$ c) $n(B \backslash A)$

Solution: a) $n(A) + n(B) = n(A \cap B) + n(A \cup B)$

$\Rightarrow 8 + 9 = 0 + n(A \cup B) \Rightarrow n(A \cup B) = 17$

b) $n(A \backslash B) = n(A) - n(A \cap B) = 8 - 0 = 8$

c) $n(B \backslash A) = n(B) - n(A \cap B) = 9 - 0 = 9$

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45. If $n(A) = 11$ and $n(B) = 13$

Find a) the maximum possible element of

i) a) $A \cup B$ ii) $A \cap B$

b) the minimum possible element of

i) $A \cup B$ ii) $A \cap B$

Solution: Given $n(A) = 11$, $n(B) = 13$

a) i) $A \cup B$ is maximum when $(A \cap B) = \varnothing$

i.e. $n(A \cap B) = 0$

$\therefore n(A \cup B) = n(A) + n(B) = 11 + 13 = 24$

ii) $n(A \cap B) = n(A) + n(B) - n(A \cup B)$

$n(A \cap B)$ is maximum when

$n(A \cup B)$ = smallest element of the given set

$\Rightarrow n(A \cup B) = 11$ $\therefore$ maximum of $n(A \cap B)$ is 13

b) To find: the minimum possible element of $A \cup B$:

$n(A \cup B) = n(A) + n(B) - n(A \cap B)$

when $n(A \cap B) = 11$ $\therefore$ minimum of $n(A \cup B) = 11+13-11=13$

$\therefore$ minimum of $A \cap B$ is 0

46. If $A \backslash B = \{1, 2\}$, and $B = \{3, 4\}$, then $A \cup B$

Solution: $(A \backslash B) \cup B = A \cup B$ (use venn diagram.

$\Rightarrow \{1, 2\} \cup \{3, 4\} = \{1, 2, 3, 4\}$

47. The simplified form of:

A. $(A \backslash B) \cup (A \cap B)$ B. $(A \backslash B) \cup B$

Solution: a) $(A \backslash B) \cup (A \cap B) = A - (A \cap B) + A \cap B$

$= A$

Explanation:
$A \backslash B = A - (A \cap B)$

b) $(A \backslash B) \cup B = A - (A \cap B) + B$

$= A + B - (A \cap B) = A \cup B$

Or

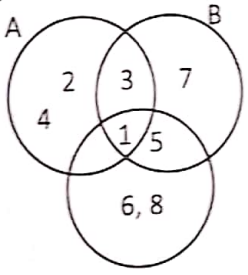
$(A \backslash B) \cup B = (A \cap B') \cup B$

$= B \cup (B' \cap A)$

$= (B \cup B') \cap (B \cup A)$

$= (U) \cap (B \cup A)$

$= A \cup B$



48. Let $A = \{1, 2, 3, 4\}$ $B = \{1, 3, 5, 7\}$ $C = \{1, 5, 6, 8\}$

Find a) $A \cap (B \cap C)$ c) $A \backslash (B \cap C)$

b) $A \cup (B \backslash C)$ d) $A \cup (B \cup C)$ e) $A \cap (B \cup C)$

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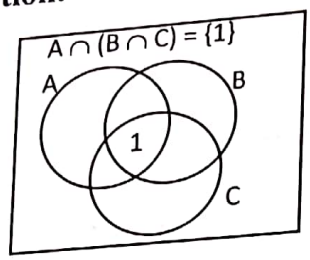
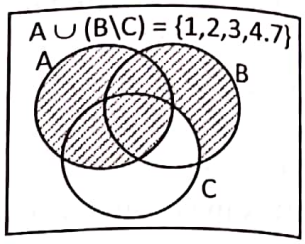
Solution:

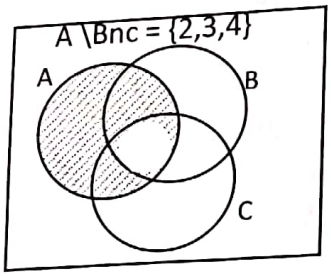
a)



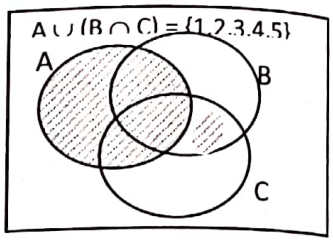
b)



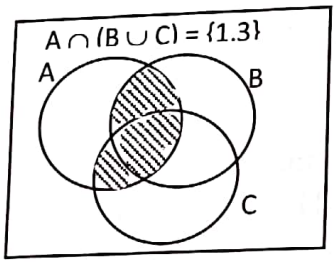
c)



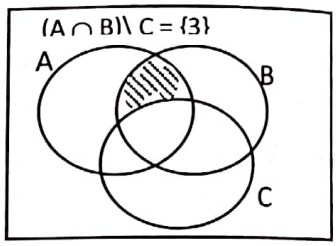
d)



e)



f)



49. If U = 30, n(A) = 12, n(B) = 13, n(A ∪ B) = 20 then
Find a) n(A′ ∩ B′) c) n(A′) d) n(A Δ B)
b) n(A′ ∪ B′) e) n(B\A)′

Solution: Here
First, we find A ∩ B

$\Rightarrow n(A) + n(B) = n(A \cap B) + n(A \cup B)$

$\Rightarrow 12 + 13 = n(A \cap B) + 20 \Rightarrow n(A \cap B) = 25 - 20 = 5$

$\therefore$ a) $n(A' \cap B') = n(U \backslash (A \cap B) = n(U) - n(A \cup B) = 30 - 20 = 10$

b) $(A' \cup B') = (A \cup B)' \Rightarrow n(A' \cup B') = n(U) - n(A \cap B) = 30 - 5 = 25$

c) $n(A') = n(U \backslash A) = n(U) - n(A) = 30 - 12 = 18$

d) $(A \Delta B) = (A \cup B) \backslash A \cap B$

$\Rightarrow n(A \Delta B) = n(A \cup B) - n(A \cap B) = 20 - 5 = 15$

e) $(B \backslash A)' = (B \cap A')' = B' \cup A$

$n(U) - n(B \backslash A) = 30 - 10 = 20$

Explanation: $(B \backslash A)' = U \backslash (B \backslash A)$
$n(B \backslash A)' = n(U) - n(B \backslash A)$
and $n(B \backslash A) = n(B) - (A \cap B)$
$= 13 - 5$
$= 8$

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50. Which one of the following sets is finite set?
   A. The set of positive composite numbers less than 1000
   B. The set of points on the number line between -1 and 1
   C. The set of positive prime numbers
   D. The set of irrational numbers between 0 and 5

Solution: A ⟵ finite B, C, D ⟵ infinite

51. If the universal set is the set of integers then which one of the following pairs of sets contains set that are not equivalent?
   A. $\{x(x+2)(3x-2)=0\}$ and $\{x: x-2=0\}$
   B. $\{x: 3x+5=0\}$ and $\{x: 2x-7=0\}$
   C. $\{x: (x-2)^2=9\}$ and $\{x: 3x-1=5\}$
   D. $\{x: 2x+3=2x-3\}$ and $\{x: 3x=2\}$

Solution: Universal set = the set of integer hence
   A. $\{x: x=-2\}$ and $\{x: x=2\}$ are equivalent
   B. $\{x: x=\varnothing\}$ and $\{x: x=\varnothing\}$ are equivalent

Note: $3x+5=0$ and $2x-7=0$ $\quad x=\frac{-5}{3}$ and $x=\frac{7}{3}$

both are not integer

$\therefore$ the solution set for $3x+5=0$ and $2x-7=0$ is $\varnothing$

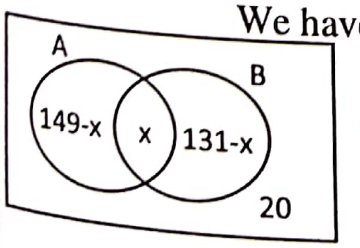
C. $\{x: x=5, 1\}$ and $\{x: x=2\}$ are not equivalent

D. $\{x: x=\varnothing\}$ and $\{x: x=\varnothing\}$ since $x=\frac{2}{3}$ is not integer

$\therefore$ They are equivalent.

52. Among the 250 regular customers of a shop, 149 of them regularly buy product A, 131 customers regularly buy product B. If 20 of the customers buy neither of the products, then what is the number of customers that regularly buy only product A?
   A. 70 B. 99 C. 83 D. 50

Solution: Let U = 250 ⟵ regular customer of shop
Let A = 149 ⟵ regular customer buy product of A
Let B = 131 ⟵ regular customer buy product of B



We have $n(A \cup B)' = 20$
$\Rightarrow n(U) - n(A \cup B) = 20$
$\Rightarrow 250 - n(A \cup B) = 20$
$\therefore n(A \cup B) = 230$
And $n(A) + n(B) = n(A \cap B) + n(A \cup B)$

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$\Rightarrow 149 + 131 = n(A \cap B) + 230 \Rightarrow n(A \cap B) = 50$

$\therefore n(A \cap B) = x = 50 \therefore A \setminus B = n(A) - n(A \cap B) = 149 - 50$

$= 99 \longleftarrow$ buy only product A

$\therefore B \setminus A = n(B) - n(A \cap B) = 131 - 50$

$= 81 \longleftarrow$ buy only product B $\therefore$ **Answer B**

Supplementary exercise

1. Let $A = \{2, 3, 4, 5\}$, $B = \{6, 8, 9, 10\}$, $C = \{7, 8, 9\}$
   a) $(A \setminus B) \setminus C$ b) $A \setminus (B \setminus C)$
2. Let $A = \{a, b, c\}$ Find all subset of A
3. Let $A = \{a, \{b, c\}\}$ Find all subset of A
4. Let $A \cup B = \{1, 2, 3, \{4\}\}$ and $A \cup C = \{1, 2, \{3, 4\}\}$
   Then Find a) $A \cup (B \cap B)$
5. Suppose A is proper subset of B, and Let $n(B) = 6$
   Then the maximum possible element of:
   a) $n(A)$ b) $n(A \cap B)$ c) $n(A \cup B)$
6. Let $A = \{1, 2, \{3, 4\}\}$, $B = \{2, 3\}$, $C = \{\{3, 4\}, 5\}$
   Find a) $A \cap B$ b) $A \cap C$ c) $B \cap C$ d) $A \cap (B \cap C)$
7. If $A \subseteq B$, then. Find $A \Delta B =$ _____
8. If $A \cap B = A \cup B$, then. Find a) $A \Delta B$ b) $A \setminus B$
9. Given $A = \{1, \{2\}, \{1,2\}\}$ and $B = \{1, \{1, 2, 3\}\}$ then find
   a) $A \cup B$ b) $A \cap B$ c. $A \setminus B$ d) $B \setminus A$
10. If $A = [-1, 1]$ and $B = [-2, 0]$, then $A \setminus B$
11. If $A \subseteq B$, then. Find a) $B' \setminus A'$ b) $A' \Delta B'$
12. Let $A \setminus C = \{1, 2, 3\}$ and $B \setminus C = \{1, 4, 6\}$ then find $(A \cup B) \setminus C$
13. the simplified form of: a) $(B \setminus C) \cup (B \cap C)$ b) $(B \setminus C) \cup C$
14. $n(U) = 12$, $n(A) = 5$, and $n(B) = 8$
    find a) $n(A')$ b) $n(B')$
15. Let $n(A \setminus B) = 6$, $n(B \setminus A) = 7$, and $n(A) = 8$, then find
    a) $n(B)$ b) $n(A \cap B)$ c) $n(A \cup B)$

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17 a) $\frac{-b}{c}$ b) $\frac{13}{6}$ c) -3 d) $\frac{-5}{2}$

18 a) $k = \frac{3 \pm 2\sqrt{5}}{2}$ b) $k = \pm 4\sqrt{3}$ 19 $k = \pm 7$

20a) 1 b) $\frac{-3}{4}$ c) $\frac{5}{2}$ d) ± 2 e) ± 2 f) $\frac{13}{4}$

21 a) $x^2 + 2x - 2$ b) $x^2 - 4x - 1$

22) 3 and 6 23) 12 and 13 24) 13 sec or 3sec

Answer key unit 3

1. a) {2, 3, 4, 5} b) {2, 3, 4, 5}
2. subset of A are ∅, {a}, {b}, {c}, {a,b}, {a,c} and {a,b,c}
3. subset of A are ∅, {a}, {a{b,c}}
4. $A \cup (B \cap C) = (A \cup B) \cap (A \cup C) = \{1, 2\}$
5. a) $n(A) = 5$ b) $n(A \cap B) = 5$ c) $n(A \cup B) = 6$
6. a) $A \cap B = \{2\}$ b) $A \cap C = \{\{3,4\}\}$ c) $B \cap C = \emptyset$ d) ∅
7. $A \Delta B = B \setminus A$
8. a) $A \Delta B = \emptyset$ b) $A \setminus B = \emptyset$
9. a) {1, {2}, {1,2},{1,2,3}} b) $A \cap B = \{1\}$
   c) $A \setminus B = \{\{2\},\{1,2\}\}$ d) $B \setminus A = \{\{1,2,3\}\}$
10. $A \setminus B = [0, 1]$
11. a) $B' \setminus A' = \emptyset$ b) $A' \Delta B' = (A' \setminus B) \cup (B' \setminus A')$
    $= A' \setminus B' \cup \emptyset$
    $= A' \setminus B'$
12. $(A \cup B) \setminus C = (A\ B)\ C' = C' \cap (A \cup B)$
    $= (C' \cap A) \cup (C' \cap B)$
    $= (A \setminus C) \cup (B \setminus C)$
    $= \{1, 2, 3, 4, 6\}$
13. a) $(A \setminus B) \cup (B \cap C) = B - (B \cap C) + B \cap C = B$
    b) $(B \setminus C) \cup C = B - (B \cap C) + C = B + C - (B \cap C) = B \cup C$
14. a) $n(A') = n(U) - n(A) = 12 - 5 = 7$, b) $n(B) = 4$
15. a) 9 b) 2 c) 15

Answer key unit 4

1. a) $R = \{(1, 1), (2, 3), (3, 5), (4, 7)\}$
   b) Domain of $R = \{1, 2, 3, 4\}$ c) Range of $R = \{1, 3, 5, 7\}$
2. a) $R = \{(1, 9), (2, 6), (3, 3)\}$
   b) Domain = {1, 2, 3} c) Range = {3, 6, 9}
3. a) $R = \{(3, 1), (4, 2)\}$
   b) Domain = {3, 4} c) Range = {1, 2}

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UNIT FOUR

Relation and function Relations

Definition: A relation is a set whose elements are ordered pairs
Example: R = {(1, 2), (3, 4)}

- Relation the way things are related.
- Relating phrase like
  "is smaller than"
  "is greater than"
  "is multiple of"
  "is factor of"
  " is father of"
  " is son of"

Definition: If A and B are sets, **a relation R** from A to B is any subset of $A \times B$ if and only if $R \subseteq A \times B$

Definition: If R is relation from A to A then R is relation on A.
i.e. $R \subseteq A \times A$.

Domain: The set of all first coordinates of elements of R.
i.e. **Domain of R** = $\{x: (x, y) \in R$ for some $y\}$

Range: The set of all 2nd coordinates of elements of R
i.e. **Range of R = {y: (x, y) ∈ R for some x}**

Illustrative example

1. Let A = {1, 2, 3, 4, 5, 6, 7, 8}
   B = {2, 4, 6, 8}. If R is relation from A to B such that
   $R = \{(x, y): y = 2x - 4\}$

Then:

i) a) List all elements of relation R.
   b) Find the domain of relation R. c) Find the range of relation R.
ii) Suppose the relation R is from B to A, then
   a) list all elements of R b) list the domain of R
   c) list the range of R.

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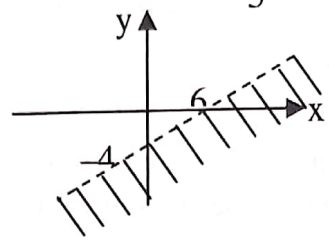
9. Sketch the graph of relation: $R = \{(x,y): 2x - 3y > 12\}$

Solution: First solve for y.

$$2x - 3y > 12 \Rightarrow -3y > -2x + 12, \Rightarrow y < \frac{2}{3}x - 4$$

$\Rightarrow R=\{(x,y): y < \frac{2}{3}x - 4\}$ ← shade **below** the boundary line



- The boundary line is broken
- To find y – intercept, set $x = 0$
  $\therefore$ y – intercept, $y = -4$
- To find x – intercept, set $y = 0$
  $\Rightarrow 2x - 0 = 12 \Rightarrow x = 6$

Graph of $R = \{(x,y): 2x - 3y > 12\}$ • Domain of $R = \{x: x \in (\infty,\infty)\}$

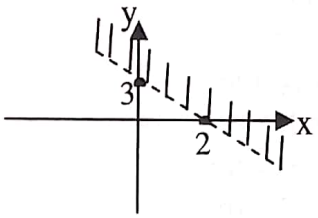
$\left\{(x, y) : y < \frac{2}{3}x - 4\right\}$ •Range of $R = \{y: y \in (\infty,\infty)\}$

10. Sketch the graph of relation: $R = \{(x,y): 3x + 2y \geq 6\}$

Solution: First solve for y.

$$\text{Thus, } 3x + 2y > 6 \Rightarrow 2y \geq -3x + 6 \Rightarrow y \geq \frac{-3}{2}x + 3$$

$\Rightarrow R=\{(x,y): y \geq \frac{-3}{2}x+3\}$



- Therefore, the solution set of R is the set of all point (x, y) that lies **above** the boundary line.
- The boundary line $y = \frac{-3}{2}x+3$ is solid.
- To find y – intercept set $x = 0 \quad \Rightarrow y = \frac{-3}{2}(0) + 3 = 3$
- To find x – intercept, set $y = 0 \qquad 3x + 2(0) = 6 \Rightarrow x = 2$

Graph of $R = \{(x,y): 3x + 2y \geq 6\}$ • Domain of $R = \{x: x \in (-\infty,\infty)\}$

$\left\{(x, y) : y \geq \frac{-3}{2}x + 3\right\}$ •Range of $R = \{y: y \in (-\infty,\infty)\}$

11. Let $R = \{(x,y): x + y \leq 4, x - y \geq -1 \text{ and } y \geq -1\}$

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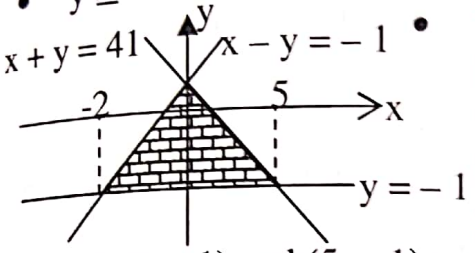
Then a) sketch the graph of R. b) domain and range of R.

Solution: First solve for y, thus

- $x + y \le 4 \Rightarrow y \le -x+4$ ⟵ shade below the boundary line
- $x-y \ge -1 \Rightarrow y- x \le 1 \Rightarrow y \le x+1$ ← **shade** below the boundary line
- $y \ge -1$ ⟵ the graph above the boundary line



- To find the domain we determine the intersection of $x - y = -1$ and $y = -1$
  $\Rightarrow x-(-1) = -1 \Rightarrow x = -2$ and
  intersection of $x+y = 4$ and
  $y = -1 \Rightarrow x - 1 = 4 \Rightarrow x = 5$

$\therefore (-2, -1)$ and $(5, -1)$ are the point where, the line $y = -1$ intersect the $x + y = -4$ and $x - y = -1$

- To find the range of R, first determine the intersection of $x - y = -1$ and $x+y = 4$

$$\Rightarrow 3 + (x + 1) = 4 \Rightarrow 2x = 3 \Rightarrow x = \frac{3}{2}, y = \frac{3}{2}+1 = \frac{5}{2}$$

$$\therefore \left(\frac{3}{2}, \frac{5}{2}\right) \text{ is the intersection of } x + y = 4 \text{ and } x - y = -1$$

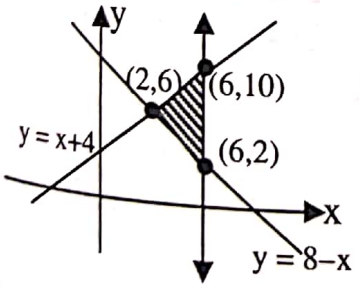
b) **Domain of R** $\{x: -2 \le x \le 5\}$ **Range of R** $= \{y: -1 \le y \le \frac{5}{2}\}$

12. Let $R = \{(x,y): y \le x + 4, y \ge 8 - x \text{ and } x \le 6\}$

a) Sketch the graph of R. b) Find the domain and range.

Solution: we known that

- $y \le x + 4$, shade below the boundary line
- $y \ge 8 - x$, shade above the boundary line
- $x \le 6$, the graph **shade** to the left of the boundary line.



- To find the domain of R we determine the intersection of $y = x + 4$ and $y = 8- x$
  $\Rightarrow 8 - x = x + 4 \Rightarrow 8 - 4 = 2x$
  $\Rightarrow 2x = 4 \Rightarrow x = 2$
- To find the range, determine the intersection of $x = 6$ and $y = 8 - x \Rightarrow y = 8 - 6 = 2$
- And $y = x + 4$ when $x = 6$, $y = 10$

b) Therefore; the domain of $R = \{x: 2 \le x \le 6\}$

- The range of $R = \{y: 2 \le y \le 10\}$

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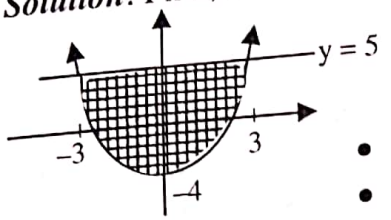
13. Let R = {(x,y): $y \geq x^2 - 4$ and $y \leq 5$} then:
a) Sketch the graph of R. b) Find the domain and range of R.

Solution: First, determine the intersection of $y = x^2 - 4$ and $y = 5$

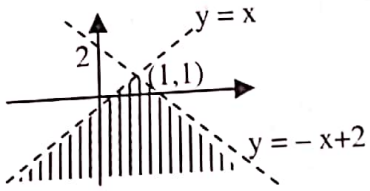
$\Rightarrow x^2 - 4 = 5 \Rightarrow x^2 = 9 \Rightarrow x = \pm 3$



- $y \geq x^2 - 4$, shade inside parabola
- $y \leq 5$, shade below the line $y = 5$
- Domain of R = {x: $-3 \leq x \leq 3$}
- Range of R = {y: $-4 \leq y \leq 5$}

14. Let R = {(x,y): $y < x$ and $x+y < 2$} which of the following is false.
A. Domain of R is $(-\infty, 1)$
B. Range of R is $(-\infty, 1)$
C. (3, –2) is solution of R
D. (–3, –1) is solution of R

Solution: Here



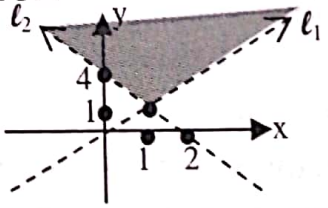
- We know that:-
- $y < x$, shade **below** the boundary line $y = x$
- $x+y < 2 \Rightarrow y < -x+2$, shade **below** the boundary line $y = -x+2$
- The intersection of $y = x$ and $y = -x+2$

$\Rightarrow x = -x+2 \Rightarrow 2x = 2 \Rightarrow x = 1, y = 1$

Thus,
A. Domain of R {x:x∈(–∞,∞)}
B. Range of R = {y:y∈(–∞,1)}
C. (3, – 2): $-2 < 3$ and $3 + -2 < 2$} True
D. (–1, –3): $-3 < -1$ and $-1 + -3 < 2$} is True ∴Answer: A

15. The shaded region (including the boundaries) for the given figure....

EGSECE



A. $y \geq x$ and $y \leq -2x+4$
B. $y \leq x$ and $y \geq -2x+4$
C. $y \geq x$ and $y \geq -2x+4$
D. $y \leq x$ and $y \leq -2x+4$

Solution: To find equation of ℓ_2, we use $y = mx + b$

- y – intercept b is 4. $\Rightarrow y = mx + 4$
- since (2,0) is on ℓ_2, so that $0 = m(2)+4$

$\Rightarrow 2m = -4 \Rightarrow m = -2$

Therefore, the equation of ℓ_2: $y = -2x+4$

Thus, the shaded region is above the boundary line, so that $y \geq$

- To find equation of ℓ_1: the line ℓ_1 is passing through (0,0) and (1,1)

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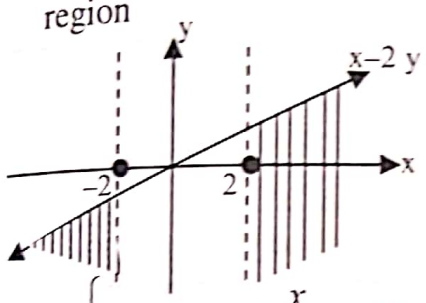
$\therefore$ equation of ℓ_1 is $y = x$.

Since the shaded region is above the boundary line y=x, therefore $y \geq x$

$\therefore R = \{(x,y): y \geq x \text{ and } y \geq -2x+4\}$ **$\therefore$ Answer: C**

16. The figure shown below is the graphical representation of a relation . . . Which one of the following relations is represented by the shaded region



A. $\left\{(x, y) : y \leq \frac{x}{2} \text{ and } x < -2\right\}$

B. $\left\{(x, y) : y \leq \frac{x}{2} \text{ and } x > 2\right\}$

C. $\left\{(x, y) : y \leq \frac{x}{2} \text{ and } |x| \geq 2\right\}$ D. $\left\{(x, y) : y \leq \frac{x}{2} \text{ and } |x| > 2\right\}$

Solution: $x - 2y = 0 \Rightarrow x = 2y \Rightarrow y = \frac{x}{2}$

- $y \leq \frac{x}{2}$, because the shaded region is below $y = \frac{x}{2}$
- $x < -2$ and $x > 2 \Rightarrow |x| > 2$

$\therefore R = \left\{(x, y) : y \leq \frac{x}{2} \text{ and } |x| > 2\right\}$. **$\therefore$ Answer: D**

Function

We known that relation is the set of ordered pair also, function is special type of a relation. Therefore, function is the set of ordered pair.

- However, every relation is not function
- But, every function is a relation.

What makes a special?

Example: $R_1 = \{(4,-2), (4,3), (1,0)\}$ it is relation but not function

$R_2 = \{(3,4),(5,7), (9,10)\}$ special types of relation: (function)

Definition: A function is a relation in which no two ordered pairs have the same first element.

Domain and Range of function

- **Domain** is the set of the first element of the ordered pairs
- **Range** is the set of 2^{nd} element of the ordered pairs.

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Example: If (3,m) and (3,n) are elements of function, what can you conclude about m and n.

Solution: m = n.

Determining whether a relation is a function or not

- *A relation is function if and only if the domain is not repeated.*

Illustrative Example

17. State whether the relation is a function. Then state the domain and the range

A. R = {(1,4),(3,6),(5,8)}
B. R = {(2,3),(2,4),(3,7)}
C. R = {(3,2),(4,3),(5,4)}
D. R = {(4,3),(0,2),(4,5)}

Solution: a) R = {(1,4),(3,6),(5,8)} is **function** because there are no ordered pairs with the same first component

- Domain = {1,3,5}, Range = {**4,6,8**}

b) R = {(2,3),(2,4),(3,7)} is **not** a **function** because (2,3) and (2,4) have the same first element

- Domain = {2,3}, Range = {3,**4,7**}

c) R = {(3,2),(4,3),(5,4)} is a **function** because no domain is repeated

- Domain = {3,4,5}, Range = {2,3,4}

d) R = {(4,3),(0,2),(4,5)} is **not function** because the domain is repeated.

- Domain = {0,4}, Range = {2,3,5}

18. State whether the relation is function. Give the domain and range.

A. A={(x,y):y is a child of x}
B. B = {(x,y):y is mother of x}
C. C={(x,y): y is a brother of x}
D. D = {x,y}:x is father of y

Solution: A) Domain = set of parents

Range = set of child (people)

A = {(x,y): y is a child of x} is not function since some parents have more than one child.

B) Domain = set of child (people)

Range = set of mothers.

B = {(x,y): y is mother of x} is a function, since each person has one natural mother,

Note: B = {(x,y): x is mother of y} is not a function. Because some mother have more than one son (daughter)

C) Domain = set of people. Range = set of boy

Not function since some people have more than one brother.

D) Exercise left for you

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19. State whether the relation is a function.
a) A = {(x,y): y is factor of x} b). B={Ix,y):x is multiple of y}
c) C = {(x,y):x is factor of y} d) D={Ix,y):y is multiple of x}

Solution: a) Not function let x = 8, then {(8,2), (8,4)...} not function.
b) Not function: Let {(6,3),(6,2)}
c) Not function d) Not function

20. State whether the relation is a function
a) R = {(x,y):y=|x|} b) R = {(x,y):x=|y|}
c) R = {(x,y):y= x^2} d) R = {(x,y):x= y^2}
e) R = {(x,y):x =4} f) R = {(x,y): y = 4}
g) R = {(x,y):y>2x+3} h) R = {(x,y): y = 2x+3}
i) R = {(x,y):x+y=6} j) R = {(x,y): x = $(y+3)^2$}

Solution: Here
a) R = {x,y}:y = |x|} is a function
b) R = {(x,y): x = |y|} is not function
Let y = -3 ⇒ x = |-3| = 3 ⇒ (3, –3)
Let y = 3 ⇒ x = |3| = 3 ⇒ (3,3)
∴ R = {(3, –3), (3,3)} is not function because the domain is repeated
c) R = {x,y}:y = x^2}is function
d) R = {(x,y): x = y^2} is not function
Let y = -3 ⇒ x = $(-3)^2$ = 9 ⇒ (9, –3)
y = 3 ⇒ x = $(3)^2$ = 9 ⇒ (9,3)
∴ Elements of R = {(9, – 3),(9,3)} is not function because the domain is repeated
e) R = {(x,y):x =4} not function
R = {(4,1), (4,2),(4,3). . . } not function

Note: R={(x,y): x=C, where C is any real number} is not function

f) R = {(x,y): y = 4} is function
g) R = {(x,y):y>2x+3}is not function
Let R = {(1,7), (1,8), (-2,0),(-2,3). . .}
h) R = {(x,y):y = 2x + 3} is function
i) R = {(x,y):x+y=6} is function
j) R = {(x,y): x = $(y+3)^2$}is not function
Let y = –1, x=r, ⇒ (4, – 1)
Let y = – 5, x = $(–5+3)^2$ = 4 ⇒(4–5)
∴R = {(4, –1(, (4 – 5)} not function

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Functional notation

- *A function f is mapping of a relation from set A to set B*, we write: A→B
- If $x \in A$ and $y \in B$, then denoted by $y = f(x)$
- $y = f(x)$ read as: "y is function of x" "f of x" "f at x"
- If $(x,y) \in f$ *means*, $f = \{(x,y): y = f(x)\}$
- If $(a,b) \in f$ *means*, $f = \{(a,b): b = f(a)\}$
- Domain of $f = \{x: x$ is the set of all possible input$\}$.
- Range off $= \{y: y$ is the set of all outputs$\}$.

Finding functional values

- $f(x)$ shows the y – value for the given x and shows that the value can be found when an x – value is known.

Example: Let $y = 3x^2 + 4$, the same as

$f(x) = 3x^2 + 4$. Replace $f(x)$ in y

- $f(2)$ means insert 2 for x

$\Rightarrow f(2) = 3(2)^2+4 = 16 \longleftarrow$ when $x = 2$, $y = 16$

$f(1) = 3(1)^2 + 4 = 7 \longleftarrow$ when $x = 1$, $y = 7$

Illustrative Example

21. Let $f(x) = \sqrt{x^2 - 4}$, evaluate

a) $f(3)$ b) $f(4)$ c) $f(-4)+f(6)$ d) $f(1)$

Solution: $f(a)$ means inset 'a' for x

a) $f(3) = \sqrt{3^2 - 4} = \sqrt{9-4} = \sqrt{5}$

b) $f(4) = \sqrt{4^2 - 4} = \sqrt{16-4} = \sqrt{12} = 2\sqrt{3}$

c) $f(-4)+f(6) = \sqrt{(-4)^2 - 4}+\sqrt{6^2 - 4} = \sqrt{12}+\sqrt{32} = 2\sqrt{3}+4\sqrt{2}$

d) $f(1) = \sqrt{(1)^2 - 4} = \sqrt{-3} \longleftarrow$ not real number this shows, $x = 1$ is not in the domain of f.

Note: Domain of $f = \{x: x^2 - 4 \geq 0\} = \{x: x < -2 \text{ or } x > 2\}$

22. If $f(x) = 3x+\sqrt{x-3}$ then find

a) $f(4)$ b) $f(7)$ c) $f(2)$ d) domain of f

Solution: Here

a) $f(4) = 3(4) + \sqrt{4-3} = 12 + \sqrt{1} = 12 + 1 = 13$

b) $f(7) = 3(7) + \sqrt{7-3} = 21 + \sqrt{4} = 21 + 2 = 23$

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c) $f(2) = 3(2) + \sqrt{2-3} = 6 + \sqrt{-1}$ ⟵ not real number because 2 is not in the domain of f.

d) we know that $\sqrt{\ }$ →defined on non – negative real number.
(Radical must be non – negative)
∴Domain of f = {x:x – 3 ≥0} = {x:x ≥3}
Range of f = {y : y≥9}.

23. Let $f(x) = \begin{cases} 2x+3 \text{ if } x<-4 \\ x^2+2 \text{ if } -4 \le x \le 5 \\ x^3 \quad \text{if} \quad x>5, \text{ then evaluate} \end{cases}$

a) f(3) b) f (– 6) c) f(7) d) f(–1) + f(–5) + f(10)

a) $f(3) = (3)^2 + 2 = 9+2 = 11$ because $f(x) = x^2 + 2$ when $-4 \le x \le 5$
b) $f(-6) = 2(-6) + 3 = -9$, we known that $f(x)\ 2x + 3$ when $x < -4$
c) $f(7) = 7^3 = 343$, we know that $f(x) = x^3$ when $x > 5$
d) $f(-1)+f(-5)+f(10) = ((-1)^2+2)+(2(-5)+3)+10^3 = 3-7+1000 = 996$

Finding the Domain of a function

- *The domain of function f* is the largest set of real numbers for which the value of f(x) is real number.
- *If x is in the domain of a function*, we say that **f is defined at x** or **f(x) exist.**
- *If x is not in the domain of* f, we say that **f is not defined at x.**

Example: Let $f(x) = \dfrac{x+3}{x^2-4}$, we see that

$$f(0) = \frac{0+3}{0-4} = \frac{-3}{4}$$

∴ f is defined at x = 0

However, $f(2) = \dfrac{2+3}{2^2-4} = \dfrac{5}{4-4} = \dfrac{5}{0}$ ← not exist (not defined)

$$f(-2) = \frac{-2+3}{4-4} = \frac{1}{0} \leftarrow \text{not defined at } x = -2$$

Thus, we conclude that f is defined at all real number except 2, and - 2
∴ The domain of f = {x:x∈IR\{2, – 2}}
That is the domain of f is the set of all real number except 2 and – 2.

24. Find the domain and range of the following

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a) $f(x) = \sqrt{4 - x}$ b) $f(x) = \sqrt{x - 4}$ c) $f(x)\ \sqrt{x + 3}$

d) $f(x) = \sqrt{x^2 - 9}$ e) $f(x) = \sqrt{25 - x^2}$ f) $f(x) = \sqrt{x^2 - 4}$

Solution: Remind that radical expression is defined on non – negative real number

a) Domain of $f = \{x: 4 - x \geq 0\} = \{x: 4 \geq x\} = \{x: x \leq 4\}$
   Range of $f = \{y: y \geq 0\}$
b) Domain of $f = \{x: x - 4 \geq 0\} = \{x: x \geq 4\}$
   Range of $f = \{y: y \geq 0\}$
c) Domain of $f = \{x: x + 3 \geq 0\} = \{x: x \geq -3\}$
   Range of $f = \{y: y \geq 0\}$
d) Domain of $f = \{x: x^2 - 9 \geq 0\} = \{x: x \leq -3 \text{ or } x \geq 3\}$
   Range of $f = \{y: y \geq 0\}$
e) Domain of $f = \{x: 25 - x^2 \geq 0\} = \{x: -5 \leq x \leq 5\}$
   Range of $f = \{y: y \geq 0\}$
f) **Exercise left for you**

25. Find the domain and range of f.

a) $f(x) = \dfrac{1}{\sqrt{x+4}}$ b) $f(x) = \dfrac{1}{\sqrt{x-3}}$ c) $f(x) = \dfrac{1}{\sqrt{x+5}}$

Solution:

a) Domain of $f = \{x: x + 4 > 0\} = \{x: x > -4\}$
   Range of $f = \{y: y \geq 0\}$
b) Domain of $f = \{x: x - 3 > 0\} = \{x: x > 3\}$
   Range of $f = \{y: y > 0\}$
c) **Exercise left for you**

26. Find the domain and range of f

a) $f(x) = 3x + 4$ b) $f(x) = |x - 2| + 3$ c) $f(x) = |x+1| - 3$

Solution:

a) Domain of $f = \{x: x \in R\}$ Range of $f = \{y: y \in R\}$
b) Domain of $f = \{x: x \in R\}$ Range of $f = \{y: y \geq 3\}$
c) Domain of $f = \{x: x \in R\}$ Range of $f = \{y: y \geq -3\}$

27. Let $f(x) = 5x - 10$, for what values of x

a) $f(x) = 100$ b) $f(x) = 25$ c) $f(x) = k$

Solution: a) $f(x) = 5x - 10$, since $f(x) = 100$
Therefore, $5x - 10 = 100$ implies $5x = 110$

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Thus, $x = \frac{110}{5} = 22$

b) $5x - 10 = 25$ because $f(x) = 25 \Rightarrow 5x = 35, \Rightarrow x = \frac{35}{5} = 7$

c) $5x - 10 = k \Rightarrow 5x = 10 + k \therefore x = \frac{10+k}{5} = 2 + \frac{1}{5}k$

28. If $f(x) = x^3 + Ax^2 + 4x - 5$ and $f(-1) = 10$. What is the value of A?

Solution: $f(-1) = (-1)^3 + A(-1)^2 + 4(-1) - 5 = 10$

$\Rightarrow A - 1 - 4 - 5 = 10 \Rightarrow A - 10 = 10$

$\therefore A = 10 + 10 = 20$

29. If $f(x) = \frac{2x - A}{3x + 4}$ and $f(1) = 6$, What is the value of A

Solution: $f(x) = \frac{2(1) - A}{3(1) + 4} = 6 \Rightarrow \frac{2 - A}{7} = 6$

$\Rightarrow 2 - A = 42$, therefore $A = -40$

Combination of Function

Here is some operation on functions. Functions like numbers, can be **added subtracted, multiplied** and **divided**.

The general definitions are given next.

- If f and g are functions, then **f + g, f – g, f.g and** f/g are the function defined by :
  - i) $(f+g)(x) = f(x) + g(x)$ ⟵ sum of functions
  - ii) $(f-g)(x) = f(x) - g(x)$ ⟵ difference of function
  - iii) $(f \cdot g)(x) = f(x) \cdot g(x)$ ⟵ product of function
  - iv) $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}, g(x) \neq 0$ ⟵ Quotients of function.

- **The domain of**: f +g , f – g and f· g is the intersection of the **domain** of **f** and **the domain** of **g**.
- **The domain of**: $\frac{f}{g}$ is the intersection of the domain of **f** and the domain of **g** but **g ≠ 0**

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Illustrative Example

30. Let $f(x) = \frac{x}{x-2}$ and $g(x) = \frac{3x}{x+4}$

i) Find, a) (f+g)(x) b) (f – g) (x) c) (f·g)(x) d) $\left(\frac{f}{g}\right)(x)$

ii) Find the domain of f + g, f – g and f·g

iii) Find the domain of $\frac{f}{g}$ and $\frac{g}{f}$

Solution: Here

i) a) $(f+g)(x) = f(x) + g(x) = \frac{x}{x-2} + \frac{3x}{x+4} = \frac{4x^2 - 2x}{(x-2)(x+4)}$

b) $(f-g)(x) = f(x) - g(x) = \frac{x}{x-2} - \frac{3x}{x+4} = \frac{-2x^2 + 10x}{(x-2)(x+4)}$

c) $(f\cdot g)(x) = f(x) \cdot g(x) = \frac{x}{x-2} \cdot \frac{3x}{x+4} = \frac{3x^2}{(x-2)(x+4)}$

d) $\frac{f}{g}(x) = \frac{f(x)}{g(x)} = \frac{x}{x-2} \cdot \frac{x+4}{3x} = \frac{x+4}{3(x-2)}$

ii) Domain f = R/{2} and Domain of g = R/{– 4}

∴ Domain of: f + g, f – g, **f · g** is the intersection of the domain of f and the domain of **g**.

∴ Domain of • f + g = IR/{2,–4} • f–g= R/{2,–4} • f · g = IR/{2,–4}

That is Domain of f + g, f – g, f · g = IR/{2, – 4} = (–∞, – 4) U(– 4,2) U(2,∞)

ii) Domain of $\frac{f}{g}$ IR/{2}, (–∞, 2) U(2,∞)

$$\frac{g}{f}(x) = \frac{3x}{x+4} \cdot \frac{x-2}{x} = \frac{3(x-2)}{x+4}, \; x \neq -4$$

∴ Domain of $\frac{g}{f}$ IR/{– 4} = (–∞,– 4) U(– 4,∞)

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31. Let $f(x) = x - 4$ and $g(x) = \sqrt{3-x}$

i) Find a) $(f+g)(x)$ b) $(f-g)(x)$ c) $(f \cdot g)(x)$

d) $\left(\frac{f}{g}\right)(x)$ e) $\left(\frac{g}{f}\right)(x)$

ii) Find the domain of each function.

Solution: The **domain of f** $= IR = (-\infty,\infty)$

The **domain of g** $= \{x: 3 - x \geq 0\} = (-\infty,3]$

$\therefore$ The intersection of domain of f and g $= (-\infty,3]$

i) Combination **ii) Domain**

a) $(f+g)(x) = f(x) + g(x) = (x-4) + \sqrt{3-x}$, **domain** $-\infty < x \leq 3$

b) $(f-g)(x) = f(x) - g(x) = (x-4) - \sqrt{3-x}$, **domain** $\{x: -\infty < x \leq 3\}$

c) $(f \cdot g)(x) = f(x) \cdot g(x) = (x-4)\sqrt{3-x}$, **domain** $\{x: -\infty < x \leq 3\}$

d) $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} = \frac{x-4}{\sqrt{3-x}}$, **domain** $\{x: -\infty < x < 3\}$

e) $\left(\frac{g}{f}\right)(x) = \frac{g(x)}{f(x)} = \frac{\sqrt{3-x}}{x-4}$, **domain** $\{x: -\infty < x \leq 3\}$

32. Let $f(x) = \sqrt{9-x^2}$ and $g(x) = \sqrt{x-1}$, Then determine the combination of the function and state their respective domain

a) $(f+g)(x)$ b) $(f-g)(x)$ c) $(f \cdot g)(x)$ d) $\left(\frac{f}{g}\right)(x)$ e) $\left(\frac{g}{f}\right)(x)$

Solution: Here

- Domain of $f = \{x: 9 - x^2 \geq 0\} = \{x: -3 \leq x \leq 3\}$
- Range of $g = \{x: x - \geq 0\} = \{x: x \geq 1\}$

$\therefore$ Domain of $(f \cap g) = \{x: 1 \leq x \leq 3\} = [1, 3]$

Combination **Domain**

a) $(f+g)(x) = \sqrt{9-x^2} + \sqrt{x-1}$ **domain** $\{x: 1 \leq x \leq 3\}$

b) $(f-g)(x) = \sqrt{9-x^2} - \sqrt{x-1}$ **domain** $\{x: 1 \leq x \leq 3\}$

c) $(f \cdot g)(x) = \left(\sqrt{9-x^2}\right)\left(\sqrt{x-1}\right)$ **domain** $\{x: 1 \leq x \leq 3\}$

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d) $\left(\frac{f}{g}\right)(x) = \frac{\sqrt{9-x}}{\sqrt{x-1}} = \sqrt{\frac{9-x^2}{x-1}}$ **domain** {x: 1< x ≤ 3}

e) $\left(\frac{g}{f}\right)(x) = \sqrt{\frac{x-1}{9-x^2}}$ **domain** {x: 1≤ x < 3}

- If P(x) = $\sqrt{x}+2$ and q(x) $\frac{1}{\sqrt[3]{x}}$, then which one of the following is true

A. (p·q)(0)=2 B. (pq)(−1)= −3 C) (p·q) (4) = 2 D) (p·q) (1) =3

Solution: A. (p·q)(0) = P(0)·g(0) doesn't exist because of (0) is not define

B. (pq)(−1)= P(−1) ·g(−1) doesn't exist because p(−1) is not defined

C. (p·q) (4) = P(4) ·g(4) = $\left(\sqrt{4}+2\right)\left(\frac{1}{\sqrt[3]{4}}\right) = \frac{4}{\sqrt[3]{4}}$

D. (p·q)(1)=P(1)· g(1)=$\left(\sqrt{1}+2\right)\left(\frac{1}{\sqrt[3]{1}}\right)$ = (1 + 2) (1) = 3 ∴ **Answer: D**

- If h{(1,2),(2,3)(3,5)} and g(x) = $\frac{3}{2}x^2 + \frac{1}{2}x$, *then* $\left(\frac{g}{h}\right)(3)$,

A. 15 B. cannot determined C. 3 D. 5

Solution: $\left(\frac{g}{h}\right)(3) = \frac{g(3)}{h(3)} = \frac{\frac{27}{2}+\frac{3}{2}}{5} = \frac{15}{5} = 3,$ ∴ **Answer: D**

33. If f = {(1,3),(2,4),(5,7),(8,9)} and g = {(1,4),(2,0),(3,5),(8,1)} then find

a) Domain of (f +g), (f · g) and $\frac{f}{g}$ b) f + g, f − g, f· g. $\frac{f}{g}$

c) $\left(2f - \frac{f}{g}\right)(8)$ d) (3f + 4g) (1)

Solution:

a) Domain of f + g and f· g are the intersection of the domain of f and domain of g.

- Domain of f = {1,2,5,8}
- Domain of g = {1,2,3,8}
- Domain of (f∩g) = {1,2,8}

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Therefore, domain of $\cdot$ (f + g); (f – g) and (f $\cdot$ g) = {1,2,8}

- Domain of $\frac{f}{g} = \{1,8\}$

b) (f+g)(1) = f(1) + g(1) = 3 +4 = 7 $\Rightarrow$ (1,7)
(f+g)(2) = f(2) + g(2) = 4 + 0 = 4 $\Rightarrow$ (2,4)
(f+g)(8) = f(8) + g(8) = 9 + 1 = 10 $\Rightarrow$ (8,10)
$\therefore$ f + g = {(1,7), (2,4), (8,10)}
(f $\cdot$ g) = {(1,f(1)$\cdot$g(1), (2, f(2)$\cdot$g(2),(8, f(8)$\cdot$g(8))
= {(1,(3)$\cdot$(4),(2, 4.0),(8, 9.1)
$\therefore$ f$\cdot$ g = {(1,12),(2,0),(8,9)}

- $\frac{f}{g} = \left\{\left(1,\frac{f(1)}{g(1)}\right),\left(8,\frac{f(8)}{g(8)}\right)\right\} = \left\{\left(1,\frac{3}{4}\right),(8,9)\right\}$
- f–g = {(1,f(1) – g(1), (2, f(2) – g(2), (8, f(8) – g(8))}
= {(1, 3 – 4), (2, 4– 0), (8, 9 – 1)}
= {(1, –1), (2, 4), (8,8)}

c) $\left(2f-\frac{f}{g}\right)(8) = 2f(8) - \frac{f(8)}{f(8)} = 2(9) - 9 = 18 - 9 = 9$

d) (3f + 4g) (1) = 3f(1) + fg(1) = 3(3) + 4(4) = 25

34. Let f(x) = 3x+2 and g(x) = 2x – 4 Find a) (2f+4g)(1)

b) $\left(3g-\frac{2f}{g}\right)(0)$ c) $f^2(-2)$

Solution: Here

a) (2f+4g)(1) =2f(1)+4(g)=2(5)+4(–2)=2

Explanation
g(0)=2(0)–4= –4
f(0)=3(0)+2= 2

b) $\left(3g-\frac{2f}{g}\right)(0) = 3g(0)\ \frac{-2f(0)}{g(0)}$

$= 3(-4)\ \frac{-2(2)}{-4} = -12 + 1 = -11$

c) $f^2(-2) = f(-2)\cdot f(-2)$
= (3(–2) +2) (3(–2)+2)
= (–4) (–4)
= 16

Explanation
f(x)= 3x + 2
f(– 2)=3–2)+2
= –4

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35. Let $f(x)=\begin{cases} x+2 \text{ if } x\le -1 \\ x^3 \text{ if } -1<x\le 1 \\ -x+3 \text{ if } x\ge 1 \end{cases}$ and $g(x)=\begin{cases} x-3 \text{ if } x\le -2 \\ -x^2 \text{ if } -2<x<3 \\ -x+4 \text{ if } x\ge 3 \end{cases}$

Find a) $(f+g)\left(\frac{-3}{2}\right)$ b) $(f+g)(2)$

Explanation
$f(x)= x+2, x<\frac{-3}{2}$
$g(x)= -x^2, -2<x<3.$

Solution: a) $(f+g)\left(\frac{-3}{2}\right)=f\left(\frac{-3}{2}\right)+g\left(\frac{-3}{2}\right)$

$$=\left(\frac{-3}{2}+2\right)+\left(-\left(\frac{-3}{2}\right)^2\right)=\frac{1}{2}\frac{-9}{4}=\frac{-7}{4}$$

a) $(f+g)(2)=f(2)+g(2)=(-2+3)+(-(2)^2)=1-4=-3$

Explanation: $f(2)=-2+3=1$ we know that $f(x)=-x+3$ if $x\ge 1$

$g(2)=-(2)^2=-4$ we know that $g(x)=-x^2$ if $-2<x<3$

36. Let $f(x)=\sqrt{x+2}$ and $g(x)=3-x$, then what is the largest set of the domain of $\frac{f}{g}$?

A. $(-\infty,\infty)$ B. $[-2,\infty)$ c) $(3,\infty)$ D. $[-2,3)\cup(3,\infty)$

Solution: $\left(\frac{f}{g}\right)(x)=\frac{f(x)}{g(x)}=\frac{\sqrt{x+2}}{3-x}$

- Domain $\left(\frac{f}{g}\right)$ = Domain $(f\cap g)$ but $g\neq 0$

  $=([-2,\infty)\cap[(-\infty,3)\cup(3,\infty)]$ since $x\neq 3$

- Domain $\left(\frac{f}{g}\right)=[-2,\infty)\cup(3,\infty)$, ∴ **Answer: D**

37. Let $f(x)=2x^2-6x$, then $f(x)+f(4)$ is equal to

A. $2x^2-6x+4$ B. $2x^2-6x+8$ C. $2x^2-6x-16$ D. $2x^2-6x-8$

Solution: $f(x)+f(4)=2x^2-6x+f(4)$

$=2x^2-6x-16$, ∴ **Answer: C**

38. If $f(x)=\sqrt{x}$, and $g(x)=\sqrt{6-x}$, then what is the domain of $(f+g)(x)$

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Solution: Domain of $f = \{x: x \geq 0\} = [0,\infty)$

Domain of $g = \{x: 6 - x\} = (-\infty,6]$

$\therefore$ Domain of $(f + g)$ = Dom $(f \cap g) = [0,6]$ $\therefore$ **Answer: B**

Graph of Linear Function

- The graph of any equation of the form $y = ax + b$ is a straight line. Such an equation is called **a linear equation**. A function defined by a linear equation is called **a linear function**.

Definition: A function is **a linear function** if and only if its domain is the set of all real numbers denoted by $f(x) = ax + b$, where a is the slope of linear function.

Note: The graph of $f(x) = ax + b$: is straight line

- A is the slope of the line
- For $a > 0$, the graph of $f(x) = ax + b$ is **increasing**
- For $a < 0$, the graph of $f(x) = ax+b$ is **decreasing**
- For $a = 0$, the graph **will be** $f(x) = b$ is **horizontal line**, it is called **constant linear function**
- If $x = 0$, then $f(0) = b \longleftarrow$ **y – intercept** where the graph crosses the y – axis
- If $f(x) = 0$, then $ax + b = 0 \Rightarrow x = \frac{-b}{a} \longleftarrow$ x – intercept

  i.e. $\left(\frac{-b}{a},0\right)$ crosses the x – axis.

To consolidate **about Linear Function** study the following table.

Function	y – intercept	x – intercept	Slope	Property
$f(x)=ax+b$	$(0,b)$	$\left(-\frac{b}{a},0\right)$	a	Increasing if $a>0$ Decreasing if $a<0$
$f(x)=-3x+5$	$(0,5)$	$\left(\frac{5}{3},0\right)$	-3	Decreasing Since $-3<0$
$f(x)4x=-3$	$(0,-3)$	$\left(\frac{3}{4},0\right)$	4	Increasing
$f(x)=-2x-6$	$(0,-6)$	$(3,0)$	-2	Decreasing
$y=\frac{1}{3}x-\frac{2}{3}$	$\left(0,-\frac{2}{3}\right)$	$(2,0)$	$\frac{1}{3}$	Increasing

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$\therefore f(x) = \frac{1}{6}x + \frac{3}{2}$, it is increasing function

- y – intercept $\left(0, \frac{3}{2}\right)$
- x–intercept (–9,0),by setting $x = \frac{-b}{a} = \left(\frac{\frac{-3}{2}}{\frac{1}{6}}\right) = -9$

b) Since f(x) = ax + b, so that

a) $\left.\begin{matrix} f(-2) = -2a + b = 7 \\ f(4) = 4a + b = -2 \end{matrix}\right\}$ *Subtracting together*

$-6a + 0 = 9 \Rightarrow a = \frac{-9}{6} = \frac{-3}{2}$

To find b, put $a = \frac{-3}{2}$ in $-2a + b = 7$

$\Rightarrow -2\left(\frac{-3}{2}\right) + b = 7 \Rightarrow b = 7 - 3 = 4$

Therefore, $f(x) = \frac{-3}{2}x + 4$, it is decreasing function

y – intercept (0,4) and x – int $\left(\frac{8}{3}, 0\right)$

c and d **exercise let for you**.

41. Let g(x) = (1 – a) x – b + 2, where a and b are positive real numbers. Which of the following is true about the graph . . . . **EGSECE 2005**

A. Its slope is – a
C. The y- intercept is 2
B. It is increasing for a < 1
D. the x – intercept is $\frac{b-2}{a-1}$

Solution: Given g(x) = (1 – a) x – b + 2 $\Leftrightarrow$ y = (1 – a)x + 2 – b

A. Slope, m = 1 – a
B. Increasing if m > 0 $\Rightarrow$ 1 – a > 0 $\Rightarrow$ a < 1
   Decreasing if m < 0 $\Rightarrow$ 1 – a < 0 $\Rightarrow$ a > 1
C. y – intercept, at x = 0 $\Rightarrow$ y = 0 + 2 – b = 2 – b

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D. x – intercept, at $y = 0 \Rightarrow 0 = (1 - a)x + 2 - b$

$\Rightarrow (1 - a)x = b - 2 \Rightarrow x = \dfrac{b-2}{1-a}$ ∴ **Answer: B**

Graph of a quadratic function

- A quadratic function is a function of the form $f(x) = ax^2+bx+c$, with $a \neq 0$.
- The domain of a quadratic function is the set of all real numbers
- The graph of all quadratic function is called **parabolas**.
- The lowest or the highest point of a parabola is called **the vertex**.
- The vertical lines passing through the vertex in each parabola, is called **the axis of symmetry**
- To graph and quadratic function $f(x) = ax^2 + bx + c$

$\Rightarrow f(x) = ax^2 + bx+c$, using completing the square, expressed as:

$$= a\left(x^2 + \frac{b}{a}x\right) + x \longleftarrow \text{factoring a from } ax^2+bx$$

$$= a\left(x^2 + \frac{b}{a}x + \frac{b^2}{4a^2}\right) + c - a\left(\frac{b^2}{4a_2}\right)$$

$$= a\left(x + \frac{b}{2a}\right)^2 + \frac{4ac-b^2}{4a},$$ based on these results, we conclude the following:

If $h = \dfrac{-b}{2a}$ and $k = \dfrac{4ac-b^2}{4a}$ then

$f(x) = ax^2 + bx + c$ is expressed as $f(x) = a(x - h)^2+k$

To graph of the equation $f(x) = a(x - h)^2 + k$, for $a \neq 0$

- Vertex: = (h,k)
- Axis of symmetry, x = h
- If $a > 0$, the graph opens upward.
- If $a < 0$, the graph open downward.
- The graph narrows as lal increase
- To obtain the graph of $f(x) = a(x- h)^2 + k$, shifting the graph of $f(x) = ax^2$
  i) h units to right if $f(x) = a(x - h)^2 + k$
  h units to left if $f(x) = a(x+ h)^2 + k$
  ii) k units up if $k > 0$

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|k| units down if $k < 0$

Let $f(x) = a(x-h)^2 + k$, for $a \neq 0$, then

1) The vertex; (h,k) is:
   i) Minimum value if $a > 0$
   ii) Maximum value of $a < 0$

If $f(x) = ax^2 + bx + c$, for $a \neq 0$, then

i) $f\left(\frac{-b}{2a}\right) = \begin{cases} \text{maximum value of f is } a < 0 \\ \text{minimum value of f is } a > 0 \end{cases}$

ii) the range of f is $\left[\frac{4ac-b^2}{4a}, \infty\right)$ if $a > 0$

$\left(-\infty, \frac{4ac-b^2}{4a}\right]$ if $a < 0$

iii) The vertex: $(h,k) = \left(\frac{-b}{2a}, \frac{4ac-b^2}{4a}\right)$

To consolidate study the following table

Function	Vertex	Axis	max/min	Range	The graph opens to
$f(x)=ax^2$	(0,0)	$x = 0$	(0,0)	$y \geq 0$ if $a>0$ $y \leq 0$ if $a<0$	Up if $a \geq 0$ down if $a \leq 0$
$f(x)=2x^2$	(0,0)	$x = 0$	min(0,0)	$[0,\infty)$	Upward
$f(x)=2x^2+3$	(0,3)	$x = 0$	min(0,3)	$[3,\infty)$	Upward
$f(x)=-2x^2+3$	(0,3)	$x = 0$	max(0,3)	$[-\infty,3)$	downward
$f(x)=(x-4)^2$	(4,0)	$x = 4$	min(4,0)	$[0,\infty)$	Upward
$f(x)=(x+3)^2-5$	(-3,-5)	$x = -3$	min(-3,-5)	$[-5,\infty)$	Upward
$f(x)=(x-2)^2+4$	(2,4)	$x = 2$	min(2,4)	$[4,\infty)$	Upward
$f(x)= -3(x-5)^2-6$	(5,-6)	$x = 5$	max(5,-6)	$[-\infty,-6)$	downward

Illustrative Example

42. Let $f(x) = 3x^2 + 24x + 50$, then
   a) Express as $f(x) = a(x-h)^2 + k$
   b) Find the vertex, axis, the minimum value and the range of function.

Solution: Before completing the square, we factor out the coefficient of from the first two terms of f(x)

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a) $f(x) = 3x^2 + 24x + 50 = 3(x^2 + 8x) + 50$
$= 3(x^2 + 8x + 16) + 50 - 48 \quad \therefore f(x) = 3(x + 4)^2 + 2$

b) Vertex = (–4,2), axis x = – 4.

- Minimum value of f is 2
- Range: = {y: y≥2} = [2,∞)

Method II. We can use $h = \frac{-b}{2a}$ *and* $k = \frac{4ac - b^2}{4a}$

$$\Rightarrow h = \frac{-24}{2(3)} = -4 \text{ and } k = \frac{4(3)(50) - (24)^2}{4(3)} = \frac{600 - 576}{12} = 2$$

$\Rightarrow f(x) = a(x - h)^2 + k = 3\ (x + 4)^2 + 2$

43. Let $f(x) = -2x^2 + 12x - 5$, then

a) Express as form $f(x) = a(x - h)^2 + k$

b) Determine: vertex, the axis, the maximum and the range.

Solution: we know that: a= –2, $h = \frac{-b}{2a}$, $k = \frac{4ac - b^2}{4a}$.

$$\Rightarrow h = \frac{-b}{2a} = \frac{-2}{2(-2)} = 3 \text{ and } k = \frac{4(-2)9 - 5) - (12)^2}{4(-2)} = -13$$

$\therefore f(x) = -2(x - 2)^2 - 13$,

b) Vertex = (3, –13) and axis, x = 3,

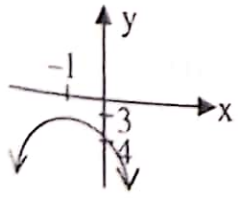
- Maximum value of f is – 13
- Range = {y: y ≤– 13}= (-∞,-13]

44. Let $f(x) = -x^2 - 2x - 4$, then

a) Express as form $f(x) = a(x - h)^2 + k$

b) Sketch the graph.

Solution: we known that: $h = \frac{-b}{2a} = \frac{2}{2(-1)} = -1$ and



$$k = \frac{4ac - b^2}{4a} = \frac{4(-1)(-4) - (-2)^2}{4(-1)} = \frac{16 - 4}{-4} = -3$$

$\therefore f(x) = -(x + 1)^2 - 3$

b) Vertex (–1,–3), axis, x = –3, range = (-∞,–3]

- The graph opens downward
- The maximum value of f is – 3

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45. Suppose there is a parabola with the vertex at (3,4) and (1, - 4) is point on that parabola.
Find the equation of parabola.

Solution: use $f(x) = a(x-h)^2 + k$, so that

$\Rightarrow f(x) = a(x-3)^2 + 4$ since vertex (h, k) = (3,4)

$\Rightarrow f(1) = a(1-3)^2 + 4 = -4 \Rightarrow 4a + 4 = -4$

$\Rightarrow 4a = -8 \Rightarrow a = -2$

$\therefore f(x) = -2(x-3)^2 + 4 = -2x^2 + 12x - 14$

46. Find the vertex, the axis, the range and maximum value of
a) $f(x) = -2x^2 + 8x$ b) $f(x) = -4x^2 + 6x$

Solution: a) $h = \dfrac{-b}{2a} = \dfrac{-8}{2(-2)} = 2$

$k = f(h) = f(2) - 2(2)^2 + 8(2) = -8 + 16 = 8$

$\therefore f(x) = -2(x-2)^2 + 8$

- Vertex (2,8), axis x = 2,
- Maximum value of, f is 8,
- Range = $(-\infty, 8]$

b) Exercise left for you

47. **Find the equation of a parabola with**
a) The vertex at (5,4) is 3 times narrower than $y = x^2$ and op downward

b) The vertex at (–3,1) that is 4 times narrower than $y = \frac{2}{3}x^2$

c) The vertex at (2,3) that is 3 times wider than $y = x^2$.

Solution: we know that: $\mathbf{f(x) = a(x-h)^2+k}$

a) $f(x) = a(x-5)^2 + 4$ since (h,k) = (5,4)
$= 3(x-5)^2 + 4$ because 3 times narrower

b) $f(x) = a(x+3)^2 + 1$, because (h,k) = (-3,1)

$= 4\left(\frac{2}{3}\right)(x+3)^2 + 1$, because 4 times narrower

$= \frac{8}{3}(x+3)^2 + 1$

c) $f(x) = a(x-2)^2 + 3$, because (h, k) = (2,3)

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$= \frac{1}{3}(x-2)^2 + 3$, because 3 times wider

48. what are the new values of b and c when the graph of $f(x) = x^2 + 4x+3$ is shifted up 2 units and to the left 3 unit (if $f(x)=x^2+bx+c$)

Solution: $f(x) = x^2 + 4x + 3$

$\Rightarrow f(x) = x^2 + 4x + 4 + 3 - 4$

$f(x) = (x + 2)^2 + 1 \longleftarrow$ add 3 to x + 2

$f(x) = (x + 5)^2 - 1 \longleftarrow$ add – 1+ 2

$f(x) = x^2 + 10x + 26$ $\quad\therefore$ b = 10, and c = 26

49. What is the equation of function, when $y= -5x^2$ shifts to the left 2 units, shifts up 3 units 4 times wider, and opens in opposite direction.

Solution: $f(x) = \frac{5}{4}(x+2)^2 + 3$, since

$-5x^2 \longrightarrow -5(x+2)^2 \longleftarrow$ shifting 2 unit to left.

$\Rightarrow f(x) = \frac{5}{4}(x+2)^2 + 3 \longleftarrow$ shifting 3 unit up and 4 times wider..

50. Let $f(x) = 4-3x-x^2$, then determine:

a) The maximum or minimum value of f.

b) The range of f.

Solution: $f(x) = 4 - 3x - x^2 = - x^2 - 3x+4$

$= -(x^2+3x)+4$

$= -\left(x^2 + 3x + \frac{9}{4}\right) + 4 + \frac{9}{4} \rightarrow \left(\frac{3}{2}\right)^2 = \frac{9}{4}$

$= -\left(x + \frac{3}{2}\right)^2 + \frac{25}{4}$

a) The maximum value is at $x = \frac{-3}{2}$ which is $\frac{25}{4}$

b) The range of $f = \{y: y \le \frac{25}{4}\} = (-\infty, \frac{25}{4}]$

51. Let $f(x)=x^2-3x$, then which of the following is not true about f(x)

A. The graph of f(x) passes through (–2,2)

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B. Its vertex is $\left(\frac{3}{2}, \frac{-9}{4}\right)$ C. Its y – intercept is 0

D. The graph of f(x) is symmetric about the line $x = \frac{3}{2}$

Solution: $f(x) = x^2 - 3x$

$$f(x) = \left(x^2 - 3x + \left(\frac{-3}{2}\right)^2\right) - \left(\frac{-3}{2}\right)^2 = \left(x^2 - 3x + \frac{9}{4}\right) - \frac{9}{4}$$

$$\therefore f(x) = \left(x - \frac{3}{2}\right)^2 - \frac{9}{4}$$

A) $f(-2) = (-2)^2 - 3(-2) = 4 + 6 = 10$

B) vertex $\left(\frac{3}{2}, \frac{-9}{4}\right)$

C) y – intercept, at $x = 0 \Rightarrow y = 0$

D) $x = \frac{3}{2}$ is axis. **∴Answer: A**

52. If a function is defined by, $y = -x^2 + 2x + 1$ then what is the range of this function? **…EGSECE 2005**

A. $\{y \in IR: y \leq 2\}$ B. $\{y \in IR: y \geq 2\}$ C) $\{y \in IR: y \leq 1\}$ D) $\{y: y \geq 1\}$

Solution: $y = -x^2 + 2x + 1 = -(x^2 - 2x) + 1$

$= -(x^2 - 2x + 1) + 1 + 1 = -(x - 1)^2 + 2$

∴ the maximum value is 2

∴ the range = $\{y: y \leq 2\}$ **∴Answer: A**

53. Let $f(x) = x^2 + bx + 1$, then what is the minimum value of f.

A. $\frac{-b^2}{2}$ B. $\frac{b^2}{2} + 1$ C. $1 - \frac{b^2}{4}$ D. $1 + \frac{b^2}{4}$

Solution: The minimum value of the function is

$$f\left(\frac{-b}{2a}\right) = f\left(\frac{-b}{2}\right) = \frac{b^2}{4} - \frac{b^2}{2} + 1 = 1 - \frac{b^2}{4}$$ **∴Answer: C**

54. An object is projected upward from the ground with an initial velocity is V_0 feet per second. Its height h(t) in feet about the ground after t second is given by $h(t) = V_0 t - 16t^2$.

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If $V_0 = 240\ \frac{ft}{\sec}$

a) What is the time it takes to attain the maximum height
b) What is the maximum height

Solution: $h(t) = 240t - 16t^2$ since $V_0 = 240\ f\frac{t}{\sec}$

- $t = \frac{-b}{2a} = \frac{-240}{2(-16)} = \frac{15}{2}\sec$

b) the maximum height will be

$$f\left(\frac{15}{2}\right) = 240\left(\frac{15}{2}\right) - 16\left(\frac{15}{2}\right)^2 = 900\,ft$$

52. The range of the function $f(x) = -4x^2+2x+1$

A. $\left(-\infty, \frac{1}{4}\right]$ B. $\left[\frac{1}{4}, \infty\right)$ C. $\left(-\infty, \frac{5}{4}\right]$ D. $\left[\frac{5}{4}, \infty\right)$

Solution: $f(x) = -4\left(x^2 - \frac{1}{2}x\right) + 1 \Leftrightarrow$

$$\Rightarrow f(x) = -4\left(x^2 - \frac{1}{2}x + \frac{1}{6}\right) + 1 + \frac{1}{4} \leftarrow \left(\frac{1}{2}\cdot\left(\frac{-1}{2}\right)\right)^2 = \frac{1}{16}$$

$$= -4\left(x - \frac{1}{4}\right)^2 + \frac{5}{4}$$

$\therefore$ The maximum value of f for all x is $y = \frac{5}{4}$

$\therefore$ the range of $f = \left\{y : y \le \frac{5}{4}\right\}$ $\therefore$ **Answer: C**

53. What is the equation of quadratic function whose vertex is (1,-5) and whose y – intercept is – 3.

Solution: vertex is (1,– 5), so that

$f(x) = a(x-h)^2 + k = a(x-1)^2 - 5, \longleftarrow h = 1, k = -5$

$f(0) = a(0-1)^2 - 5 = -3 \longleftarrow$ y – intercept is – 3

$= a - 5 = -3 \Rightarrow a = -3 + 5 = 2$

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$\therefore f(x) = 2(x-1)^2 - 5 = 2x^2 - 4x - 3$

Note: the minimum value of f is -5

54. Find the standard equation of quadratic function that has:
   a) Vertex (1,5) passing through (2,–3)
   b) Vertex (–2,4) passing through (0,2)

Solution: $f(x) = a(x-h)^2 + k$, vertex is (1,5)

$\Rightarrow f(x) = a(x-1)^2 + 5$ since $f(x) = -3$

$\Rightarrow f(2) = a(2-1)^2 + 5 \Leftrightarrow a + 5 = -3 \Rightarrow a = -8$

$\therefore f(x) = -8(x-1)^2 + 5 = -8x^2 + 16x - 3$

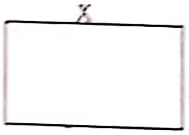
Note the maximum value of f is 5.

- The range of f is $(-\infty, 5]$

b) Exercise left for you.

55. A farmer has 200m of fence to enclose a rectangular field. Fin largest area that can be enclose.

Solution: let x = length and y = width of rectangle



- Perimeter. P: $2x + 2y = 200$

  $\Rightarrow x + y = 100 \Rightarrow y = 100 - x.$
- Area $A = xy$

  $A(x) = x(100 - x) = 100x - x^2 = -x^2 + 100x$

The maximum value occurs at $x = \dfrac{-b}{2a} = \dfrac{-100}{2(-1)} = 50$

$\therefore$ The largest area that can enclosed by 200meters of fence if

$f(50) = -(50)^2 + 100(50) = -2500 + 5000 = 2500m^2$

56. The sum of two positive numbers is 18. What is the maximum pro of number

Solution: let x and y be two number sum $x + y = 18 \Rightarrow y = 18 - x$ pro

P: xy = maximum $\quad \therefore P(x) = x(18 - x) = 18x - x^2 = -x^2 + 18x$

max. at $x = \dfrac{-b}{2a} = \dfrac{-18}{-2} = 9$

then $y = 18 - x = 18 - 9 = 9$

$\therefore xy = (9)(9) = 81 \longleftarrow$ maximum product

Supplementary Exercise on Unit Four

1. Let $A = \{0, 1, 2, 3, 4, 5, 6, 7, 8\}$

   $B = \{1, 3, 5, 7\}$. If R is relation from A to B such that $R = \{(x, y)$

   $y = 2x - 1\}$

   a) list elements of R b) list the domain of R c) list the range of R

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2. Let A = the set of natural number. If R is relation from A to A such that $R = \{(x, y): 3x + y = 12\}$
   a) List elements of R b) List the domain of R c) List the range of
3. Let $A = \{1, 2, 3, 4\}$ and $R = \{(x, y): y = x - 2\ x \in A, y \in A\}$
   Find a) Relation R b) domain of R c) range of R
4. Let $R = \{(x, y): x \geq -1, x + y \leq 4, \text{ and } x + 2y \geq 2$
   Find a) domain b) Range
5. Let $R = \{(x, y): y < 6, \text{ and } y \geq x^2 - 3\}$
   Find a) domain b) Range
6. Determine whether each of the following relations is a function or not, and give reasons for those that are not function
   a) $R_1 = \{(x, y): x = |y - 1|\}$ b) $R_2 = \{(x, y): y = |2x|\}$
   c) $R_3 = \{(x, y): x = y^2 + 3\}$ d) $R_4 = \{(x, y): y \leq x + 3$
   e) $R_5 = \{(x, y): y = 5\}$ f) $R_6 = \{(x, y): x = 0\}$
7. In each of the following find g(x) if:
   a) $f(x) = 2x + 3$ and $(f + g)(x) = x^2 - 1$
   b) $f(x) = 4x - \frac{2}{3}$ and $(g - f)(x) = \frac{1}{2}x + 1$
   c) $f(x) = \frac{1}{2}x^2 + 1$ and $(f - g)(x) = 2x - \frac{1}{2}$
8. Find the domain of f + g
   a) $f(x) = \sqrt{x+4}$ and $g(x) = \frac{1}{x+4}$ b) $f(x) = \sqrt{3-x}$ and $g(x) = \sqrt{x+1}$
9. If $f = \{(1,2),(3,-5),(4,7), \left(\frac{1}{2}, \frac{3}{2}\right)\}$ $g = \{(1,-4), (4,-4), (\frac{1}{2}, \frac{1}{2}), (5, 8)$ then
   a) f + g b) f – g c) g – f d) f·g
10. Find the domain of $g(x) = \frac{\sqrt{x-2}}{x-3}$
11. Find the domain of $f(x) = \sqrt{x-2} + \sqrt{x} - 2$
12. Determine whether each of the following is increasing or decreasing.
    a) $f(x) = 3 - 4x$ c) $x + 6y - 10 = 2$
    b) $f(x) = 2x + 6$ d) $2x - 5y + 6 = 0$
13. Find vertex, axis of symmetry, range, p standard equation $(a(x-h)^2+k$ of the following.
    a) $f(x) = 4x^2 + 6x + 1$ b) $f(x) = -x^2 + 4x - 8$

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$= (C' \cap A) \cup (C' \cap B)$

$= (A \setminus C) \cup (B \setminus C)$

$= \{1, 2, 3, 4, 6\}$

13. a) $(A \setminus B) \cup (B \cap C) = B - (B \cap C) + B \cap C = B$

b) $(B \setminus C) \cup C = B - (B \cap C) + C = B + C - (B \cap C) = B \cup C$

14. a) $n(A') = n(U) - n(A) = 12 - 5 = 7$, b) $n(B) = 4$

15. a) 9 b) 2 c) 15

Answer key unit 4

1. a) $R = \{(1, 1), (2, 3), (3, 5), (4, 7)\}$

b) Domain of $R = \{1, 2, 3, 4\}$ c) Range of $R = \{1, 3, 5, 7\}$

2. a) $R = \{(1, 9), (2, 6), (3, 3)\}$

b) Domain $= \{1, 2, 3\}$ c) Range $= \{3, 6, 9\}$

3. a) $R = \{(3, 1), (4, 2)\}$

b) Domain $= \{3, 4\}$ c) Range $= \{1, 2\}$

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4. a) domain = {x: $-1 \le x \le 6$}
b) Range = {y: $-2 \le y \le 5$}
5. a) domain = {x: $-3 < x < 3$}
b) Range = {y: $-6 \le y < 3$}
6. a) R_1 is not function.
b) R_2 is function.
d) R_4 is not function.
c) R_3 is not function.
e) R_5 is function.
f) R_6 is not function

7. a) $g(x) = x^2 - 2x - 4$

b) $g(x) = \frac{9}{2}x + \frac{1}{3}$

c) $g(x) = \frac{1}{2}x^2 - 2x + \frac{3}{2}$

8. a) Domain = {x: $x > -4$}
b) Domain = {x: $-1 \le x \le 3$}
9. a) Domain f = {1, 3, 4, ½}
Domain g = {1, 4, ½, 5}
a) $f + g$ = {(½, 2), (1, -2), (4, 3)}
b) $f - g$ = {(½, 1), (1, 6), (4, 11)}
c) $g - f$ = {(½, -1), (1, -6), (4, -11)}
d) $f \cdot g = \left\{\left(\frac{1}{2}, \frac{3}{4}\right), (1, -8), (4, -28)\right\}$

10. domain = {x: $x > 2$ and $x \notin 3$} = $(2, 3) \cup (3, \infty)$
11. domain = {x: $x \ge 2$}
12. a) f is decreasing
b) f is increasing
c) decreasing
d) increasing

13. a) $vertex\left(\frac{-3}{4}, \frac{5}{4}\right)$, $axis$, $x = \frac{-3}{4}$

Equation: $f(x) = 4\left(x + \frac{3}{4}\right)^2 + \frac{5}{4}$, $range\left[\frac{5}{4}, \infty\right)$

b) $f(x) = -(x - 2)^2 - 4$
- vertex (2, -4)
- axis, $x = 2$
- range = {y: $y \le -4$}

Answer key unit 5

1. Mean = 3.6, mode = 1, median = 3
2. mean = 1.05, mode = 0, median = 0
3. $x = 5$ and $y = 3$
4. 5.4
5. a) mean = 8
b) $\frac{5}{2}$
6. 13
7. Mean $\bar{x} = 2$, mode is 1 and 2, median = 2
8. variance = 2, S.D = $\sqrt{2}$
9. a) variance = ¼
b) $\text{var}(\sigma^2) = \frac{25}{4}$
c) $\delta = \sqrt{\frac{25}{4}} = \frac{5}{2}$
10. variance = $M(x^2) - (m(x))^2 = 19 - 9 = 10$
∴ Standard deviation = $\sqrt{10}$

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UNIT FIVE

Statistics

Statistics: is the science dealing with

- collecting data.
- Organizing data
- Presentation of data
- Analysis of data
- Interpretation of data

Collecting data: obtaining measurement or count.

Organizing data – edited, classified and tabulated.

Presentation of data - data presented by bar graph, histogram, pictograms, frequency polygon

Analysis of data – high light information for decision making

Interpretation – final image about the data or

- final decision about the data
- make judgment about eh population

statistics may be classified into two parts

- *Descriptive statistics*
- *Inferential statistics*

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UNIT SEVEN

Introduction to Polynomial

A term is a number or the product of a number and variable raised to powers. Example, $-5x$, $4a^2b$, $\frac{1}{4}y^6$, 0.2

The numerical coefficient or coefficient of a term is its numerical factor.

Definition of a polynomial function

Definition: A polynomial in x is a finite sum of terms of the form $P(x) = a_n x^n + a_{n-1}x^{n-1} + \ldots + a_1 x + a_0$, with $a_n \neq 0$ is called polynomial function invariable x of degree n.

Where n is non-negative integer (0, 1, 2, 3, … n)

Note: i) $a_n, a_{n-1}, a_{n-2}, \ldots a_1, a_0$ are called the coefficients of the polynomial

ii) a_n is leading coefficient of the polynomial and $a_n x^n$ is leading term.

iii) a_0 is the constant term.

Degree of a polynomial

· The degree of a polynomial is the greatest degree of any term of the polynomial.

Domain of polynomial

The domain of polynomial function is all real number.

Note: Constant numbers polynomial function of degree 0

Example: $6, \pi, \frac{3}{4}, \ldots$ etc are polynomial function of degree 0

- 0 is three only polynomial function that has no degree.

Illustrative example

1. Which of the following are polynomial function?
   Find: i) the degree ii) leading coefficient iii) constant term

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a) $x^5 - 3x^3 + 2x^2 + 5x + 6$

b) $P(x) = \sqrt{(x+4)^2}$

c) $P(x)=2x^{-3}+4x^2+x+4$

d) $P(x) = \dfrac{x^2+3}{x^2+3}$

e) $P(x) = \dfrac{x^2-3}{x^2-3}$

f) $P(x) = \dfrac{x^2}{x^2}$

g) $P(x) = \dfrac{x+3}{x+3}$

o) $\dfrac{(x-2)(x^2+1)}{x^2+1}$

q) $P(x) = \log_4^3$

s) $P(x) = \cos 23^0$

u) $P(x) = 3^x + 4x^2$

h) $P(x) = \pi x^4 - \sqrt{3}x^3 + \sqrt{2}$

i) $P(x) = \sqrt{5x} + x + b$

j) $P(x) = \dfrac{6x^2 + 4 - 5x^4}{8} - x^3$

k) $P(x) = \dfrac{x}{3x}$

l) $P(x) = \sqrt{6}$

m) $P(x) = 0$

n) $P(x) = \dfrac{2}{3} - 4\pi$

p) $P(x) = x^{\frac{1}{2}} + 5x + x^2$

r) $P(x) = \log_4^x$

t) $P(x) = \cos x + 2$

w) $P(x) = \sqrt{(x^2+3)^2}$

***Solution*:**

a) Polynomial function, degree 5, leading coefficient 1

b) $P(x) = \sqrt{(x+4)^2} = |x+4|$, absolute value function

∴ It is not a polynomial function.

c) $P(x) = 2x^{-3} + 4x^2 + x + 4$.

It is not polynomial because the term $2x^{-3}$ has negative exponent.

d) $P(x) = \dfrac{x^2+3}{x^2+3} = 1$, polynomial, degree 0 leading coefficient 1.

e) $P(x) = \dfrac{x^2-3}{x^2-3} = 1, \quad x \neq \pm\sqrt{3}$

- It is not a polynomial, because the domain is not real number.

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f) $P(x)=\frac{x^2}{x^2}=1,\ x\neq 0,$

$\therefore$ It is not a polynomial , because the domain is not real number

g) $P(x)=\frac{x+3}{x+3}=1,\ x\neq -3$ $\therefore$ It is not a polynomial

h) $P(x)=\pi x^4-\sqrt{3}x^3+\sqrt{2}$

- It is polynomial
- Degree 4
- Leading coefficient π
- Constant term 2

i) $P(x)=\sqrt{5x}+x+6$

- It is not polynomial because $\left(\sqrt{5x}\right)=(5x)^{\frac{1}{2}}$

j) $P(x)=\frac{6x^2+4-5x^4}{8}-x^3=\frac{-5}{8}x^4-x^3+\frac{3}{4}x^2+\frac{\log_5^2}{8}$

$\therefore$ It is polynomial

- Degree 4

- leading coefficient $\frac{-5}{8}$

- constant term $\frac{\log_5^2}{8}$

k) $P(x)=\frac{x}{3x}=\frac{1}{3},\ x\neq 0$

It is not a polynomial

l) $P(x)=\sqrt{6}$ we can write $P(x)=\sqrt{6}x^0$

- It is polynomial
- Degree 0

- leading coefficient $\sqrt{6}$

- constant term $\sqrt{6}$

m) $P(x)=0$, we can write $P(x)=0x^0+0x^{n-1}+0x^{n-2}+\ldots$

- It is a polynomial

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- It has no degree
- leading coefficient 0

n) $P(x)=\frac{2}{3}-4\pi$

- It is a polynomial
- It has degree 0
- leading coefficient $\frac{2}{3}-4\pi$

o) $\frac{(x-2)(x^2+1)}{x^2+1}=x-2$

- It is a polynomial
- It has degree 1
- leading coefficient 1
- constant term -2

p) $P(x)=x^{\frac{1}{2}}+5x+x^2$

- It is not polynomial

q) $P(x)=\log_4^3$

- It is a polynomial of constant function
- has degree 0
- leading coefficient $\log_4^3$

r) $P(x)=\log_4^x$

It is not polynomial
It is logarithm function

s) $P(x)=\cos 23^0$

- It is a polynomial function
- Degree 0
- leading coefficient $\cos 23^0$
- constant term $\cos 23^0$

t) $P(x)=\cos x+2$

It is not polynomial (Trigonometric function)

u) $P(x)=3^x+4x^2$

It is not polynomial

w) $P(x)=\sqrt{(x^2+3)^2}=|x^2+3|=x^2+3$

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- It is polynomial
- Degree 2
- leading coefficient 1
- constant term 3

2. Which of the following is a polynomial expression …(EHEECE)

A. $\sqrt{(x^2+x-1)^2}$ C. $3x^{-4}+2x^{-2}-x^{-1}+4$

B. $4\sqrt{\pi}-1$ D. $\dfrac{x^2-1}{x+1}$

Solution: A. $\sqrt{(x^2+x-1)^2} = |x^2+x-1|$. Not polynomial

B. $4\sqrt{\pi}-1$ is constant polynomial

C. not polynomial because of negative exponent

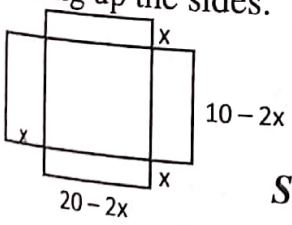
D. $\dfrac{x^2-1}{x+1}, x \neq -1$ not polynomial ∴ **Answer B**

3. Which of the following is a polynomial expression?

A) $x^2-3x+\sin x$ C) $\dfrac{2+\pi}{1+\pi^2}$

B. $\dfrac{4x^3+12x^2-x}{\pi x}$ D) $2-9x^{\frac{2}{3}}+7x^{\frac{5}{2}}+3x^{-1}$ ∴**Answer C**

4. An open box is made from piece of card board having dimension 20cm by 10cm by cutting equal square of length xcm from the corners, and turning up the sides.

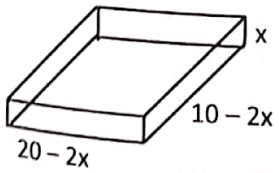


Find a) volume of the box
b) the domain of v
c) v(4)

Solution: a) $v(x) = (20-2x)(10-2x)x$

b) the domain of $v = \{x: 0<x<5\}$

c) $v(4) = (12)(2)(4) = 96\text{cm}^3$



5. Find the degree of each of the following polynomial.

a) $4x^6-3x^2+5x+7$ b) $x-7x^2+4-x^3$ c) x^4-3x^5+2

d) $2x+3$ e) π^2-3 f) 0 g) 8

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Solution: **Remember that**. *Degree of polynomial is the highest exponent of the term in one variable*

a) The highest power term is $4x^6$.
   - The exponent in this term is 6. So **the degree** is 6.

b) The highest power tem is $-x^3$
   - The exponent in this term is 3. So the **degree** is 3

c) The highest power term is x^4.
   - The exponent in this term is 4. So the **degree** is 4.

d) The highest power term is 2x.
   - The exponent in this term is 1. So the **degree** is 1.

e) The highest power term is $\pi^4 - 3$. (constant polynomial)
   - The exponent in this term is 0. So that **degree** is 0..

f) 0 is called **zero polynomial**.
   0 is the only polynomial that has **no degree**.

g) The highest power term is 8.
   - The exponent in this term is 0. So the **degree** is 0

6. In each of the following polynomial functions are **over integers,** over **rational, over real** numbers.

   a) $3x^4 - 4x^3 + 2x - 7$ b) $3x^4 - \frac{4}{3}x^3 + 2x - 7$ c) $\sqrt{2}x^3 - \frac{1}{2}x^2 + \pi$

Solution:

a) $3x^4 - 4x^3 + 2x - 7$; Here the coefficients of the polynomials are 3, −4, 2 and − 7 which are all integer.
   So that it is **polynomial over integer.**

b) The coefficient of the polynomials are 3, $\frac{-4}{3}$, 2 and − 7 which are rational number
   ∴ it is **polynomial function over rational number**

c) The coefficient of the polynomials are $\sqrt{2}, -\frac{1}{2}$ and π which are real number.
   ∴ it is polynomial function **over real number**.

Operation on Polynomial Function

A. Adding and Subtracting Polynomials

- Polynomials are added and subtracted by combining like terms.
  - **Like terms** contains the same variable factors and exponent

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Example a) $7x^4$ and $3x^4$ are like terms
b) $3x^2$ and x^5 are not like terms

The degree of the sum of two polynomial is less than or equal to the degree of the polynomial with higher degree.

Let f(x) and g(x) be two polynomial then.

i) $f(x) + g(x)$ ⟵ sum of two polynomial

ii) $f(x) - g(x)$ ⟵ the difference of two polynomial

Illustrative Example

7. Let $f(x) = x^4 - 3x^3 + 2x+6$ and $g(x) = 2x^3 + 4x - 8$ Then find
a) $f(x) + g(x)$ b) $f(x) - g(x)$ c) The degree of (f+g) and (f – g)

Solution: Here

a) $f(x)+g(x) = (x^4-3x^3+2x+6) + (2x^3 + 4x - 8)$
$= x^4 + (-3x^3 + 2x^3)+(2x+4x) + (6 - 8)$ ← **grouping like terms**
$= x^4+(-3+ 2)x^3+(2+4)\ x+ (6 - 8)$ ← **adding, their coefficient**
$\therefore f(x) + g(x) = x^4 - x^3 + 6x - 2$

- The highest power term is x^4. The exponent in this term is 4.
So the **degree** f + g is 4

b) $f(x) - g(x) = (x^4-3x^3+2x+6) - (2x^3 + 4x - 8)$
$= x^4- 3x^3 + 2x+6-2x^3 -4x+8$ ← **Removing brackets**
$= x^4+(-3x^3-2x^3)+(2x-4x)+ (6+8)$ ← **Adding like terms**
$= x^4+(-3-2)x^3+(2-4)x+14$
$= x^4-5x^3-2x+14$

c) the degree of f + g, and f – g is 4.

8. Let $f(x) = 4x^3 - 2x^2 + 1$ and $g(x) = 4x^3 + x^2 - 1$, then find
a) $f(x) + g(x)$ and degree of $f(x) + g(x)$
b) $f(x) - g(x)$ and degree of $f(x) - g(x)$

Solution:

a) $f(x) + g(x) = (4x^3-2x^2+1) + (4x^3 + x^2 - 1)$
$= (4x^3+4x^3)+(-2x^2+x^2)+(1-1)$ ← Grouping like term
$= (4 + 4)x^3 + (-2 + 1)x^2 + 0$
$= 8x^3 - x^2$
$f(x) + g(x) = 8x^3 - x^2$

- The highest power term is $8x^3$. The exponent in this term is 3.
So the **degree** of f + g is 3

b) $f(x) - g(x) = (4x^3 - 2x^2 + 1) - (4x^3 + x^2 - 1)$
$= 4x^3 - 2x^2+1- 4x^3 - x^2 +1$ ← **Removing brackets**

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$= (4x^3 - 4x^2) + (-2x^2 - x^2 + (1+1)$
$= 0 + -3x^2 + 2$
$\therefore f(x) - g(x) = -3x^2 + 2.$

The highest power term is $-3x^2$. The exponents in this term is 2. So the **degree** of $f(x) - g(x)$ is 2

9. Let $f(x) = 5x^3 - 4x^2 + x + 3$ and $g(x) = -3x^3 + x^2 - x + 1$
Then find $3f(x) + 5g(x)$ and the degree

Solution: Here

a) $3f(x) + 5g(x) = 3(5x^3 - 4x^2 + x + 3) + 5(-3x^3 + x^2 - x + 1)$
$= (15x^3 - 12x^2 + 3x + 9) + (-15x^3 + 5x^2 - 5x + 5)$
$= (15x^3 - 15x^3) + (-12x^2 + 5x^2) + (3x - 5x) + (9 + 5)$
$= 0 + -7x^2 - 2x + 14$

$\therefore 3f(x) + 5g(x) = -7x^2 - 2x + 14$ of degree 2.

10. What should be added to $x^4 - x^2 + x + 2$ to get $x^2 + x + 4$

Solution: Let $f(x) = x^4 - x^2 + x + 2$ and
Let $r(x) = x^2 + x + 4$ and $g(x) = ?$

$\Rightarrow f(x) + g(x) = r(x)$
$\Rightarrow g(x) = r(x) - f(x) = (x^2 + x + 4) - (x^4 - x^2 + x + 2)$
$= x^2 + x + 4 - x^4 + x^2 - x - 2$
$= x^4 + (x^2 + x^2) + (x - x) + (4 - 2)$

$\therefore g(x) = -x^4 + 2x^2 + 2$

Multiplication of two polynomials

- *The product of two polynomial f(x) and g(x) is found by multiplying each term of f(x) by every term of g(x) and finally adding the results.*
- **The degree of the product of two non zero polynomials** is *the sum of the degrees of the two polynomials in the product.* That is let the degree of $f(x)$ is **n** and let the degree of $g(x)$ is **m** then the degree of $f(x).g(x)$ is **n + m.**

Illustrative Example

11. Multiply each of the following polynomial
a) Multiply: $2x^3 - x^2 + x - 2$ by $3 - 2x$
b) Multiply: $2x^3 - x + 1$ by $x^2 - 2x + 3$

Solution: a) $(3 - 2x)(2x^3 - x^2 + x - 2)$
$= 3(2x^3 - x^2 + x - 2) - 2x(2x^3 - x^2 + x - 2)$
$= 6x^3 - 3x^2 + 3x - 6 - 4x^4 + 2x^3 - 2x^2 + 2x$
$= (-4x^4 + 0x^4) - (6x^3 + 2x^3) + (-3x^2 - 2x^2) + (3x + 2x) - 6$
$= -4x^4 + 8x^3 - 5x^2 + 5x - 6$

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b) $(2x^3-x+1)(x^2-2x+3)=2x^3(x^2-2x+3)-x(x^2-2x+3)+1(x^2-2x+3)$

$= 2x^5 - 2x^4 + 6x^3 - x^3 + 2x^2 - 3x + x^2 - 2x + 3$

$= 2x^5 - 2x^4 + (6x^3 - x^3) + (2x^2 + x^2) + (-3x - 2x) + 3$

$= 2x^5 - 2x^4 + 5x^3 + 3x^2 - 5x + 3.$

12. Find the coefficient of x^3 and x^5 in the product of $(2x^3-x^2+x-1)(x^3-x^2-3)$

Solution: $2x^3(x^3-x^2-3) - x^2(x^3-x^2-3) + x(x^3-x^2-3)+1(x^3-x^2-3)$

$= 2x^6-2x^5-6x^3-x^5+x^4+3x^2+x^4-x^3-3x+x^3-x^2-3$

$= 2x^6+(-2x^5-x^5)+(x^4+x^4)+(-6x^3-x^3+x^3)+(3x^2-x^2)-3x-3$

$= 2x^6-3x^5+2x^4-6x^3+2x^2-3x-3$

So terms are $-3x^5$ and $-6x^3$

$\therefore$ Coefficient of x^5 is -3

$\therefore$ Coefficient of x^3 is -6

13. Multiply: (x^m+2) by (x^n-3)

Solution: $(x^m+2)(x^n-3) = x^m(x^n-3) + 2(x^n-3)$

$= x^{m+n} - 3x^m + 2x^n - 6$

14. Multiply a) (x^2+2x+4) by $x-2$ b) (x^2-3x+9) by $(x+3)$

Solution: a) $(x^2+2x+4)(x-2) = x(x^2+2x+4) - 2(x^2+2x+4)$

$= x^3 + 2x^2 + 4x - 2x^2 - 4x - 8$

$= x^3 - 8$

b) $(x+3)(x^2-3x+9) = x(x^2-3x+9) + 3(x^2-3x+9)$

$= x^3 - 3x^2 + 9x + 3x^2 - 9x + 27$

$= x^3 + 27$

Multiplying polynomials vertically

- Another convenient method for multiplying polynomials is *to multiply vertically similar to the way we multiply real numbers.*

Illustrative Example

15. Find the product:
    a) Multiply: $2x^2+5$ by x^2-3x+4
    b) Multiply: $2x^2-3x+4$ by x^2+5x-2

Solution: First we arrange the polynomial in a vertical column. Then we multiply each term of the second polynomial by each term of the first polynomial.

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a) $x^2 - 3x + 4$

$2x^2 + 0x + 5$ ⟵ insert with o coefficient for missing term

$5x^2 - 15x + 20$ ⟵ Multiply $x^2 - 3x + 4$ by 5

$0x + 0x^2 + 0x + 0$ ⟵ Multiply $x^2 - 3x + 4$ by 0x

$2x^4 - 6x^3 + 8x^2 + 0 + 0$ ⟵ Multiply $x^2 - 3x + 4$ by $2x^2$

$2x^4 - 6x^3 + 13x^2 - 15x + 20$ ⟵ Adding vertically like terms

b) $2x^2 - 3x + 4$

$x^2 + 5x - 2$

$-4x^2 + 6x - 8$ ⟵ Multiply $2x^2 - 3x + 4$ by (-2)

$10x^3 - 15x^2 + 20x + 0$ ⟵ Multiply $2x^2 - 3x + 4$ by 5x

$2x^4 - 3x^3 + 4x^2 + 0 + 0$ ⟵ Multiply $2x^2 - 3x + 4$ by x^2

$2x^4 + 7x^3 - 15x^2 + 26x - 8$ ⟵ Adding vertically like term.

Polynomial Division theorem

Let P(x) and D(x) are **polynomials** such that $D(x) \neq 0$, then there exist unique polynomials Q(x) and R(x) such that

$P(x) = D(x) \cdot Q(x) + R(x)$ where either **$R(x) = 0$** or the degree of R(x) less than the degree of **D(x)**

i) $\underbrace{P(x)}_{dividend} = \underbrace{D(x)}_{Divisor} \cdot \underbrace{Q(x)}_{Quotient} + \underbrace{R(x)}_{Remainder}$

ii) $\frac{P(x)}{D(x)} = Q(x) + \frac{R(x)}{D(x)}$

Note: If R(x) = 0 then D(x) is a factor of P(x) (P(x) (divides exactly in to D(x))

Method of Division

- The procedure for dividing two polynomials is similar to the procedure for dividing two integers.

Example: Divide 842 by 15

```
      56  ⟵ quotient
15 | 842  ⟵ dividend
     -75  ⟵ 5 × 15 (subtract)
      92
     -90  ⟵ 6.15 (subtract)
       2  ⟵ Remainder
```

Illustrative Example

16. In each of the following, divide the polynomial P(x) by D(x) and find the quotient and the remainder in which case D(x) is a factor of P(x).

a) $P(x) = 2x^3 + 9x^2 - x - 8$, $D(x) = 2x + 3$

b) $P(x) = x^3 - 2x^2 + 3x - 18$, $D(x) = x - 3$

Solution:

a) $2x+3 \;|\; 2x^3 + 9x^2 - x - 8$

$x^2 + 3x - 5$

$2x^3 + 9x^2 - x - 8 \leftarrow \frac{2x^3}{2x} = x^2$ multiply by $2x+3$

$-(2x^3 + 3x^2)$ ⟵ subtract

$6x^2 - x - 8 \leftarrow \frac{6x^2}{2x} = 3x$ and $(3x)(2x+3)\ 6x^2 + 9x$

$-(6x^2 + 9x)$ ⟵ subtract

$-10x - 8 \leftarrow \frac{-10x}{2x} = -5x$ *and* $(-5)(2x+3)$

$-(-10x - 15)$ ⟵ subtract

$7 \leftarrow$ Remainder

Therefore, quotient: $q(x) = x^2 + 3x - 5$; remainder = 7

- Since the remainder is not zero, hence D(x) is not factor of P(x)

$$\frac{P(x)}{D(x)} = \frac{2x^3 + 9x^2 - x - 8}{2x + 3} = x^2 + 3x - 5 + \frac{7}{2x+3}$$

b) $x - 3 \;|\; x^3 - 2x^2 + 3x - 8$

$x^2 + x + 6$

$x^3 - 2x^2 + 3x - 8 \leftarrow$ observe that: $\frac{x^3}{x} = x^2$

$-(x^3 - 3x^2)$ ⟵ subtract

$x^2 + 3x - 18 \leftarrow \frac{x^3}{x} = x$

$-(x^2 - 3x)$ ⟵ subtract

$6x - 18 \leftarrow \frac{6x}{x} = 6$

$-(6x - 18)$ ⟵ subtract

$0 \leftarrow$ Remainder

∴ quotient: $q(x) = x^2 + x + 6$ and remainder = 0. Since the remainder is 0, hence $x - 3$ is factor of $x^3 - 2x^2 + 3x - 18$

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17. In each of the following pairs of polynomials. Find the quotient g(x) and r(x) that satisfy **polynomial division theorem** in which case P(x) is divisible by D(x)
   a) $P(x) = x^3 + 1$ and $D(x) = x + 1$
   b) $P(x) = x^3 + 8$ and $D(x) = x^2 - 2x + 4$.
   c) $P(x) = x^4 + 2x^3 + 3x^2 + 9x - 1$ and $D(x) = x^2 + 2x$.

Solution: **Remember that** polynomial division theorem

$$P(x) = D(x)\cdot g(x) + r(x)$$

a) $x + 1 \;\big|\; x^3 + 0x^2 + 0x + 1$ ⟵ $x^3+1 = x^3 + 0x^2 + 0x + 1$

$-(x^3 - x^2)$ ⟵ subtracting

$-x^2 + 0x + 1$ ⟵ $\dfrac{-x^2}{x} = -x$

$-(-x^2 - x)$ ⟵ subtracting

$x + 1$ ⟵ $\dfrac{x}{x} = 1$

$-(x + 1)$ ⟵ subtracting

0 ⟵ Remainder

∴ quotient: $g(x) = x^2 - x + 1$

∴ Remainder: $R(x) = 0$

Since the remainder is 0, hence $x^3 + 1$ is exactly divisible by $x + 1$. We say that $x + 1$ is factor of $x^3 + 1$, $\dfrac{P(x)}{D(x)} = \dfrac{x^3+1}{x+1} = x^2 - x + 1$

b) $x^2 - 2x + 4 \;\big|\; x^3 + 0x^2 + 0x + 8$ ⟵ $x^3+8 = x^3 + 0x^2 + 0x + 8$

$-(x^3 - 2x^2 + 4x)$ and $\dfrac{x^3}{x^2} = x$

$2x^2 - 4x + 8$ ⟵ $\dfrac{2x^2}{x^2} = 2$

$-(2x^2 - 4x + 8)$

0 ⟵ Remainder

∴ the quotient: $q(x) = x + 2$

∴ the remainder: $R(x) = 0$

∴ x^3+8 is exactly divisible by x^2-2x+4, since the remainder 0.

c) $x^2 + 2x \;\big|\; x^4 + 2x^3 + 3x^2 + 9x - 1$ ⟵ $\dfrac{x^4}{x^2} = x^2$

$-(x^4 - 2x^3)$

$3x^2 + 9x - 1$ ⟵ $\dfrac{3x^2}{x^2} = 3$

$-(3x^2 + 6x)$

$3x - 1$ ⟵ Remainder

$$\Rightarrow \frac{P(x)}{D(x)} = \frac{x^4 + 2x^3 + 3x^2 + 9x - 1}{x^2 + 2x} = x^2 + 3 + \frac{3x-1}{x^2+2x}$$

∴ the quotient: $q(x) = x^2 + 3$

∴ the remainder: $R(x) = 3x - 1$

∴ $x^4 + 2x^3 + 3x^2 + 9 - 1$ is not exactly divisible by $x^2 + 2x$ because the remainder is R(x) is not zero. Hence D(x) is not factor of P(x).

$$\frac{P(x)}{D(x)} = \frac{x^4 + 2x^3 + 3x^2 + 9x - 1}{x^2 + 2x} = x^2 + 3 + \frac{3x-1}{x^2+2x}$$

18. Divide: $\dfrac{x^{3n} + 5x^{2n} + 12x^n + 18}{x^n + 3}$

Solution: Here

$x^n + 3 \;\big|\; x^{3n} + 5x^{2n} + 12x^n + 18$ ⟵ $\dfrac{x^{3n}}{x^n} = x^{2n}$

$-(x^{3n} + 3x^{2n})$ ⟵ Subtracting

$2x^{2n} + 12x^n + 18$ ⟵ $\dfrac{2x^{2n}}{x^n} = 2x^n$

$-(2x^{2n} + 6x^n)$ ⟵ Subtracting

$6x^n + 18$ ⟵ $\dfrac{6x^n}{x^n} = 6$

$-(6x^n + 18)$

0 ⟵ Remainder

∴ the quotient: $q(x) = x^{2n} + 2x^n + 6$

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$\therefore$ the remainder: R(x) = 0

The synthetic Division

- The procedure for dividing a polynomial by a binomial of the form x – c can be condensed by a method called **synthetic division.**

Illustrative Example

19. Dividing a) $2x^3 - x^3 - 5x^2 + 3x + 6$ by x – 4

Solution: take the zeros of x – 4 which is 4 and write the coefficient of polynomial in descending power of x.

- **First multiply 2x4 = 8 ← write under then next coefficient.**

4	2	–1	–5	3	6 ←	coefficient of polynomial
	↓	8	28	92	380	
	2	7	23	95	386 ← P(4)	•Multiply 2x4 = 8
	|	|	|		↑	• Add (–1+8) = 7
	↓	↓	↓		Remainder	•Multiply 7x4 = 28
	$2x^3$	x^2	x			•Multiply 23x4 = 92

$$\therefore \frac{2x^4 - x^3 - 5x^2 + 3x + 6}{x - 4} = 2x^3 + 7x^2 + 23x + 95 + \frac{386}{x-4}$$

Note that: P(4) = 386 ← Remainder

20. Divide $3x^4 + x^3 - 5x^2 - 5x^2 - 3x - 20$ by (x – 2)

Solution: Coefficient of: $3x^4 + x^3 - 5x2 - 3x - 20 = 3,1,-5,-3,-20$ write the zero of x – 2 which is 2.

- **First multiply 2x3, then write under the next coefficient**

2	3	1	–5	–3	–20 ←	coefficient of polynomial
	↓	6	14	18	30	• Multiply 2x3=6
	3	7	9	15	10 ← Remainder	• Add: 1+6 = 7

$$\therefore \frac{3x^4 + x^3 - 5x^2 - 3x - 20}{x - 2} = 3x^3 + 7x^2 + 9x + 15 + \frac{10}{x-2}$$

21. Divide: $x^2 - 8$ by (x – 2)

Solution: Coefficient of $x^3 - 8 = x^3 + 0x^2 + 0x - 8$ is 1 0 0 – 8, and the Zeros of x – 2 = 2

2	1	0	0	–8	• Add vertical terms
	↓	2	4	8	• Multiply diagonal term by 2
	1	2	4	0 ←	Remainder P(2) = 0

$$\therefore \frac{x^3 - 8}{x - 2} = x^2 + 2x + 4$$

22. Find the value of k for which the polynomial

a) $2x^3 + 9x^2 - x - k$ is exactly divisible by 2x + 3

b) $2x^3 2x^3 - 7x^2 - x + k$ is exactly divisible by 2x + 3.

Solution: Here

x^2+3x-5

a) $2x+3 \;\big|\; 2x^3 + 9x^2 - x - k$ ← $\frac{2x^3}{2x} = x^2$

$-(2x^3 + 3x^2)$

$6x^2 - x - k$ ← $\frac{6x^2}{2x} = 3x$

$-(6x^2 + 9x)$

$-10x - k$ ← $\frac{-10x}{2x} = -5$

$-(-10x - 15)$

$15 - k$

If the dividend is exactly divisible by the divisor, then the remainder must be zero.

$\therefore 15 - k = 0 \Rightarrow k = 15$

b) Exercise left for you

23. What must be added to $x^4 + x^3 - x^2 - 3x - 2$ so that the polynomial so obtained is exactly divisible by $x^2 + 2$

Solution: Here

$x^2 + x - 3$

$x^2+2 \;\big|\; x^4 + x^3 - x^2 - 3x - 2$ ← $\frac{x^4}{x^2} = x^2$

$-(x^4 + 2x^2)$

$x^2 - 3x^2 - 3x - 2$ ← $\frac{x^3}{x^2} = x$

$-(x^3 + 2x)$

$-3x^2 - 5x - 2$ ← $\frac{-3x^2}{x^2} = -3$

$-(-3x^2 - 6)$

$-5x+4$ ← Remainder

Hence we must add – (–5x + 4) = 5x – 4 to dividend

i.e. $x^4 + x^3 - x^2 - 3x - 2 + (5x - 4) = x^4 + x^3 - x^2 + 2x - 6$ is exactly divisible by $x^2 + 2$.

Theorem on Polynomials

There are two, these are

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A. The remainder theorem
B. The factor theorem

A. The Remainder theorem

If a polynomial P(x) is divided by (x – c) then the **remainder is P(c)**, i.e
Remainder: R = P(c)

$$\boxed{P(x) = (x - c)q(x) + P(c)}$$

Illustrative Example

24. Let $P(x) = x^3 - 3x^2 + x - 5$, Find the remainder (R), when P(x) is divided by;
    a) x – 1 b) x + 2 c) 2x – 1 d) x + 1

Solution: Using the remainder theorem

a) Divisor, x–1; Then $R = P(1) = 1^3 - 3 + 1 - 5 = -6$ ← Remainder
b) Divisor, x–2. Then $R = P(-2) = -8 - 3(4) - 2 - 5 = -27$ ← Remainder
c) Divisor, 2x–1. Then $R = P\left(\frac{1}{2}\right) = \frac{1}{8} - \frac{3}{4} + \frac{1}{2} - 5 = \frac{-41}{8}$ ← Remainder
d) Divisor, x+1. Then $R = P(-1) = -1 - 3 - 1 - 5 = -10$ ← Remainder

25. Find the remainder (R) when $P(x) = (x - 2)^2(2x+1)$ is divided by:
    a) x– 1 b) x+3 c) 2x – 1 d) x

Solution: using the remainder theorem

a) Remainder (R) = $P(1) = (1 - 2)^2 (2 + 1) = 3$ ⟵ Remainder
b) $R = P(-3) = (-3-2)^2(-3(2)+1) = -125$ ⟵ Remainder
c) $R = P\left(\frac{1}{2}\right) = \left(\frac{1}{2} - 2\right)^2\left(2\left(\frac{1}{2}\right)+1\right) = \frac{9}{2}$ ⟵ Remainder
d) $R = P(0) = (0 - 2)^2 (0 + 1) = 4$ ⟵ Remainder

26. Find the value of k such that: $P(x) = x^3 + 2x^2 - 3kx - 10$ is divided by x + 3 has a remainder of 8

Solution: $P(-3) = 8 \Leftrightarrow (-3)^3 + 2(-3)^2 - 3k(-3) - 10 = 8$

$\Rightarrow -27 + 18 + 9k - 10 = 8 \Rightarrow 9k - 19 = 8$

$\Rightarrow 9k = 27$, therefore $k = \frac{27}{9} = 3$

27. When the polynomial $P(x) = ax^3 - x^2 + bx + 3$ is divided by x – 1 and x – 2, the remainder are 4 and 9 respectively. Find the value of a and b

Solution: the zeros of x – 1 is 1 so that P(1) = 4

$\Rightarrow P(1) = a - 1 + b + 3 = 4$

$\Rightarrow a + b = 4 - 3 + 1 = 2 \ldots\ldots\ldots\ldots (*)$

- $P(2) = 8a - 4 + 2b + 3 = 9$
  $\Rightarrow 8a + 2b = 9 - 3 + 4 = 10 \ldots\ldots\ldots\ldots (**)$

From (*) and (**) solving together

$$\begin{cases} a + b = 2 \\ 8x + 2b = 10 \end{cases} \Rightarrow -\begin{cases} 8a + 8b = 16 \\ 8x + 2b = 10 \end{cases}$$
$$6b = 6 \Rightarrow b = 1$$

$\therefore a = 2 - b = 2 - 1 = 1$

23. $P(x) = A(x - 1)^2 + B(x - 2)$ where A and B are constant when P(x) is divided by (x + 3) and (x – 3) the remainder are 58 and 10 respectively. Find the value of A and B.

Solution: The zeros of x + 3 is – 3, so that P(–3) = 58

$\Rightarrow P(-3) = A(-3-1)^2 + B(-3-2) = 58$

$\Rightarrow 16A - 5B = 58 \ldots\ldots\ldots (*)$

The zeros of x – 3 is 3, so. That P(3) = 10

$\Rightarrow P(3) = A(3-1)^2 + B(3-2) = 10$

$\Rightarrow 4A + B = 10 \ldots\ldots\ldots (**)$

From (*) and (**), solving together

$$\begin{cases} 16A - 5B = 58 \\ 4A + B = 10 \end{cases} \Rightarrow -\begin{cases} 16A - 5B = 58 \\ 16A + 4B = 40 \end{cases}$$
$$-9B = 18 \Rightarrow B = -2$$

$$\therefore A = \frac{10 - B}{4} = \frac{10 - (-2)}{4} = \frac{12}{4} = 3$$

24. If the polynomial $P(x) = x^3 + x^2 - 4x + 5$ and $g(x) = x^3 + 3x - 7$ have the same remainder when they are divided by (x – a)

Find the value of 'a'

Solution: P(a) = g(a) because they have the same remainder.

$\Rightarrow a^3 + a^2 - 4a + 5 = a^3 + 3a - 7$
$\Rightarrow a^2 - 4^2 - 3a + 5 + 7 = 0$ ⟵ cancelling (a^3)
$\Rightarrow a^2 - 7a + 12 = 0 \Rightarrow (a - 3)(a - 4) = 0$
$\therefore a = 3$ or $a = 4$

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25. The polynomial P(x) divided by x + 3 and x – 1 the remainder are 4 and 3 respectively. Find the remainder when P(x) is divided by (x+3)(x – 1)

Solution:

Remark. If P(x) is divided by (x – a) (x – b), then the remainder will be linear in x, which is of the form Ax + B. thus P(x) = (x – a)(x – b) Q(x) + Ax + B

That is the remainder will be linear equation because P(x) divided by Degree 2

Therefore, $P(x) = (x+3)(x-1)Q(x) + (Ax+B)$

$\Rightarrow P(-3) = (-3+3)(-3-1)Q(-3) + (-3A+B) = 4$

$\Rightarrow 0 + -3A + B = 4 \Rightarrow -3A+B = 4 \ldots\ldots (*)$

$\Rightarrow P(1) = (1+3)(1-1)Q(1) + (A+B) = 3$

$\Rightarrow 0 + A + B = 3 \ldots\ldots\ldots\ldots\ldots (**)$

From (*) and (**) we have

$$-\begin{cases} -3A + B = 4 \\ A + B = 3 \end{cases}$$

$$-4A = 1 \Rightarrow A = \frac{1}{4} \text{ and } B = 3 - A = 3 - \left(\frac{1}{4}\right) = \frac{13}{4}$$

Therefore, the remainder will be $-\frac{1}{4}x - \frac{13}{4}$

26. **Find the value of K.**
   a) If $8x^3 - 4x^2 - 7x + k$ is divided by (x – 1) The remainder is 5
   b) If $x^3 - kx^2 + 2x + 5$ is divided by (x + k) The remainder is 7

Solution:

a) Exercise left for you.

b) $P(-k) = 7 \Leftrightarrow -k^3 + k(-k)^2 + 2(-k) + 5 = 7$

$\Rightarrow -k^3 + k^3 - 2k + 5 = 7$

$\Rightarrow -2k + 5 = 7 \Rightarrow -2k = 2$

Therefore, k = –1

31(b) Let P and g be polynomials given by $P(x) = 6x^5 + 4x^4 - 2x^3$ and $g(x) = 3x^2 + 2x - 1$. For $x \neq -1$ and $x \neq \frac{1}{3}$, *then* $\frac{P(x)}{g(x)}$ is equal to

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A. $2x^3$ B. $3x^2$ C. $3x^2 + 2x + 1$ D. $2x^3 + 1$

Solution: $\frac{P(x)}{g(x)} = \frac{6x^5 + 4x^4 - 2x^3}{3x^2 + 2x - 1} = \frac{2x^3(3x^2 + 2x - 1)}{3x^2 + 2x - 1} = 2x^3$

∴Answer: A

- One immediate consequence of the remainder theorem is the **factor theorem**

Factor theorem

For each polynomial P(x), of degree n≥1.

- If P(a) = 0, the x – a is factor of P(x).

Conversely, if (x – a) is a factor of P(x), then P(a) = 0

Illustrative Example

27. Let $f(x) = x^3 - 9x^2 + 26x - 24$ use factor theorem to determine whether
   a) x – 1 is a factor of f(x). b) x – 2 is a factor of f(x).
   c) x + 1 is a factor of f(x).

Solution: $f(x) = x^3 - 9x^2 + 26x - 24$

a) **If x - 1 is a factor of f(x), then f(1) must be zero**

$\Rightarrow f(x) = 1^3 - 9(1)^2 + 26(1) - 24 = -6 \neq 0$

Therefore, x – 1 is not factor of f(x).

b) If x – 2 is a factor of f(x), then f(2) must be zero

$\Rightarrow f(2) = (2)^3 - 9(2)^2 + 26(2) - 24 = 0$

Therefore, x – 2 is a factor of f(x).

c) $f(-1) = (-1)^3 - 9(-1)^2 + 26(-1) - 24 = -42 \neq 0$

Therefore, x + 1 is not a factor of f(x).

28. Find the value of k such that:
   a) x + 2 is a factor of $P(x) = kx^3 + x^2 + k^2x + 3k^2 + 11$.
   b) x – 1 is a factor of $P(x) = k^2x^3 - 4kx + 3$.
   c) x + 2 is a factor of $P(x) = k^2x^4 - 2kx + 1$.

Solution:

a) Let $P(x) = kx^3 + x^2 + k^2x + 3k^2 + 11$

Since x + 2 is a factor of P(x), we must have P(–2) = 0

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$\Rightarrow k(-2)^3 + (-2)^2 + k^2(-2) + 3k^2 + 11 = 0$

$\Rightarrow -8k - 2k^2 + 3k^2 + 15 = 0 \Leftrightarrow k^2 - 8k + 15 = 0$

$\Rightarrow (k-3)(k-5) = 0$

Therefore, $k = 3$ or $k = 5$

b) $P(1) = k^2 - 4k + 3 = 0$

$\Rightarrow (k-3)(k-1) = 0$

Therefore, $k = 3$ or $k = 1$

c) **Exercise left for you**

29. In the following two polynomials, find the value of **a** if $(x + a)$ is a factor of each of the following.

a) $x^3 + ax^2 - 2x + a + 4$ b) $x^4 - a^2x^2 + 3x - a + 3$

c) $x^6 + ax^5 + x^4 + ax^3 + 3x - a + 2$ d) $x^5 - a^2x^3 + 2x + a + 1$

Solution: a) Let $P(x) = x^3 + ax^2 - 2x + a + 4$

Since $(x + a)$ is factor of $P(x)$ we must have $P(-a) = 0$

$\Rightarrow -a^3 + a(-a)^2 + 2(a) + a + 4 = 0$

$\Rightarrow -a^3 + a^3 + 3a + 4 = 0 \Leftrightarrow 3a + 4 = 0$

Therefore, $a = \dfrac{-4}{3}$

b) Let $P(x) = x^4 - a^2x^2 + 3x - a + 3$

Since $(x + a)$ is factor of $P(x)$ we must have $P(-a) = 0$

$\Rightarrow (-a)^4 - a^2(a^2) + 3a - a + 3 = 0$

$\Rightarrow a^4 - a^4 + 2a + 3 = 0 \Leftrightarrow 2a + 3 = 0$

c) **Therefore,** $a = \dfrac{-3}{2}$ and d **exercise left for you**

30. Find the number **a** and **k** such that $x + 2$ is a factor of $P(x) = ax^3 - 2x^2 + kx - 18$ and $P(-1) = 8$

Solution: since $x + 2$ is factor of $P(x) \Rightarrow P(-2) = 0$

$\Rightarrow P(-2) = a(-2)^3 - 2(-2)^2 + k(-2) - 18 = 0$

$\Rightarrow -8 - 2k = 26 \ldots\ldots\ldots\ldots\ldots\ldots$ (*)

$\therefore P(-1) = a(-1)^3 - 2(-1)^2 + k(-1) - 18 = 8$

$\Rightarrow -a - k = 28 \ldots\ldots\ldots\ldots\ldots\ldots$ (**)

From (*) and (**) we have

$$\begin{array}{r} \left\{\begin{array}{l} 4a + k = -13 \\ a + k = -28 \end{array}\right. \\ \hline 3a = 15 \Rightarrow a = 5 \end{array}$$

31. Find h and k so that the polynomial $x^3 - hx^2 - 13x + k$ has $(x - 1)$ and $(x - 3)$ as factor.

Solution: Let $f(x) = x^3 - hx^3 - 13x + k$. since $x - 1$ and $x + 3$ are factors of $f(x)$,

Therefore, $f(1) = 0$ and $f(-3) = 0$

$\Rightarrow f(1) = 1 - h - 13 + k = 0 \Leftrightarrow k - h = 12 \ldots\ldots\ldots$ (*)

$\Rightarrow f(-3) = -27 - 9h + 39 + k = 0$

$\Rightarrow k - 9h = -12 \ldots\ldots\ldots\ldots\ldots\ldots\ldots$ (**)

From (*) and (**) we have

$$\begin{array}{r} \left\{\begin{array}{l} k - h = 12 \\ k - 9h = -12 \end{array}\right. \\ \hline 8h = 24 \Rightarrow h = 3, k = 15 \end{array}$$

32. Find the value of **m** and **n** in the polynomial $2x^2 + mx^2 + nx - 14$ such that $(x - 1)$ and $(x + 2)$ are its factor.

Solution: **Exercise left for you (Tip!)**

33. For what value of **k** is the polynomial $2x^4 + 3x^3 + 2kx^2 + 3x + 6$ exactly **divisible** by $(x + 2)$

Solution: Let $P(x) = 2x^4 + 3x^3 + 2kx^2 + 3x + 6$

Since **P(x)** is divisible **(x+2)** $\Rightarrow P(-2) = 0$

$\Rightarrow 2(-2)^4 + 3 + (-2)^3 + 2k(-2)^2 + 3(-2) + 6 = 0$

$\Rightarrow 8k + 32 - 24 - 6 + 6 = 0$

$\Rightarrow 8k - 8 = 0 \Rightarrow 8k = 8$

Therefore, $k = 1$

34. For what value of **k**, such that the denominator divide the numerator exactly.

a) $\dfrac{x^4 - 5x^3 - kx^2 + 18k + 18}{x - 3}$ b) $\dfrac{5x^3 - x^2 + 4x + k}{1 - 5k}$

Solution: Let $P(x) = x^4 - 5x^3 - kx^2 + 18k + 18$. Since $(x - 3)$ divides $P(x)$ exactly, $\Rightarrow P(3) = 0$

$\Rightarrow P(3) = 3^4 - 5(3)^3 - k(3)^2 + 18k + 18 = 0$

$\Rightarrow 81 = 135 - 9k + 18k + 18 = 0 \Rightarrow 9k = 36$

Therefore, k = 4

b) **Exercise left for you (TiP!)** (Answer, $k = \dfrac{4}{5}$)

35. Find the value of **k** and **m** if $(x+2)$ is a common factor of

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$f(x)= x^4+kx^3+2x^2-20x+m$ and $g(x) = x^4+(k-5)x^3+(k-3)x^2+m$

Solution: $x+2$ is a factor of $f(x)$ and $g(x)$

$\Rightarrow f(-2) = 16-8k+8+40+m = 0$

$\Rightarrow m-8k = -64$ .....(*)

$\Rightarrow g(-2) = 16+(k-5)(-8)+(k-3)(4)+m = 0$

$\Rightarrow -8k+40\ 4k-12+m = -16$

$\Rightarrow m-4k = -44$ .....(**)

From (*) and (**) we have

$$\begin{array}{l} m-8k=-64 \\ -\left\{ m-4k=-44 \right. \\ \hline -4k=-20 \Rightarrow k=5 \end{array}$$

Therefore, $k = 5$ and $m - 24$

36. Find a polynomial function of degree 3 such that $f(1) = 24$ and $x-2, x+1$ and $x-3$ are factors of the polynomial

Solution: Let $f(x) = k(x-2)(x+1)(x-3)$ and $f(1) = 24$

$\Rightarrow f(1) = k(1-2)(1+1)(1-3) = 24$

$\Rightarrow 4k = 24$, **Therefore**, $k = \dfrac{24}{4} = 6$

So that , $f(x) = 6(x-2)(x+1)(x-3) = 6x^3-24x^2+6x+36$

37. **Use the factor theorem to verify the following statement.**

a) $x-c$ is a factor of x^n-c^n, for every positive integer '**n**'

b) $x+c$ is a factor of x^n-c^n, for every positive **even** integer '**n**'

c) $x+c$ is a factor of x^n+c^n, for every positive **odd** integer '**n**'

Solution: a) Let $P(x) = x^n-c^n$. By factor theorem $P(x)$ is divisible by $x-c$, only when $P(c) = 0$

$\Rightarrow P(c) = c^n-c^n = 0$, for all positive integer

$\therefore x-c$ is factor of x^n-c^n, for all positive integer

b) Let $P(x) = x^n-c^n$. By factor theorem $P(x)$ is exactly divisible by $x+c$, only when $P(-c) = 0$

$\Rightarrow P(-c) = (-c)^n-c^n = \begin{cases} 0 \text{ if n is even} \\ -2c^n \text{ if n is odd} \end{cases}$

$\therefore x+c$ is a factor of x^n-c^n, only when n is even positive integer.

c) Let $f(x) = x^n+c^n$. By factor theorem $f(x)$ is exactly divisible by $x+c$, only when $f(-c) = 0$

$\Rightarrow f(-c) = (-c)^n+c^n = 0$, if n is odd positive integer

Zeros of a polynomial function

- If f is a polynomial function and c is a real number for which $f(c) = 0$ then c is called **zero of f.**
  - a) c is an x – intercept of the graph of f.
  - b) $(x-c)$ is a factor of f

Illustrative Example

38. Find the zeros of function

a) $f(x) = \dfrac{x}{4}+\dfrac{x+1}{3}-\dfrac{x+2}{3}$ b) $f(x) = \dfrac{x}{4}-\dfrac{x+2}{3}-\dfrac{x+2}{3}$

Solution: The expression is the same as

a) $f(x) = 0 \Rightarrow \dfrac{x}{4}+\dfrac{x+1}{3}-\dfrac{x+2}{3} = 0$

$\dfrac{3x+4x+4}{12} = \dfrac{x+2}{3}$

$\dfrac{7x+4}{12} = \dfrac{4x+8}{12}$ ⟵ Multiplying by 4

$\Rightarrow 7x+4 = 4x+8 \Rightarrow 7x-4x = 8-4$

$3x = 4$, **therefore,** $x = \dfrac{4}{3}$ is the zeros of $f(x)$

b) $f(x) = 0 \Rightarrow \dfrac{x}{4}-\dfrac{x+2}{3}-\dfrac{x+2}{3} = 0 \Rightarrow \dfrac{x}{4}-\dfrac{x+2}{3} = \dfrac{x+2}{3}$

$\Rightarrow \dfrac{3x-4x-8}{12} = \dfrac{4x+8}{12} \Rightarrow -x-8 = 4x+8$

$\Rightarrow -5x = 16 \Rightarrow x = \dfrac{-16}{5}$

Therefore, $x = \dfrac{-16}{5}$ is the zero of $f(x)$.

39. Find the zeros of function

a) $f(x) = x^2-9$

b) $f(x) = x^4-2$

c) $f(x) = x^2+9$

d) $f(x) = x^4-5x^2+4$

e) $f(x) = x^3-5x+4$

f) $f(x) = 6x^4-5x^2-4$

Solution: Here

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a) $f(x)=0 \Rightarrow x^2-9=0 \Rightarrow (x-3)(x+3)=0$

$\therefore x=3$ or $x=-3$

$\therefore$ the zeros of $f(x)=x^2-9$ are -3 and 3.

b) $f(x)=0 \Rightarrow x^4-2=0 \Rightarrow (x^2-\sqrt{2})(x^2+\sqrt{2})=0$

$\Rightarrow x^2-\sqrt{2}=0 \Rightarrow x=-\sqrt[4]{2}$ or $x=\sqrt[4]{2}$

Therefore, the real zeros of $f(x)=x^4-2$ are $-\sqrt[4]{2}$ and $\sqrt[4]{2}$.

c) $f(x)=x^2+9$ has no real zeros that makes $f(x)=0$ because $f(x) \geq 9$ for all x

d) $f(x)=0 \Rightarrow x^4-5x^2+4=0 \Rightarrow (x^2)^2-5x^2+4=0$

Let $U=x^2 \Rightarrow U^2-5U+4=0$

$\Rightarrow (U-4)(U-1)=0 \Rightarrow U=4$ or $U=1$

Since $U=x^2$, so that $x^2=4$ or $x^2=1$

$\therefore x=0, \pm 2$ or $x=\pm 1$

Therefore, $-2, 2, -1$ and 1 are zeros of $f(x)$.

e) $f(x)=0 \Rightarrow x^3-5x+4=0$, since $x=1$ is zeros of $f(x)$.

To find all other zeros divide x^3-5x+4 by $x-1$

Using synthetic division: it is same as x^3+0x^2-5x+4

Coefficient of x^3-5x+4 is $1, 0, -5, 4$

1	1	0	−5	4
	↓	1	1	−4
	1	1	−4	0

$\Rightarrow x^2+x-4=0$, using quadratic formula

$$x=\frac{1\pm\sqrt{1-4(1)(-4)}}{2(1)}=\frac{-1+\sqrt{17}}{2} \text{ or } \frac{-1-\sqrt{17}}{2}$$

Therefore, $1, \dfrac{-1-\sqrt{17}}{2}$ and $\dfrac{-1+\sqrt{17}}{2}$ are zeros of f.

f) $f(x)=0 \Rightarrow 6x^4-5x^2-4=0 \Rightarrow 6(x^2)^2-5x^2-4=0$

Let $U=x^2$. Then $6U^2-5U-4=0$

$\Rightarrow 6U^2-8U+3U-4=0$

$2U(3U-4)+1(3U-4)=0$

$(2U+1)(3U-4)=0$

$2U+1=0$ or $3U-4=0$

$U=\dfrac{-1}{2}$ or $U=\dfrac{4}{3}$

Since $U=x^2$, so that $x^2=\dfrac{4}{3}$ or $x^2=\dfrac{-1}{2}$

$$x=\pm\sqrt{\frac{4}{3}}=\pm\frac{2}{\sqrt{3}}=\frac{\pm 2\sqrt{3}}{3}$$

Therefore, the real zeros of $f(x)$ are $\dfrac{-2\sqrt{3}}{3}$ and $\dfrac{2\sqrt{3}}{3}$

20. Find a polynomial function $f(x)$ of degree 3, that has the indicated zeros satisfying the given condition.

a) $-1, 2, 3$ and $f(-2)=80$ c) $-5, 2, 4$ and $f(3)=-24$

b) $0, -4, 3$ and $f(2)=-36$ d) $1, 2-\sqrt{3}, 2+\sqrt{3}$ and $f(2)=-3$

Solution: we know that if c is zeros of f, then $x-c$ is a factor of f. this means that $f(c)=0$

a) $x+1, x-2$, and $x-3$ are factor of f.

$\therefore f(x)=k(x+1)(x-2)(x-3)$, for $k \neq 0$

Since $f(-2)=80 \Rightarrow k(-1)(-2-2)(-2-3)=80$

$\Rightarrow -20k=80 \Rightarrow k=-4$

$\therefore f(x)=-4(x+1)(x-2)(x-3)=-4x^3+8x^2-4x-24$.

b) $x, x+4$, and $x-3$ are factor of f. as a result,

$f(x)=kx(x+4)(x-3)$, since $f(2)=-36$

$$\Rightarrow 2k(6)(-1)=-36 \Rightarrow k=\frac{-36}{-12}=3$$

$\therefore f(x)=3x^3+3x^2-36$

c) $x+5, x-2$ and $x-4$ are factor of f. as a result

$f(x)=k(x+5)(x-2)(x-4)$, since $f(3)=-24$

$\Rightarrow k(8)(1)(-1)=-24 \Rightarrow k=3$.

$\therefore f(x)=3x^3-3x^2-66x+120$.

d) $(x-1), (x-(2-\sqrt{3}))$ and $(x-(2+\sqrt{3}))$ are factor of f.

$\therefore f(x)=k(x-1)(x-(2-\sqrt{3}))(x-(2+\sqrt{3}))$, since $f(2)=-3$

$k(1)(\sqrt{3})(-\sqrt{3})=-3 \Rightarrow k=1$

$\therefore f(x)=x3-9x^2+5x-1$

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41. Find a polynomial function of degree 3.
   a) If P(–2) = P(0) = P(2) = 0 and P(3) = 15
   b) If P(–3) = P(0) = P(3) = 0 and P(–4) = –28

Solution: a) P(–2) = 0 ⇒ x + 2 is factor of P(x)
P(0) = 0 ⇒ x is factor of P(x)
P(2) = 0 ⇒ x – 2 is factor of P(x)
As a result, P(x) = k(x) (x+2) (x – 2), since P(3) = 15
⇒ k(3)(5)(1) = 15 ⇒ k = 1
∴ P(x) = $x^3 - 4x$
b) x+3, x, and x – 3 are factor of P(x)
∴ P(x) = kx(x+3) (x–3), since P(–4) = –28
⇒ –4k(–1) (–7) = –28 ⇒ k = 1
∴ P(x) = $x^3 - 9x$

Zeros and their multiplicities

If the same factor (x – c) occurs more than once, then c is called repeated or **Multiple zero of f.**

Definition: If $(x - c)^k$ is a factor of a polynomial f and $(x - c)^{k+1}$ is not factor of f, then c is called **a zero of multiplicity k of f**

To illustrate,
Let f(x) = $(x - 2)^3 (x + 4)^2 (x + 5)x^7$ Find all zero of f(x) and state its multiplicity

Solution: **0,** 2, –4 and –5 are zeros of f(x)
- The factor x – 2 occurs **3 times** in f(x)
  ∴ 2 has multiplicity 3
- The factor x + 4 occurs **2 times** in f(x)
  ∴ –4 has multiplicity 2
- The factor x + 5 occurs **one times** in f(x)
  ∴ –5 has multiplicity 1
- The factor x occurs **7 times** in f(x)
  ∴ 0 has multiplicity 7

Illustrative Example

42. Find all zeros of f(x) and state its multiplicity
   a) f(x) = $(x+2)(x-1)^3(x-4)^5$
   b) f(x) = $x^3 (x+\sqrt{2}-1)^4$
   c) f(x) = $x^2(2x+1)(3x-2)^3$
   d) f(x) = $(x^2 - 9)(x^2 + 6x+9)$
   e) f(x) = $(4x^2 - 3)^2$
   f) f(x) = $x^3 + 6x^2 + 9x + 4$

Solution:
a) f(x) = $(x+2) (x-1)^3 (x-4)^5$, all the zeros are, –2, 1, 4
- –2 has multiplicity 1
- 1 has multiplicity 3
- 4 has multiplicity 5

b) f(x) = $x^3 (x+\sqrt{2}-1)^4$, all the zeros are, 0, and $-\sqrt{2}+1$
- 0 has multiplicity 3
- $-\sqrt{2}+1$ has multiplicity 4

c) f(x) = $x^2 (2x+1) (3x-2)^3$, the zeros are, 0, $\frac{-1}{2}$ *and* $\frac{2}{3}$
- 0 has multiplicity 2
- $\frac{-1}{2}$ has multiplicity 1
- $\frac{2}{3}$ has multiplicity 3

d) f(x) = $(x^2-9)(x^2 + 6x+9) = (x-3)(x+3)(x+3)^2 = (x - 3)(x+3)^3$ the zeros of f are, 3 and –3
- 3 has multiplicity 1
- –3 has multiplicity 3

e) f(x) = $(4x^2 - 3) = (2x-\sqrt{3})^2 (2x+\sqrt{3})^2$, the zeros are, $\frac{\sqrt{3}}{2}$ *and* $\frac{-\sqrt{3}}{2}$
- $\frac{-\sqrt{3}}{2}$ has multiplicity 2
- $\frac{\sqrt{3}}{2}$ has multiplicity 2

f) f(x) = $x^3 + 6x^2 + 9x + 4 = (x+1)(x^2 + 5x + 4) = (x+1)(x+1)(x+4)$ The zeros are, – 1 and – 4
- –1 has multiplicity 2
- – 4 has multiplicity 1

43. Find a polynomial function of degree 4 with leading coefficient 1 such that both –4 and 3 are zeros of multiplicity 2.

Solution: f(x) = $(x + 4)^2(x - 3)^2$

44. Find a polynomial P(x) of degree 5 such that 1 and 2 are the zeros of multiplicity 2 and 3 respectively, and P(–1) = –54

Solution: P(x) = $k(x - 1)^2 (x - 2)^3$, since P(–1) = –54
⇒ $k(-1-1)^2(-1-2)^3 = -54$

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$\Rightarrow -108k = -54 \Rightarrow k = \frac{1}{2}$ $\therefore P(x) = \frac{1}{2}(x-1)^2(x-2)^3$.

The location theorem:

- *If f is a polynomial function such that f(a) and f(b) have opposite signs then there is a real number* **c** *between* **a** *and* **b** *such that f(c) = 0*

- The location theorem makes no statement about f if f(a) and f(b) are **both positive** or **both negative**.

We conclude, the function has no roots between **a** *and* **b**.

45. Use the location theorem to verify that f(x) has a zero between a and b
   a) $f(x) = 2x^3 - 3x^2 - 16x + 24$; $a = -3$, $b = -2$
   b) $f(x) = 12x^3 - 16x^2 - 3x - 4$
      i) $a = -3$, $b = -2$
      ii) $a = -2$, $b = -1$
      iii) $a = -1$, $b = 0$
      iv) $a = 0$, $b = 1$

Solution: a) $f(x) = 2x^3 - 3x^2 - 16x + 25$

- When $a = -3 \Rightarrow f(-3) = a\,2(-27) - 3(9) - 16(3) + 24 = -9$
- When $b = -2 \Rightarrow f(-2) = 2(-8) - 16(-2) + 24 = 28$

Since f(−3) is negative and (−2) is positive

Therefore, the function f has root (zero) between −3 and −2

b) $f(x) = 12x^3 + 16x^2 - 3x - 4$, using table of values

x	−3	−2	−1	0	1
f(x)	−175	−30	3	−4	21

This table of values show that, the function has:

i) No roots between −3 and − 2, **because** f(−3) and f(−2) both negative
ii) f has roots between − 2 and − 1 **because** f(−2) and f(−1) have positive sign.
iii) f has roots between −1 and 0
iv) f has roots between 0 and 1

46. If f is a polynomial function given by $f(x) = 2x^3 + 4x^2 + 5x - 9$, then this function has root between _____. . . . . **EGSECE 2005**
   A. 1 and 2 B. −1 and 0 C. −1 and −2 D. 0 and 1

Solution: $f(x) = 2x^3 + 4x^2 + 5x - 9$

A) $f(1) = 2 + 4 + 5 - 9 = 2$ and $f(2) = 16 + 16 + 10 - 9 = 33$
$\Rightarrow$ f has no roots between 1 and 2

B) $f(-1) = -2 + 4 - 5 - 9 = -12$ and $f(0) = -9$
$\Rightarrow$ f has no roots between −1 and 0

C) $f(-1) = -12$ and $f(-2) = -16 + 16 - 10 - 9 = -19$
$\Rightarrow$ f has no roots between −1 and −2

D) $f(0) = 0 + 0 + 0 - 9 = -9$ and $f(1) = 2$
Therefore, f has roots between 0 and 1 **∴Answer: D**

47. which of the following interval may not contain a zero (root) of $f(x) = x^4 - 7x^2 + 10$
   A. (−2, −1) B. (−1,0) C. (1,2) D. (2,3)

Solution: $f(x) = x^4 - 7x^2 + 10$

A) $f(-2) = 16 - 28 + 10 = -2$ and $f(-1) = 1 - 7 + 10 = 4$
Therefore, f has roots between − 2 and − 1

B) $f(-1) = 4$ and $f(0) = 10$
$\Rightarrow$ f has no roots between − 1 and 0

C) $f(1) = 1 - 7 + 10 = 4$ and $f(2) = 16 - 28 + 10 = -2$
Therefore, f has roots between 1 and 2

D) $f(2) = -2$ and $f(3) = 18 - 63 + 10 = 28$
Therefore, f has roots between 2 and 3 **∴Answer: B**

48. If f is a polynomial function given by $f(x) = 2x^3 - 3x^2 - 16x + 24$, then this function has a root (zeros) between _____
   A. −2 and −1 B. −1 and 0 C. 0 and 1 D. 1 and 2

Solution:

A. $f(-2) = 28$ and $f(-1) = 35 \Rightarrow$ f has no roots between −2 and −1
B. $f(-1) = 35$ and $f(0) = 24 \Rightarrow$ f has no roots between −1 and 0
C. $f(0) = 24$ and $f(1) = 7 \Rightarrow$ f has no roots between 0 and 1
D. $f(1) = 7$ and $f(2) = -4$ **Therefore** f has roots between 1 and 2

∴Answer: D

49. Locate the zeros (roots) of f(x) between successive integers.
   a) $f(x) = 8x^3 + 4x^2 - 2x - 1$, for $-2 \le x \le 2$
   b) $f(x) = 2x^3 - x^2 - 6x + 3$, for $-2 \le x \le 2$

Solution: a) $f(x) = 8x^3 + 4x^2 - 2x - 1$ using table of value

x	−2	−1	0	1	2
f(x)	−55	−3	−1	9	75

This table of values show that, f has roots **between 0 and 1**

b) Exercise left for you.

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Rational Root Test

Here, the result, called **rational roots test**, provides information about the rational zeros (roots) of a polynomial **with integer coefficients.**

Theorem: Rational root test

Let $f(x)\ a_n x^n + a_{n-1} x^{n-1} + \ldots . a_1 x + a_0$ is a polynomial **with integer coefficients.**

If $\frac{P}{q}$ is a rational number, in its lowest terms is **the zeros of f(x)**, then

i) P is factor of $\mathbf{a_0}$.

ii) q is factor of $\mathbf{a_n}$.

In short: All the possible rational zeros $\left(\frac{P}{q}\right)$

$$= \frac{\text{factors of constant term } a_0}{\text{factors of the leading coefficient } a_n}$$

Illustrative Example

50. In each part, find i) all possible rational zeros.
ii) all the rational zeros

a) $f(x) = x^3 - 6x^2 + 12x - 8$
b) $f(x) = x^3 + x^2 - 4x - 4$
c) $f(x) = x^3 - 7x + 6$
d) $f(x) = x^3 + 3x^2 - 4x - 12$
e) $f(x) = 2x^3 + 11x^2 - 7x - 6$
f) $f(x) = 12x^3 - 32x^2 + 25x - 6$
g) $f(x) = 3x^4 + 14x^3 + 14x^2 - 8x - 8 = 0$
h) $f(x) = x^3 - 4x - 2$
i) $f(x) = x^3 + 3x^2 - 4x + 6$

Solution: If $\frac{P}{q}$ is to be a rational zero of f(x), by rational root (zero) theorem **p** = factor of $\mathbf{a_0}$ and **q** = factor of $\mathbf{a_n}$

- Since all the coefficients are integers

a) Factor of P: $\pm 1, \pm 2, \pm 4, \pm 8$ ⟵ **factor of 8**
Factor of q: ± 1

i) All possible rational zeros $\frac{P}{q}$ are $\pm 1, \pm 2, \pm 4, \pm 8$.

(all these are candidates to be elected for zeros of these 8 possible candidates for rational zeros, at most 3 candidates can be the winner (the zeros) of f

ii) **Using factor theorem:**
$f(-1) = -1 - 6 - 12 - 8 = -27 \neq 0$
$\therefore x = -1$ is not zeros of f(x)
$f(2) = 8 - 24 + 24 - 8 = 0$
Therefore, $x = 2$ is the only rational zeros

b) $f(x) = x^3 + x^2 - 4x - 4$
Factor of $4 = \pm 1, \pm 2, \pm 4$ and factor of $1 = \pm 1$

i) all possible zeros $\frac{P}{q} = \frac{\text{factor of } 4}{\text{factor of } 1}$ $\pm 1, \pm 2, \pm 4$

ii) **Using factor theorem** of these 6 possible, at most 3 can be zeros of f(x)
$\Rightarrow f(1) = 1 + 1 - 4 - 4 = -6 \neq 0$ implies $x = 1$ is not the zeros of f(x)
$\Rightarrow f(-1) = 0, f(-2) = 0$ and $f(2) = 0$
Therefore, all the rational zeros are $-1, -2$ and 2

c) $f(x) = x^3 - 7x + 6$,
factor of $6 = \pm 1, \pm 2, \pm 3, \pm 6$ and factor of $1 = \pm 1$

i) all possible rational zeros $= \frac{\text{factor of } 6}{\text{factor of } 1} = \pm 1, \pm 2, \pm 3, \pm 6$

These are candidates for the zeros of f(x)
- $f(-1) = -1+7+6 = 12 \pm 0$ implies $x = -1$ is not zeros of f(x)
- $f(1) = 1-7+6 = 0 \Rightarrow x = 1$ is the zeros of f(x)

To find the other zeros divide $f(x) = x^3 - 7x + 6$ by $x - 1$, we get $x^2 + x - 6$, that is $x^3 - 7x + 6 = (x - 1)(x^2 + x - 6) = (x-1)(x - 2)(x + 3)$

ii) **Therefore,** 1, 2 and -3 are all the rational zeros of f(x)

d) $f(x) = x^3 + 3x^2 - 4x - 12$

i) All possible rational zeros $\left(\frac{P}{q}\right) = \frac{\text{factor of } 12}{\text{factor of } 1}$
$= \pm 1, \pm 2, \pm 3, \pm 4, \pm 6, \pm 12$

These are candidates for the zeros of f(x)
- $f(-1) \neq 0, f(1) = -12 \neq 0$
- however, $f(-3) = -27 + 27 + 12 - 12 = 0$
$\Rightarrow x = -3$ the zeros of f(x)

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- **To find the other zeros.** Divide $f(x) = x^3 + 3x^2 - 4x - 12$ by factor $x + 3$, we get

$x^3 + 3x^2 - 4x - 12 = (x+3)(x^2 - 4) = (x+3)(x-2)(x+2)$

ii) Therefore, $-3, -2$ and 2 are the rational zeros of $f(x)$

e) $f(x) = 2x^3 + 11x^2 - 7x - 6$

i) All possible rational zeros $\left(\frac{P}{q}\right) = \frac{factor\ of\ 6}{factor\ of\ 2} = \frac{\pm1, \pm2, \pm3, \pm6}{\pm1, \pm2}$

$= \pm1, \pm\frac{1}{2}, \pm2, \pm\frac{3}{2}, \pm3, \pm6$

These are candidates for the zeros of $f(x)$

- $f(-1) = -2 + 11 + 7 - 6 = 10 \neq 0 \Rightarrow x = -1$ is not the zeros
- $f(1) = 2 + 11 - 7 - 6 = 0 \Rightarrow x = 1$ is the zeros of $f(x)$

To find the other zeros, divide $f(x)$ by $x - 1$ we get, $2x^2 + 13x + 6$, that is

$2x^3 + 11x^2 - 7x - 6 = (x - 1)(2x^2 + 13x + 6)$

$= (x - 1)(2x+1)(x+6)$

ii) Therefore, all the rational zeros are $1, \frac{-1}{2}$ and -6

f) $f(x) = 12x^3 - 32x^2 + 25x - 6$

- factor of 6 are $\pm1, \pm2, \pm3, \pm6$
- factor of 12 are, $\pm1, \pm2, \pm3, \pm4, \pm6, \pm12$

i) All possible rational zeros $\left(\frac{P}{q}\right) = \frac{factor\ of\ 6}{factor\ of\ 12} =$

$\pm1, \pm2, \pm3, \pm6, \pm\frac{1}{2}, \pm\frac{1}{3}, \pm\frac{1}{4}, \pm\frac{1}{6}, \pm\frac{1}{12}, \pm\frac{2}{3}, \pm\frac{3}{2}, \pm\frac{3}{4}$.

There are 24 candidates for the zeros of $f(x)$.

- $f\left(\frac{1}{2}\right) = 0$, $f\left(\frac{2}{3}\right) = 0$, and $f\left(\frac{3}{2}\right) = 0$

ii) Therefore, $\frac{1}{2}, \frac{2}{3}$ and $\frac{3}{2}$ are all the rational zeros of $f(x)$

g) $f(x) = 3x^4 + 14x^3 + 14x^2 - 8x - 8$

- factor of $8 = \pm1, \pm2, \pm4, \pm8$
- factor of $3 = \pm1, \pm3$

i) All the possible rational zeros $\left(\frac{P}{q}\right) = \frac{factor\ of\ 8}{factor\ of\ 3} =$

$\pm1, \pm2, \pm4, \pm8, \pm\frac{1}{3}, \pm\frac{2}{3}, \pm\frac{4}{3}, \pm\frac{8}{3}$ let see by substitution:

$f(-2) = 0 \Rightarrow x = -2$ is the zeros of $f(x)$

$f\left(\frac{-2}{3}\right) = 0 \Rightarrow x = \frac{-2}{3}$ is the zeros of $f(x)$ by

$(x + 2)\left(x + \frac{2}{3}\right)$, we get $3x^2 + 6x - 6$; that is

$3x^4 + 14x^3 + 14x^2 - 8x - 8 = (x+2)\left(x + \frac{2}{3}\right)(3x^2 + 6x - 6)$

ii) Therefore, -2 and $\frac{-2}{3}$ are the rational zeros of $f(x)$

h) $f(x) = x^3 - 4x - 2$

- factor of 2 are $\pm1, \pm2$
- factor of 1 are ±1

i) All the possible rational zeros $\left(\frac{P}{q}\right) = \frac{factor\ of\ 2}{factor\ of\ 1} = \pm1, \pm2$

Substitute each of these number for x we obtain.

- $f(1) = 1 - 4 - 2 = -5$, $f(-1) = -1 + 4 - 2 = 1$
- $f(-2) = -8 + 8 - 2 = -2$ and $f(2) = 8 - 8 - 2 = -2$

since $f(\pm1) \neq 0$ and $f(\pm1) \neq 0$

therefore, f(x) has no rational zeros.

i) **Exercise left for you**

51. Find all rational zeros of $f(x) = x^3 + \frac{11}{6}x^2 - \frac{1}{2}x - \frac{1}{3}$

Solution: we want to find the value of x that make $f(x) = 0$

$\Rightarrow x^3 + \frac{11}{6}x^2 - \frac{1}{2}x - \frac{1}{3} = 0$

To make the coefficient integer multiply both side by 6.

$\Rightarrow 6x^3 + 11x^2 - 3x - 2$

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i) all possible rational zero of $f(x) = \frac{factor\ of\ 2}{factor\ of\ 6}$

$= \frac{1\pm,\pm2}{\pm1,\pm2,\pm3,\pm6} = 1, \pm2, \pm\frac{1}{2}, \pm\frac{1}{3}, \pm\frac{1}{6}, \pm\frac{2}{3}, \pm1, \pm2, \pm3, \pm6,$

- $f\left(-\frac{1}{2}\right) = 0$, and $f\left(\frac{2}{3}\right) = 0$

Therefore, $\frac{-1}{2}$ and $\frac{2}{3}$ are the rational zeros of f(x)

52. Find all rational zeros of $f(x) = \frac{2}{3}x^5 + \frac{7}{9}x^4 - x^3 - \frac{5}{3}x^2$

Solution: First make the coefficient integers

- $\frac{2}{3}x^5 + \frac{7}{9}x^4 - x^3 - \frac{5}{3}x^2 = 0$

Multiply both side by 9

$\Rightarrow 6x^5 + 7x^4 - 9x^3 - 15x^2 = 0$

$\Rightarrow x^2(6x^3 + 7x^2 - 14x - 15) = 0$

$\Rightarrow x^2(x+1)(6x^2 + x - 15) = 0$

$\Rightarrow x^2(x+1)(2x-3)(3x+5) = 0 \Rightarrow x=0\ x=-1, x = \frac{3}{2}$ or $x = \frac{-5}{3}$

53. What are the zeros of the polynomial function

$f(x) = x^3 + 3x^2 - x - 3$

A. -2, a and 2 C. $-2, -1$ and 3

B. -3. 2 and 3 D. $-3, -1$ and 1

Solution: **Using rational root test**

- Factor of 3: $\pm1, \pm3$
- Factor of 1: ±1
- $\frac{factor\ of\ 3}{factor\ of\ 1} = \pm1, \pm3$

$\Rightarrow f(1) = 1 + 3 - 1 - 3 = 0 \Rightarrow 1$ is zeros of f.

$\Rightarrow f(-1) = -1 + 3 + 1 - 3 = 0 \Rightarrow -1$ is zeros of f.

$\Rightarrow f(3) = 27 + 27 - 3 - 3 = 48 \pm 0$

- $F(-3) = -27 + 27 + 3 - 3 = 0 \Rightarrow -3$ is zeros of f

Therefore, $-3, -1$, and 1 are the zeros of f ∴**Answer: D**

Method II: $x^3 + 3x^2 - x - 3 = 0$

$\Rightarrow x^2(x+3) - 1(x+3) = 0$

$\Rightarrow (x^2 - 1)(x+3) = 0 \Rightarrow x = 1, -1, -3.$

54. Let f be a polynomial function in x. If the degree of f(x) is greater than or equal to 1, then what will be the remainder when f(x) is divided by x + 3 . . . .**EGSECE 2005**

A. f(–3) B. f(3) C. x – 3 D. 3

Solution: **Remainder** the remainder theorem

- If f(x) is divided by x + 3, the remainder will be f(– 3)

∴**Answer: A**

- **Factorization of some polynomial expression**

Illustrative Example

55. Factorize each of the following polynomial

a) $x^3 + y^3$ b) $x^3 - y^3$ c) $x^3 + 8$ d) $x^3 - 8$

e) $x^3 + 1$ f) $x^3 - 1$ g) $x^6 - 64$ h) $x^6 - 7x^3 - 8$

Solution: Here

a) $x^3 + y^3 = (x+y)(x^2 - xy + y^2)$

b) $x^3 - y^3 = (x-y)(x^2 + xy + y^2)$

c) $x^3 + 8 = x^3 + 2^3 = (x+y)(x^2 - 2x + 4)$

d) $x^3 - 8 = x^3 - 2^3 = (x-2)(x^2 + 2x + 4)$

e) $x^3 + 1$ ⟵ **Exercise left for you**

f) $x^3 - 1 = x^3 - 1^3 = (x-1)(x^2 x + x + 1)$

g) $x^6 - 64 = (x^3 - 8)(x^3 + 8)$

$= (x-2)(x^2 + 2x + 4)(x+2)(x^2 - 2x + 4)$

$= (x-2)(x+2)(x^2 + 2x + 4)(x^2 - 2x + 4)$

h) $x^6 - 7x^3 - 8 = (x^3 + 1)(x^3 - 8)$

$= (x+1)(x^2 + x + 1)(x-2)(x^2 + 2x + 4)$

$= (x+1)(x-2)(x^2 - x + 1)(x^2 + 2x + 4)$

Graph of polynomial function

- The graph of a polynomial function is **a pictorial representation of polynomial**
- The graph of a polynomial function is **smooth continuous curve**

Explanation: The term **smooth** means does not have a sharp corner

- The term **continuous curve** means without break or hole **or jump**

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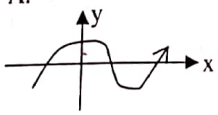
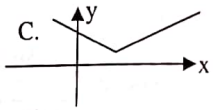
- The graph of a polynomial function has: **No gaps, No jumps and** **hole**

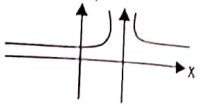
Illustrative Example

56. Which one of the following is graph of polynomial function

A. 

B.

C. 

D. 

Explanation:

B. The graph has **jumps**, so that not polynomial graph.

C. The graph has **sharp corner** , so that not polynomial function graph

D. The graph has **gap**, so that not polynomial function graph.

∴Answer

The x – and y – intercepts of a polynomial function.

i) An **x – intercepts** is a point where a curve crosses the x – ax

- The curve crosses the x – axis at any point where y = f(x) = 0
- x – intercepts of curve is the zeros of function.

ii) A **y – intercepts** is a point where a curve crosses the y – axis

- The curve crosses the y– axis at any point where x= 0

Illustrative Example

57. Find the x – and y – intercepts of f(x).

a) $f(x) = x^4 - 3x^2 - 4$ b) $f(x) = x^4 - 10x^2 + 9$

c) $f(x) = x^3 + 8$ d) $f(x)\ x^3 + x^2 - 6x$

e) $f(x)=(x+2)\ (x-3)\ (2x+7)$ f) $f(x) = x(x-1)\ (5x-2)$

Solution:

a) **To find x – intercepts**: set: $f(x) = 0$

$\Rightarrow x^4 - 3x^2 - 4 = 0 \Leftrightarrow (x^2 - 4)\ (x^2 + 1) = 0$

$\Rightarrow x^2 - 4 = 0$, however $x^2 + 1 \neq 0$ for all x

$\therefore (2,0)$ and $(-2,0)$ are x – intercepts

- **To find y – intercepts**: set $x = 0$

$\Rightarrow oy = 0 - 0 - 4 = -4$ $\therefore (0,-4)$ y – intercepts

b) $f(x) = 0 \Rightarrow x^4 - 10x^2 + 9 = 0 \Rightarrow (x^2 - 1)(x^2 - 9) = 0$

$\Rightarrow x^2 - 1 = 0$ or $x^2 - 9 = 0$

$\therefore (1,0), (-1,0), (-3,0)$ and $(3,0)$ are x – intercepts

- When $x = 0 \Rightarrow f(x) = x^4 - 0^2 + 9 = 9$

$\therefore (0,9)$ is y – intercepts

c) **x – intercepts**, but $f(x) = 0 \Rightarrow x^3 + 8 = 0 \Rightarrow (x+2)(x^2 - 2x + 4)\ x + 2$

$= 0$ however, $x^2 - 2x + 4 \neq 0$ for all x

$\therefore (-2,0)$ is the x – intercepts

- **y – intercept**, when $x = 0$

$f)x_- = 0 + 8 = 8 \Rightarrow (0,8)$ – y – intercepts

d) **x – intercepts**, set $f(x) = 0$

$\Rightarrow x^3 + x^2 - 6x = 0 \Rightarrow x(x^2 + x - 6) = 0$

$\Leftrightarrow x(x+3)(x-2) = 0 \Rightarrow x = 0, x = -3, x = 2$

$\therefore (0,0), (-3,0)$ and $(2,0)$ are x – intercepts

e) **Exercise left for you**

f) **Exercise left for you**

Note: • The graph of a polynomial function:

i) With n – **degree**, meets (crosses) the x – axis at **most n** - times

ii) **Crosses the y – axis** exactly once

iii) has at most **n – 1 turning point**

- Polynomial function of **even – degree** may not cross the x – axis

Example a) $f(x) = x^2 + 3$, $x^2 + 2x + 4$, . . . does not crosses the x – axis at all

- The graph is completely lies **above the x – axis**

b) $f(x) = x^4 + 5$, $x^4 + x + 10$ . . . etc does not crosses the x – axis

- The graph lies **above the x – axis**

c) $f(x) = -x^4 - 4, -x^4 - 5, -x^6 - 1$ . . . etc does not crosses the x – axis

- The graph lies **below the axis**

Far left and far right behavior of polynomial graph

- *The graph of polynomial function may have several up and down fluctuations*; however the graph of every polynomial function will eventually increase or decrease without bound as the graph **moves far to the left or far to the left** from the origin.

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- The **leading term $a_n x^n$** is said to be **dominate** polynomial function $P(x) = a_n x^n + a_{n-1} x^{n-1} \ldots + a_1 x + a_0$ as |x| become large, because the absolute value of $a_n x^n$ will be much larger than the absolute value of any other term.

In general, the behavior of the graph of a polynomial function as **x – decrease** without bound to the left or **as x – increase** without bound to the right can be determined:

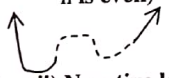
- **By its degree (even or odd)**
- **By its leading coefficient**

Let $P(x) = a_n x^n + a_{n-1} x^{n-1} + \ldots a_1 x + a_0$

- Dominant term of P(x) is $a_n x^n$
- The graph of P(x) eventually rises or falls in the following way

There are four possible cases

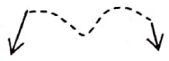
Case i) Positive leading coefficient and even degree ($a_n > 0$ and n is even)



- The graph **rises** to the left and to the right.

Example. $P(x) = x^2, x^4 + 6x, 5x^6 + 3x, \ldots$ etc

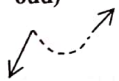
Case ii) Negative leading coefficient and even degree ($a_n < 0$ and n is even)



- The graph falls (move down) to the left and to the right

Example. $-x^2 + 3, -x^4 + 7x, -6x^4 + 5x^3, \ldots$ etc

Case iii) Positive leading coefficient and Odd degree ($a_n > 0$ and n is odd)



- The graph falls to the left and rise to the right

Example. $x^3, 5x^3 + 4x^2, 8x^5 + 6, \ldots$ etc

Case iv) Negative leading coefficient and Odd degree ($a_n < 0$ and n is odd)

- The graph rises to the left and the Falls to the right

Example. $-x^3, -6x^3 + 3x^2 + 3x^2, \ldots$ etc

Illustrative Example

58. Use the leading coefficient test to determine the left and right behavior of the graph of each polynomial function.

a) $f(x) = -4x^3 + 16x$ b) $f(x) = 2x^4 - 10x^2 + 8$ c) $f(x) = 2x^5 - 2x$

Solution:

a) because **the degree is odd** and the **leading coefficient is negative**, the graph rises to the left and falls to the right

b) because **the degree is even** and the **leading coefficient is positive**, the graph rises up to the left and to the right

c) Because the degree is odd and the leading coefficient is positive, the graph falls to the left and rise to the right.

64. Describe the far-left and far-right behavior of a polynomial function.

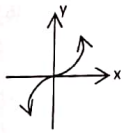
a) $f(x) = x^3$ b) $f(x) = -x^5$

c) $f(x) = x^3 - x$ d) $f(x) = -2x^3 + x^2 + 7x - 6$

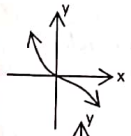
e) $f(x) = 5x^4$ f) $f(x) = x^4 - 3x^2 - 4$

g) $f(x) = -x^4 + 8x^2 + 9$

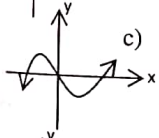
Solution:



a) Because $a_n = 1$ is positive and degree n = 3 is odd the graph of $f(x) = x^3$ falls to the left and rises to its right.



b) Because $a_n = -1$ is negative and degree n = 5 is odd the graph of $f(x) = -x^5$ rises to the left and falls to its right.



c) Because $a_n = 1$ is positive and degree n = 3 is odd the graph of $f(x) = x^3 - x$ falls to the left and rises to its right.

d) Because $a_n = -2$ is negative and degree n = 3 is odd the graph of $f(x) = -2x^3 + x^2 + 7x - 6$ rises to the left and falls to right

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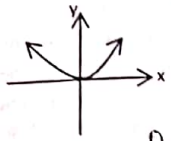
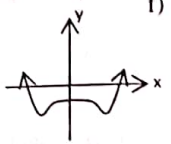
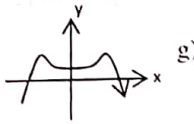


e) Because $a_n = 5$ is positive and degree n = 4 is even the graph of $f(x) = 5x^4$ rises to its left and far right.



f) Because $a_n = 1$ is positive and degree n = 4 is even the graph of $f(x) = x^4 - 3x^2 - 4$ rises to its far left and far-right.
- x-intercept, (-2, 0) and (2, 0)
- y-intercept (0, -4)
- has 3 turning point



g) Because $a_n = -1$ is negative and degree n = 4 is even the graph of $f(x) = -x^4 + 8x^2 + 9$ falls to its far left and far-right.

65. Which one of the following is not true about the graph of the polynomial function $f(x) = 2x^5 - 7x^3 - 2x - 1$.
A. As we move far from the origin, it rises upward to the right and falls down ward to the left.
B. It has most 4 turning point and intersects the x – axis atб most 5 times.
C. It passes through the origin.
D. It is a smooth and continuous curve.

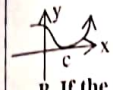
Solution: Here, $f(x) = 2x^5 - 7x^3 + 2x - 1$
A. Because $a_n = 5$ is positive and n = 5 is odd ***the graph of f goes down to its far left and rises upward to the right***
B. Because, n = 5, the graph of f has 5 – 1 = 4turning point and the graph crosses the x – axis at most n = 5 times
C. When x = 0, y = 2(0) – 5(0) + 2(0) –1 = –1
Therefore, the graph does not pass through (0,0)
D. The graph of polynomial function is smooth and continuous curve
∴ **Answer: C**

Local maximum and local minimum

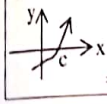
- The points at which a graph changes direction are called turning **points**.
- If the turning point is peak, then it is called **a local maximum**
- If the turning point is valley, then it is called **a local minimum**

- *A point of f that is either maximum point or minimum point is called* **a local extremum point of f.**

Multiplicity and x – intercepts
Let $(x - c)^k$ is factor of **f:**
A. If the exponent k is even, then
i) the graph of f **touches** the x – axis at x = c
ii) sign of f(x) does not change from one side to the other side of c.



iii) the graph has **local extremum** at x = c
B. If the exponent is odd, then:
i) the graph of f **crosses** the x – axis at x = c
ii) sign of f(x) changes from one side to the other side of x = c.

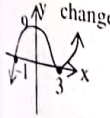


iii) the graph does not have a **local extremum** at x = c

Illustrative Example

66. Determine where the graph:
i) touches, crosses the x – axis
ii) has a local extremum.
iii) below the x – axis, Iv) above the x – axis
a) $f(x) = (x + 1)(x-3)^2$ b) $f(x) = (x+3)(x-2)^4(x-1)^3$
c) $f(x) = x^2(x-3)(x+4)$ d) $f(x) = -x^2(x^2-1)(x+1)$

Solution:
a) $f(x) = (x + 1)(x - 3)^2$
i) **The exponent of (x+1) is odd,**
Therefore, the graph crosses the x – axis at x = –1
The exponent of (x–3) is even,
Therefore, the graph touches the x – axis at x = 3, (not crosses the x – axis)
ii) The graph has **no local extremum** at x = –1 because f(x) has sign change near x = –1



- the graph **has local extremum** at x = 3 since no sign change of f(x) near x = 3, that is local minimum at x = 3

iii) **the y – intercept** , set x = 0, $f(0) = (0 + 1)(0 - 3)^2 = 9$

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- **the x – intercept** $x = -1$ and $x = 3$
- $f(x) \le 0$, when $x \le -1$,,

$\Rightarrow$ the graph lies below the x – axis on $(-\infty, -1]$

iv) **$f(x) \ge 0$, when $x \ge -1$**

$\therefore$ The graph lies above the x – axis on $[-1, \infty)$

b) $f(x) = (x+3)(x-2)^4(x-1)^3$

i) The exponent of (x+3) and (x–1) is odd,

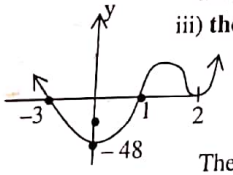
$\therefore$ the graph crosses the x – axis at $x = -3$ and $x = 1$

The exponent of (x–2) is even,

$\therefore$ the graph touches (but not cross) at $x = 2$

ii) The graph has **no local extremum** at $x = -3$ and $x = 1$, since sign change near –3 and 1

- the graph **has local extremum** at $x = 2$



iii) **the space y – intercept** , set $x = 0$

$\Rightarrow f(0) = (0+3)(0-2)^4(0-1)^3 = -48$

- **the x – intercept** are $x = -3, 2$ and 1
- **$f(x) \le 0$, when $-3 \le x \le 1$, i.e**

The graph lies below the x – axis

- $f(x) \ge 0$ when $x \le -3$ or $x \ge 1$, i.e.

The graph lies above the x – axis

c) **Exercise left for you.**

d) $f(x) = -x^2(x^2-1)(x+1)$

$= -x^2(x-1)(x+1)(x+1) = -x^2(x-1)(x+1)^2$

i) The **exponent of (–x) and (x+1) are even**

$\therefore$ the graph touches the x – axis at x=0 and x= –1(but not cross)

- **the exponent of (x–1) is odd,**

$\therefore$ the graph crosses the x – axis at $x = 1$

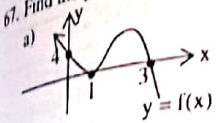
ii) The graph has **no local extremum** at $x = 0$ and $x = -1$

the graph has **no local extremum** at $x = 1$, since it has no sign change near $x = 1$,

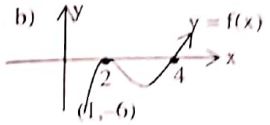
- **Because the degree is odd and the leading coefficient is negative, then the graph rises up to the left and falls down to the right**

67. Find the polynomial function of degree 3 whose graph is given below.

a)



b)



Solution: Here

a) Since $x = 1$ and $x = 3$ are x – intercepts of $f(x)$ and 1 is zeros of multiplicity 2.

$\therefore f(x) = k(x-1)^2(x-3)$

Since the y – intercept is 4 $\Rightarrow f(0) = 4$

$\Leftrightarrow f(0) = k(0-1)^2(0-3) = -3k = 4 \Rightarrow k = \dfrac{-4}{3}$

$\therefore f(x) = \dfrac{-4}{3}(x-1)^2(x-3)$

b) Since $x = 2$ and $x = 4$ are x – intercepts of $f(x)$ with multiplicity of 2 and 1 respectively, so that $(x-2)^2$ and $x-4$ are factor of $f(x)$

$\therefore f(x) = k(x-2)^2(x-4)$

Since (1,–6) on the graph of $f(x)$ so that $f(1) = -6$

$\Rightarrow f(1) = k(1-2)^2(1-4) = -6 \Rightarrow -3k = -6 \Rightarrow k = 2$

$\therefore f(x) = 2(x-2)^2(x-4) = 2x^3 - 16x^2 + 40x - 32$

68. Which of the following function touches but never crosses the x – axis

A. $f(x) = 1 - x^3$ C. $f(x) = (x^2-1)^2$

B. $f(x) = x^4 - 16$ D. $f(x) = x - x^5$

Solution: Here

A. $f(x) = 1 - x^3 = (1-x)(x^2+x+1)$

The exponent of $(1-x)$ is odd.

$\therefore$The graph of f crosses x – axis at $x = 1$

B. $f(x) = x^4 - 16 = (x^2-4)(x^2+4) = (x-2)(x+2)(x^2+4)$

$\therefore$at $x = 2$ and $x = -2$ crosses the x – axis

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C. $f(x) = (x^2-1)^2 = (x^2-1)(x^2-1) = (x^2-1)^2 (x+1)^2$

The exponent of $x-1$ and $x+1$ is even

∴ The graph touches x – axis at $x = 1$ and $x = -1$ but not crosses.

∴ **Answer: C**

69. Which of the following is true about $f(x) = x^5(x-1)^4(x^2+3)$

A. O is a root of $f(x)$ crosses the x – axis at $x = 1$

B. The graph of $f(x)$ crosses the x – axis at $x = 1$

C. The graph of $f(x)$ crosses the x – axis only at $x = 0$

D. $f(x)$ has three distinct real root

Solution: i) 0 is root of multiplicity 5

∴ the graph crosses x – axis at $x = 0$

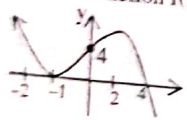
ii) 1 is root of multiplicity 4

∴ the graph touches x – axis at $x = 1$ but not crosses

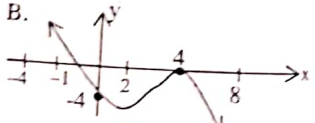
∴ **Answer: C**

70. From the graphs given below, which one is the possible graph of the polynomial function $f(x) = (x+1)^2 (x-4)$?

A.

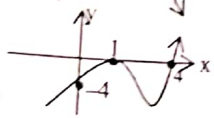


B.



C.

D.



Solution: A) $f(x) = k(x+1)^2 (x-4)$, since $f(0) = 4$

$\Rightarrow k(0+1)(0-4) = 4 \Rightarrow -4k = 4 \Rightarrow k = -1$

$\therefore f(x) = -(x+1)^2 (x-4)$

B) $f(x) = k(x+1)(x-4)^2$, since $f(0) = -4$

$\Rightarrow k(0+1)(0-4)^2 = 16k = -4 \Rightarrow k = \dfrac{-1}{4}$

$\therefore f(x) = \dfrac{-1}{4}(x+1)(x-4)^2$

C) $f(x) = k(x+1)^2 (x-4)$, since $f(0) = -4$

$\Rightarrow k(0+1)^2 (0-4) = -4 \Rightarrow -4k = -4 \Rightarrow k = 1$

$\therefore f(x) = (x+1)^2 (x-4)$

D) $(x-1)^2 (x-4)$

∴ **Answer: C**

Solved Problem

71. The polynomial $P(x)$ is divided by $x-4$ and $x+2$ the remainders are remainders are 8 and 4 respectively.

Find the remainder when $P(x)$ is divided by $(x-4)(x+2)$

Solution: The remainder will be linear equation $Ax + B$ because $P(x)$ divided by Degree 2.

$P(x) = Q(x) D(x) + R(x)$

$\Rightarrow P(x) = Q(x)(x-4)(x+2) + (Ax + B)$

$\Rightarrow P(4) = Q(4)(4-4)(4+2) + 4x + B = 8$

$\Rightarrow 4A + B = 8 \ldots\ldots\ldots\ldots\ldots\ldots (*)$

$\Rightarrow P(-2) = Q(-2)(-2-4)(-2+2) + (-2A + B) = 4$

$\Rightarrow 0 - 2A + B = 4 \ldots\ldots\ldots\ldots\ldots (**)$

From (*) and (**) solving together

$$\begin{cases} 4A + B = 8 \\ -2A + B = 4 \end{cases}$$

$6A = 4 \Rightarrow A = \dfrac{4}{6} = \dfrac{2}{3}$ *and* $B = 8 - \dfrac{8}{3} = \dfrac{16}{3}$

∴ The remainder will be $R(x) = \dfrac{2}{3}x + \dfrac{16}{3}$

72. When $2x^3 - 5x^2 + 4x - 8$ is divided by $ax^2 + bx + c$ the quotient is $x-2$ and the remainder is $bx - a$. Find the value of a,b and c

Solution: Let $P(x) = 2x^3 - 5x^2 + 4x - 8$

Let $D(x) = ax^2 + bx + c$

Let $R(x) = bx - a$

Let $Q(x) = x - 2$

Using division algorithm:

$P(x) = Q(x) D(x) + R(x)$

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$\Rightarrow 2x^3 - 5x^2 + 4x - 8 = (x-2)(ax^2 + bx + c) + bx - a$

$= ax^3 + bx^2 + cx - 2ax^2 - 2bx - 2x + bx - a$

$= ax^3 + (b-2a)x^2 + (c-2b+b)x - 2c - a$

Then comparing the coefficient:

- $ax^3 = 2x^3 \Rightarrow a = 2$ and $(b-2a)x^2 = -5x^2$

  $\Rightarrow b - 2a = -5 \Rightarrow b - 4 = -5 \Rightarrow b = -5 + 4 = -1$
- $(c - 2b + b)x = 4x \Rightarrow c - b = 4 \Rightarrow c = 3$

Therefore, $a = 2$, $b = -1$ and $c = 3$

$\therefore D(x) = 2x^2 - x + 3$ and $R(x) = -x - 2$

73. If $f(x) = 3x^3 - kx^2 + x - 5k$, find a number k such that the graph of f contains the point $(-1, 4)$

Solution: since $f(-1) = 4 \Rightarrow 3(-1) - k - 1 - 5k = 4$

$$\Rightarrow -6k = 8 \Rightarrow k = \frac{-8}{6} = \frac{-4}{3}$$

74. If $f(x) = kx^3 + x^2 - kx - 4$, find a number k such that the graph of f contains the point $(k, 12)$.

Solution: we know that $f(k) = 12$

$\Rightarrow k^4 + k^2 - k^2 - 4 = 12 \Rightarrow k^4 - 4 = 12$

$\Rightarrow k^4 = 16 \Rightarrow k = \pm 2$

75. What number should odd to **f** when $f(x) = 2x^3 - 3x^2 + 4x - 5$;

a) Is exactly divisible by $x - 2$

b) Is divided by $x - 2$, the remainder 9

Solution: Let k be the adding number, so that

a) $f(x) = 2x^3 - 3x^2 + 4x - 5 + k$

$\Rightarrow f(2) = 0 \Leftrightarrow 16 - 12 + 8 - 5 + k = 0$

$\Rightarrow 7 + k = 0 \Rightarrow k = -7$

$\therefore f(x) = 2x^3 - 3x^2 + 4x - 12$

b) $f(2) = 9 \Rightarrow 16 - 12 + 8 - 5 + k = 9$

$\Rightarrow k + 7 = 9 \Rightarrow k = 2$

$\therefore f(x) = 2x^3 - 3x^2 + 4x - 3$

76. If $x + 2$ and $x - 3$ are factors of the polynomial $P(x)$, then which of the following is not necessarily true.

A. x is factor of $P(x + 3)$ C. x is a factor of $P(x - 2)$

B. $x + 1$ is a factor of $P(x - 3)$ D. $x + 4$ is a factor of $P(x + 2)$

Solution: Let $P(x) = (x + 2)(x - 3) f(x+3) = x(x + 5) f(x+3)$

A) $P(x+3) = (x + 5)(x+3 - 3) f(x+3)$

$\therefore x$ and $x + 5$ are factor of $P(x + 3)$

B) $P(x-3) = (x - 3+2)(x-3 - 3) f(x-3)$

$= (x - 1)(x - 6) f(x - 3)$

$\therefore x - 1$ and $x - 6$ are factor of $P(x - 3)$

C) $P(x-2) = (x - 2+2)(x-2 - 3) f(x-2)$

$= (x)(x - 5) f(x - 2)$

$\therefore x$ and $x - 5$ are a factor of $P(x - 2)$

D) $P(x+2) = (x + 2+2)(x+2 - 3) f(x+2)$

$= (x + 4)(x - 1) f(x + 2)$

$\therefore x + 4$ and $x - 1$ are a factor of $P(x + 2)$ **$\therefore$ Answer: B**

77. Find the remainder if the polynomial

$P(x) = 3x^{101} + 5x^{200} + 4x^{97} + 8x^{48} + 6$ is divided by $x + 1$

Solution: $P(-1) = 3(-1)^{101} + 5(-1)^{200} + 4(-1)^{97} + 8(-1)^{48} + 6$

$= -3 + 5 - 4 + 8 + 6$

$= 12 \longleftarrow$ Remainder.

78. If the zeros of a polynomial $P(x)$ are 0 and -1, then which of the following are the zeros of $P(x^2 - 6x + 8)$? . . . **EGSEC**

A. 0, –1 and 2 C. 0, –1 and 3

B. 0, –1, 2 and 4 D. 2,3, and 4

Solution: Let $P(x) = x \cdot (x+1)\, g(x)$

$\Rightarrow P(x^2 - 6x + 8) = (x^2 - 6x + 8)(x2 - 6x + 8 + 1) g(x^2 - 6x + 8)$

$= (x - 4)(x - 2)(x - 3)^2 g(x^2 - 6x + 8)$

$\therefore x = 4$, $x = 2$, and $x = 3$ are zeros of $P(x)$ **$\therefore$ Answer: D**

79. For what number k is $x + 1$ a factor of $f(x) = 2x^6 - kx^3 + 5x - 1$ ...UEE

A. 6 B. 4 C. –4 D. –6

Solution: since $x + 1$ is factor of $f(x)$

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$\Rightarrow f(-1) = 0 \Rightarrow 2 - k(-1) + 5(-1) - 1 = 0$

$\Rightarrow k - 4 = 0 \Rightarrow k = 4,$ ∴ **Answer: B**

80. Which of the following is not a factor of the polynomial $P(x) = x^3 - \pi x^2 - x + \pi$

B. $x - 1$ B. $x + \pi$ C. $x + 1$ D. $x - \pi$

Solution: A. $P(1) = 1^3 - \pi - 1 + \pi = 0$

∴ $x - 1$ is factor of $P(x)$

B. $P(-\pi) = -\pi^3 - \pi^3 + \pi + \pi = -2\pi^3 + 2\pi \neq 0$

∴ $x + \pi$ is not factor of $P(x)$ ∴ **Answer: B**

81. For which of the following does its graph falls to the left and rises up to the right

A. $P(x) = (x-3)^2 (x+4)^2$ C. $(x^2 - 4)(3x^3 + 1)$

B. $P(x) = (1 - x^2)(x + 2)$ D. $(-x^2 + 1)(x+1)^2$

Solution: Here

A. Degree of $P(x)$ is $2 + 2 = 4$

- Even degree with positive leading coefficient.

$\Rightarrow$ the graph rises to the left and to the right

B. Degree of $P(x)$ is $2 + 1 = 3$ and negative leading coefficient.

- Odd degree with negative leading coefficient.

$\Rightarrow$ the graph rises to the left and falls to right

C. Degree of $P(x)$ is $2 + 3 = 5$.

- Odd degree with negative leading coefficient.

$\Rightarrow$ the graph falls to left and rises to the right

D. Even degree with negative leading coefficient

$\Rightarrow$ the graph falls to left and right ∴ **Answer: C**

82. In each of the following function.

i) Which term has the greatest influence on function value when $|x|$ is large?

ii) Where the graph falls down and rising up.

a) $f(x) = -4x^3 + 7x^2 - 8x + 6$ b) $f(x) = 5x^3 + 7x^2 - 8x + 6$

c) $f(x) = 3x^4 + 2x^3 + 6$ d) $f(x) = -8x^4 + 7x^3$

Solution: a) The shape of the graph of $f(x) = -4x^3 + 7x^2 - 8x + 6$ is similar to the graph of $-4x^3$.

i) $-4x^3$ has the greatest influence

ii) As $x \longrightarrow -\infty$, $f(x) \longrightarrow \infty$ (the graph rises up on left)

As $x \longrightarrow \infty$, $f(x) \longrightarrow -\infty$ (the graph falls down on right)

Explanation: $x \longrightarrow -\infty$ (Read as x goes to negative infinity

- $x \to \infty$ means x goes to positive infinity
- $f(x) \to \infty$ means the value of f(x) increase to without bound

b) The shape of the graph of $f(x) = 5x^3 + 7x^2 - 8x + 6$ is similar (resemble) to the graph of $5x^3$

i) $5x^3$ has the greatest influence

ii) **Positive leading coefficient with odd degree**

a) As $x \to -\infty$, $f(x) \longrightarrow -\infty$

$\Rightarrow$ The graph falls to the left.

- As $x \to \infty$, $f(x) \to \infty$

$\Rightarrow$ The graph raises on the right

c) The shape of the graph of $f(x) = 3x^4 + 2x^3 + 6$ is similar to the graph of $3x^4$

i) $3x^4$ has the greatest influence.

ii) **Remember that**: A function with

Positive leading coefficient with even degree

the graph is rising up to left and right.

d) **Exercise left for you**

Supplementary Exercise

1. Divide by long division

a) $\dfrac{24x^2 - x - 8}{3x - 2}$ b) $\dfrac{x^4 - 3x^2 + 2}{x^2 - 1}$ c) $\dfrac{x^4 + x^2 - x^2 + 2x}{x^2 + 2x}$

d) $\dfrac{6x^3 - 19x^2 + 16x - 4}{x - 2}$ e) $\dfrac{2x^4 + 4x^3 - 5x^2 + 3x - 2}{x^2 + 2x - 3}$

2. Divide: a) $\dfrac{x^{3n} + 9x^{2n} + 27x^n + 27}{x^n + 3}$ b) $\dfrac{x^{3n} - 3x^{2n} + 5x^n - 6}{x^n - 2}$

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3. Find the remainder using remainder theorem $f(c) = k$,
   a) $f(x) = x^{99} + 4x^{97} + 3$, $c = -1, 1$
   b) $f(x) = 2x^2 + 3x + 1$, $c = -1$
   c) $f(x) = 3x^3 - x^2 + 5x - 4$, $c = 2$
   d) $f(x) = x^4 + 3x^2 - 12$, $c = -2$
4. a) Find all values of k, when $f(x) = (k-7)x^3 - x^2 + kx - 1$ divided by $x + 2$ the remainder is -29
   b) Find all value of k and n, when $f(x) = kx^3 + nx^2 - 3x - 1$ is divided by $x - 2$ and $x + 1$ the remainders are 25 and 6 respectively
   c) Find all values of a and b x+1 if factor of $f(x)=bx^3 +3x^2+3x +a$ and divided by $x + 2$ the reminder is 20
5. In each of the following, find a number k satisfying the given condition
   a) $x + 3$ is factor of $k^2x^3 + x^2 - 2x + 12k$
   b) $x - 2$ is factor of $x^4 + x^3 + kx^2 - x + (k + 13)$
   c) $x+3$ is factor of $2x^4 + 7x^3 - 4x^2 - (k^2 + 9)x - k^2$
6. Find the zeros of the given polynomial and state the multiplicity of each zero
   a) $f(x) = x^2(x + 2)(2x - 5)^3$ b) $f(x) = x(x + 1)^4(3x - 7)$
   c) $f(x) = (x^2 - 3)^2$
7. Find a polynomial $f(x)$ of degree 3, that has the indicated zeros and satisfies the given condition
   a) -1, 2, 3 and $f(-2) = 80$ b) -5, 2, 4 and $f(3) = -24$
   c) -4, 3, 0 and $f(2) = -36$
8. Find a polynomial $f(x)$ of degree 7 such that -2 and 2 are zeros of multiplicity 2, 0 is multiplicity 3 and $f(-1) = 27$
9. Find the rational zeros of the following function
   a) $f(x) = 2x^3 + 3x^2 - 8x + 3$ b) $f(x) = 6x^3 - 4x^2 + 3x - 2$
10. Use the leading coefficient test to determine as x takes values far to the right and far to the left.
    a) $f(x) = -2x^3 + 6x^2 - \frac{9}{2}x$ c) $f(x) = x^5 - 6x^4 + 3x^2 + 9$
    b) $f(x) = 12x^3 - 32x^2 + 3x + 5$ d) $f(x) = 8 - x^3$
    e) $f(x) = x^6 + 5x^3 + x^2 + 10$ f) $f(x) = -x^4 + 8x^3 + 3x^2 + 6$
11. Power of $(x - c)$ theorem to determine whether the graph of f crosses x-axis or touch but does not cross the x-axis.
    a) $f(x) = (x - 1)(x + 1)(x - 3)$ c) $f(x) = (2x - 3)^4(x - 1)^7$
    b) $f(x) = -(x - 3)^2(x - 7)^5$ d) $f(x) = x^3 - 6x^2 + 9x$

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3. a) 1

4. a) $\overrightarrow{OB}$ b) $\overrightarrow{OD}$ c) $\overrightarrow{OD}$ d) $\overrightarrow{DC}$ e) $\overrightarrow{OD}$

Answer key unit 7

1. a) $8x+5+\dfrac{2}{3x-2}$ b) $x^2-2,\ x\neq\pm 1$ c) $x^2-x+1,\ x\neq 0,-2$

b) $6x^2-7x+2$ e) $2x^2+1+\dfrac{x+1}{x^2+2x-3}$

2. a) $x^{2n}+6x^n+9$ b) $x^{2n}-x^n+7+\dfrac{8}{x^n-2}$

3. a) -2, 8 b) 0 c) 26 d) 16

4. a) $k=8$ b) $k=2,\ n=4$ c) $a=-2$ and $b=-2$

5. a) $k=1$ b) $k=-7$ c) $k=\pm 3\sqrt{2}$

6. a) 0 is zeros of multiplicity 2
   -2 is zeros of multiplicity 1
   b) 0 is zeros of multiplicity 1
   -1 is zeros of multiplicity 4

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$\frac{5}{2}$ is zeros of multiplicity 3 $\frac{7}{3}$ is zeros of multiplicity 2

c) $-\sqrt{3}$ is zeros of multiplicity 2

$\sqrt{3}$ is zeros of multiplicity 2

7. a) $f(x) = -4(x + 1)(x - 2)(x - 3)$ c) $f(x) = 3(x(x + 4)(x - 3))$

b) $f(x) = -\frac{1}{3}(x + 5)(x - 2)(x - 4)$

8. $f(x) = 3x^3(x - 2)^2(x + 2)^2$ 9. a) $x = 1, ½, -3$ b) $x = \frac{2}{3}$

10. a) Rises to the left and falls to the right. d) rises to the left and falls to the right
b) falls to the left and rises to the right e) rises to the left and the right
c) falls to the left and rises to the right f) falls to the left and the right.

11. a) crosses at (-1, 0),(1, 0) and (3, 0)
b) crosses at (7, 0), touch but does not cross at (3, 0)
c) crosses at (1, 0), touch but does not cross at $\left(\frac{3}{2},0\right)$
d) crosses at (0, 0), touch but does not cross at (3, 0)

Answer key Unit 8

1. a) $243^{\sqrt{2}}$ b) $2^{-\sqrt{3}}$ c) 1 d) $3\sqrt[4]{3}$

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UNIT EIGHT

Integer Exponent

There are many mathematical situation in which we multiply a number by itself repeatedly. Writing such expressions in exponential form provide a short hand method for representing this repeated multiplication of the same factor.

For example, a) $\underbrace{3\times3\times3\times3\times3\times3}_{} = 3^6$ ← exponent, ← base

- 6 indicates the number of factor in the product
  3 appears as a factor 6 times

b) $3\times3\times3\times\ldots\times3 = 3^n$
3 appears as a factor n times.

c) $\underbrace{a\times a\times a\times\ldots\times a}_{} = a^n$ ← exponent, ← base

a appears as a factor n times.

Zero exponent If a is any nonzero real number then $a^0 = 1$
Note the symbol 0^0 is undefined.

Illustrative example

1. Evaluate each expression
   a) $12^0 = 1$ b) $(-6)^0 = 1$ c) $-6^0 = -(6^0) = -1$

Negative exponent For any natural number **n** and any non zero real number **a**, $a^{-n} = \frac{1}{a^n}$

Illustrative example

2. Write the following expression with only positive exponent.

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a) $2^{-3} = \frac{1}{2^3} = \frac{1}{8}$

b) $6^{-1} = \frac{1}{6^1} = \frac{1}{6}$

c) $3^{-2} = \frac{1}{3^2} = \frac{1}{8}$

d) $5z^{-3} = 5\left(\frac{1}{z^3}\right) = \frac{5}{z^3}$

e) $-4^{-2} = -\frac{1}{4^2} = \frac{-1}{16}$

f) $-a^{-m} = -\frac{1}{a^m}$

g) $(-a)^{-m} = \frac{1}{(-a)^m}$

h) $\left(\frac{8}{3}\right)^{-2} = \left(\frac{3}{8}\right)^2 = \frac{9}{72}$

o) $\left(\frac{1}{3^{-4}}\right)^{-1} = (3^4)^{-1} = 3^{-4} = \frac{1}{3^4}$

Note: For $(-4)^{-2}$ the base is -4 but for -4^{-2} the base is 4

3. Evaluate a) $3^{-1} + 4^{-4}$

Since $3^{-1} = \frac{1}{3}$ and $4^{-1} = \frac{1}{4}$

$\therefore 3^{-1} + 4^{-1} = \frac{1}{3} + \frac{1}{4} = \frac{4}{12} + \frac{3}{12} = \frac{7}{12}$

b) $5^{-1} - 2^{-1} = \frac{1}{5} - \frac{1}{2} = \frac{2}{10} - \frac{5}{10} = \frac{-3}{10}$

c) $a^{-1} + b^{-1} = \frac{1}{a} + \frac{1}{b} = \frac{a+b}{ab}$

The product rule for exponent If m and n are natural numbers and any real number, then $\mathbf{a^m \cdot a^n = a^{m+n}}$

Illustrative example

4. Apply the product rule for exponents in each case

a) $3^4 \cdot 3^7 = 3^{4+7} = 3^{11}$

b) $5^3 \cdot 5 = 5^{3+1} = 5^4$

c) $y^3 \cdot y^8 \cdot y^2 = y^{3+8+2} = y^{13}$

d) $\left(\sqrt{x+y}\right)(x+y) = (x+y)^{\frac{1}{2}}(x+y)^1 = (x+y)^{\frac{1}{2}+1} = (x+y)^{\frac{3}{2}}$

e) $(5y^2)(-3y^4) = 5(-3)y^2y^4 = -15y^{2+4} = -15y^6$

f) $(5x^3y)(2x^5y^2) = 10x^{3+5}y^{2+1} = 10x^8y^3$

g) $\left(6^{\sqrt{2}}\right)\left(6^{\sqrt{8}}\right) = 6^{\sqrt{2}+\sqrt{8}} = 6^{\sqrt{2}+2\sqrt{2}} = 6^{3\sqrt{2}} = 216^{\sqrt{2}}$

h) $\sqrt[3]{x^2}\sqrt[4]{x} = x^{\frac{2}{3}} \cdot x^{\frac{1}{4}} = x^{\frac{2}{3}+\frac{1}{4}} = x^{\frac{11}{12}}$

Quotient Rule for exponents If $a \neq 0$ and m and n are integer then

$$\frac{a^m}{a^n} = a^{m-n}$$

Illustrative example

5. Apply the quotient rule for exponent

a) $\frac{3^7}{3^2} = 3^{7-2} = 3^5$

b) $\frac{x-1}{\sqrt{x-1}}, \quad x \neq 1$

$$= \frac{x-1}{(x-1)^{\frac{1}{2}}} = (x-1)^{1-\frac{1}{2}} = (x-1)^{\frac{1}{2}} = \sqrt{x-1}$$

c) $\frac{3^{4x+6}}{3^{5x+1}} = 3^{4x+6-5x-1} = 3^{-x+5}$

d) $\frac{\sqrt{x^3}}{\sqrt[3]{x^2}} = \frac{x^{\frac{3}{2}}}{x^{\frac{2}{3}}} = x^{\frac{3}{2}-\frac{2}{3}} = x^{\frac{5}{6}}$

Power Rules for example If a and **b** are real numbers and **m** and **n** are integers then

- $(a^m)^n = a^{mn}$
- $(ab)^n = a^n b^n$
- $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$

6. Use power rule in each case

a) $(x^8)^3 = x^{8 \cdot 3} = x^{24}$

b) $(3y)^4 = 3^4y^4 = 81y^4$

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c) $\left(\frac{2}{3}\right)^4 = \frac{2^4}{3^4} = \frac{16}{81}$

7. Simplify each of the following expression using one or more of the l of exponents

a) $8^{\frac{-2}{3}}$ b) $\left(\frac{5^3}{2^6}\right)^{-\frac{1}{3}}$ c) $\sqrt[3]{64a^{12}b^6}$

d) $\frac{(x^{-2})^{-3}(x^{\frac{-1}{3}})^9}{\left(x^{\frac{1}{2}}\right)^{-3}\left(x^{\frac{-3}{2}}\right)^5}$ e) $\sqrt[a]{a^{a^2-a}}$ f) $\frac{\left(3^{\sqrt{2}}\right)^2 \times 9^{-\sqrt{3}}}{3^{-\sqrt{12}}}$

g) $(0.125)^{\frac{1}{3}}(0.25)^{-\frac{1}{2}}$ h) $\frac{a^{-2}-b^{-2}}{a^{-1}-b^{-1}}$

Solution: a) $8^{\frac{-2}{3}} = (2^3)^{\frac{-2}{3}} = 2^{-2} = \frac{1}{2^2} = \frac{1}{4}$

b) $\left(\frac{5^3}{2^6}\right)^{-\frac{1}{3}} = \frac{5^{-1}}{2^{-2}} = \frac{4}{5}$

c) $\sqrt[3]{64a^{12}b^6} = (64a^{12}b^6)^{\frac{1}{3}}$

$= (64)^{\frac{1}{3}}(a^{12})^{\frac{1}{3}}(b^6)^{\frac{1}{3}}$

$= 4a^4b^2$

d) $\frac{(x^{-2})^{-3}(x^{\frac{-1}{3}})^9}{\left(x^{\frac{1}{2}}\right)^{-3}\left(x^{\frac{-3}{2}}\right)^5} = \frac{x^6x^{-3}}{x^{\frac{-3}{2}}x^{\frac{-15}{2}}} = \frac{x^{6+(-3)}}{\left(-\frac{3}{2}-\frac{15}{2}\right)}$

$= \frac{x^3}{x^{-9}} = x^{3-(-9)} = x^{12}$

e) $\sqrt[a]{a^{a^2-a}} = \left(a^{a^2-a}\right)^{\frac{1}{a}}$

$= \left(a^{a(a-1)}\right)^{\frac{1}{a}} = a^{a-1}$

f) $\frac{\left(3^{\sqrt{2}}\right)^2\left(9^{-\sqrt{3}}\right)}{3^{-\sqrt{12}}} = \frac{3^{2\sqrt{2}} \cdot 3^{-2\sqrt{3}}}{3^{-2\sqrt{3}}} = 3^{2\sqrt{2}+2\sqrt{3}-2\sqrt{3}} = 9^{\sqrt{2}}$

g) $(0.125)^{\frac{1}{3}}(0.25)^{-\frac{1}{2}} = (10^{-3}\times 125)^{\frac{1}{3}}(10^{-2}\times 25)^{-\frac{1}{2}}$

$= (10^{-1})(5^1)(10(5^{-1}))$

$= (10^0)(5^{1-1})$

$= (10^0)(5^0) = 1$

h) $\frac{a^{-2}-b^{-2}}{a^{-1}-b^{-1}} = \frac{(a^{-1})^2-(b^{-1})^2}{a^{-1}-b^{-1}}$

$= \frac{(a^{-1}-b^{-1})(a^{-1}+b^{-1})}{a^{-1}-b^{-1}} = a^{-1}+b^{-1}$

8. Simplify

a) $(4^{-1})^4 \times 2^5 \times \left(\frac{1}{16}\right)^3 \times (8^{-2})^5 \times (64^2)^3$

Solution: a) $2^{-8}\times 2^5 \times 2^{-12} \times 2^{-30} \times 2^{36}$

$= 2^{-8+5-12-30+36}$

$= 2^{-9} = \frac{1}{2^9} = \frac{1}{512}$

9. Simplify

a) $\sqrt[3]{-0.000027}$ b) $\sqrt[6]{0.000064x^{18}}$

Solution: Here

a) $\sqrt[3]{-0.000027} = \sqrt[3]{27\times 10^{-6}}$

$= \sqrt[3]{3^3\times(10^{-2})^3}$ ← **Explanation** $a^n \cdot b^n = (ab)^n$

$= \sqrt[3]{(3\times 10^{-2})^3}$

$= 3\times 10^{-2}$

b) $\sqrt[6]{0.000064x^{18}} = \sqrt[6]{2^6\times 10^6 \cdot (x^3)^6}$

Explanation
- $10^{-6} = (10^{-1})^6$
- $\sqrt[n]{a^n} = |a|$ if n is even

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$= (2)(10^{-1})(|x^3|)^6$

$= 0.2|x^3|$

Remember that. $\sqrt[n]{a}$

- Is real number if $a < 0$ and n is odd
- Is not real number if $a < 0$ and n is even.

Example a) $\sqrt{-9}, \sqrt[4]{-7}, \sqrt[6]{-9}$ . . . etc is not real number.

b) $\sqrt[3]{-2}, \sqrt[5]{-32}, \sqrt[9]{-12}$ . . . etc is real number.

Comparing the exponents

A. Let $a > b > 1$;

- i) if $a^x > b^x$, then $x > 0$
- ii) if $a^x < b^x$, then $x < 0$

Example: $3^x > 2^x$ when $x > 0$

$3^x < 2^x$ when $x < 0$

B. Let $0 < a < b < 1$

- i) if $a^x < b^x$, then $x > 0$
- ii) if $a^x > b^x$, then $x < 0$

Example: $\left(\frac{1}{3}\right)^x < \left(\frac{1}{2}\right)^x$ *when* $x > 0$

Let $x = 4$

$$\left(\frac{1}{3}\right)^4 < \left(\frac{1}{2}\right)^4 \Rightarrow \frac{1}{3^4} < \frac{1}{2^4}$$

- $\left(\frac{1}{3}\right)^x > \left(\frac{1}{2}\right)^x$, *when* $x < 0$,

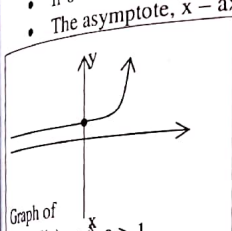
Exponential Function and their graph

Definition: For $a > 0$, $a \neq 1$ and for all real number x, $f(x) = a^x$ is called an **exponential function** with base **a**

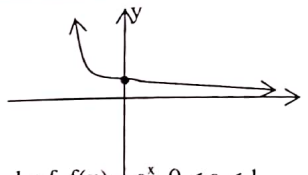
Note these facts about the graphs of exponential functions $f(x) = a^x$

- Domain = $(-\infty, \infty)$ = the set of all real numbers
- Range = $(0, \infty)$ = the set of positive real numbers
- No – x – intercept
- y – intercept.

- The graph of f contains the point (a, 1) (1, a), $\left(-1, \frac{1}{a}\right)$.
- If $a > 1$, then $f(x) = a^x$ is **increasing** function (**growth functions**)
- If $0 < a < 1$, then $f(x) = ax$ is **decreasing function** (decay function)
- The asymptote, x – axis (y = 0)



Graph of
- $f(x) = a^x$, $a > 1$
- $f(x) = 2^x, 3^x, 4^x, \ldots$



Graph of, $f(x) = a^x$, $0 < a < 1$

Example, $f(x) = \left(\frac{1}{2}\right)^x, \left(\frac{1}{3}\right)^x, \left(\frac{2}{3}\right)^x$

- a^{-x} is the mirror image of a^x on y – axis.
- The graph of $\left(\frac{1}{2}\right)^x$ is the mirror image of 2^x on y – axis.
- $\left(\frac{1}{3}\right)^x$ is the mirror image of 3^x on y – axis

The larger the base of exponential function the more, quickly it grows for positive x and more quickly approaches zero for negative values of x

Example. a) $6^x > 4^x$ on $x > 0$, since $6 > 4$

- $4^x > 6^x$ on $x < 0$

b) $\left(\frac{1}{2}\right)^x > \left(\frac{1}{3}\right)^x$ on $x > 0$, since $\frac{1}{2} > \frac{1}{3}$

$\left(\frac{1}{2}\right)^x < \left(\frac{1}{3}\right)^x$ on $x < 0$

For any positive real number a and b

- $a^x = a^y \Leftrightarrow x = y$, if $a \neq 1$
- $a^x < a^y \Leftrightarrow x < y$ for $a > 1$
- $a^x < a^y \Leftrightarrow x > y$ for $0 < a < 1$

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- $a^x < b^x \Leftrightarrow a < b$ for $x > 0$
- $a^x > b^x \Leftrightarrow a < b$ for $x < 0$

Illustrative Example

10. Graph the function using the same set of axes and on different set of axes.

a) $f(x) = 3^x$ and $g(x) = 4^x$

b) $f(x) = \left(\frac{1}{3}\right)^x$ and $g(x) = \left(\frac{1}{4}\right)^x$

c) $f(x) = 3^x$ and $g(x) = \left(\frac{1}{3}\right)^x$

d) $f(x) = 4^x$ and $g(x) = \left(\frac{1}{4}\right)^x$

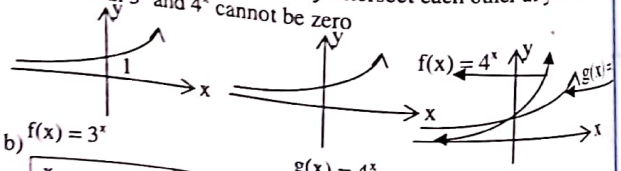
Solution: Here

a) To draw the graph considers the following table.

x	−2	−1	0	1	2
$f(x) = 3^x$	$\frac{1}{9}$	$\frac{1}{3}$	1	3	9
$g(x) = 4^x$	$\frac{1}{16}$	$\frac{1}{4}$	1	4	16

From the table, we conclude that both $f(x) = 3x$ and $g(x) = 4^x$ are increasing.

- For $a < 0$, $3^x > 4^x$. $\Rightarrow$ The graph of 3^x lies above 4^x.
- For $x > 0$, $3^x < 4^x$.
  $\Rightarrow$ The graph of 3^x lies below 4^x.
- For $x = 0$, $3^x = 4^x = 1$, they intersect each other at $y = 1$
- Both 3^x and 4^x cannot be zero



b) $f(x) = 3^x$ $g(x) = 4^x$

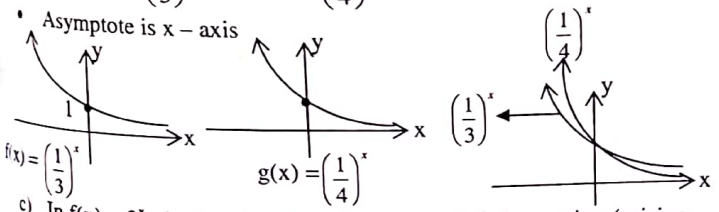
x	−2	−1	0	1	2
$f(x) = \left(\frac{1}{3}\right)^x$	9	3	1	$\frac{1}{3}$	$\frac{1}{9}$

$f(x) = \left(\frac{1}{4}\right)^x$	16	4	1	$\frac{1}{4}$	$\frac{1}{16}$

- From the table, we conclude that as x – increase, both value of $f(x) = \left(\frac{1}{3}\right)^x$ and $g(x) = \left(\frac{1}{4}\right)^x$ are decreasing so that the graph falls from left to right.
- For $x < 0$, $\left(\frac{1}{4}\right)^x > \left(\frac{1}{3}\right)^x$

  $\Rightarrow$ The graph of $\left(\frac{1}{4}\right)^x$ lies above the graph of $\left(\frac{1}{3}\right)^x$
- For $x > 0$, $\left(\frac{1}{4}\right)^x < \left(\frac{1}{3}\right)^x$

  $\Rightarrow$ the graph of $\left(\frac{1}{4}\right)^x$ lies below the graph of $\left(\frac{1}{3}\right)^x$.
- For $x = 0$, $\left(\frac{1}{3}\right)^x = \left(\frac{1}{4}\right)^x = 1$, they intersect each other at $y = 1$
- Both $f(x) = \left(\frac{1}{3}\right)^x$ and $g(x) = \left(\frac{1}{4}\right)^x$ cannot be zero and negative
- Asymptote is x – axis



c) In $f(x) = 3^x$, the base $3 > 1$, so that the graph is increasing (raising up from left to right).

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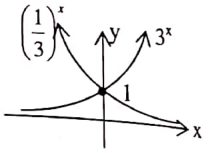
- In $g(x)=\left(\frac{1}{3}\right)^x$, the base $\left(\frac{1}{3}\right)<1$, so that the graph is decreasing (falling down from left to right

- For $x<0$, $\left(\frac{1}{3}\right)^x > x^3$

- For $x>0$ $\left(\frac{1}{3}\right)^x < 3^x$



- *The graph of* $\left(\frac{1}{3}\right)^x$ *is the mirror image of* 3^x *on* $y-axis$

- if $x<0$, $\left(\frac{1}{3}\right)^x > 1$ and $0<3^x<1$

- if $x>0$, $3^x>1$ and $0<\left(\frac{1}{3}\right)^x<1$

d) Exercise left for you.

11. Let $f(x)=\left(\frac{1}{2}\right)^x$ and $g(x)=\left(\frac{2}{3}\right)^x$ investigate, for what values of x.

a) $\left(\frac{1}{2}\right)^x > \left(\frac{2}{3}\right)^x$ b) $\left(\frac{1}{2}\right)^x < \left(\frac{2}{3}\right)^x$

c) $\left(\frac{1}{2}\right)^x = \left(\frac{2}{3}\right)^x$ d) $\left(\frac{1}{2}\right)^x < 0$

e) $\left(\frac{2}{3}\right)^x > 0$ f) $\left(\frac{1}{2}\right)^x = 0$

Solution:

a) For $x<0$, $\left(\frac{1}{2}\right)^x > \left(\frac{2}{3}\right)^x$, let $x=-1 \Rightarrow \left(\frac{1}{2}\right)^{-1} > \left(\frac{2}{3}\right)^{-1} = 2 > \frac{3}{2}$

b) For all $x>0 \Rightarrow x\in(0,\infty)$. $\left(\frac{1}{2}\right)^x < \left(\frac{2}{3}\right)^x$ let $x=2 \Rightarrow$

$\left(\frac{1}{2}\right)^2 < \left(\frac{2}{3}\right)^2 = \frac{1}{4} < \frac{4}{9}$

c) For $x=0$, $\left(\frac{1}{2}\right)^x = \left(\frac{2}{3}\right)^x = 1$

d) No x – value that makes $\left(\frac{1}{2}\right)^x < 0$ $\therefore S.S = \phi$

e) For all $x\in(-\infty,\infty)$, $\left(\frac{2}{3}\right)^x > 0$

f) No x – values that makes $\left(\frac{1}{2}\right)^x = 0$

Graphing Exponential Functions using Transformation

- Study the following equal function relation

a) • $\left(\frac{1}{3}\right)^x = 3^{-x}$, • $\left(\frac{1}{5}\right)^x = 5^{-x}$ • $-2^x = -(2^x)$

Illustrative Example

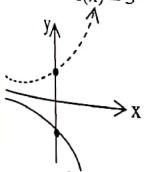
12. Determine the domain, range, asymptote, and sketch the graph

a) $f(x)=-3^x$ b) $f(x)=-3^{-x}$

Solution: $f(x)=-3^x$ is the mirror image of 3^x on x – axis

- the domain $=(-\infty,\infty)$
- the range $=(-\infty,0)$

$f(x)=3^x$



- asymptote, x – axis
- No – x – intercept
- y – intercept, when $x=0$, $y=-1$
- $f(x)=3^x$ is increasing but -3^x decreasing For $x<0 \Rightarrow -1<-3^x<0$
- For $x>0 \Rightarrow -\infty<-3^x<-1$

b) We have $f(x)=-3^{-x}=(3^{-1})^x=\left(\frac{1}{3}\right)^x$

- The graph of $f(x)=-3^{-x}$ is the mirror image of 3^{-x} on x – axis.
- The domain $=(-\infty,\infty)$

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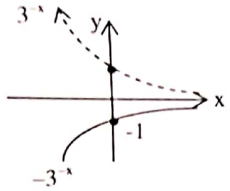


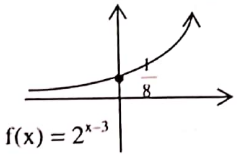
- The range = $(-\infty,0)$
- asymptote, x – axis
- No – x – intercept
- y – intercept, y = –1
- $f(x) = 3^{x}$ is decreasing but -3^{x} is increasing
- For $x > 0 \Rightarrow -1 < -3^{x} < 0$
- For $x < 0 \Rightarrow -3^{x} < -1$

13. Determine the domain, range asymptote intercepts, and sketch the graph.

a) $f(x) = 2^{x-3}$ b) $f(x) = 2^{3-x}$

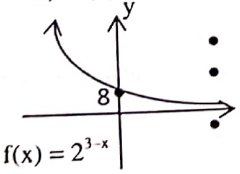
Solution: we have $f(x) = 2^{x-3} = 2^{x} \cdot 2^{-3} = \frac{1}{8}(2^{x})$

- since the base 2 > 1, the graph rises up from left to right (that is increasing)



- Domain = $(-\infty,\infty)$, Range = $(-\infty,0)$
- asymptote, x – axis
- No – x – intercept
- y – intercept, when $x = 0 \Rightarrow f(0) = \frac{1}{8}(1) = \frac{1}{8}$
- For $x > 0$, $f(x) > \frac{1}{8}$
- For $x < 0$, $0 < f(x) < \frac{1}{8}$

b) $f(x) = 2^{3-x} = 2^{3} \cdot 2^{-x} = (8)(2^{-x}) = 8\left(\frac{1}{2}\right)^{x}$



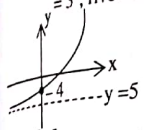
- Domain = $(-\infty,\infty)$, Range = $(-\infty,0)$
- asymptote, x – axis
- since the base $\left(\frac{1}{2}\right) < 1$, the graph falls down from left to right. (i.e. decreasing function)
- No – x – intercept
- y – intercept, when $x = 0 \Rightarrow f(0) = 8$

14. Determine the domain, the range, asymptote, intercepts and sketch

a) $f(x) = 3^{x} - 5$ b) $f(x) = 3^{x} + 5$

Solution: Because the base 3 > 1, the graph of $f(x) = 3^{x} - 5$ is rising up from left to right (i.e. increasing function)

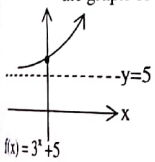
- The graph of $f(x) = 3^{x} - 5$ is obtained by shifting the graph of $y = 3^{x}$, five unit own ward



- The domain = $(-\infty,\infty)$
- The range = $(-5, \infty)$
- Asymptote, the line y = –5
- y – intercept, $f(0) = 1 - 5 = -4$
- x – intercept, $f(x) = 0 \Rightarrow 3^{x} = 5 \Rightarrow x = \log_{3}^{5}$
- For $x > 0$, $f(x) > -4$
- For $x < 0$, $-5 < f(x) < -4$

b) Since the base 3 > 1, so that the graph will rise up from left to right.

- We can obtain the graph of $y = 3^{x} + 5$ by shifting vertically the graph of $y = 3^{x}$, five units up.

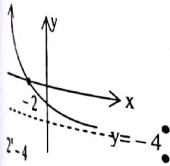


- Domain = $(-\infty,\infty)$
- Range = $(5, \infty)$
- Asymptote, the line y = 5
- y – intercept, $f(0) = 3^{0} + 5 = 1 + 5 = 6$
- No – x – intercept
- For all $x \in (-\infty,0)$, $5 < f(x) < 6$
- For all $x \in (0,\infty)$, $f(x) > 6$

15. Determine the domain, the range, asymptote, the intercepts and sketch

a) $f(x) = 2^{-x} - 4$ b) $f(x) = -2^{-x} + 4$

Solution: we have $2^{-x} = \left(2^{-1}\right)^{x} = \left(\frac{1}{2}\right)^{x}$



- since the base $\left(\frac{1}{2}\right) < 1$, the graph falls down from left to right
- 2^{-x} is the mirror image of 2^{x} on y – axis
- The graph of $f(x) = 2^{-x} - 4$ is obtained by shifting $y = 2^{-x}$ four unit down.
- Domain = $(-\infty,\infty)$
- Range = $(-4, \infty)$

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- x – intercept, $2^{-x} - 4 = 0 \Rightarrow 2^{-x} = 2^2 \Rightarrow x = -2$
- y – intercept, when $x = 0 \Rightarrow y = 1 - 4 = -3$
- Asymptote, the line $y = -4$
- For $x > 0$, $-4 < f(x) < -3$
- For $x < 0$, $f(x) > -3$

b) Exercise left for you

16. Let $f(x)$ and $g(x)$ be exponential functions given by $f(x) = a^x$, $g(x) = b^x$ for $a > 1$, and $b > 1$, then which one of the following statements is true
A) $f(x) > g(x)$, for all $x > 0$, if $a > b$ B) $f(x) > g(x)$, for all $x>0$ if $a<b$
C) $f(x) < g(x)$, for all $x > 0$ if $a > b$ D) $f(x) < g(x)$, for all $x<0$ if $a<b$

Solution: since the base $a > 1$ and $b > 1$,

Assume $a > b$, let $a = 3$ and $b = 2$

$\Rightarrow f(x) = 3^x$ and $g(x) = 2^x$,

In this case

- When $x > 0$, $3^x > 2^x \Rightarrow f(x) > g(x)$, $a > b$, $x > 0$
- When $x < 0$, $3^x < 2^x \Rightarrow f(x) < g(x)$, $a > b$, $x < 0$

Assume $a < b$, let $a = 2$ and $b = 3$

$\Rightarrow f(x) = 2^x$ and $g(x) = 3^x$,

In this case

- When $x > 0$, $2^x < 3^x \Rightarrow f(x) < g(x)$, $a < b$, $x > 0$
- When $x < 0$, $2^x > 3^x \Rightarrow f(x) > g(x)$, $a < b$, $x < 0$

$\therefore$ Answer: A

17. Let $f(x) = \left(\frac{3}{4}\right)^x$ *and* $g(x) = \left(\frac{4}{2}\right)^x$. Which one of the following statements about $f(x)$ and $g(x)$ is true.
A. If $x < 0$, then $f(x) < g(x)$
B. If $x > 0$, then $f(x) > g(x)$
C. If $x < 0$, then $g(x) < 1$ and if $x > 0$, then $f(x) < 1$
D. If $x < 0$, then $g(x) > 1$ and if $x > 0$, then $f(x) > 1$

Solution: $f(x) \left(\frac{3}{4}\right)^x$ *and* $g(x) = \left(\frac{4}{3}\right)^x$

A) $\left(\frac{3}{4}\right)^x > \left(\frac{4}{3}\right)^x$, for all $x < 0$, let $x = -1$

$\Rightarrow f(x) > g(x)$ for all $x < 0$

B) $\left(\frac{3}{4}\right)^x < \left(\frac{4}{3}\right)^x$, for all $x > 0$, let $x = 1$

$\Rightarrow f(x) < g(x)$, for all $x > 0$

C) $\left(\frac{3}{4}\right)^x < 1$, for all $x > 0$, let $x = 1,2,3,\ldots$

$\Rightarrow f(x) < 1$, for all $x > 0$.

- $\left(\frac{3}{4}\right)^x > 1$, for all $x < 0$, let $x = -2, -1$

$\Rightarrow f(x) > 1$, for all $x < 0$.

Similarly $\left(\frac{4}{3}\right)^x > 1$, for all $x > 0$, let $x = 1, 2, 3, \ldots$

$\Rightarrow g(x) > 1$, for all $x > 0$.

D) $0 < \left(\frac{4}{3}\right)^x < 1$, for all $x < 0$, let $x = -1, -2, \ldots$.

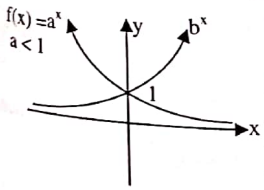
$\Rightarrow 0 < g(x) < 1$, for all $x < 0$.

Answer: C

18. Let $f(x) = a^x$, for $0 < a < 1$
$g(x) = b^x$, for $b > 1$, then for what value of x
a) $f(x) > g(x)$ b) $f(x) < g(x)$ c) $f(x) > 1$
d) $0 < f(x) < 1$ e) $g(x) > 1$ f) $0 < g(x) < 1$ g) $f(x) = g(x)$

Solution: consider the graph.

- Since the base $a < 1$, so that the graph of $f(x) = a^x$ is falling down from left to right
- Since the base $b > 1$, so that the graph of $g(x) = b^x$ is rising up from left to right



We can observe from the graph
a) For $x < 0 \Rightarrow f(x) > g(x)$
b) For all $x < 0 \Rightarrow f(x) < g(x)$
c) For all $x < 0 \Rightarrow f(x) > 1$
d) For all $x > 0 \Rightarrow 0 < f(x) < 1$
e) For all $x > 0 \Rightarrow 0 < g(x) > 1$
f) For all $x < 0 \Rightarrow 0 < g(x) < 1$
g) For $x = 0 \Rightarrow f(x) = g(x) = 1$

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- **Finding an equation of the exponential function satisfying prescribed conditions**

Illustrative Example

19. Find an exponential function of the form
   a) $f(x) = ba^x$, that has y – intercept 8, passes through P(3,1)
   b) $f(x) = ba^x$, that has y – intercept 6, pass through $P\left(2, \frac{3}{32}\right)$
   c) $f(x) = ba^{-x} + c$ that has horizontal asymptote $y = -2$, y – intercept 16, and x – intercept 2

Solution: Here

a) $f(x) = ba^x$ since y – intercept $= 8 \Rightarrow f(0) = 8$

$\Rightarrow b(a)^0 = 8 \Rightarrow b = 8$

To find a, since $P(3,1) \in f \Rightarrow f(x) = 1$

$\Rightarrow b(a)^3 = 1 \Leftrightarrow 8a^3 = 1$

$\Rightarrow a^3 = \frac{1}{8} \Rightarrow a = \frac{1}{2}$

$\therefore f(x) = 8\left(\frac{1}{2}\right)^x$ ⟵ required function

b) Exercise left for you.

c) $f(x) = ba^{-x} + c$, since $y = -2$ is asymptote

$\Rightarrow f(x) = ba^{-x} - 2$ and y – intercept $= 16 \Rightarrow f(0) = 16$

$\Rightarrow b(a^0) - 2 = 16 \Rightarrow b - 2 = 16 \Rightarrow b = 18$

Since x – intercept is 2, $\Rightarrow f(2) = 0$

$\Rightarrow 18(a^{-x}) - 2 = 0 \Rightarrow (a^{-2}) - 2 = 0$

$\Rightarrow a^{-2} = \frac{2}{18} = \frac{1}{9} \Rightarrow \frac{1}{a^2} = \frac{1}{9}$

$\therefore a = \pm 3$, since **a** must be positive

$\therefore f(x) = 18(3^{-x}) - 2 = 18\left(\frac{1}{3}\right)^x - 2$

The exponential function with base 'e'

The number 'e' arises in the investigation of many physical phenomena, such as for measuring population growth and decay.

The number 'e' is defined as the number that the expression

$\left(1+\frac{1}{n}\right)^n = e$ as $n \longrightarrow \infty$

Where, n is positive inter

$e \approx 2.71828$.

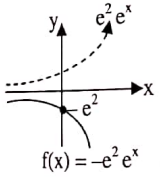
$2 \le \left(1+\frac{1}{n}\right)^n < e$

The natural exponential function f is defined by $f(x) = e^x$, for every real number x.

Illustrative Example

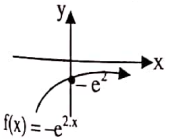
20. Determine, the domain, the range, the y – intercept and describe the behavior.
   a) $f(x) = -e^{x+2}$
   b) $f(x) = -e^{2-x}$
   c) $f(x) = e^x$
   d) e^{-x}

Solution: a) $f(x) = -e^{x+2} = -(e^{x+2}) = -(e^2 . e^x) = -e^2 e^x$



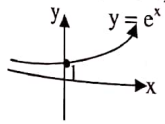
- Domain = $(-\infty,\infty)$
- Range = $(-\infty,0)$
- y – intercept, $x = 0 \Rightarrow y = -e^2$
- The graph lines completely below the x – axis
- For all $x < 0$, $-e^2 < f(x) < 0$
- For all $x > 0$ $f(x) < -e^2$.
- It is decreasing function

b) $f(x) = -e^{2-x} = -(e^{2-x}) = -(e^2 e^{-x}) = -e^2 e^{-x}$



- Domain = $(-\infty,\infty)$
- Range = $(-\infty,0)$
- y – intercept, $y = -e^2$
- For all $x < 0$, $f(x) < -e^2$
- For all $x > 0$, $-e^2 < f(x) < 0$
- It is increasing function.

c) Since the base $e > 1$, then the graph will rise as x – increase



- Domain = $(-\infty,\infty)$
- Range = $(-\infty,0)$
- For all $x > 0$, $e^x > 1$

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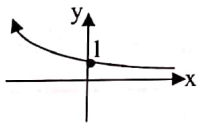
- For all $x > 0$, $0 < e^x < 1$
- y – intercept, y = 1
- It is increasing function.

d) $f(x) = e^{-x} = (e^{-1})^x = \left(\frac{1}{e}\right)^x$ since $\frac{1}{e} < 1$, then the graph will fall from left to right



- Domain = $(-\infty,\infty)$
- Range = $(0,\infty)$
- For $x > 0$, $0 < e^x < 1$
- For $x < 0$, $e^x > 1$
- y – intercept, y = 1
- Decreasing function.

Solving exponential function

Exponential equations are equations where the variable is in the exponent

- We can solve many exponential equations by writing in terms of **the same** base of each side.

Example, a) $8^x = 4$, to solve, first express each side with common base

$\Rightarrow 2^{3x} = 2^2 \Leftrightarrow 3x = 2 \Rightarrow x = \frac{2}{3}$

Often we use equality property for exponential equations

i) $a^x = a^y$, if and only if $x = y$

ii) $a^x - b^x$, if and only if $a = b$, $x \neq 0$

Illustrative Example

21. Solve each of the following equation.

a) $3^{2x} = 27^{3x+1}$

b) $9^{(x^2)} = 3^{3x+2}$

c) $5^{(x^2)} = 25^x$

d) $2^{(x^2)} = 4^{(1-3x)}$

e) $9^{2x} \cdot \left(\frac{1}{3}\right)^{x+2} = 27.(3^x)^{-2}$

f) $8^x = 2^{(\sqrt[3]{3x})}$

g) $e^{(x^2)} = (e^x)^4 \cdot \left(\frac{1}{e^3}\right)$

h) $8^x \cdot (5^x)^3 = 10\sqrt{10}$

i) $\left(\frac{2}{5}\right)^{2x+1} = \left(\frac{25}{4}\right)^{x+2}$

j) $(e^4)^x \cdot (e^4)^2 \cdot e^{x^2} = e^{12}$

Solution: First make the base the same to each side.

a) $3^{2x} = 27^{3x+1} \Leftrightarrow 3^{2x} = (3^3)^{3x+1}$

$\Rightarrow 3^{2x} = 3^{9x+3}$

$\Rightarrow 2x = 9x + 3 \Rightarrow -7x = 3$

$\therefore x = \frac{-3}{7}$

Explanation:
$a^x = a^y \Leftrightarrow x = y$

b) $9^{(x^2)} = 3^{3x+2} \Leftrightarrow 3^{2x^2} = 3^{3x+2}$ ⟵ for the same base

$\Rightarrow 2x^2 = 3x + 2$

$\Rightarrow 2x^2 - 3x - 2 = 0 \Leftrightarrow (2x + 1)(x - 2) = 0$

$\therefore x = \frac{-1}{2}$ or $x = 2$

c) $5^{(x^2)} = 25^x \Leftrightarrow 5^{x^2} = 5^{2x} \Leftrightarrow x^3 = 2x$

$\Rightarrow x^3 - 2x = 0$

$\Rightarrow x(x^3 - 2) = 0$

$\Rightarrow x = 0$ or $x = \sqrt{2}, -\sqrt{2}$

d) $2^{(x^2)} = 4^{(1-3x)} \Leftrightarrow 2^{x^2} = 2^{2-6x}$

$\Rightarrow x^2 = 2 - 6x$

$\Rightarrow x^2 + 6x - 2 = 0$ ⟵

$\therefore x = -3 + \sqrt{11}$ or $x = -3 - \sqrt{11}$

Explanation:
Use: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

e) $9^{2x} \cdot \left(\frac{1}{3}\right)^{x+2} = 27 \cdot (3^x)^{-2}$

$\Rightarrow 3^{4x} \cdot 3^{-x-2} = 3^{3-2x}$ ⟶

$\Leftrightarrow 3^{3x-2} = 3^{3-2x}$ ⟵

$\Rightarrow 3x - 2 = 3 - 2x \Rightarrow 5x = 5$

$\therefore x = 1$

Explanation:
$a^x \cdot a^y = a^{x+y}$
$a^x = a^y \Leftrightarrow x = y$

f) $8^x = 2^{(\sqrt[3]{3x})} \Leftrightarrow 2^{3x} = 2^{\sqrt{3x}}$

$\Rightarrow 3x = \sqrt[3]{3x}$ ⟵

$\Rightarrow 27x^3 = 3x$

$\Rightarrow 27x^3 - 3x = 0$

$\Rightarrow 3x(9x^2 - 1) = 0$

$\therefore x = 0$ or $9x^2 - 1 = 0$

$\therefore x = 0$ or $x = \pm\frac{1}{3}$

$\sqrt[3]{m} = n$
$\Rightarrow n^3 = m$

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g) $e^{(x^2)} = (e^x)^4 \cdot \left(\frac{1}{e^3}\right) \Leftrightarrow e^{x^2} = e^{4x} \cdot (e^{-3})$

$\Rightarrow e^{x^2} = e^{4x-3} \Leftrightarrow x^2 = 4x - 3$

$\Leftrightarrow x^2 - 4x + 3 = 0$

$\Rightarrow (x-3)(x-1) = 0$

$\therefore x = 3$ or $x = 1$

h) $8^x \cdot (5^x)^3 = 10\sqrt{10} \Rightarrow 2^{3x} \cdot 5^{3x} = 10^{1+\frac{1}{2}}$

$\Rightarrow (10)^{3x} = 10^{\frac{3}{2}} \Rightarrow 3x = \frac{3}{2} \Rightarrow x = \frac{1}{2}$

i) $\left(\frac{2}{5}\right)^{2x+1} = \left(\frac{25}{4}\right)^{x+2} \Leftrightarrow \left(\frac{2}{5}\right)^{2x+1} = \left(\frac{5}{2}\right)^{2x+4}$

$\Rightarrow \left(\frac{2}{5}\right)^{2x+1} \Leftrightarrow \left(\frac{2}{5}\right)^{-2x-4} \leftarrow \left(\frac{a}{b}\right) = \left(\frac{b}{a}\right)^{-1}$

$\Leftrightarrow 2x + 1 = -2x - 4 \Rightarrow 4x = -5 \Rightarrow x = \frac{-5}{4}$

j) **Exercise left for you**

22. Solve each of the following equation

a) $(8^x)(27^x) = 6\sqrt[3]{6}$ c) $4^x = 2\sqrt{2}$

b) $(9^x)(25^x) = \sqrt[4]{15}$ d) $\left(\sqrt[3]{6}\right)^x = \sqrt[5]{36}$

Solution: Here we use: $a^x \cdot b^x = (ab)^x$

a) $(8^x)(27^x) = 6\sqrt[3]{6} \Leftrightarrow (2^{3x})(3^{3x}) = 6^{1+\frac{1}{3}}$

$\Rightarrow (2\cdot 3)^{3x} = 6^{\frac{4}{3}} \Leftrightarrow 6^{3x} = 6^{\frac{4}{3}}$

$\Rightarrow 3x = \frac{4}{3} \Rightarrow x = \frac{4}{9}$

b) $(9^x)(25^x) = \sqrt[4]{15} \Leftrightarrow (3^{2x})(5^{2x}) = 15^{\frac{1}{4}}$

$\Leftrightarrow (15)^{2x} = 15^{\frac{1}{4}} \Leftrightarrow 2x = \frac{1}{4} \quad \therefore x = \frac{1}{8}$

c) **Exercise left for you**

d) **Exercise left for you**

23. Solve

a) $3^{|x|} = 9.3^{-|x|}$ b) $8^{|x+1|} = 16^{|2x-3|}$

Solution: Here

a) $3^{|x|} = 3^2 \cdot 3^{-|x|} \Leftrightarrow 3^{2-|x|} \Leftrightarrow |x| = 3 - |x|$

$\Rightarrow |x| + |x| = 2 \Rightarrow 2|x| = 2$

$|x| = 1$, therefore, $x = 1$ or $x = -1$

b) $8^{|x+1|} = 16^{|2x-3|} \Leftrightarrow 2^{3|x+1|} = 2^{4|2x-3|}$

$\Leftrightarrow 3|x+1| = 4|2x-3|$

$\Rightarrow 3x + 3 = \pm(8x - 12)$

$\Rightarrow 3x + 3 = 8x - 12$ and $3x + 3 = -8x + 12$

$\Rightarrow -5x = -15$ and $11x = 9$

$\therefore x = 3$ and $x = \frac{9}{11}$

Explanation:
$|x| = |y|$
$\Leftrightarrow x = \pm y$

24. Solve each of the following.

a) $10^x + 10^{x-1} = 0.11$ c) $2^{2x+2} = 9(2^x) - 2$

b) $9^x = 4(3^x) - 3$ d) $5^x + 5^{x-1} = \frac{6}{25}$

Solution:

a) $10^x + 10^{x-1} = 0.11 \Leftrightarrow 10^x + 10^x \cdot 10^{-1} = 0.11$

$\Leftrightarrow 10^x \left(1 + \frac{1}{10}\right) = 0.11$

$\Rightarrow 10^x \left(\frac{11}{10}\right) = 0.11$

$\Rightarrow 10^x = (0.11)\frac{(10)}{11} = \frac{1}{10} \Rightarrow = 10^x = 10^{-1}$

$\therefore x = -1$

b) $9^x = 4(3^x) - 3 \Leftrightarrow (3^x)^2 - 4(3^x) + 3 = 0$ Let $U = 3^x$.

$\Leftrightarrow U^2 - 4U + 3 = 0 \Rightarrow (U-3)(U-1) = 0$

$U = 3$ or $U = 1$, since $U = 3^x$

$\Rightarrow 3^x = 3$ or $3^x = 1$

$\therefore x = 1$ or $x = 0$

c) $2^{2x+2} = 9(2^x) - 2 \Leftrightarrow (2^x)^2 \cdot 4 - 9(2^x) + 2 = 0$

Let $U = 2^x$.

$\Leftrightarrow 4U^2 - 9U + 2 = 0 \Rightarrow (4U-1)(U-2) = 0$

$\Rightarrow 4U - 1 = 0$ or $U - 2 = 0$

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$\Rightarrow U = \frac{1}{4}$ or $U = 2$, since $U = 2^x$

$\Rightarrow 2^x = \frac{1}{4}$ or $2^x = 2$

$\Rightarrow 2^x = 2^{-2}$ or $2^x = 2$

$\therefore x = -2$ or $x = 1$

d) $5^x + 5^{x-1} = \frac{6}{25} \Leftrightarrow \left(1+\frac{1}{5}\right) = \frac{6}{25}$

$\Rightarrow 5^x\left(\frac{6}{5}\right) = \frac{6}{25}$

$5^x = \frac{6}{25}\cdot\frac{5}{6} = \frac{1}{5} \Leftrightarrow 5^x = 5^{-1} \therefore x = -1$

25. What is the solution set of the equation $4^{2x+2} = 4^{2x-1} + 126$.

A) $\left\{\frac{3}{4}\right\}$ B) {6} C) { } D) {3}

Solution: $4^{2x+2} = 4^{2x-1} + 126 \Leftrightarrow 16.(4^{2x}) - 4^{2x}.\left(\frac{1}{4}\right) = 126$

$\Rightarrow 4^{2x}\left(16 - \frac{1}{4}\right) = 126$

$4^{2x}\left(\frac{63}{4}\right) = 126 \Rightarrow \frac{4^{2x}}{4} = \frac{126}{63} = 2$

$\Rightarrow 4^{2x-1} = 2^1 \Leftrightarrow 2^{4x-2} = 2^1 \Leftrightarrow 4x - 2 = 1$

$\Rightarrow 4x = 3 \Rightarrow x = \frac{3}{4}$ ∴ Answer: A

26. What is the solution set of $25^{1-x} = \sqrt{125}$

A) $\frac{1}{2}$ B) $\frac{1}{4}$ C) $\frac{3}{2}$ D) $\frac{-1}{2}$

Solution: **Exercise left for you.**

27. The solution set of the equation $\left(\sqrt{5}^{3}\right)\left(5^{x^2-1}\right) = 0.2$

Solution: $\left(\frac{1}{5^{\frac{1}{2}x}}\right)5^{x^2-1} = 5^{-1} \Rightarrow 5^{x^2+\frac{1}{2}x-1} = 5^{-1}$

$\Leftrightarrow x^2 + \frac{1}{2}x - 1 = -1 \Rightarrow x^2 + \frac{1}{2}x = 0$

$\Rightarrow x\left(x + \frac{1}{2}\right) = 0 \Rightarrow x = 0$ or $x = -\frac{1}{2}$

28. If $(10^{21})\left(\sqrt{10^{18}}\right) = \left(\sqrt{5}\sqrt{2}\right)^x$, then the value of x is

Solution: $10^{21}\cdot 10^{0.9} = \left(\sqrt{10}\right)^x \Leftrightarrow 10^{2.1+0.9} = 10^{\frac{1}{2}x}$

$\Rightarrow 10^3 = 10^{\frac{1}{2}x} \Leftrightarrow \frac{1}{2}x = 3 \Rightarrow x = 6$

29. If $8^x = 16^y$, then $\frac{x}{y} =$ _____

Solution: $\Rightarrow 2^{3x} = 2^{4y} \Leftrightarrow 3x = 4y$

$\Rightarrow \frac{3x}{y} = 4 \Rightarrow \frac{x}{y} = \frac{4}{3}$

30. What is the solution set of $(0.1)^y = \left(\sqrt[3]{10}\right)^2$

A. $\left\{\frac{-3}{2}\right\}$ B. $\left\{\frac{2}{3}\right\}$ C) $\left\{\frac{-2}{3}\right\}$ D) $\left\{\frac{1}{3}\right\}$

Solution: $(0.1)^y = \left(\sqrt[3]{10}\right)^2 \Leftrightarrow 10^{-y} = 10^{\frac{2}{3}}$

$\Rightarrow -y = \frac{2}{3} \Rightarrow y = \frac{-2}{3}$ **∴Answer: C**

31. a) If $4^x = 7$, what does 4^{-2x} equal?

Solution: $4^{-2x} = (4^x)^{-2} = (7)^{-2} = \frac{1}{49}$

b) If $3^{-x} = 2$, what does 3^{2x} equal

Solution: $3^{2x} = \frac{1}{3^{-2x}} = \frac{1}{(3^{-x})^2} = \frac{1}{2^2} = \frac{1}{4}$

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32. If $16\cdot(3^x) = 81(2^x)$, then x is equal to
A. 4 B. 1 C. 3 D. 2

Solution: $16\cdot(3^x) = 81(2^x) \Rightarrow \frac{1}{81} = \frac{2^x}{3^x} \Leftrightarrow \left(\frac{2}{3}\right)^4 = \left(\frac{2}{3}\right)^x \Rightarrow x = 4$

∴ Answer: A

33. Which of the following is true value of x, in $\left(\frac{125}{27}\right)^x = \sqrt[3]{0.6}$....

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1. $\frac{-1}{3}$ B. 1 C. $\frac{1}{9}$ D. $\frac{-1}{9}$

Solution: $\left(\frac{5}{3}\right)^{3x} = \left(\frac{6}{10}\right)^{\frac{1}{3}} \Leftrightarrow \left(\frac{3}{5}\right)^{-3x} = \left(\frac{3}{5}\right)^{\frac{1}{3}}$

$\Rightarrow -3x = \frac{1}{3} \Rightarrow = \frac{-1}{9}$

∴ Answer: D

34. What is the solution set of $(9^{2x-1})\,(16^{x-0.5})\,5) = \sqrt[3]{36}$

Solution: $(3^2)^{2x-1}\,(2^{4(x-0.5)}) = \sqrt[3]{36}$

$\Rightarrow 3^{4x-2} \cdot 2^{4x-2}\, 6^{\frac{2}{3}}$

$\Rightarrow (6)^{4x-2} = 6^{\frac{2}{3}}$ → **Explanation:** $a^m \cdot b^m = (ab)^m$

$\Leftrightarrow 4x - 2 = \frac{2}{3} \Rightarrow 4x = \frac{2}{3} + 2 = \frac{8}{3}$

$\Rightarrow x = \frac{8}{3} \times \frac{1}{4} = \frac{2}{3}$

35. The solution set of the equation $(2)^{x^2} = \frac{2}{8^{x+1}}$ is
A. $\{1, -1\}$ B. $\{-2, -1\}$ $\{-1\}$ D. $\{2\}$

Solution: $(2)^{x^2} = \frac{2}{8^{x+1}} \Rightarrow 2^1 \cdot (2^{-3x-3})$

$\Rightarrow 2^{x^2} = 2^{-3x-3+1} \Leftrightarrow 2^{x^2} = 2^{-3x-2}$

$\Leftrightarrow x^2 = -3x - 2 \Leftrightarrow x^2 + 3x + 2 = 0$

$\Rightarrow (x+2)(x+1) = 0$

$\therefore x = -2$, or $x = -1$ ∴ **Answer: B**

Logarithms

- Logarithm is the **inverse** of exponents we know that:
$2^3 = 8 \longleftarrow$ **exponential form**.
In logarithmic form: $\log_2^8 = 3$ we **read as** logarithm of 8 to base 2 is 3.

> • **The logarithm of** x to base **b** is denoted by $m = \log_b^x$, $b \neq 1$, $b > 0$, read as "m is the logarithm of x to base b.
> **Definition:** $m = \log_b^x$ if and only if $b^m = x$ for $x > 0$, and $b > 0$, $b \neq 1$

- $\log_2^8$ means "to what power of 2 is 8"

Illustrative Example

36. Change to logarithmic form and express in statement form
a) $4^3 = 64$ b) $2^{-4} = \frac{1}{16}$ c) $3^5 = 243$
d) $a^6 = 24$ e) $m^k = n$ f) $e^t = 8$

Solution: we use the fact that $\log_b^x = m \Leftrightarrow b^m = x$

a) $4^3 = 64$ is the same as $\log_4^{64} = 3$
- Read **the logarithm of** 64 to base 4 is 3

b) $2^{-4} = \frac{1}{16}$ is the same as $\log_2^{\frac{1}{16}} = -4$
- **Read** "-4 is the logarithm of $\frac{1}{16}$ to base 2"

c) $3^5 = 243$ is the same as $\log_3^{243} = 5$
- **Read** "5 is the logarithm of 243 to base 3"

d) $a^6 = 24$ is the same as $\log_a^{24} = 6$
- **Read** "5 is the logarithm of 24 to base a"

e) $m^k = n$ is the same as $\log_m^n = k$
- **Read** "k is the logarithm of n to base m"

f) Exercise left for you.

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37. Change to exponential form

a) $\log_{13}^{169} = 2$ b) $\log_4^x = 3$ c) $\log_2^{\frac{1}{8}} = -3$

d) $\log_4^{\frac{1}{16}} = -2$ e) $\log_a^n = y$

Solution: Here

a) $\log_{13}^{169} = 2 \Leftrightarrow 13^2 = 169$

b) $\log_4^x = 3 \Leftrightarrow 4^3 = x$

c) $\log_2^{\frac{1}{8}} = -3 \Leftrightarrow 2^{-3} = \frac{1}{8}$

d) $\log_4^{\frac{1}{16}} = -2 \Leftrightarrow 4^{-2} = \frac{1}{6}$

e) $\log_a^n = y \Leftrightarrow a^y = x$

Finding the value of logarithm

- **To evaluate** $\log_b^m$ we let the exponent x such that $b^x = m$

  i.e we set $\log_b^m = x \Leftrightarrow b^x = m$

Illustrative Example

38. Find the exact value

a) $\log_3^{81}$ b) $\log_{\sqrt{2}}^{\sqrt[3]{16}}$ c) $\log_{\sqrt[3]{9}}^{\sqrt[4]{27}}$ d) $\log_{100}^{0.1}$

Solution: In each case, we let the exponent x for given $\log_b^m$. So that:

a) $\log_3^{81} = x \Leftrightarrow 3^x = 81 \Rightarrow 3^x = 3^4$

$\Rightarrow x = 4$, therefore, $\log_3^{81} = 4$

b) $\log_{\sqrt{2}}^{\sqrt[3]{16}} = x \Leftrightarrow \sqrt{2}^x = \sqrt[3]{16} \Rightarrow 2^{\frac{1}{2}x} = 2^{\frac{4}{3}}$

$\Leftrightarrow \frac{1}{2}x = \frac{4}{3} \Rightarrow x = \frac{8}{3}$

$\therefore \log_{\sqrt{2}}^{\sqrt[3]{16}} = \frac{8}{3}$

c) $\log_{\sqrt[3]{9}}^{\sqrt[4]{27}} = x \Leftrightarrow \left(\sqrt[3]{9}\right)^x = \sqrt[4]{27}$

$\Leftrightarrow 3^{\frac{2}{3}x} = 3^{\frac{3}{4}} \Rightarrow \frac{2}{3}x = \frac{3}{4}$

$\Rightarrow x = \left(\frac{3}{4}\right)\left(\frac{3}{2}\right) = \frac{9}{8}$ $\therefore \log_{\sqrt[3]{9}}^{\sqrt[4]{27}} = \frac{9}{8}$

d) $\log_{100}^{0.1} = x \Leftrightarrow 100^x = 0.1 \Leftrightarrow 10^{2x} = 10^{-1}$

$\Leftrightarrow 2x = -1 \Rightarrow x = \frac{-1}{2}$ $\therefore \log_{100}^{0.1} = \frac{-1}{2}$

39. Simplified form of $\log_{\frac{1}{2}}^{8\sqrt[3]{2}}$

A. $\frac{20}{3}$ B. $\frac{10}{3}$ C. $\frac{-10}{3}$ D. $\frac{3}{10}$

Solution: Here

Let $\log_{\frac{1}{2}}^{8\sqrt[3]{2}} = x \Leftrightarrow \left(\frac{1}{2}\right)^x = 8\sqrt[3]{2}$

$\Rightarrow 2^{-x} = 2^3 \cdot 2^{\frac{1}{3}} = 2^{3+\frac{1}{3}} = 2^{\frac{10}{3}}$

$\Leftrightarrow 2^{-x} = 2^{\frac{10}{3}} \Rightarrow -x = \frac{10}{3}$

$\Rightarrow x = \frac{-10}{3}$ **$\therefore$ Answer: C**

40. Find x, if:

a) $\log_x^{\sqrt{2}} = \frac{1}{6}$ b) $\log_{5x}^5 = \frac{1}{2}$

c) $\log_{x-1}^3 = 2$ d) $\log_x^{\frac{1}{4}} = -2$

Solution:

a) $\log_x^{\sqrt{2}} = \frac{1}{6}$ $\Leftrightarrow x^{\frac{1}{6}} = \sqrt{2} \Rightarrow x = \left(\sqrt{2}\right)^6 = 2^3 = 8$

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b) $\log_{5x}^{5} = \frac{1}{2} \Leftrightarrow (5x)^{\frac{1}{2}} = 5x = 25 \therefore x = \frac{25}{5} = 5$

c) $\log_{x-1}^{3} = 2 \Leftrightarrow (x-1)^2 = 3 \Leftrightarrow x - 1 = \sqrt{3} \Rightarrow x = 1 + \sqrt{3}$

d) $\log_{x}^{\frac{1}{4}} = -2 \Leftrightarrow x^{-2} = \frac{1}{4} \Rightarrow x^{-2} = 2^{-2} \Leftrightarrow x = 2$

41. Simplify: $\log_{\frac{1}{2}}^{4} + \log_{\sqrt[3]{\frac{1}{2}}}^{4} - \log_{\frac{1}{27}}^{\sqrt[4]{243}} + \log_{\frac{1}{\sqrt{2}}}^{\sqrt[3]{128\sqrt{2}}}$

Solution: Here

- Let $\log_{\frac{1}{2}}^{4} = x \Leftrightarrow \left(\frac{1}{2}\right)^{x} = 4 \Rightarrow 2^{-x} = 2^2 \Leftrightarrow x = -2$
- Let $\log_{\sqrt[3]{\frac{1}{2}}}^{4} = m \Leftrightarrow \left(\sqrt[3]{\frac{1}{2}}\right)^{m} = 4$

  $\Leftrightarrow 2^{\frac{-m}{3}} = 2^2 \Rightarrow \frac{-m}{3} = 2$

  $\Rightarrow m = -6$
- Let $\log_{\frac{1}{27}}^{\sqrt[4]{243}} = k \Leftrightarrow \left(\frac{1}{27}\right)^{k} = \sqrt[4]{243} \Leftrightarrow k = \frac{-5}{12}$
- Let $\log_{\frac{1}{\sqrt{2}}}^{\sqrt[3]{128\sqrt{2}}} = n \Leftrightarrow \left(\frac{1}{\sqrt{2}}\right)^{n} = \sqrt[3]{128\sqrt{2}}$

  $\Leftrightarrow 2^{\frac{-1}{2}n} = \left(2^7 \cdot 2^{\frac{1}{2}}\right)^{\frac{1}{3}} = \left(2^{\frac{15}{2}}\right)^{\frac{1}{3}}$

  $\Leftrightarrow 2^{\frac{-1}{2}n} = 2^{\frac{1}{3}} \Rightarrow \frac{-1}{2}n = \frac{5}{2}$

  $\therefore n = -5$

$\therefore x + m - k + n = -2 - 6 - \left(\frac{-5}{12}\right) + -5 = \frac{-151}{12}$

42. The value of $\log_{8}^{4^5}$ is equal to _____

Solution: let $\log_{8}^{4^x} = x \Rightarrow 8^x = 4^5$

$\Leftrightarrow \left(2^3\right)^x = \left(2^2\right)^5 \Leftrightarrow 2^{3x} = 2^{10}$

$\Leftrightarrow 3x = 10 \Rightarrow x = \frac{10}{3}$

Therefore, $\log_{8}^{4^x} = \frac{10}{3}$

Properties of logarithms

A. Change of Base formula

Let a,b, and x be positive real numbers such that $a \neq 1$, and $b \neq 1$.

The $\log_a^x$ can be converted to a different base using any of the following formula

Base b	Base 10	Base 'e'
$\log_a^x = \frac{\log_b^x}{\log_b^a}$	$\log_a^x = \frac{\log_{10}^x}{\log_{10}^a}$	$\log_a^x = \frac{\ell nx}{\ell na}$

Example a) $\log_{27}^{81} = \frac{\log_3^{81}}{\log_3^{27}} = \frac{4}{3}$

Note: • $\log_a^x = \frac{\log_x^x}{\log_x^a} = \frac{1}{\log_x^a}$ • $\log_a^b = \frac{\log_b^b}{\log_b^a} = \frac{1}{\log_b^a}$

• $\left(\log_a^x\right)\left(\log_b^a\right) = \log_b^x$

B. Properties of logarithm

For x > 0, y > 0, and $k \in R$

1. $\log_a^{xy} = \log_a^x + \log_a^y$
2. $\log_a^{\left(\frac{x}{y}\right)} = \log_a^x - \log_a^y$,
3. $\log_b^{\left(x^k\right)} = k \log_b^x$
4. $\log_{\left(b^k\right)}^x = \frac{1}{k}\log_b^x$

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5. $b^{\log_b^x} = x$

Example (property 1)

- $\log^{(8)(32)} = \log_2^{256} = 8$
- $\log^{(8)(32)} = \log_2^8 + \log_2^{32} = 3 + 5 = 8$

Example (property 2)

- $\log_2^{\left(\frac{256}{8}\right)} = \log_2^{32} = 5$
- $\log_2^{\left(\frac{256}{8}\right)} = \log_2^{256} - \log_a^8 = 8 - 3 = 5$

Illustrative Example

43. Express in terms of logarithm x,y, and z.

a) $\log_4^{(xy)} = \log_4^x + \log_4^y$ ⟵ by property 1

b) $\log_4^{\sqrt[3]{z}} = \log_4^{\left(z^{\frac{1}{3}}\right)} = \frac{1}{3}\log_4^z$ ⟵ by property 3

c) $\log_3^{\left(\frac{xz}{y}\right)} = \log_3^{xz} - \log_3^y$

$= \log_3^x + \log_3^z - \log_3^y$

44. Express as single logarithm.

a) $\frac{1}{3}\log_a^{(x^2-1)} - \log_a^y - 4\log_a^z$

b) $z\log_b^x - \log_b^y$

Solution: we apply the laws of logarithm.

a) $\frac{1}{3}\log_a^{(x^2-1)} - \log_a^y - 4\log_a^z$

$= \log_a^{(x^2-1)^{1/3}} - -\log_a^y - \log_a^{z^4}$

$= \log_a^{\sqrt[3]{x^2-1}} - \left(\log_a^y + \log_a^{z^4}\right)$

$= \log_a^{\sqrt[3]{x^2-1}} - \left(\log_a^{yz^4}\right) = \log_a^{\frac{\sqrt[3]{x^2-1}}{yz^4}}$

b) $z\log_b^x - z\log_b^y = z\left(\log_b^x - \log_b^y\right) = z\log_b^{\left(\frac{x}{y}\right)^z}$

45. Evaluate:

a) $\log_{100}^{\sqrt[4]{0.0001}}$ b) $\log_9^{\sqrt[5]{81}}$ c) $\log_{\sqrt[3]{5}}^{125}$

Solution: Here: **Remember** $\log_b^b = 1$

a) $\log_{100}^{\sqrt[4]{0.0001}} = \log_{100}^{(0.0001)^{\frac{1}{4}}} = \frac{1}{4}\log_{100}^{0.0001}$

$= \frac{1}{4}\log_{100}^{100^{-2}} = \frac{-4}{4}\log_{100}^{100} = -1$

b) $\log_9^{\sqrt[5]{81}} = \log_9^{81^{\frac{1}{5}}} = \frac{1}{5}\log_9^{81}$

$= \frac{1}{5}\log_9^{(9^2)} = \frac{2}{5}\log_9^9 = \frac{2}{5}(1) = \frac{2}{5}$

c) $\log_{\sqrt[3]{5}}^{125} = \log_{5^{\frac{1}{3}}}^{125} = 3\log_5^{125}$

$= 3\log_5^{5^3} = (3)(3)\log_5^5 = 9(1) = 9$

46. Evaluate:

a) $5^{1-\log_{125}^4}$ b) $\left(\log_{\sqrt[3]{3}}^{81}\right) + 3^{1+\log_3^4}$

c) $2^{\log_{2\sqrt{2}}^{16}}$ d) $4^{\log_{16}^7}$

Solution: Here

a) $5^{1-\log_{125}^4} = \left(5^1\right)\left(5^{-\log_{(5^3)}^4}\right)$

$= \left(5^1\right)\left(5^{\frac{-1}{3}\log_5^4}\right)$ ⟵ from $\log_{b^k}^c = \frac{1}{k}\log_b^c$

$= \left(5^1\right)\left(5^{\log_5^{(4)^{-\frac{1}{3}}}}\right) = (5)\left(\frac{-4}{3}\right) = \frac{-20}{3}$

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b) $\log_{3\sqrt{3}}^{81} = \log \dfrac{81}{\sqrt{27}} = \dfrac{\log_3^{81}}{\log_3^{\sqrt{27}}} = \dfrac{4\log_3^3}{\frac{3}{2}\log_3^3} = \dfrac{8}{3}$

$3^{1+\log_3^4} = 3^1 \cdot 3^{\log_3^4} = (3)(4) = 12$

$\therefore \log_{3\sqrt{3}}^{81} + \left(3^{1+\log_3^4}\right) = \dfrac{8}{3} + 12 = \dfrac{44}{3}$

c) $2^{\log_{2\sqrt{2}}^{15}} = 2^{\log_{\sqrt{8}}^{15}} = 2^{2\log_8^{15}}$

$= 2^{2\log_{2^3}^{15}}$

$= 2^{\frac{2}{3}\log_2^{15}}$

$= 2^{\log_2^{(15)^{\frac{2}{3}}}} = 15^{\frac{2}{3}} = \sqrt[3]{15^2} = \sqrt[3]{225}$

d) $4^{\log_{16}^7} = 4^{\log_{4^2}^7} = 4^{\frac{1}{2}\log_4^7} = 4^{\log_4^{\sqrt{7}}} = \sqrt{7}$

47. **Evaluate:**

a) $\log_3^{4\sqrt{27}} + \log_3^{6\sqrt{3}} - \log_3^8$

Solution: a) $\log_3^{4\sqrt{27}} + \log_3^{6\sqrt{3}} - \log_3^8$

$= \log_3^{\left(\frac{4\sqrt{27}\times 6\sqrt{3}}{8}\right)} = \log_3^{\frac{24\times\sqrt{81}}{8}} = \log_3^{27} = 3$

48. $a^{\log_{3\sqrt{a}}^k}$ is equal to . . . **(Arbon)**

A. $\dfrac{k}{3}$ B. 3K C. $\sqrt[3]{k}$ D. K^3

Solution: $a^{\log_{3\sqrt{a}}^k} = a^{3\log_a^k} = a^{\log_a^{k^3}} = k^3$ **∴ Answer: D**

49. $5^{1-\log_5^3}$ is equal to . . . .

A. $\dfrac{5}{3}$ B. 2 C. 25 D. –15

Solution: $5^{1-\log_5^3} = \left(5^1\right)\left(5^{-\log_5^3}\right) = (5)(3^{-1}) = \dfrac{5}{3}$ **∴ Answer: A**

50. Simplify . . . . **Arbon**

a. $\log_1^2 \cdot \log_4^3 \cdot \log_5^4 \cdot \log_6^5 \cdot \log_7^6 \cdot \log_8^7$

Solution: using the base law formula the expression is the same as:

$\dfrac{\log_a^2}{\log_a^1} \cdot \dfrac{\log_a^3}{\log_a^4} \cdot \dfrac{\log_a^4}{\log_a^5} \cdot \dfrac{\log_a^5}{\log_a^6} \cdot \dfrac{\log_a^6}{\log_a^7} \cdot \dfrac{\log_a^7}{\log_a^8}$

$= \dfrac{\log_a^2}{\log_a^8} = \log_8^2 = \dfrac{1}{\log_2^8} = \dfrac{1}{3}$

51. If $\log_a^x = m, \log_a^y = n$, then **evaluate**

a) $\log_a^{x^2y}$ b) $\log_a^{\frac{x}{y^2}}$ c) $\log_a^{\frac{x^3}{\sqrt{y}}}$ d) $\log_a^{\frac{x^2}{y^3}}$

Solution: using the laws of logarithm

a) $\log_a^{x^2y} = \log_a^{x^2} + \log_a^y = 2\log_a^x + \log_a^y = 2m + n$

b) $\log_a^{\frac{x}{y^2}} = \log_a^x - \log_a^{y^2}$

$= \log_a^x - 2\log_a^y$

$= m - 2n$

c) $\log_a^{\frac{x^3}{\sqrt{y}}} = \log_a^{x^3} - \log_a^{y^{\frac{1}{2}}} = 3\log_a^x - \dfrac{1}{3}\log_a^y = 3m - \dfrac{1}{3}n$

d) $\log_a^{\frac{x^2}{y^3}} = \log_a^{x^2} - \log_a^{y^3} = 2\log_a^x - 3\log_a^y = 2m - 3n$

52. If $\log_a^3 = x$ and $\log_a^2 = y$, when $\log_a^{\frac{\sqrt[3]{2}}{\sqrt{3}}}$ is equal to. . . .

A. $\dfrac{1}{6}(3x + 2y)$ B. $\dfrac{1}{6}(2y - 3x)$

C. $\dfrac{1}{6}(3y - 2x)$ D. $\dfrac{1}{6}(2x - 3y)$

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Solution: $\log_a^{\frac{\sqrt{2}}{\sqrt{3}}} = \log_a^{2^{\frac{1}{2}}} - \log_a^{3^{\frac{1}{2}}} = \frac{1}{3}\log_a^2 - \frac{1}{2}\log_a^3$

$= \frac{1}{3}y - \frac{1}{2}x = \frac{1}{6}(2y - 3x)$, **∴ Answer: C**

53. If $\log_a^4 = x, \log_a^3 = y$ *and* $\log_a^b = z$, *then* $\log_b^{36}$ is equal to

A. $\frac{2y+x}{z}$ B. $\frac{y^2+x}{z}$ C. z(2y + x) D. $\frac{z}{y^2+x}$

Solution: $\log_b^{36} = \frac{\log_a^{36}}{\log_a^b} = \frac{\log_a^{(4.33)}}{\log_a^b} = \frac{\log_a^4 + \log_a^3 + \log_a^3}{\log_a^b}$

$= \frac{x+y+y}{z} = \frac{x+2y}{z}$ **∴ Answer: A**

54. **Given**, that $\log_2^5 = 2.322$ and $\log_2^3 = 1.585$, then **evaluate**:

a) $\log_2^{40}$ b) $\log_2^{25}$ c) $\log_2^{15}$ d) $\log_2^{60}$ e) $\log_2^{3\sqrt{0.6}}$

Solution: Here

a) $\log_2^{40} = \log_2^{(5.8)} = \log_2^5 + \log_2^8 = \log_2^5 + \log_2^{2^3}$

$= \log_2^5 + 3\log_2^2 = 2.322 + 3 = 5.322$

b) $\log_2^{25} = \log_2^{5^2} = 2\log_2^5 = 2(2.322) = 4.644$

c) $\log_2^{15} = \log_2^{(3.5)} = \log_2^3 + \log_2^5 = 1.585 + 2.322 = 5.907$

d) $\log_2^{60} = \log_2^{(53.2^2)} = \log_2^5 + \log_2^3 + 2\log_2^2$

$= 2.322 + 1.585 + 2 = 5.907$

e) $\log_2^{3\sqrt{0.6}} = = \frac{1}{3}\log_2^{\frac{(2\times3)}{10}} = \frac{1}{3}\log_2^{\frac{(2\times3)}{2\times5}}$

$= = \frac{1}{3}\log_2^{\left(\frac{3}{5}\right)} = \frac{1}{3}l\left(og_2^3 - \log_2^5\right)$

$= \frac{1}{3}(1.585 - 2.322) = \frac{1}{3}(-0.737) = -0.246$

55. If $\log_{10}^2 = 0.3010$ and $\log_{10}^3 = 0.4771$, then **find**

a) $\log_2^{\sqrt{3}}$ b) $\log_3^5$ c) $\log_3^{0.02}$ d) $\log_2^{0.003}$

Solution: **Remember that**: $\log_a^b = \frac{\log_{10}^b}{\log_{10}^a}$

a) $\log_2^{\sqrt{3}} = \frac{\log_{10}^{\sqrt{3}}}{\log_{10}^2} = \frac{\frac{1}{2}\log_{10}^3}{\log_{10}^2} = \frac{\frac{1}{2}(0.4771)}{0.3010} = 0.7925$

b) $\log_4^5 = \frac{\log_{10}^5}{\log_{10}^4} = \frac{\log_{10}^{10/2}}{2\log_{10}^2} = \frac{\log_{10}^{10} - \log_{10}^2}{2\log_{10}^2}$

$= \frac{1-0.3010}{2(0.3010)} = \frac{0.699}{0.6020} \approx 1.16$

c) $\log_3^{0.02} = \frac{\log_{10}^{0.02}}{\log_{10}^3} = \frac{\log_{10}^{2/100}}{2\log_{10}^2} = \frac{\log_{10}^5 - \log_{10}^{100}}{2\log_{10}^3}$

$= \frac{0.3010-2}{0.4771} = \frac{-1.699}{0.4771} \approx -3.56$

d) $\log_2^{0.003} = \frac{\log_{10}^{0.003}}{\log_{10}^2} = \frac{\log_{10}^{3/1000}}{\log_{10}^2} = \frac{\log_{10}^3 - \log_{10}^{1000}}{\log_{10}^2}$

$= \frac{0.4771-3}{0.3010} \approx -8.4$

56. If $\log_4^a = 2.4$, then $\log_{\sqrt{a}}^{64}$ is equal to:

A. 1.25 B. 2.5 C. 0.4 D. 3.6

Solution: Express in terms of base 4

$\log_{\sqrt{a}}^{64} = \frac{\log_4^{64}}{\log_4^{\sqrt{a}}} = \frac{\log_4^{4^3}}{\frac{1}{2}\log_4^a}$

$= \frac{3}{\frac{1}{2}(2.4)} = \frac{6}{2.4} = 2.5$, **therefore,** **∴ Answer: B**

Explanation:

- $\log_y^x = \frac{\log_a^x}{\log_a^y}$

57. Given that $\log_{10}^5 = b$, then $\log_5^4$ is equal to

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A. $2\left(\frac{b}{b-1}\right)$ B) $2\left(\frac{b}{1-b}\right)$ C) $2\left(1-\frac{1}{b}\right)$ D. $2\left(\frac{1}{b}-1\right)$

Solution: Express in terms of base 10.

$$\log_5^4 = \frac{\log_{10}^4}{\log_{10}^5} = \frac{2\log_{10}^2}{b} = \frac{2\left(\log_{10}^{10/5}\right)}{b}$$

$$= \frac{2\left(\log_{10}^{10} - \log_{10}^5\right)}{b} = \frac{2(1-b)}{b} = 2\left(\frac{1}{b}-1\right)$$

58. If $\log_a^2 = x$, then $\log_{\sqrt{2a}}^8$ is equal to

A. $\frac{x^2}{x+1}$ B. $\frac{\sqrt{x}}{3x+3}$ C. $\frac{6x}{x+1}$ D. $\frac{3x}{x+2}$

Solution: Express in terms of the base a

$$\Rightarrow \log_{\sqrt{2a}}^8 = \frac{\log_a^8}{\log_a^{\sqrt{2a}}} = \frac{3\log_a^2}{\frac{1}{2}\log_a^{2a}} = \frac{2\log_a^2}{\frac{1}{2}\left(\log_a^2 + \log_a^a\right)}$$

$$= \frac{6\log_a^2}{(x+1)} = \frac{6x}{x+1}, \textbf{ therefore,} \quad \textbf{Answer: C}$$

59. **Given that**: $\log_{10}^2 = b$, then $\log_4^{0.05}$ is equal to

A. $-b-\frac{1}{2}$ B. $-(1+4b)$ C. $\frac{-(b+1)}{2b}$ D. $\frac{b-2}{2b}$

Solution: $\log_4^{0.05} = \frac{\log_{10}^{0.05}}{\log_{10}^4} = \frac{\log_{10}^{\frac{5}{100}}}{2\log_{10}^2}$

$$= \frac{\log_{10}^5 - \log_{10}^{100}}{2\log_{10}^2}$$

$$= \frac{\log_{10}^{10/2} - \log_{10}^{100}}{2\log_{10}^2}$$

Explanation:

$\log_{10}^5 = \log_{10}^{10/2} = \log_{10}^{10} - \log_{10}^2$

$= 1 - \log_{10}^2$

$= \frac{1-b-2}{2b} = \frac{-(1+b)}{2b}$, **therefore,** **Answer: C**

60. Suppose $\log_7^2 = a$ and $\log_7^5 = b$. therefore, $\log_7^{0.002}$

A. $-3a-3b$ B. $-2a-3b$ C. $3a+3b$ D. $2a+3b$

Solution: $\log_7^{0.002} = \log_7^{(2\times10^{-3})} = \log_7^2 + \log_7^{10^{-3}}$

$= \log_7^2 - 3\log_7^{10}$

$= \log_7^2 - 3\left(\log_7^2 + \log_7^5\right)$

Explanation:

$\log_7^{10} = \log_{70}^{(2\times5)}$

$= \log_7^2 = \log_7^5$

$= a + b$

$= a - 3(a+b) = -2a - 3b$, **therefore,** **Answer: B**

61. Let $a > 1$. If $\log_a^8 = 0.09$ and $\log_a^{4\sqrt{9}} = 0.04$ then $\log_a^6 \ldots$

A. 0.11 B. 0.05 C. 0.13 D. 0.17

Solution: We have

- $\log_a^8 = \log_a^{2^3} = 3\log_a^2 = 0.09$

  $\Rightarrow \log_a^2 = 0.03$

- $\log_a^{4\sqrt{9}} = \log_a^{9^{\frac{1}{4}}} = \frac{1}{4}\log_a^9 = \frac{2}{4}\log_a^3 = \frac{1}{2}\log_a^3$

- $\log_a^6 = \log_a^{(2\times3)} = \log_a^2 + \log_a^3$

  $= 0.03 + 0.08 = 0.11$, **therefore,** **Answer: A**

62. Let $a > 1$, If $\log_a^9 = x$ and $\log_a^{3\sqrt{5}} = y$ then $\log_a^{75}$ is equal

A. $2x+3y$ B. $3x+2y$ C. $6y+\frac{x}{2}$ D. $3y+x$

Solution: $\log_a^9 = x \Rightarrow \log_a^{3^2} = x$

$$\Rightarrow 2\log_a^3 = x \Rightarrow \log_a^3 = \frac{x}{2}$$

Similarly, $\log_a^{3\sqrt{5}} = \log_a^{5^{1/3}} = \frac{1}{3}\log_a^5 = y$

$$\Rightarrow \log_a^5 = 3y$$

Hence, $\log_a^{75} + \log_a^{(25\times3)} = \log_a^{(5^2\times3)}$

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$= 2\log_a^5 + \log_a^3 = 2(3y) + \frac{x}{2} = 6y + \frac{x}{2}$, ∴ Answer: C

63. Let $a^{0.35} = 2$, $a^{0.55} = 3$, $a^{0.82} = 5$, (where $a > 0$) If $a^x = \sqrt[10]{\frac{60}{a}}$, then the value of x

A. 107 B. 10.7 C. 0.107 D. $\sqrt[10]{1.07}$

Solution: Here

$a^{0.35} = 2 \Leftrightarrow \log_a^2 = 0.35$ $a^{0.55} = 3 \Leftrightarrow \log_a^3 = 0.55$

$a^{0.82} = 5 \Leftrightarrow \log_a^5 = 0.82$

Hence, $a^x = \sqrt[10]{\frac{60}{a}} \Leftrightarrow x = \log_a^{\sqrt[10]{\frac{60}{a}}} = \frac{1}{10}\left(\log_a^{\frac{60}{a}}\right) = \frac{1}{10}\left(\log_a^{60} - \log_a^a\right)$

$= \frac{1}{10}\left(\log_a^{(2\times2\times5\times3)} - 1\right) = \frac{1}{10}\left(2\log_a^2 + \log_a^5 + \log_a^3 - 1\right)$

$= \frac{1}{10}(2(0.35) + 0.82 + 0.55 - 1)$

$= \frac{1}{10}(0.7 + 0.82 + 0.55 - 1) = 0.107.$ ∴ Answer: C

64. Suppose, $m = a^y$, $n = a^z$ and $n = m^t$. then t is equal to

A. $n\log_a^m$ B. $\frac{\log_a^n}{\log_a^m}$

B. $(\log_a^m)(\log_m^n)$ D. $(\log_a^m)(\log_a^n)$

Solution: $n = m^t \Leftrightarrow \log_m^n = t$,

- $m = a^y \Leftrightarrow \log_a^m = y$ and $n = a^z \Leftrightarrow \log_a^n = z$

$\Leftrightarrow t = \log_m^n = \frac{\log_a^n}{\log_a^m} = \frac{z}{y}$ **therefore,** **Answer: B**

65. a) If $\log_a^x = 4$, determine each of the value

a) $\log_a^{\frac{1}{a}}$ b) $\log_a^{\frac{1}{x}}$ c) $\log_{\sqrt[3]{a}}^x$ d) $\log_a^{\sqrt{x}}$

Solution: Here

a) $\log_a^{\frac{1}{a}} = \log_{a^{-1}}^a = -\log_a^a = -4$

b) $\log_a^{\frac{1}{x}} = \log_a^{x^{-1}} = -\log_a^x = -4$

c) $\log_{\sqrt[3]{a}}^x = \log_{a^{\frac{1}{3}}}^x = 3\log_a^x = 3(4) = 12$

d) $\log_a^{\sqrt{x}} = \log_a^{x^{\frac{1}{2}}} = \frac{1}{2}\log_a^x = \frac{1}{2}(4) = 2$

Common logarithm

- The logarithm in base 10 is called

Common – logarithm, denoted by $y = \log_{10}^x$

- Common logarithm is usually written without indicating its base.

Example i) $\log_{10}^x = \log^x$ ii) $\log_{10}^{18} = \log^{18}$

iii) $\log_{10}^{100} = \log^{100}$

Computation of common logarithm

To illustrative consider the following.

i) $10^0 = 1 \Leftrightarrow \log_{10}^1 = 0$ iii) $10^2 = 100 \Leftrightarrow \log_{10}^{100} = 2$

ii) $10^1 = 10 \Leftrightarrow \log_{10}^{10} = 1$ iii) $10^3 = 1000 \Leftrightarrow \log_{10}^{1000} = 3$

Illustrative Example

66. Evaluate:

a) $\log^{0.1}$ b) $\log^{0.01}$ c) $\log^{0.001}$ d) $\log^{\sqrt[3]{100}}$ e) $\log^{\sqrt{0.001}}$

Solution: Here we know that $\log^x = \log_{10}^x$

a) $\log_{10}^{0.1} = \log_{10}^{10^{-1}} = -1\log_{10}^{10} = -1$

b) $\log_{10}^{0.01} = \log_{10}^{10^{-2}} = -2\log_{10}^{10} = -2(1) = -2$

c) $\log_{10}^{0.001} = \log_{10}^{10^{-3}} = (-3)\log_{10}^{10} = (-2)(1) = -3$

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d) $\log_{10}^{\sqrt[3]{100}} = \log_{10}^{10^{\frac{2}{3}}} = \frac{2}{3}\log_{10}^{10} = \frac{2}{3}(1) = \frac{2}{3}$

e) $\log_{10}^{\sqrt{0.001}}\ \log_{10}^{10^{-3/2}} = \frac{-3}{2}\log_{10}^{10} = \frac{-3}{2}$

67. **Evaluate:**

a) $\log^{(100\times\sqrt[3]{10})}$ b) $\log^{\frac{\sqrt[3]{100}}{\sqrt[4]{10}}}$

c) $\log^{(10^3\times\sqrt[3]{10})}$ d) $\log\left(\frac{10^{n+1}}{10^{1-n}}\right)$

Solution: Here, **use laws of logarithm**

a) $\log^{(100\times\sqrt[3]{10})} = \log_{10}^{100} + \log_{10}^{10^{\frac{1}{3}}} = 2 + \frac{1}{3} = \frac{7}{3}$

b) $\log_{10}^{\left(\frac{\sqrt[3]{100}}{\sqrt[4]{10}}\right)} = \log_{10}^{100^{\frac{1}{3}}} \div \log_{10}^{10^{1/4}} = \frac{1}{3}(2) - \frac{1}{4} = \frac{5}{12}$

c) $\log_{10}^{(10^3\times\sqrt[3]{10})} = \log_{10}^{10^3} + \log_{10}^{10^{1/5}} = 3 + \frac{1}{5} = \frac{16}{5}$

d) $\log_{10}^{\left(\frac{10^{n+1}}{10^{1-n}}\right)} = \log_{10}^{10^{n+1}} - \log_{10}^{10^{1-n}} = 3 + 1 - (1-n) = 2n$

68. Given that $\log 3 = 0.4771$, then, **evaluate:**

a) $\log 300$ b) $\log 0.03$ c) $\log 0.81$

Solution: **Using laws of logarithm.**

a) $\log_{10}^{300} = \log_{10}^{(3\times10^2)} = \log_{10}^{3} + \log_{10}^{10^2}$
$= 0.4771 + 2 = 2.4771$

b) $\log_{10}^{0.03} = \log_{10}^{(3\times10^{-2})} = \log_{10}^{3} + \log_{10}^{10^{-2}}$
$= 0.4771 - 2 = -1.5229$

c) $\log_{10}^{0.81} = \log_{10}^{(81\times10^{-2})} = \log_{10}^{81} + \log_{10}^{10^{-2}}$
$= \log_{10}^{3^4} - 2\log_{10}^{10} = 4\log_{10}^{3} - 2$
$= 4(0.4771) - 2 = 1.9084 - 2 = -0.0916$

69. Given that $\log 2 = 0.3010$, then, **evaluate:**

a) b) $\log 500$ c) $\log 0.8$

Solution: Here

a) $\log_2^5 = \frac{\log^5}{\log^2} = \frac{\log^{10/2}}{\log^2}$

$= \frac{0.699}{0.3010} = 2.322$

Explanation:

$\log^5 = \log^{10/2}$
$= \log_{10}^{10} - \log_{10}^{2}$
$= 1 - 0.3010 = 0.699$

b) $\log_{10}^{500} = \log_{10}^{(5\times10^2)} = \log_{10}^{5} + \log_{10}^{10^2}$
$= 0.699 + 2 = 2.699 \longleftarrow \log 5 = \log^5 = \log^{10/2}$
$= \log_{10}^{10} - \log_{10}^{2}$

c) $\log_{10}^{0.8} = \log_{10}^{(8\times10^{-1})} = \log_{10}^{8} + \log_{10}^{10^{-1}}$
$= \log_{10}^{2^3} + -1\log_{10}^{10}$
$= 3\log 2 - 1$
$= 3(0.3010) - 1$
$= -0.097$

70. Given that $\log_4^x = m$, then $\log_4^{x^3}$ is equal to

A. $3m + 1$ B. $m^3 + 1$ C. $3m$ D. m^3

Solution: $\log_4^{4x^3} = \log_4^4 + \log_4^{x^3} =$

$= 1 + 3\log_4^x = 1 + 3m$, **therefore** **Answer: A**

Table of logarithm

- Common logarithms of numbers between 1 and 10 can be found correct to four decimal places in the **tables**.

If we write in scientific notation $\mathbf{x = mx\ 10^C}$, where $0 \le m < 10$, then the logarithm of **x** can be read from a common logarithm table.

- The logarithm of **m** is called **mantisa of logarithm**
- C is called **the characteristics of logarithm.**

Therefore, the logarithm of a number has two parts these are

i) **The mantisa of a common logarithm** is the non – negative decimal part of logarithm.

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ii) **The characteristics of common logarithm** is the integral part of logarithm.

Illustrative Example

71. **Suppose that** $\log 2.63 = 0.4200$ then Find the common logarithm of
a) 263 b) 26300 c) 0.0263

Solution: Using scientific notation

a) $263 = 2.63 \times 10^2$ ⟵ scientific notation

$\Rightarrow \log 263 = \log(2.63\times 10^2) = \log_{10}^{2.63} + \log_{10}^{10^2}$

$= (\log 2.63) + 2 = 0.4200 + 2 = 2.4200$

Here, 2 is called characteristics of log 263

0.4200 is called mantisa of log 263.

b) $26300 = 2.63 \times 10^4$ ⟵ scientific notation

$\Rightarrow \log 26300 = \log(2.63\times 10^4) = \log_{10}^{2.63} + \log_{10}^{10^4}$

$= 0.4200 + 4 = 4.4200$

Here, 4 is called characteristics of log 26300

0.4200 is called mantisa of log 26300

c) $0.0263 = 2.63 \times 10^{-2}$ ⟵ scientific notation

$\Rightarrow \log 0.0263 = \log(2.63\times 10^{-2}) = \log_{10}^{2.63} + \log_{10}^{10^{-2}}$

$= 0.4200 + (-2)$

Here, -2 is characteristics of log 0.0263

Finding the characteristics and mantisa of the logarithm

To find the characteristics and mantisa of logarithm, first write in scientific notation. $x = m \times 10^C$

m is the mantisa and C is characteristics

Illustrative Example

72. **Identify the characteristics** and mantisa of logarithm of each of the following

a) 6478.52 b) 0.000302 c) 5.5 d) 0.00001

e) $\frac{1}{8}$ f) 8 g) $\frac{3}{40}$ h) 1000

i) 3610 j) 406 k) $\frac{2}{5}$ l) $\frac{1}{2}$

Solution: first write in scientific notation

a) $6478.52 = 6.47852 \times 10^3$ ⟵ scientific notation

$\therefore$ the characteristics is 3

$\therefore$ the mantisa is log6.47852

b) $0.000302 = 3.02 \times 10^{-4}$ ⟵ scientific notation

$\therefore$ characteristics is -4

$\therefore$ mantisa of logarithm is log(3.02)

c) $5.5 = 5.5 \times 10^0$ ⟵ scientific notation

$\therefore$ characteristics is 0 $\therefore$ mantisa of log5.5

d) $0.00001 = 1 \times 10^{-5}$ ⟵ scientific notation

$\therefore$ characteristics is -5 $\therefore$ mantisa is log 1 = 0

e) $\frac{1}{8} = 0.125 = 1.25 \times 10^{-1}$ ⟵ scientific notation

$\therefore$ characteristics is -1 $\therefore$ mantisa is log 1.25

f) $8 = 8 \times 10^0$ ⟵ scientific notation

$\therefore$ characteristics is 0 $\therefore$ mantisa is log 8

g) $\frac{3}{40} = 0.075 = 7.5 \times 10^{-2}$ ⟵ scientific notation

$\therefore$ characteristics is -2 $\therefore$ mantisa is log 7.5

h) $1000 = 1.0 \times 10^3$ ⟵ scientific notation

$\therefore$ characteristics is 3 $\therefore$ mantisa is log 1 which is 0

i) $3610 = 3.61 \times 10^3$ ⟵ scientific notation

$\therefore$ characteristics is 3 $\therefore$ mantisa is log 3.61

j) 406 ⟵ Exercise left for you

k) $\frac{2}{5} = 0.4 = 4.0 \times 10^{-1}$, $\therefore$ chxs is -1, mantisa is log4

l) $\frac{1}{2}$ ⟵ Exercise left for you

73. Suppose logx =-3.6696, then **determine**

a) mantisa of log x b) characteristics of log x.

Solution: Here

We know that mantisa is non – negative decimal part and characteristic is the integer part of logarithm. So that

log x = – 3.6696 = –3 + 0.6696

since the decimal part – 0.6696 is negative so that **add** 1 and **take off** 1 in order to make positive decimal

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$\Rightarrow -3.6696 = -0.6696 + (-3)$
$= 1 + -0.6696 + (-3) - 1$
$= (1.0000 - 0.6696 + (-3-1)$
$= 0.3304 + (-4)$

Therefore, $\log x = 0.3304 + (-4)$

a) mantisa of log x is 0.3304 b) characteristic of log x is -4

74. If $\log x = -2.4618$, then the characteristics of log x is.
A. 0.4618 B. –2 C) –3 D. 0.5382

Solution: $\log x = -2.4618 = -2 + -0.4618$ since there is no negative mantisa, so that add 1 and take off 1

$\Rightarrow -2.4618 = -2 + -0.4618$
$= (-2-1) + (1 + -0.4618)$
$= (1.0000 - 0.4618 + (-2-1)$
$= 0.5382 + (-3)$
$\Rightarrow \log x = 0.5382 + (-3)$
$\therefore 0.5382$ is mantisa of log x
$\therefore -3$ is characteristics of log x $\therefore$ **Answer: C**

75. **Given log 5** = 0.6990, the mantissa of log 0.05 is
A. 0.3010 B. –2 C. 0.6990 D. –1

Solution: $\log(0.05) = \log(5\times10^{-2}) = \log_{10}^{5} + \log_{10}^{10^{-2}} = 0.6990 + (-2)$
Therefore, mantissa is 0.6990 and characteristics is –2 $\therefore$ **Answer: C**

76. If logx = –0.7321, then which of the following is true?
A. $X = 10^{0.2679}$ C. the characteristic of logx is 1
B. x > 0.1 D. The mantissa of log x is 0.7321

Solution: **Remember** that log x is base 10 logarithm

$\Rightarrow \log x = -0.7321 \Leftrightarrow \log_{10}^{x} = -0.7321$

- To make the decimal part non – negative add 1 and take off 1.

$\Rightarrow \log_{10}^{x} = -1 + 1 - 0.7321 = -1 + 0.2679$
$\Rightarrow \log_{10}^{x} = -1 + 0.2679$

Therefore, characteristics of log x is – 1
mantissa is 0.2679
Since $\log x = -1 + 0.2679$
$\Rightarrow x = 10^{-1+0.2679} = 10^{-0.7321} \Leftrightarrow x > 0.1 \Rightarrow 10^{-0.7321} > 10^{-1}$.
Since $-0.7321 > -1$ **Therefore , Answer: D**

77. If $10^{0.3462} = 2.219$, then which of the following is not true
A. Log 221.9 = 2.3462 C. $\log\sqrt[3]{2219} = 1.1154$
B. Log 0.002219 = –26538 D. log0.2219=–0.6538

Solution: Remember that $\log x = y \Leftrightarrow 10^{y} = x$. so that

$10^{0.3462} = 2.219 \Rightarrow \log_{10}^{2.219} = 0.3462$

A. $\log_{10}^{2.219} = \log_{10}^{(2.219\times10)} = \log_{10}^{2.219} + \log_{10}^{10^2}$
$= 0.3462 + 2 = 2.3462$

B. $\log_{10}^{0.002219} = \log_{10}^{(2.219\times10^{-3})} = \log_{10}^{2.219} + \log_{10}^{10^{-3}}$
$= 0.3462 + (-3) = -2.6538$

C. $\log\sqrt[3]{2219} = \log\ (2219)^{\frac{1}{3}} = \frac{1}{3}\log 2219 =$

$= \frac{1}{3}\log(2.219\times10^{3}) = \frac{1}{3}\left(\log_{10}^{2.219} + \log_{10}^{10^3}\right)$

$= \frac{1}{3}(0.3462 + 3) = 1.1154$

D. $\log 0.2219 = \log_{10}^{(2.219\times10^{-1})} = \log 2.219 + (-1) =$
$= 0.3462 + (-1) = -0.6538$ $\therefore$ All are true

78. If log 20 = 1.3010, then which of the following is not true
A. $\log\frac{1}{20} = -1.3010$ C. log40 = 2.6020
B. log 4000 = 3.6020 D. log 0.2 = 0.6990

Solution:

$\log 20 = \log(2\times10) = \log 2 + \log 10 = 1.3010$
$\Rightarrow \log 2 + 1 = 1.3010 \Rightarrow \log 2 = 0.3010$. Thus,

A) $\log\frac{1}{20} = \log 1 - \log 20 = 0 - (1.3010) = -1.3010$

B) $\log 4000 = \log(2^2\times10^3) = 2\log 2 + 3$
$= 2(0.3010) + 3 = 0.6020 + 3 = 3.6020.$

C) $\log 40 = \log(2^2\times10) = 2\log 2 + \log 10.$
$= 2(0.3010) + 1 = 1.6020.$

D) $\log 0.2 = \log(2\times10^{-1}) = \log 2 + \log(10^{-1}).$
$= \log 2 + (-1) = 0.3010 + (-1) = -0.6990.$

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79. If log50 = 1.699, then **evaluate**:

a) $\log\left(\frac{2}{25}\right)$ b) log(0.2) c) log 0.025

d) $\log_2^{50}$ e) $\log_2^{10}$ f) $\log_4^{50}$

Solution: Here

We know that: log50 = log(5× 10) = 1.699

$\Rightarrow$ log 50 = log5 + log 10 $\Rightarrow$ log 5 = 0.699

a. $\log\left(\frac{2}{25}\right) = \log 2 - \log 25 = \log\left(\frac{10}{5}\right) - 2\log 5$

$= \log 10 - \log 5 - 2\log 5 = 1 - 3\log 5$

$= 1 - 3(0.699) = -1.097$

b. $\log(0.2) = \log\left(\frac{2}{10}\right) = \log\left(\frac{1}{5}\right) = \log 1 - \log 5$

$= 0 - (0.6990) = -0.6990$

c. $\log(0.025) = \log(25\times10^{-3})\ \log(5^2\times 10^{-3})$

$= 2\log 5 + \log 10^{-3} = 2(0.6990) + (-3) = -1.602$

d. $\log_2^{50} = \frac{\log_2^{50}}{\log_{10}^{2}} = \frac{1.6990}{0.3010} \approx 5.6445 =$

e. $\log_2^{10} = \frac{\log 10}{\log 2} = \frac{1}{0.3010} \approx 3.322$

f. $\log_4^{50} = \frac{\log 50}{\log 4} = \frac{\log 50}{2\log 2} \approx \frac{1.6990}{2(0.3010)} \approx 2.822$

78 b) If logx = 1.3909 **find** log10x

Solution: log10x = log10 + logx = 1 + 1.3909

78 c) If $10^{1.7782} = 60$ **find** $\log\sqrt[3]{60}$

Solution: Since $\log_{10}^{60} = 1.7782$

$\Rightarrow \log\sqrt[3]{60} = \frac{1}{3}\log 60 = \frac{1}{3}(1.7782) = 0.5927$

78 d) If $A = 10x^2$, then log A = _____

Solution: take the logarithm to both side:

$\therefore$ Answer I

$= \log A = \log 10x^2 = \log 10 + \log x^2$

$\therefore \log A = 1 + 2\log x.$

Given that log2 = 0.3010 and log 3 = 0.4771 then $\log\left[(0.027)^{-\frac{2}{3}} \times \sqrt{0.125}\right]$ is equal to

A. −2.5942 B. 0.5943 + (−2) C. 0.5943 D. −1.4057

Solution: first we can simplify

$(0.027)^{\frac{-2}{3}} = \left(3^3\times10^{-3}\right)^{\frac{-2}{3}} = \left(3^{-2}\times10^{2}\right)$, thus

$\log\left[(0.027)^{-\frac{2}{3}} \times \left(5^3\times10^{-3}\right)^{\frac{1}{2}}\right]$

$= \log\left[3^{-2}\times10^{2}\times5^{\frac{3}{2}}\times10^{\frac{-3}{2}}\right]$

$= -2\log 3 + \log_{10}^{2} + \frac{3}{2}\log 5 + \log 10^{-3/2}$

$= -2(0.4771) + 2 + \frac{3}{2}(\log 10 - \log 2) + \left(-\frac{3}{2}\right)$

$= -0.9542 + 2 + \frac{3}{2} - \frac{3}{2}(0.3010) - \frac{3}{2} = 0.5943$ $\therefore$ **Answer C**

80. If log80 = 1.9030, then **evaluate** $\log\sqrt{\frac{1}{8}}$

Solution: we know that log 80 = log(8×10)

$\Rightarrow \log 8 + \log 10 = 1.9030$

$\Rightarrow \log 8 + 1 = 19030 \Rightarrow \log 8 = 0.9030$

Hence $\log\sqrt{\frac{1}{8}} = \frac{1}{2}\log\frac{1}{8} = \frac{1}{2}(\log 1 - \log 8)$

$= \frac{1}{2}(0 - 0.9030) = 1 - 0.4515$

80 b) If log2.5 = 0.3979 and log4.9 = 0.6902 then

$\log\left(\frac{625\times7^3}{\sqrt{4900}}\right)$ is

A. 2.4860 B. 1.9860 C. 3.4860 D. 2.9860

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Solution: $\log\left(\frac{625\times 7^3}{\sqrt{4900}}\right) = \log\left(\frac{25^2\times 7^3}{70}\right)$

$= \log(25^2 \times 4.9)$

$= 2\log(25) + \log 4.9$

$= 2\log(2.5 \times 10) + \log 4.9$

$= 2(\log 2.5+1) + \log 4.9$

$= 2(0.3979 + 1) + 0.6902$

$= 2.7958 + 0.6902$

$= 3.4860,$ ∴ **Answer C**

80 c) If $\log_{10}^{21} = x$, then **evaluate** $\log_{10}^{0.021}$

Solution: we know that $0.021 = 21\times 10^{-3}$

$\Rightarrow \log_{10}^{0.021} = \log_{10}^{(21\times 10^{-3})} = \log_{10}^{21} + \log_{10}^{10^{-3}}$

$= x - 3$

g. If $\log^{0.0057} = -2.2441$, then $\log\sqrt[3]{57}$ is equal to

A. -0.7480 B. 0.5853 C. 2.2677 D. 1.7559

Solution: $\log 0.0057 = \log(57\times 10^{-4}) = \log 57 + \log_{10}^{-4}$

$= \log 57 + (-4) = -2.2441$

$\Rightarrow \log 57 = -2.2441 + 4 = 1.7559$

Thus, $\log\sqrt[3]{57} = \log 57^{\frac{1}{3}} + \frac{1}{3}\log 57$

$= \frac{1}{3}(1.7559) = 0.5853,$ ∴ **Answer B**

Antilogarithm

Suppose $\log x = 0.3010 \Rightarrow 10^{0.3010} = x$

$\log x = 0.4771 \Rightarrow 10^{0.4771} = x$

$\log x = y \Rightarrow 10^{y} = x,$

Then the process of finding x from the body of logarithm table is called **antilogarithm.**

- **Remark**: Antilog(logx) = x

Nation i) Antilog (log x) = Antilog (0.3010)

$\Rightarrow$ x = antilog (0.3010) = 2

That is $\log x = 0.3010 \Leftrightarrow x = 10^{0.3010} = 2$

ii) Antilog (logx) = Antilog(0.4771)

$\Rightarrow$ x = antilog (0.4771) = 3

That is $\log x = 0.4771 \Leftrightarrow x = 10^{0.4771} = 3$

Illustrative Example

i) Suppose $\log 2 = 0.3010 \Rightarrow 10^{0.3010} = 2$

$\log 3 = 0.4771 \Rightarrow 10^{0.3010} = 3$

$\log 6 = 0.7781 \Rightarrow 10^{0.7781} = 6$

$\log 1.5 = 0.1761 \Rightarrow 10^{0.1761} = 1.5$

$\log 5 = 0.6990 \Rightarrow 10^{0.699} = 5$, then

Evaluate:

a) Antilog 3.3.3010 b) antilog 1.4771 c) antilog2.7781

d) antilog 3.1761 e) antilog 2.6990 f) antilog(–0.6990)

g) antilog(–2.2219) h) antilog(–1.5229)

Solution: Here, let antilog (m) = x $\Leftrightarrow$ logx = m $\Rightarrow$ $x = 10^m$

a) Antilog 3.3010 = x $\Leftrightarrow$ logx = 3.3010 $\Rightarrow x = 10^{3.3010}$

$\Rightarrow x = 10^{0.3010+3} = \left(10^{0.3010}\right)\left(10^3\right) = 2\left(10^3\right) = 2000.$

∴ Antilog3.3010 = 2000

b) Antilog (1.4771) = x $\Leftrightarrow$ logx = 1.4771 $\Rightarrow x = 10^{1.4771}$

$\Rightarrow x = 10^{0.4771+1} = \left(10^{0.4771}\right)\left(10^1\right) = 3(10) = 30.$

∴ Antilog1.4771 = 30

c) Antilog (2.7781) = x $\Leftrightarrow$ logx = 2.7781 $\Rightarrow x = 10^{2.7781}$

$\Rightarrow x = 10^{0.7781+2} = \left(10^{0.7781}\right)\left(10^2\right) = 6(100) = 600.$

∴ Antilog (2.7781 = 600

d) Antilog (3.1761) ⟵ **exercise left for you**

e) Antilog (2.6990) = x $\Leftrightarrow$ log x = 2.6990

$\Rightarrow x = 10^{0.6990+2} = (10^{0.6990})(10^2) = 5(100) = 500$

f) Antilog (–0.6990) = x $\Leftrightarrow$ log x = – 0.699

Since there is no negative mantissa from the table of logarithms.

Hence, add(1) and take off (1)

$\Rightarrow -0.6990 = 1 + (-0.6990) + (-1) = 0.3010 + (-1)$

$\Rightarrow \log x = 0.3010 + (-1) \Rightarrow x = 10^{0.3010} + (-1)$

$\Rightarrow x = (10^{0.3010})(10^{-1}) = (2)(10^{-1}) = 0.2$

∴ antilog ((–0.6990) = 0.2

Here, characteristics of log x = –0.6990 is (–1) and mantissa is 0.3010.

g) Antilog (–2.2219) = x $\Leftrightarrow$ logx = –2.2219

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$\Rightarrow \log x = -2 + -0.2219$

To make positive, add (1) and take off (1)

$\Rightarrow \log x = -2 + (-1) + (1 + -0.2219) = -3 + 0.7781$

$x = (10^{-3})(10^{0.7781}) = (10^{-3})(6) = 0.006$

$\therefore$ antilog $(-2.2219) = 0.006$

- Here, characteristics of $\log x = -2.2219$ is -3 and mantissa is 0.7781

h) Antilog $(-1.5229) = x \Leftrightarrow \log x = -1.5229$

$\Rightarrow \log x = -1 + -0.5229$

To make positive: add 1 and take off 1

$\Rightarrow \log x = (-1 + (-1)) + (1 + -0.5229) = -2 + 0.4771$

$\Rightarrow x = (10^{-2})(10^{0.4771}) = (10^{-2})(3) = 0.03$

$\therefore$ antilog $(-1.5229) = 0.03$.

82. **Find**:

a) Antilog (-4) b) antilog (0) c) antilog(3)

d) antilog (-2) e) antilog (2) f) antilog (-10)

Solution:

a) Antilog$(-4) = x \Leftrightarrow \log x = -4 \Rightarrow x = 10^{-4} = 0.0001$

b) Antilog$(0) = x \Leftrightarrow \log x = 0 \Rightarrow x = 10^{0} = 1$

c) Antilog $3 = x \Leftrightarrow \log x = 3 \Rightarrow x = 10^{3} = 1000$.

d) Antilog $(-2) = x \Leftrightarrow \log x = -2 \Rightarrow x = 10^{-2} = 0.01$

e) Antilog $(2) = x \Leftrightarrow \log x = 2 \Rightarrow x = 10^{2} = 100$.

f) Antilog $(-10) = x \Leftrightarrow \log x = -10 \Rightarrow x = 10^{-10} = 0.0000000001$

Computation with logarithms

Logarithms are used for computation: such as to multiplying, to dividing, to find the nth root . . . etc.

Illustrative Example

83. Compute the following using logarithm.

a) $x = \dfrac{\sqrt[6]{27} \times 243}{\sqrt[3]{81}}$ b) $x = \dfrac{\sqrt[5]{16} \times \sqrt[3]{32}}{\sqrt[9]{64}}$

Solution: To compute take the logarithm both side of expression.

a) Take the logarithm of base 3.

$$\Rightarrow \log_3^x = \log_3^{\left(\frac{\sqrt[6]{27}\times 243}{\sqrt[3]{81}}\right)} = \log_3^{\sqrt[6]{27}} + \log_3^{243} - \log_3^{\sqrt[3]{81}}$$

$$= \frac{1}{6}\log_3^{27} + 5\log_3^3 - \frac{4}{5}\log_3^3$$

$$= \frac{1}{6}(3) + 5 - \frac{4}{5} = \frac{1}{2} + 5 - \frac{4}{5} = \frac{47}{10} = 4.7$$

$\Rightarrow \log_3^x = 4.7 \Leftrightarrow x = 3^{4.7} = \sqrt[10]{3^{47}}$

b) Take the logarithm of base 2

$$\log_2^x = \log_3^{\left(\frac{\sqrt[5]{16}\times\sqrt{32}}{\sqrt[9]{64}}\right)}$$

$$= \frac{1}{5}\log_2^{16} + \frac{1}{3}\log_2^{32} - \frac{1}{9}\log_2^{64}$$

$$= \frac{4}{5}\log_2^2 + \frac{5}{3}\log_2^2 - \frac{6}{9}\log_2^2$$

$$= \frac{4}{5} + \frac{5}{3} - \frac{6}{9} = \frac{9}{5}$$

$$\Rightarrow \log_2^x = \frac{9}{5} \Leftrightarrow x = 2^{\frac{9}{5}} = \sqrt[5]{2^9} \quad \therefore x = 2\sqrt[5]{16}$$

84. Given that $\log 3.5 = 0.5441$, $\log 7.5 = 0.8751$, $\log 2.04 = 0.3096$ and $x = \dfrac{350\times 750}{204}$

Then $\log x$ is equal to:

A. 2.1192 B. 11.0960 C. 6.1096 D. 3.1096

Solution: Take the logarithm of base 10.

$$\Rightarrow \log x = \log\left(\frac{350\times 750}{204}\right) = \log\left(\frac{3.5\times 10^2 \times 7.5\times 10^2}{2.04\times 10^2}\right)$$

$$= \log\left(\frac{3.5\times 7.5\times 10^2}{2.04}\right)$$

$= \log 3.5 + \log 7.5 + \log 10^2 - \log 2.04$

$= 0.5441 + 0.8751 + 2 - 0.3096$

$= 3.1096$, $\therefore$ **Answer D**

85. If $\log 3.5 = 0.5440$ and $\log 2.07 = 0.316$ then the value of $x = 35^{3/2}$ equals _____

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A. 211 B. 218 C. 207 D. 227

Solution: $x = 35^{3/2} \Leftrightarrow \log x = \log 35^{3/2}$

$$\Rightarrow \log x = \frac{3}{2}\log 35 = \frac{3}{2}[\log(3.5\times 10)]$$

$$= \frac{3}{2}[\log 3.5 + \log 10]$$

$$= \frac{3}{2}(0.5440 + 1) = 2.316$$

$\Rightarrow \log_{10}^{x} = 2.316 \Leftrightarrow x = 10^{2.316} = (10^{0.316})(10^{2})$

$\Rightarrow x = (2.07)(100) = 207 \longleftarrow \log 2.07 = 0.316$

∴ Answer C

86. If $\log 3.09 = 0.49$ and $\log 1.99 = 0.2985$ then the value of $x = (3.09)^{2.65}$ is _____

A. 11.18 B. 15.03 C. 19.9 D. 21.015

Solution: $x = (3.09)^{2.65}$

$\Rightarrow \log x = 2.65\log 3.09$

$= 2.65(0.49) = 1.2985$

$\Rightarrow \log_{10}^{x} = 1.2985$

$\Rightarrow x = 10^{1.2985} = 10^{1+0.2985}$

$\Rightarrow x = (10^{1})(10^{0.2985}) \longleftarrow \log 1.99 = 0.2985$

$\therefore x = (10)(1.99) = 19.9,$ ∴ Answer C

Natural logarithm

- The base '**e**' logarithm are called **natural logarithm**. It is denoted by $\log_e^x = \ln x,\ x > 0$
  - $\ln 1 = \log_e^1 = 0$
  - $\ln e = \log_e^e = 1$

Illustrative Example

87. **Evaluate:**

a) $\ell n\sqrt[5]{e}$ b) $\ell n(e^{4x})$ c) $\ell n\sqrt[3]{e^2}$

b) $\ell n\left(\frac{e^m}{e^n}\right)$ e) $\ell n(e^2)$ f) $e^{\ell n 2}$

g) $e^{\ell nx - \ell ny}$ h) $e^{2+\ell n 3}$ i) $e^{x+\ell n 2}$

Solution: Here

a) $\ell n\sqrt[5]{e} = \ell n e^{1/5} = \frac{1}{5}\ell ne = \frac{1}{5}(1) = \frac{1}{5}$

b) $\ell n(e^{4x}) = (4x)\ell ne = 4x(1) = 4x$

c) $\ell n\sqrt[3]{e^2} = \ell n e^{2/3} = \frac{2}{3}\ell ne = \frac{2}{3}(1) = \frac{2}{3}$

d) $\ell n\left(\frac{e^m}{e^n}\right) = \ell n(e^{m-n}) = (m-n)\ell ne = m-n$

e) $\ell n(e^2) = 2\,\ell ne = 2(1) = 2$

f) $e^{\ell n 2} = e^{\log_e^2} = 2$

g) $e^{\ell nx - \ell ny} = e^{\log_e^x - \log_e^y} = \frac{e^{\log_e^x}}{e^{\log_e^y}} = \frac{x}{y}$

h) $e^{2+\ell n 3} = (e^2)\cdot e^{\ell n 3} = (e^2)(3) = 3e^2$

i) $e^{x+\ell n 2} = (e^2)e^{\ell n 2} = 2e^x$

88. The simplified form of $e^{\frac{1-\ell n\left(\frac{1}{2}\right)}{2}}$ is equal to _____

A. $\sqrt{2e}$ B. $\frac{1}{2}\sqrt{e}$ C. $(\ln 2)\sqrt{e}$ d) $\frac{1}{2}$

Solution: $e^{\frac{1-\ell n\left(\frac{1}{2}\right)}{2}} = e^{\frac{1-[\ell n 1 - \ell n 2)}{2}} = e^{\frac{1-(0-\ell n 2)}{2}} = e^{\frac{1+\ell n 2}{2}} = \left(e^{\frac{1}{2}}\right)\left(e^{\frac{1}{2}\ell n 2}\right)$

$= \left(e^{\frac{1}{2}}\right)e^{\ell n\sqrt{2}} = \sqrt{e}\sqrt{2} = \sqrt{2e}$

∴ Answer A

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89. If $\ln 2 = 0.7$ and $e^{-5y+1} = 0.25$, then the value of y

A. -0.08 B. 0.24 C. 0.27 D. 0.48

Solution: $e^{-5y+1} = 0.25 \Leftrightarrow \log_e^{0.25} = -5y + 1$

$\Rightarrow \ln(0.25) = -5y + 1$

$\Rightarrow \ln\left(\frac{1}{4}\right) = -5y + 1 \longleftarrow \ln\left(\frac{1}{4}\right) = \ln 1 - \ln 4$

$\Rightarrow -\ln 4 = -5y + 1 \longleftarrow$ subtracting (1)

$\Rightarrow -2\ln 2 - 1 = -5y \longleftarrow$ dividing by (–5)

$\Rightarrow y = \frac{2\ell n 2 + 1}{5} = \frac{2(0.7) + 1}{5} = \frac{2.4}{5} = 0.48$

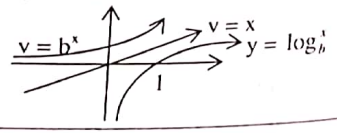
90. If $\ln 4 = 1.4$ and $e^{2k} = 64$, the **evaluate K**

Solution: $e^{2k} = 64 \Leftrightarrow \log_e^{64} = 2k \Rightarrow \ln 64 = 2k \Rightarrow \ln 4^3 = 2k$

$\Rightarrow 3\ln 4 = 2k \quad \therefore k = \frac{3\ell n 4}{2} = \frac{3(1.4)}{2} = 2.1$

Graphs of Logarithmic function

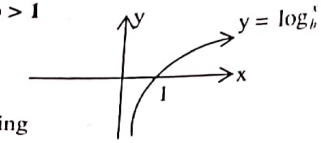
- The function $y = \log_b^x$ $x > 0$, $b > 0$, and $b \neq 1$ is called **logarithmic function.**
- The logarithm function $y = \log_b^x$ is the inverse of exponential function $y = b^x$
- The graph of $y = \log_b^x$ and $y = b^x$ are **reflections** of each other about the line $y = x$.



- **A. Graph of $y = \log_b^x$**

i) The graph of $y = \log_b^x$, if $b > 1$

- Domain = {x: x > 0}
- Range = {y: y ∈ IR}
- x-intercept (1,0)
- The function is increasing
- If $0 < x < 1$, then $\log_b^x < 0$ (negative)

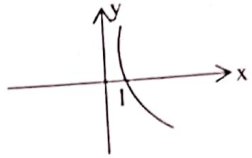


- If $x > 1$, then $\log_b^x > 0$ (positive value)
- y – axis (x = 0) is asymptote.

For example, $y = \log_2^x$, $y = \log_{\frac{1}{2}}^x$, $y = \log_{10}^x$ . . .etc . are satisfied the above condition.

ii) The graph of $y = \log_b^x$, if $0 < b < 1$

- Domain = {x: x > 0}
- Range = {y: y ∈ IR}
- x-intercept is at x = 1
- The function is decreasing



- For $0 < x < 1$, $y = \log_b^x$ is positive

  i.e. $\log_b^x > 0$

- For $x > 1$, $y = \log_b^x$ is negative

  i.e. $\log_b^x < 0$

- y – axis (x = 0) is vertical asymptote.

For example, $y = \log_{\left(\frac{1}{2}\right)}^x$, $y = \log_{\left(\frac{2}{3}\right)}^x$, $y = \log_{\left(\frac{1}{5}\right)}^x$ . . .etc, are satisfies the above condition.

Illustrative Example

91. Compare and contrast the graph of the following.

a) $h(x) = \log_3^x$ and $f(x) = \log_{\left(\frac{1}{3}\right)}^x$

b) $g(x) = \log_{\left(\frac{2}{3}\right)}^x$ and $k(x) = \log_{\left(\frac{3}{2}\right)}^x$

c) $g(x) = \log_{10}^x$ and $k(x) = \log_{\left(\frac{1}{10}\right)}^x$

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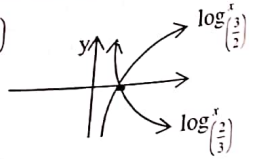
a)

$h(x) = \log_3^x$	$f(x) = \log_{\left(\frac{1}{3}\right)}^x$
• Domain = {x: x > 0} • Range = {y: y ∈IR} • x – intercept, x = 1 • Since the base 3 > 1 ⇒ the graph is **increasing** • For 0 < x < 1, $\log_3^x$ is **negative** (i.e. $\log_3^x < 0$) • For x > 1, $\log_3^x$ is **positive** (i.e. $\log_3^x > 0$)	• Domain = {x: x > 0} • Range = {y: y ∈IR} • x – intercept, x = 1 • Since the base $\frac{1}{3} < 1$ ⇒ the graph is decreasing • For 0 < x < 1, $\log_{\frac{1}{3}}^x$ is **positive** (i.e. $\log_{\left(\frac{1}{3}\right)}^x > 0$) • For x > 0, $\log_{\frac{1}{3}}^x$ is negative i.e. $\log_{\left(\frac{1}{3}\right)}^x < 0$
Graph of $h(x) = \log_3^x$	Graph of $h(x) = \log_3^x$

One of them is obtained by reflecting the other about **x – axis**

b) Both $g(x) = \log_{\left(\frac{2}{3}\right)}^x$ and $k(x) = \log_{\left(\frac{3}{2}\right)}^x$

- Domain = {x:x > 0}
- Range = {y:y∈IR}
- x – intercept, x = 1



However, the base of g(x), $\frac{3}{2} > 1$, therefore

- $g(x) = \log_{\left(\frac{2}{3}\right)}^x$ is **decreasing** but $k(x) = \log_{\left(\frac{3}{2}\right)}^x$ **is increasing**
- for $0 < x < 1$, $\log_{\left(\frac{2}{3}\right)}^x$ **is positive** but $\log_{\left(\frac{3}{2}\right)}^x$ **is negative**

- for x > 1, $\log_{\left(\frac{2}{3}\right)}^x$ **is negative** but $\log_{\left(\frac{3}{2}\right)}^x$ **is positive**
- one of them is obtained by reflecting the other about x – axis.

: Exercise left for you.

92. Compare and contrast the graph of the following.

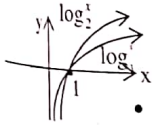
a) $f(x) = \log_2^x$ and $g(x) = \log_3^x$

b) $f(x) = \log_{\left(\frac{1}{2}\right)}^x$ and $g(x) = \log_{\left(\frac{2}{3}\right)}^x$

c) $f(x) = \log_2^x$ and $g(x) = \log_{\left(\frac{3}{2}\right)}^x$

Solution:

a) Both $f(x) \log_2^x$ and $g(x) = \log_3^x$ have



- Domain = {x: x > 0}
- x – intercept, x = 1
- Range = {y: y > 0}
- base 2 > 1 and 3 > 1

⇒ f(x) and g(x) increasing function.

- $\log_2^x > 0$ and $\log_3^x > 0$ for x > 1
- $\log_2^x < 0$ and $\log_3^x < 0$ for $0 < x < 1$

However, $\log_2^x > \log_3^x$ for x > 1 $\log_2^x < \log_3^x$ for $0 < x < 1$

b) Both $f(x) = \log_{\left(\frac{1}{2}\right)}^x$ and $g(x) = \log_{\left(\frac{2}{3}\right)}^x$

- Have the domain positive real number
- Have the range is all real number
- Have x – intercept at x = 1
- Have positive value for $0 < x < 1$
- Have negative value for x > 1
- Are decreasing function

However, $\log_{\left(\frac{2}{3}\right)}^x > \log_{\left(\frac{1}{2}\right)}^x$ for $0 < x < 1$ $\log_{\left(\frac{2}{3}\right)}^x < \log_{\left(\frac{1}{2}\right)}^x$ for x > 1

c) **Exercise left for you.**

93. Compare and contrast the graph of each of the following.

a) $f(x) = \log_3^x$ and $g(x) = \log_3^{-x}$ $f(x) = \log_2^x$ and $g(x) = \log_2^{-x}$

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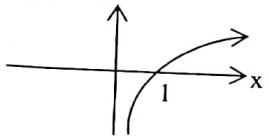
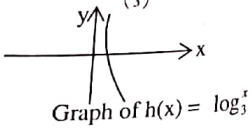
Solution:

a)

$h(x) = \log_3^x$	$f(x) = \log_{\left(\frac{1}{3}\right)}^x$
• Domain = {x: x > 0} • Range = {y: y ∈IR} • x – intercept. • Since the base 3 > 1 ⇒ the graph is **increasing** • For 0 < x < 1, $\log_3^x$ is **negative** (i.e. $\log_3^x < 0$) • For x > 1, $\log_3^x$ is **positive** (i.e. $\log_3^x > 0$)	• Domain = {x: x > 0} • Range = {y: y ∈IR} • x – intercept, x = 1 • Since the base $\frac{1}{3} < 1$ ⇒ the graph is **decreasing** • For 0 < x < 1, $\log_{\left(\frac{1}{3}\right)}^x$ is **positive** (i.e. $\log_{\left(\frac{1}{3}\right)}^x > 0$) • For x > 0, $\log_{\frac{1}{3}}^x$ is **negative** i.e. $\log_{\left(\frac{1}{3}\right)}^x < 0$
 Graph of $h(x) = \log_3^x$	 Graph of $h(x) = \log_3^x$

One of them is obtained by reflecting the other about **x – axis**

b) Exercise left for you

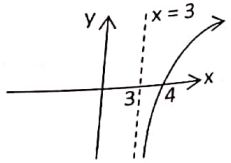
95. Discuss the behavior and sketch the graph of the following.

a) $f(x) = \log_2^{(x-3)}$ b) $g(x) = \log_3^{4+x}$

c) $h(x) = \log_2^{2-x}$ d) $k(x) = \log_2^{(x+1)}$

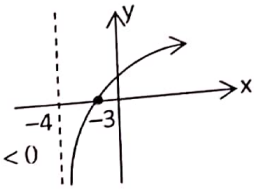
Solution: Here

a)
- Domain of f = {x:x > 3}
- Range of f = {y: y∈IR}
- x – intercept x = 4
- Asymptote, x = 4
- For 3 < x < 4, $\log_2^{(x-3)} < 0$

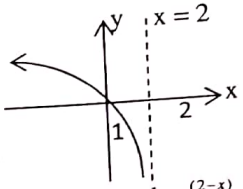


- For x > 4, $\log_2^{(x-3)} > 0$
- Increasing function

b) $g(x) = \log_3^{(4+x)}$
- Domain of f = {x:x > –4}
- Range of f = {y: y∈IR}
- x – intercept x = –3
- Asymptote, x = –4
- For –4 < x < –3, $\log_2^{(x+4)} < 0$
- For x > –3, $\log_2^{(x+4)} > 0$
- Increasing function

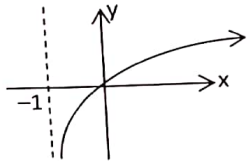


c) $h(x) = \log_3^{(2-x)}$
- Domain = {x: x < 2}
- Range = {y: y∈IR}
- x – intercept, x = 1
- Asymptote, x = 2
- For 1 < x < 2, h(x) < 0
- For x < 1, h(x) > 0
- Decreasing function



$h(x) = \log_3^{(2-x)}$

d) $k(x) = \log_2^{(x+1)}$
- Domain = {x: x >–1}
- Range = {y: y∈IR}
- x – intercept, x = 1
- Asymptote, at x = –1
- For –1 < x < 0, k(x) < 0
- For x > 0, k(x) > 0
- Increasing function



$K(x) = \log_2^{(x+1)}$

Relationship between $y = b^x$ and $y = \log_b^x$
- $y = b^x$ is the inverse of $y = \log_b^x$
- One of them is obtained by reflecting the other on the line y=x
- The domain of $y = b^x$ is the same as the range of $y = \log_b^x$.
- The range of $y = b^x$ is the same as the domain of $y = \log_b^x$.

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Illustrative Example

96. Compare and contrast the graph.

a) $f(x) = 2^x$ and $g(x) = \log_2^x$ b) $f(x) = \left(\frac{1}{4}\right)^x$ and $g(x) = \log_{\left(\frac{1}{4}\right)}^x$

c) $f(x) = e^x$ and $g(x) = \ln x$

Solution: Here

a)
- Domain of $f(x)$ = Range of $g(x) = (-\infty, \infty)$
- Range of $f(x)$ = domain of $g(x) = (0, \infty)$
- $f(x) = 2^x$ is the inverse of $g(x) = \log_2^x$
- $f(x) = 2^x$ is **the mirror image** of $g(x) = \log_2^x$ along the line $y = x$.
- Both $f(x) = 2^x$ and $g(x) = \log_2^x$ are increasing.

b) Exercise left for you.

c)
- $f(x) = e^x$ is inverse of $g(x) = \ln x$
- $f(x) = e^x$ is the mirror image of $g(x) = \ln x$ along the line $y = x$.
- Both are increasing

97. If $a > 1$, then for what value of x is $\log_a^x < 0$?

A. $(1, \infty)$ B. $(0,1)$ C. $(0, \infty)$ D. $(-\infty, 0)$

Solution: Since the base $a > 1$, the graph rise from left to right (i.e increasing)

- For $0 < x < 1$, $\log_a^x < 0$ • For $x > 1$, $\log_a^x > 1$, ∴ **Answer B**

98. If $f(x) = \log_a^x > 0$, for $0 < x < 1$, then which of the following is true about b.

A. $0 < b < 1$ B. $b > 0$ C. $b > 1$ D. $1 < b < 10$

Solution: Here

- If the base $b > 1$, then $f(x) = \log_b^x$ is increasing
- If the base $0 < b < 1$, then $f(x) = \log_b^x$ is decreasing.

$\Rightarrow \log_b^x > 0$, if $0 < x < 1$,

$\Rightarrow \log_b^x < 0$, if $x > 0$ ∴ **Answer A**

99. If $y = a^x$ and $y = \log_a^x$ for $a > 0$ and $a \neq 1$ are given, then along which line shall. We reflect the graph of one of them in order to get the graph of the other?

A. the line $y = x$ B. The line $y = -x$ C) the x-axis D. y-axis

Solution: **Answer A**

Solving exponential equation using logarithm

Remember that $a^x = m \Leftrightarrow \log_a^m = x$

$$\Rightarrow x = \log_a^m = \frac{\log_{10}^m}{\log_{10}^a}$$

Illustrative Example

100. Solve:

a) $2^x = 15$ b) $3^x = 5$ c) $4^x = 10$

Solution: Here

a) $2^x = 15 \Leftrightarrow \log_2^{15} = x$

$$\Rightarrow x = \log_2^{15} = \frac{\log_2^{15}}{\log_{10}^2}$$

$$\Rightarrow x = \frac{\log(1.5 \times 10)}{\log 2} = \frac{\log\left(\frac{3}{2}\right) + \log 10}{\log 2}$$

$$\Rightarrow x = \frac{\log 3 - \log 2 + 1}{\log 2} = \frac{0.4771 - 0.3010 + 1}{0.3010} = 3.907$$

b) $3^x = 5 \Leftrightarrow x = \log_3^5 = \frac{\log 5}{\log 3}$

$$\Rightarrow x = \frac{\log\left(\frac{10}{2}\right)}{\log 3} = \frac{\log 10 + \log 2}{\log 3}$$

$$\Rightarrow x = \frac{1 - 0.3010}{0.4771} = 1.465$$

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c) $4^x = 10 \Leftrightarrow x = \log_4^{10} = \dfrac{\log 10}{\log 4} = \dfrac{1}{2\log 2}$

$\Rightarrow x = \dfrac{1}{2(0.3010)} = 1.66$

101. The solution set of $3^{x+4} = 8^{x-2}$ is equal to _____

Solution: $3^{x+4} = 8^{x-2} \Leftrightarrow \log 3^{x+4} = \log 8^{x-2}$

$\Rightarrow (x+4)\log 3 = (x-2)\log 8$

$\Rightarrow x\log 3 + 4\log 3 = x\log 8 - 2\log 8$

$\Rightarrow x\log 3 + x\log 8 = -4\log 3 - 2\log 8$

$\Rightarrow x(\log 3 - x\log 8) = -(4\log 3 + 2\log 8)$

$\therefore x = -\left(\dfrac{4\log 3 + 2\log 8}{\log 3 - \log 8}\right)$

Logarithmic equations

- Equations that involve logarithms of expressions containing variables are called **logarithmic equations.**
- Logarithmic equation can be solved using the properties of logarithm.
  - i) $\log_b^x = \log_b^y$ if and only if $x = y$
  - ii) $\log_b^{xy} = \log_b^x + \log_b^y$
  - iii) $\log_b^{(x/y)} = \log_b^x - \log_b^y$
- **Domain of logarithm function**-is the set of all positive real number.
- **Universe of logarithm function = Domain of logarithm.**
- **The solution must be in the domain (universe)**

Illustrative Example

102. State the universe and solve for x.

a) $\log_4^{(x-5)} = 2$ b) $\log_2^{(2x+3)} = 3$

c) $\log_5^{(3x-2)} = 2$ d) $\log_{\sqrt{2}}^{(x^2+1)} = 4$ $\log_{\sqrt{2}}^{(x^2+1)} = 4$

Solution: Remember that logarithm function is valid when the expression is positive.

a) $\log_4^{(x-5)}$ is valid when $x - 5 > 0 \Rightarrow x > 5$

Universe/ Domain = {x: x > 5} = (5,∞)

$\log_4^{(x-5)} = 2 \Leftrightarrow x - 5 = 4^2$ ⟵ **Exponential form**

$\Rightarrow x = 16 + 5 = 21 \in (5, \infty)$

$\therefore x = 21$

b) $\log_2^{(2x+3)} = 3$ is valid when $2x + 3 > 0 \Rightarrow \left\{x : x > \dfrac{-3}{2}\right\}$

Therefore, universe $U = \left(\dfrac{-3}{2}, \infty\right)$ ⟵ domain

$\Rightarrow \log_2^{(2x+3)} = 3 \Leftrightarrow 2x+3 = 2^3$ ⟵ **Exponential form**

$\Rightarrow 2x = 8 - 3 = 5$

$\therefore x = \dfrac{5}{2}$, since $\dfrac{5}{2} \in \left(\dfrac{-3}{2}, \infty\right)$

c) $\log_5^{(3x-2)}$ is valid when $3x - 2 > 0 \Rightarrow \left\{x : x > \dfrac{2}{3}\right\}$

Therefore, universe $U = \left(\dfrac{2}{3}, \infty\right)$ ⟵ domain.

$\Rightarrow \log_5^{(3x-2)} = 2 \Leftrightarrow 3x - 2 = 5^2$ ← **Exponential form**

$\Rightarrow 3x - 2 = 25 \Rightarrow 3x = 27$

$\therefore x = 9$, since $9 \in \left(\dfrac{2}{3}, \infty\right)$

d) Universe, $U = \{x: x^2 + 1 > 0\} = (-\infty, \infty)$

$\Rightarrow \log_{\sqrt{2}}^{(x^2+1)} = 4 \Leftrightarrow \sqrt{2}^4 = x^2 + 1$ ← **Exp. Form.**

$\Rightarrow x^2 + 1 = 4$

$\Rightarrow x^2 = 3 \quad \therefore x = -\sqrt{3}$, and $x = \sqrt{3}$

103. Solve each part:

a) $\log_{(4x)}^{144} = 2$ b) $\log_{(3x)}^{324} = 2$

Solution:

a) $\log_{(4x)}^{144} = 2 \Leftrightarrow (4x)^2 = 144$ ⟵ **Exponential form**

$\Rightarrow 16x^2 = 144 \Rightarrow x^2 = \dfrac{144}{16} = 9 \quad \Rightarrow x = \pm 3$

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Since the base of logarithmic function cannot be negative.

$\therefore$ The only S.S is $x = 3$

b) $\log_{(3x)}^{324} = 2 \Leftrightarrow (3x)^2 = 324 \Rightarrow 9x^2 = 324 \Rightarrow x^2 = 36$

$\Rightarrow x = \pm 6$

$\therefore$ The only S.S is $x = 6$, since the base of log cannot be negative.

104. In each part state the universe and solve.

a) $\log_2^{(x^2-9)} = 2$ b) $\log_2^{x^2-7x} = 3$

c) $\log_{10}^{(2x^2+5x+97)} = 2$ d) $\log_2^{(x^2+6x+8)} = 3$

Solution: Here

a) $\log_2^{(x^2-9)}$ is valid when $x^2 - 9 > 0 \Rightarrow (-\infty,-3)U(3,\infty)$

Therefore, universe, $U = (-\infty,-3)U(3,\infty)$ ⟵ domain

$\Rightarrow \log_2^{(x^2-9)} = 2 \Leftrightarrow x^2 - 9 = 2^2$ ⟵ **Exponential form**

$\Rightarrow x^2 = 4 + 9 = 13 \Rightarrow x = \pm\sqrt{13}$

$\therefore x = \sqrt{13}$ or $x = -\sqrt{13}$

b) Universe: $U = \{x: x^2 - 7x \geq 0\} = \{x: x \in (-\infty,0)\ U(7,\infty)\}$

$\Rightarrow \log_2^{x^2-7x} = 3 \Leftrightarrow x^2 - 7x = 2^3$

$\Rightarrow x^2 - 7x = 8 \quad \Rightarrow x^2 - 7x - 8 = 0$

$\Rightarrow (x - 8)(x + 1) = 0 \quad \therefore x = 8$ or $x = -1$

c) Universe, $U = \{x: 2x^2 + 5x + 97 > 0\} = (-\infty,\infty)$

$\Rightarrow \log_{10}^{2x^2+5x+97} = 2 \Leftrightarrow 2x^2 + 5x + 97 = 10^2$

$\Rightarrow 2x^2 + 5x + 97 = 100$

$\Rightarrow 2x^2 + 5x + -3 = 0$

$\Rightarrow (2x - 1)(x + 3) = 0 \therefore x = \frac{1}{2}$ or $x = -3$

d) **Exercise left for you.**

105. Solve:

a) $\log(\log x) = 0$ b) $\log(\log x) = -1$

c) $\log_2^{(\log_2^{4x})} = 3$ d) $\log_2^{(\log_2^{(5x-3)})} = 1$

Solution: Here

a) $\log(\log x) = 0 \Leftrightarrow \log x = 10^0$ ⟵ Express log as exponential

$\Rightarrow \log x = 1 \Leftrightarrow x = 10^1$

$\therefore x = 10$

b) $\log(\log x) = -1 \Leftrightarrow \log x = 10^{-1} = 0.1$

$\Rightarrow x = 10^{0.1} = \sqrt[10]{10}$

c) $\log_2^{(\log_2^{4x})} = 3$ ⟵ Exercise left for you

d) $\log_2^{(\log_2^{(5x-3)})} = 1 \Leftrightarrow \log_2^{(5x-3)} = 2$

$\Rightarrow 5x - 3 = 2^2 \Leftrightarrow 5x - 3 = 4$

$\Rightarrow 5x = 7 \qquad \therefore x = \frac{7}{5}$

106. Solve

a. $e^{\ln(x-1)} = 4$ b) $e^{\ln(4-x)} = 2x+1$ c) $x+1 = e^{\ln 4x}$

d) $e^{\ln(x+4)} = 10 - x$ e) $e^{\ln(x-3)} = 9 - 2x$ f) $3x + 7 = e^{\ln(x+10)}$

Solution: Here

Remember that $\ln x = \log_e^x$

a) $e^{\ln(x-1)} = 4 \Leftrightarrow e^{\log_e^{(x-1)}} = 4 \Leftrightarrow x - 1 = 4 \Rightarrow x = 1 + 4 = 5$

b) $e^{\ln(4-x)} = 2x+1 \Rightarrow e^{\log_e^{4-x}} = 2x+1 \Leftrightarrow 4 - x = 2x + 1$

$\Rightarrow 4 - 1 = 2x + x = 3x$

$\therefore x = 1$

c) $x + 1 = e^{\ln 4x} \Leftrightarrow x + 1 = 4x \qquad \Rightarrow 3x = 1$

$\therefore x = \frac{1}{3}$

d) $e^{\ln(x+4)} = 10 - x \Leftrightarrow x + 4 = 10 - x$

$\Rightarrow 3x = 6 \quad \therefore x = 3$

e) Exercise left for you f) Exercise left for you

Solving logarithmic equation using the property:

i) $\log x + \log y = \log xy$, for $x > 0, y > 0$

ii) $\log x - \log y = \log\left(\frac{x}{y}\right)$, for $x > 0, y > 0$

- The solution must in the universe/ domain.

Illustrative Example

107. In each part state the universe and solve

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a) $\log x + \log(x+15) = 3$

b) $\log_2^{(x+1)} + \log_2^{(x-1)} = 3$

c) $\log_2^{x-3} + \log_2^{(x-5)} = 3$

d) $\log_{\sqrt{3}}^{(x+1)} + \log_2^{x+3} = 3$

e) $\log(x^2-16) - \log(x+4) = 1$

f) $\log^{8x^3} - \log^{2x} = 2$

g) $\log_5^{(x+3)} - \log_5^{x} = 2$

h) $\log_6^{(x+3)} - \log_5^{(x-2)} = 1$

Solution:

a) $\log x + \log(x+15)$ is valid when $x > 0$ and $x + 15 > 0$
so that both $\log x$ and $\log(x+15)$ is valid for $x > 0$

Therefore, Universe $U = \{x: x > 0\} = (0, \infty)$

$\Rightarrow \log x + \log(x+15) = 2 \Leftrightarrow \log x\,(x+15) = 2$

$\Rightarrow x(x+15) = 10^2$

$\Rightarrow x^2 + 15x - 100 = 0$

$\Rightarrow (x-5)(x+20) = 0$

$\Rightarrow x = 5$ or $x = -20$

But $x = -20$ is not an element of universe

Hence $x = -20$ is not solution

$\therefore$ The solution is $\{x: x = 5\}$

b) **Universe** $U = \{x: x+1 > 0$ and $x - 1 > 0\}$

$= (-1, \infty) \cap (1, \infty)$

$\therefore U = (1, \infty) \longleftarrow$ Domain

$\Rightarrow \log_2^{(x+1)} + \log_2^{(x-1)} = 3 \Leftrightarrow \log_2^{(x+1)(x-1)} = 3$

$\Rightarrow (x+1)(x-1) = 2^3 = 8$

$\Rightarrow x^2 - 1 = 8 \Rightarrow x^2 = 9$

$\Rightarrow x = \pm 3$

Since $x = -3$, makes both $\log_2^{(x+1)}$ *and* $\log_2^{(x-1)}$ negative.

$\therefore x = 3$ is the only solution

c) **Universe** $U = \{x: x - 3 > 0$ and $x - 5 > 0\}$

$= \{x: x > 3$ and $x > 5\}$

$= (3, \infty) \cap (5, \infty)$

$= (5, \infty) \longleftarrow$ Domain

$\log_2^{(x-3)} + \log_2^{(x-5)} = 3 \Leftrightarrow \log_2^{(x-3)(x-5)} = 3$

$\Rightarrow (x-3)(x-5) = 2^3 = 8$

$\Rightarrow x^2 - 8x + 15 = 8 \Leftrightarrow x^2 - 8x + 7 = 0$

$\Rightarrow (x-7)(x-1) = 0$

$\Rightarrow x = 7$ or $x = 1$, since $1 \notin (5, \infty)$

$\therefore$ The solution is $x = 7$, but not $x = 1$

d) Universe $U = \{x: x + 1 > 0$ and $x + 3 > 0.\}$

$= \{x: x > -1$ and $x + 3 > 0\}$

$= (-1, \infty) \cap (-3, \infty)$

$= (-1, \infty) \longleftarrow$ Domain

$\Rightarrow \log_{\sqrt{3}}^{(x+1)} + \log_{\sqrt{3}}^{(x+3)} = 2 \Leftrightarrow \log_{\sqrt{3}}^{(x+1)(x+3)} = 2$

$\Rightarrow (x+1)(x+3) = \sqrt{3}^2 \Rightarrow x^2 + 4x + 3 = 3$

$\Rightarrow x^2 + 4x = 0 \Rightarrow x(x+4) = 0$

$\Rightarrow x = 0$ or $x = -4$

However, $x = -4 \notin (-1, \infty)$

$\therefore$ The solution is $x = 0$

e) Universe, $U = \{x: x^2 - 16 > 0$ and $x + 4 > 0\}$

$= \{x: x > 4\} = (4, \infty) \longleftarrow$ Domain

$\Rightarrow \log^{(x^2-16)} - \log^{(x+4)} = 1 \longleftarrow \log\left(\frac{x}{y}\right) = \log x - \log y$

$\Rightarrow \log\left(\frac{x^2-16}{x+4}\right) = 1 \Leftrightarrow \frac{(x-4)(x+4)}{x+4} = 10^1$

$\Rightarrow x - 4 = 10 \Rightarrow x = 14 \in (4, \infty)$

$\therefore x = 14$

f) Universe, $U = \{x: x > 0\} = (0, \infty)$

$\Rightarrow \log 8x^3 - \log 2x = 2 \longleftarrow \log\left(\frac{x}{y}\right) = \log x - \log y$

$\Rightarrow \log\left(\frac{8x^3}{2x}\right) = 2 \Rightarrow \frac{8x^3}{2x} = 10^2$

$\Rightarrow 4x^2 = 100 \Rightarrow x^2 = 25 \quad \Rightarrow x = 5$ or $x = -5$

However, $x = -5 \notin (0, \infty)$

$\therefore$ The solution is $x = 5$

g) Universe $U = \{x: x + 3 > 0$ and $x > 0\}$

$= \{x: x > -3$ and $x > 0\} \longleftarrow$ **Intersection**

$= \{x: x > 0\} = (0, \infty) \longleftarrow$ Domain

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$\Rightarrow \log_5^{(x+3)} - \log_5^{x} = 2 \longleftarrow$

Explanation:

$\log x - \log y = \log\left(\frac{x}{y}\right)$

$\Rightarrow \log_5^{\left(\frac{x+3}{x}\right)} = 2$

$\Rightarrow \frac{x+3}{x} = 5^2 \Leftrightarrow \frac{x+3}{x} = 25$

$\Rightarrow x + 3 = 25x \Rightarrow 3 = 25x - x$

$\Rightarrow 24x = 3 \Rightarrow x = \frac{3}{24} = \frac{1}{8} \in (0,\infty) \quad \therefore x = \frac{1}{8}$

h) **Exercise left for you.**

108. Solve:

a) $10^{2\log x} = 5$ b) $3^{4\log_3^{(x-5)}} = 16$

Solution: Here

a) $\Leftrightarrow x^2 = 5 \longleftarrow$

$\Rightarrow x = \sqrt{5}$ or $x = -\sqrt{5}$

Explanation:

- $C^{\log_c^A} = A$

However, $x = -\sqrt{5}$ is not the domain

$\therefore$ The solution is $x = \sqrt{5}$

b) $3^{4\log_3^{(x-5)}} = 16 \Leftrightarrow (x-5)^4 = 16$

$\Rightarrow (x-5)^4 = 2^4 \Rightarrow |x-5| = 2$

$\Rightarrow x = 7$ or $x = 5 - 2 = 3$

However, $x = 3$ makes $x - 5$ negative.

$\therefore$ The solution is $x = 7$

109. Solve:

a) $\log_5^{\left[2+\log_3^{(x+5)}\right]} = 0$ b) $\log_3^{\left[2+\log_3^{(x+8)}\right]} = 0$

Solution: Here

a) $\log_5^{\left[2+\log_3^{(x+5)}\right]} = 0 \Leftrightarrow 2 + \log_5^{(x+5)} = 5^0 = 1$

$\Rightarrow \log_5^{(x+5)} = 1 - 2 = -1$

$\Rightarrow x + 5 = 5^{-1} \Rightarrow x + 5 = \frac{1}{5}$

$\Rightarrow x = \frac{1}{5} - 5 = \frac{-24}{5}$

b) $\log_3^{\left[2+\log_3^{(x+8)}\right]} = 0 \Rightarrow 2 + \log_3^{(x+8)} = 3^0 = 1$

$\Rightarrow \log_3^{(x+8)} = 1 - 2 = -1$

$\Rightarrow \log_3^{(x+8)} = 1 \Leftrightarrow x + 8 = 3^{-1} = \frac{1}{3}$

$x + 8 = \frac{1}{3} \quad x = \frac{1}{3} - 8 \quad x = \frac{-23}{3}$

110. The solution set of $\log_3^{\left[2+\log_3^{(x+3)}\right]} = 0$ is

a) $\left\{\frac{1}{3}\right\}$ B. $\left\{\frac{-8}{3}\right\}$ C. $\{-1\}$ D. $\{\ \}$

Solution: Here

$\log_3^{\left[2+\log_3^{(x+3)}\right]} = 0 \Leftrightarrow \log_3^{(x+3)} = 3^0 = 1$

$\Rightarrow \log_3^{(x+3)} = 1 - 2 = -1 \Rightarrow x + 3 = 3^{-1} = \frac{1}{3}$

$\Rightarrow x = \frac{1}{3} - 3 = \frac{-8}{3}$, therefore, **Answer B.**

111. $\log_{0.01}^{10x} - \log_{10}^{x} = 1$ is equal to

A. 10 B. 0.01 C. 0.1 D. 100

Solution: $\log_{0.01}^{10x} - \log_{10}^{x} = 1 \Leftrightarrow \frac{\log_{10}^{10}}{-2} - \log_{10}^{x} = 1$

$\log_{0.01}^{10x} = \frac{\log_{10}^{10x}}{\log_{10}^{0.01}} - \frac{\log_{10}^{10}}{-2}$

$\Rightarrow \log_{0.01}^{10x} - \log_{10}^{x} = 1 \Leftrightarrow \frac{\log_{10}^{10}}{-2} - \log_{10}^{x} = 1$

$\Rightarrow \frac{\log_{10}^{10x} + 2\log_{10}^{x}}{-2} = 1 \Rightarrow \log_{10}^{10x} + \log_{10}^{x^2} = -2$

$\Rightarrow \log_{10}^{(10x)(x^2)} = -2 \Leftrightarrow 10x^3 = 10^{-2}$

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$$\Rightarrow x^3 = \frac{10^{-2}}{10} = 10^{-3} \Rightarrow x^3 = \left(\frac{1}{3}\right)^3$$

b) $\Rightarrow x = \frac{1}{10} = 0.1$, ∴ **Answer C**

112. Solve:

a) $16^{1+3\log_4^x} = 2$ b) $27^{1-2\log_3^x} = 9$ c) $9^{1-2\log_3^x} = 4$

Solution: Recall that $a^{x+y} = a^x . a^y$

$$\Rightarrow 16^{1+3\log_4^x} = 2 \Leftrightarrow \left(16^1\right)\left(16^{3\log_4^x}\right) = 2$$

$$\Rightarrow 16^{3\log_4^x} = \frac{2}{16} = \frac{1}{8} = 2^{-3}$$

$$\Rightarrow \left(2^4\right)^{3\log_4^x} = 2^{-3}$$

$$\Rightarrow 2^{12\log_4^x} = 2^{-3}$$

$$\Rightarrow 12\log_4^x = -3 \Leftrightarrow \log_4^x = -\frac{1}{4}$$

$$\Rightarrow x = 4^{\frac{1}{4}} = \left(2^2\right)^{-\frac{1}{4}} = 2^{-\frac{1}{2}} = \frac{1}{\sqrt{2}}$$

b) $27^{1-2\log_3^x} = 9 \Leftrightarrow \left(27^1\right)\left(27^{-2\log_3^x}\right) = 9$

$$\Rightarrow 27^{-2\log_3^x} = \frac{9}{27} = \frac{1}{3} = 3^{-1}$$

$$\Rightarrow \left(3^3\right)^{-2\log_3^x} = 3^{-1} \Leftrightarrow 3^{-6\log_3^x} = 3^{-1}$$

$$\Rightarrow -6\log_3^x = -1 \longleftarrow a^x = a^y \Leftrightarrow x = y$$

$$\Rightarrow \log_3^x = \frac{1}{6} \Rightarrow x = 3^{\frac{1}{6}} = \sqrt[6]{3}$$

c) Exercise left for you. (Answer $x = \sqrt{\frac{3}{2}}$)

113. Solve:

a) $\log_3^x + \log_x^3 = \frac{5}{2}$ b) $\log_x^8 - \log_8^x = \frac{5}{6}$

Solution: Recall that $\log_b^a = \frac{1}{\log_a^b}$

$$\Rightarrow \log_3^x + \log_x^3 = \frac{5}{2} \Leftrightarrow \log_3^x + \frac{1}{\log_3^x} = \frac{5}{2}$$

$$\Rightarrow \frac{\left(\log_3^x\right)\left(\log_3^x\right)+1}{\log_3^x} = \frac{5}{2}$$

$$\Rightarrow \left(\log_3^x\right)^2 + 1 = \frac{5}{2}\log_3^x, \text{ let } \log_3^x = y$$

$$\Rightarrow y^2 + 1 = \frac{5}{2}y \Leftrightarrow y^2 - \frac{5}{2}y + 1 = 0$$

$$\Rightarrow 2y^2 - 5y + 2 = 0 \Leftrightarrow (2y-1)(y-2) = 0$$

$$\Rightarrow y = \frac{1}{2} \text{ or } y = 2$$

But $\log_3^x = y \Leftrightarrow \log_3^x = \frac{1}{2}$ or $\log_3^x = 2$

$\Rightarrow x = \sqrt{3}$ or $x = 3^2 = 9$

∴ S.S = $\{\sqrt{3}, 9\}$

b) $\log_x^8 - \log_8^x = \frac{5}{6} \Leftrightarrow \log_x^8 - \frac{1}{\log_x^8} = \frac{5}{6}$

$$\Rightarrow \left(\log_x^8\right)\left(\log_x^8\right) - 1 = \frac{5}{6}\log_x^8$$

$$\Rightarrow \left(\log_x^8\right)^2 - 1 = \frac{5}{6}\log_x^8, \textbf{ let } \log_x^8 = y$$

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$\Rightarrow y^2 - 1 = \frac{5}{6}y \Leftrightarrow y^2 - \frac{5}{6}y - 1 = 0$

$\Rightarrow 6y^2 - 5y - 6 = 0$

$\Rightarrow 6y^2 - 9y + 4y - 6 = 0 \Leftrightarrow 3y(2y-3) + 2(2y-3) = 0$

$\Rightarrow (3y+2)(2y-3) = 0$

$\Rightarrow y = \frac{-2}{3}$ or $y = \frac{3}{2}$

Since $\log_x^8 = y \Rightarrow \log_x^8 = \frac{-2}{3}$ or $\log_x^8 = \frac{3}{2}$

$\Rightarrow x^{-2/3} = 8$ or $x^{3/2} = 8$

$\Rightarrow x = (8)^{-3/2}$ or $8^{2/3}$

$\Rightarrow x = \frac{1}{512}$ or $x = 4$

114. The solution set of $\log_2^x + \log_4^{x^2} = 2$ is

A. {2} B. {−2} C. {4} D. {−2, 2}

Solution: Recall that $\log_2^x = \frac{\log_4^x}{\log_4^2} = 2\log_4^x$

$\Rightarrow \log_2^x + \log_4^{x^2} = 2 \Leftrightarrow 2\log_4^x + \log_4^{x^2} = 2$

$\Rightarrow 2\log_4^x + 2\log_4^x = 2$

$\Rightarrow 4\log_4^x = 2 \Rightarrow \log_4^x = \frac{2}{4} = \frac{1}{2}$

$\Rightarrow x = 4^{\frac{1}{2}} = \sqrt{4} = 2,$ **∴ Answer: A**

115. The solution set of $\log(x+10) + \log(x-5) = 2$.

A. {10,15} B. {10, −15} C. {10} D. {−10,5}

Solution: universe U = {x: x + 10 > 0 and x − 5 > 0}

= {x: x > − 10 and x > 5}

= (−10,∞) ∩ (5, ∞)

= (5, ∞)

$\Rightarrow \log(x+10) + \log^{(x-5)} = 2 \Leftrightarrow \log(x+10)(x-5) = 2$

$\Rightarrow (x+1)(x-5) = 10^2 = 100$ ⟵ exponential form

$\Rightarrow x^2 + 5x - 50 = 100$

$\Rightarrow x^2 + 5x - 150 = 0$

$\Rightarrow (x-10)(x+15) = 0$

$\Rightarrow x = 10$ or $x = -15$

However, $x = -15 \notin (5, \infty)$ **∴ Answer: C**

∴ x = 10,

116. The solution set of the equation $\log_3^{(6x+3)} - \log_3^{(x-2)} = 2$ **(Exercise left for you)**

A. {7} B. ϕ C. $\left\{\frac{4}{5}\right\}$ D. 4 **∴ Answer: A**

117. The solution of the equation $\log_2^{(x-2)} + \log_2^x = 3$

A. $1+\sqrt{7}$ B. −2 and 4 C. $1-\sqrt{7}$ and $1+\sqrt{7}$ D. 4

Solution: Here

Universe, U = {x: x − 2 > 0 and x > 0}

= {x: x > 2 and x > 0} ⟵ intersection

= {x: x > 2}

$\Rightarrow \log_2^{(x-2)(x)} = 3 \Rightarrow (x-2)(x) = 8 \Rightarrow x^2 - 2x = 8 \Rightarrow x2 - 2x - 8 = 0$

$\Rightarrow (x-4)(x+2) = 0 \Rightarrow x = 4$ or $x = -2$

However, $x = -2 \notin (2, \infty)$

∴ S.S x = 4, **∴ Answer: D**

Solving logarithmic equation of the form $\log_a^x = \log_a^y \Leftrightarrow x = y$

Illustrative Example

118. Solve:

a) $\log_{10}^{(2x+3)} - \log_{10}^x = \log_{10}^4$

b) $\log_4^{(x+2)} + \log_4^{(x-3)} = \log_4^{4x}$

c) $\log_2^x + \log_2^{(x+5)} = \log_2^{(x+4)}$

d) $\ln(x-5) = \ln(x-3) - \ln(x+3)$

e) $\log_3^x + \log_3^{(x^2-8)} = \log_3^8$

f) $\log\sqrt[3]{x} = \sqrt{\log x}$

g) $2\ln x = 3\ln 4$

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h) $\log_4^{(3x-2)} = \log_2^{(x-2)}$

i) $\ln x + \ln(x+1) = \ln 3 + \ln 4$

j) $\log(x-4) - \log(3x-10) = \log\left(\frac{1}{x}\right)$

Solution:

a) $\log_{10}^{(2x+3)} - \log_{10}^{4} = \log_{10}^{4}$ ⟵ $\log x - \log y = \log\left(\frac{x}{y}\right)$

$\Rightarrow \log_{10}^{\left(\frac{2x+3}{3}\right)} = \log_{10}^{4}$

$\Rightarrow \frac{2x+3}{x} = 4$ ⟵ **the same base 10**

$\Rightarrow 2x + 3 = 4x \Leftrightarrow 4x - 2x = 3$

$\Rightarrow 2x = 3 \Rightarrow x = \frac{3}{2}$

b) $\log_4^{(x+2)} + \log_4^{(x-3)} = \log_4^{4x}$

$\Rightarrow \log_4^{(x+2)(x-3)} = \log_4^{4x}$ ⟵ **Explanation**

• **$\log x + \log y = \log xy$**

$\Rightarrow (x+2)(x-3) = 4x$ ⟵ the same base 4

$\Rightarrow x^2 - x - 6 = 4x \Rightarrow x^2 - 5x - 6 = 0$

$\Rightarrow (x-6)(x+1) = 0$ $x = 6$ or $x = -1$

However, $x = -1$, makes both $x - 3$ and $4x$ Negative

$\therefore$ The solution is $x = 6$

c) $\log_2^{x} + \log_2^{(x+5)} = \log_2^{(x+4)}$

$\Rightarrow \log_2^{x(x+5)} = \log_2^{(x+4)} \Leftrightarrow x(x+5) = x + 4$

$\Rightarrow x^2 + 5x - x - 4 = 0 \Rightarrow x^2 + 4x - 4 = 0$

$\Rightarrow x = \frac{-4 \pm \sqrt{16 - 4(1)(-4)}}{2} = \frac{-4 \pm \sqrt{32}}{2} = -2 \pm 2\sqrt{2}$

$\Rightarrow x = -2 + 2\sqrt{2}$ or $x = -2 - 2\sqrt{2}$

But $x = -2 - 2\sqrt{2}$ makes x and $x + 4$ negative

$\therefore$ The solution is $x = -2 + 2\sqrt{2}$

d) $\ln(x-5) = \ln(x-3) - \ln(x+3)$

$\Rightarrow \ln(x-5) = \ln\left(\frac{x-3}{x+3}\right)$

$\Rightarrow x - 5 = \frac{x-3}{x+3} \Rightarrow (x-5)(x+3) = x - 3$

$\Rightarrow x^2 - 2x - 15 = x - 3$

$\Rightarrow x^2 - 3x - 18 = 0 \Leftrightarrow (x-6)(x+3) = 0$

$\Rightarrow x = 6$ or $x = -3$

However, $x = -3$ is makes $x - 5$, $x - 3$ and $x + 3$ negative or zero

$\therefore$ The solution is $x = 6$

e) $\log_3^{1} + \log_3^{(x^2-8)} = \log_3^{8x}$

$\Rightarrow \log_3^{x(x^2-8)} = \log_3^{8x} \Leftrightarrow x(x^2 - 8) = 8x$

$\Rightarrow x^3 - 8x = 8x \Rightarrow x^3 - 16x = 0$

$\Rightarrow x(x^2 - 16) = 0 \Rightarrow x = 0$ or $x^2 - 16 = 0$

$\Rightarrow x = 0$ or $x = 4$ or $x = -4$

However, $x = 0$ and $x = -4$ makes x and $8x$ negative and zero

$\therefore$ The solution is $x = 4$

f) $\log \sqrt[3]{x} = \sqrt{\log x} \Leftrightarrow \log^{\frac{1}{3}} = \sqrt{\log x}$

$\Rightarrow \log^{\frac{1}{3}} = (\log x)^{\frac{1}{2}} \Leftrightarrow \left(\log x^{\frac{1}{3}}\right) = \log x$

$\Rightarrow \left(\frac{1}{3}\log x\right)^2 = \log x$, let $\log x = y$

$\Rightarrow \left(\frac{1}{3}y\right)^2 = y \Leftrightarrow \frac{1}{9}y^2 = y$

$\Rightarrow y^2 = 9y \Leftrightarrow y^2 - 9y = 0$

$\Rightarrow y(y-9) = 0 \Rightarrow y = 0$ or $y = 9$

But $\log x = y \Rightarrow \log x = 0$ or $\log x = 9$

$\Rightarrow x = 10^0$ or $x = 10^9$

$\therefore$ The solution set is $\{1, 10^9\}$

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g) $2\ln x = 3\ln 4 \Leftrightarrow \ln x^2 = \ln 4^3$

$\Rightarrow x^2 = 4^3 = 64$

$\Rightarrow x = 8$ or $x = -8$

However, $x = -8$ makes x negative

$\therefore$ The solution is $x = 8$

h) $\log_4^{(3x-2)} = \log_2^{(x-2)}$

$\Rightarrow 3x - 2 = 4^{\log_2^{(x-2)}}$ ⟵ Exponential form.

$\Rightarrow 3x - 2 = \left(2^2\right)^{\log_2^{(x-2)}}$ ⟵ $(a^x)^y = a^{xy}$

$= 2^{2\log_2^{(x-2)}} = 2^{2\log_2^{(x-2)^2}}$

$(3x - 2) = (x - 2)^2 \Leftrightarrow x^2 - 4x + 4 = 3x - 2$

$\Rightarrow x^2 - 7 + 6 = 0$

$\Rightarrow (x - 6)(x - 1) = 0 \Rightarrow x = 6$ or $x = 1$

However, $x = 1$ makes $x - 2$ negative

$\therefore$ The solution is $x = 6$

i) $\ln x + \ln(x + 1) = \ln 3 + \ln 4$

$\Rightarrow \ln(x)(x+1) = \ln(3.4)$

$\Rightarrow \ln x(x+1) = \ln 12$

$\Rightarrow x(x+1) = 12 \Leftrightarrow x^2 + x = 12$

$\Rightarrow x^2 + x - 12 = 0 \Leftrightarrow (x - 3)(x+4) = 0$

$\Rightarrow x = 3$ or $x = -4$

However, $x = -4$ makes x and $x + 1$ negative

$\therefore$ The solution is only $x = 3$

j) $\log(x - 4) - \log(3x - 10) = \log\left(\frac{1}{x}\right)$

$\Rightarrow \log\left(\frac{x-4}{3x-10}\right) = \log\left(\frac{1}{x}\right)$

$\Rightarrow \frac{x-4}{3x-10} = \frac{1}{x} \Leftrightarrow x^2 - 4x = 3x - 10$

$\Rightarrow x^2 - 7x + 10 = 0 \Leftrightarrow (x - 5)(x - 2) = 0$

$\Rightarrow x = 5$ or $x = 2$

However, $x = 2$ makes x and $x - 4$ and $3x - 10$ negative

$\therefore$ The solution is only $x = 5$

119. Solve

a) $\log_x^a = \log_a^x$ b) $\log_x^4 = \log_4^x$

Solution: $\log_x^a = \frac{\log_a^a}{\log_a^x} = \frac{1}{\log_a^x}$

$\Rightarrow \log_x^a = \log_a^x \Leftrightarrow \frac{1}{\log_a^x} = \log_a^x$

$\Rightarrow \left(\log_a^x\right)\left(\log_a^x\right) = 1 \Leftrightarrow \left(\log_a^x\right)^2 = 1$

$\Rightarrow \log_a^x = \pm 1 \Rightarrow x = a^1$ or $x = a^{-1}$

$\therefore$ The solution set is $\left\{a, \frac{1}{a}\right\}$

b) $\log_x^4 = \log_4^x \Leftrightarrow \log_4^x = \frac{1}{\log_4^x}$

$\Rightarrow \left(\log_x^4\right)\left(\log_x^4\right) = 1$

$\Rightarrow \left(\log_x^4\right)^2 = 1 \Rightarrow \log_4^x = \pm 1$

$\Rightarrow x = 4^1$ or $x = 4^{-1}$

$\therefore$ The solution set is $\left\{4, \frac{1}{4}\right\}$

120. Solve

a) $\ln x = 2$ b) $\ln 3x = 4$ c) $\ln(ex) = 3$

Solution: Here, Recall that $\ln x = \log_e^x$

a) $\ln x = 2 \Leftrightarrow \log_e^x = 2 \Rightarrow x = e^2$

b) $\ln 3x = 4 \Leftrightarrow \log_e^{3x} = 4 \Rightarrow 3x = e^4 \Rightarrow x = \frac{e^4}{3}$

c) $\ln ex = 3 \Leftrightarrow \log_e^{ex} = 3 \Rightarrow ex = e^3 \Rightarrow x = e^2$

121. Solve:

a) $\log_{0.3}^{0.027} = \left(\frac{1}{81}\right)^x$ b) $\left(\frac{27}{8}\right)^x = \log_7^{\sqrt[3]{49}}$

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Solution: Here

a) $\log_{0.3}^{0.027} = y \Leftrightarrow (0.3)y = (0.3)^3 \Rightarrow y = 3$

$\Leftrightarrow 3 = \left(\frac{1}{81}\right)^x \Rightarrow 3^1 = 3^{-4x} \Rightarrow -4x = 1$

$\therefore x = -\frac{1}{4}$

b) $\left(\frac{27}{8}\right) = \log_7^{\sqrt[3]{49}} \Leftrightarrow \left(\frac{3}{2}\right)^{3x} = \log_7^{7^{\frac{2}{3}}}$

$\Rightarrow \left(\frac{3}{2}\right)^{3x} = \frac{2}{3} \Leftrightarrow \left(\frac{2}{3}\right)^{-3x} = \left(\frac{2}{3}\right)^1$

$\Rightarrow -3x = 1$

$\therefore x = -\frac{1}{3}$

122. Solve:

a) $e^{2y} + 2e^y - 15 = 0$ b) $3^{2x} - 2(3^x) - 8 = 0$

c) $\frac{\log(7x-12)}{\log x} = 2$ d) $\frac{\log(3x-8)}{\log(x-2)} = 2$

e) $\frac{\log_0^8}{\log x} = 3$ f) $\frac{2\log(x+4)}{\log 3} = 4$

Solution: Here

a) $e^{2y} + 2e^y - 15 = 0 \Leftrightarrow (e^y)^2 + 2e^y - 15 = 0$

$\Rightarrow (e^y - 3)(e^y + 5) = 0$

$\Rightarrow e^y - 3 = 0$ or $e^y + 5 = 0$

$\Rightarrow e^y = 3$ or $e^y = -5$

$\Rightarrow y = \log_e^3 = \ln 3$ but $e^y = -5$ is not defined

$\therefore$ The solution is $y = \ln 3$

b) $3^{2x} - 2(3^x) - 8 = 0 \Leftrightarrow (3^x)^2 - 2(3^x) - 8 = 0$

$\Rightarrow (3^x - 4)(3^x + 2) = 0$

$\Rightarrow 3^x - 4 = 0$ or $3^x + 2 = 0$

$\Rightarrow 3^x = 4$ or $3^x = -2$

$\Rightarrow x = \log_3^4$ but $3^x = -2$ is not defined

$\therefore$ The solution is $x = \log_3^4$

c) $\frac{\log(7x-12)}{\log x} = 2 \Leftrightarrow \log_x^{7x-12} = 2$

$\Rightarrow x^2 = 7x - 12$

$x^2 - 7x + 12 = 0$

$(x - 3)(x - 4) = 0$ $\therefore x = 3$ or $x = 4$

d) $\frac{\log(3x-8)}{\log(x-2)} = 2 \Leftrightarrow \log_{(x-2)}^{(3x-8)} = 2$

$\Rightarrow 3x - 8 = (x - 2)^2$

$\Rightarrow x^2 - 4x + 4 = 3x - 8$

$\Rightarrow x^2 - 4x - 3x + 4 + 8 = 0$

$\Rightarrow x^2 - 7x + 12 = 0$

$\Rightarrow (x - 3)(x - 4) = 0$

$\therefore x = 3$ or $x = 4$

e) **Exercise left for you**

f) $\frac{2\log(x+4)}{\log 3} = 4 \Leftrightarrow 2\log_3^{(x+4)} = 4$

$\Rightarrow \log_3^{(x+4)^2} = 4$

$\Rightarrow (x+4)^2 = 3^4 = 81$

$\Rightarrow x + 4 = \pm 9$

$\Rightarrow x = 9 - 4 = 5$ or $x + 4 = -9$

$\Rightarrow x = 5$ or $x = -13$

However, $x = -13$ is not in the domain

$\therefore$ The solution is $x = 5$

123. Solve

a) $\log_5^{(6^{3(4x-3)})} - \log_5^{(36^{3x+3})} = 0$ b) $\log_3^{(4^{3x+1})} - \log_3^{(8^{x+1})} = 0$

Solution: Here **Recall that** $a^x - a^y = 0 \Leftrightarrow a^x = a^y$

a) $\log_5^{(6^{3(4x-3)})} - \log_5^{(36^{3x+3})} = 0$

$\Rightarrow \log_5^{(6^{3(4x-3)})} = \log_5^{(36^{3x+3})}$

$\Rightarrow 6^{3(4x-3)} = 36^{3x+3}$

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a) $\log_5^{(6^{3(4x-3)})} - \log_5^{(36^{3x+3})} = 0$

$\Rightarrow \log_5^{(6^{3(4x-3)})} = \log_5^{(36^{3x+3})}$

$\Rightarrow 6^{3(4x-3)} = 36^{3x+3}$

$\Rightarrow 6^{12x-9} = 6^{6x+6} \Leftrightarrow 12x - 9 = 6x+6$

$\Rightarrow 12x - 6x = 6 + 9 = 15$

$6x = 16 \Rightarrow x = \frac{15}{6} = \frac{3}{2}$ ∴ The solution is $x = \frac{3}{2}$

b) $\log_3^{(4^{3x+1})} - \log_3^{(8^{x+1})} = 0 \Rightarrow \log_3^{4^{3x+1}} = \log_3^{8^{x+1}}$

$\Rightarrow 4^{3x+1} = 8^{x+1} \Leftrightarrow 2^{6x+2} = 2^{3x+3}$

$\Rightarrow 6x + 2 = 3x + 3.$

$\Rightarrow 6x - 3x = 3 - 2 \Rightarrow 3x = 1 \Rightarrow x = \frac{1}{3}$

Using Definition
- $\log_a^r = \log_a^s$
- $\Rightarrow x = y$

Application of logarithms

Compound interest.

- The **rate of interest (r)** expressed as a percent, is the amount charged for the use of **the principal (A)** for a given period of time, usually. Annually, semiannually, quarterly.
- We define the term **payment period** as follows:
  - Annually **means** once per year.
  - Semi-annually **means** Twice per year.
  - Quarterly **means** four times per year.
  - Monthly **means** 12 times per year.

Compound interest formula:

- The amount **'A'** after **t** years due to a principal **'P'** invested at an annual interest rate **'r'** compound **'n'** times per year is:

$$A = P\left(1+\frac{r}{n}\right)^{nt} \text{ where}$$

- **A = the future value**
- **P = preset value**
- **r = the interest rate**
- **n = number of compounding per year**
- **t = number of year**

e) how long does it take to double investment compounded semiannually

f) how long does it take to triple investment, **compounded semi-annually.**

Solution: We use $A = P\left(1+\frac{r}{n}\right)^{nt}$

r = 10%, P = 1000birr, t = 4 year.

a) **Annually compounding: n =1,** $A = P\left(1+\frac{r}{n}\right)^{nt}$, since t = 4

$\Rightarrow A = 1000(1+0.1)^4 = 1000(1.1)^4 = 1{,}464.10$birr.

b) **Semi-annually (x = 2):** $A = P\cdot\left(1+\frac{r}{2}\right)^{(2)(4)}$

$\Rightarrow A = 1000\,(1+0.05)^8 = 1{,}477.45$birr

c) **Quarterly means n = 4,** $A = P\left(1+\frac{r}{4}\right)^{(4)(4)}$, since t = 4

$\Rightarrow A = 1000\,(1+0.025)^{16} = 1484.50$birr

d) **Monthly means n = 12,** $A = P\left(1+\frac{r}{12}\right)^{(4)(12)}$, since t = 4

$\Rightarrow A = 1000\,(1+0.083)^{48} = 1489.35$birr.

e) If the investment is to be doubled then $A = 2P = 2(1000) = 2000$

$\Rightarrow A = P\left(1+\frac{r}{n}\right)^{nt}$,

$\Rightarrow 2000 = 1000\,(1+0.05)^{2t}$ since compounded semiannually n = 2

$\Rightarrow 2000 = 1000\,(1.05)^{2t}$

$\Leftrightarrow \frac{2000}{1000} = (1.05)^{2t}$

$\Leftrightarrow 2 = (1.05)^{2t}$ ⟵ **take the logarithm of both side.**

$\Rightarrow \log 2 = 2t \log(1.05)$

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$$\Rightarrow 2t = \frac{\log 2}{\log(1.05)} = \frac{0.3010}{0.0212} = 14.19 \approx 14.2$$

$$\Rightarrow t = \frac{14.2}{2} \approx 7.1 \approx 7\ year$$

f) If the investment is to be tripled compounded semi-annually.
Then, A = 3P = 3(1000) = 3,000

$\Rightarrow A = P\left(1+\frac{r}{n}\right)^{nt}$, since compounded semiannually, n = 2

$$\Rightarrow 3000 = 1000\left(1+\frac{0.1}{2}\right)^{2t}$$

$$\Rightarrow \frac{3,000}{1000} = (1.05)^{2t} \Leftrightarrow 3 = (1.05)^{2t}$$

$$\frac{3,000}{1000}(1.05)^{2t} \Leftrightarrow 3 = (1.05)^{2t}$$

$\log 3 = 2t \log(1.05)$

$$\Rightarrow 2t = \frac{\log 3}{\log 1.05} = \frac{0.4771}{0.0212} = 22.504$$

$\Rightarrow t = 11.25$ year

125. For how many years must 1000birr be invested at 12% compounded annually in order for the investment to grow to 3000 birr.

Solution: Use $A = P\left(1+\frac{r}{n}\right)^{nt}$

$\Rightarrow 3000 = 1000\ (1 + 0.12)^t$ since n = 1

$$\Rightarrow \frac{3000}{1000} = (1.12)^t \Leftrightarrow 3 = (1.12)^t$$

$$\Rightarrow t = \frac{\log 3}{\log(1.12)} = \frac{0.4771}{0.0492} \approx 9.69\ year$$

$\Rightarrow t = 10$ years

Doubling and Tripling Time for an investment

- If the investment is to be doubled then A = 2P
- If the investment is to be tripled then A = 3P.

Illustrative Example

126. i) How long will it take for an investment to double the initial investment if invested at the rate of 6% compound
Use log2 = 0.3010, log1.06 = 0.0253 and log (1.03) = 0.0128
a) Annually b) semiannually
ii) How long will it take to triple at this rate?
(Use log3 = 0.4771)

Solution: we use $A = P\left(1+\frac{r}{n}\right)^{nt}$

i) Since, A = 2P ⟵ doubling investment
a) **Annually**, means n = 1, $2P = P(1+0.06)^t$

$\Rightarrow 2 = (1.06)^t$

$\Rightarrow \log 2 = t \log(1.06)$

$$\Rightarrow t = \frac{\log 2}{\log(1.06)} = \frac{0.3010}{0.0253} \approx 11.89\ year$$

$\Rightarrow$ It will take about 12years to doubled

b) **Semi-annually** means, n = 2 and A = 2P

$$\Rightarrow 2P = P\left(1+\frac{0.06}{2}\right)^{2t} = P(1.03)^{2t}$$

$\Rightarrow 2 = (1.03)^{2t} \Leftrightarrow \log 2 = 2t \log(1.03)$

$$\Rightarrow 2t = \frac{\log 2}{\log 1.03} = \frac{0.3010}{0.0128}$$

$\Rightarrow t = 11.75$year.

ii) If it is to be tripled then A = 3P

a) **annually**, means n = 1 and A = 3P

$$\Rightarrow 3P = P\left(1+\frac{0.06}{2}\right)^{2t} = P(1.06)^t$$

$\Rightarrow 3 = (1.06)^t \Leftrightarrow \log 3 = t \log(1.06)$

$$\Rightarrow t = \frac{\log 3}{\log(1.06)} = \frac{0.4771}{0.0253} = 18.85 \text{ year}$$

$\Rightarrow t = 19$ year.

b) **Semia-nnually** means, n = 2 and A = 3P

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$$\Rightarrow 3P = P\left(1+\frac{0.06}{2}\right)^{2t} = P(1.03)^{2t}$$

$$\Rightarrow 3 = (1.03)^{2t} \Leftrightarrow \log 3 = 2t\log(1.03)$$

$$\Rightarrow 2t = \frac{\log 3}{\log(1.03)} = \frac{0.4771}{0.0128} = 37.27$$

$$\Rightarrow t = \frac{37.27}{2} = 18.63 year$$

127. If 1000 Birr is invested at 8% compounded quarterly, how long does it take to double the initial investment?
(**Use: log2 = 0.3010, log (1.02) = 0.0086**).

Solution: $A = P\left(1+\frac{r}{n}\right)^{nt}$, Since A = 2P and n = 4

$$\Rightarrow 2P = P\left(1+\frac{0.08}{4}\right)^{4t}$$

$$\Rightarrow 2 = (1.02)^{4t} \Leftrightarrow \log 2 = 4t\log(1.02)$$

$$\Rightarrow 4t = \frac{\log 2}{\log(1.02)} = \frac{0.3010}{0.0086} = 35$$

$$\Rightarrow t = \frac{35}{4} = 8.75 \text{ year}$$

$$\Rightarrow t \approx 9 \text{ years.}$$

128. If some amount of money is invested at the rate of 12% compounded semi-annually, how long does it take to doubled the initial investment.
(**Use**: log2 = 0.3, log(1.02) = 0.009, log1.03 = 0.013 and log1.06 = 0.025

A. $\frac{150}{13}$ years B. 6 years C. 8 years D. $\frac{50}{3}$ years

Solution: $A = P\left(1+\frac{r}{n}\right)^{nt}$, since A = 2P and n = 2

$$\Rightarrow 2P = P\left(1+\frac{0.12}{2}\right)^{2t} \Leftrightarrow 2 = (1.06)^{2t}$$

$$\Rightarrow \log 2 = 2t\log(1.06) \Leftrightarrow 2t = \frac{\log 2}{\log 1.06}$$

$$\Rightarrow 2t = \frac{0.3010}{0.025} = 12.04$$

$$\Rightarrow t = \frac{12.04}{2} = 6.02 \text{ years}$$

$\Rightarrow$ t = 6 years. **$\therefore$Answer B**

129. If 1000Birr is invested today at a rate of 4% compounded semiannually, what will be the amount after one year?
A. 2080.80 birr B. 1040.40Birr C. 1400Birr D. 2080Birr

Solution: $A = P\left(1+\frac{r}{n}\right)^{nt}$, r = 0.04 = 4%, n = 2, t = 1

$$\Rightarrow A = 1000\left(1+\frac{0.04}{2}\right)^{(2)(1)} = 1000(1.02)^2 = 1040.40$$

Answer: B

Population Growth

- The total population after t years is given by the formula:
  $A_t = P(1 + r)^t$, where P is current population.

Illustrative Example

130. **Population growth**: The population of a certain town is increasing at the rate of 5% per year. If s the present number of population is 100,000 Find the population after:
a) t year b) 4 years c) 10 years

Solution: $A_t = P(1 + r)^t$, **r = 5% = 0.05**

a. $\Rightarrow A_t = 100{,}000(1 + 0.05)^t = 100{,}000(1.05)^t$
b. $A_4 = 100{,}000(1.05)^4 = 121{,}550.$
c. $A_{10} = 100{,}000(1.05)^{10} = 162, 889.$

131. From the above example,
how long does it take the population
(**use log2.4 = 0.3802, log 1.05 = 0.0212 and log2 = 0.3010**).
a) to be 240,000 b) to be doubled

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Solution: Here, we use $A_t = P(1 + r)^t$

$r = 5\%$, $P = 100{,}000$ and $A_t = 240{,}000$

a) $240{,}000 = 100{,}000\,(1 + 0.05)^t = 100{,}000(1.05)^t$

$$\Rightarrow (1.05)^t = \frac{240{,}000}{100{,}000} = 2.4$$

$$\Rightarrow t\log(1.05) = \log 2.4 \Leftrightarrow t = \frac{\log 2.4}{\log 1.05} = \frac{0.3802}{0.0212}$$

$$\Rightarrow t = 17.93$$

Therefore, in needs 18 years to be come 240,000.

b) If the population is double, then $A_t = 2P$

$\Rightarrow A_t = P(1 + r)^t \Leftrightarrow 2P = P(1 + r)^t = P(1.05)^t$

$\Rightarrow 2 = (1.05)^t \Leftrightarrow t\log(1.05) = \log 2$

$$\Rightarrow t = \frac{\log 2}{\log 1.05} = \frac{0.3010}{0.0212} = 14.19 \text{ years}$$

$\Rightarrow t \approx 14$ years

$\therefore$ The population become double at the end of 14 years

132. If the present number of population of town A is 6.24×10^8 and town B is 1.36×10^9 growth at the rate of 6% and 3% per year respectively

When will town A and town B have the same population?

(Use: log 1.03 = 0.0128, log 2.17 = 0.3365

Solution: use $A = P(1 + r)^t$

a) **For town A:** $6.24 \times 10^8(1+0.06)^t$ ⟵ after t year

For town B: $1.36 \times 10^9(1 + 0.03)^t$ ⟵ after 't' year

$\Rightarrow$ **town A = town B**

$\Rightarrow 6.24 \times 10^8\,(1 + 0.06)^t = 1.36 \times 10^9\,(1 + 0.03)^t$

$$\Rightarrow \frac{(1.06)^t}{(1.03)^t} = \frac{1.36\times10^9}{6.24\times10^8} = 2.17$$

$$\Rightarrow \left(\frac{1.06}{1.03}\right)^t = 2.17 \Leftrightarrow$$

$\Rightarrow (1.03)^t = 2.17 \Leftrightarrow t\log(1.03) = \log(2.170$

$$\Rightarrow t = \frac{\log 2.17}{\log 1.03} = \frac{0.3365}{0.0128} = 26.28 \text{ year}$$

$\therefore$The population of town A and town B are equal after 26 year.

133. **Growth of bacteria**: A certain culture initially contain 1000 bacteria and grow at the rate 5% every hour. Find the number of bacterial after a) n hours b) 2 hours c) 10 hours

Solution: bacteria growth at the end of hours is given by

$A = P(1 + \text{rate})^n$, $P = 1000$, $r = 5\% = 0.05$

a) $A_n = P(1 + 0.05)^n = 1000(1.05)^n$ ⟵ after n – years

b) $A_2 = 1000\,(1.05)^2 = 1102$ ⟵ after 2 hours.

b) $A_{10} = 1000\,(1.05)^{10} = 1628$ ⟵ after 10 hours.

134. **Depreciation of value**: If the car. Purchased by 240,000 Birr depreciates at the rate of 20% per year. What will be the value at the end of a) t year b) 8 years c) 10 years

Solution: The depreciation value at the end of **t** years is given by

$V_t = V_o\,(1 - \text{rate})^t$, $Vo = 240{,}000$, $r = 20\% = 0.2$

a) **at the end of 't' years,** $V_t = 240{,}000\,(1 - 0.2)^t$

$\Rightarrow V_t = 240{,}000\,(0.8)^t$

b) **at the end of 8 years,** $V_8 = 240{,}000\,(0.8)^8 = 40{,}265$birr

c) **at the end of 10 years,** $V_{10} = 240{,}000\,(0.8)^{10} = 25{,}769$birr

II. The pH of a solution

The other application of logarithm function is to find the **PH** of solution

- The pH of a solution with a hydrogen ion concentration of $[H^+]$ in moles per liter is given by **$pH = -\log[H^+]$**

Note: The solution is said to be

i) Acidic if $pH < 7$

ii) Base if $pH > 7$ iii) neutral if $pH = 7$

Illustrative Example

130. Find the pH of each liquid

a) Orange juice with $[H^+] = 2.8 \times 10^{-4}$ mole per liter

b) Milk with $[H^+] = 3.97 \times 10^{-7}$ mole per liter

Explanation: Here

a) $pH = -\log[H^+] = -\log(2.8\times 10^{-4}) \approx 3.6$

$\therefore$ **The orange juice has a pH of 3.6**

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b) $pH = -\log[H^+] = -\log(3.97\times 10^{-7}) \approx 6.4$

∴ **The milk has a pH of 6.4**

131. If the pH of rain water is 5.6 **what is the hydrogen – ion concentration $[H^+]$ of the rain water (in mole per liter)**

Solution: $pH = -\log[H^+]$

$\Rightarrow 5.6 = -\log[H^+] \Rightarrow \log[H^+] = -5.6$

$\Rightarrow$ Antilog $(\log[H^+]$ = antilog $[-5.6] = 2.51\times 10^{-6}$

$\Rightarrow [H^+] = 2.51 \times 10^{-6}$ mole per liter

131. (A) Given: $\log 5 = 0.6990$

Solve a) $10^{2.6990} = x$ b) $\log x = -1.3010$

Solution: we have $\log 5 = 0.6990 \Rightarrow 10^{0.6990} = 5$

a) $10^{2.6990} = (10^2)(10^{0.6990}) = x \Leftrightarrow x = 5(100) = 500$

b) $\log x = -1.3010 \longleftarrow$ since no negative mantissa

$\Rightarrow \log x = -1.3010 + (2) + (-2) = 0.6990 + (-2)$

$\Rightarrow x = 10^{0.6990}(10^{-2}) = 5(10^{-2}) = 0.05$

132. **The Richter scale**: On the Richter scale the magnitude R of an earthquake of intensity I is given by $\mathbf{R = \log\left(\frac{I}{I_0}\right)}$, where I_0 is a certain minimum intensity

a) If the intensity of an earth quake is $1000I_0$, **Find R.**

b) Express I in terms of R and I_0.

Solution: a) $R = \log\left(\frac{I}{I_0}\right) = \log\left(\frac{1000I_0}{I_0}\right)^0$

$= \log 1000 = 3$

b) $R = \log\left(\frac{I}{I_0}\right) \Leftrightarrow \left(\frac{I}{I_0}\right) = 10^R \Rightarrow I = 10^R.I_0$

Solved Problem

133. Which of the following is true about the exponential function $y = a^x$, $a > 0$ and $a \neq 1$.

A. If $a > 1$ and $x < 0$, then $y > 1$

B. If $0 < a < 1$ and $x > 0$, then $y > 1$

C. If $a > 1$ and $x > 0$, then $y > 1$

D. If $0 < a < 1$ and $x < 0$, then $y > 1$

Explanation: consider the following table

Case	$x > 0$	$x < 0$	
i) If $a > 1$ Eg. $y = 2^x$	$y > 1$	$0 < y < 1$	Increasing function
ii) If $0 < a < 1$ Eg. $y = \left(\frac{1}{2}\right)^x$	$0 < y < 1$	$y > 1$	decreasing function

∴ **Answer: D**

134. Which of the following is **not true** about the function $g(x) = 1 + e^x$

A. Its range is (e, ∞) c) for $x < 0$, $1 < g(x) < 2$

B. Its y – intercept is 2 d) It is increasing

Explanation:

A. The range of g is $(1,\infty)$

B. Y – intercept, which $x = 0$ is $1 + 1 = 2$

C. For $x < 0$, $1 < f(x) < 2$ **Answer: A**

D. It is increasing because the base $e > 1$

135. The inverse of the function $f(x) = \left(\frac{1}{3}\right)^x$

A. $g(x) = \log_3^{(-x)}$ C) $g(x) = \log_3^x$

B. $g(x) = -\log_3^x$ D) $g(x) = (\log_3^x)^{-1}$

Explanation: Remember that the inverse of $y = b^x$:

Case 1 if $b > 1$ then the inverse of $y = b^x$ is $y = \log_b^x$

Case 2 if $0 < b < 1$, then the inverse of $y = b^x$ is $y = -\log_b^x$

Since $\frac{1}{3} < 1$, then inverse of $f(x) = \left(\frac{1}{3}\right)^x$ is $g(x) = -\log_3^x$,

Answer: B

Note • $y = \left(\frac{1}{2}\right)^x$ is the inverse of $y = -\log_2^x$

• $y = 3^x$ is the inverse of $y = \log_3^x$

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136. If $\log_2^x = \frac{1}{3}\log_2^8 - 2\log_2^{\sqrt{3}}$, then the value of x

A. $\frac{2}{3}$ B: $\frac{\sqrt{3}}{2}$ C. $\frac{2}{\sqrt{3}}$ D. $\frac{3}{2}$

Solution: $\log_2^x = \log_2^{\sqrt[3]{8}} - 2\log_2^{\sqrt{3}} \Leftrightarrow \log_2^x = \log_2^{\left(\frac{\sqrt[3]{8}}{3}\right)}$

$\Leftrightarrow x = \frac{\sqrt[3]{8}}{3} = \frac{2}{3}$, **Answer: A**

137. If $a = \left(\frac{1}{4}\right)^{-x}$ and $b = \left(\frac{1}{8}\right)^{y}$, then **Evaluate**, $\log_2^{ab}$

Solution: $a = \left(\frac{1}{4}\right)^{-x} = 2^{2x} \Leftrightarrow \log_2^a = 2x$

$b = \left(\frac{1}{8}\right)^{y} \Leftrightarrow b = 2^{-3y} \Leftrightarrow \log_2^b = -3y$

So that $\log_2^{ab} = \log_2^a + \log_2^b = 2x - 3y$

138. If $\log_{\left(\frac{1}{3}\right)}^a = x$ and $\log_9^b = y$ then **find** $\log_3^{ab}$

Solution: $\log_{\frac{1}{3}}^a = x \Leftrightarrow -\log_3^a = x \Rightarrow \log_3^a = -x$ and $\log_9^b = y$

$\Leftrightarrow \log_{3^2}^b = \frac{1}{2}\log_3^b = -x + 2y$

139. If $\log x = a$ and $\log y = b$, **then evaluate**

a) $\log_y^x$ b) $\log_x^{xy}$ c) $\log_y^{yx}$

Solution: Recall that $\log_y^x = \frac{\log x}{\log y}$

a) $\log_y^x = \frac{\log x}{\log y} = \frac{a}{b}$

b) $\log_x^{xy} = \frac{\log xy}{\log x} = \frac{\log x + \log y}{\log x} = \frac{a+b}{a} = 1 + \frac{b}{a}$

c) Exercise left for you

140. If $m = C^a$ and $n = C^b$, then **Evaluate** a) $\log_c^{\sqrt[3]{m}}$ b) $\log_{\sqrt{m}}^n$

Solution: we have $m = C^a \Leftrightarrow \log_c^m = a$ and $n = c^b \Leftrightarrow \log_c^n = b$

Therefore, $\log_c^{\sqrt[3]{m}} = \frac{\frac{1}{3}\log_c^{\sqrt[3]{m}}}{\log_c^n} = \frac{\frac{1}{3}\log_c^m}{\log_c^n} = \frac{a}{3b}$

b) $\log_{\sqrt{m}}^n = \frac{\log_c^n}{\log_c^{\sqrt{m}}} = \frac{\log_c^n}{\frac{1}{2}\log_c^m} = \frac{2b}{a}$

141. Find the **characteristics and mantissa**

a) $\log\left(\frac{1}{400}\right)$ b) $\log\left(\frac{1}{8000}\right)$ c) $\log\left(\frac{1}{2000}\right)$

Solution:

a) $\log\left(\frac{1}{400}\right) = \log(0.0025) = \log(2.5 \times 10^{-3})$

$\therefore$ Characteristics is -3 $\therefore$ mantissa $\log(2.5)$

b) $\log\left(\frac{1}{8000}\right) = \log(0.000125) = \log(1.25 \times 10^{-4})$

$\therefore$ Characteristics is -4 $\therefore$ mantissa is a $\log(1.25)$

c) **Exercise left for you**

142. Solve

a) $\ln(3x) - \ln(x-1) = \ln 4$ b) $\log_5^{x^3} = \log_5^{16x}$

c) $16^{\log_4^x} = 18$ d) $\log^{\sqrt{x^3-17}} = \frac{1}{2}$

e) $\log(5x-1) = 2 + \log(x-2)$ f) $1 + \log(3x-1) = \log(2x+1)$

Solution:

a) $\ln\left(\frac{3x}{x-1}\right) = \ln 4 \Leftrightarrow \frac{3x}{x-1} = 4 \Rightarrow 3x = 4x - 4$

$\Rightarrow 4x - 3x = 4 \Rightarrow x = 4$

b) $\log_5^{x^3} = \log_5^{16x} \Leftrightarrow x^3 = 16x \Rightarrow x^3 - 16x = 0$

$\Rightarrow x(x^2 - 16) = 0 \Rightarrow x = 0$ or $x = 4$ or $x = -4$

$\therefore$ The solution is $x = 4$ only

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c) $16^{\log_4^x} = 18 \Leftrightarrow 16^{\log_4^x} = 16^{\log_{16}^{18}} \Leftrightarrow \log_4^x = \log_{16}^{18}$

$\Rightarrow \log_4^x = \frac{1}{2}\log_4^{18}$ ⟵ **Explanation:**

$$\log_{16}^{18} = \log_{4^2}^{18} = \frac{1}{2}\log_4^{18}$$

$\Rightarrow \log_4^x = \log_4^{\sqrt{18}} \Leftrightarrow x = \sqrt{18} = 3\sqrt{2}$

d) $\log_{10}^{\sqrt{x^3-17}} = \frac{1}{2} \Leftrightarrow 10^{\frac{1}{2}} = \sqrt{x^3-17}$

$\Leftrightarrow x^3 - 17 = 10 \Rightarrow x^3 = 27 \quad \therefore x = 3$

e) $\log^{(5x-1)} = 2 + \log^{(x-2)}$, since $\log^{100} = 2$

$\Leftrightarrow \log^{(5x-1)} \log^{100} + \log^{(x-2)}$

$\Rightarrow \log^{(5x-1)} = \log^{100(x-2)}$

$= \Leftrightarrow 5x - 1 = 100x - 200 \Leftrightarrow -95x = -199, \; x = \frac{199}{95}$

f) Exercise left for you

140. **What is the time** it takes for money invested at an annual rate of r to double if the money is compounded continuously? **(Use $A = Pe^{rt}$)**

A. $\ln\left(\frac{r}{2}\right)$ B. $\ln\sqrt{2r}$ C. $\ln\frac{\ell nr}{2}$ D. $\frac{\ell n2}{r}$

Explanation: $A = Pe^{rt}$, since $A = 2P$

$\Rightarrow 2P = p(e^{rt}) \Leftrightarrow 2 = e^{rt} \Rightarrow \ln 2 = \ln e^{rt}$

$\Rightarrow \ln 2 = rt\,(\ln e) \Rightarrow \ln 2 = rt \therefore t = \frac{\ell n2}{r}$, **Answer: D**

141. The graph of 3^{2x-1} intersect with the graph of 5^{x+2} at ____

Explanation: At the point of intersection

$$3^{2x-1} = 5^{x+2} \Leftrightarrow \frac{3^{2x}}{3} = 5^x.(25)$$

$\Rightarrow \frac{3^{2x}}{5^x} = (3)(25)$ ⟵ **Take natural logarithm**

$\Rightarrow \ln\left(\frac{3^{2x}}{5^x}\right) = \ln(3.5^2)$

$\Rightarrow \ln 3^{2x} - \ln(5^x) = \ln 3 + 2\ln 5$

$\Rightarrow 2x\ln 3 - x\ln 5 = \ln 3 + 2\ln 5$

$\Rightarrow x(2\ln 3 - \ln 5) = \ln 3 + 2\ln 5$

$\Rightarrow x = \frac{\ell n3 + 2\ell n5}{2\ell n3 - \ell n5}$ ⟵ **intersection point.**

142. Determine the interval for x such that

a) $\log x = 0.0212$ c. $\log x = 4.3874$

b) $\log x = -0.3010$ d. $\log x = -3.6021$

Explanation: $\log x = m \Leftrightarrow x = 10^m$

a) $\log x = 0.0212 \Leftrightarrow x = 10^{0.0212}$

$\Rightarrow 10^0 < 10^{0.0212} < 10^1 \Leftrightarrow 1 < x < 10$

$\therefore$ The solution set is $\{x: 1 < x < 10\}$

b) $\log x = -0.3010 \Leftrightarrow x = 10^{-0.3010}$

we know that $-0.3010 = (-0.3010 + 1) - 1$

$= 0.6990 + (-1)$

$\Rightarrow x = 10^{-0.3010} = 10^{0.6990+(-1)} = (10^{0.6990})(10^{-1})$

$\Rightarrow 10^{-1} < (10^{-0.6990})(10^{-1}) < 10^0$

$\therefore$ The solution set is $\{x: 0.1 < x < 1\}$

c) $\log x = 4.3874 \Leftrightarrow x = 104.3874 \quad \Rightarrow 10^4 < 10^{4.3874} < 10^5$

$\therefore$ The solution set is $\{x: 10{,}000 < x < 100{,}000\}$

d) $\log x = -3.6021$ the same as $\log x = 0.3979 + (-4)$

$\Rightarrow x = 10^{0.3979}(10^{-4}) \Leftrightarrow 10^{-4} < 10^{-3.602} < 10^{-3}$

$\therefore$ The solution set is $\{x: 10^{-4} < x < 10^{-3}\} = (x: 0.0001 < x < 0.001\}$

143. **Simplify:** $\frac{\log_{10}^{\left(\frac{8}{27}\right)} - \log_{10}^{\left(\frac{64}{27}\right)}}{\log_{10}^{24} - \log_{10}^{6}} \; \frac{\log_{10}^{\left(\frac{8}{27}\right)} - \log_{10}^{\left(\frac{64}{27}\right)}}{\log_{10}^{24} - \log_{10}^{6}}$

Solution: **We known that** $\log x - \log y = \log\left(\frac{x}{y}\right)$

$$\Leftrightarrow \frac{\log_{10}^{\left(\frac{8}{27}\cdot\frac{27}{54}\right)}}{\log_{10}^{\left(\frac{24}{6}\right)}} = \frac{\log_{10}^{\left(\frac{8}{64}\right)}}{\log_{10}^{4}} = \log_4^{\frac{1}{8}} \longleftarrow \frac{\log k}{\log m} = \log_m^k$$

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$$= -\log_4^8 = \frac{-3}{2}\log_2^2 = \frac{-3}{2}$$

147. **Evaluate,** $\dfrac{\log\left(\frac{25}{24}\right)+\log\left(\frac{216}{75}\right)}{\log 6 - \log 54}$

***Solution*: we know that logx + log y = logxy**

$$\Leftrightarrow \frac{\log\left(\frac{25}{24}\cdot\frac{216}{75}\right)}{\log\left(\frac{6}{54}\right)} = \frac{\log\left(\frac{9}{3}\right)}{\log\left(\frac{1}{9}\right)} = \frac{\log 3}{\log\left(\frac{1}{9}\right)} = \log_{\frac{1}{9}}^3$$

$$\Rightarrow -\frac{1}{2}\log_3^3 = \frac{-1}{2}$$

147. A machine purchased for 200,000 birr depreciate at the rate of 5% per annum which is calculated on its value. If the machine was bought at the beginning of the year 2002 E.C, then what will be its value at the end of 2006 E.C

(Use log2 = 0.3010, log 9.5 = 0.9777 log 1.55 = 0.1895).

A. Birr 185,000 C. Birr 175,000

B. Birr 155,000 D. Birr 135,000

***Solution*:** $V(t) = V_0(1 - \text{rater})^t$, $t = 5$, $r = 5\% = 0.05$

$= 200{,}000\,(1 - 0.05)^5$ ⟵ **taking log both side**

$\Rightarrow \log V(t) = \log(200{,}000) + \log(0.95)$

$= \log(2\times10^5) + 5\log(9.5\times10^{-1})$

$= \log 2 + 5 + 5(\log 9.5 + (-1))$

$= 0.3010 + 5 + 5(0.9777) + -5$

$\log V(t) = 5.1895$

$\Rightarrow V(t) = 10^{5.1895} = (10^5)(10^{0.1895})$

$= (10^5)(1.55) = 155000,$ **∴ Answer: B**

148. Let $f(x) = -3^x + 1$, then

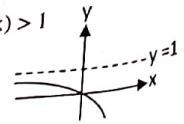
a) For what value of x,

i) $0 \le f(x) < 1$ ii) $f(x) < 0$ iii) $f(x) > 1$

b) What is the range of f

***Solution*:** we know that: $-3^x + 1 = -(3^x) + 1$

a) i) $0 \le f(x) < 1$ if $x \le 0$



ii) $f(x) < 0$ if $x > 0$

iii) No x value that makes $f(x) > 1$ for all x value $f(x) < 1$

b) the range = $(-\infty, 1)$

149. Solve a) $\log_x^2 = \log_{16}^x$ b) $\log_x^{36} = \log_6^x$

Solution: we know that: $\log_y^x = \dfrac{\log_c^x}{\log_c^y}$

$$\Rightarrow \log_x^2 = \frac{\log_2^x}{\log_x^{16}} \Leftrightarrow \log_x^2 = \frac{\log_2^x}{4} \Leftrightarrow 4\log_x^2 = \log_2^x$$

$$\Rightarrow 4\log_x^2 = \frac{1}{\log_x^2} \Leftrightarrow \left(\log_x^2\right)^2 = \frac{1}{4}$$

$$\Rightarrow x = 4 \text{ or } x = -\frac{1}{4}$$

b) **Exercise left for you**

150. If $4^{-x} = \dfrac{1}{32}$, then $\log_{\left(\frac{4}{25}\right)}^x =$ _____

A. 2 B) $-\dfrac{1}{2}$ C) $\dfrac{5}{2}$ D) $\dfrac{-5}{2}$

Solution: $4^{-x} = \dfrac{1}{32} \Leftrightarrow 2^{-2x} = 2^{-5} \Leftrightarrow x = \dfrac{5}{2}$

Let $\log_{\left(\frac{4}{25}\right)}^5 = y \Leftrightarrow \left(\dfrac{4}{25}\right)^y = \dfrac{5}{4} = y$

$$\Rightarrow \left(\frac{5}{2}\right)^{-2y} = \left(\frac{5}{2}\right)^1 \Leftrightarrow -2y = 1 \Rightarrow y = \frac{-1}{2}$$ **∴ Answer: B**

151. **Determine** the difference between

a) Log763 and log0.763

b) Log6480 and log 6.48

c) Log25.4 and log 2.54

***Solution*:**

a) $\log 763 - \log(0.763) = \log 763 - \log(763 \times 10^{-3})$

$= \log 763 - \log 763 + 3 = 3$

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b) $\log 6480$ and $\log 6.48 = \log(6.48\times10^3) - \log 6.48$
$= \log 6.48 + 3 - \log 6.48 = 3$

c) **Exercise left for you**

152. The solution set of $10^{y+3} = e^{y+6}$ in terms of natural logarithms.

A. $\frac{6+3\ell n10}{\ell n10-1}$ B. $\frac{6-3\ell n10}{\ell n10-1}$ C. $\frac{1+3\ell n10}{\ell n10-1}$ D. $\frac{6-3\ell n10}{\ell n10+1}$

Solution: Take natural log both side

$$\Leftrightarrow 10^y \cdot 10^3 = e^y \cdot e^6 \Rightarrow \left(\frac{10}{e}\right)^y = \frac{e^6}{10^3}$$

$$\Rightarrow \ln\left(\frac{10}{e}\right)^y = \ell n\left(\frac{e^6}{10^3}\right) \Rightarrow y = \frac{6-3\ell n10}{\ell n10-1}, \quad \therefore \text{Answer: B}$$

153. Suppose you invested 5000 birr into an account where the interest was compounded semiannually. If after 4 years you had 8000birr. what was the interest rate? **(Use log 1.6 = 0.2040, log (1.06) = 0.0256)**

Explanation: $A = p\left(1+\frac{r}{n}\right)^{nt}$, t = 4, n = 2 (semiannually)

$$\Rightarrow 8000 = 5000\left(1+\frac{r}{2}\right)^{2t} \Rightarrow \frac{8}{5} = \left(1+\frac{r}{2}\right)^8$$

$$\Rightarrow \log\left(\frac{8}{5}\right) = 8\log\left(1+\frac{r}{2}\right)$$

$$\Rightarrow \log\left(1+\frac{r}{2}\right) = \frac{\log(1.6)}{8} = \frac{0.2040}{8} = 0.0256$$

$$\Rightarrow \text{antilog}\left(\log\left(1+\frac{r}{2}\right)\right) = \text{antilog}\,(0.0256)$$

$$\Rightarrow 1+\frac{r}{2} = 1.06 \longleftarrow \text{since } \log 1.06 = 0.0256$$

$$\Rightarrow \frac{r}{2} = 1.06 - 1 = 0.06 \Rightarrow r = 0.12$$

Therefore, the interest rate, r = 12%

1. Let $f(x) = \log_a^x$
   a) On what condition f is increasing.
   b) In what condition **f** is decreasing
   c) Discuss all the case
      i) $f(x) > 0$ ii) $f(x) < 0$

Explanation:

a) $f(x) = \log_a^x$ is increasing if $a > 1$

b) $\log_a^x$ is decreasing if $0 < a < 1$

c) case(i) **if a > 1**
   i) $f(x) < 0$ when $0 < x < 1$
   $f(x) > 0$ when $x > 1$
   Case (ii) **if $0 < a < 1$**
   ii) $f(x) < 0$ when $x > 1$
   $f(x) > 0$ when $0 < x < 1$

Supplementary Exercise

1. Simplify each of the following

a) $3^{\sqrt{2}} \times 3^{\sqrt{32}}$

b) $\frac{2^{\sqrt{3}} \times 4^{-\sqrt{3}} \times 8^{\sqrt{3}}}{2^{\sqrt{27}}}$

c) $\frac{3^{\sqrt{5}} \times 3^{\sqrt{20}}}{27^{\sqrt{5}}}$

d) $\sqrt[4]{3} \times \sqrt[4]{3} \times \sqrt[4]{3} \times \sqrt[4]{3} \times \sqrt[4]{3}$

2. Solve each of the following

a) $4^x = \frac{1}{16}$

b) $3^{2-x} = \sqrt[4]{27}$

c) $\left(\frac{3}{4}\right)^x = \sqrt[5]{\frac{64}{27}}$

d) $\left(\frac{1}{3}\right)^{x+2}\left(\frac{1}{9}\right)^x = 3$

e) $\sqrt[3]{128} = 4^{2x}$

f) $\left(\frac{2}{3}\right)^{2x-1} = 1$

g) $4^x - 6(2^x) - 16 = 0$

3. Solve each of the following
   a) $1 < b^x < 2$, for $0 < b < 1$

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b) $b^x > 2$, for $0 < b < 1$
c) $0 < b^x < 1$, for $b > 1$
d) $b^x > 1$ for $b > 1$
e) $b^x > 0$ for $b \neq 1$, $b > 0$
f) $4^x < 3^x$
g) $(¼)^x > (½)^x$

4. Identify the following features of the graph
i) Asymptote ii) Intercept
iii) Range iv) Increasing or decreasing
a) $y = 5^{-x} - 3$ c) $y = 5^x - 3$ e) $y = 4^{x-2}$
b) $y = \left(\frac{3}{2}\right)^{-x} + 2$ d) $y = \left(\frac{3}{2}\right)^{2} + 2$ f) $y = -3^{-x} + 2$

5. Find
a) $\log_{\sqrt{27}}^{9}$ b) $\log_{10}^{0.01}$ c) $\log_{2}^{(\log_4 \sqrt{2})}$ d) $\log_{16}^{8}$
e) $\log_{100}^{\sqrt[3]{100}}$ f) $\log_{100}^{\sqrt[3]{0.1}}$

6. Identify the characteristic and mantissa of the logarithm of each of the following
a) 0.000301 b) 406 c) 2.2 d) 5170
e) 0.25 f) 6 g) 43

7. Solve the logarithm equation
a) $\log_2^{(x-3)} = 4$ d) $\ln x = 3$
b) $\log_4^x - \log_4^{(x-1)} = \frac{1}{2}$ e) $\log_{10}^{x^2} = 6$
c) $\log_3^x + \log_3^{(x^2-8)} = \log_3^{8x}$ f) $\ln(x+4) = 1$

8. Let $\log_b^2 = A$, $\log_b^3 = B$ and $\log_b^5 = C$, then find
a) $\log_b^6$ b) $\log_b^{1.5}$ c) $\log_2^5$ d) $\log_5^6$ e) $\log_b^{0.006}$

9. Find the domain of each of the following
a) $f(x) = \log_3^{(4-x)}$ b) $f(x) = \log_2^{(4-x^2)}$

10. **Simplify**: a) $e^{\frac{3-\ln\left(\frac{3}{2}\right)}{2}}$ b) $3^{4-\log_3^6}$

c) $8^{(1-2\log_2^3)}$ d) $\dfrac{\log\left(\frac{18}{19}\right) - \log\left(\frac{54}{19}\right)}{\log 36 - \log 4}$

11. Given $\log_5^3 = 0.6826$ and $\log_5^6 = 1.1133$
Find the following
a) $\log_5^2$ b) $\log_5^9$ c) $\log_4^{25}$ d) $\log_{36}^5$

12. If $8^{-1} = \frac{1}{64}$, then $\log_{32}^x =$ _____

13. If 2000 Birr is invested at 6% compounded quarterly, in how many years will the investment
use **log1.5 = 0.1761, log(1.015) = 0.0086**
a) to reach 3000 birr b) to double

14. Solve (Use **log2 = 0.3010, log 3 = 0.4741**)
a) $4^x = 18$ b) $6^x = 24$ c) $4^{3x+1} = 6^{x+2}$

15. Solve: a) $\log_3^x = \log_x^{81}$ b) $\log_x^2 = \log_{64}^x$

16. Given log2.62 = 0.4183, log 5.6 = 0.7482
The find a) antilog 0.4183 b) antilog (2.4183)
c) antilog (–2.5817) d) antilog (– 2.518)
e) log 262 f) log (0.00262) g) log(0.56)

17. **Evaluate:**
a) $\log_2^{(x+\sqrt{x^2-8})} + \log_2^{(x-\sqrt{x^2-8})}$

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$\frac{7}{3}$ is zeros of multiplicity 2

$\sqrt{3}$ is zeros of

7. a) $f(x) = -4(x+1)(x-2)($ c) $f(x) = 3(x(x+4)(x-3))$

b) $f(x) = -\frac{1}{3}(x+5)(x-2)(x-4)$

8. $f(x) = 3x^3(x-2)^2(x+2)^2$ 9. a) $x = 1, \frac{1}{2}, -3$ b) $x = \frac{2}{3}$

10. a) Rises to the left and falls to the right. d) rises to the left and falls to the right
b) falls to the left and rises to the right e) rises to the left and the right
c) falls to the left and rises to the right f) falls to the left and the right.

11. a) crosses at (-1, 0),(1, 0) and (3, 0)
b) crosses at (7, 0), touch but does not cross at (3, 0)
c) crosses at (1, 0), touch but does not cross at $\left(\frac{3}{2}, 0\right)$
d) crosses at (0, 0), touch but does not cross at (3, 0)

Answer key Unit 8

1. a) $243^{\sqrt{2}}$ b) $2^{-\sqrt{3}}$ c) 1 d) $3\sqrt[4]{3}$

2. a) $x = -2$, b) $x = \frac{5}{4}$ c) $x = \frac{-5}{3}$ d) $x = -1$ e) $x = \frac{7}{6}$ f) $x = \frac{1}{2}$ g) $x = 3$

3. a) For all $x > 0$ d) for all $x > 0$
b) for all $x < 0$ d) for all real number $x \in (-\infty, \infty)$
c) For all $x < 0$ f) for all $x < 0$ g) for all $x \in (-\infty, 0)$

4. a) i) $y = -3$ ii) $(0, -2)$ iii) $(-3, \infty)$ iv) decreasing
b) i) $y = 2$ ii) $(0, 3)$ iii) $(2, \infty)$ iv) decreasing
c) i) $y = -3$ ii) $(0, -2)$ iii) $(-3, \infty)$ iv) Increasing
d) i) $y = 2$ ii) $(0, 3)$ iii) $(2, \infty)$ iv) Increasing
e) i) $y = 0$ ii) $(0, \frac{1}{16})$ iii) $(0, \infty)$ iv) Increasing
f) i) $y = 2$ ii) $(0, 1)$ iii) $(-\infty, 2)$ iv) Increasing

5. a) $\frac{4}{3}$ b) -2 c) -2 d) $\frac{3}{4}$ e) $\frac{1}{5}$ f) $\frac{-1}{6}$

6. Characteristics:- a) -4 b) 2 c) 0 d) 3 e) -1 f) 0 g)1
Mantissa: a) log3.01 b) log4.06 c) log2.2 d) log5.17 e) log f) log6

7. a) $x = 19$, b) $x = 2$ c) $x = 4$ d) $x = e^3$
e) $x = 10^3$ f) $x = e - 4$

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8. a) A + B b) B – A C) $\frac{C}{A}$ D) $\frac{A+B}{C}$ E) -2A +B–3C

9. a) $(-\infty, 4)$ b) (-2, 2)

10. a) $\sqrt{\frac{2}{3}e^3}$ b) $\frac{27}{2}$ c) $\frac{8}{729}$ d) $-\frac{1}{2}$

11. a) 0.4307 b) 1.3652 c) 2.322 d) 0.449

12. a) $\frac{1}{5}$ 13) a) 5 years b) 9 years

14. a) $\log_4^x = \log 18 \Leftrightarrow x \log 4 = 2\log 3 + \log 2$

$$\Rightarrow x = \frac{2\log 3 + \log 2}{2\log 2} = \frac{2(0.4771) + 0.3010}{2(0.3010)} \approx 2.09$$

a) $x = \frac{3\log 2 + \log 3}{\log 2 + \log 3}$ c) x = 0.92

15. a) x = 9 or x = $\frac{1}{9}$ b) $\log_x^2 = \frac{\log_x^2}{6} \Leftrightarrow 6\log_x^2 = \frac{1}{\log_x^2}$

$$\Rightarrow \left(\log_x^2\right)^2 = \frac{1}{6} \Rightarrow x = 2^{\sqrt{6}} \text{ or } x = 2^{-\sqrt{16}}$$

16. a) 2.62 b) 262 c) 0.00262 d) 0.056 e) 2.4183
f) –2.5817 g) – 0.2518

17. a) $\log_2^8 = 3$

Answer key Unit 9

1. a) $\left(\frac{-4}{7}, \infty\right)$ b) (–6, 2)

2. a) $(-\infty,-2) \cup (-2,\infty)$ b) $(-\infty,-2) \cup (2,\infty)$
c) $(-\infty,-3) \cup [4,\infty)$ d) $(-\infty,0) \cup (0,1) \cup (1,\infty)$

3. a) $IR \setminus \left\{-\frac{1}{2}\right\}$ b) IR c) $IR \setminus \{-2,2\}$ d) IR e) $IR \setminus \{-2, -1\}$

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UNIT NINE

Solving Inequalities

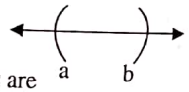
Using interval notation

Set of real numbers may be described by **interval notation.**

➢ Let a and b represent two real numbers with a < b.

i) A closed interval, denoted by **[a,b]** consists of set of all real numbers x for which {x: a ≤ x ≤ b}.
(Equivalent to a ≤ x and x≤ b)

ii) An open interval, denoted by **(a,b)**, consists of the set of all real numbers x, for which {x: a < x < b}.
(Equivalent to a < x and x < b).
Graphically.



iii) half – open, half – closed, intervals are
1) (a,b] consists of the set of all real numbers x for which {x:a < x ≤ b}
2) [a, b) consists of the set of all real number x for which {x:a ≤ x < b}

- The symbol **(∞) read as "infinity"** is not real number, but used to indicate **unboundedness** in the positive direction
- The symbol **(-∞)** read as **"negative infinity'** also not real number, but used to indicate **unboundedness** in the negative direction:

Using the symbols ∞ and –∞

i) **[a, ∞)** *is the same as* set of all x for which {x: a≤ x < ∞}
- **Equivalent to** {x: x ≥ a}

Graphically

ii) **(a, ∞)** *is the same as* set of all x for which {x: a< x < ∞)
- **Equivalent to** {x: x > a}

ii) **(-∞,a]** *is the same as* {x: –∞< x ≤ a)
- **Equivalent to** {x: x ≤ a}

Remember that: Two inequalities linked by a connective" **and "or"**
"or"

i) **"And"** means that both inequalities must be satisfied at the same time.
- This relationship is sometime called the **intersection**, where both inequalities hold true.

ii) **"Or"** means that either inequality must be satisfied, not necessarily both. (This relationship is called **union**).

Illustrative Example

1. Solve and write each using interval notation.

a) $3-2x<5$

b) $4x+7\geq 2x-3$

c) $-5<3x-2<1$

d) $-1\leq\frac{3-5x}{2}\leq 9$

Solution: Here

a) $3-2x<5$ **Explanation:**

$\Rightarrow 3-3-2x<5-3$ ⟵ subtract 3 from both side

$\Rightarrow -2x<2$

$\Rightarrow \frac{-2x}{-2}>\frac{2}{-2}$ ⟵ Dividing both side by (–2)

$\Rightarrow \{x: x>-1\}=(-1,\infty)$

b) $4x+7\geq 2x-3$ **Explanation**

$\Rightarrow 4x-2x+7-7\geq 2x-3-7$ ← subtract 2xand 7 from each side

$\Rightarrow 2x\geq -10$

$\Rightarrow \{x: x\geq -5\}=[-5,\infty)$

c) $-5<3x-2<1$ **Explanation:**

$\Rightarrow -5+2<3x-2+2<1+2$ ⟵ add (2) from each part

$\Rightarrow -3<3x<3$ ⟵ Divide each part by 3.

$\Rightarrow \{x: -1<x<1\}=(-1,1)$

d) $-1\leq\frac{3-5x}{2}\leq 9$ **Explanation:**

$\Rightarrow 2(-1)\leq 2\left(\frac{3-5x}{2}\right)\leq 2(9)$ ⟵ Multiply each part by 2 to

$\Rightarrow -2\leq 3-5x\leq 18$ remove denominator Subtract

$\Rightarrow -2-3\leq 3-3-5x\leq 18-3$ ⟵ 3from each part to isolate x

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$\Rightarrow -5 \le 5x \le 15$

$\Rightarrow \frac{-5}{-5} \ge \frac{-5x}{-5} \ge \frac{15}{-5}$

$\Rightarrow 1 \ge x \ge -3$ ⟵ Reverse the order

$\{x: -3 \le x \le 1\} = [-3,1]$

2. **Find a and b**

a) If $-3 < x < 2$, then $a < x - 6 < b$.

b) If $0 < x < 4$, then $a < 2x + 3 < b$

Solution: a) change the middle part from x to x – 6

It is given $-3 < x < 2$ so that:

$\Rightarrow -3 - 6 < x - 6 < 2 - 6$ ⟵ Subtract 6 to each part

$\Rightarrow -9 < x - 6 < -4$ $\therefore a = -9$ and $b = -4$

b) Change the middle part x to 2x +3

$0 < x < 4$

$\Rightarrow 0 < 2x < 8$ ⟵ Multiply each part by (2)

$\Rightarrow 0 + 3 < 2x + 3 < 8 + 3$ ⟵ Add 3 to each part

$3 < 2x + 3 < 11$ $\therefore a = 3$ and $b = 11$

3. If we describe by interval notation $\{x: 4x - 6 < x < 7\}$

A) ϕ B) (2 7) C) $(-\infty,7)$ D) $(-\infty,2)$

Solution: $4x - 6 < x < 7$ is the same as

$4x - 6 < x$ and $x < 7$

$\Rightarrow 4x - x < 6$ and $x < 7$ $\Rightarrow 3x < 6$ and $x < 7$

$\Rightarrow x < 2$ and $x < 7$, the intersection of both inequalities

$\{x: x < 2\}$ that is all x in the interval $(-\infty,2)$ **∴Answer: D**

4. **Describe each of the following using** intervals:

a) $\{x: x \in R/\{-1\}\}$ b) $\{x: x \in R$ and $x \notin [-2,1)\}$

c) $\{x: x \in R$ and x (–3,4)$\}$ d) $\{x: x \in R/(-\infty,-1) \cup (3,\infty)$

Solution:

a) $\{x: x \in IR/\{-1\}\} = (-\infty,-1) \cup (-1,\infty)$

b) $\{x: x \in R$ and $x \notin [-2,1)\} = (-\infty,-2) \cup [1,\infty)$

c) $\{x: x \in R$ and x $(-3,4)\} = (-\infty,-3) \cup [4,\infty)$

d) $\{x: x \in R/(-\infty,-1) \cup (3,\infty) = [-1,3]$

5. Let $M = \{x \in R: x \notin (-1,1]\}$. Which one of the following intervals is the correct description of set M?

A. $(-\infty,1) \cup (1,\infty)$ C. $(-\infty,-1] \cup (1,\infty)$

B. $(-1,1]$ D. $(-\infty,1] \cup (-1,\infty)$

Solution: $\{x: x \notin (-1,1]\}$ means the set of all real number x, except between –1 and 1, inclusive 1. i.e,

$= (-\infty,-1] \cup (1,\infty)$ **∴Answer: C**

5. Which one of the following is the interval notation of set M, where $M = \{x \in R: x < -1 \text{ or } x \ge 2\}$. . . EGSECE 2005

Solution: $(-\infty,-1) \cup [2,\infty)$

Equation involving absolute value

Definition: If x is an algebraic expression and **a** is positive real number then

i) $|x| = \begin{cases} x \text{ if } x \ge 0 \\ -x \text{ if } x < 0 \end{cases}$

ii) $|x| = a$ **is equivalent to** $x = a$ or $x = -a$

iii) $|x| = a$ has **no *Solution*** if $a < 0$

Example: $|x| = -6$, has no ***Solution*** because $|x| \ge 0$, for all $x \in IR$.

Illustrative Example

6. Solve

a) $|2x + 4| = 8$ b) $|3x - 1| = -2$ c) $|x^2 - 8| = 1$

d) $|1 - 4x| + 8 = 13$ e) $|x^2 + 3x - 2| = 2$ f) $4 - |2x| = 3$

Solution: **Remember that** $|x| = a \Leftrightarrow x = a$ Or $x = -a$

a) $|2x + 4| = 8 \Leftrightarrow 2x + 4 = 8$ Or $2x + 4 = -8$

$\Rightarrow 2x = 4$ Or $2x = -12$ $\therefore x = 2$ Or $x = -6$

b) $|3x - 1| = -2$ has no ***Solution*** because **absolute value is non – negative**

c) $|x^2 - 8| = 1 \Leftrightarrow x^2 - 8 = 1$ Or $x^2 - 8 = -1$

$\Rightarrow x^2 = 9$ Or $x^2 = 7$ $\therefore x = -3,3, -\sqrt{7}, \sqrt{7}$

d) $|1 - 4x| + 8 = 13 \Leftrightarrow |1 - 4x| = 5$

$\Leftrightarrow 1 - 4x = 5$ Or $1 - 4 = -6$

$\Rightarrow -4x = 4$ Or $-4x = -6$ $\therefore x = -1$ Or $x = \frac{3}{2}$

e) $|x^2 + 3x - 2| = 2 \Leftrightarrow x^2 + 3x - 2 = 2$ Or $x^2 + 3x - 2 = -2$

$\Rightarrow x^2 + 3x - 4 = 0$ Or $x^2 + 3x = 0$

$\Rightarrow (x - 1)(x + 4) = 0$ Or $x(x + 3) = 0$ $\therefore x = -4, -3, 0, 1$

f) $4 - |2x| = 3 \Leftrightarrow -|2x| = -1 \Leftrightarrow |2x| = 1$

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$\Rightarrow 2x = 1$ Or $2x = -1 \therefore x = \frac{1}{2}$ Or $x = \frac{-1}{2}$

7. **Solve:**

a) $|x + 2| = |2x - 6|$ b) $|x + 1| = 2x - 3$ c) $|x^3| = x$

Solution: $|x+2| = |2x - 6| \Leftrightarrow x + 2 = 2x - 6$ Or $x + 2 = -2x + 6$

$\Rightarrow -x = -8$ Or $3x = 4 \quad \therefore x = 8$ Or $x = \frac{4}{3}$

b) $|x+1| = 2x - 3 \Leftrightarrow x + 1 = 2x - 3$ Or $x + 1 = -2x + 3$

$\Rightarrow -x = -4$ Or $3x = 2 \quad \therefore x = 4$ Or $x = \frac{2}{3}$

However, $2x - 3 \geq 0 \Rightarrow x \geq \frac{3}{2}$

Since $x = \frac{2}{3}$ makes $2x - 3$ negative

$\therefore$ The ***Solution*** is $x = 4$ only.

c) $|x^3| = x \Leftrightarrow x^3 = x$ Or $x^3 = -x$ **Provided that $x \geq 0$**

$\Rightarrow x^3 - x = 0$ Or $x^3 + x = 0$ Provided that $x \geq 0$

$\Rightarrow x(x^2-1)=0$ Or $x(x^2+1)=0$, Provided that $x \geq 0$

$\therefore$ S.S = {0,1}

Inequality involving Absolute value

If a is positive number and if x is an algebraic expression, then

i) $|x| \leq a$ **is the same as** $-a \leq x \leq a$

ii) $|x| < a$ **is the same as** $-a < x < a$

Geometrically: If $|x| < a$, then all points in the graph are **Less than a units from 0**

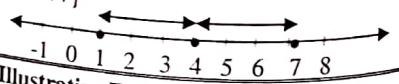
- If $|x - 4| < 3$, then all points in the graph are **less than 3** unit from 4

$\Rightarrow |x - 4| < 3 \Leftrightarrow -3 < x - 4 < 3 \quad \Rightarrow -3 < x - 4$ and $x - 4 < 3$

$\Rightarrow -3 + 4 < x$ and $x < 3 + 4 \quad \Rightarrow 1 < x$ and $x < 7$

$\therefore$ S.S = {x: 1 < x < 7}

Geometrically:



Illustrative Example

8. Solve an inequality involving absolute value

a) $|x| < 3$ b) $|x| < 0$ c) $|x| \leq 0$ d) $|x - 4| < 0$

e) $|x - 4| \leq 0$ f) $|x^2+x - 6| \leq 0$ g) $|x^2 + 3x + 1| < 0$

h) $|x^2 - 9| \leq 0$ i) $|x^2 - 5x + 6| \leq 0$

Solution: Use $|x| < a \Leftrightarrow -a < x < a$.

a) $|x| < 3 \Leftrightarrow -3 < x < 3$

b) $|x| < 0 \Leftrightarrow -0 < x < 0 \Leftrightarrow 0 < x < 0$ is false

$\therefore$ S.S = ϕ (absolute value is never less than 0

c) $|x| \leq 0 \Leftrightarrow 0 \leq x \leq 0$ is true if $x = 0 \quad \therefore$ S.S = {0}

d) $|x - 4| < 0$, the value of $|x - 4|$ will never less than 0

$\Rightarrow |x - 4| < 0 \Leftrightarrow -0 < x - 4 < 0$

$\Rightarrow 4 < x < 4$ is false $\quad \therefore$ S.S = ϕ

e) $|x - 4| \leq 0 \Leftrightarrow -0 \leq x - 4 \leq 0$

$\Rightarrow 4 \leq x \leq 4$ is true when $x = 4 \therefore$ S.S = {x: x = 4}

f) $|x^2+x - 6| \leq 0 \Leftrightarrow 0 \leq x^2 + x - 6 \leq 0$

It is true if $x^2 + x - 6 = 0$

$\Rightarrow (x - 2)(x + 3) = 0 \quad \therefore$ S.S = {x: x = 2, –3}

g) $|x^2 + 3x + 1| < 0$ is false, since

absolute value is never less than 0. $\therefore$ S.S = ϕ

h) $|x^2 - 9| \leq 0 \Leftrightarrow 0 \leq x^2 - 9 \leq 0$

It is true if $x^2 - 9 = 0$

$\Rightarrow (x + 3)(x - 3) = 0 \quad \therefore$ S.S = {–3, 3}

i) Exercise left for you

9. Solve:

a) $|2x + 4| \leq 5$ b) $|1 - 5x| < 4$ c) $|2x| < -1$

d) $|x^3 - 1| \leq 7$ e) $|x^5| \leq 32$ f) $|3x - 4| < -6$

Solution: **Use $|x| \leq a \Leftrightarrow -a \leq x \leq a$**

a) $|2x + 4| \leq 5 \Leftrightarrow -5 \leq 2x + 4 \leq 5$

$\Rightarrow -5 \leq 2x + 4$ and $2x+4 \leq 5$

$\Rightarrow -9 \leq 2x$ and $2x \leq 1$

$\Rightarrow -\frac{9}{2} \leq x$ and $x \leq \frac{1}{2} \quad \therefore$ S.S = $\{x: -\frac{9}{2} \leq x \leq \frac{1}{2}\}$

b) $|1 - 5x| < 4 \Leftrightarrow -4 < 1 - 5x < 4$

$\Rightarrow -4 - 1 < -5x$ and $-5x < 4 - 1$

$\Rightarrow -5 < -5x$ and $-5x < 3$

$\Rightarrow 1 > x$ and $x > -\frac{3}{5} \quad \Rightarrow -\frac{3}{5} < x < 1 \therefore$ S.S = $\{x: -\frac{3}{5} < x < 1\}$

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c) $|2x| < -1$ is false, because there is no number whose absolute value is less than -1 $\therefore$ S.S $= \phi$

d) $|x^3 - 1| \le 7 \Leftrightarrow -7 \le x^3 - 1 \le 7$

$\Rightarrow -7+1 \le x^3 \le 7+1$

$\Rightarrow -6 \le x^3 \le 8 \Leftrightarrow \sqrt[3]{-6} < x \le \sqrt[3]{8}$ $\therefore$ S.S $= \{x: \sqrt[3]{-6} \le x \le 2\}$

e) $|x^5| \le 32 \Leftrightarrow -32 \le x^5 \le 32$

$\Rightarrow \sqrt[5]{-32} \le x \le \sqrt[5]{32}$ $\therefore$ S.S $= \{x: -2 \le x \le 2\}$

f) $|3x - 4| < -6$ is false, $\therefore$ S.S $= \phi$

Definition: If $a > 0$ and x is an algebraic expression, then:

i) $|x| > a$ **is equivalent to** $x < -a$ Or $x > a$

ii) $|x| \ge a$ **is equivalent to** $x \le -a$ Or $x \ge a$

Geometrically: $|x| > a$ means all points on the graph are more than a unit from the origin.

Example a) $|x - c| > k$ *means all points on the graph are more than k unit from the point C*

b) $|x - 2| > 3$ means all points on the graph are more than 3 unit from the point 3.

Illustrative Example

10. **Solve each inequality**

a) $|x| > 3$ b) $|2x + 6| > 4$ c) $\left|\frac{2}{3}x - \frac{1}{9}\right| > \frac{1}{3}$

d) $|6 - 2x| + 3 > 8$ e) $|x^3 - 1| > 2$ f) $|2x - 5| > -2$

Solution: **Use**: $|x| > a \Leftrightarrow x < -a$ Or $x > a$

a) $|x| > 3 \Leftrightarrow x < -3$ Or $x > 3$

$\therefore$ S.S $= \{x: x < -3$ Or $x > 3\} = (-\infty, -3) \cup (-3, -\infty)$

b) $|2x + 6| > 4 \Leftrightarrow 2x + 6 < -4$ Or $2x + 6 > 4$

$\Rightarrow 2x < -10$ Or $2x > -2$

$\Rightarrow x < -5$ Or $x > -1$

$\therefore$ S.S $= \{x: x < -5;$ Or $x > -1\} = (-\infty, -5) \cup (-1, \infty)$

c) $\left|\frac{2}{3}x - \frac{1}{9}\right| > \frac{1}{3} \Leftrightarrow \frac{2}{3}x - \frac{1}{9} < \frac{-1}{3}$ Or $\frac{2}{3}x - \frac{1}{9} > \frac{1}{3}$

$\Rightarrow \frac{2}{3}x < \frac{-1}{3} + \frac{1}{9}$ Or $\frac{2}{3}x > \frac{1}{3} + \frac{1}{9}$

$\Rightarrow (9)\frac{2}{3}x < 9\left(\frac{-1}{3} + \frac{1}{9}\right)$ Or $9\left(\frac{2}{3}x\right) > 9\left(\frac{1}{3} + \frac{1}{9}\right)$

$\Rightarrow 6x < -3 + 1$ Or $6x > 3 + 1$

$\Rightarrow x < \frac{-1}{3}$ Or $x > \frac{2}{3}$

$\therefore$ S.S $= \left\{x: x < \frac{-1}{3} \text{ Or } x > \frac{2}{3}\right\} = \left(-\infty, \frac{-1}{3}\right) \cup \left(\frac{2}{3}, \infty\right)$

d) $|6 - 2x| + 3 > 8 \Leftrightarrow |6 - 2x| > 5$

$\Rightarrow 6 - 2x < -5$ Or $6 - 2x > 5$

$\Rightarrow -2x < -5 - 6$ Or $-2x > 5 - 6$

$\Rightarrow x > \frac{11}{2}$ Or $x < -\frac{1}{2}$

$\therefore$ S.S $= \left\{x: x < \frac{-1}{2} \text{ Or } x > \frac{11}{2}\right\} = \left(-\infty, \frac{-1}{2}\right) \cup \left(\frac{11}{2}, \infty\right)$

e) $|x^3 - 1| > 2 \Leftrightarrow x^3 - 1 < -2$ Or $x^3 - 1 > 2$

$\Rightarrow x^3 < -1$ Or $x^3 > 3$

$\Rightarrow x < \sqrt[3]{-1}$ Or $x > \sqrt[3]{3}$

$\therefore$ S.S $= \{x: x < -1$ Or $x > \sqrt[3]{3}\ \} = (-\infty, -1) \cup (\sqrt[3]{3}, \infty)$

Special case of Absolute value

11. Solve each inequality

a) $|x| > 0$ b) $|x| \ge 0$ c) $|x - 1| > 0$ d) $|x - 1| \ge 0$

e) $|x + 2| > 0$ f) $|x + 2| \ge 0$ g) $|x^2 - 9| > 0$ h) $|x^2 + 5x + 6| > 0$

Solution: Here

a) $|x| > 0$ is true for all $x \in R \setminus \{0\}$

$\therefore$ S.S $= (-\infty, 0) \cup (0, -\infty)$

b) $|x| \ge 0$ is true for all real number x

$\therefore$ S.S $= (-\infty, \infty)$

c) $|x - 1| > 0$ is true for all real number x, except $x = 1$

$\therefore$ S.S $= \{x: x \ne 1\} = (-\infty, 1) \cup (1, \infty)$

d) $|x - 1| \ge 0$ is true for all $x \in R$

$\therefore$ S.S $= (-\infty, \infty)$

e) $|x + 2| > 0$ is true for all $x \ne -2$

$\therefore$ S.S $= (-\infty, -2) \cup (-2, \infty)$

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f) $|x+2| \geq 0$ is true for all $x \in R$ $\therefore$ S.S = $(-\infty, \infty)$

g) $|x^2 - 9| > 0$ is true for all $x \neq 3, -3$ $\therefore$ S.S = $(-\infty, \infty)$

h) **Exercise left for you**

12. Solve each **special absolute value inequality**

a) $|x| > x$ b) $|x| \geq x$ c) $|x-1| > x-1$

d) $|x-1| \geq x-1$ e) $|x+3| > x+3$ f) $|x+3| \geq x+3$

Solution: Here

a) $|x| > x$ is true for all $x < 0$ $\therefore$ S.S = $(-\infty, 0)$

b) $|x| \geq x$ is true for all $x \in R$ $\therefore$ S.S = R = $(-\infty, \infty)$

c) $|x-1| > x-1$ is true for all $x < 1$ $\therefore$ S.S = $(-\infty, 1)$

d) $|x-1| \geq x-1$ is true for all $x \in R$ $\therefore$ S.S = $(-\infty, \infty)$ = R

e) $|x+3| > x+3$ is true for all $x < -3$ $\therefore$ S.S = $(-\infty, -3)$

f) $|x+3| \geq x+3$ is true for all $x \in R$ $\therefore$ S.S = $(-\infty, \infty)$

13. Solve each **special absolute value**

a) $|x| = x$ b) $|x| = -x$ c) $|x+2| = x+2$ d) $|x+2| = -x-2$

e) $|x-2| = 2-x$ f) $|x-3| = x-3$ g) $|x-4| = -x+4$ h) $|2x-3| = 3-2x$

Solution: **The explanation is here**

a) $|x| = x$, S.S = $\{x: x \geq 0\} = [0, \infty)$

b) $|x| = -x$, S.S = $\{x: x \leq 0\} = (-\infty, 0]$

c) $|x+2| = x+2$, S.S = $\{x: x \geq -2\} = [-2, \infty)$

d) $|x+2| = -x-2 = -(x+2)$, S.S. $\{x: x \leq -2\} = (-\infty, -2]$

e) $|x-2| = 2-x$, S.S = $\{x: x \leq 2\} = (-\infty, 2]$

f) $|x-3| = x-3$, S.S = $\{x: x \geq 3\} = [3, \infty)$

14. **Solve:**

a) $|x| > -2$ b) $|x+4| > -7$ c) $|3x-6| > -10$

Solution: Since $|x| \geq 0$ for all $x \in IR$.

a) $|x| > -2$ is true for all $x \in R$ $\therefore$ S.S = R = $(-\infty, \infty)$

b) $|x+4| > -7$ is true for all $x \in R$ $\therefore$ S.S = R = $(-\infty, \infty)$

c) $|3x-6| > -10$ is true for all $x \in R$

14. b) **Solve each of the following absolute value inequality**

a) $|x^2 - 9| = x^2 - 9$ c) $|x^2 - 1| = x + 1$

b) $|x^2 - 9| = 9 - x^2$ d) $|x^2 - 4| = x + 2$

Explanation:

a) It is valid if $x^2 - 9 \geq 0 \Rightarrow (-\infty, -3] \cup [3, \infty)$

$|x^2 - 9| = x^2 - 9$ is true for all real number $x \in R \setminus (-3, 3)$

$\Rightarrow$ S.S = $R \setminus (-3, 3) = (-\infty, -3] \cup [3, \infty)$

b) $|x^2 - 9| = 9 - x^2$ is valid if $9 - x^2 \geq 0 \Rightarrow [-3, 3]$

$\therefore$ S.S = $\{x: -3 \leq x \leq 3\} = [-3, 3]$

c) $|x^2 - 1| = x + 1$ is valid if $x + 1 \geq 0 \Rightarrow x \geq -1$

$\Leftrightarrow x^2 - 1 = x + 1$ Or $x^2 - 1 = -x - 1$

$\Rightarrow x^2 - x - 2 = 0$ Or $x^2 = -x$

$\Rightarrow (x-2)(x+1) = 0$ Or $x^2 + x = 0$ $\therefore$ S.S = $\{-1, 0, 2\}$

d) **Exercise left for you (Tip!)**

Systems of linear inequality in two variable

The *Solution* of system of inequalities.

- The *Solution*s of the system of inequalities are the ***Solution***s common to all inequalities in the system.
- The graph of system of inequalities consists of the points corresponding to the ***Solution***s.

Illustrative example

5. Solve each of the following system of linear inequalities:

i) graphically

ii) in interval notation

iii) Domain and range

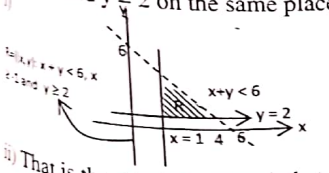
a) $\begin{cases} x + y < 6 \\ x \geq 1 \\ y \geq 2 \end{cases}$ b) $\begin{cases} y - x \leq 4 \\ y + x \geq 1 \\ x \leq 3 \end{cases}$ c) $\begin{cases} x + y \leq 4 \\ 2x - y \leq 4 \end{cases}$ d) $\begin{cases} x + y > 2 \\ x \leq 2 \end{cases}$

Solution: (a) Here, to determine the set of points whose coordinates (x, y) satisfy all three inequality, at the same time.

First draw each boundary line, $x + y = 6$

$x = 1$ and $y = 2$ on the same place

i)



we see that the ***Solution***s of the system correspond to the points below the line $x + y = 6$ and above (and on) the line $y = 2$ and to the right of (and on) $x = 2$

ii) That is the set of (x, y) such that $x \in [1, 4)$ and $y \in [2, 6-x)$

or $R = \{(x, y): 1 \leq x < 4 \text{ and } 2 \leq y < 6 - x\}$

iii) Domain = $\{x: 1 \leq x < 4\}$

Range = $\{2 \leq y < 5\}$

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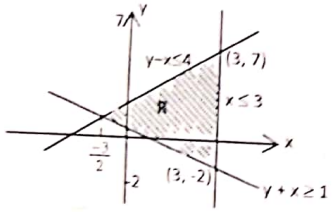
b) i) the boundary $y - x = 4$, $y + x = 1$ and $x = 3$

- The graph of $y - x \le 4$ consists of point below (and on) the line
- The graph of $y + x \ge 1$ consists of point above the line
- The graph of $x \le 3$ consists of point to the left (and on) the line.



- To find the intersection point of

$$\begin{cases} y - x = 4 \\ y + x = 1 \end{cases}$$

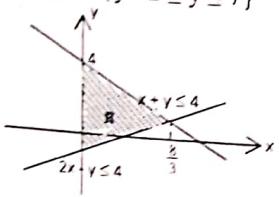
$$2y = 5 \Rightarrow y = \frac{5}{2}, \quad x = \frac{-3}{2}$$

$R = \{(x, y): y - x \le 4,\ y + x \ge 1 \text{ and } x \le 3\}$

ii) ***Solution*** set $= \{(x, y): \frac{-3}{2} \le x \le 3 \text{ and } 1 - x \le y \le 4 + x\}$

iii) Domain $= \left\{x: \frac{-3}{2} \le x \le 3\right\}$

Range $= \{y: -2 \le y \le 7\}$



To sketch the graph of $x + y = 4$

Put $x = 0$ in $x + y = 4$, then $y = 4$ and put $y = 0$ in $x + y = 4$, then $x = 0$ using the test point $(0, 0)$ we see that the ***Solution***s of the system correspond to the point below (and on) the lien $x + y = 4$ and above (and on) the line $2x - y = 4$

ii) the ***Solution*** set in the interval:

$S = \{(x, y): x \le \frac{8}{3} \text{ and } 2x - 4 \le y \le 4 - x\}$

iii) Domain $= \left\{x: x \le \frac{8}{3}\right\}$

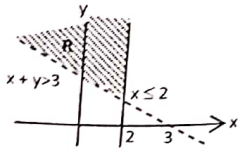
Range $= (-\infty, \infty)$

d) $R = \{(x, y): x + y > 3 \text{ and } x \le 2\}$

ii) in interval notation:

$R = \{(x, y): x \le 2 \text{ and } y > 3 - x\}$

$= (-\infty, 2] \times (1, \infty)$

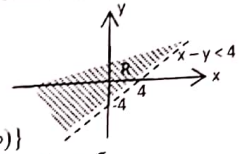


16. Find the ***Solution*** set of the system:

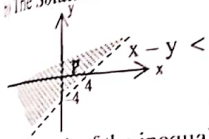
a) $\begin{cases} x + y > 3 \\ x + y < -1 \end{cases}$ b) $\begin{cases} x - y < 4 \\ x - y < 6 \end{cases}$ c) $\begin{cases} x + y < 3 \\ x + 2y > 3 \end{cases}$ d) $\begin{cases} x - y > 4 \\ x - y < 6 \end{cases}$

Solution: It is clear that the system has no ***Solution***s because $x + y$ can not be both less -1 and greater than 3

$\therefore$ the ***Solution*** set is $\varnothing$



b) The ***Solution*** set $R = \{(x, y): x \in R \text{ and } y \in [4 - x, \infty)\}$



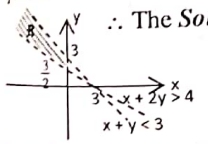
let $x = -1$, then $y = -1 - 4 = -5$

$-1 - 5 < 4$ True

$\therefore$ S.S $= \{(x,y): y > x - 4\}$

c) The graph of the inequality $x + y < 3$ is the half – plane that lies below the line $x + y = 3$

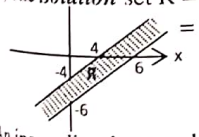
- The graph of the inequality $x + 2y > 3$ is the half plane that lies above the line $x + 2y = 3$.
- The intersection of these two half plane is an infinite wedge that has a vertex at $(3, 0)$.
- This unbounded region is the ***Solution*** set has an infinite number of points.



$\therefore$ The ***Solution*** set $R = \{(x, y): x \in (-\infty, 3) \text{ and } y \in \left[\frac{3 - x}{2}, 3 - x\right]$

d) The ***Solution*** set $R = \{(x, y): x \in R \text{ and } y \in (x - 6, x - 4)\}$ or

$= \{(x, y): -\infty < x < \infty \text{ and } x - 6 < y < x - 4\}$



Quadratic Inequalities

An inequality that can be reduced to any one of the following forms:

$ax^2 + bx + c \le 0$ or $ax^2 + bx + c < 0$

$ax^2 + bx + c \ge 0$ or $ax^2 + bx + c > 0$

Where a, b and c are constants and $a \ne 0$ is called a quadratic inequality .

The position of the curve relative to the x-axis depends on the nature of the roots of $ax^2 + bx + c = 0$

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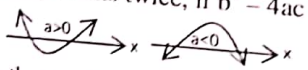
The roots of the equation are the zeros of the function, i.e. the value of x where y = 0 which is where the curve cuts the x-axis.
The values are given by

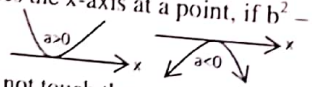
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Note: $f(x) = ax^2 + bx + c$

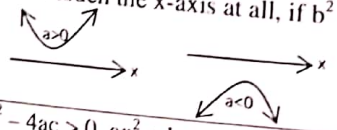
i) crosses the x-axis twice, if $b^2 - 4ac > 0$



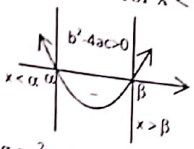
ii) touches the x-axis at a point, if $b^2 - 4ac = 0$



iii) does not touch the x-axis at all, if $b^2 - 4ac < 0$

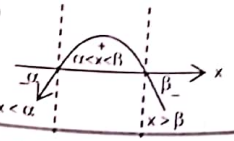


A. If $b^2 - 4ac > 0$, $ax^2 + bx + c = 0$ has two real roots α and β
When $a > 0$ (parabola opens up ward)
i) $ax^2 + bx + c < 0$ for $\alpha < x < \beta$
ii) $ax^2 + bx + c > 0$ for $x < \alpha$ or $x > \beta$



Note: Solving $ax^2 + bx + c < 0$ means finding the interval on which the function lies below the x-axis.

Solving $ax^2 + bx + c > 0$ means finding the interval on which the function lies above the x-axis.
When $a < 0$ (parabola opens downward)
i) $ax^2 + bx + c < 0$ for $x > \beta$ or $x < \alpha$
ii) $ax^2 + bx + c > 0$ for $\alpha < x < \beta$



Illustrative example

17. Solve each of the following inequalities

a) $(x - 1)(x + 2) > 0$

e) $\left(1 + \frac{x}{4}\right)(3 - x) < 0$

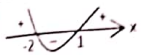
b) $(1 + x)(3 - 2x) \geq 0$

f) $(3 + x)(3 - x) > 0$

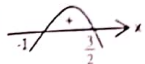
c) $(5 - x)\left(1 - \frac{1}{3}x\right) \leq 0$

d) $(x - 3)(x + 3) > 0$

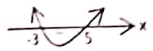
Solution:

a) 

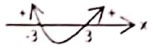
$\therefore (x - 1)(x + 2) > 0$ for $x < -2$ or $x > 1$

b) 

$\therefore (1+x)(3-2x) \geq 0$ for $-1 \leq x \leq \frac{3}{2}$

c) 

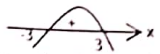
$\therefore (5 - x)\left(1 - \frac{1}{3}x\right) \leq 0$ for $3 \leq x \leq 5$

d) 

$\therefore (x-3)(x + 3) > 0$ for $x < -3$ or $x > 3$

e)

$\therefore \left(1 + \frac{x}{4}\right)(3-x) < 0$ for $x < -4$ or $x > 3$

f) 

$\therefore (3 + x)(3 - x) > 0$ for $-3 < x < 3$

18. Solve each of the following quadratic inequalities

a) $x^2 - x - 6 \geq 0$

d) $-2x^2 + x + 3 > 0$

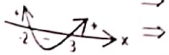
b) $3x^2 + 4x + 1 > 0$

e) $25x^2 - \frac{1}{16} < 0$

c) $x^2 - 5x + 6 < 0$

Solution: $x^2 - x - 6 = (x - 3)(x + 2)$

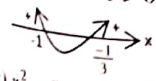
$\Rightarrow x^2 - x - 6 \geq 0$ means $(x - 3)(x + 2) \geq 0$



$\Rightarrow x - 3 = 0$ at $x = 3$ and $x + 2 = 0$ at $x = -2$

$\therefore x^2 - x - 6 \geq 0 \Rightarrow (x-3)(x+2) \geq 0$ for $x \leq -2$ or $x \geq 3$

b) $3x^2 + 4x + 1 > 0 \Leftrightarrow 3x^2 + 3x + x + 1 > 0 = 3x(x+1) + 1(x+1) > 0$



$\Rightarrow (3x + 1)(x + 1) > 0$ for $x < -1$ or $x > \frac{-1}{3}$

c) $x^2 - 5x + 6 < 0 \Leftrightarrow (x - 3)(x - 2) < 0$

$\therefore (x - 3)(x - 2) < 0$ for $2 < x < 3$

d) $-2x^2 + x + 3 > 0 \Rightarrow -2x^2 + 3x - 2x + 3 > 0 \Rightarrow -x(2x - 3) - 1(2x - 3) > 0$

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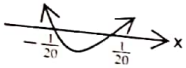
$\Rightarrow (-x-1)(2x-3)>0$

$\therefore (-x-1)(2x-3)>0$ for $-1<x<\frac{3}{2}$

e) $25x^2-\frac{1}{16}<0 \Leftrightarrow \left(5x-\frac{1}{4}\right)\left(5x+\frac{1}{4}\right)<0$



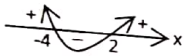
$\therefore \left(5x-\frac{1}{4}\right)\left(5x+\frac{1}{4}\right)<0$ for $\frac{-1}{20}<x<\frac{1}{20}$

19. Solve each of the following quadratic inequality

a) $x^2+2x\geq 8$ c) $(x-1)^2>2x^2-2x$

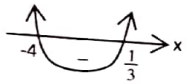
b) $3x^2+11x<4$ d) $x-1\leq x^2+2$

Solution: $x^2+2x-8\geq 0 \Leftrightarrow (x-2)(x+4)\geq 0$



$\therefore (x-2)(x+4)\geq 0$ for $x\leq -4$ or $x\geq 2$

b) $3x^2+11x<4 \Leftrightarrow 3x^2+11x-4<0 \Rightarrow 3x^2+12x-x-4<0$

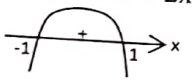


$\Rightarrow 3x(x+4)-1(x+4)<0$

$\Rightarrow (3x-1)(x+4)<0$

$\therefore (3x-1)(x+4)<0$ for $-4<x<\frac{1}{3}$

c) $(x-1)^2>2x^2-2x \Leftrightarrow (x-1)^2-2x^2+2x>0$



$\Rightarrow x^2-2x+1-2x^2+2x>0$

$\Rightarrow -x^2+1>0$

$\Rightarrow 1-x^2>0 \Leftrightarrow (1-x)(1+x)>0$

$\therefore 1-x^2>0$ for $-1<x<1$

d) $(x-1)\leq x^2+2 \Leftrightarrow x-1-x^2-2\leq 0$

$\Rightarrow -x^2+x-3\leq 0$

Using completing the square

$-(x^2-x+3)<0 \Leftrightarrow -\left(x^2-x+\left(\frac{1}{2}\right)^2+3-\frac{1}{4}\right)$

$\Leftrightarrow -\left(x^2-x+\frac{1}{4}+\frac{11}{4}\right) \Leftrightarrow -\left(\left(x-\frac{1}{2}\right)^2+\frac{11}{4}\right)$

$-\left(x-\frac{1}{2}\right)^2-\frac{11}{4}<0$ for all x

$\therefore$ *Solution* set is $(-\infty,\infty)$

Solving quadratic inequalities using the sign chart method

Illustrative example

20. Solve each of the following inequalities using sign charts

a) $x^2-\frac{3}{4}x<0$ b) $x^2+5x>0$ c) $3x^2+11x>4$

Solution: Factorize $x^2-\frac{3}{4}x=x\left(x-\frac{3}{4}\right)$

Draw assign chart, and write the zeros in increasing order

$x\left(x-\frac{3}{4}\right)=0$ if $x=0$ or $x=\frac{3}{4}$

	$-\infty$	0	$\frac{3}{4}$	∞
x	-	+	+	
$x-\frac{3}{4}$	-	-	+	
$x\left(x-\frac{3}{4}\right)$	+	-	+	

From the sign chart, you can conclude that

$x\left(x-\frac{3}{4}\right)<0$ for each $x\in\left(0,\frac{3}{4}\right)$

$x\left(x-\frac{3}{4}\right)>0$ for each $x\in(-\infty,0)\cup\left(\frac{3}{4},\infty\right)$

b) $x^2+5x>0 \Leftrightarrow x(x+5)>0$

$\therefore x(x+5)>0$ for $x\in(-\infty,-5)\cup(0,\infty)$

$x(x+5)<0$ for $x\in(-5,0)$

c) $3x^2+11x>4 \Leftrightarrow 3x^2+11x-4>0 \Rightarrow (3x-1)(x+4)>0$

	$-\infty$	-4	$\frac{1}{3}$	∞
$3x-1$	-	-	+	
$x+4$	-	+	+	
$(3x-1)(x+4)$	+	-	+	

$\therefore (3x-1)(x+4)>0$ for $x\in(-\infty,-4)\cup(\frac{1}{3},\infty)$

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$\therefore (3x-1)(x+4) < 0$ for $x \in (-4, \frac{1}{3})$

Solving perfect square inequalities

The form $(x+1)^2, (x+4)^2, (x-5)^2, \ldots$ etc is perfect square expression.

Illustrative example

21. Solve each of the following inequalities

a) $x^2-6x+9>0$ b) $x^2-6x+9\geq 0$ c) $x^2-6x+9<0$

d) $x^2-6x+9\leq 0$ e) $-x^2+4x-4<0$ f) $-x^2+4x-4\geq 0$

g) $-x^2+4x-4\leq 0$ h) $-x^2+4x-4>0$

Solution: $x^2-6x+9=0 \Leftrightarrow (x-3)(x-3)>0 \Rightarrow (x-3)^2>0$

Since $(x-3)^2$ is always non-negative

$\therefore (x-3)^2>0$ for all $R\backslash\{3\} = (-\infty, 3)\cup(3, \infty)$

b) $x^2-6x+9\geq 0 \Leftrightarrow (x-3)^2\geq 0$

$\therefore (x-3)^2\geq 0$ for every real number x

$\therefore$ ***Solution*** set is $(-\infty,\infty)$

c) $x^2-6x+9=0 \Rightarrow (x-3)^2$ is never less than 0

$\therefore (x-3)^2<0$ is $\varnothing$

d) $x^2-6x+9\leq 0 \Leftrightarrow (x-3)^2\leq 0$ is true for $x=3$

$\therefore (x-3)^2\leq 0$ for $x=3$

e) $-x^2+4x-4<0 \Leftrightarrow -(x-2)^2$ is less than 0 for every x except at $x=2$

$\therefore$ the ***Solution*** set is $(-\infty, 2)\cup(2, \infty)$

f) $-x^2+4x-4\geq 0 \Leftrightarrow -(x-2)^2\geq 0$ is true only at $x=2$

$\therefore$ the ***Solution*** set is $\{x: x=2\}$

g) $-x^2+4x-4\leq 0 \Leftrightarrow -(x-2)^2\leq 0$ is true for every value of x

$\therefore$ the ***Solution*** set is $(-\infty, \infty)$

h) $-x^2+4x-4>0 \Leftrightarrow -(x-2)^2>0$ is false $\therefore$ the ***Solution*** set is $\varnothing$

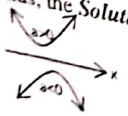
Solving quadratic inequality not factorizable into linear factors.

If $b^2-4ac<0$, then $ax^2+bx+c=0$ has no real roots and the curve $y=ax^2+bx+c$ does not meet the x-axis.

Note: $b^2-4ac<0$ is the indicator for ax^2+bx+c is not factorized.

In this case, either $ax^2+bx+c>0$ for all values of x or $ax^2+bx+c<0$ for all values of x.

Thus, the ***Solution*** set of $ax^2+bx+c>0$ is

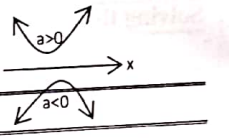


i) for $a>0$, is $(-\infty, \infty)$

ii) for $a<0$ is $\varnothing$

and the ***Solution*** set of $ax^2+bx+c<0$

i) for $a>0$ is $\varnothing$ ii) for $a<0$, is $(-\infty, \infty)$



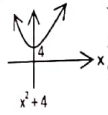
Illustrative example

22. Solve each of the following quadratic inequalities

a) $x^2+4>0$ b) $x^2+1<0$ c) $x^2+4x+7>0$

d) $3x^2-x+1<0$ e) $-x^2+4x-6<0$

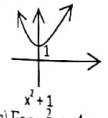
Solution: we know that the graph of x^2+4 is lies above the x-axis for all values of x



So that $x^2+4>0$ for all $x\in(-\infty, \infty)$

Therefore, the ***Solution*** set $S = (-\infty, \infty)$

b) For $x^2+1<0$ is false because, the parabola opens upward and does not meet the x-axis



$\therefore$ ***Solution*** set $= \{\}$ or $\varnothing$

c) For x^2+4x+7, $a=1$, $b=4$, $c=7$ and $b^2-4ac = 4^2-4(1)(7)$

$= 16-28 = -12<0$

Hence x^2+4x+7 cannot factorized

Take test point, $x=0$, then $0^2+4(0)+7=7$

So, $x^2+4x+7>0$ for all $x\in(-\infty, \infty)$

$\therefore$ the solution set $S.S = (-\infty, \infty)$

d) For $3x^2-x+1$, $a=3$ $b=-1$, $c=1$ and $b^2-4ac = (-1)^2-4(3)(1)$

$= -11<0$

Hence, $3x^2-x+1$ cannot factorized and the parabola opens upward and the graph of $3x^2-x+1$ lies above the x-axis.

Thus, $3x^2-x+1>0$ for all value of x

However $3x^2-x+1<0$ is false, take test point $x=0$,

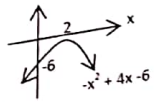
then $0-0+1<0$ is false $\therefore$ ***Solution*** set $S.S = \{\}$ or $\varnothing$

e) For $-x^2+4x-6<0$

$a=-1$, $b=4$, $c=-6$

$b^2-4ac = 4^2-4(-1)(-6)$

$= 16-24 = -8<0$



Hence $-x^2+4x-6$ cannot factorized

So, take test point $x=0$

Then $-0+0-6=-6<0$ true $\therefore$ The ***Solution*** set $S.S = (-\infty, \infty)$

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Solving the value of k which is true for all values of x

Illustrative example

23. For what value(s) of k is
a) $2x^2 + x + k > 0$ for each real number x
b) $kx^2 + 6x + 1 > 0$ for each real number x
c) $x^2 + kx + 2k > 0$ for each real number x
d) $kx^2 + 4x + k \geq 0$ for all x
e) $x^2 - 5x + k < 0$ only for $x \in (2, 3)$

Solution: Now we set $b^2 - 4ac < 0$, then

For $2x^2 + x + k$, $a = 2$, $b = 1$, $c = k$ and $b^2 - 4ac < 0$

$\Rightarrow 1 - 4(2)(k) < 0 \Rightarrow 1 - 8k < 0$

$\Rightarrow 1 < 8k$

$\therefore k > \frac{1}{8}$

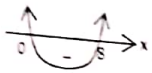
b) For $kx^2 + 6x + 1$, $a = k$, $b = 6$, $c = 1$ and we set $b^2 - 4ac < 0$

$\Rightarrow 6^2 - 4(k)(1) < 0$

$\Rightarrow 36 < 4k$ $\therefore$ the ***Solution*** set S.S = {k: k < 9}

C For $x^2 + kx + 2k$, $a = 1$, $b = k$, $c = 2k$ and

$\Rightarrow b^2 - 4ac < 0$



$\Rightarrow k^2 - 4(1)(2k) < 0$

$\Rightarrow k^2 - 8k < 0 \Rightarrow k(k - 8) < 0$

$\therefore$ the ***Solution*** set S.S = {k: 0 < k < 8}

d) For $kx^2 + 4x + k$, $b = 4$, $c = k$ and

$\Rightarrow b^2 - 4ac \leq 0 \Rightarrow 16 - 4(k)(k) \leq 0 \Rightarrow 16 \leq 4k^2$

$\Rightarrow 4 \leq k^2 \Leftrightarrow k^2 \geq 4 \Rightarrow |k| \geq 2$

$\therefore$ the ***Solution*** set S.S = {k: k ≥ 2 or k ≤ -2}

$= k \in (-\infty, -2] \cup [2, \infty)$

e) put x = 2 in $x^2 - 5x + k < 0$

then $(2)^2 - 5(2) + k < 0$

$\Rightarrow 4 - 10 + k < 0$

$\Rightarrow -6 + k < 0$ $\therefore k < 6$

$\therefore$ the ***Solution*** set S.S = {k: k < 6}

The character of the roots

The quadratic equation $ax^2 + bx + c = 0$ is determined by the discriminant $b^2 - 4ac$

Assuming a, b, c are rational numbers then
i) if $b^2 - 4ac > 0$, it has *two distinct real root*
ii) if $b^2 - 4ac = 0$, it has *only one real root*
iii) if $b^2 - 4ac < 0$, it has *no real root.*

Illustrative example

24. What must the value of k, each equation subject to the given condition?
i) two distinct real root
ii) one real root
iii) no real root
a) $x^2 - 3x + 1 = k(x - 3)$ b) $x^2 + kx + 8 = 0$
c) $kx^2 + (k + 1)x + k = 0$ d) $(3k - 4)x^2 + 2kx - 1 = 0$

Solution: a) Rewriting the equation in standard form

$x^2 - (k + 3)x + 3k + 1 = 0$ and here, $a = 1$

$b = -(k + 3)$ and $c = 3k + 1$

i) For two distict real roots

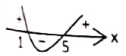
$b^2 - 4ac > 0$ i.e. $b^2 > 4ac$

Hence, $(k + 3)^2 > 4(1)(3k + 1)$

$\Rightarrow k^2 + 6k + 9 > 12k + 4$

$\Rightarrow k^2 - 6k + 5 > 0$

$\therefore (k - 5)(k - 1) > 0$



$\therefore$ the ***Solution*** set, S.S = {k: k < 1 or k > 5}

$= k \in (-\infty, 1) \cup (5, \infty)$

ii) For one real roots, $b^2 - 4ac = 0$

$\Rightarrow (k - 5)(k + 1) = 0$ $\Rightarrow k = 5$ or $k = -1$

iii) For no real roots, $b^2 - 4ac < 0$

$\Rightarrow (k - 5)(k - 1) < 0$

$\therefore$ the ***Solution*** set, S.S = {k: 1 < k < 5} $= k \in (1, 5)$

b) Exercise left for the reader

c) $kx^2 + (k + 1)x + k = 0$ $a = k$, $b = k + 1$, $c = k$

i) For two distinct real root, $b^2 - 4ac > 0$

hence, $(k + 1)^2 - 4(k)(k + 1) > 0$

gives $k^2 + 2k + 1 - 4k^2 - 4k > 0$ or $-3k^2 - 2k + 1 > 0$

$\Rightarrow -3k^2 - 3k + k + 1 > 0$

$\Rightarrow -3k(k + 1) + 1(k + 1) > 0$ $\Rightarrow (-3k + 1)(k + 1) > 0$

$\therefore$ ***Solution*** set S.S = $\left\{k : -1 < k < \frac{1}{3}\right\}$

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ii) For one real root, $b^2 - 4ac = 0$

$\Rightarrow (-3k + 1)(k + 1) = \quad \Rightarrow k = -1 \text{ or } k = \frac{1}{3}$

iii) For no real root, we set $b^2 - 4ac < 0$

$\Rightarrow (-3k + 1)(k + 1) < 0$

$\therefore$ ***Solution*** set $S.S = \left\{k : k < -1 \text{ or } k > \frac{1}{3}\right\}$

$= k \in (-\infty, -1) \cup \left(\frac{1}{3}, \infty\right)$

d) $(3k - 4)x^2 + 2kx - 1 = 0$

$a = 3k - 4, \ b = 2k, \ c = -1$

i) For two distinct real root, set $b^2 - 4ac > 0$

Hence $4k^2 - 4(3k - 4)(-1) > 0$

$\Rightarrow 4k^2 + 12k - 16 > 0 \quad \Rightarrow k^2 + 3k - 4 > 0 \Rightarrow (k - 1)(k + 4) > 0$

$\Rightarrow k < -4 \text{ or } k > 1$

$\therefore$ ***Solution*** set, $S.S = \{k: k < -4 \text{ or } k > 1\} \quad = k \in (-\infty, -4) \cup (1, \infty)$

Application

25. A rocket is fired straight upward from the ground level with an initial velocity of 128km/hr. After t second, its distance above the ground level is given by $S(t) = 128t - 16t^2$. For what time interval is the rocket *more than 240km* above the ground . . . A/g/64/189

Solution: $s(t) > 140km$

$\Rightarrow -16t^2 + 12\,8t > 140 \longleftarrow$ Divide both side by -16

$\Rightarrow t^2 - 8t < -15$

$\Rightarrow t^2 - 8t + 15 < 0 \Leftrightarrow (t - 3)(t - 5) < 0$

$\Rightarrow \{t: 3 < t < 5\}$

26. A shop owner selling x pairs of shoes per day has a profit (in birr) expressed by the equation $P(x) = -x^2 + 25x - 10$. If the owner needs to obtain a profit of more than 140birr per day, then the number of pairs of shoes the shop owner makes should the range between . . . .

EGSECE

A. 10 and 15
B. 5 and 10
C. 12 and 10
D. 5 and 15

Solution: $P(x) > 140$

$\Rightarrow -x^2 + 25x - 10 > 140 \Leftrightarrow -x^2 + 25x - 150 > 0$

$\Rightarrow x^2 - 25x + 150 < 0 \Leftrightarrow (x - 15)(x - 10) < 0$

$\therefore \{x: 10 < x < 15\}$, $\therefore$ **Answer A**

27. **Determine the number**

a) If the sum of a non – negative number and its square is less than 30

Solution: right here,

$\Rightarrow x + x^2 < 30 \Leftrightarrow x^2 + x - 30 < 0$

$\Rightarrow (x - 5)(x + 6) < 0$

$\Rightarrow -6 < x < 5$, since it is non – negative $\quad \therefore \{x: 0 \le x < 5\}$

28. A manufacturer determines that the profit obtained from selling x – units of a certain item in Birr is $P(x) = 200x - 0.25x^2$

a) How many units must be produced to secure a profit

b) At how many units level will there be no profit or no loss

Solution: a) To get profit, we set $P(x) > 0$

$\Leftrightarrow 200x - 0.25x^2 > 0 \Rightarrow x(200 - 0.25x) > 0$

$\Rightarrow x > 0 \text{ and } 200 - 0.25x > 0$

$\Rightarrow x > 0 \text{ and } 800 - x > 0$

$\Rightarrow 0 > x \text{ and } 0 < x < 800 \quad \therefore S.S = \{x: 0 < x < 800\}$

b) No profit or no loss, we set $P(x) = 0, \ x \neq 0$

$\Rightarrow 200x - 0.25x^2 = 0$

$\Rightarrow 200x - \frac{1}{4}x^2 = 0 \Rightarrow 800x = x^2$

$\Rightarrow x^2 - 800x = 0 \Leftrightarrow x(x - 800) = 0 \quad \therefore x = 800$

Solved Problem

27 b) Solve each of the following system of inequality and **Describe the domain and range)**

a) $\begin{cases} x - y \le 3 \\ x - y \ge 1 \end{cases}$ b) $\begin{cases} x + y \ge 4 \\ x + y \ge 2 \end{cases}$ c) $\begin{cases} 2x + y > 3 \\ x - y \ge 1 \end{cases}$

d) $\begin{cases} x > 0 \\ y > 0 \\ x + y \le 5 \end{cases}$ e) $\begin{cases} y + x > 0 \\ y - x > 0 \\ y \ge 4 \end{cases}$ f) $\begin{cases} x - y < 2 \\ x - y < -1 \end{cases}$

Solution:

a) $\begin{cases} x - y \le 3 \\ x - y \ge 1 \end{cases} \Rightarrow \begin{cases} x - 3 \le y \\ x - 1 \ge y \end{cases} \Rightarrow \begin{cases} x - 3 \le y \\ y \le x - 1 \end{cases} \Leftrightarrow x - 3 \le y \le x - 1$

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$\therefore$ S.S = $\{(x,y): -\infty < x < \infty$ and $x - 3 \le y \le x - 1\}$

$\therefore$ Domain = R and range = R

b) $\begin{cases} x + y \ge 4 \\ x + y \ge 2 \end{cases} \Rightarrow \begin{cases} y \le -x + 4 \\ y \ge -x + 2 \end{cases} \Leftrightarrow \begin{cases} 4 - x \ge y \\ y \ge 2 - 2 \end{cases}$

$\therefore$ S.S = $\{(x,y): -\infty < x < \infty$ and $2 - x \le y \le 4 - x\}$

$\therefore$ Domain = R and range = R

c) $\begin{cases} 2x + y > 3 \\ x - y \ge 1 \end{cases} \Rightarrow \begin{cases} y > 3 - 2x \\ x - 1 \ge y \end{cases} \Leftrightarrow \begin{cases} 3 - 2x < y \\ y \le x - 1 \end{cases}$

Solving: $\begin{cases} 2x + y > 3 \\ x - y \ge 1 \end{cases}$

$3x = 4 \Rightarrow x = \frac{4}{3}$

So, $x > \frac{4}{3}$, $3 - 2x < y \le x - 1$

S.S = $\{(x,y): \frac{4}{3} < x < \infty$ and $3 - 2x < y \le x - 1\}$

$\therefore$ Domain = $\left(x : x > \frac{4}{3}\right) = \left(\frac{4}{3}, \infty\right)$

Range = IR

d) $x + y \le 5 \Rightarrow y \le 5 - x$, since $y > 0 \Rightarrow 0 < y$

When $y = 0 \Rightarrow x = 5$

When $x = 0 \Rightarrow y = 5$

$\therefore$ S.S = $\{(x, y): 0 < x \le 5$ and $0 < y \le 5 - x\}$

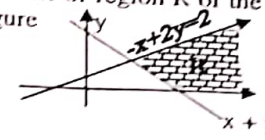
- Domain = $\{x: 0 < x \le 5\}$
- Range = $\{y: 0 < y \le 5\}$

e) $\begin{cases} y + x > 0 \\ y - x > 0 \end{cases} \Rightarrow \begin{cases} y > -x \\ y > x \end{cases}$ and $y \le 4$

$\therefore$ S.S = $\{(x,y) - 4 < x < 4$ and $0 < y \le 4\}$

f) **Exercise left for yoy.**

29. The ***Solution*** of region R of the inequality in the shaded region shown in the figure



- Which one of the following is the correct description of R.

A. $R = \{(x,y): -x + 2y \le 2, x + y \ge 3, x \ge 0\}$

B. $R = \{(x,y): -x + 2y \le 2, x + y \ge 3, x \ge 0, y \ge 0\}$

C. $R = \{(x,y): -x + 2y \le 2, x + y \ge 3, y \ge 0\}$

D. $R = \{(x,y): -x + 2y \le 2, x + y \le 3, x \ge 0\}$

Solution:

- The shaded region bounded

i) **below** the line $-x + 2y = 2$, written by $-x + 2y \le 2$

ii) **above** the line $x + y = 3$, and above x – axis written by $-x + y \ge 3$ and $y \ge 0$

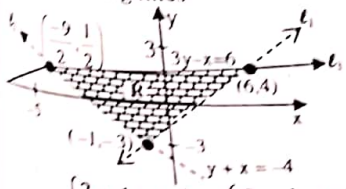
$\therefore R = \{(x,y): -x + 2y \le 2, x + y \ge 3, y \ge 0\}$ $\therefore$ **Answer: C**

Note that , The intersection of the two line is

$\begin{cases} -x + 2y = 2 \\ x + y = 3 \end{cases} \Rightarrow \begin{array}{r} -x + 2y = 2 \\ +\ x + y = 3 \\ \hline -3x = -4 \Rightarrow x = \frac{4}{3} \end{array}$

Therefore, **Domain of R** = $\left\{x : x \ge \frac{4}{3}\right\}$ **Range R** = $\{y: y \ge 0\}$

30. As shown on the figure below the shaded region is formed by three intersecting lines



Which one of the following system of linear inequalities algebraically described the **shaded region R . . . (EGSECE 2005)**

A. $\begin{cases} 3y \le x + 6 \\ y > -x - 4 \\ y > x - 2 \end{cases}$ B. $\begin{cases} 3y \le x - 6 \\ -y > x + 4 \\ y > x - 2 \end{cases}$ C. $\begin{cases} 3y \le x - 6 \\ y > -x + 4 \\ y \ge -x + 2 \end{cases}$ D. $\begin{cases} 3y \le x + 6 \\ y > x - 4 \\ y < -x + 2 \end{cases}$

Solution:

The shaded region R is bounded

i) below the solid line ℓ_3: $3y - x = 6$, so that $3y - x \le 6 \Rightarrow 3y \le x + 6$

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ii) above the broken line ℓ_2: $y + x = -4$ and ℓ_1

Equation of ℓ_1 is given by $y - y_1 = m(x - x_1)$

Since (6,4) and (−1, −3) on ℓ_1.

$\therefore m = \dfrac{-3-4}{-1-6} = \dfrac{-7}{-7} = 1$, let $(x_1, y_1) = (6,4)$

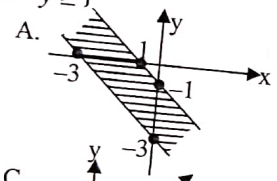
ℓ_1: $y - 4 = 1(x - 6) \Leftrightarrow y = x - 2 \Rightarrow y > x - 2$

$\therefore R = \{(x,y): 3y \leq x+6,\ y + x > -4 \text{ and } y > x - 1\}$

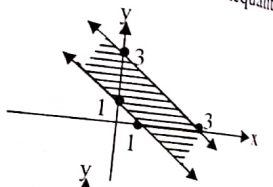
31. Which one of the following shaded regions is the graphical representation of the ***Solutions*** et for the system of linear inequalities

$$\begin{cases} x - y \leq 3 \\ x - y \geq 1 \end{cases}$$

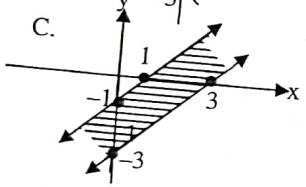
A.



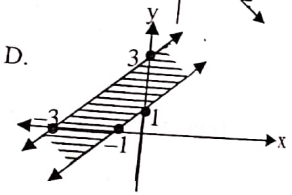
B.



C.



D.



Explanation: Here to sketch the graph

i) $x - y \leq 3$ **is the same as** $x - 3 \leq y \Rightarrow y \geq x - 3$

ii) $x - y \geq 1$ **is the same as** $x - 1 \geq y \Rightarrow y \leq x - 1$

Now, the ***Solution*** set of the system lies

i) **above** the solid line $y = x - 3$ which is the same as $y \geq x - 3$

ii) **below** the solid line $y = x - 1$ which is the same as $y \leq x - 1$

Answer: C

Note that: A is **described by** $\begin{cases} y + x \leq -1 \\ y + x \geq -3 \end{cases}$

Graph C is **described by** $\begin{cases} y + x \geq 1 \\ y + x \leq 3 \end{cases}$

Graph D is **described by** $\begin{cases} y - x > 1 \\ y - x \leq 3 \end{cases}$

$\therefore R = \{(x,y): y \leq 2x,\ 2 \leq y \leq 6 \text{ and } 2x + 3y \leq 30\}$

Find the elements of R that makes

a. y − x a minimum

b. y − x a maximum

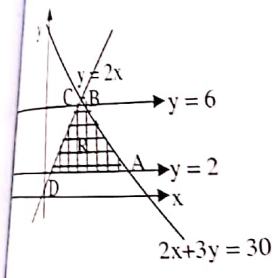
c. y − x a maximum

d. 3x + 8x a maximum

Solution: Here

Maximum minimum property

If a, b are real numbers a and b and all (x, y) pairs in a given closed region then the maximum and minimum of R = ax + by is **at the vertices of** the region



i) **vertex A** is the intersection of $\begin{cases} y = 2 \\ 2x + 3y = 30 \end{cases}$

$\Rightarrow 2x + 6 = 30 \Rightarrow 2x = 24$

$\therefore$ vertex A = (12,2)

Vertex B is the intersection of $\begin{cases} y = 6 \\ 2x + 3y = 30 \end{cases} \Rightarrow 2x = 30 - 18$

$\Rightarrow x = 6 \quad \therefore$ Vertex B = (6,6)

Vertex C is the intersection of $\begin{cases} y = 2x \\ y = 6 \end{cases}$

$\therefore$ Vertex C = (3,6)

Vertex D is the intersection of $\begin{cases} y = 2x \\ y = 2 \end{cases} \Rightarrow 2x = 2 \Rightarrow x = 1$

$\therefore$ Vertex D = (1,2)

R	At vertex A (12,2)	At vertex B(6,6)	At vertex C(3,6)	At vertex D(1,2)
y − x	2−12=− 10	6 − 6 = 0	6 − 3 = 3	2 − 1 = 1
2x + 5y	2(12)+5(2)=34	2(6)+5(6)= 24	2(3)+6(5)=36	2(1)+5(2)=12
3x + 8y	3(12)+8(2)=52	3(6)+8(6)= 66	3(3)+6(8)=57	3(1)+8(2)=19

Therefore, from the table

a) y − x is minimum at vertex (12,2) = − 10

b) y − x is maximum at vertex (3,6) = 3

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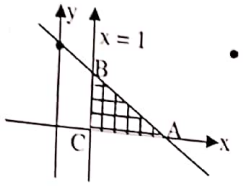
c) $2x + 5y$ is minimum at vertex $(1,2) = 12$
$2x + 5y$ is maximum at vertex $(6,6) = 42$

d) $3x + 8y$ is maximum at vertex $(6,6) = 66$

33. Let $R = \{(x,y): y + 4x \le 6, x \ge 1\}$ and $y \ge 0\}$.
What element of **R** makes $x - y$ maximum.

A) $(1,2)$ B. $(1,0)$ C) $\left(\frac{1}{2},4\right)$ D. $\left(\frac{3}{2},0\right)$

Solution:



- **Vertex A** is the intersection of $\begin{cases} y = 0 \\ y + 4x = 6 \end{cases}$

$\Rightarrow 4x = 6 \Rightarrow x = \frac{3}{2}$ $\therefore$ Vertex A $= \left(\frac{3}{2},0\right)$

Vertex B is the intersection of $\begin{cases} x = 1 \\ y + 4x = 6 \end{cases}$

$\Rightarrow y+4(1) = 6 \Rightarrow y = 2$ $\therefore$ Vertex B $= (1,2)$

Vertex C is the intersection of $\begin{cases} x = 1 \\ y = 0 \end{cases}$

$\therefore$ Vertex C $= (1,0)$

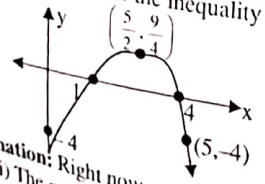
Hence, at A, $\left(\frac{3}{2},0\right)$: $R = x - y = \frac{3}{2} \longleftarrow$ maximum

at B $(1,2)$: $R = x - y = 1 - 2 = -1 \longleftarrow$ minimum

at C $(1,0)$: $R = x - y = 1 - 0 = 1$

34. If the figure below is the graph of the function $y = x^2 + 5x - 4$, what is the ***Solution*** set of the inequality $-x^2 + 5x - 4 \le 0$

Answer: D



A. $(-\infty,1)\cup(4,\infty)$

B. $\left(\frac{5}{2},\infty\right)$

C. $[1,4]$

D. $(-\infty,1]\cup[4,\infty)$

Explanation: Right now, we see that:

i) The graph of $-x^2 + 5x - 4$ **lies above the x – axis**

for $x \in [1,4]$ that is $-x^2 + 5x - 4 \ge 0$

ii) The graph of $-x^2 + 5x - 4$ **lies below the x – axis**
for $x \le 1$ or $x \ge 4$ that is
$-x^2 + 5x - 4 \le 0$, $\therefore$ **Answer: D**

35. Find the ***Solution*** set of the following inequality:

a) $9x^2 - 6x + 1 < 0$ c. $9x^2 - 6x + 1 > 0$

b) $9x^2 - 6x + 1 \le 0$ d. $9x^2 - 6x + 1 \ge 0$

Solution: $9x^2 - 6x+1$ is same as $(3x - 1)^2$ therefore,

a) $9x^2 - 6x + 1 < 0$ means $(3x - 1)^2 < 0$
Since $(3x - 1)^2$ is always non – negative
$\Rightarrow (3x - 1)^2 < 0$ is false $\therefore$ S.S $= \phi$

b) $9x^2 - 6x + 1 \le 0 \Leftrightarrow (3x - 1)^2 \le 0$ is true for $x = \frac{1}{3}$

$\therefore$ S.S $= \left\{\frac{1}{3}\right\}$

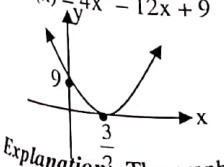
c) $9x^2 - 6x + 1 > 0 \Leftrightarrow (3x - 1)^2 > 0$ is true for all $IR\backslash\left\{\frac{1}{3}\right\}$

Because when $x = \frac{1}{3}$, $(3x - 1)^2 < 0$.

$\therefore$ S.S $= \left(-\infty,\frac{1}{3}\right)\cup\left(\frac{1}{3},\infty\right)$

d) $9x^2 - 6x + 1 \ge 0$ is true for every real number x
$\therefore$ S.S $= IR = (-\infty,\infty)$

36. The figure shown below is the graph of a quadratic function given by $f(x) = 4x^2 - 12x + 9$



Using this graph find the ***Solution*** set of the inequality $4x^2 - 12x + 9 \le 0$

A. { } B. $(-\infty,0]$ C. $\left\{\frac{3}{2}\right\}$ D. IR

Explanation: The graph of $f(x) = 4x^2 - 12+ 9 = (2x - 3)^2$ touch the x – axis at $x = \frac{3}{2}$ and lies above the x – axis

So that $4x^2 - 12x + 9 \ge 0$ for all real number x

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However $4x^2 - 12x + 9 \le 0 \Leftrightarrow (2x-3)^2 \le 0$ is true only for $x = \frac{3}{2}$

37. What is the ***Solution*** set of the inequality $4x^2 + 4x + 1 > 0$ ∴ Answer: C

A. $IR\setminus\left\{\frac{-1}{2}\right\}$ B. $\left(\frac{-1}{2},\frac{1}{2}\right)$ C. $\left(-\infty,\frac{-1}{2}\right)\cup\left(\frac{1}{2},\infty\right)$ D. IR

Explanation: $4x^2 + 4x + 1 > 0 \Leftrightarrow (2x+1)^2 > 0$ for all x

except at $x = \frac{1}{2}$, ∴ **Answer: A**

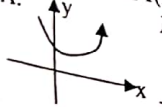
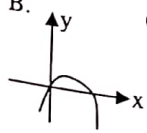
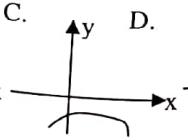
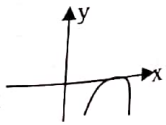
38. **For what values of k, the graph of f(x)**
i) does not crosses the x – axis
ii) crosses the x – axis
a) $f(x) = x^2 + kx + 4$ b) $f(x) = kx^2 - 2x + k$

***Explanation*: Remember that**
The graph of $f(x) = ax^2 + bx + c$
i) does not cross the x – axis if $b^2 - 4ac < 0$
ii) crosses the x – axis if $b^2 - 4ac \ge 0$

a) $f(x) = x^2 + kx + 4$, $a = 1$, $b = k$, $c = 4$
i) $b^2 - 4ac < 0 \Rightarrow k^2 - 4(1)(4) < 0$
$\Rightarrow k^2 - 16 < 0 \Leftrightarrow -4 < k < 4$
ii) $b^2 - 4ac \ge 0 \Rightarrow k^2 - 16 \ge 0 \Rightarrow k^2 \ge 16$
S.S = $\{k: k \le -4 \text{ or } k \ge 4\} = (-\infty, -4]\cup[4,\infty)$

b) $f(x) = kx^2 - 2x + k$, with $a = k$, $b = -2$, $c = k$
i) $b^2 - 4ac < 0 \Rightarrow (-2^2) - 4(k)(k) < 0$
$\Rightarrow 4 - 4k^2 < 0 \Leftrightarrow 4 < 4k^2 \Rightarrow 1 < k^2$
$\Rightarrow k^2 > 1$ if $k < -1$ or $k > 1$
∴ S.S = $\{k: k < -1 \text{ or } k > 1\} = (-\infty,-1)\cup(1,\infty)$
ii) $b^2 - 4ac \ge 0 \Rightarrow 4 - 4k^2 \ge 0 \Rightarrow 1 \ge k^2$
$\Rightarrow k^2 \le 1$ if $-1 \le k \le 1$
S.S = $\{k: -1 \le k < 1\}$

39. Let $a \ne 0$ and $b^2 - 4ac \le 0$, then which of the following graphs cannot be the graph of $\mathbf{f(x) = ax^2 + bx + c}$

A.  B.  C.  D. 

Explanation: A. The graph does not cross the x – axis and completely lies above the x – axis
∴ $b^2 - 4ac < 0 \Rightarrow ax^2 + bx + c > 0$ for all x

B. The graph crosses the x – axis
∴ $b^2 - 4ac \ge 0$

C. The graph does not crosses the x – axis and completely the graph below the x – axis
∴ $b^2 - 4ac < 0 \Rightarrow ax^2 + bx + c < 0$ for all real number x

D. The graph touch the x – axis (but not cross) and the graph lies below the x – axis
∴ $b^2 - 4ac \le 0 \Rightarrow ax^2 + bx + c \le 0$, ∴ **Answer: D**

40. For what a values of k, the quadratic equation $kx^2 + 2x - k = 0$, has
i) two real ***Solution*** ii) No real ***Solution***

Explanation: use $b^2 - 4ac > 0$ two ***Solution***
$b^2 - 4ac < 0$ if No ***Solution***:

i) $kx^2 + 2x - k = 0$ has two ***Solution*** if
$b^2 - 4ac > 0 \Rightarrow (2)^2 - 4(k)(-k) > 0$
$\Rightarrow 4 + 4k^2 > 0$ for all real number k.
∴ For $k = IR = (-\infty,\infty)$ has two ***Solution***.
ii) $kx^2 + 2x - k = 0$ has no real ***Solution*** if
$b^2 - 4ac < 0 \Rightarrow 4 + 4k^2 < 0$ is false
because $4 + 4k^2 > 0$ for all values of k
∴ $k = \phi$

41. If $-1 < a < 1$, then what is the number of ***Solution***s of $ax^2 + 2x + a = 0$
A. 2 B. 0 C. infinite D. 1

Solution: use $b^2 - 4ac \Rightarrow (2)^2 - 4(a)(a) = 4 - 4ac^2$
If it has two ***Solution*** then we use $b^2 - 4ac > 0$
$\Rightarrow 4 - 4a^2 > 0 \Rightarrow 1 - a^2 > 0 \Leftrightarrow a^2 < 1$
∴ S.S = $\{a: -1 < a < 1\}$, for two ***Solution*** ∴ **Answer: A**

42. **What are the cases of coefficient if the Solution set of**
a) $ax^2 + bx + c < 0$ is the set of all real number
b) $ax^2 + bx + c > 0$ is the set of all real number
c) $ax^2 + bx + c < 0$ is empty set
d) $ax^2 + bx + c > 0$ is empty set

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Explanation:

a) $ax^2 + bx + c < 0$ for all $x \in (-\infty,\infty)$
When $a < 0$ and $b^2 - 4ac < 0$.
i.e. the graph lies completely below the x – axis
∴ In this case S.S = R = $(-\infty,\infty)$

b) $ax^2 + bx + c > 0$ for each $x \in (-\infty,\infty)$
When $a > 0$ and $b^2 - 4ac < 0$.
i.e. the graph open upward and lies above the x – axis

c) $ax^2 + bx + c < 0$ is empty set
When $a > 0$ and $b^2 - 4ac <$

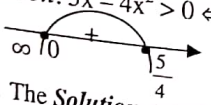
d) $ax^2 + bx + c > 0$ is empty set
When $a < 0$ and $b^2 - 4ac < 0$

43. The ***Solution*** set of $5x - 4x^2 > 0$ is

A. $\left(0,\frac{5}{4}\right)$ C. $\left(0,\frac{4}{5}\right)$

B. $(-\infty,0)\cup\left(\frac{5}{4},\infty\right)$ D. $(-\infty,0)\cup\left(\frac{4}{5},\infty\right)$

Solution: $5x - 4x^2 > 0 \Leftrightarrow x(5 - 4x) > 0$



$\Rightarrow \{x: 0 < x < \frac{5}{4}\}$, ∴ **Answer: A**

44. The ***Solution*** set of the inequality $2x^2 + 5x - 3 < 0$

A. $\left[-3,\frac{1}{2}\right]$ C. $\left(\frac{1}{2},3\right)$

B. $(-\infty,-3)\cup\left(\frac{1}{2},\infty\right)$ D. $\left(-3,\frac{1}{2}\right)$

Solution: $2x^2 + 5x - 3 < 0 \Leftrightarrow 2x^2 + 6x - x - 3 < 0$

$\Rightarrow 2x(x + 3) - (x + 3) < 0$
$\Rightarrow (2x - 1)(x + 3) < 0$
∴ S.S = $\{x: -3 < x < \frac{1}{2}\}$, ∴ **Answer: D**

45. The ***Solution*** set of the inequality
a) $(x - 2)^2 \le 4 - x$ b) $(x - 1)^2 \ge x + 1$

Solution: Here

a) $(x-2)^2 \le 4-x \Leftrightarrow x^2-4x+4 \le 4-x \Rightarrow x^2-3x \le 0$

$\Rightarrow x(x-3) \le 0 \Leftrightarrow 0 \le x \le 3$
∴ S.S = $\{x: 0 \le x \le 3\} = [0,3]$

b) $(x-1)^2 \ge x+1 \Leftrightarrow x^2-2x+1 \ge x+1 \Rightarrow x^2-3x \ge 0$
$\Rightarrow x(x-3) \ge 0 \Leftrightarrow x \le 0$ Or $x \ge 3$
∴ S.S = $\{x: x \le 0$, Or $x \ge 3\} = (-\infty,0)\cup(3,\infty)$

46. The *Solution* set of the inequality $(x - 2)^2 \le 2 - x$ is
A. [1,2] B. $(-\infty,1]\cup[2,\infty)$ C. [1, 3] D. $(-\infty,1]\cup[3,\infty)$

Solution: $(x - 2)^2 \le 2 - x \Leftrightarrow x^2 - 4x + 4 \le 2 - x$
$\Rightarrow x^2 - 3x + 2 \le 0 \Leftrightarrow (x - 2)(x - 1) \le 0$
∴ S.S = $\{x: 1 \le x \le 2\}$ ∴ **Answer: A**

47. Determine the set of x that belongs, for $a < 2$ in each of the following expression.
a) $(a - 2)x > (x^2 - 4)$ b) $(a - 2)x < (x^2 + a - 6)$

Explanation: If $a < 2$, then $a - 2$ is negative when you divide by negative, you have to change the inequality

a) $(a-2)x > (x^2-4) \Leftrightarrow x < \frac{(a-2)(a+2)}{a-2}$

$\Rightarrow x < a + 2$
∴ The set of all x belongs to $(-\infty, a + 2)$

b) $(a - 2)x < (x^2 + a - 6) \Leftrightarrow (a - 2)x > (a - 2)(a+3)$

$\Rightarrow x > \frac{(a-2)(a+3)}{a-2} \Rightarrow x > a + 3.$

∴ The set of all x belongs to $(a + 3,\infty)$

48. If $a < 1$ and $(a - 1)x > (a^2 - 1)$ then to which of the following sets does x – belong . . . **EGSECE**
A. $(a + 1, \infty)$ B. $(-\infty, 1)$ C. ϕ D. $(-\infty, a + 1)$
Explanation: **Exercise left for you** **(Answer: D)**

49. Suppose the equation $x^3 + 2x^2 + kx = 0$ has exactly one real root. Then the set of value of k is . . . **EGSECE**
A. $\{k|k > 1\}$ B. $\{k|k \le 1\}$ C. $\{k|k \ge 1\}$ D. $k|k < 1\}$
Solution: $x^3 + 2x^2 + kx = 0 \Leftrightarrow x(x^2 + 2x + k) = 0$
We set, $x^2 + 2x + k$ to have no *Solution*
$\Rightarrow b^2 - 4ac < 0 \Rightarrow (2)^2 - 4(1)k < 0$
$\Rightarrow 4 - 4k < 0 \Rightarrow 4 < 4k \Rightarrow 1 < k$

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$\therefore$ S.S = {k:k > 1}

50. For what value of b the quadratic equation $2x^2 + bx + 3 = 0$ has two distinct real root.

$\therefore$ **Answer: A**

A. $B < -2\sqrt{6}$

B. $B > -2\sqrt{6}$

C. $b < -2\sqrt{6}$ Or $b > 2\sqrt{6}$

D. $b > -2\sqrt{6}$ Or $b < 2\sqrt{6}$

Solution: set $b^2 - 4ac > 0$

$\Rightarrow b^2 - 4(2)(3) > 0 \Rightarrow b^2 - 24 > 0$

$\Rightarrow b^2 > 24 \Leftrightarrow |b| > \sqrt{24}$

$\Rightarrow |b| > 2\sqrt{6}$

$\therefore$ S.S = {b:b < $-2\sqrt{6}$ Or b > $2\sqrt{6}$ }, $\therefore$ **Answer: C**

51. The ***Solution*** set of the inequality $|x + 1| < 2x$ is

A. {x: x ≥ 1}

B. B. $\{x: x > \frac{-1}{3}\}$

C. {x: x > 1}

D. $\left\{x : x < \frac{1}{3} \cup x > 1\right\}$

Solution: $|x + 1| < 2x \Leftrightarrow -2x < x + 1 < 2x$

$\Rightarrow -2x < x + 1$ and $x + 1 < 2x$

$\Rightarrow -1 < 3x$ and $1 < x$

$\Rightarrow x > \frac{-1}{3}$ and $x > 1$ ⟵ the intersection

$\therefore$ S.S = {x: x > 1}, $\therefore$ **Answer: C**

52. **Determine the *Solution* set of each of the following**

a) $|2 - 5x| > -7$

b) $|4x + 3| > -6$

c) $|2 - 5x| < -7$

d) $|4x + 3| < -6$

Solution: **Remember that $|ax + b| \geq 0$ for all x.**

a) $|2 - 5x| > -7$ is true for all real number x

$\therefore$ S.S = IR = $(-\infty, \infty)$

b) $|4x + 3| \geq 0 \Rightarrow |4x + 3| > -6$ is true for all x

$\therefore$ S.S = IR = $(-\infty, \infty)$

c) $|2 - 5x| \geq 0 \Rightarrow |2 - 5x| < -7$ is false

$\therefore$ S.S = ϕ

d) $|4x + 3| < -6$ is false

$\therefore$ S.S = ϕ

53. **What are condition (cases) for**

i) $|a+b| = |a| + |b|$

ii) $|a + b| < |a| + |b|$

Explanation:

i) $|a+b| = |a| + |b|$ either a < 0 and b < 0 or a > 0 and b > 0 (i.e. if both 'a' and 'b' have **the same** sing)

ii) $|a+b| < |a| + |b|$, for both 'a' and 'b' have **opposite** sign.

Review Exercise to Unit Nine

1. **Solve and write each of the following using interval notation:**

a) $\frac{1}{3}x + 2 > \frac{1}{2} - 2x$ b) $-1 < 3 - 2x < 9 - x$

2. Describe each of the following using intervals:

a) {x: x∈R\{−2}} c) {x: x∈R and x∉(−3,4)}

b) {x∈R: x∉[−2,2)} d) {x: x∈R\{0,1}}

3. Solve each of the following absolute value inequality.

a) $|2x+1| > 0$ b) $|2x + 1| \geq 0$ c) $|x^2 - 4| > 0$

d) $|x^2 - 4| \geq 0$ e) $|x^2 + 3x + 2| > 0$ f) $|4x + 6| > -5$

g) $|x^2 + 6x| > -8$

4. Solve each of the following absolute value inequality.

a) $|3x-2| \leq 0$ b) $|3x - 2| < 0$ c) $|4 - x^2| \leq 0$

d) $|x| < -3$ e) $|x + 1| < -10$ f) $|9 - x^2| \leq -1$

5. Solve:

a) $|2x + 3| > 4$ b) $|3x + 5| < 8$ c) $|x^3 + 1| < 7$

6. Solve

a) $|x-4| = 4 - x$

b) $|3 - x| = x - 3$

c) $|3 - x| = x - 3$

d) $|x^2 - 9| = x - 3$

e) $|x^2 - 1| = x^2 - 1$

e) $|x^2 - 1| = 1 - x^2$

7. Find the ***Solution*** set of the following inequality. **(describe the domain and range)**

a) $\begin{cases} x + y > 2 \\ x + y < 6 \end{cases}$ b) $\begin{cases} y \geq -4x + 6 \\ y < x - 4 \end{cases}$ c) $\begin{cases} x + y > 1 \\ y - x < 1 \\ x < 4 \end{cases}$

8. Solve each of the following inequality

a) $x^2 < x$ b) $(x - 1)^2 < x - 1$ c) $(x + 2)^2 > x + 4$

d) $(2 - x)(x - 6) > 0$ e) $2x^2 \leq 1 - x$ f) $(x - 2)^2 \leq 4$

g) $(x - 3)^2 - 1 \geq 0$ h) $x^2 > 6x - 9$ i) $-1(x - 4)^2 + 6 > 0$

9. **Solve each of the following inequality**

a) $x^2 + 8x + 16 > 0$ b) $4x^2 - 12x + 9 > 0$ c) $9x^2 + 24x + 16 > 0$

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d) $x^2 - 6x + 9 \geq 0$ e) $x^2 - 4x + 4 \geq 0$ f) $9x^2 - 6x + 1 \geq 0$

g) $-x^2 + 4x - 4 < 0$ h) $-x^2 + 4x - 4 \geq 0$ i) $-x^2 + 6x - 4 > 0$

10. **Solve each of the following inequality.**

a) $x^2 + 12x + 36 \leq 0$ b) $4x^2 - 12x + 9 \leq 0$ c) $9x^2 + 24x + 16 \leq 0$

d) $-9x^2 - 12x - 4 \leq 0$ e) $-9x^2 + 12x - 4 < 0$ f) $4x^2 + 4x + 1 < 0$

11. Solve each of the following inequality

a) $x^2 + 2x + 3 > 0$ b) $4x^2 + 4x + 3 > 0$

c) $2x^2 + 3x + 2 > 0$ d) $-3x^2 + 2x - 1 < 0$

e) $-3x^2 + 2x - 1 > 0$ f) $3x^2 - 4x + 2 < 0$

g) $3x^2 - 4x + 2 > 0$ h) $5x^2 - 6x + 3 > 0$

12. **What must the value of k be so that:**

a) $kx^2 - 2x + 1 \leq 0$ for each real number x.

b) $kx^2 + kx + 2 \geq 0$ for each real number x.

c) $kx^2 - 4x + k < 0$ for each real number x.

d) $kx^2 - kx + 1 > 0$ for each real number x.

13. **For what value of k does each of the following have**

i) one real root ii) two real root iii) No real root

a) $x^2 + kx + 8 = 0$ b) $x^2 + (3 - k)x + 9 = 0$ c) $kx^2 + 2x + k = 0$

14. **Determine the set of x that belongs for**

$a < 3$, in each of the following expression

a) $(a - 3)x < (a^2 - 9)$ b) $(a - 3)x > (a^2 - 5a + 6)$

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8. a) A + B b) B – A C) $\frac{C}{A}$ D) $\frac{A+B}{C}$ E) -2A +B–3C

9. a) $(-\infty, 4)$ b) (-2, 2)

10. a) $\sqrt{\frac{2}{3}e^3}$ b) $\frac{27}{2}$ c) $\frac{8}{729}$ d) $-\frac{1}{2}$

11. a) 0.4307 b) 1.3652 c) 2.322 d) 0.449

12. a) $\frac{1}{5}$ 13) a) 5 years b) 9 years

14. a) $\log_4^x = \log 18 \Leftrightarrow x \log 4 = 2\log 3 + \log 2$

$$\Rightarrow x = \frac{2\log 3 + \log 2}{2\log 2} = \frac{2(0.4771) + 0.3010}{2(0.3010)} \approx 2.09$$

a) $x = \frac{3\log 2 + \log 3}{\log 2 + \log 3}$ c) x = 0.92

15. a) x = 9 or x = $\frac{1}{9}$ b) $\log_x^2 = \frac{\log_x^2}{6} \Leftrightarrow 6\log_x^2 = \frac{1}{\log_x^2}$

$$\Rightarrow \left(\log_x^2\right)^2 = \frac{1}{6} \Rightarrow x = 2^{\sqrt{6}} \text{ or } x = 2^{-\sqrt{16}}$$

16. a) 2.62 b) 262 c) 0.00262 d) 0.056 e) 2.4183
f) –2.5817 g) – 0.2518

17. a) $\log_2^8 = 3$

Answer key Unit 9

1. a) $\left(\frac{-4}{7}, \infty\right)$ b) (–6, 2)

2. a) $(-\infty,-2) \cup (-2,\infty)$ b) $(-\infty,-2) \cup (2,\infty)$
c) $(-\infty,-3) \cup [4,\infty)$ d) $(-\infty,0) \cup (0,1) \cup (1,\infty)$

3. a) $IR \setminus \left\{-\frac{1}{2}\right\}$ b) IR c) $IR \setminus \{-2,2\}$ d) IR e) $IR \setminus \{-2,-1\}$

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f) IR g) IR

4. a) $\left\{\frac{2}{3}\right\}$ b) ϕ c) $\{-2,2\}$ d) ϕ e) ϕ f) ϕ

5. a) $\left(-\infty,\frac{-7}{2}\right)\cup\left(\frac{1}{2},\infty\right)$ b) $\left(\frac{-13}{3},1\right)$ c) $\left(-2,\sqrt[1]{6}\right)$

6. a) $\{x: x \le 4\}$ b) $\{x: x \ge 3\}$ c) $\{-3, 2, 4\}$ d) $\{3\}$
e) IR\ $(-1, 1)$ f) $[-1, 1]$

7. a) $\{(x,y): -\infty < x < \infty$ and $z - x < y < 6 - x\}$
- Domain = IR and range = IR

b) $\{(x,y): 2 < x < \infty$ and $6 - 4x \le y < x - 4\}$
- Domain= $(2, \infty)$ and range = IR

c) $\{(x,y): 0 < x < 4$ and $1 - x < y < x + 1\}$
- Domain = (0,4) and range = (–3,5)

8. a) (0,8) b) (1,2) c) $(-\infty,-3) \cup (0, \infty)$ d) (2,6) e) $\left[-1,\frac{1}{2}\right]$ f) [0,4]
g) $(-\infty,2) \cup [4, \infty)$ h) IR\{3} i) $\left(-\sqrt{6}+4,4+\sqrt{6}\right)$

9. a) IR\{4} b) $IR \setminus \left\{\frac{3}{2}\right\}$ c) $IR \setminus \left\{\frac{-4}{3}\right\}$ d) IR e) IR f) IR
g) IR \{2} h) {2} i) ϕ

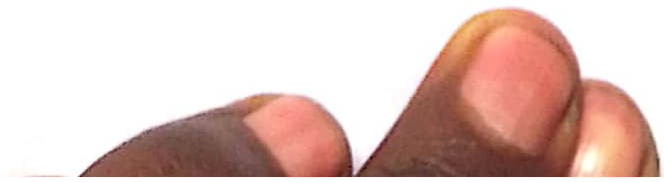
10. a) $\{-6\}$ b) $\left\{\frac{3}{2}\right\}$ c) $\left\{\frac{-4}{3}\right\}$ d) IR e) IR\ $\left\{\frac{2}{3}\right\}$ f) IR\ $\left\{\frac{-1}{2}\right\}$

11. a) IR b) IR c) IR d) Ir e) ϕ f) ϕ g) IR h) IR

12. a) $\{k: k \ge 1\}$ b) $\{k: 0 < k \le 8\}$
c) $\{k: k < -2, k > 2\}$ d) $\{k: 0 < k < 4\}$

13. a) i) $\{k: k = \pm 4\sqrt{2}\}$ ii) $\{k: k < -4\sqrt{2}, k > 4\sqrt{2}\}$
iii) $k: -4\sqrt{2} < k < 4\sqrt{2}$
b) i) $\{k: k = 9, -3\}$ ii) $\{k: k < -3 \text{ Vk} > 9\}$ iii) $\{k: -3 < k < 9\}$
c) i) $\{k: k = -1$ Or $k = 1\}$ ii) $\{k: k < -1$ Or $k > 1\}$ iii) $\{k: -1 < k < 1\}$

14. a) $(a + 3, \infty)$ b) $(-\infty, a - 2)$



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UNIT TEN

Coordinate Geometry

Distance Between Two points

- The distance **d** between points $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ is given by the formula.

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Illustrative Example

1. Find the distance between the points
   a) $P_1(2,5)$ and $P_2(5, 9)$ b) $P_1(4,2)$ and $P_2(8,0)$
   c) $P_1(0,x)$ and $P_2(x, 0)$ d) $P_1(x,-y)$ and $P_2(-y,x)$
   e) $P_1(a,4a)$ and $P_2(4a, a)$ f) $P_1(2,a)$ and $P_2(a,2)$

Solution: Let d be the distance between points.

a) $d = \sqrt{(5-2)^2 + (9-5)^2} = \sqrt{9+16} = 5$unit

b) $d = \sqrt{(8-4)^2 + (0-2)^2} = \sqrt{16+4} = 2\sqrt{5}$ unit

c) $d = \sqrt{(8-0)^2 + (0-x)^2} = \sqrt{x^2 + x^2} = \sqrt{2x^2} = \sqrt{2}\,|x|$ unit

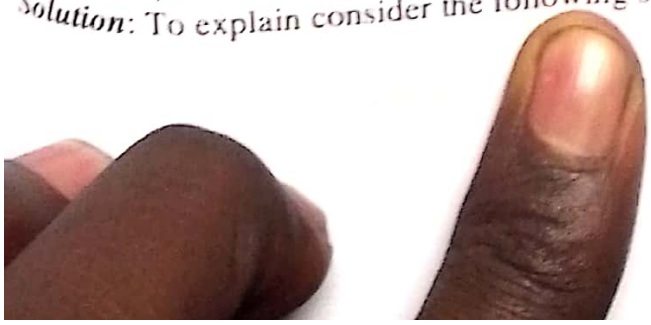
d) $d = \sqrt{(x+y)^2 + (x+y)^2} = \sqrt{2(x+y)^2} = 2|x+y|$ unit

e) $d = \sqrt{(4a-a)^2 + (a-4a)^2} = \sqrt{9a^2 + 9a^2} = 3\sqrt{2}\,|a|$ unit

f) $d = \sqrt{(a-2)^2 + (2-a)^2} = \sqrt{2(2-a)^2} = \sqrt{2}\,|a-2|$ unit

2. If A(1,5) and B(4, 2) are two opposite vertices of a square, then
   Find a) the length of the diagonal of square
   b) the area of the square
   c) the coordinate of the other vertices

Solution: To explain consider the following square



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e) $3x - 2y = 0$ and $3x + 2y = 0$

f) $4x + 5y = -1$ and $8x + 10y = 3$

Solution: we know that the slope of $Ax + By + C = 0$ is $\frac{-A}{B}$

a) Slope of $3x - y + 5 = 0$ is $\frac{-3}{-1} = 3$

Slope of $x + 3y - 1$ is $\frac{-1}{3}$ provided that $3\left(\frac{-1}{3}\right) = -1$

Therefore, they are perpendicular

b) Slope of $3x - 4y + 1 = 0$ is $\frac{-3}{-4} = \frac{3}{4}$

Slope of $4x - 3y + 1 = 0$ is $\frac{-4}{-3} = \frac{3}{4}$ implies $\left(\frac{3}{4}\right)\left(\frac{4}{3}\right) = -1$

Therefore, neither perpendicular nor parallel

c) Slope of $3x + 2y = 6$ is $\frac{-3}{2}$

Slope of $6x + 4y = 10$ is $\frac{-6}{4} = \frac{-3}{2}$ implies $\frac{-3}{2} = \frac{-3}{2}$.

Therefore, they are parallel.

d) **Exercise left for you (Tip!)**

e) Slope of $3x - 2y = 0$ is $\frac{3}{2}$ and slope of $3x + 2y = 0$ is $\frac{-3}{2}$

$\Rightarrow \frac{3}{2} \neq \frac{-3}{2}$, therefore, neither parallel nor perpendicular

f) **Exercise left for you (Tip!)**

Solved problem

48. For what values of k, so that the line $8x + ky - 6 = 0$ will be
   a) Parallel to the line $4x + 6y + 7 = 0$
   b) Parallel to the line $kx + 2y = 5$
   c) Perpendicular to the line $4x + 5y + 1 = 0$
   d) Perpendicular to the line $-2x + ky = 3$

Solution: For $\ell 1 \| \ell_2$ we set $m_1 = m_2$

For $\ell 1 \perp \ell_2$ we set $m_1 \cdot m_2 = -1$

We know that slope of $8x + ky - 6 = 0$ is $\frac{-8}{k}$

a) Slope of $4x + 6y + 7 = 0$ is $\frac{-4}{6} = \frac{-2}{3}$

Since, parallel implies $\frac{-4}{6} = \frac{-8}{k}$ provide that $\frac{2}{3} = \frac{8}{k}$

$\Rightarrow 2k = 24$ implies $k = 12$

b) Slope of $kx + 2y = 5$ is $\frac{-k}{2}$, since parallel to $8x + ky = 6$

$\Rightarrow \frac{-k}{2} = \frac{-8}{k}$ implies $k^2 = 16$

Therefore, $k = 4$ or $k = -4$

c) Slope of $4x + 5y + 1 = 0$ is $\frac{-4}{5}$, since perpendicular to $8x + ky = 6$

$\Rightarrow \left(\frac{-4}{5}\right)\left(\frac{-8}{k}\right) = -1$ implies $-5k = 32$

Therefore, $k = \frac{-32}{5}$

d) Slope of $2x+ky=3$ is $\frac{-2}{k}$ since perpendicular to $8x + ky = 6$

$\Rightarrow \left(\frac{2}{k}\right)\left(\frac{-8}{k}\right) = -1$ implies $\frac{-16}{k^2} = -1$

$\Rightarrow k^2 = 16$ provided that $k = -4$ or $k = 4$

49. Given that the equation of a line are ℓ_1: $(6 - k)x + y = -1$ and ℓ_2: $8x + ky = 7$

Find the value of k so that:

a) ℓ_1 is parallel to ℓ_2 b) ℓ_1 is perpendicular to ℓ_2

Solution: explained as slope of ℓ_1 is $\frac{-(6-k)}{1} = k - 6$

Slope of ℓ_2 is $\frac{-8}{k}$

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a) $\ell_1 \| \ell_2$ implies slope of ℓ_1 = slope of ℓ_2

$\Leftrightarrow k - 6 = \dfrac{-8}{k}$ provided that $k^2 - 6k \neq -8$

$\Rightarrow k^2 - 6k + 8 = 0$ implies $(k-2)(k-4) = 0$

Therefore, $k = 2$ or $k = 4$

b) $\ell_1 \perp \ell_2$ implies $m_1 \cdot m_2 = -1 \Leftrightarrow (k-6)\left(\dfrac{-8}{k}\right) = -1$

$\Rightarrow -8k + 48 = -k$ implies $7k = 48$

Therefore, $k = \dfrac{48}{7}$

50. If the lines $y = kx + 8$ and $ky = 9x$ are parallel then what is the set of value of k.

A. $\left\{\dfrac{1}{3}, \dfrac{-1}{3}\right\}$ B. $\{3, -3\}$ C. $\{0,9\}$ D. $\{2, -2\}$

Solution: Slope of $y = kx + 8$ is k

Slope of $ky = 9x$ is $\dfrac{9}{k}$ since parallel

$\Rightarrow k = \dfrac{9}{k}$ implies $k^2 = 9$, $k = 3, -3$, **∴ Answer B**

51. Which of the following lines is parallel to the line $5x - 2y = 0$.

A. $-5x - 2y = 1$ B. $-5x + 2y = 6$ C. $y = \dfrac{-5}{2}$ D. $2x + 5y = -4$

Solution: slope of $5x - 2y = 0$ is $\dfrac{5}{2}$

A. Slope of $-5x - 2y = 1$ is $-\left(\dfrac{-5}{-2}\right) = \dfrac{-5}{2}$ ⟵ not parallel

B. Slope of $-5x + 2y = 6$ is $-\left(\dfrac{-5}{2}\right) = \dfrac{5}{2}$ ⟵ parallel

Therefore, Answer: B

52. The equation of the line that passes through the point $(2, -1)$ and perpendicular to the line $2x + 4y + 3 = 0$ is given by:

A. $-2x + y - 4 = 0$ B. $x + 2y = 0$ C. $-2x + y + 5 = 0$ D. $x - 2y + 2 = 0$

Solution: The equation of line through (x_0, y_0) is parallel to $Ax + By + C = 0$ is given by $A(y - y_0) - B(x - x_0) = 0$

$\Rightarrow (2, -1)$: $2x + 4y + 3 = 0$ is given by: $2(y+1) - 4(x-2) = 0$

$\Rightarrow -2x + y + 5 = 0$ **therefore, Answer C**

53. What is the equation of a line whose x – intercept and y – intercept are 1 and – 2 respectively?

A. $x - y + 2 = 0$ B. $-2x + y - 1 = 0$ C. $x - 2y - 1 = 0$ D. $2x - y - 2 = 0$

Solution: we know that the intercept form $\dfrac{x}{a} + \dfrac{y}{b} = 1$

$\Rightarrow \dfrac{x}{1} + \dfrac{y}{-2} = 1$ implies $-2x + y = -2 \leftarrow$ (Divided by (-1))

$\Rightarrow 2x - y - 2 = 0$, **Therefore,** **Answer: D**

54. If a line with x – intercept 4 and y – intercept – 6 is given, then its slope is equal to _____

A. $\dfrac{2}{3}$ B. $\dfrac{3}{2}$ C. $\dfrac{-2}{3}$ D. $\dfrac{-3}{2}$

Solution: we use $\dfrac{x}{a} + \dfrac{y}{b} = 1 \longleftarrow$ slope $\dfrac{-b}{a}$

$\Rightarrow \dfrac{x}{4} + \dfrac{y}{-6} = 1 \longleftarrow$ slope, $m = \left(\dfrac{-6}{4}\right) = \dfrac{-3}{2}$ **∴ Answer: B**

55. Given that A(3,8), B(5,2k), C(2,4) and D(1,2) so that for what values of k line through AB is perpendicular to line CD.

A. $\dfrac{7}{2}$ B. -6 C. 2 D. $\dfrac{-7}{2}$

Solution: Slope of $AB = \dfrac{2k-8}{5-2} = \dfrac{2(k-4)}{2} = k - 4$

Slope of $CD = \dfrac{2-4}{1-2} = \dfrac{-2}{-1} = 2$, since $AB \perp CD$

$\Rightarrow (k-4)(2) = -1$ implies $2k - 8 = -1$

$\Rightarrow 2k = 7$, implies $k = \dfrac{7}{2}$, **∴ Answer: A**

56. A triangle has vertices at A(– 3,3), B(1,5) and C(3, – 1) what are the coordinates of the point at which the median intersects . . . EGSECE 2003

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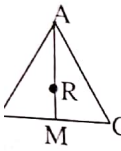
A. $\left(\frac{5}{3},\frac{4}{3}\right)$ B. $\left(\frac{1}{3},\frac{7}{3}\right)$ C. $\left(\frac{-4}{3},\frac{8}{3}\right)$ D. $\left(\frac{1}{3},\frac{11}{3}\right)$

olution: First determine the mid – point BC = M = $\left(\frac{3+1}{2},\frac{-1+5}{2}\right)$



$\Rightarrow$ M = (2,2) and AR: RM = 2:1

$\Rightarrow R(x_0, y_0) = \left(\frac{(1)(-3)+2(2)}{2+1},\frac{(1)(3)+2(2)}{2+1}\right) = \left(\frac{1}{3},\frac{7}{3}\right)$

$\therefore$ Answer: B

7. Determine the value of K for which the line $(k-3)x + (k^2-4)y + 2k^2 + 7k + 5 = 0$ is
   a) Parallel to $2x + 3y = 6$
   b) Parallel to the x – axis
   c) Parallel to the y – axis
   d) Passes through the origin
   e) Parallel to $4x + .y = 8$

Solution: Slope is $\left(\frac{k-3}{k^2-4}\right) = \frac{3-k}{k^2-4}$

a) Slope of $2x + 3y = 6$ is $\frac{-2}{3}$ since parallel

$\Rightarrow \frac{3-k}{k^2-4} = \frac{-2}{3}$ implies $9 - 3k = -2k^2 + 8$

$\Rightarrow 2k^2 - 3k + 1 = 0$ implies $(2k-1)(k-1) = 0$

Therefore, $k = 1$ or $k = \frac{1}{2}$.

b) Parallel to the x – axis means horizontal line

$\Rightarrow$ Slope = 0, implies $\frac{3-k}{k^2-4} = 0$

$\Rightarrow 3 - k = 0$, therefore $k = 3$

c) Parallel to the y – axis, mean vertical line

$\Rightarrow$ No – slope (undefined) implies $k^2 - 4 = 0$

$\Rightarrow (k-2)(k+2) = 0$, therefore, $k = -2$, or $k = 2$

d) Exercise left for you.

e) **Exercise left for you (Tip)**

58. Determine the set of points on x – axis that is equidistant from A(2, – 4) and B(3,5)

Solution: The set of point on x – axis is P(x,0)

$\Rightarrow$ PA = PB implies $(x-2)^2 + (0+4)^2 = (x-3)^2 + (x-5)^2$

$\Rightarrow x^2 - 4x + 4 + 16 = x^2 - 6x + 9 = 25$

$\Rightarrow -4x + 6x = 34 - 20 = 14$ implies $2x = 14$

$\Rightarrow x = 7$, therefore, the coordinate point (7,0)

59. If an equilateral triangle has two vertices A(– 4, 0) and B(4,0), what are the coordinate of third vertex

Solution: Let P(x,y) be a point

$\Rightarrow$ PA = PB = AB = 8

$\Leftrightarrow (x+4)^2 + (y-0)^2 = (x-4)^2 + (y-0)^2$

$\Leftrightarrow x^2 + 8x + 16 + y^2 = x^2 - 8x + 16 + y^2$

$\Rightarrow 8x + 8x = 0$ implies $x = 0$

However PA = AB = 8

$\Rightarrow \sqrt{(x+4)^2 + (y-0)^2} = 8$ since x = 0

$\sqrt{(0+4)^2 + y^2} = 8$ implies $16 + y^2 = 64$

$\Rightarrow y^2 = 64 - 16 = 48$, therefore, $y = \pm\sqrt{48} = \pm 4\sqrt{3}$

Therefore, the coordinate value $(0, 4\sqrt{3})$ or $(0, -4\sqrt{3})$

60. What is the set of value of K if the point (k,4) is 10 units distance from (0, – 2)

A. {4, – 4} B. {2, – 2} C. {8} D. {–8,8}

Solution: Distance between (k,4) and (0, – 2) is 10

$\Rightarrow \sqrt{(k-0)^2 + (4-(-2)^2} = 10$ implies $\sqrt{k^2 + 36} = 10$

$\Rightarrow k^2 + 36 = 100$ implies $K^2 = 100 - 36 = 64$

Therefore, K = 8 Or K = – 8 $\therefore$ **Answer: D**

61. Which of the following, the line through P(– 8, – 1) and Q(2,4) is perpendicular to the line through _____

A. R(–2,6) and (4,9)
B. R(1,5) and S(–2, – 1)
C. R(–2,9) and (3, – 1)
D. R(6,0) and S(8,1)

Solution: If PQ $\perp$ RS, then (slope of PQ) (Slope or RS) = – 1

Slope of PQ = $\frac{4-(-1)}{2+8} = \frac{5}{10} = \frac{1}{2}$, provided that

Slope of RS must be – 2, hence,

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A. Slope of RS $= \frac{9-6}{4-(-2)} = \frac{3}{6} = \frac{1}{2}$ implies RS//PQ

B. Slope of RS $= \frac{1-5}{-2-1} = \frac{-6}{-3} = 2 \leftarrow$ neither perpendicular nor parallel to PQ

C. Slope of RS $= \frac{-1-9}{3-(-2)} = \frac{-10}{5} = -2 \leftarrow$ perpendicular to PQ

D. Slope of RS $= \frac{1-0}{8-6} = \frac{1}{2} \leftarrow$ parallel to PQ ∴ **Answer: C**

65. What is the equation of the line with the indicated intercepts
a) x – intercept – b and y – intercepts 2b point on the line (–3,10)
b) x – intercept a and y – intercept 4a and (2, – 7) point on the line.

Solution: Remember the intercept form $\frac{x}{a} + \frac{y}{b} = 1$

$\Rightarrow \frac{x}{-b} + \frac{y}{2b} = 1$ implies $\frac{-2x+y}{2b} = 1$

$\Rightarrow -2x + y = 2b$ implies $2b = 16 \Rightarrow b = 8$

Therefore $y - 2x = 16$ is the desired equation

b) Exercise left for you.

66. What is the set of all points (x,y) for which the distance from (x,y) to (x,4) is 5.

Solution: $\sqrt{(x-3)^2 + (y-4)^2} = 5$ implies $(x-3)^2 + (y-4)^2 = 25$

Review Exercise on Unit 10

1. Find the distance between the pairs of the point.
   a) (1,2) and (– 1,4)
   b) (a,3a) and (3a,a)
   c) (4,a) and (a,4)
   d) $(-\sqrt{2}, \sqrt{6})$ and $(-\sqrt{2}, 4\sqrt{6})$
2. A(1,2) and B(–2,3) are two opposite vertices of square, then determine
   a) The length of a diagonal
   b) The area of square
   c) The coordinate of the other vertices
3. If an equilateral triangle has two vertices (0, – 2) and (0,2), what are the coordinates of the third vertex.
4. Triangle ABC has vertices A(2,3), B(6,8) and C(7, – 1).

   a) Find the length of sides, AB, BC and AC
   b) State what type of triangle.
5. Find the set of all points:
   a) On the x – axis that are 8 units from (– 2,4)
   b) On the y – axis that are 5 unit from (–4,3)
6. For what value of k is the distance between
   a) A(k,1) and B(– 2, k) is 3
   b) A(–1,k) and B(k + 1, – 4) is $\sqrt{74}$
7. a) Find the point on the y – axis is that is equidistant from A(1,2) and B(3, – 1)
   b) Find the point on the x – axis that is equidistant from A(3,1) and B(2, – 3)
8. Find the coordinates of the points R that divides the line segment AB with given end points.
   a) A(1,3) and B(– 4,6) in the ration 2:3
   b) A(3,–1) and B(2,5) in the ration 3:7
9. A line segment has end – points P(–3,4) and Q (7, – 6) and it is divided in to three equal parts. Determine the coordinate of the point that trisect the segment
   a) Near to Q(7, – 6) b) near to P(–3,4)
10. Triangle ABC has vertices A(– 1, 3), B(4,6) and C(3, – 1). Find
    a) The length of medians
    b) The mid – point of AB, AC and BC
    c) The coordinate point at which median intersect
11. If P(– 1,2) and Q(5, – 3) are points of plane so that Q is the mid – point of PR, then what are the coordinate point of R.
12. Define slope of a straight line.
13. Using slope concept determine a point Q from a line through P(1,6) that has
    a) Slope $\frac{2}{3}$ b) slope $\frac{-2}{3}$
14. Find slope of a line passing through the points.
    a) (– 3,2) and (5, – 4) c) (2,5) and (– 7,5)
    b) (4, – 1) and (–6, –3) d) (– 3,2) and (– 3,5)
15. For what value of k such that the line passing through A(7k, 6k – 8) and B(4k – 4, k^2) has

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a) Slope zero b) No slope c) slope $\frac{1}{5}$

16. Define collinear point.
17. For what values K, the points are collinear
    a) (1, – 1),(3,5) and (k, – 10) b) (–2,1) (–3,k) and (–k,7)
18. Determine the measure of angle of triangle with vertices A(3,2),B(5,2) and (3,4)
19. Find a general form of an equation for the line through the indicated point and slope.
    a) A(5, – 3); slope – 2 b) A(4, – 5); slope 3.
20. Find an equation of the line L containing the point.
    a) (3, – 4) and (– 2,6) b) (1,4) and (3, – 5)
21. Determine the slope and the equation of the line with the indicated intercept.
    a) x – intercept 4 and y – intercept – 6
    b) x – intercept – 3 and y – intercept 2
    c) x – intercept – 6 and y – intercept 8
    d) x – intercept (a,0) and y – intercept (0, 3a), point on the line (5,9)
22. Find equation of indicated line with given point.
    a) Through (1,3) that is parallel to $3x + 4y = -24$
    b) Through (2, – 1) that is parallel to $x + y = 10$
    c) Through (1,2) that is perpendicular to $x + y = 4$
    d) Through $\left(\frac{1}{2}, \frac{-1}{3}\right)$ that is parallel to $6x + 2y + 5 = 0$
    e) Through $\left(\frac{1}{2}, \frac{-1}{3}\right)$ that is perpendicular to $6x + 2y + 5 = 0$
23. For what value (s) of **m** for the line $(m-3)x+(2m+1)y+4m+3=0$ is
    a) Parallel to the line $4x + y = 6$
    b) Perpendicular to the line $3x + 5y - 1 = 0$
    c) Parallel to the x – axis
    d) Parallel to the y – axis

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Answer key Unit 10

1. a) $2\sqrt{2}$ unit b) $2\sqrt{2}$ lal unit c) $\sqrt{2}$ l4 – al unit d) $6\sqrt{2}$ Unit
2. a) $\sqrt{10}$ unit b) 5 sq. unit c) (0,4) and (– 1,1)
3. $(4\sqrt{3},0)$ or $(-4\sqrt{3},0)$
4. a) AB = $\sqrt{41}$, AC = $\sqrt{41}$ and BC = $\sqrt{82}$ b) Isosceles triangle
5. a) $(x+2)^2 + (0-4)^2 = 64$ implies $(x+2)^2 = 64 - 16 = 48$

   $\Rightarrow x + 2 = \pm 4\sqrt{3}$ implies $(-2 \pm 4\sqrt{3}, 0)$

   a) {(0,6), (0,0)}
6. a) $k = \pm\sqrt{2}$ b) $k = -3 \pm 6$
7. a) $\left(0, \frac{-5}{6}\right)$ b) $\left(\frac{-5}{6}, 0\right)$
8. a) $R(x_0,y_0) = \left(\frac{3.1+2.(-4)}{5}, \frac{3.3+2.6}{5}\right) = \left(-1, \frac{21}{5}\right)$ b) (2.7,0.8)
9. a) near to Q = $\left(\frac{11}{3}, \frac{-8}{3}\right)$ b) near to P = $\left(\frac{1}{3}, \frac{2}{3}\right)$
10. a) $\sqrt{\frac{41}{2}}, \sqrt{\frac{65}{2}}$ and $5\sqrt{2}$ b) $\left(\frac{3}{2}, \frac{9}{2}\right)$,(1,1) and $\left(\frac{7}{2}, \frac{5}{2}\right)$ c) $\left(2, \frac{8}{3}\right)$
11. $\left(\frac{-1+x}{2}, \frac{2+y}{2}\right) = (5, -3)$ implies R(x,y) = (11, – 8)
12. Try your self
13. a) $\frac{\Delta y}{\Delta x} = \frac{-2}{3}$ implies $\frac{y-6}{x-1} = \frac{2}{3} \Rightarrow$ y–6=2 and x – 1 = 3, Q(8,4)

    b) $\frac{\Delta y}{\Delta x} = \frac{-2}{3}$ implies y – 6= – 2and x – 1 = 3 $\Rightarrow$ Q(4,4)
14. a) $m = \frac{-3}{4}$ b) $m = \frac{1}{5}$ c) m = 0 d) No – slope
15. a) k = 2 or k = 4 b) $k = \frac{-4}{3}$
16. Try to define by yourself.

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17. a) $k = -2$ b) $k = -1$

18. $45^0, 90^0, 45^0$ 19. a) $y = -2x + 7$ b) $y = 3x - 17$

20. a) $y = -2x + 2$ b) $9x + 2y = 17$

21. a) slope $-\left(\frac{-6}{4}\right) = \frac{3}{2}$ and $\frac{x}{4} + \frac{y}{-6} = 1$ implies $-3x + 2y = -12$

b) slope $\frac{2}{3}$ and $\frac{x}{-3} + \frac{y}{2} = 1$ implies $2x - 3y = -6$

c) slope $\frac{3}{4}$ and $\frac{x}{-6} + \frac{y}{8} = 1$ implies $-4x + 3y = 24$

d) slope -3 and $3x + y = 24$

22. a) $3(x-1) + 4(y-3) = 0$ implies $3x + 4y - 15 = 0$

b) $1(x-2) + 1(y+1) = 0$ implies $x + y - 1 = 0$

c) $1(y-2) - 1(x-1) = 0$ implies $y - x - 1 = 0$

d) $6(x - \frac{1}{2}) + 2(y + \frac{1}{3}) = 0$ implies $6x + 2y = \frac{7}{3}$

e) $6(y + \frac{1}{3}) - 2(x - \frac{1}{2}) = 0$ implies $-2x + 6y + 3 = 0$

23. a) $\frac{3-m}{2m+1} = -4 \Rightarrow m = -1$ b) $m = \frac{4}{13}$

c) $\frac{3-m}{2m+1} = 0$ implies $m = 3$ d) $2m + 1 = 0 \Rightarrow m = \frac{-1}{2}$

Answer key Unit 11

1) a) III quadrant b) first quadrant c) quadrantal
d) III quadrant e) III quadrant f) IV quadrant

2) a) $\frac{\pi}{6}$ b) $\frac{4}{3}\pi$ c) $\frac{5}{6}\pi$ d) $\frac{7}{6}\pi$ e) -4π f) $\frac{3}{2}\pi$

3) a) 15^0 b) -30^0 c) 120^0 d) 150^0 e) -600^0

4) a) $\sin\theta = \frac{2}{\sqrt{5}}, \cos\theta = \frac{1}{\sqrt{5}}, \tan\theta = \frac{1}{2}$

$\theta = \frac{-3}{}$ $= \frac{-4}{5}, \tan\theta = \frac{3}{4}$

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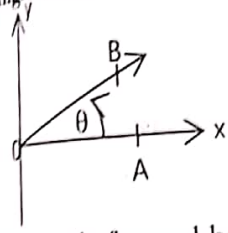
UNIT ELEVEN

Trigonometric Function

Basic terminologies

Angle: angle is the amount of rotation about fixed point 0.



- $\overrightarrow{OA}$ (initial position) is called the initial side of θ.
- $\overrightarrow{OB}$ (terminal position) is called the terminal side of θ.
- The angle formed by a ray rotating anti-clock wise is **positive angle.**
- The angle formed by a ray rotating clock wise direction is **Negative angle.**

Angles in standard position

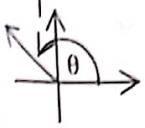
An angle in the coordinate plane is said to be in standard position, if

1. its vertex is **the origin,** and
2. its initial side lies on the positive x-axis

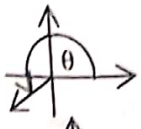
Quadrant angle

If the terminal side lies on

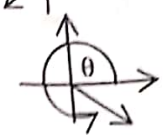
1. 1st quadrant then it is called 1st quadrant angle. For example, 30^0, 50^0, 60^0, … -300^0, … etc



2. 2nd quadrant then it is called 2nd quadrant angle. For example 120^0, 110^0, 150^0, -210^0, -240^0, …, etc.



3. 3rd quadrant then it is called 3rd quadrant angle. Like, 190^0, 240^0, -120^0, -150^0, … etc



4. 4th quadrant then it is called 4th quadrant angle. For example, 300^0, 330^0, 315^0, 219^0, -60^0, -30^0, … etc

Quadrantal angle

- If the terminal side of an angle concide with x-axis or the y-axis then the angle is called **quadrantal angle.**

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a) $\sin\theta = 1$ b) $\cos\theta = 1$, c) $\tan\theta = 1$
d) $\sin\theta = -1$ e) $\cos\theta = -1$ f) $\tan\theta = -1$

Solution: a) $\sin\theta = 1$, if $\theta = \frac{\pi}{2}$

b) $\cos\theta = 1$ if $\theta = 0$

c) $\tan\theta = 1$ if $\theta = \frac{\pi}{4}$

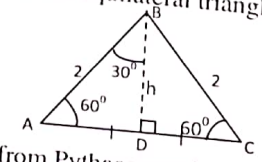
d) $\sin\theta = -1$, if $\theta = \frac{3\pi}{2}$

e) $\cos\theta = -1$ if $\theta = \pi$

f) $\tan\theta = -1$ if $\theta = \frac{3\pi}{4}$

Special Trigonometric values for 30^0, and 60^0

Consider the equilateral triangle with side length 2 unit



- The altitude of equilateral triangle bisects the angle and the side, i.e.

$\angle ABD = 30^0$ *and* $\overline{AD} = \overline{DC} = 1$

Now, from Pythagorean identity

$$(\overline{AD})^2 + (\overline{BD})^2 = (\overline{AB})^2 \Rightarrow (1)^2 + h^2 = 2^2 \Rightarrow h^2 = 4 - 1$$

$$\Rightarrow h = \sqrt{3}$$

Thus from ΔADB, we get

$\sin 30^0 = \frac{1}{2}$ *and* $\sin 60^0 = \frac{\sqrt{3}}{2}$ $\cos 30^0 = \frac{\sqrt{3}}{2}$ *and* $\cos 60^0 = \frac{1}{2}$

$\tan 30^0 = \frac{1}{\sqrt{3}}$ *and* $\tan 60^0 = \frac{\sqrt{3}}{1}$

Trigonometric value of Negative angle

For any angle θ

$\sin(-\theta) = -\sin\theta$

$\cos(-\theta) = \cos\theta$

$\tan(-\theta) = -\tan\theta$

$\therefore \sin(-30^0) = -\sin 30^0 = -\frac{1}{2}$ $\sin(-60^0) = -\sin 60^0 = -\frac{\sqrt{3}}{2}$

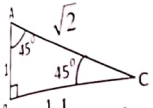
$\cos(-30^0) = \cos 30^0 = \frac{\sqrt{3}}{2}$ $\cos(-60^0) = \cos 60^0 = \frac{1}{2}$

$\tan(-60^0) = -\tan 60^0 = -\sqrt{3}$

10. Find the exact value of

a) $\sin 45^0$ b) $\cos\left(\frac{-\pi}{4}\right)$ c) $\tan 45^0$

Solution: Consider isosceles right angle triangle with side length of the two legs 1 unit.



$(AB)^2 + (BC)^2 = AC^2$

$1^2 + 1^2 = AC^2$

$\Rightarrow AC = \sqrt{1+1} = \sqrt{2}$

$\therefore \sin 45^0 = \frac{1}{\sqrt{2}}$ *and* $\cos 45^0 = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$

$\tan 45^0 = 1$ $\cos\left(\frac{-\pi}{4}\right) = \cos\frac{\pi}{4} = \frac{\sqrt{2}}{2}$

Sign of the sine, cosine and tangent function

If f is a function, it is often useful to know which values of θ makes $f(\theta)$ positive, which makes $f(\theta)$ negative, and which make $f(\theta)$ zero. Recall that the zeros of a function are the values of θ for which $f(\theta) = 0$.

Now, the signs of the $\sin\theta$, $\cos\theta$, and $\tan\theta$ functions in all of the four quadrants can be summarized as below

$\frac{\pi}{2} < \theta < \pi$	y	$0 < \theta < \frac{\pi}{2}$
Quadrant II		Quadrant I
(x, y): (-, +)		(x, y): (+,+)
$\sin\theta$ is (+)		$\sin\theta$ is (+)
$\cos\theta$ is(–)		$\cos\theta$ is (+)
$\tan\theta$ is (–)		$\tan\theta$ is (+)
$\pi < \theta < \frac{3}{2}\pi$		$\frac{3}{2}\pi < \theta < 2\pi$
Quadrant III		Quadrant IV
(x, y): (–,–)		(x,y): (+,–)
$\sin\theta$ is (–)		$\sin\theta$ is (–)
$\cos\theta$ is (–)		$\cos\theta$ is (+)
$\tan\theta$ is (+)		$\tan\theta$ is (–)

Illustrative Example

11. Solve the following inequality for $\theta \in (0, 2\pi)$

a. $\sin\theta < 0$ c. $\tan\theta > 0$ e. $\cos\theta > 0$

b. $\cos\theta < 0$ d. $\sin\theta > 0$ f. $\tan\theta < 0$

Solution: $\sin\theta < 0$ means in which quadrant $\sin\theta$ is negative.

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a) $\sin\theta < 0$ if $\theta \in \left(\pi, \frac{3}{2}\pi\right) \cup \left(\frac{3}{2}\pi, 2\pi\right)$

b) $\cos\theta < 0$ if $\theta \in \left(\frac{\pi}{2}, \pi\right) \cup \left(\pi, \frac{3}{2}\pi\right)$

c) $\tan\theta > 0$ if $\theta \in \left(0, \frac{\pi}{2}\right) \cup \left(\pi, \frac{3\pi}{2}\right)$

d) $\sin\theta > 0$ if $\theta \in \left(0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right)$

e) $\cos\theta > 0$ if $\theta \in \left(0, \frac{\pi}{2}\right) \cup \left(\frac{3}{2}\pi, 2\pi\right)$

f) $\tan\theta < 0$ if $\theta \in \left(\frac{\pi}{2}, \pi\right) \cup \left(\frac{3}{2}\pi, 2\pi\right)$

12. Solve the following inequality
   a. $\tan\theta > 0$ and $\cos\theta < 0$
   b. $\sin\theta > 0$ and $\cos\theta < 0$
   c. $\cos\theta > 0$ and $\tan\theta < 0$
   d. $\sin\theta < 0$ and $\tan\theta > 0$

Solution:

a) $\tan\theta > 0$ and $\cos\theta < 0$
$\Rightarrow \tan\theta > 0$ in the 1^{st}, 3^{rd} quadrant
$\cos\theta < 0$ in the 2^{nd} and 3^{rd} quadrant

So that the intersection is 3^{rd} quadrant i.e. $\tan\theta > 0$ and $\cos\theta < 0$ is $\theta \in \left(\pi, \frac{3\pi}{2}\right)$

b) $\sin\theta > 0$ if the terminal side of θ lies in {I, II} quadrant
$\cos\theta < 0$ if the terminal side of θ lies in {II, III} quadrant
$\therefore \sin\theta > 0$ and $\cos\theta < 0$ in quadrant II i.e. $\theta \in \left(\frac{\pi}{2}, \pi\right)$

c) $\cos\theta > 0$ if the terminal side of θ lies in quadrant {I, IV}
$\tan\theta < 0$ if the terminal side of θ lies in quadrant {II, IV}
$\therefore \cos\theta > 0$ and $\tan\theta < 0$ in quadrant IV i.e. $\theta \in \left(\frac{3}{2}\pi, 2\pi\right)$

d) **Exercise for the reader.**

13. Determine the sign of:
   a. $\cos 268, \sin 268$ b. $\sin\frac{25\pi}{36}, \cos\frac{25\pi}{36}$ c. $\tan(-280^0)$

d. $\sin(-815^0)$, $\cos(-815^0)$, $\tan(-850^0)$,

Solution:

a) $\cos 268^0$, $180 < 268 < 270^0$
$\therefore 268$ lies in quadrant III
$\therefore \cos 268 < 0$ $\sin 268 < 0$

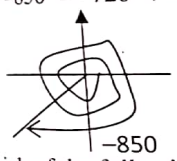
b) $\frac{\pi}{2} < \frac{25\pi}{36} < \pi$

$\therefore \frac{25\pi}{36}$ lies in quadrant II

$\therefore \sin\frac{25\pi}{36} > 0$ and $\cos\frac{25\pi}{36} < 0$

c) $\tan(-280^0)$, -280^0 lies in quadrant I $\therefore \tan(-280^0) > 0$

d) $-850^0 = -720^0 + -130^0$



the terminal side of -850^0 lies in quadrant III
$\therefore \sin(-850^0) < 0$
$\cos(-850^0) < 0$
$\tan(-850^0) > 0$

14. Which of the following pairs of values are equal
   a) $\sin(-45^0)$ and $\sin 45^0$
   b) $\sin\left(\theta - \frac{\pi}{2}\right)$ and $\cos 45^0$
   c) $\cos(-45^0)$ and $\cos 45^0$
   d) $\cos\left(\theta - \frac{\pi}{2}\right)$ and $\cos\left(\frac{\pi}{2} - \theta\right)$
   e) $\sin(\theta - 180^0)$ and $\sin(180^0 - \theta)$
   f) $\cos(\theta - 180^0)$ and $\cos(180^0 - \theta)$
   g) $\sin(\theta - 360^0 k)$ and $\sin(360^0 k - \theta)$, $k \in z$
   h) $\cos(\theta - 360^0 k)$ and $\cos(360^0 k - \theta)$, $k \in z$
   i) $\tan(\theta - 180^0)$ and $\tan(180^0 - \theta)$

Solution:

a) $\sin(-45^0) = -\sin 45^0 \neq \sin 45^0$
$\therefore \sin(-45^0) \neq \sin 45^0$ (because $\sin(-\theta) = -\sin\theta$)

b) $\sin\left(\theta - \frac{\pi}{2}\right) = -\sin\left(\frac{\pi}{2} - \theta\right) = -\cos\theta$ and $\sin\left(\frac{\pi}{2} - \theta\right) = \cos\theta$

$\therefore \sin\left(\theta - \frac{\pi}{2}\right) \neq \sin\left(\frac{\pi}{2} - \theta\right)$ i.e. $-\cos\theta \neq \cos\theta$

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c) $\cos(-45) = \cos 45$ because $\cos(-\theta) = \cos\theta$

d) $\cos\left(\theta - \frac{\pi}{2}\right) = \cos\left(\frac{\pi}{2} - \theta\right)$

e) $\sin(\theta - 180^0) = -\sin(180^0 - \theta)$

f) $\cos(\theta - 180^0) = \cos(180^0 - \theta)$ $\quad \therefore \sin(\theta - 180^0) \neq \sin(180^0 - \theta)$

g) $\sin(\theta - 360^0 k) = \sin(360^0 k - \theta)$

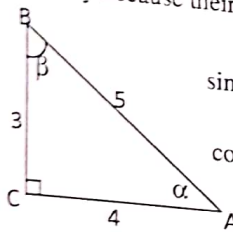
h) $\cos(\theta - 360^0 k) = \cos(360^0 k - \theta)$

i) $\tan(\theta - 180^0) = -\tan(180^0 - \theta)$ $\quad \therefore \tan(\theta - 180^0) \neq \tan(180^0 - \theta)$

Note $\sin\theta = \sin(\theta - 360^0 k)$, $k \in z$

Complementary angles relation between a trigonometric function and its co-function.

If a and b are the acute angles in the right triangle ABC, then they are complementary because their sum is 90^0



More over from the figure

$\sin\alpha = \frac{3}{5} = \cos\beta$

$\cos\alpha = \frac{4}{5} = \sin\beta$ $\qquad \tan\alpha = \frac{3}{4} = \cot\beta$

$\csc\alpha = \frac{5}{3} = \sec\beta$

Any trigonometric function of an acute angle is equal to the co function of its complementary angle. That is if $0 \leq \theta \leq 90^0$, then

$\sin\theta = \cos(90^0 - \theta^0)$	$\csc\theta = \sec(90^0 - \theta^0)$
$\cos\theta = \sin(90^0 - \theta^0)$	$\sec\theta = \csc(90^0 - \theta^0)$
$\tan\theta = \cot(90^0 - \theta^0)$	$\cot\theta = \tan(90^0 - \theta^0)$

Illustrative example

15. In Each of the following, find the degree measure of θ

a) $\cot 43^0 = \tan\theta$

b) $\sin 44^0 = \cos\theta$

c) $\sin 2\theta = \cos 3\theta$

d) $\tan 3\theta = \cot(\theta - 10^0)$

e) $\sin 38 = \cos 2\theta$

f) $\sec 20^0 = \csc(\theta^0 - 30^0)$

g) $\frac{\sin\theta}{\cos 55^0} = 1$

***Solution*:**

a) $\cot 43^0 = \tan\theta$ if $43 + \theta = 90$, then $\theta = 90 - 43 = 47^0$

b) $44^0 + \theta = 90^0 \Leftrightarrow \theta = 90^0 - 44^0 = 46^0$

c) $\sin 2\theta = \cos 3\theta \Leftrightarrow 2\theta + 3\theta = 90^0 \Rightarrow \theta = 18^0$

d) $\tan 3\theta = \cot(\theta - 10^0) \Leftrightarrow 3\theta + (\theta - 10^0) = 90^0$

$\Rightarrow 4\theta - 10 = 90^0 \quad \Rightarrow 4\theta - 100 \quad \therefore \theta\ 25^0$

e) $\sin 38^0 = \cos 2\theta \Leftrightarrow 38^0 + 2\theta = 90^0$

$\Rightarrow 2\theta = 90^0 - 38^0 = 52^0 \quad \therefore \theta = 26$

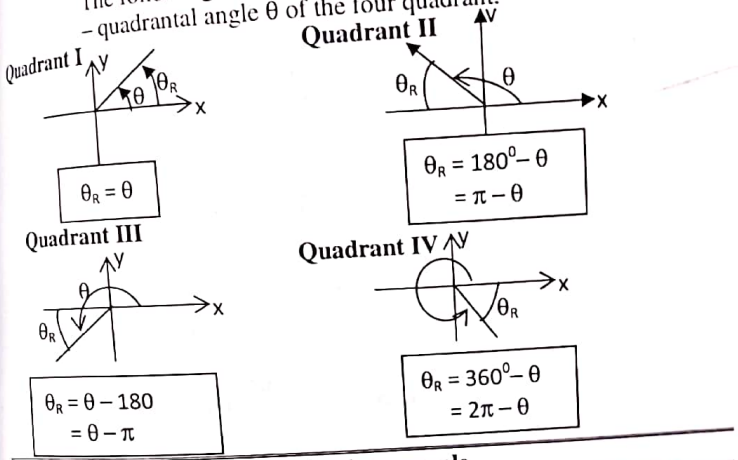
f) $\sec 20^0 = \csc(\theta^0 - 30^0) \Leftrightarrow 20^0 + \theta - 30^0 = 90^0 \Rightarrow \theta = 100$

g) $\frac{\sin\theta}{\cos 55^0} = 1 \Rightarrow \sin\theta = \cos(55^0) \Leftrightarrow \theta + 55^0 = 90^0 \Rightarrow \theta = 35^0$

Reference angle (θ_R)

Definition: Let θ be a non – quadrantal angle in standard position. The reference angle for θ is the acute angle θ_R formed by the terminal side of θ and the x-axis.

The following figure illustrate the reference angle θ_R for a non – quadrantal angle θ of the four quadrant.



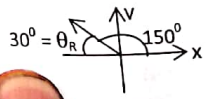
Illustrative example

16. Find the reference angle θ_R for θ if:

a) $\theta = 135^0$ b) $\theta = 190^0$ c) $\theta = 240^0$ d) $\theta\ 315^0$

e) $\theta = -240^0$ f) $\theta = \frac{5\pi}{6}$ g) $\theta = 840^0$ h) $\theta = 4$

***Solution*:**



a) The angle $\theta = 150^0$ is in quadrant II, and hence $\theta_R = 180^0 - 150^0 = 30^0$

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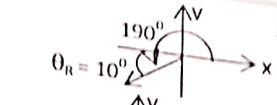
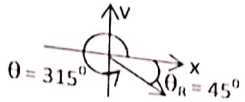
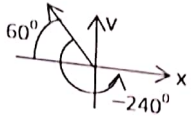


b) The angle $\theta = 190^0$ is in quadrant III.
$\therefore \theta_R = 190^0 - 180^0 = 10^0$

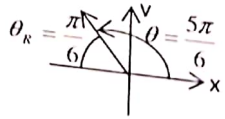
c) The angle $\theta = 240^0$ is in quadrant III
$\therefore \theta_R = 240^0 - 180^0 = 60^0$



d) The angle $\theta = 315^0$ is in quadrant IV
$\therefore \theta_R = 360^0 - 315^0 = 45^0$

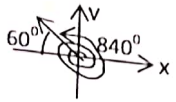


e) The angle between 0^0 and 360^0 that is co-terminal with $\theta = -240^0$ is
$-240^0 + 360^0 = 120^0$ which is in quadrant II
$\theta_R = 180^0 - 120^0 = 60^0$

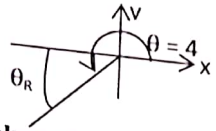


f) The angle $\theta = \frac{5\pi}{6}$ is in quadrant II, we have

$\theta_R = \pi - \frac{5\pi}{6} = \frac{\pi}{6}$



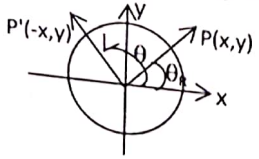
g) The angle $\theta = 840^0$ is in quadrant II
$840^0 - 720^0 = 120^0$
$\therefore \theta_R = 180^0 - 120^0 = 60^0$



h) Since $\pi < 4 < \frac{3\pi}{2}$, the angle $\theta = 4$ is in quadrant III
$\therefore \theta_R = 4 - \pi$

Values of trigonometric functions of θ and its reference angle θ_R.

a) Suppose θ is a quadrant II angle



We use the y – axis as axis of symmetry

$\sin\theta = \frac{y}{r}$, $\sin\theta_R = \frac{y}{r} \Rightarrow \sin\theta = \sin\theta_R$

$\cos\theta = \frac{-x}{r}$, $\cos\theta_R = \frac{x}{r} \Rightarrow \cos\theta = -\cos\theta_R$

Note $\theta_R = 180 - \theta$

$\tan\theta = \frac{y}{-x}$, $\tan\theta_R = \frac{y}{x} \Rightarrow \tan\theta = -\tan\theta$

Observe that:
$\sin\theta = \sin(180^0 - \theta)$
$\cos\theta = -\cos(180^0 - \theta)$
$\csc\theta = \csc(180^0 - \theta)$
$\sec\theta = -\sec(180^0 - \theta)$
$\tan\theta = -\tan(180^0 - \theta)$

Suppose θ is a quadrant III angle Note $\theta_R = \theta - 180^0$
$\sin\theta = -\sin(\theta - 180^0)$
$\cos\theta = -\cos(\theta - 180^0)$
$\csc\theta = -\csc(\theta - 180^0)$
$\sec\theta = -\sec(\theta - 180^0)$
$\tan\theta = -\tan(\theta - 180^0)$

Illustrative example

17. Express the sine, cosine and tangent function of each of the following interms of their reference angle

a. $\theta = 105^0$ b. $\theta = 225^0$ c. $\theta = \frac{5\pi}{6}$ d. $\theta = -300$

e. $\theta = 390^0$ f. $\theta\ 315^0$ g. $\theta = 2.5$

Solution: a) The angle $\theta = 105^0$ and its reference angle
$\theta_R = 180^0 - 105^0 = 75^0$ $\sin 105^0 = \sin 75^0$
$\cos 105^0 = -\cos 75^0$ $\tan 105^0 = -\tan 75^0$

b) $\theta_R = \theta - 180^0 \Rightarrow \theta_R = 225^0 - 180^0 = 45^0$

$\therefore \sin 225^0 = -\sin 45^0 = -\frac{\sqrt{2}}{2}$ $\cos 225^0 = -\cos 45^0 = \frac{-\sqrt{2}}{2}$

$\tan 225^0 = \tan 45^0 = 1$

c) $\theta_R = \pi - \theta \Rightarrow \theta_R = \pi - \frac{5\pi}{6} = \frac{\pi}{6}$

$\therefore \sin\frac{5\pi}{6} = \sin\frac{\pi}{6} = \frac{1}{2}$

$\cos\frac{5\pi}{6} = -\cos\frac{\pi}{6} = -\frac{\sqrt{3}}{2}$

$\tan\frac{5\pi}{6} = -\tan\frac{\pi}{6} = -\frac{1}{\sqrt{3}}$

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d) $-300^0 + 360^0 = 60^0$ $\sin(-300^0) = -\sin 300^0 = -(-\sin 60^0) = \frac{\sqrt{3}}{2}$

$\cos(-300^0) = \cos 300^0 = \cos 60^0 = \frac{1}{2}$

$\tan(-300^0) = -\tan 300^0 = -(-\tan 60^0) = \sqrt{3}$

e) $390^0 = 360^0 + 30^0$ $\therefore \sin 390^0 = \sin 30^0 = \frac{1}{2}$

$\cos 390^0 = \cos 30 = \frac{\sqrt{3}}{2}$

f) $\theta_R = 360^0 - \theta \Rightarrow 360^0 - 315^0 = 45^0$ $\therefore \sin 315^0 = -\sin 45^0 = -\frac{\sqrt{2}}{2}$

$\cos 315^0 = \cos 45^0 = \frac{\sqrt{2}}{2}$ $\tan 315^0 = -\tan 45^0 = -1$

g) $\theta_R = 3.14 - 2.5 = 0.64$, since the terminal side of the angle $\theta = 2.5$ is in quadrant II $\Rightarrow 360^0 - 315^0 = 45^0$

$\therefore \sin 2.5 = \sin 0.64$ $\cos 2.5 = -\cos 0.64$ $\tan 2.5 = -\tan 0.64$

18. Find the value of
   a. $\tan 140^0$, if $\tan 40 = k$
   b. $\cos 143^0$, if $\cos 37^0 = 0.7986$
   c. $\sin 136^0$, if $\cos 46 = k$
   d. $\tan 138^0$, if $\tan 42 = 0.9004$
   e. $\cos 260^0$, if $\cos 80 = 0.1736$
   f. $\cot 104$, if $\cot 76 = 0.2493$.

Solution:

a) The angle $\theta = 140^0$ is in quadrant II $\therefore \theta_R = 180^0 - 140^0 = 40^0$

$\therefore \tan 140^0 = -\tan(180^0 - 140^0) = -\tan 40^0 = -k$

b) $\theta_R = 180^0 - 143^0 = 37^0$

$\therefore \cos 143^0 = -\cos 37^0 = -(0.786) = -0.7986$

c) $\theta_R = 180^0 - 136^0 = 44^0$

Note $\cos 46^0 = \sin 44^0 = k$ $\therefore \sin 136^0 = \sin 44^0 = \cos 46^0 = k$

d) $\tan 138^0 = -\tan(180^0 - 138^0) = -\tan 42^0 = -0.9004$

e) $\theta = 260^0$ lies in quadrant III

$\therefore \theta_R = \theta - 180^0 \Rightarrow 260^0 - 180^0 = 80^0$

$\therefore \cos 260^0 = -\cos 80^0 = -0.1736$

f) $\theta = 104$, $\theta_R = 180^0 - 104^0 = 76$

$\cot(104) = -\cot 76 = -0.2493$

Co-terminal angles *are angles in standard position that have a common terminal side*

Given an angle θ, all angles which are co-terminal with θ are given by the formula $\theta \pm 360^0 k$, where $k = 1,2,3,\ldots$ or $\theta \pm 2\pi k$, where $k = 1,2,3,\ldots$

Illustrative example

19. If the given angle is in standard position find one positive co-terminal angles and one negative co-terminal angels

a. 120^0	c. 240^0	e. 620^0	g. 570^0
b. -30^0	d. -150^0	f. $\frac{5\pi}{6}$	h. $\frac{11\pi}{4}$

We use $\theta \pm 360^0$

Solution: $120^0 + 360^0 = 480^0$

$120^0 - 360^0 = -240^0$

$\therefore 120^0$, 480^0 and -240^0 are co-terminal angle

b) -30^0, $-30^0 + 360^0$, $-30^0 - 360^0$ i.e.

-30^0, -330^0 and -390^0 are co-terminal angle

c) 240^0, 600^0 and -120^0 are co-terminal angle

d) -150^0, 210^0 and -510^0 are co-terminal angle

e) 620^0, 260^0 and -100^0 are co-terminal angle

f) $\frac{5\pi}{6}, \frac{17\pi}{6}$ and $\frac{-7\pi}{6}$ are co-terminal angle

g) 570^0, 210^0 and -150^0 are co-terminal angle

h) $\frac{11}{4}\pi, \frac{3\pi}{4}$ and $\frac{19\pi}{4}$ are co-terminal angle

20. Find co-terminal angle between 0 and 2π for each of the following

a) $\frac{-127\pi}{4}$ b) $\frac{-143\pi}{6}$ c) $\frac{-207\pi}{5}$ d) $\frac{55\pi}{3}$ e) $\frac{144\pi}{7}$

Solution: To find co-terminal angle between 0 and 2π to that of given angle is Add multiple of 2π i.e. $\{2\pi, 4\pi, 6\pi, 8\pi, \ldots$ etc$\}$ until you get the required angle

a) $\frac{-127\pi}{4} + 32\pi = \frac{-127\pi + 128\pi}{4} = \frac{\pi}{4} \leftarrow$ required angle

b) $\frac{-143\pi}{6} + 24\pi = \frac{-143\pi + 144\pi}{6} = \frac{\pi}{6} \leftarrow$ required angle

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c) $\frac{-207\pi}{5} + 42\pi = \frac{-207\pi + 210\pi}{5} = \frac{3}{5}\pi \leftarrow$ required angle

d) $\frac{55\pi}{3} - 18\pi = \frac{55\pi - 54\pi}{3} = \frac{\pi}{3}$

e) $\frac{144\pi}{7} - 20\pi = \frac{144\pi - 140\pi}{7} = \frac{4\pi}{7}$

Trigonometric values of co-terminal angle

- co-terminal angles have the same trigonometric values

That is if θ is any angle, we know that θ and $\theta \pm 360^0 k$, or $\theta \pm 2\pi k$ are co-terminal angles thus we have

1) $\sin\theta = \sin(\theta \pm 360^0) = \sin(\theta \pm 720^0) = \ldots$
2) $\cos\theta = \cos(\theta \pm 360^0 k)$, for $k = 1,2,3,\ldots$
3) $\tan\theta = \tan(\theta \pm 360^0 k)$, for $k = 1,2,3,\ldots$
   $= \tan(\theta \pm 180^0 k)$
4) $\csc\theta = \csc(\theta \pm 360^0 k)$, for $k = 1,2,3,\ldots$
5) $\sec\theta = \sec(\theta \pm 360^0 k)$, for $k = 1,2,3,\ldots$
6) $\cot\theta = \cot(\theta \pm 360^0 k)$, for $k = 1,2,3,\ldots$

Illustrative example

21. Find the exact value of each expression

a. $\cos\left(\frac{-3}{4}\pi\right)$ b. $\csc\frac{2\pi}{3}$ c. $\tan\frac{-11\pi}{6}$

d. $\sin\left(\frac{-23}{4}\pi\right)$ e. $\sin\left(\frac{137\pi}{6}\right)$ f. $\cos 780^0$

g. $\cos\left(\frac{-22\pi}{3}\right)$ h. $\sin 1395^0$ i. $\cos 1395^0$

o. $\sec\left(\frac{-10\pi}{3}\right)$ p. $\sin(-135^0)$

Solution:

a) Note, $\cos(-\theta) = \cos\theta$

$$\therefore \cos\left(\frac{-3}{4}\pi\right) = \cos\frac{3}{4}\pi$$

Since angle $\theta = \frac{3}{4}\pi$ is in quadrant II

$$\therefore \theta_R = \pi - \theta \Rightarrow \theta_R = \pi - \frac{3\pi}{4} = \frac{\pi}{4}$$

$$\therefore \cos\frac{3}{4}\pi = -\cos\left(\pi - \frac{3}{4}\pi\right) = -\cos\frac{\pi}{4} = -\frac{-\sqrt{2}}{2}$$

b) The angle $\theta = \frac{2}{3}\pi$ is in quadrant II

$$\therefore \csc\left(\frac{2}{3}\pi\right) = \csc\left(\pi - \frac{2}{3}\pi\right) = \csc\frac{\pi}{3} = \frac{2}{\sqrt{3}}$$

c) The angle between 0^0 and 2π that is co-terminal with $\theta = \frac{-11\pi}{6}$ is

$$\frac{-11\pi}{6} + 2\pi = \frac{\pi}{6} \therefore \tan\left(\frac{-11\pi}{6}\right) = \tan\frac{\pi}{6} = \frac{1}{\sqrt{3}}$$

d) The angle between 0^0 and 2π that is co-terminal with $\frac{-23}{4}\pi$ is

$$\frac{-23\pi}{4} + 6\pi = \frac{\pi}{4} \therefore \sin\left(\frac{-23\pi}{4}\right) = \sin\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}}{2}$$

e) The angle between 0^0 and 2π that is co-terminal with $\frac{137\pi}{6}$ is

$\frac{137\pi}{6} - 22\pi = \frac{5\pi}{6}$ But $\frac{5\pi}{6}$ is in quadrant two

$$\therefore \theta_R = \pi - \frac{5\pi}{6} = \frac{\pi}{6}$$

$$\therefore \sin\left(\frac{137\pi}{6}\right) = \sin\left(\frac{5\pi}{6}\right) = \sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$$

f) The angle between 0^0 and 360^0 that is co-terminal with 780^0 is

$780^0 - 2(360^0) =$ $\therefore \cos 780^0 = \cos 60^0 = \frac{1}{2}$

g) The angle between 0 and 2π that is co-terminal with $\frac{-22\pi}{3}$ can obtain by adding multiple of 2π i.e

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$$\frac{-22\pi}{3}+8\pi=\frac{2}{3}\pi$$

And the angle $\frac{2}{3}\pi$ is in quadrant two

$$\therefore\theta_R=\pi-\theta\Rightarrow\pi-\frac{2\pi}{3}=\frac{\pi}{3}$$

$$\therefore\cos\left(\frac{-22\pi}{3}\right)=\cos\frac{2}{3}\pi=-\cos\frac{\pi}{3}=-\frac{1}{2}$$

Not that $\cos\left(\frac{-22\pi}{3}\right)=\cos\left(\frac{22\pi}{3}\right)=-\cos\frac{\pi}{3}=-\frac{1}{2}$

h) The angle between 0^0 and 360^0 that is co-terminal with 1395^0 can obtain by subtracting multiple of 360^0 like $360^0, 720^0, 1080^0, 1440^0 \ldots$

$\therefore 1395^0 - 1080^0 = 315^0$

The angle 315^0 is lies in 4^{th} quadrant

$\therefore$ The smallest positive acute angle is

$\theta_R = 360^0 - \theta \Rightarrow 360^0 - 315^0 = 45^0$

$$\therefore \sin 1395^0 = \sin 315^0 = -\sin 45^0 = \frac{-\sqrt{2}}{2}$$

$$\therefore \sin 1395^0 = \cos 315^0 = \cos 45^0 = \frac{\sqrt{2}}{2}$$

o) The angle between 0 and 2π that is co-terminal with $\frac{-10\pi}{3}$ is obtain by adding the multiple of 2π, like, $2\pi, 4\pi, 6\pi, 8\pi, \ldots$

$$\therefore \frac{-10\pi}{3}+4\pi=\frac{2}{3}\pi$$

Since the angle $\theta=\frac{2}{3}\pi$ is in quadrant II

$$\therefore\theta_R=\pi-\theta\Rightarrow\theta_R=\pi\,\frac{-2}{3}\pi=\frac{\pi}{3}$$

$$\therefore\sec\left(\frac{-10\pi}{3}\right)=\sec\left(\frac{2}{3}\pi\right)=-\sec\frac{\pi}{3}=-2$$

p) The angle between 0^0 and 360^0 that is co-terminal with $\theta=-135^0$ is obtained by adding 360^0

$\therefore -135^0 + 360^0 = 225$

$$\therefore \sin(-135^0) = \sin 225 = -\sin(225 - 180^0) = -\sin 45^0$$

$$= \frac{-\sqrt{2}}{2}$$

The graph of sine function

Let $y = \sin\theta$

I) Find a) the domain b) the range c) the θintercept (the zeros)
d) the y-intercept e) Investigate the behavior

i) at $(*) = \frac{\pi}{2}, \frac{3\pi}{2}$,

ii) as θ increase from 0 to $\pi/2$ from $\frac{\pi}{2}\to\pi$ from $\pi\to\frac{3\pi}{2}$

from $\frac{3}{2}\pi\to 2\pi$

II) Sketch the graph

Solution: a) the domain of since function is the set of all real number

b) the range of since function is $\{y : -1 \le y \le 1\}$

- the maximum value is 1, at $\theta = \pi/2$
- the minimum value is -1, at $\theta = \frac{3\pi}{2}$

c) the θ intercept, the graph cross the θaxis i.e. $\sin\theta = 0$ is at $\theta = \pi k$ ($\theta = \pi, 2\pi, 3\pi, \ldots$)

d) the y-intercept is (0,0)

e) at $\theta = \frac{\pi}{2}$, $y = 1$ and at $\theta = \frac{3\pi}{2}$, then $y = -1$

ii) as θ increase from $0 \to \pi/2$ y increase $0 \to 1$

as θ increase from $\pi/2 \to \pi$

then sine function decrease $1 \to 0$

as θ increase from $\pi \to \frac{3\pi}{2}$

then sine function decrease $0 \to -1$

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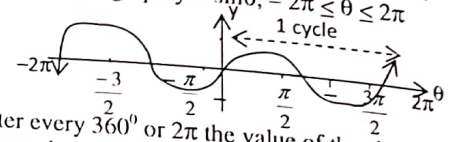
as θ increase from $\frac{3\pi}{2} \to 2\pi$

then sine function increase $-1 \to 0$

So that the graph $y = \sin\theta$, $-2\pi \le \theta \le 2\pi$



Note, after every 360^0 or 2π the value of the sine function repeat themselves. This means $\sin\theta = \sin(\theta \pm 360^0) = \sin(\theta \pm 2(360^0))$...

i.e. $\sin\theta = \sin(\theta \pm 360^0 k)$ were k = 1,2,3...

- A function that repeats its values at regular interval is called a periodic function.
- Therefore, 360^0 or 2π is called the period of the sine function.
- Any part of the graph that corresponds to one period (2π) is one cycle of the graph of $y = \sin\theta$

The Graph of the cosine function

Let y = cosθ

I) Find a) the domain b) the range c) θ, that makes $\cos\theta = 0$

d) the y-intercept e) θ where $\cos\theta = 1$ and $\cos\theta = -1$

II) Describe the behavior as θ increase from:

i) $0 \to \frac{\pi}{2}$ ii) $\frac{\pi}{2} \to \pi$ iii) $\pi \to \frac{3}{2}\pi$ iv) $\frac{3}{2}\pi \to 2\pi$

III) Sketch the graph

Solution: a) the domain is the set of all real number

b) the range is $\{y: -1 \le y \le 1\}$

c) $\cos\theta = 0$ when $\theta = \frac{\pi}{2}k$, for k is odd integer

i.e. $\left\{\frac{-3\pi}{2}, \frac{-\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}...\right\}$ d) y-intercept (0,1)

e) $\cos\theta = 1$ if $\theta = -4\pi, -2\pi, 0, 2\pi, 4\pi, ...$

and $\cos\theta = -1$ if $\theta = ... -3\pi, -\pi, \pi, 3\pi, ...$

II) i) As θ increase from $0 \to \frac{\pi}{2}$ cosθ decrease $1 \to 0$

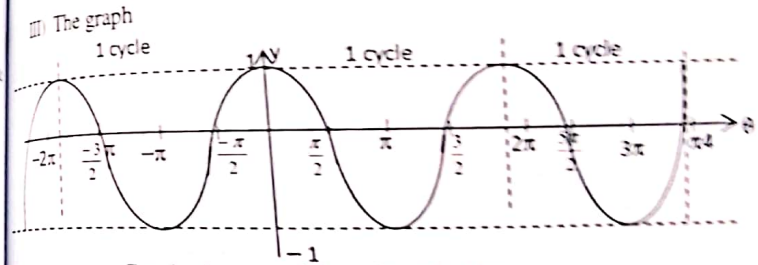
ii) As θ increase from $\frac{\pi}{2} \to \pi$ then cosθ decreased $0 \to -1$

iii) As θ increase from $\pi \to \frac{3}{2}\pi$ then cosθ increase $-1 \to 0$

iv) As θ increase from $\frac{3}{2}\pi \to 2\pi$ then cosθ increase $0 \to 1$

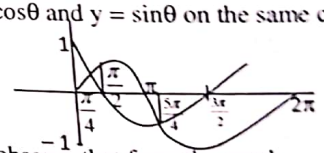
- The period of cosine function is 2π or 360^0

III) The graph



Graph of $y = \cos\theta$ for $-2\pi \le \theta \le 4\pi$

The graph of $y = \cos\theta$ and $y = \sin\theta$ on the same coordinate plane



Note: we can observe that from the graph

- $\cos\theta > \sin\theta$, if $\theta \in \left[0, \frac{\pi}{4}\right] \cup \left[\frac{5\pi}{4}, 2\pi\right]$
- $\sin\theta > \cos\theta$, if $\theta \in \left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$
- $\sin\theta = \cos\theta$ at $\theta = \frac{\pi}{4}, \frac{5\pi}{4}$
- both sinθ and cosθ decreasing at the same time, when $\theta \in \left[\frac{\pi}{2}, \pi\right]$

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$$\Rightarrow \sqrt{3} = \frac{AB}{7cm}$$

$$\Rightarrow AB = 7\sqrt{3}cm$$

ii) To find AC, use

$$\cos 60^0 = \frac{AB}{BC} \Rightarrow \frac{1}{2} = \frac{7}{AC}$$

$$\Rightarrow AC = 14$$

$\therefore$ The length of the side of ΔABC, will be 7cm, $7\sqrt{3}$ cm, 14cm

26. In fig. if $m(<B) = 90^0$, $m(<BDC) = 90^0$

$\overline{AB} = 8cm$ *and* $\overline{AD} = 4cm$ then

Find a) length $\overline{BC}, \overline{DC}$

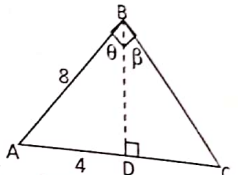
b) measure of all acute angle

***Solution*:**

$$\sin\theta = \frac{AD}{AB} \qquad = \frac{4}{8} = \frac{1}{2}$$

$\therefore \theta = 30^0$ i.e. $\sin 30^0 = \frac{1}{2}$

$\therefore \beta = 60^0$



To find $\overline{BD}$, use $\cos 30^0 = \frac{\overline{BD}}{AB} \Rightarrow \frac{\sqrt{3}}{2} = \frac{BD}{8} \Rightarrow \overline{BD} = 8\left(\frac{\sqrt{3}}{2}\right) = 4\sqrt{3}$

To find BC, since $\beta = 60^0$

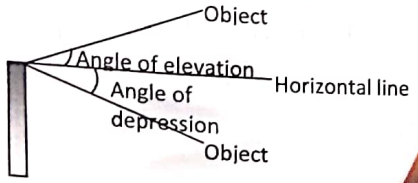
Now, use $\cos 60^0 = \frac{\overline{BD}}{BC} \Rightarrow \frac{1}{2} = \frac{4\sqrt{3}}{BC}$

$\Rightarrow BC = 8\sqrt{3}$

$\therefore$ To find DC,

Use $\tan 60^0 = \frac{DC}{BD} \Rightarrow \sqrt{3} = \frac{DC}{4\sqrt{3}} \Rightarrow \overline{BD} = \left(4\sqrt{3}\right)\left(\sqrt{3}\right) = 12$

B) Angle of elevation and depression

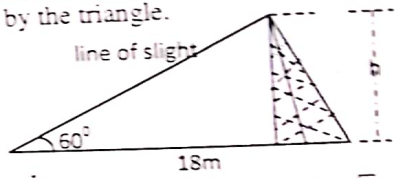


- As illustrated in fig. if an observer at point x sights an object, then the angle that the line of sight makes with the horizontal line l is the angle if elevation of the object, if the object is above the horizontal line, or the angle of depression of the object, if the object is below the horizontal line.

Illustrative example

1. From a point on level ground 18m from the base of a tower, the angle of elevation of the top of the tower is 60^0.
Find the height of the tower

***Solution*:** If we let h denote the height of the tower, then the given facts are represented by the triangle.



$$\tan 60^0 = \frac{h}{18m} \Rightarrow h = (18)\tan 60^0 = 18\sqrt{3}m$$

$\therefore$ The tower is $18\sqrt{3}$ m high

28. When the sun is 30^0 above the horizon, how long is the shadow cast by building 40 meter high?

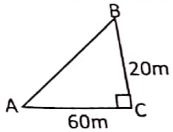
***Solution*:**

$m(< A) = 30^0, \overline{BC} = 40m$ then $\tan 30^0 = \frac{BC}{AC}$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{40m}{AC} \Rightarrow AC = 40\sqrt{3}cm$$

$\therefore$ The length of the shadow is $40\sqrt{3}cm$.

29. A tree 20 meters tall casts shadow 60 meters long. Find the angle of elevation of the sun.



***Solution*:** Let <A be angle of elevation

Then $\overline{CB}$ = 20m, and $\overline{AC}$ = 60m, then

$$\tan A = \frac{BC}{AC} = \frac{20}{60} = \frac{1}{3}$$

$$\therefore A = \tan^{-1}\left(\frac{1}{3}\right)$$

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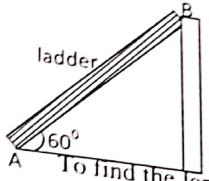
30. A ladder leans against the side of a building with its foot 4 meter from the building flow far from the ground is the top of the ladder and how long is the ladder if it makes an angle of 60^0 with the ground?

Solution: Let $\overline{AC}$ is the distance of the foot of the ladder from the building and AB is the length of the ladder, then



$$\tan A = \frac{BC}{AC}$$

Thus $CB = AC \tan 60^0$

$= (4)\sqrt{3}$ m

$\therefore$ the top the ladder is $4\sqrt{3}$ m above the ground.

To find the length of the ladder we have

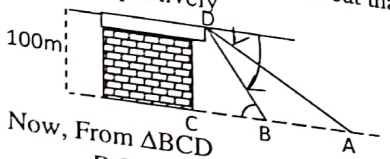
$$\cos 60^0 = \frac{AC}{AB} \Rightarrow \frac{1}{2} = \frac{4}{AB} \Rightarrow AB = 8m$$

Thus the ladder is 8m long.

31. From the top of building 100 meters above sea level. If the angle of depression of the boat changes from 30^0 to 45^0 during the period of observation.

How far a part are the boats.

Solution: Let A and B be the position of the boat that correspond to the 30^0 to 45^0 angle respectively



Now, From ΔBCD

$$\tan B \Rightarrow \tan 45^0 = \frac{DC}{CB} = \frac{100}{BC} \quad \overline{BC} = \frac{100}{\tan 45^0} = \frac{100}{1} = 100 \text{ meter}$$

From ΔDAC

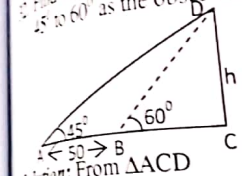
$$\tan A = \frac{DC}{AC} \Rightarrow \tan 30^0 = \frac{100}{AC}$$

$$\Rightarrow \overline{AC} = \frac{100}{\tan 30^0} = \frac{100}{\frac{1}{\sqrt{3}}} = 100\sqrt{3} \text{ meter}$$

But $\overline{AB} = \overline{AC} - BC = 100\sqrt{3} - 100 = 100(\sqrt{3} - 1)$ meter

$\therefore$ The distance between the two boat is $100(\sqrt{3} - 1)$ meter

32. Find the height of tree if the angle of elevation of its top changes from 45^0 to 60^0 as the observer advances 50 meters toward its base.



Solution: Let A,B be two point from which the top of the tree is sighted such that AB = 50meter let C and D represent the foot and the top of tree respectively.

From ΔACD

$$\tan 45^0 = \frac{h}{AC} = \frac{h}{AB + BC} \Rightarrow 1 = \frac{h}{50 + BC}$$

$\Leftrightarrow h = 50 + \overline{BC}$ ..... (1)

And From ΔBCD

$$\tan 60^0 = \frac{h}{BC} \Rightarrow \sqrt{3} = \frac{h}{BC}$$

$$\Rightarrow h = \sqrt{3}BC \Rightarrow BC = \frac{h}{\sqrt{3}} \quad \ldots\ldots (2)$$

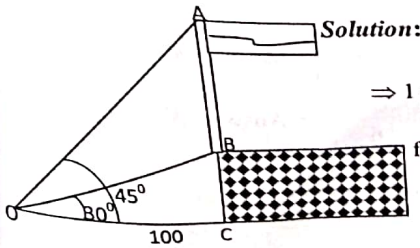
From (1) and (2) $h = 50 + \frac{h}{\sqrt{3}}$

Solving for h

$$\Rightarrow h - \frac{h}{\sqrt{3}} = 50 \Rightarrow h\left(1 - \frac{1}{\sqrt{3}}\right) = 50 \Rightarrow h\left(\frac{\sqrt{3} - 1}{\sqrt{3}}\right) = 50$$

$\therefore$ The height of tree is $\frac{50\sqrt{3}}{\sqrt{3} - 1}$ meter

33. From a point 100 meter in front of public library the angles of elevation to the base of flagpole and the top of the flag pole are 30^0 and 45^0 respectively. The flag pole is mounted on the front of the library's roof. Find the height of the flag pole.



Solution: from $\Delta OAC \tan 45^0 = \frac{AC}{OC} = \frac{AC}{100m}$

$$\Rightarrow 1 = \frac{AC}{100m} \Rightarrow AC = 100 \text{ and}$$

from $\Delta OBC \tan 30^0 = \frac{BC}{OC} = \frac{BC}{100m}$

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$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{BC}{100} \Rightarrow BC = \frac{100}{\sqrt{3}}$$

Now, $\therefore AC = AB + BC$

$$\Rightarrow AB = AC - BC = 100m - \frac{100}{\sqrt{3}} = 100\left(1 - \frac{1}{\sqrt{3}}\right)m$$

$\therefore$ the height of the flagpole is $100\left(1 - \frac{1}{\sqrt{3}}\right)m$

Solved Problem

34) Le A and B be co-terminal angles in standard position. If $m(\hat{B}) = \frac{-23\pi}{3}$ radians, and $0 < m(\hat{A}) < 2\pi$ then what is the measure of angle A in radians.

A. $\frac{-2}{3}\pi$ B) $\frac{-4}{3}\pi$ C) $\frac{\pi}{3}$ D. $\frac{5}{3}\pi$

Solution: The angle between 0 and 2π that is co-terminal with $\hat{B} = \frac{-23\pi}{3}$ is obtained by adding multiple of 2π, like, $2\pi, 4\pi, \pi 6, 8\pi, \ldots$

$\therefore \frac{-23\pi}{3} + 8\pi = \frac{-23\pi + 24\pi}{3} = \frac{\pi}{3}$ $\therefore$ **Answer. C**

35) Which of the following angles is not co-terminal with 60^0?

A. $\frac{7}{6}\pi$ B) -660^0 C) 780^0 D. $\frac{-5}{3}\pi$

Solution: An angle co-terminal with 60^0 is obtained by adding multiple of 2π, or 360^0

$\therefore$ A) $\frac{\pi}{3} + 2\pi = \frac{7\pi}{3}$

B) $60^0 - 2(360^0) = -660^0$

C) $60^0 + 2(360^0) = 780^0$

D) $\frac{\pi}{3} - 2\pi = \frac{-5}{3}\pi$

60^0 or $\frac{\pi}{3}$ is not co-terminal with $\frac{7\pi}{6}$ $\therefore$ **Answer. A**

36) If an angle θ is in standard position and its terminal side lies in quadrant IV on the line whose equation is $V = -3x$, then what is the value of $\sin\theta$.

A. $\frac{-\sqrt{10}}{10}$ B. $\frac{\sqrt{10}}{10}$ C. $\frac{3\sqrt{10}}{10}$ D. $\frac{-3\sqrt{10}}{10}$

Solution: Let $x = 1$, then $y = -3$

$\therefore r^2 = 1^2 + (-3)^2 = 10$

$\therefore r = \sqrt{10}$ $\sin\theta = \frac{-3}{\sqrt{10}} = \frac{-3\sqrt{10}}{10}$ $\therefore$ **Answer. D**

37) Which of the following **is not equal** to $\sin\frac{\pi}{13}$? . . . EGSEC

A. $\frac{1}{\csc\frac{\pi}{13}}$ B. $\sin\frac{11}{26}\pi$ C. $\sin\frac{12}{13}\pi$ D. $\cos\frac{11}{26}\pi$

Solution: A) $\frac{1}{\csc\frac{\pi}{13}} = \sin\frac{\pi}{13}$ B) $\sin\frac{\pi}{13} = \sin\left(\pi - \pi/13\right) = \sin\frac{12}{13}\pi$

C) $\sin\theta = \cos\left(\frac{\pi}{2} - \theta\right)$ $\sin\frac{\pi}{13} = \cos\left(\frac{\pi}{2} - \frac{\pi}{13}\right) = \cos\frac{11}{26}\pi$

$\therefore$ **Answer. B**

38) Which of the following is not equal with the rest?

A. $\sin 215^0$ B. $-\cos 55^0$ C. $-\sin 145^0$ D. $-\cos 125^0$

Solution: Reference angle $\theta R = 215^0 - 180^0 = 35$

$\therefore$ A) $\sin 215^0 = -\sin 35^0 = -\cos(55^0)$

B) complementary angle relation

$\cos(-55^0) = \cos 55^0 = \sin 35^0$

C) $-\sin 145^0 = -(\sin(180^0 - 145^0)$

D) $-\cos 125^0 = -(\cos(180^0 - 125^0) = \cos 55$ $\therefore$ **Answer. D**

39) In which of the following intervals are both the sine and cosine functions are decreasing

A. $\left(\frac{\pi}{4}, \frac{5\pi}{4}\right)$ B. $\left(\frac{\pi}{2}, \frac{5\pi}{4}\right)$ C. $\left(\frac{\pi}{2}, \frac{3\pi}{2}\right)$ D. $\left(\frac{\pi}{2}, \pi\right)$

Solution: Refer the graph on the same coordinate plane

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both increasing on $\left(\frac{3\pi}{2}, 2\pi\right)$

- $\sin\theta$ and $\cos\theta$ both decreasing no $\left(\frac{\pi}{2}, \pi\right)$

40) Which of the following is equal to $\sin 280^0$?
A. $\sin 80^0$ B. $\cos(-10^0)$ C. $-\cos 80^0$ D. $-\cos 10^0$

Solution: $\sin 280^0 = -\sin(360^0 - 280^0) = -\sin 80^0 = -\cos 10^0$ ∴ Answer: D

41) What is the simple name for $\frac{\tan(t+\pi)}{\sin(t+2\pi)}$?
A) cost B) csct C) sect D) cott

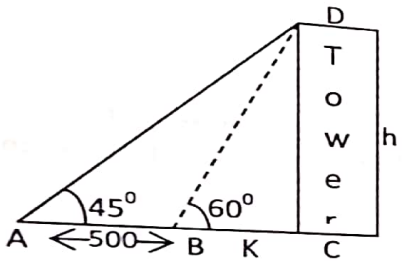
Solution: tangent is periodic at π sine is periodic at 2π

$\therefore \frac{\tan(t+\pi)}{\sin(t+2\pi)} = \frac{\tan t}{\sin t} = \frac{\sin t}{(\cos t)\sin t} = \sec t$ ∴ Answer C

42) Two points on a level ground are 500m a part. The angles of elevation from these points to the top of a tower are 45^0 and 60^0 respectively. If the two points and the base of the tower lie on a straight line, as shown by the figure below what is the height of the tower in meters

A) $\frac{500\sqrt{2}}{\sqrt{2}-1}$ B) $\frac{500\sqrt{3}}{\sqrt{3}-1}$ C) 600 D) $\frac{500}{\sqrt{3}-1}$

Solution: Let BC be K and height of tower be h. Then, $\tan 45^0 = \frac{h}{500+k}$ and $\tan 60^0 = \frac{h}{k}$



$\Rightarrow 1 = \frac{h}{500+k}$ and $\sqrt{3} = \frac{h}{k}$

$\Rightarrow h = 500m + k$ and $h = \sqrt{3}k$

$\Rightarrow k = h - 500m$

$\Rightarrow h = \sqrt{3}(h - 500m)$

$\Rightarrow h - \sqrt{3}h = -500\sqrt{3}$

$h(1 - \sqrt{3}) = -500\sqrt{3}$ $\therefore h = \frac{500\sqrt{3}}{\sqrt{3}-1}$

∴ Answer: B

43) $\sin(-430^0)$ is equal to:
A) $-\sin 20^0$ B) $\cos 20^0$ C) $\sin 70^0$ D) $-\cos 20^0$

Solution: $\sin(-430^0) = \sin(-430 + 360^0) = \sin(-70^0) = -\sin 70^0$
$= -\cos 20^0$ ∴ Answer: D

44) If x is an acute angle and $\tan x = 3$ then what is the value of $\cos x$
A) $\sqrt{10}$ B) $\frac{3}{5}$ C) $\frac{\sqrt{10}}{10}$ D) $\frac{1}{\sqrt{5}}$

Solution: we know $\sin^2 x + \cos^2 x = 1$
Dividing both side by $\cos^2 x$
$\Rightarrow \tan^2 x + 1 = \sec^2 x$
$\Rightarrow 3^2 + 1 = \sec^2 x$

$\Rightarrow \sec x = \sqrt{10} \Rightarrow \cos x = \frac{1}{\sqrt{10}} = \frac{\sqrt{10}}{10}$ ∴ Answer: C

45) Let θ be an angle in standard position and a point P(x, y) is on the terminal side of θ. If $x > 0$ and $y = 0$ then the cose cant of θ is equal to
A) $\frac{\sqrt{x^2+y^2}}{x}$ B) $\frac{x}{\sqrt{x^2+y^2}}$ C) $\frac{y}{\sqrt{x^2+y^2}}$ D) $\frac{\sqrt{x^2+y^2}}{y}$

Solution: $x^2 + y^2 = r^2$
$\Rightarrow r = \sqrt{x^2+y^2}$

$\sin\theta = \frac{y}{\sqrt{x^2+y^2}} \Rightarrow \csc\theta = \frac{\sqrt{x^2+y^2}}{y}$ ∴ Answer: D

46) Which of the following angles is reference angle to -880^0
A) -20^0 B) 20^0 C) 200^0 D) -160^0

Solution: first find angle between 0^0 and 360^0 co-terminal with -880^0 is
$-880^0 + 1080^0 = 200^0$
$\theta_R = 200^0 - 180^0 = 20^0$ ∴ Answer: B

47) Which of the following is a positive number?
A) $\cos 250^0$ B) $\cos 170^0$ C) $\cos(-280)^0$ D) $\cos(-260)^0$

Solution: A) $\cos 250^0 = -\cos(70^0)$
B) $\cos 170^0 = -\cos(180^0 - 170^0) = -\cos 10^0$
C) $\cos(-280^0) = \cos 280^0 = \cos(360^0 - 280^0) = \cos 80^0$
D) $\cos(-260^0) = \cos 260^0 = -\cos(260^0 - 180^0) = -\cos 80^0$

∴ Answer: C

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48) If $\cos\theta = \frac{5}{6}$ and $0 < \theta < \frac{\pi}{2}$, then $\csc^2\theta$ is equal to

A) $\frac{36}{11}$ B) $\frac{36}{25}$ C) $\frac{\sqrt{11}}{6}$ D) $\frac{11}{36}$

Solution: $\sin^2\theta + \cos^2\theta = 1$

$\Rightarrow \sin^2\theta + \frac{25}{36} = 1$

$\Rightarrow \sin^2\theta = 1 - \frac{25}{36} = \frac{36-25}{36} = \frac{11}{36}$

$\Rightarrow \sin^2\theta = 1 - \frac{\sqrt{11}}{6} \Rightarrow \csc\theta = \frac{6}{\sqrt{11}}$

$\Rightarrow \csc^2\theta = \frac{36}{11}$ ∴ **Answer: A**

49) The exact value of $\csc\frac{11}{3}\pi$ . . . EGSECE 2006

A. $\frac{-2\sqrt{3}}{3}$ B. $\frac{2\sqrt{3}}{3}$ C. -2 D. 2

Solution: $\csc\frac{11}{3}\pi = \frac{1}{\sin\frac{11}{3}\pi} = \frac{1}{\sin\left(\frac{11}{3}\pi - 2\pi\right)} = \frac{1}{\sin\frac{5\pi}{3}}$

$= \frac{1}{\sin\frac{5}{3}\pi} = \frac{1}{-\sin\left(\frac{5}{3}\pi - \pi\right)} = \frac{1}{-\sin\frac{2\pi}{3}} = \frac{1}{\frac{-\sqrt{3}}{2}}$

$= \frac{-2}{\sqrt{3}} = \frac{-2\sqrt{3}}{3}$ ∴ **Answer: A**

50) If θ is the measure of an acute angle, then which of the following is a correct trigonometric identity

A. $\cot\theta = \tan\left(\frac{\pi}{2} + \theta\right)$ C. $\sin\theta = \cos\left(\frac{\pi}{2} - \theta\right)$

B. $\cos\theta = \sin\left(\frac{\pi}{2} + \theta\right)$ D. $\tan\theta = \tan\left(\frac{\pi}{2} - \theta\right)$

Solution: A) $\cot\theta = \tan\left(\frac{\pi}{2} - \theta\right)$ C) $\sin\theta = \cos\left(\frac{\pi}{2} - \theta\right)$

B) $\cos\theta = \sin\left(\frac{\pi}{2} - \theta\right)$ D) $\tan\theta = \cot\left(\frac{\pi}{2} - \theta\right)$

∴ **Answer: C**

51) Which one of the following is **NOT** equal to $\cos 72^0$?

A. $\cos 432^0$ B. $\sin 18^0$ C. $\sin 792^0$ D. $\cos(-648^0)$

Solution: we know that: $\cos\theta = \cos(\theta \pm 360^0 k)$, $k \in Z$

$\Rightarrow \cos 72^0 = \cos(72^0 \pm 360^0 k)$, where $k = 0,1,2,3\ldots$

- If $k = 1$; $\cos 72 = \cos(72 + 360^0) = \cos 432^0$
- If $k = 2$; $\cos 72 = \cos(72^0 + 720^0) = \cos 792^0 \neq 792^0$.
- $\cos 72^0 = \sin 18^0$, because $72^0 + 18^0 = 90^0$
- If $k = 2$; $\cos 72^0 = \cos(72^0 - 720^0) = \cos(-648^0)$

∴ **Answer: A**

52) $\csc\left(\frac{3\pi}{4}\right)$ is equal to

A. $\frac{\sqrt{2}}{2}$ B. $\frac{-\sqrt{2}}{2}$ C. $\frac{2}{\sqrt{3}}$ D. $\sqrt{2}$

Solution: $\csc\left(\frac{3\pi}{4}\right) = \csc\left(\pi - \frac{3\pi}{4}\right) = \csc\frac{\pi}{4} = \frac{2}{\sqrt{2}} = \sqrt{2}$

∴ Answer: D

53) Suppose $\cos\theta = 0.8$ and $\csc\theta < 0$, then $\tan\theta$ is equal to . . . EGSEC 97/05

A. $\frac{3}{5}$ B. $\frac{-3}{4}$ C. $\frac{4}{5}$ D. $\frac{4}{-3}$

Solution: θ lies on 4th quadrant

∴ $\tan\theta < 0$

Thus, we have $\cos^2\theta + \sin^2\theta = 1$

$\Rightarrow \frac{64}{100} + \sin^2\theta = 1$

$\Rightarrow \sin^2\theta = 1 - \frac{64}{100} = \frac{36}{100}$

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$\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{-0.6}{0.8} = \frac{-3}{4}$

act value of $\cos\frac{19\pi}{6}$ is . . . EGSECE

A. $\frac{-1}{2}$ B. $\frac{-\sqrt{3}}{2}$ C. 0.866 D. −0.866

∴ Answer: B

Solution: The angle between 0 and 2π that is co-terminal with $\frac{19\pi}{6}$ obtained by subtracting multiple of 2π, like $2\pi, 4\pi, 6\pi$

$\therefore \frac{19\pi}{6} - 2\pi = \frac{7\pi}{6}$

The reference angle, $\theta_R = \frac{7\pi}{6} - \pi = \pi/6$

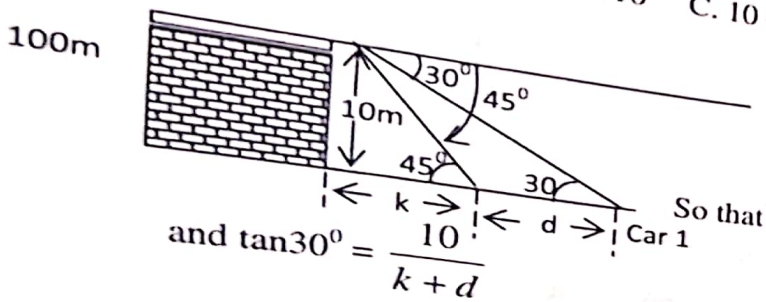
$\therefore \cos\frac{19\pi}{6} = \cos\frac{7\pi}{6} = -\cos\left(\frac{7\pi}{6} - \pi\right) = -\cos\frac{\pi}{6}$

$= \frac{-\sqrt{3}}{2}$ ∴ Answer: B

55) From the top of a 10m high building, a man observed two cars at angle of depression 30^0 and 45^0 due east. How far apart are the two cars in meters?... EGSEC 97/05

A. $10\sqrt{3} - 10$ B. $10\sqrt{3} + 10$ C. $10 + \frac{10}{\sqrt{3}}$ D. $10 - \frac{10}{\sqrt{3}}$

Solution:



Let d be the distance between two car.

So that $\tan 45^0 = \frac{10}{k} \Leftrightarrow k = 10$

and $\tan 30^0 = \frac{10}{k+d}$

$\Rightarrow \frac{1}{\sqrt{3}} = \frac{10}{10+d} \Leftrightarrow 10 + d = 10\sqrt{3}$

$\Rightarrow d = 10\sqrt{3} - 10$ ∴ Answer: A

56) A positive angle of θ radians $0 < \theta < 2\pi$, is in standard position. If this angle is co-terminal with an angle of $\frac{-203\pi}{3}$ radian in standard position, what is the value of θ . . .

A. $\frac{\pi}{3}$ B. $\frac{2}{3}\pi$ C. $\frac{5}{3}\pi$ D. $\frac{7}{3}\pi$

Solution: Note $\frac{-203\pi}{3} = -67\frac{2}{3}\pi$

$\therefore \frac{-203\pi}{3} + 68\pi = \frac{\pi}{3}$ ∴ Answer: A

57) A man standing on a bridge looks at a tower which is at a distance of 12meters from the bridge. If the angle of elevation from the eye level of the man of the tower is 45^0 and the angle of depression to the toot of the tower is 30^0, then what is the height of the tower in meters. . . EGSECE 95/03

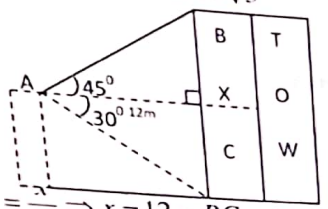
A. $24\sqrt{3}$ B. $48\sqrt{3}$ C. $16\sqrt{3}$ D. $4(3+\sqrt{3})$

Solution:

Given $\overline{AC} = 12$m

Required $\overline{BD}$

From ΔABC



$\therefore \tan 45^0 = \frac{BC}{AC} \Rightarrow 1 = \frac{x}{12} \Rightarrow x = 12 = BC$ and from ΔACD, we have

$\tan 30^0 = \frac{CD}{AC} = \frac{y}{12}$

$\Rightarrow \frac{1}{\sqrt{3}} = \frac{y}{12} \Leftrightarrow y = \frac{12}{\sqrt{3}} = 4\sqrt{3}$

$\therefore BD = \overline{BC} + \overline{CD} = x + y = 12 + 4\sqrt{3} = 4(3+\sqrt{3})$ ∴ Answer: D

58) Which of the following pairs of equations has no solution

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A. Cost = –1 and sinθ = 1

B. sint = $\frac{5}{13}$ and sect = $\frac{13}{12}$

C. cost = $\frac{8}{17}$ and tant = $\frac{15}{8}$

D. sint = $\frac{1}{\sqrt{2}}$ and cost = $\frac{1}{\sqrt{2}}$

Solution:

A) we use $\cos^2 t + \sin^2 t = 1$

$\Rightarrow (-1)^2 + (1)^2 \quad \Rightarrow 2 \neq 1 \quad \therefore$ has ***Solution***

B) $\cos^2 t + \sin^2 t = 1$

$\Rightarrow \frac{144}{169} + \frac{25}{169} = \frac{169}{169} = 1$, Note sect = $\frac{13}{12}$ then cost = $\frac{12}{13}$

$\therefore$ has no ***Solution***

C) $\cos^2 t + \sin^2 t = 1$ Dividing by $\cos^2 t$, Not cost = $\frac{8}{17} \Rightarrow$ sect = $\frac{17}{8}$

$\Rightarrow 1 + \tan^2 t = \sec^2 t$

$\therefore \sec^2 t - \tan^2 t = 1$

$\therefore \left(\frac{17}{8}\right)^2 - \left(\frac{15}{8}\right)^2 = 1 \quad \therefore$ It has ***Solution***

D) $\sin^2 t + \cos^2 t = 1$

$\Rightarrow \left(\frac{1}{\sqrt{2}}\right)^2 + \left(-\frac{1}{\sqrt{2}}\right)^2 = 1 \quad \Rightarrow \frac{1}{2} + \frac{1}{2} = 1 \quad \therefore$ has ***Solution***

Thus, **Answer A**

59) The exact value of $\tan\left(\frac{8}{3}\pi\right)$ is . . . EGSECE 2000/08

A. $\frac{-1}{\sqrt{3}}$ B. $-\sqrt{3}$ C. $\sqrt{3}$ D. $\frac{1}{\sqrt{3}}$

Solution: $\tan\left(\frac{8}{3}\pi\right) = \tan\left(\frac{8}{3}\pi - 2\pi\right) = \tan\frac{2}{3}\pi = \left(-\tan\frac{\pi}{3}\right) = -\sqrt{3}$

$\therefore$ Answer: B

60) What is the reference angle of -575^0 . . . GGSECE 2000

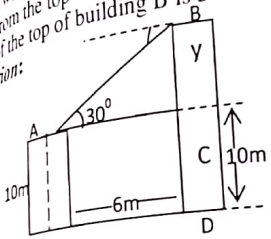
A. 35^0 B. 55^0 C. 145^0 D. 75^0

Solution: The angle between 0^0 and 360^0 that is co-terminal with -575^0 can find by $-575^0 + 720^0 = 145^0$

$\therefore \theta_R = 180^0 - 145^0 = 35^0 \quad \therefore$ **Answer: A**

62) Two buildings A and B on a level around are 6 meters apart. As seen from the top of building A that is 10 meters high, the angle of elevation of the top of building B is 30^0. How high is building B in meters?

Solution:



from ΔABC

$\tan 30^0 = \frac{y}{AC} = \frac{y}{6}$

$\Rightarrow \frac{1}{\sqrt{3}} = \frac{y}{6} \Leftrightarrow y = \frac{6}{\sqrt{3}} = 2\sqrt{3}$

$\therefore$ BD = BC + CD = $2\sqrt{3} + 10$

$\therefore$ **Answer: B**

63) What is the minimum value of sec x in the interval $\left(\frac{-\pi}{2}, \frac{3\pi}{2}\right)$

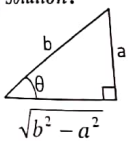
A. $\sqrt{2}$ B. 1 C. No minimum value D. –1

Solution: the minimum value of secx in $\left(\frac{-\pi}{2}, \frac{\pi}{2}\right)$ is 1 but No minimum value of secx in $\left(\frac{-\pi}{2}, \frac{3\pi}{2}\right)$ **$\therefore$ Answer: C**

63) If $\sin\theta = \frac{a}{b}$ where θ is an acute angle, express

A. tanθ B. $\frac{1}{\sin^2\theta} + \frac{1}{\cos^2\theta}$

Solution:



a) $\tan\theta = \frac{a}{\sqrt{b^2 - a^2}}$

b) $\frac{1}{\sin^2\theta} + \frac{1}{\cos^2\theta} = \frac{\sin^2\theta + \cos^2\theta}{\sin^2\theta\cos^2\theta}$

$\Rightarrow \frac{1}{\sin^2\theta\cos^2\theta} = \csc^2\theta \sec^2\theta$

$= \left(\frac{b}{a}\right)^2 \left(\frac{b^2 - a^2}{b^2}\right) = \frac{b^2 - a^2}{a^2}$

64) Given that $\tan\theta = \frac{2t}{1 - t^2}$ and that θ is an acute angle Find a) sinθ

b) cosθ

Solution:

$AC^2 = AB^2 + BC^2$

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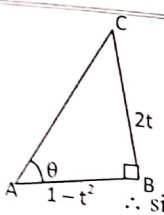


$= (1-t^2)^2 + (2t)^2$
$= 1 - 2t^2 + t^4 + 4t^2$
$= t^4 + 2t^2 + 1$
$= (t^2+1)^2$ $\therefore AC = t^2+1$

$\therefore \sin\theta = \frac{BC}{AC} = \frac{2t}{t^2+1}$ $\therefore \cos\theta = \frac{AB}{AC} = \frac{1-t^2}{1+t^2}$

65) Simplify $(1+\tan^2 x)(1-\sin^2 x)$

Solution: $\left(1+\frac{\sin^2 x}{\cos^2 x}\right)(\cos^2 x)\cos^2 x + \sin^2 x = 1$

66) If $\tan\theta = \frac{m}{\eta}$, then Find a) $\frac{1}{\tan(90^0-\theta)}$ b) $\frac{1}{\tan(\theta-90^0)}$

Solution: Note $\tan\theta = \cot(90^0-\theta)$ $\therefore$ a) $\tan\theta = \frac{m}{\eta} = \cot(90^0-\theta)$

b) $\frac{1}{\tan(\theta-90^0)} = \cot(\theta-90^0) = -\cot(90^0-\theta)$

$\therefore \frac{1}{\tan(\theta-90^0)} = \frac{-1}{\tan(90^0-\theta)} = -\cot(90^0-\theta) = -\frac{m}{\eta}$

67) If $\sin 31^0 = k$, then Find a) $\sin 59$ b) $\cos 31$ c) $\sin 211$ d) $\cos 211$

Solution: we use $\sin^2 31 + \cos^2 31 = 1$

$\Rightarrow k^2 + \cos^2 31 = 1 \Rightarrow \cos^2 31^0 = 1-k^2 \therefore \cos 31^0 = \sqrt{1-k^2}$

Note $\sin 31^0 = \cos(59^0) = k$

From, $\sin^2 59^0 + \cos^2(59^0) = 1$

$\Rightarrow \sin^2 59 + k^2 = 1 \Rightarrow \sin 59^0 = \sqrt{1-k^2}$

c) $\sin 211^0 = -\sin(211^0 - 180^0) = -\sin 31^0 = -k$

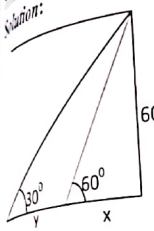
d) $\cos 211 = -\cos(211^0 - 180^0) = -\cos 31^0 = -\sqrt{1-k^2}$

68. If from the top of a hill 600m above the sea level the angle of depressions of two boats in line with the observer were found to be 30^0 and 60^0, then the distance between the two boats was

A) $400\sqrt{3}$ m B) 400m C) $300\sqrt{3}$ m D) $200\sqrt{3}$ m

67.

Solution:



$\tan 60^0 = \frac{600}{x} \Leftrightarrow x = \frac{600}{\tan 60^0} = \frac{600}{\sqrt{3}}$

$\therefore x = 200\sqrt{3}$ and $\tan 30^0 = \frac{600}{x+y}$

$\Rightarrow \frac{1}{\sqrt{3}} = \frac{600}{200\sqrt{3}+y}$

$200\sqrt{3} + y = 600\sqrt{3}$ $\therefore y = 600\sqrt{3} - 200\sqrt{3}$

$\therefore y = 400\sqrt{3} \leftarrow$ distance between boats **$\therefore$ Answer A**

Supplementary exercise

1. Each of the following angle, indicate to which quadrant they belong.
   a) 215^0 b) 1140^0 c) -900^0 d) -480^0
   e) 620^0 f) 350^0
2. convert each of the following degrees to radius
   a) 30^0 b) 240^0 c) 150^0 d) 210^0 e) -720^0 f) 270^0
3. convert each of the following radians to degree
   a) $\frac{\pi}{12}$ b) $\frac{-\pi}{6}$ c) $\frac{2\pi}{3}$ d) $\frac{5\pi}{6}$ e) $\frac{-10n}{3}$
4. Find the sine, cosine and tangent value of function of θ, if θ is in standard position and its terminal side contains the given point p(x,y)
   a) p(1,2) b) p(–4,–3) c) p(2, –2) d) $p\left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$
5. Find the angle θ in standard position, if the terminal side contains the given point p(x,y)
   a) p(4, 0) b) p(–4,0) c) p(0–6) d) p(0, –6)
6. Which of the following pairs are equal
   a) $\sin(60^0)$ and $\sin(-60^0)$ d) $\sin(\theta-80^0)$ and $\sin(80^0-\theta)$
   b) $\cos(-80^0)$ and $\cos 80^0$
   c) $\cos\left(\theta-\frac{\pi}{2}\right)$ *and* $\cos\left(\frac{\pi}{2}-\theta\right)$ e) $\tan\left(\theta-\frac{\pi}{2}\right)$ *and* $\tan\left(\frac{\pi}{2}-\theta\right)$
7. Determine the sign of:
   a) $\cos 268^0$ b) $\sin(-280^0)$ c) $\tan(-815^0)$

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8. Find any two co-terminal angles (one of them positive the other negative) for each of the following
   a) 80^0 b) 120^0 c) 230^0 d) -38^0 e) -100^0
9. Find co-terminal angle between 0 and 2π for each of the following
   a) $\frac{-113\pi}{4}$ b) $\frac{-201\pi}{5}$ c) $\frac{-131\pi}{5}$ d) $\frac{-17\pi}{3}$ e) $\frac{31\pi}{5}$ f) $\frac{73\pi}{3}$
10. Find the exact value of each of the following
    a) $\sin 30^0$ and $\sin(-330^0)$ b) $\cos\frac{121\pi}{6}$ c) $\sin(-405^0)$
11. Find the reference angle θ_R for θ if
    a) $\theta = -575^0$ b) -902^0 c) 570^0
12. Find the interval in which $y = \sin\theta$ is increasing on $[-2\pi, 2\pi]$
13. using the Pythagorean identity, find the value of the other five trigonometric function if
    a) $\sin\theta = \frac{-3}{5}$, and θ is in quadrant IV
    b) $\sin\theta = \frac{-1}{\sqrt{2}}$, and θ is in quadrant III
    c) c) $\tan\theta = \frac{1}{2}$, and θ is in quadrant I

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45^0, 90^0, 45^0 19. a) $y = -2x + 7$ b) $y = 3x - 17$

$y = -2x + 2$ b) $9x + 2y = 17$

a) slope $-\left(\frac{-6}{4}\right) = \frac{3}{2}$ and $\frac{x}{4} + \frac{y}{-6} = 1$ implies $-3x + 2y = -12$

b) slope $\frac{2}{3}$ and $\frac{x}{-3} + \frac{y}{2} = 1$ implies $2x - 3y = -6$

c) slope $\frac{3}{4}$ and $\frac{x}{-6} + \frac{y}{8} = 1$ implies $-4x + 3y = 24$

d) slope -3 and $3x + y = 24$

a) $3(x - 1) + 4(y - 3) = 0$ implies $3x + 4y - 15 = 0$

b) $1(x - 2) + 1(y + 1) = 0$ implies $x + y - 1 = 0$

c) $1(y - 2) - 1(x - 1) = 0$ implies $y - x - 1 = 0$

d) $6(x - \frac{1}{2}) + 2(y + \frac{1}{3}) = 0$ implies $6x + 2y = \frac{7}{3}$

e) $6(y + \frac{1}{3}) - 2(x - \frac{1}{2}) = 0$ implies $-2x + 6y + 3 = 0$

3. a) $\frac{3-m}{2m+1} = -4 \Rightarrow m = -1$ b) $m = \frac{4}{13}$

c) $\frac{3-m}{2m+1} = 0$ implies $m = 3$ d) $2m + 1 = 0 \Rightarrow m = \frac{-1}{2}$

Answer key Unit 11

1) a) III quadrant b) first quadrant c) quadrantal
d) III quadrant e) III quadrant f) IV quadrant

2) a) $\frac{\pi}{6}$ b) $\frac{4}{3}\pi$ c) $\frac{5}{6}\pi$ d) $\frac{7}{6}\pi$ e) -4π f) $\frac{3}{2}\pi$

3) a) 15^0 b) -30^0 c) 120^0 d) 150^0 e) -600^0

4) a) $\sin\theta = \frac{2}{\sqrt{5}}, \cos\theta = \frac{1}{\sqrt{5}}, \tan\theta = \frac{1}{2}$

b) $\sin\theta = \frac{-3}{5}, \cos\theta = \frac{-4}{5}, \tan\theta = \frac{3}{4}$

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c) $\sin\theta = \frac{-1}{\sqrt{2}}, \cos\theta = \frac{1}{\sqrt{2}}, \tan\theta = -1, \theta = 315^0 \pm 360^0 k, k \in Z$

d) $\sin\theta = \frac{1}{\sqrt{2}}, \cos\theta = \frac{1}{\sqrt{2}}, \cos\theta = \frac{-1}{\sqrt{2}}, \tan\theta = -1, \theta = 315^0 \pm 360^0 k, k \in Z$

5) a) $\theta = 0^0 \pm 360^0$k, k∈Z, b) $\theta = 180^0 \pm 360^0$k, k∈Z,
c) $\theta = 90^0 \pm 360^0$k, k∈Z, d) $\theta = 270^0 \pm 360^0$k, k∈Z,

6) a) not equal b) equal c) equal d) not equal e) not equal

7) a) Negative b) positive c) Negative

8) a) $440^0, -280^0$ b) $480^0, -240^0$ d) $322^0, -398^0$
c) $-130^0, 590^0$ e) $260^0, -460^0$

9) a) $\frac{7}{4}\pi$ b) $\frac{9\pi}{5}$ c) $\frac{\pi}{6}$ d) $\frac{\pi}{3}$ e) $\frac{7\pi}{6}$ f) $\frac{\pi}{3}$

10) a) $\frac{1}{2}$ b) $\frac{\sqrt{3}}{2}$ c) $\frac{-\sqrt{2}}{2}$

11) a) 35^0 b) 2^0 c) 30^0

12) a) increasing on $\left[-2\pi, \frac{-3\pi}{2}\right] \cup \left[\frac{-\pi}{2}, \frac{\pi}{2}\right] \cup \left[\frac{3\pi}{2}, 2\pi\right]$

decrease on $\left[\frac{-3\pi}{2}, \frac{\pi}{2}\right] \cup \left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$

13) a) $\sin\theta = \frac{-3}{5}, \cos\theta = \frac{4}{5}, \tan\theta = \frac{-3}{4}$

$\csc\theta = \frac{-5}{3}, \sec\theta = \frac{5}{4}, \cot\theta = \frac{-4}{3}$

b) $\sin\theta = \frac{-1}{\sqrt{2}}, \cos\theta = \frac{-1}{\sqrt{2}}, \tan\theta = 1$

$\csc\theta = -\sqrt{2}, \sec\theta = -\sqrt{2}, \cot\theta = 1$

c) $\sin\theta = \frac{2}{\sqrt{5}}, \cos\theta = \frac{1}{\sqrt{5}}, \tan\theta = \frac{1}{2}$

$\csc\theta = \frac{\sqrt{5}}{2}, \sec\theta = \sqrt{5}, \cot\theta = 2$

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UNIT TWELVE

Geometry and Measurement

Definition: *A polygon is simple closed figure* formed by the union of three or more line segments such that:

1. the line segment that have a common end points are non – collinear.
2. each line segment intersects only at their end – point

- The line segments are called **the sides of polygon**
- The end points of the sides are called **the vertices**

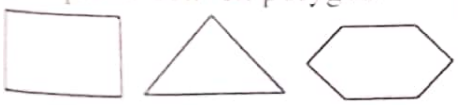
Convex – polygon is a polygon such that no line containing a side of the polygon contains a point in the interior of the polygon.

- *Each interior angle of convex polygon is* ***less than*** 180^0.

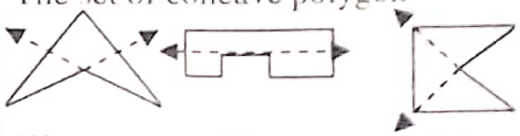
Concave – polygon is a polygon that is not convex, or (non – convex).

- *One or more of interior angle of concave polygon is greater than* 180^0.

Example of convex polygon



The set of concave polygon



We can extend line segment that contains the interior of the polygon

Polygons can be classified according to the number of sides

- A polygon with **three sides** is called **triangle**
- A polygon with **four sides** is called **quadrilateral**
- A polygon with **five sides** is called **pentagon**
- A polygon with **six sides** is called **Hexagon**
- A polygon with **seven sides** is called **heptagon**
- A polygon with **eight sides** is called **octagon**
- A polygon with **nine sides** is called **nonagon**
- A polygon with **ten sides** is called **decagon**

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b) $4500^0 = (n-2)180^0 \Rightarrow n-2 = \frac{4500^0}{180^0} = 25 \Rightarrow n = 25+2 = 27$ side no

diagonal from one vertex is $27-3 = 24$

c) $5400^0 = (n-2)180^0 \Rightarrow n-2 = \frac{5400^0}{180^0} = 30$ triangle formed

$\Rightarrow n = 30+2 = 32$ side d = b − 3 = 32 − 3 = 19 ⟵ diagonal from one vertex.

5. Determine the measure of the interior angle of indicated polygon:
   a) Triangle with measure of $(x+15)^0$, $(3x-15)^0$ and $(2x-30)^0$
   b) Pentagon with measure of x^0, $2x^0$, $(x+50)^0$, $(x-10)^0$ and $(x+40)^0$
   c) Hexagon with measure of $(x+15)^0$, $(3x+5)^0$, $(2x+80)^0$, $(x+40)^0$ and $(x+20)^0$

Solution: Here

a) n = 3, sum = $(n-2)180^0$ implies $(3-2)180^0 = 180^0$

$\Rightarrow (x+15+3x-15+2x-30)^0 = 180^0 \Rightarrow 6x = 210^0$

$\Rightarrow x = 35^0$, therefore, $50^0, 90^0, 40^0$ are required.

b) n = 5, $S = (5-2)180^0 = 540^0$ ⟵ sum of all interior angle.

$\Rightarrow x^0 + 2x^0 + (x+30)^0 + (x-10)^0 + (x+40)^0 = 540^0$

$\Rightarrow 6x^0 + 60^0 = 540^0$ implies $6x^0 = 480^0 \quad \therefore x = 80^0$

Therefore, $80^0, 160^0, 110^0, 70^0$ and 120^0 are interior angle of pentagon.

c) **Exercise left for you**

- **The sum of the measures of the exterior angle of any polygon taking one angle at each vertex is 360^0**

6. Determine the sum of the measures of the exterior angle of the following polygon.
   a) triangle b) quadrilateral c) hexagon.

Explanation: If $e_1, e_2, e_3, \ldots, e_n$ are the measure of the exterior angles of n – sided polygon. then $e_1 + e_2 + e_3 + \ldots + e_n = 360^0$

a) 360^0 b) 360^0 c) 360^0

Regular Polygon

A polygon is called **regular polygon** if all its sides are equal and all its angles are equal.

- Any regular polygon of **n – sides** has:

  i) *Measure of each interior angle* $i = \frac{(n-2)180^0}{n}$

ii) *Measure of each central angle* $= \frac{360^0}{n}$

iii) *Measure of each exterior angle* $e = \frac{360^0}{n}$

Illustrative Example

Find i) the measure of each interior angle (i)

ii) The measure of each exterior angle (e)

iii) the measure of each central angle of a regular polygon with

a) 8 sides b) 10 sides c) 18 sides d) 24 side

Solution: we use: $\left(\frac{n-2}{n}\right)180^0$ ⟵ each interior angle

$\left(\frac{360^0}{n}\right)$ ⟵ each exterior and central angle

a) n = 8, $\Rightarrow i = \left(\frac{8-2}{8}\right)180^0 = 135^0$ ⟵ each interior angle

ii) $e = \left(\frac{360^0}{8}\right) = 45^0$ ⟵ each exterior angle

iii) $0 = \left(\frac{360^0}{8}\right) = 45^0$ ⟵ each central angle

b) $n = 10 \Rightarrow i = \left(\frac{10-2}{10}\right)180^0 = 144^0$ ⟵ each interior angle

ii) $e = \left(\frac{360^0}{10}\right) = 36^0$ ⟵ each exterior angle

iii) $0 = \left(\frac{360^0}{10}\right) = 36^0$ ⟵ each central angle

c) $n = 18 \Rightarrow i = \frac{(18-2)180}{18}\ 160^0 = 144^0$ ⟵ each interior angle

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ii) $e = \left(\frac{360^0}{18}\right) = 20^0$ ⟵ each exterior angle

iii) $0 = \left(\frac{360^0}{18}\right) = 20^0$ ⟵ each central angle

d) **Exercise left for you**

8. Find the number of sides of regular polygon if each interior angle is
a) 168^0 b) 165^0 c) 156^0

Solution: a) $\left(\frac{n-2}{n}\right)180^0 \Rightarrow 168^0$ implies $180^0n - 360^0 = 168^0n$

$\Rightarrow 180^0n - 168^0n = 360^0 \Rightarrow 12n^0 = 360^0$, $\therefore n = 30$ side

b) $\left(\frac{n-2}{n}\right)180^0 = 165^0$ implies $180^0n - 360^0 = 165^0n$

$\Rightarrow 180^0n - 165^0n = 360^0$ implies $15^0n = 360^0 \Rightarrow n = \frac{360^0}{15}$ $\therefore n = 24$ side

c) **Exercise left for you (Tip)**

9. Find the number of sides of a regular polygon if each exterior angle is
a) 5^0 b) 24^0 c) 15^0 d) 164^0 e) 16^0

Solution: Let e be exterior angle

Explanation: For n, – *sides regular polygon each exterior angle*

$e = \frac{360^0}{n} \Leftrightarrow \frac{360^0}{e} = n$

a) $n = \frac{360^0}{e} = \frac{360^0}{5^0} = 72$ sides b) $n = \frac{360^0}{24^0} = 15$ sides

c) **Exercise left for you**

d) $n = \frac{360^0}{164^0} = 2.19$, It is impossible because n = 2.19 is not natural number.

e) $n = \frac{360^0}{16^0} = 22.5$, not possible because n = 22.5 is not natural number

Properties of Regular polygon

A Symmetric properties of regular polygon

A plane figure can be divided exactly into two identical parts by a line

If the line ℓ is called **line of symmetry**.

A figure is called **symmetric** if it has at least one line of symmetry.

Regular polygons are symmetric.

A regular n – side polygon has **n – lines of symmetry**

Illustrative Example

How many lines of symmetry can have?

a) Circle b) Isosceles triangle c) Scalene triangle
d) Rectangle e) parallelogram f) Rhombus

Solution

a) Circle has infinite line of symmetry
b) Isosceles triangle has **one line of symmetry**
c) Scalene triangle has **no line of symmetry**.
d) Rectangle has **two line of symmetry**.
e) Parallelogram has at least two line of symmetry.
f) Rhombus has **four** line of symmetry

How many lines of symmetry can have?

a) Equilateral triangle d) Regular hexagon
b) Square e) regular heptagon
c) Regular pentagon f) regular – n – side polygon

Solution

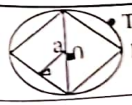
Explanation: n – sided regular polygon has **n – lines of symmetry**

a) n = 3, therefore **three** lines of symmetry
b) n = 4, therefore **four** lines of symmetry
c) n = 5, therefore **five** lines of symmetry
d) n = 6, regular hexagon has **six** lines of symmetry
e) n = 7, regular hexagon has **seven** lines of symmetry
f) A regular – n – sides polygon has **n** lines of symmetry

Note A regular polygon of **odd number of sides** has every line of symmetry, passes through the vertex.

B. Inscribed Regular Polygon

An inscribed polygon is a polygon all of whose sides are chords of a circle



The circle is called **circumscribed** about the polygon.

Inscribe polygon is the same term as circumscribed circle

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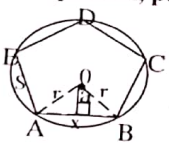
Regular polygons have special relation to circle.

- A regular polygon can always be inscribed in a circle.

Apothem, perimeter, and area of regular polygon.



- **An apothem of regular polygon** is segment from its center perpendicular to one of its sides.
- The distance **a** is an apothem.
- **A central angle** of a regular polygon is an angle between two radii (r) given by $m(<AOB)=\frac{360^0}{n}$, where n is the number of sides.

i) Apothem of regular polygon

Consider the above inscribed pentagon we know that $m(<AOB) = \frac{360^0}{n}$ implies that $m(<AOX) = \frac{180^0}{n}$ ⟵ Apothem bisects the central angle

- We know that $\cos\left(\frac{180^0}{n}\right) = \frac{a}{r} = \frac{\text{apothem}}{\text{radius}}$

$\Rightarrow a = r\cos\left(\frac{180^0}{n}\right)$ ⟵ apothem regular polygon

Therefore, apothem of regular polygon with **n – sides** and radius **r** is given by $\mathbf{a} = r\cos\left(\frac{180^0}{n}\right)$

ii) The length of side 'S' of regular polygon.

Consider the above inscribed pentagon.

$m(<AOX) = \frac{180^0}{n}$, so that $\sin\left(\frac{180^0}{n}\right) = \frac{Ax}{r}$

since $Ax = \frac{1}{2}AB = \frac{1}{2}S \leftarrow AB = S$

$\Rightarrow \sin\left(\frac{180^0}{n}\right) = \frac{\frac{1}{2}s}{r}$ implies, $\frac{1}{2}S = r\sin\left(\frac{180^0}{n}\right)$

Therefore, $S = 2r\sin\left(\frac{180^0}{n}\right)$ ⟵ side length of regular polygon

iii) Perimeter (P) of a regular polygon.

Perimeter = AB + BC + CD + DE + EA . . . .
= S + S + S + S + ... = nS

$\therefore P = nS = 2nr\sin\left(\frac{180^0}{n}\right)$

iv) Area of regular n – sided polygon

Consider the above inscribed polygon Let area of a polygon be A

Area: $A = a(\Delta AOB + \Delta BOC + \Delta COD + \ldots)$

$= \frac{1}{2}r.r\sin\left(\frac{360^0}{n}\right) + \frac{1}{2}r.r\sin\left(\frac{360^0}{n}\right) + \ldots .$

$\therefore A = \frac{1}{2}nr^2\sin\left(\frac{360^0}{n}\right)$ ⟵ **Area of regular n – sided – polygon with radius (r).**

Method II

Area: $A = \frac{1}{2}ap$ ⟵ **Area of regular polygon with perimeter p and apothem (a)**

Illustrative Example

12. **Find** i) the length of side
ii) the apothem
iii) the area, of the regular polygon inscribed in a circle of radius r = $\sqrt{3}$ cm of the following regular polygon.
a) triangle b) square c) hexagon

Solution: since the **inscribed** polygon is regular, so that

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Use: $S = 2r\sin\left(\frac{180^0}{n}\right)$ ⟵ side length

$a = r\cos\left(\frac{180^0}{n}\right)$ ⟵ apothem

$A = \frac{1}{2}ap$ or $A = \frac{1}{2}nr^2\sin\left(\frac{360^0}{n}\right)$

a) $n = 3$, $r = \sqrt{3}$.

i) $S = 2(\sqrt{3})\sin\left(\frac{180^0}{3}\right) = 2\sqrt{3}\left(\frac{\sqrt{3}}{2}\right) = 3$cm ⟵ side length

ii) $a = r\cos\left(\frac{180^0}{n}\right) = \sqrt{3}\cos\left(\frac{180^0}{3}\right) = \sqrt{3}\left(\frac{1}{2}\right) = \frac{\sqrt{3}}{2}$ ⟵ apothem

iii) $A = \frac{1}{2}ap = \frac{1}{2}\left(\frac{\sqrt{3}}{2}\right)(3(3)) = \frac{9\sqrt{3}}{4}$ cm²

Method II $A = \frac{1}{2}nr^2\sin\left(\frac{360^0}{n}\right) = \frac{1}{2}(3)(\sqrt{3})^2\left(\frac{\sqrt{3}}{2}\right) = \frac{9\sqrt{3}}{4}$ cm²

b) Exercise left for you

c) Regular hexagon, $n = 6$, $r = \sqrt{3}$

i) $S = 2r\sin\left(\frac{180^0}{n}\right) = 2(\sqrt{3})\sin\left(\frac{180^0}{6}\right) = \sqrt{3}$ cm

ii) $a = r\cos\left(\frac{180^0}{n}\right) = \sqrt{3}\left(\cos\left(\frac{180^0}{6}\right)\right) = (\sqrt{3})\left(\frac{\sqrt{3}}{2}\right) = \frac{3}{2}$ cm

iii) perimeter $(P) = ns = (6)(\sqrt{3}) = 6\sqrt{3}$.

$\therefore A = \frac{1}{2}ap = \frac{1}{2}\left(\frac{3}{2}\right)6\sqrt{3}$ cm² $= \frac{9\sqrt{3}}{2}$ cm²

13. The side length of **inscribed** regular polygon is 10 unit long find the area of each of the following polygon

a) hexagon b) triangle c) square

Solution: To find r use the relation $S = 2r\sin\left(\frac{180^0}{n}\right)$

To find Area use, $A = \frac{1}{2}nr^2\sin\left(\frac{360^0}{n}\right)$

a) regular hexagon: $n = 6$ and $S = 10$cm

$\Rightarrow S = 2r\sin\left(\frac{180^0}{n}\right)$ implies $10 = 2r\sin\left(\frac{180^0}{6}\right)$

$\Rightarrow 2r\left(\frac{1}{2}\right) = 10$cm implies $r = 10$cm.

$\therefore A = \frac{1}{2}nr^2\sin\left(\frac{360^0}{n}\right) = \frac{1}{2}(6)(10)^2\sin 60^0 = 150\sqrt{3}$ cm².

b) triangle, $n = 3$ and $S = 10$cm, since $S = 2r\sin\left(\frac{180^0}{n}\right)$

$\Rightarrow 10 = 2r\sin\left(\frac{180^0}{3}\right)$ implies $2r\left(\frac{\sqrt{3}}{2}\right) = 10 \Rightarrow r = \frac{10}{\sqrt{3}} = \frac{10\sqrt{3}}{3}$

Area $A = \frac{1}{2}nr^2\sin\left(\frac{360^0}{n}\right) = \frac{1}{2}(3)\left(\frac{10}{\sqrt{3}}\right)^2\sin 120^0 = 25\sqrt{3}$ cm².

c) **Exercise left for you (Tip)**

14. The apothem of inscribed regular polygon is 6cm.
Find i) the radius (r)
ii) the area of regular polygon of each of the following
a) square b) triangle c) hexagon

Solution: **To find r**: Use the relation: $a = r\cos\left(\frac{180^0}{n}\right)$

To find Area A, Use $A = \frac{1}{2}nr^2\sin\left(\frac{360^0}{n}\right)$

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a) $n = 4$, $a = 6$, since $a = r\cos\left(\frac{180^0}{n}\right)$

i) $6 = r\cos\left(\frac{180^0}{4}\right)$ implies that $r\cos 45^0 = 6$

$\Rightarrow r = \frac{6}{\cos 45^0} = (6)(\sqrt{2}) = 6\sqrt{2}$ cm ← radius of circle

ii) Area: $A = \frac{1}{2}nr^2\sin\left(\frac{360^0}{n}\right) = \frac{1}{2}(4)(6\sqrt{2})^2\sin 90^0 = 144\text{cm}^2$

b) **Exercise left for you**

c) Regular hexagon: $n = 6$, $a = 6$cm, since $a = r\cos\left(\frac{180^0}{n}\right)$

i) $6 = r\cos\left(\frac{180^0}{6}\right)$ implies $6 = r\cos 30^0$

$\Rightarrow r\left(\frac{\sqrt{3}}{2}\right) = 6$ implies $r = \frac{12}{\sqrt{3}} = 4\sqrt{3}$ ← radius

ii) Area: $A = \frac{1}{2}nr^2\sin\left(\frac{360^0}{n}\right) = \frac{1}{2}(6)(4\sqrt{3})^2\sin 60^0$

$\therefore A = 72\sqrt{3}\text{ cm}^2$ ← Area of hexagon inscribed

5. The area of inscribed regular hexagon is $18\sqrt{3}\text{ cm}^2$
Find a) the radius of circle b) the side length
c) the apothem d) the perimeter

***lution*: To find r,** use the relation $A = \frac{1}{2}nr^2\sin\left(\frac{360^0}{n}\right)$

a) $n = 6$, Area, $A = 18\sqrt{3}\text{ cm}^2$, since $A = \frac{1}{2}nr^2\sin\left(\frac{360^0}{n}\right)$

$\Rightarrow 18\sqrt{3}\text{ cm}^2 = \frac{1}{2}(6)r^2(\sin 60^0)$ implies $3r^2\left(\frac{\sqrt{3}}{2}\right) = 18\sqrt{3}$

$\Rightarrow r^2 = 12$ implies $r = \sqrt{12} = 2\sqrt{3}$ cm

b) $S = 2r\sin\left(\frac{180^0}{n}\right)$ implies $S = 2(4\sqrt{3})\left(\frac{1}{2}\right) = 4\sqrt{3}$

c) Since $A = \frac{1}{2}ap$ implies $2A = ap$ and $p = ns$.

$\Rightarrow a = \frac{2A}{P} = \frac{2(18\sqrt{3})}{(6)(4\sqrt{3})} = \frac{3}{2}$ cm

d) **Exercise left for you**

6. If a regular polygon of 12 sides with radius 4 units long is given, then its area is equal to

A. 24sq unit B. 48sq unit C. $24\sqrt{3}$ sq. unit D. $12\sqrt{3}$ sq. unit

Solution: we know that $A = \frac{1}{2}nr^2\sin\left(\frac{360^0}{n}\right)$

$\Rightarrow A = \frac{1}{2}(12)(4)^2\sin\left(\frac{360^0}{12}\right) = (6)(16)\left(\frac{1}{2}\right) = 48\text{sq. unit}$ **Answer: B**

7. Find the area of **inscribed** regular polygon with indicated side of polygon and radius

a) triangle with radius $r = 2$ unit.
b) $n = 12$ with radius $r = 3$ unit.
c) $n = 6$, with side length, $S = 8$ unit.
d) $n = 5$ with side length, $S = 4$ unit

Explanation: Area, $A = \frac{1}{2}nr^2\sin\left(\frac{360^0}{n}\right)$

a) $n = 3$, $r = 2$ implies, $A = \frac{1}{2}(3)(2)^2\sin(120^0) = 3\sqrt{3}$ sq. unit

b) **Exercise left for you (Tip!)**

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c) **To find r,** Use, $S = 2r \sin\left(\frac{180^0}{n}\right)$

$\Rightarrow 8 = 2r \sin\left(\frac{180^0}{6}\right)$ implies $r = 8$

Area, $A = \frac{1}{2} nr^2 \sin\left(\frac{360^0}{n}\right) = \frac{1}{2}(6)(64)\frac{(\sqrt{3})}{2} = 96\sqrt{3}$ sq. u

d) **Exercise left for you (Tip!)**

18. The perimeter of **inscribed** a regular polygon is 24 unit
Find i) the radius ii) the apothem
iii) the area of each of the following polygon
a) Triangle b) quadrilateral c) hexagon

Explanation: perimeter = ns $\Rightarrow s = \frac{P}{n}$

a) Regular triangle, n = 3, p = 24 implies $S = \frac{24}{3} = 8$

i) $S = 2r \sin\left(\frac{180^0}{n}\right)$ implies $8 = 2(r) \sin 60^0$

$\Rightarrow r = \frac{8}{2\sin 60^0} = \frac{8}{2\left(\frac{\sqrt{3}}{2}\right)} = \frac{8}{\sqrt{3}}$ unit

ii) Apothem: $a = r \cos\left(\frac{180^0}{n}\right) = \frac{8}{\sqrt{3}}(\cos 60^0) = \frac{4}{\sqrt{3}}$

iii) Area: $A = \frac{1}{2} nr^2 \sin\left(\frac{360^0}{n}\right)$

$= \frac{1}{2}(3)\left(\frac{8}{\sqrt{3}}\right)^2\left(\frac{\sqrt{3}}{2}\right) = 16\sqrt{3}$ sq. unit

Method II. $A = \frac{1}{2} ap = \frac{1}{2}\left(\frac{4}{\sqrt{3}}\right)24 = 16\sqrt{3}$ sq. unit

Regular quadrilateral, n=4, P=24 unit implies, $S = \frac{24}{4} = 6$ unit

i) $S = 2r \sin\left(\frac{180^0}{n}\right)$ implies $6 = 2r \sin 45^0$

$\Rightarrow r = \frac{6}{\sqrt{2}} = 3\sqrt{2}$ unit.

ii) Apothem: $a = r \cos\left(\frac{180^0}{n}\right) = 3\sqrt{2} \cos 45^0 = 3\sqrt{2}\left(\frac{\sqrt{2}}{2}\right) = 3$ sq. unit

iii) Area: $A = \frac{1}{2} ap = \frac{1}{2}(3)(24) = 36$ squ. unit

c) **Exercise left for you**

If ΔABC is an equilateral triangle inscribed in a circle.
Find the area of an equilateral triangle inscribed in a circle with radius
a) r = 1 unit b) 2 unit c) $\sqrt{2}$ unit d) $\sqrt{6}$ unit

Solution: Use $A = \frac{1}{2} nr^2 \sin\left(\frac{360^0}{n}\right)$ n

Where n = 3, implies $A = \frac{3r^2\sqrt{3}}{4} = \frac{3\sqrt{3}}{4} r^2$

a) $r = 1 \Rightarrow A = \frac{1}{2}(3)(1)^2 \sin(120^0) = \frac{3\sqrt{3}}{4}$ sq. unit

b) r = 2unit **therefore,** $A = \frac{(3)\sqrt{3}(2)^2}{4} = 3\sqrt{3}$ sq. unit

c) $r = \sqrt{2}$ unit, therefore, $A = \frac{3\sqrt{3}}{4} r^2 = \frac{3\sqrt{3}(\sqrt{2})^2}{4} = \frac{3\sqrt{3}}{2}$ sq. unit

d) $r = \sqrt{6}$ ⟵ Exercise left for you.

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20. If ABCD is quadrilateral inscribed in circle '0' such that $\overline{DC} // \overline{AB}$, then which of the following is necessarily true

A. ABCD is square C. ABCD is trapezium

B. ABCD is a rhombus D. ABCD is a rectangle

Explanation: Regular polygon is inscribed in a circle,

therefore, it is square **Answer: A**

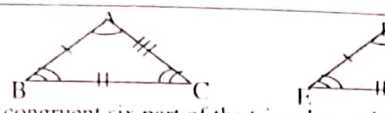
Congruence and Similarity

- *Two geometric figures are said to be congruent if they have exactly the same shape and size.*
- To line segments AB and CD are congruent if AB = CD
- Two angles are congruent if and only if their measures are equal.

A. Congruency of Triangle

- Triangles that fit exactly are called **congruent triangle**
- Congruent triangles are triangles that have the same size and the same shape.

If two triangles are congruent; their corresponding sides and angles must be congruent.



The congruent six part of the triangle are the three corresponding angle and the three corresponding size of ΔABC and ΔDEF denote ΔABC ≡ ΔDEF

These are:

$$\left.\begin{matrix}\overline{AB} \equiv \overline{DE} \\ \overline{BC} \equiv \overline{EF} \\ \overline{AC} \equiv \overline{DF}\end{matrix}\right\} \text{corresponding sides} \qquad \left.\begin{matrix}\angle A \equiv \angle D \\ \angle B \equiv \angle E \\ \angle C \equiv \angle F\end{matrix}\right\} \text{corresponding angle}$$

Note: "The order in which the vertices" match determine a correspondence between the sides and angles of two triangles.

Four Basic principles of Congruent triangle

These are: i) S SS (three side)

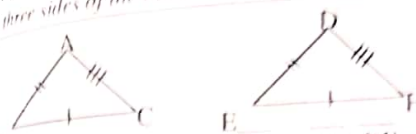
ii) A S A (two angle and included side)

iii) S A S (two side and included angle)

iv) R H S (Right angle hypotenuse and a side)

i) Side – side – side congruence (S S S)

Two triangles are congruent if three sides of one triangle are equal to the three sides of the other triangle

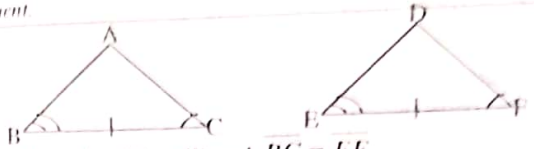


Given: Δ ABC and ΔDEF, $\overline{AB} = \overline{DE}$, $\overline{BC} = \overline{EF}$ and $\overline{AC} = \overline{DF}$

Result: Δ ABC = ΔDEF, ∴ ∠A = ∠D, ∠B = ∠E, ∠C = ∠F

ii) ASA (Angle Side – Angle)

If two angels are the included side of one triangle are congruent to two angels and the included side of other triangle ***each to each*** *the triangles are congruent*



Given: ∠B = ∠E, ∠C = ∠F, and $\overline{BC} = \overline{EF}$

Result: Δ ABC = ΔDEF, by ASA

iii) AAS (Side – Angle - Side)

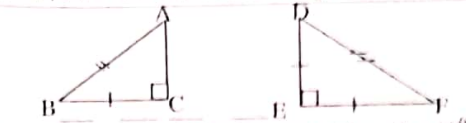
Two triangles are congruent if two sides and included angle of one triangle are equal to the corresponding two sides and included angle of the other triangle.

Given: $\overline{AB} = \overline{DE}$, $\overline{AC} = \overline{DF}$ and ∠A = ∠D

Result: Δ ABC = ΔDEF by SAS

iv) RHS (Right – Angle – hypotenuse and aside)

If two right angle triangles have two sides of the one is equal to two sides of the other, each to each then the triangles are congruent



Given: $\overline{AB} \equiv \overline{DF}$, $\overline{BC} \equiv \overline{DE}$ and ∠C = ∠E = 90°

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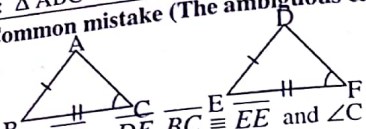
Result: $\Delta ABC \equiv \Delta DEF$ by RHS.

Common mistake (The ambiguous case)



Given: $\overline{AB} \equiv \overline{DE}, \overline{BC} \equiv \overline{EE}$ and $\angle C \equiv \angle F$

Result: Δ ABC not congruent to ΔDEF because the angle is not included between the sides.

In what case $\Delta ABC \equiv \Delta DEF$? (Explain:)

Answer: Case 1: Tf the given angle of triangle right angle, $\Delta ABC \equiv \Delta DEF$

Case 2: If $\angle A \equiv \angle D$,

Illustrative Example

21. Given that $\Delta ABC \equiv \Delta PQR$, which of the following is necessarily true?

A. $\overline{AC} \equiv \overline{PQ}$ C. $\overline{AB} \equiv \overline{QR}$

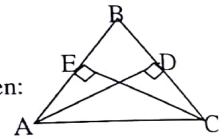
B. $\angle BAC = \angle QRP$ D. $\angle BCA = \angle PRQ$

Explanation: $\Delta ABC \equiv \Delta PQR$ provided that

$$\left.\begin{matrix}\angle A \equiv \angle P \\ \angle B \equiv \angle Q \\ \angle C \equiv \angle R\end{matrix}\right\} \text{and} \left.\begin{matrix}\overline{AB} \equiv \overline{PQ} \\ \overline{AC} \equiv \overline{PR} \\ \overline{BC} \equiv \overline{QR}\end{matrix}\right.$$

∴ Answer: A

22. In ΔABC, if $\overline{AB} \equiv \overline{BC}$, $\overline{AD} \perp \overline{BC}$, $\overline{CE} \perp \overline{AB}$, then:



Prove: $\overline{AD} \equiv \overline{CE}$

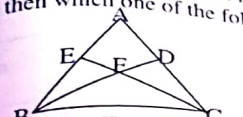
Explanation: **Two column proof**

Statements	**Reasons**
1. $\overline{AB} \equiv \overline{BC}$	← Given
2. $\angle BAC \equiv \angle BCA$	← Isosceles triangle theorem
3. $\overline{AC} \equiv \overline{CA}$	← common side
4. $\angle AEC \equiv \angle CDA$	← Given (Right angle triangle)
5. $\angle CAD \equiv \angle ACE$	← from 2 and 3.

Therefore, $\Delta ACD \equiv \Delta CAE$, by ASA

$\therefore \overline{AD} \equiv \overline{CE}$

23. Given ΔABC as shown in the figure below if $\overline{AB} \equiv \overline{AC}$ and $\overline{AE} \equiv \overline{AD}$, then which one of the following is correct?



A. $\Delta ABD \equiv \Delta ACE$
B. $\Delta EBC \equiv \Delta DCB$
C. $\Delta EBF \equiv \Delta DCF$
D. All of the above

Explanation: Two column proof

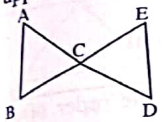
1) $\overline{AB} \equiv \overline{AC}$ ← Given
2) $\overline{AE} \equiv \overline{AD}$ ← Given
3) $\angle A \equiv \angle A$ ← common vertex

Therefore, $\Delta ABD \equiv \Delta ACE$ by SAS

4) $\overline{BD} \equiv \overline{CE}$ ← from the above
5) $\overline{BE} \equiv \overline{CD}$ ← since $\overline{AB} \equiv \overline{AC}$ and $\overline{AE} \equiv \overline{AD}$
6) $\overline{BC} \equiv \overline{CB}$ ← common side

Therefore, $\Delta BCD \equiv \Delta CBE$ by SSS Provided that $\Delta EBC \equiv \Delta DCB$ ∴ **Answer:** D

24. In the following figure $\overline{AD}$ and $\overline{BE}$ bisect each other at C. The appropriate theorem used to prove that $\Delta ABC \equiv \Delta DEC$ is



A. AAS B. ASA C. SAS D. SSS

Explanation: $\overline{AC} \equiv \overline{CD}$ ← Given

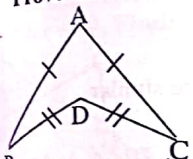
$\overline{BC} \equiv \overline{CE}$ ← Given

$m(\angle ACB) \equiv m(\angle DCE)$ ← V.o. angle

$\therefore \Delta ACB \equiv \Delta DCE$ by SAS, **∴ Answer:** C

25. In the following figure $\overline{AB} \equiv \overline{AC}$, $\overline{BD} \equiv \overline{DC}$

Prove that: $m(\angle ABD) \equiv m(\angle ACD)$



***Explanation*:**

1) $\overline{AB} \equiv \overline{AC}$ ← Given
2) $\overline{BD} \equiv \overline{CD}$ ← Given
3) $\overline{AD} \equiv \overline{AD}$ ← common side (construction)

Therefore, $\Delta ABD \equiv \Delta ACD$ by SSS

Provided that $m(\angle ABD) \equiv m(\angle ACD)$

B. Similar Figure

When figures have the **same shape** but are **different sizes**, they are called **similar figure**

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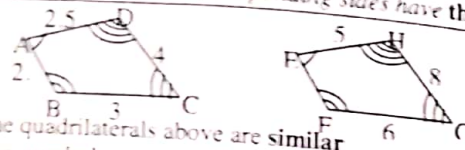
Similar polygons

Two polygons are **similar** *if and only if their corresponding angles are congruent and their corresponding sides have* **the same ratio.**



The quadrilaterals above are **similar**

The symbol ~ means '**is similar to**'. Quadrilateral ABCD is similar to quadrilateral EFGH, is denoted by ABCD ~ EFGH

Provided that

i) $\angle A \equiv \angle E$, $\angle B \equiv \angle F$, $\angle C \equiv \angle G$ and $\angle D \equiv \angle H$

ii) $\frac{AB}{EF} = \frac{BC}{FG} = \frac{CD}{GH} = \frac{DA}{HE} = \frac{2.5}{5} = \frac{1}{2}$

- The ratio of corresponding sides of two similar polygon is called the **scale factor.**
- The scale factor of ABCD to EFFH is $\frac{1}{2}$
- The scale factor of EFGH top ABCD is 2.

Similarity of Triangles *Two triangles are similar,* if

i) their corresponding sides are **proportional**

ii) their corresponding angles are equal.

Note: similar triangles should always be named so that the order of letters indicates the correspondence between the two figure

Thus, Δ ABC ~ ΔDEF are similar means that $\angle A \equiv \angle D$, $\angle B \equiv \angle E$, $\angle C \equiv \angle F$ and that $\frac{BC}{EF} = \frac{CA}{FD} = \frac{AB}{DE}$

Identifying similar Triangles

How do you know whether two triangles are similar?

There are also tests to determine whether two triangles are similar

These are: i) AA - similarity (Angle – Angle)

ii) SSS – Similarity (side – side – side)

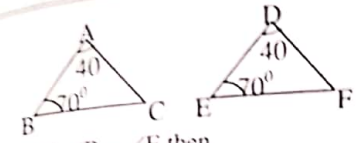
iii) SAS – similarity (side – angle – side)

i) AA – similarity

- *If two angles of one triangle are congruent to two angles of another triangle, then the triangles are similar*

28

Example

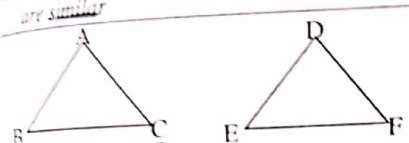


Given If $\angle A = \angle D$, $\angle B = \angle E$ then

ΔABC ~ ΔDEF by AA – similarity

ii) SSS – similarity

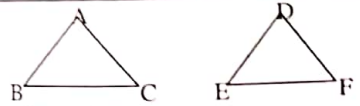
The corresponding sides of two triangles are proportional, then the triangles are similar



If $\frac{AB}{DE} = \frac{AC}{DF} = \frac{BC}{EF}$ then ΔABC ~ ΔDEF by SSS – similarity

iii) SAS – similarity

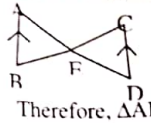
- *If two sides of a triangle are proportional to the two corresponding sides of another triangle and the included angles are congruent, then the triangles are* **similar.**



If $\frac{AB}{DE} = \frac{AC}{DF}$ and $\angle A = \angle D$ then ΔABC ~ ΔDEF

Illustrative Example

26. Given that $\overline{AB} // \overline{CD}$, AB = 4 unit, AE = 3x + 4 CD = 8 and ED = x + 12, **Find: x**



Explanation: since $\overline{AB} // \overline{CD}$ imply $\angle BAE \equiv \angle CDE$ and $\angle ABE \equiv \angle DCE$, because they are interior angles.

Therefore, ΔABE ~ Δ DCE by AA

Provided that $\frac{AB}{DC} = \frac{AE}{DE} \Leftrightarrow \frac{4}{8} = \frac{3x+4}{x+12}$

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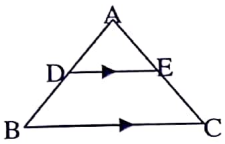
$\Rightarrow x + 12 = 6x + 8$ implies $-5x = -4$ Therefore, $x = \frac{4}{5}$

Parallel Lines and Proportional parts

A. Triangle proportionality

If a line is parallel to one side of a triangle and intersects the other two sides in two distinct points, then it separates these sides into segment of proportional length.



If DE//BC, then $\frac{DB}{AD} = \frac{EC}{AE}$

To prove:

1) DE//BC ⟵ Given
2) $\angle ADE \equiv \angle ABC$ ⟵ corresponding
3) $\angle AED \equiv \angle ACB$ ⟵ corresponding angle

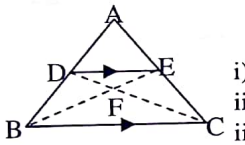
Therefore, $\Delta ABC \sim \Delta ADE$ by AA – similarity

Provided that: $\frac{AB}{AD} = \frac{AC}{AE}$

$\Rightarrow \frac{AD + DB}{AD} = \frac{AE + EC}{AE}$ implies $1 + \frac{DB}{AD} = 1 + \frac{EC}{AE}$

Therefore, $\frac{DB}{AD} = \frac{EC}{AE}$

27. Given that DE//BC, $\overline{AD} = 5cm$, $\overline{DE} = 4cm$ $\overline{DF} = 2cm$, $\overline{FE} = 3cm$ and $\overline{FC} = 5cm$



Find a) $\overline{DB}$ b) $\overline{FB}$ c) $\overline{BC}$

Explanation:

i) DE//BC ⟵ Given
ii) $\angle DEF \equiv \angle CBF$ ⟵ alternate interior
iii) $\angle EDF \equiv \angle BCF$ ⟵ alternate interior

Therefore, $\Delta DEF \sim \Delta CBF$; consequently

$\frac{DE}{CB} = \frac{EF}{BF} = \frac{DF}{CF}$ implies $\frac{4}{CB} = \frac{3}{BF} = \frac{4}{10} = \frac{2}{5}$ ←scale factor

$\Rightarrow \frac{4}{CB} = \frac{2}{5}$ implies $\overline{CB} = \frac{20}{2} = 10cm$

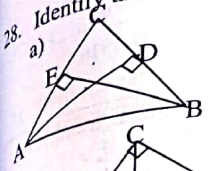
Similarly; $\Delta ADE \sim \Delta ABC$ implies $\frac{AD}{AB} = \frac{DE}{BC}$

$\Rightarrow \frac{5}{5 + BD} = \frac{4}{10}$ implies $25 = 10 + 2\overline{BD} \Rightarrow BD = \frac{15}{2}$

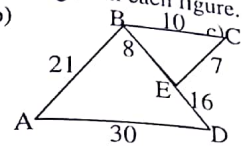
Thus, a) $\overline{DB} = \frac{15}{2} cm$ a) $\overline{BF} = \frac{15}{2} cm$ c) $\overline{BC} = 10cm$

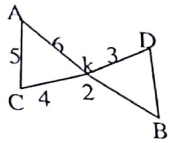
28. Identify the similar triangles in each figure.

a)

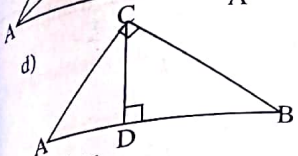


b)





d)



Solution:

a) 1) $m(\angle ADC) = m(\angle BEC)$⟵ Given right angle
2) $m(\angle C) = m(\angle C)$ ⟵ common vertex

Therefore, $\Delta ADC \sim \Delta BEC$ by A.A similarity

Similarly: $m(\angle AEF) = m(\angle BFD)$ ← vertically opposite angle

Therefore, $\Delta AEF \sim \Delta BFD$ by A.A – similarity

$\therefore \Delta ADC \sim \Delta BEC$, $\Delta AEF \sim \Delta BFD$ and $\Delta ABE \sim \Delta BAD$

b) We observe that $\frac{7}{21} = \frac{10}{30} = \frac{8}{8 + 16}$

Provided that, $\Delta CEB \sim \Delta ABD$ by SSS

c) We observe that $\frac{3}{6} = \frac{2}{4} = \frac{1}{2}$ implies

$\Delta DKB \sim \Delta AKC$ by SAS – similarity $\Rightarrow \Delta DBK \sim \Delta ACK$

d) i) $\Delta ACB \sim \Delta ABC$ ⟵ by A.A – similarity

ii) $\Delta ABC \sim \Delta BAC \sim \Delta CAD$ by – A.A similarity

consequently: $CD^2 = AD \cdot DB$ and $AC^2 = AD \cdot AB$

29. Determine whether each statement is true or false

a. All similar triangles are congruent
b. All congruent triangles are similar

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c. Every two equilateral triangles are congruent.
d. Every two equilateral triangles are similar.
e. Every two circles are similar
f. Every two squares are similar.

Solution: a) False b) True c) false d) True e) True f) true

Ratio of perimeters and Areas of similar Figure

Here we are going to see the way how to relate the side length, the perimeter and areas of similar triangle (polygon).

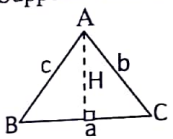
Let P_1 = perimeter of first triangle/ polygon
P_2 = perimeter of 2nd triangle/ polygon
A_1 = Area of first triangle/ polygon
A_2 = Area of 2nd triangle/ polygon

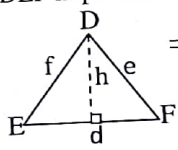
If the ratio of the lengths of the corresponding sides of two similar triangle or polygon is **k**, then:

i) **the ratio of their perimeter is given by** $\frac{P_1}{P_2}=\frac{S_1}{S_2}=k$

ii) **the ratio of their area is given by** $\frac{A_1}{A_2}=\left(\frac{S_1}{S_2}\right)^2=k^2$

Suppose: $\Delta ABC \sim \Delta DEF$ implies that $\frac{AB}{DE}=\frac{BC}{EF}=\frac{AC}{EF}=k$





$\Rightarrow a+b+c=P_1$ and $f+d+e=P_2$

$\Rightarrow \frac{a+b+c}{d+e+f}=\frac{P_1}{P_2}$

We know that $\frac{\text{area of } \Delta ABC}{\text{area of } \Delta DEF}=\frac{\frac{1}{2}(H)BC}{\frac{1}{2}(h)EF}=\left(\frac{H}{h}\right)\left(\frac{BC}{EF}\right)$

$\therefore \frac{\text{area of } \Delta ABC}{\text{area of } \Delta DEF}=(k)(k)=k^2 \rightarrow \frac{H}{h}=\frac{BC}{EF}=k$

To sumup: Area ratio = (side length ratio)^2 = (perimeter ratio)^2

Illustrative Example

30. Two triangles are similar. A side of one is 3unit long. The corresponding side of the other is 4 unit long.

Find a) the ratio of their perimeter
b) the ratio of their area

Solution: Here, we use $\frac{P_1}{P_2}=\frac{S_1}{S_2}=k \Rightarrow S_1=3, S_2=4$

a) $\frac{P_1}{P_2}=\frac{S_1}{S_2}=\frac{3}{4}$ ← the perimeter ratio.

b) $\frac{\text{area of } first}{\text{area of } second}=\frac{A_1}{A_2}=\left(\frac{S_1}{S_2}\right)^2=\left(\frac{3}{4}\right)^2=\frac{9}{16}$

31. Find the ratio of the area of two similar triangles if the ratio of two corresponding side is

a) $2:\sqrt{3}$ b) 2:5 c) 1:5 d) a:3a

Explanation: Area – ratio = (side length ratio)^2

a) Area ratio = $\left(\frac{2}{\sqrt{3}}\right)^2=\frac{4}{3}$ b) Area ratio = $\left(\frac{2}{5}\right)^2=\frac{4}{25}$

c) Area ratio = $\left(\frac{1}{5}\right)^2=\frac{1}{25}$ d) Area ratio = $\left(\frac{a}{3a}\right)^2=\frac{1}{9}$

32. The area of two similar triangles are in the ratio of 25:16.
Find a) the side length of the larger triangle if the corresponding side of the smaller triangle has length 80cm.
b) the perimeter of smaller if the perimeter of the larger is 125cm

Solution: we have $\frac{A_1}{A_2}=\left(\frac{S_1}{S_2}\right)^2=\frac{25}{16}$

$\Rightarrow \frac{S_1}{S_2}=\sqrt{\frac{25}{16}}=\frac{5}{4}$ it is same as $\frac{P_1}{P_2}=\frac{5}{4}$

a) $\frac{S_1}{S_2}=\frac{5}{4} \Rightarrow S_1=\frac{5}{4}(80)=100\text{cm}$ ← length of the larger

b) $\frac{P_1}{P_2}=\frac{5}{4} \Rightarrow \frac{125}{P_1}=\frac{5}{4}$ implies $P_2=125\times\frac{4}{5}=100\text{cm}$

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33. Two polygons are similar. A side of one is 2cm long. The corresponding side of the other is 5cm. the area of the first is $16cm^2$.
Find the area of the second polygon.
Explanation: we know that
Area ratio = (side length ratio)^2

$$\Rightarrow \frac{A_1}{A_2} = \left(\frac{S_1}{S_2}\right)^2 \text{ implies } \frac{16}{A_2} = \left(\frac{2}{5}\right)^2 = \frac{4}{25}$$

$$\Rightarrow A_2 = 16 \times \frac{25}{4} = 100cm^2 \longleftarrow \text{area of } 2^{nd} \text{ polygon}$$

34. The sides of a polygon have length 6,10,12,18 and 30cm. the perimeter of similar polygon is 120cm.
Find a) the lengths of the sides of smaller polygon
b) the ratio of their areas of smaller polygon to larger.
Explanation: Here
Let $P_1 = 6 + 10 + 12 + 18 + 34 = 80cm$
Let $P_2 = S_1 + S_2 + S_3 + S_4 + S_5 = 120cm$, so that

a) We have $\frac{P_1}{P_2} = \frac{80}{120} = \frac{2}{3} \longleftarrow$ ratio (scale factor)

$$\Rightarrow \frac{6}{S_1} = \frac{10}{S_2} = \frac{12}{S_3} = \frac{18}{S_4} = \frac{34}{S_5} = \frac{2}{3} \longleftarrow \text{provided that}$$

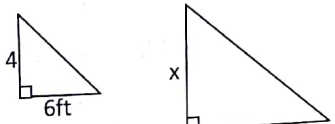
$$\Rightarrow \frac{6}{S_1} = \frac{2}{3} \text{ implies } S_1 = 6 \times \frac{3}{2} = 9cms \leftarrow \text{length of small polygon}$$

b) Area ratio = (side length ratio)$^2 = \left(\frac{2}{3}\right)^2 = \frac{4}{9}$

35. A telephone pole casts a 33 – foot shadow Near by a girl 4 feet tall casts a 6 – foot shadow **Find the height of the pole**
Explanation:

$$\frac{x}{33} = \frac{4}{6} = \frac{2}{3}$$

$$\Rightarrow x = \frac{2}{3}(33) = 22ft$$



Therefore, the pole is 22 feet tall.

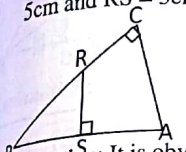
36. ΔABC and ΔDEF are similar and the area of $\Delta ABC = 12$ sq. unit. If DE is twice of AB then the area of $\Delta DEF =$
A. 6sq. unit B. 24 sq. unit C) 36 sq. unit D. 48 sq. unit
Explanation: we have $\overline{DE} = 2\overline{AB}$

$$= \frac{area\ of\ \Delta DEF}{\text{area of } \Delta ABC} = \left(\frac{DE}{AB}\right)^2 \text{ since, DE} = 2AB$$

$$\Rightarrow \frac{area\ of\ \Delta DEF}{12} = \left(\frac{2AB}{AB}\right)^2$$

$\Rightarrow$ area of $\Delta DEF = 12(2)^2 = 48$sq. unit, ∴**Answer: D**

37. In the figure below; $\overline{RS} \perp \overline{AB}$ and $\overline{BC} \perp \overline{AC}$ with AC = 15cm, BR = 5cm and RS = 3cm



Based on the information given above find the length of $\overline{AS}$
A. 16cm B. 24cm C. 21cm D. 25cm
Explanation: It is obvious $\Delta BSR \sim \Delta BCA$ by AA

$$\Rightarrow \frac{BR}{BA} = \frac{SR}{AC} \text{ implies } \frac{5cm}{BA} = \frac{3}{15} \Rightarrow BA = 25cm$$

To find BC: in ΔACB; $(BC)^2 + (AC)^2 = (AB)^2$
$\Rightarrow (BC)^2 + (15)^2 = (25)^2$ implies $(BC)^2 = (25)^2 - (15)^2$
Therefore, $BC = \sqrt{400} = 20cm$

To find $\overline{BS}$ $\frac{BS}{BC} = \frac{SR}{CA}$ implies $\frac{BS}{20} = \frac{1}{5}$

$$BS = 20\left(\frac{1}{5}\right) = 4cm$$

To find AS; $\overline{BS} + \overline{AS} = \overline{AB}$ implies $\overline{AS} + \overline{AB} - \overline{AS}$
$\Rightarrow AS = 25 - 4 = 21cm$, ∴Answer: C

Special segments theorem on Triangles

Triangles have four types of **special segment**
These are i) Median of a triangle ii) Altitude of a triangle
iii) Angle bisector of a triangle iv) side bisector of a triangle

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33. Two polygons are similar. A side of one is 2cm long. The corresponding side of the other is 5cm. the area of the first is $16cm^2$.

Find the area of the second polygon.

Explanation: we know that

Area ratio = (side length ratio)^2

$$\Rightarrow \frac{A_1}{A_2} = \left(\frac{S_1}{S_2}\right)^2 \text{ implies } \frac{16}{A_2} = \left(\frac{2}{5}\right)^2 = \frac{4}{25}$$

$$\Rightarrow A_2 = 16 \times \frac{25}{4} = 100cm^2 \longleftarrow \text{area of } 2^{nd} \text{ polygon}$$

34. The sides of a polygon have length 6,10,12,18 and 30cm. the perimeter of similar polygon is 120cm.

Find a) the lengths of the sides of smaller polygon

b) the ratio of their areas of smaller polygon to larger.

Explanation: Here

Let $P_1 = 6 + 10 + 12 + 18 + 34 = 80cm$

Let $P_2 = S_1 + S_2 + S_3 + S_4 + S_5 = 120cm$, so that

a) We have $\frac{P_1}{P_2} = \frac{80}{120} = \frac{2}{3} \longleftarrow$ ratio (scale factor)

$$\Rightarrow \frac{6}{S_1} = \frac{10}{S_2} = \frac{12}{S_3} = \frac{18}{S_4} = \frac{34}{S_5} = \frac{2}{3} \longleftarrow \text{provided that}$$

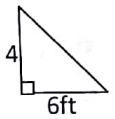
$$\Rightarrow \frac{6}{S_1} = \frac{2}{3} \text{ implies } S_1 = 6 \times \frac{3}{2} = 9cms \leftarrow \text{length of small polygon}$$

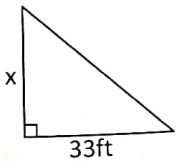
b) Area ratio = (side length ratio)$^2 = \left(\frac{2}{3}\right)^2 = \frac{4}{9}$

35. A telephone pole casts a 33 – foot shadow Near by a girl 4 feet tall casts a 6 – foot shadow **Find the height of the pole**

Explanation:

$$\frac{x}{33} = \frac{4}{6} = \frac{2}{3}$$





$$\Rightarrow x = \frac{2}{3}(33) = 22ft$$

Therefore, the pole is 22 feet tall.

36. ΔABC and ΔDEF are similar and the area of $\Delta ABC = 12$ sq. unit. If DE is twice of AB then the area of ΔDEF =

A. 6sq. unit B. 24 sq. unit C) 36 sq. unit D. 48 sq. unit

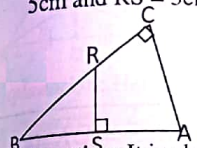
Explanation: we have $\overline{DE} = 2\overline{AB}$

$$= \frac{area\ of\ \Delta DEF}{\text{area of } \Delta ABC} = \left(\frac{DE}{AB}\right)^2 \text{ since, DE} = 2AB$$

$$\Rightarrow \frac{area\ of\ \Delta DEF}{12} = \left(\frac{2AB}{AB}\right)^2$$

$\Rightarrow$ area of $\Delta DEF = 12(2)^2 = 48$sq. unit, ∴**Answer: D**

37. In the figure below; $\overline{RS} \perp \overline{AB}$ and $\overline{BC} \perp \overline{AC}$ with AC = 15cm, BR = 5cm and RS = 3cm



Based on the information given above find the length of $\overline{AS}$

A. 16cm B. 24cm C. 21cm D. 25cm

Explanation: It is obvious $\Delta BSR \sim \Delta BCA$ by AA

$$\Rightarrow \frac{BR}{BA} = \frac{SR}{AC} \text{ implies } \frac{5cm}{BA} = \frac{3}{15} \Rightarrow BA = 25cm$$

To find BC: in ΔACB; $(BC)^2 + (AC)^2 = (AB)^2$

$\Rightarrow (BC)^2 + (15)^2 = (25)^2$ implies $(BC)^2 = (25)^2 - (15)^2$

Therefore, $BC = \sqrt{400} = 20cm$

To find $\overline{BS}$ $\frac{BS}{BC} = \frac{SR}{CA}$ implies $\frac{BS}{20} = \frac{1}{5}$

$$BS = 20\left(\frac{1}{5}\right) = 4cm$$

To find AS; $\overline{BS} + \overline{AS} = \overline{AB}$ implies $\overline{AS} + \overline{AB} - \overline{AS}$

$\Rightarrow$ AS = 25 – 4 = 21cm, ∴Answer: C

Special segments theorem on Triangles

Triangles have four types of **special segment**

These are i) Median of a triangle ii) Altitude of a triangle

iii) Angle bisector of a triangle iv) side bisector of a triangle

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Point of concurrency

- Three or more than three lines are said to be **concurrent** if they pass through the same point. This common point called *point of concurrency*

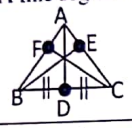
Special segment (line) of a triangle, which are *median, altitude, angle bisector and side bisector of a triangles are* **concurrent** at a point.

- The intersection point of:

1. Median is called **centraoid**
2. Altitude is called **orthocenter.**
3. Perpendicular side bisector is called **cercum-center**
4. Angle bisector is called **in-center**

Median of a triangle

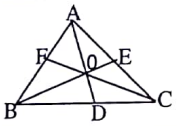
- A line segment joining a vertex of a triangle to the mid – point of the opposite side is called **a median of triangle.**



- Every triangle will have three medians
- The three medians of a triangle pass through the same point i.e. they are **concurrent**
- The point of intersection of the medians of a triangle is called **the centriod** of a triangle.

Length of Median of a triangle

The median of a triangle are meet (concurrent) at a point $\frac{2}{3}$ of the distance from any vertex to the mid – point of the opposite side



- The median $\overline{AD}, \overline{BE}$ *and* $\overline{CF}$ pass through 0.

$\therefore$ 0 is the centroid divides each median, $\overline{AD}, \overline{BE}$ *and* $\overline{CF}$ in the ratio 2:1

Provided that $\overline{AO} = 2\overline{DO}$ implies $\overline{AO} : \overline{DO} = 2:1$

$\overline{BO} = 2\overline{EO}$ implies $\overline{BO} : \overline{OE} = 2:1$

$\overline{CO} = 2\overline{FO}$ implies $\overline{CO} : \overline{OF} = 2:1$

Similarly, $\overline{AO} = \frac{2}{3}\overline{AD}$ and $\overline{DO} = \frac{1}{3}\overline{AD}$

Explanation

$\overline{AD} = \overline{AO} + \overline{OD} = 2\overline{DO} + \overline{OD}$

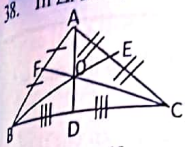
$\overline{AD} = 3\overline{DO}$ $\quad \overline{DO} = \frac{1}{3}\overline{AD}$

$\overline{BO} = \frac{2}{3}\overline{BE}$ and $\overline{OE} = \frac{1}{3}\overline{BE}$

$\overline{CO} = \frac{2}{3}\overline{CF}$ and $\overline{OF} = \frac{1}{3}\overline{CF}$

Illustrative Example

38. In ΔABC, the medians $\overline{AD}$, $\overline{BE}$ and $\overline{CF}$ pass through '0' if $\overline{BO} = 6, \overline{AD} = 7.5$ and $\overline{FO} = 4cm$



Find a) $\overline{AO}$ b) $\overline{BE}$ c) $\overline{OD}$ d) $\overline{OC}$ e) $\overline{CF}$ f) $\overline{OE}$

Explanation: Here

a) $\overline{AO} = \frac{2}{3}\overline{AD}$ implies $\overline{AO} = \frac{2}{3}(7.5) = \frac{15}{3} = 5cm$

b) $\overline{BO} = \frac{2}{3}\overline{BE}$ implies $\overline{BE} = \frac{3}{2}\overline{BO} = \frac{3}{2}(6) = 9cm$

c) $\overline{OD} = \frac{1}{3}\overline{AD}$ implies $\overline{OD} = \frac{1}{3}(7.5) = 2.5cm$

d) We know that $\overline{OC} : \overline{FO} = 2:1$

$\Rightarrow \frac{CO}{OF} = \frac{OC}{FO} = \frac{2}{1}$ implies $\frac{\text{OC}}{4\text{cm}} = \frac{2}{1} \Rightarrow \overline{OC} = 8cm$

e) $\overline{CF} = \overline{CO} + \overline{OF} = 8cm + 4cm = 12cm$

f) We know that $\overline{BO} : \overline{OE} = 2:1$

$\Rightarrow \frac{BO}{OE} = \frac{2}{1}$ implies $\frac{6}{\text{OE}} = 2 \Rightarrow \overline{OE} = 3cm$

39. In ΔABC, the medians $\overline{AD}$ and $\overline{BE}$ pass through '0' if $\overline{AO} = 16$ and $\overline{OE} = 7$

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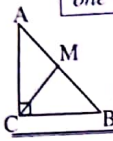
Find : $\overline{AD}, \overline{OD}, \overline{BO}$ and BE

Explanation: Exercise left for you **(Tip!)**

Note: The medians of equilateral triangle are equal to each other.

The median of right triangle

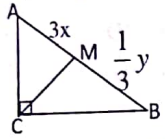
The length of the median to the hypotenuse of a right triangle equals one – half the length of the hypotenuse.



If $\overline{CM}$ is the median of right triangle ABC to hypotenuse $\overline{AB}$, then $\overline{CM} = \frac{1}{2}\overline{AB}$ that is $\overline{CM} \equiv \overline{AM} \equiv \overline{BM} = \frac{1}{2}\overline{AB}$

Illustrative Example

40. Find x and y in each part:

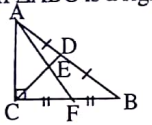


If ΔABC is a right triangle M is the mid – point of AB, $\overline{AM} = 3x$, $\overline{MB} = \frac{1}{3}y$ and $\overline{CM} = 20$

Explanation: $\overline{AM} = \overline{CM}$ implies $3x = 20$

$\Rightarrow x = \frac{20}{3}$ and $\frac{1}{3}y = 20 \Rightarrow y = 60$

41. If ΔABC is a right triangle, D is the mid – point of $\overline{AB}$, F is the mid – point of $\overline{CB}, \overline{AB} = 30\text{cm}$



Find $\overline{CE}$ *and* $\overline{ED}$

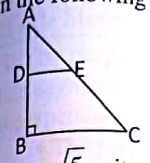
Explanation: $\overline{CD}$ is the median to hypotenuse $\overline{AB}$ *and* $\overline{AD} = CD = DB = \frac{1}{2}AB$

$\Rightarrow \overline{CD} = \frac{1}{2}AB = \frac{1}{2}(30) = 15\text{cm}$ $\overline{CE} = \frac{2}{3}(\overline{CD}) = \frac{2}{3}(15) = 10cm$

$\overline{ED} = \frac{1}{3}\overline{CD} = \frac{1}{3}(15) = 5cm$

- Note that $\overline{CE} = \frac{2}{3}CD = \frac{2}{3}AD = \frac{2}{3}BD$, since $\overline{CE} = AD = DB$

42. In the following figure ΔABC is a right triangle D is the mid – point of $\overline{AB}$ and E is the mid – point of $\overline{AC}$. If $\overline{BD} = 5\text{unit}$, and $\overline{BC} = 20\text{unit}$ then $\overline{BE}$ is equal.



A. $\sqrt{5}$ unit B. $5\sqrt{3}$ unit C. $5\sqrt{10}$ unit D. $10\sqrt{5}$ unit

Explanation: $\overline{AE} \equiv \overline{BE} \equiv \overline{EC} = \frac{1}{2}\overline{AC}$

$AD = 5\text{unit} \Rightarrow AB = 10$ unit,

We know that: $AB^2 + BC^2 = AC^2$

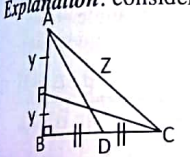
$\Rightarrow 10^2 + 20^2 + AC^2 \Rightarrow \overline{AC} = \sqrt{500} = 10\sqrt{5}$ unit

$\therefore \overline{AE} = \overline{EC} = \overline{BE} = \frac{1}{2}\overline{AC} = \frac{1}{2}\left(10\sqrt{5}\right) = 5\sqrt{5}$ unit.

40 b) If the medians of a right – angled triangle that are drawn from the vertices of the acute angles are $3\sqrt{3}$ cm and $4\sqrt{3}$ cm. then the length of the hypotenuse is

A. 75cm B. 15cm C. $2\sqrt{15}$ cm D. $\sqrt{15}$ cm

Explanation: consider the right triangle let the median from vertex:



i) A to $\overline{BC}$ *is* $\overline{AD} = 3\sqrt{3}$ cm

ii) C to $\overline{AB}$ *is* $\overline{CF} = 4\sqrt{3}$ cm

Let $\overline{BD} = \overline{DC} = x$ and $BC = 2x$

Let $\overline{AF} = \overline{FB} = y$ and $AB =$

In right triangle, ΔCBF; $(BC)^2 + (BF)^2 = (FC)^2$

$\Rightarrow (2x)^2 + y^2 = (4\sqrt{3})^2$ implies $4x^2 + y^2 = 48$ . . . .(*)

In right triangle, ΔABD; $(AB)^2 + (BD)^2 = (AD)^2$

$\Rightarrow (2x)^2 + x^2 = (3\sqrt{3})^2$ implies $4y^2 + x^2 = 27$ . . . .(**)

From (*) and (**), we have

$$4\begin{cases}4x^2 + y^2 = 48 \\ x^2 + 4y^2 = 27\end{cases} \Rightarrow -\begin{cases}4x^2 + y^2 = 48 \\ \underline{4x^2 + 16y^2 = 108}\end{cases}$$

$$0 - 15y^2 = -60 \Rightarrow y^2 = \frac{60}{15} = 4$$

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$\therefore y = 2$

To find x; put $y = 2$, in $4x^2 + y^2 = 48$

$\Rightarrow 4x^2 + (2)^2 = 48$, implies $4x^2 = 44$

To find z; in right triangle ΔABC,

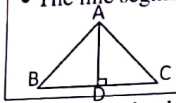
$AB^2 + BC^2 = AC^2 =$ implies $(2y)^2 + (2x)^2 = z^2$

$\Rightarrow 4y^2 + 4x^2 = z^2$ implies $4(2)^2 + 44 = z^2$

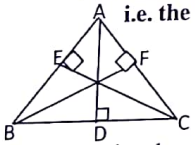
$\Rightarrow z^2 = 60$ therefore $z = 2\sqrt{15}$, $\therefore$ Answer: C

Altitude of a triangle

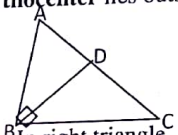
- The line segment from a vertex of a triangle perpendicular to the opposite side is called **an altitude of the triangle.**
- Every triangle will have three altitudes.



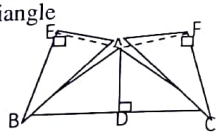
- In triangle, the three altitude pass through the same point, this common point is called **ortho-center** of triangle
- The three altitude of a triangle are concurrent, i.e. (intersect at a point)
- The three altitude of:
  - i) **an acute – angled triangle lies inside**
    (i.e. orthocenter lies inside the triangle)
  - ii) a right – angled triangle lies **the vertex** of right angled. **It is the same meaning as.**
    the orthocenter of right triangle is the vertex of right angle
  - iii) an obtuse – triangle **lies outside** the triangle.
    i.e. the orthocenter lies outside triangle



In acute triangle ΔABC, the altitude $\overline{AC}, \overline{CE}$ and $\overline{BF}$ are intersect inside ΔABC



In right triangle ΔABC, the altitude, $\overline{AB}, \overline{BC}$ and $\overline{BD}$ are are intersect



In obtuse triangle ΔABC, the altitude $\overline{AB}$ and CF are outside ΔABC

Bisector of triangle

There are two bisector of a triangle these are

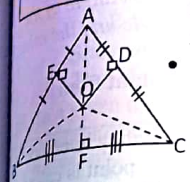
i) side bisector of a triangle

ii) Angle bisector of a triangle

Note, A segment that divides another segment into two equal segments is called **side bisector.**

i) The perpendicular bisector of a side of triangle

- A line segment that passes through the mid – point of a side of a triangle and perpendicular to that side.
- The perpendicular bisector of a side of triangle is **equidistant** from the three vertices of a triangle



- Every triangle has three perpendicular bisector and they intersect each other at common point
- The perpendicular bisectors of the sides $\overline{AB}, \overline{AC}$ and $\overline{BC}$ of a triangle ΔABC intersect at **O**, this common point is called **cercum center** of the triangle
- The point **O** is equidistant from the three vertices A,B, C of ΔABC because **O** lies on the perpendicular bisector of $\overline{AB}, \overline{BC}$ and $\overline{AC}$

Therefore, $\overline{OA} = \overline{OC} = \overline{OB} \longleftarrow$ radius

- If we draw circle with center 0 and radius OA, it will pass through B and C

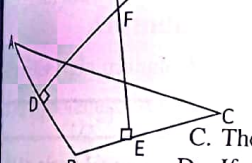
The three perpendicular side bisector of triangle for:

i) an – acute triangle lies *inside the triangle*

ii) an – obtuse triangle lies *outside the triangle.*

iii) right – triangle lies on **hypotenuse** of triangle

13. If in ΔABC as shown in the figure $\overline{FD}$ and $\overline{FE}$ are perpendicular bisector of $\overline{AB}$ and $\overline{BC}$ respectively then which of the following is **not true**



A. $\overline{AF} \equiv \overline{CF}$

B. The circle with center F and radius $\overline{BF}$ passes through all the vertices of ΔABC

C. The perpendicular bisector of $\overline{AC}$ passes through F

D. If <ABC is right angle, then 'F' lies on AC

Explanation: perpendicular bisector of side of a triangle are equidistant from 3 – vertices ABC. i.e. $\overline{AF} \equiv \overline{FC} \equiv BF$

- **F is called the circum center of triangle ABC**
- If we draw circle F is center with radius $\overline{BF}$, passes through A,B and C

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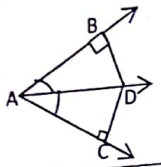


- If B = 90^0, then F files on $\overline{AC}$ ∴ All are true

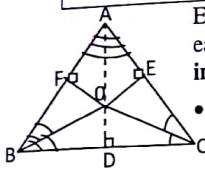
Angle bisector *divides an angle into two equal angle. Any point on the bisector of an angle is equidistant from the sides of the angle,*

i.e $\overline{BD} \equiv \overline{DC}$

ii) Angle bisector of a triangle

An angle bisector of a triangle is a segment that bisects an angle of the triangle and has one end – point at a vertex of the triangle and the other end – point at another point on the triangle.

- The angle bisector of a triangle is **equidistant** from the side of a triangle.



Every triangle has three angle bisector and they intersect each other at common point, this common point is called **in center.**

- In triangle ABC, $\overline{AD}, \overline{BE}$ and $\overline{CF}$ are angle bisector, they intersect at single point '**0**' such that $\overline{FO} = \overline{OE}$ and $\overline{OD}$

To prove: $\overline{OF} = \overline{OE} = \overline{OD}$

1) m(<BOD) ≡ (<BOF) ⟵ Definition
2) $\overline{BO} \equiv \overline{OB}$ ⟵ common side

∴ ΔBOD ≡ ΔBOF

∴ OD = OF

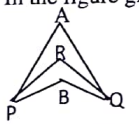
Similarly ΔCOD ≡ ΔCOE

∴ $\overline{OD} = OE$

Therefore, $\overline{OF} \equiv \overline{OE} \equiv \overline{OD}$

Illustrative Example

44. In the figure given below, $\overline{RP}, \overline{RQ}$ are the bisectors of equal angle <APB and <AQB respectively. If = $\overline{RP}$ = RQ, prove that A,R,B, lies on a straight line.



Explanation: Two prove: A,R,B lies on a straight

1. <APB ≡ <AQB ⟵ Given

2. $\overline{AP} \equiv \overline{AQ}$ ⟵ Definition of isosceles Δ.

∴ A is equidistant from two point P and Q

3. $\overline{RP} \equiv \overline{RQ}$ ⟵ Given

∴ R is equidistant from two point P and Q

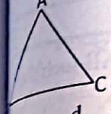
4. <BPQ ≡ <BQP ⟵ from 1,2,3
5. $\overline{PB} \equiv$ BQ ⟵ from 4

∴ B is equidistant from P and Q

Since, A,R, and B are equidistant from the point P and Q.

Therefore, A,R and B lies on straight line.

45. In a given triangle ABC, describe, the point **P,Q,R and F**, in terms of bisector of triangle, in the following statement



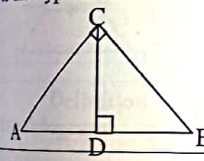
a. a point P on $\overline{BC}$ such that equidistant from $\overline{AB}$ and $\overline{AC}$
b. a point P on $\overline{AB}$ such that equidistant from B and C
c. a point Q on $\overline{AB}$ such that equidistant from A and C
d. a circle passes through the three vertices of triangle A,B and C then the point **F** the center of the circle lies at the point where
e. a point 'R', the center of circle inscribed in ΔABC

Explanation:

a) **a point P lies on the bisector of an angle** of ΔABC
b) A point P lies on the perpendicular side bisector of ΔABC
c) **Exercise left for you (Tip !)**
d) the perpendicular side bisector of ΔABC meet
e) the point R lies on the bisectors of the angle of ΔABC

Altitude theorem

Consider right triangle ΔABC with altitude $\overline{CD}$ drawn from the right angle C to the hypotenuse $\overline{AB}$.



- **Altitude theorem** describes a special relationship between
  i) three right triangle ΔABC, ΔACD and ΔACD.
  ii) the length of segment.

In right triangle ABC the altitude $\overline{CD}$ is drawn to the hypotenuse $\overline{AB}$; so that

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i) $\frac{AD}{DC} = \frac{CD}{DB}$ *the same as* $(CD)^2 = AD.DB$

ii) $\frac{AD}{AC} = \frac{CA}{AB}$ *the same as* $(AC)^2 = AD.AB$

iii) $\frac{DA}{BC} = \frac{CB}{AB}$ *the same as* $(BC)^2 = AB.DB$

To prove: From the above figure, right triangle ΔABC, ΔACD, and ΔCBD

1) $m(<BAC) \equiv m(<CAD) \equiv m(<BCD)$
2) $m(<ABC) \equiv m(<ACD) \equiv m(<CBD)$
3) $m(<ACB) \equiv m(<ADC) \equiv m(<CDB)$

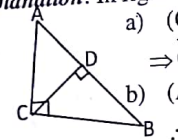
$\therefore \Delta ABC \sim \Delta CBD$ by AA similarity

Illustrative Example

46. In right triangle ABC with, $\overline{CD}$ is the altitude to hypotenuse $\overline{AB}$ such that AD = 16cm and BD = 9cm

Find a) $\overline{CD}$ b) $\overline{AC}$ c) BC

Explanation: In right triangle ABC with altitude CD to hypotenuse AB



a) $(CD)^2 = (AD)(DB)$ implies $(CD)^2 = (16)(9)$

$\Rightarrow \overline{CD} = \sqrt{16.9} = (4)(3) = 12cm$

b) $(AC)^2 = (AD)(AB)$ implies $(AC)^2 = (16)(25)$

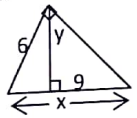
$\therefore \overline{AC} = \sqrt{(16)(25)} = (4)(5) = 20cm$

c) $(BC)^2 = (BD)(BA)$ implies $(BC)^2 = (9)(25)$

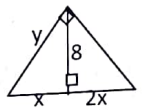
$\therefore \overline{BC} = \sqrt{(9)(25)} = (3)(5) = 15cm$

47. **Find the value of x and y**

a)



b)



Solution: a) $6^2 = (x - 9)x$ implies $x^2 - 9x = 36$

$\Rightarrow x^2 = 9x - 36 = 0$ implies $(x + 9)(x - 12) = 0$

Therefore, x = 12

To find y: $y^2 = (9)(12 - 9) = 27$

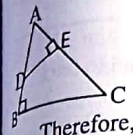
Therefore, $y = \sqrt{27} = 3\sqrt{3}$ cm

b) $8^2 = (x)(2x)$ implies $2x^2 = 64$

$\therefore x^2 = 32$, therefore, $x = \sqrt{32} = 4\sqrt{2}$

To find y ⟵ **exercise left for you**

48. In the figure given below, if triangle ABC is right angle at B and $\overline{DE} \perp AC$, then prove that $\frac{AD}{AC} = \frac{DE}{BC}$



Explanation: To proof

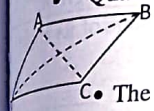
i. $<A \equiv <A$ ⟵ common vertex of ΔDAE and ΔCAB

ii. $<B \equiv <E$ ⟵ Definition of right angle

Therefore, ΔADE ~ ΔACB by A.A similarity

Provided that $\frac{AD}{AC} = \frac{DE}{CB}$ ⟵ by CPST

Quadrilaterals



- **Quadrilaterals** is any four – sided closed figure.
- A figure represents a quadrilateral ABCD
- The point A,B,C and D are called **vertices**.
- The two line segment $\overline{AC}$ and $\overline{BD}$ are called **diagonals**

• **Adjacent side** are two side of quadrilateral that have common end – point

Example, $\overline{AD}$ and $\overline{AB}$ are a pair of adjacent side with common end–point A.

• The line segment $\overline{AB}$ and $\overline{CD}$ are called **a pair of opposite side**

Special quadrilaterals

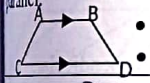
There are five these are:

Trapezium, parallelogram, rhombus, rectangle and square

• Trapezium is special quadrilaterals special – in what?

Trapezium

A quadrilateral is **trapezium** if it has exactly one pair of opposite sides parallel.



- The parallel sides AB and CD are called **bases**.
- The non – parallel sides AC and BD are called **the legs**.

Definition

Parallelogram is special quadrilaterals with both pairs of opposite sides parallel.

• Properties of a parallelogram and tests for a quadrilaterals to be a parallelogram are stated in the following theorem.

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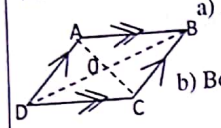
A quadrilateral is parallelogram if any of the following properties hold true.



a) Both pair of opposite sides are parallel (i.e. $\overline{AB} \parallel \overline{CD}$ and $\overline{AD} \parallel \overline{BC}$)

b) Both pairs of opposite sides are congruent. That is

i) $\overline{AB} \equiv \overline{CD}$ ii) $\overline{AD} \equiv \overline{BC}$

c) Both pairs of opposite angle are equal. That is

(i) $<A = <C$ ii) $<B = <D$

d) The diagonal bisect each other.

That is (i) $\overline{AO} \equiv \overline{OC}$ ii) $\overline{BO} \equiv \overline{OD}$.

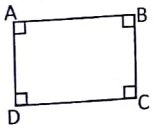
Special parallelograms

- *Special parallelograms are rectangle, Rhombus and square*
- *Rectangle, square and rhombus have all the properties of parallelogram.*

Definition: A – **rectangle A rectangle** is a parallelogram with one of its angle is right angle.

- **Rectangle is** equiangular parallelogram

Properties of a rectangle



i) A rectangle has all properties of a parallelogram

ii)Diagonal of rectangles have equal length and bisect each other.

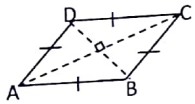
iii) All the four angles are right angle.

Rhombus

Definition: A – rhombus is a parallelogram with four congruent side.

- A rhombus is an equilateral parallelogram.

Properties of a rectangle



i) A rhombus has all the properties of parallelogram

ii) All the sides are congruent that is

$\overline{AD} \equiv \overline{AB} \equiv \overline{DC} \equiv \overline{CB}$

ii) The diagonals of a rhombus are perpendicular.

iv) The diagonals of a rhombus bisect its angle.

Note: The diagonals of a rhombus from **four congruent triangle**

Square

Definition: A square is a rectangle with congruent adjacent sides.

- A square is both a rectangle and a rhombus

Properties of square

i) A square has all the properties of the rhombus and the rectangle

ii) The diagonals are equal and perpendicular

Mid – point of Quadrilateral, parallelogram Rhombus, Rectangles, square

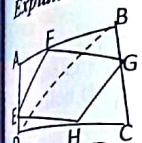
Note: *Line joining the mid – points of any two sides of a triangle is parallel to the third side.*

Illustrative Example

49. The mid – point of the sides of a quadrilateral are also quadrilateral. What kind of a quadrilateral is it

A. square B. rhombus C. rectangle D. parallelogram

Explanation: It is parallelogram



To Proof: EFGH is parallelogram

Let ABCD be quadrilateral

Let E,F,G and H are mid – point of side $\overline{AD}, \overline{AB}, \overline{BC}$ and CD

1. Draw $\overline{BD}$ ⟵ a segment can be drawn from any two point.
2. E, F, G, H are mid – point ⟵ Given
3. EF // BD and HG// BD ⟵ Mid – point definition of triangle EF $= \frac{1}{2}$ BD and GH $= \frac{1}{2}$ BD

Therefore, EF //GH and $\overline{EF} \equiv \overline{GH}$

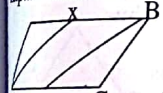
Therefore, EFGH is parallelogram, ∴ Answer: D

50. ABCD is a parallelogram, x is mid – points of $\overline{AB}$, and y is the mid-points of $\overline{DC}$.

What kind of quadrilateral AxCy?

A. rectangle B. rhombus C) parallelogram D. square

Explanation:



1) ABCD is parallelogram ⟵ given
2) AB = DC ⟵ opposite side of parallelogram
3) $Ax = \frac{1}{2}AB$, $yc = \frac{1}{2}DC$ ⟵ x and y are mid – point of AB and DC
4) AX = YC ⟵ From 1 and 2

∴ AX//YC ⟵ from 4

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∴ AXCY is parallelogram, because quadrilateral is parallelogram if its one pair of opposite sides are **parallelogram** and equal.

Therefore, Answer C

51. The mid – point of the sides of a parallelogram are vertices of a quadrilateral. What kind of a quadrilateral.

Explanation:

- D is equidistant from G and E
- F is equidistant from G and E

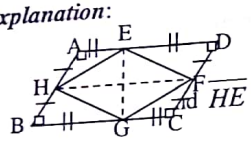
$\Rightarrow \overline{DF} \perp$ GE, since, if a point is equidistant from the end of a line segment, then the point is on perpendicular side bisector.

∴ DEFG is rhombus

52. The mid – points of the sides of a rectangle are the vertices of a quadrilateral. **What kind of a quadrilateral?**

A. Square B. rectangle C. rhombus D. parallelogram

Explanation:



- E is equidistant from H and F
- G is equidistant from H and F

$\overline{HE} \equiv \overline{EF} \equiv \overline{EF} \equiv$ EH

∴ EFGH is rhombus, **Answer: C**

∴ Quadrilateral formed is rhombus

53. The mid – point of the sides of a rhombus are the vertices of a quadrilateral **What kind of a quadrilateral?**

Explanation: Quadrilateral formed by joining the mid – points of side of a rhombus is **rectangle.**

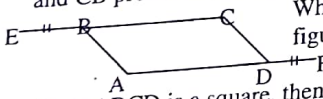
54. What kind of quadrilateral formed by joining the mid – points of side of

A) Quadrilateral B) Parallelogram C) rhombus D) rectangle

Explanation: *quadrilateral formed by joining:*

A) *The mid-point of side of quadrilateral is* **parallelogram**
B) *The mid – point of side of parallelogram is* **rhombus**.
C) *The mid – point of side of rhombus is* **rectangle**
D) *The mid – point of side of rectangle is* **rhombus**

55. In the figure, below ABCD is a parallelogram with $\overline{AD}$ produced to **F** and CB produced to **E** such that $\overline{DF} \equiv \overline{BE}$



Which one of the following is true about this figure . . . **EGSECE 2005**

A) If ABCD is a square, then AECF is also square.

B) $\Delta ABE \equiv \Delta DEF$ C) AECF is a parallelogram D) $\overline{DE} \neq \overline{BF}$

Explanation:

$\overline{BC} \equiv \overline{AD}$ implies $\overline{EC} \equiv \overline{AF}$ and $\overline{EC} // \overline{AF}$

∴ AECF is parallelogram

∴ **Answer C**

If in a quadrilateral the two diagonal intersect at right angle, then the quadrilateral is necessarily

a) a square c) a rhombus
b) a rectangle d) parallelogram with adjacent side congruent

Explanation: parallelogram with adjacent side congruent are rhombus and square both are intersect at right angle, **∴ Answer D**

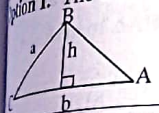
Area of Triangle and parallelogram

The amount of the surface enclosed by the boundary lines of the figure is called **the area of the figure.**

Area of Triangle

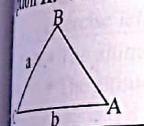
We have four option:

Option I. The area of a triangle equals **one – half** the product of the length of a side and the length of the altitude to that side. In short area, is given by $A = \frac{1}{2}bh$



Option II. The area of ant triangle with two adjacent sides **a** and **b**, and angle (c) included between these sides is given by

$A = \frac{1}{2}$ ab sin< c

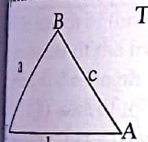


Option III. Heron's formula

The area of ant triangle with side length a, b and c, units long is given by

$A = \sqrt{S(S-a)(S-b)(S-C)}$ where

$S = \frac{a+b+c}{2}$ ⟵ Semi- perimeter



Illustrative Example

In each of the following problems, find the area of ΔABC with sides a,b,c opposite the vertices A,B,C respectively, if

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a) a = 8 unit, b = 7 unit, m(<C) = 30^0
b) a = 6 unit, b = 4 units, m(<C) = 60^0
c) a = 5 unit, b = 4 units, sin(C) = $\frac{3}{5}$

Solution: Area is given by. $A = \frac{1}{2}ab \cdot \sin < c$

a) a = 8, b=7 <C = 30^0, Area, $A = \frac{1}{2}(8)(7)\sin 30^0 = 14$ sq. unit

b) Exercise left for you (Tip)

c) A = 5 unit, b = 4 unit, sin (c) = $\frac{2}{5}$

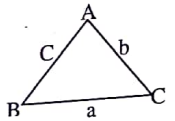
$\Rightarrow A = \frac{1}{2}ab \sin(c) = \frac{1}{2}(5)(4)\left(\frac{3}{5}\right) = 6$ sq. unit.

58. The three sides of a triangle, ΔABC are $\overline{AB} = 6$cm $\overline{BC} = 5$cm and $\overline{AC} = 7$cm

Find a) its area b) the measure of A,B,C
c) the length of the altitude from each vertex.

Explanation: use the Heron's formula to determine the area.

a = 5cm, b = 7cm, c = 6cm



Semi – perimeter $S = \frac{a+b+c}{2}$

$\Rightarrow S = \frac{5+7+6}{2} = \frac{18}{2} = 9$cm

a) Area, $A = \sqrt{S(S-a)(S-b)(S-C)} = \sqrt{9(9-5)(9-7)(9-6)} = 6\sqrt{6}$ sq. unit

b) **To find, Sin (A)**: area of Δ ABC, $A = \frac{1}{2}bc\sin(A)$

Since angle A is included between side b and c

$\Rightarrow 6\sqrt{6} = \frac{1}{2}(7)(6)\sin(A)$ implies $\sin(<A) = \frac{2\sqrt{6}}{7}$

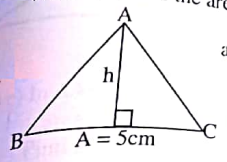
To find, sin (B), Use, area = $\frac{1}{2}$(a)(c) sin (B) , since angle B is included betwe side a and c

$\Rightarrow 6\sqrt{6} = \frac{1}{2}(5)(6)\sin(< B) \Rightarrow \sin(<B) = \frac{2\sqrt{6}}{5}$

To find, sin (<C) ⟵ Exercise left for you

c) The altitude drawn from vertex A to side BC is let h.

We know that the area is still the same $6\sqrt{6}$ sq. unit, so that area of $\Delta ABC = \frac{1}{2}BC(h)$



$\Rightarrow 6\sqrt{6} = \frac{1}{2}(5)h$ implies $h = \frac{12\sqrt{6}}{5}$ cm

Therefore, the altitude drawn from vertex A is h = $\frac{12\sqrt{6}}{5}$ cm

- **To find the altitude drawn from vertex B.**
  Let H be the altitude from vertex B to $\overline{AC}$
- Area of $\Delta ABC = \frac{1}{2}(AC)(H)$

$\Rightarrow 6\sqrt{6} = \frac{1}{2}(7)H$ implies $H = \frac{12\sqrt{6}}{7}$ cm

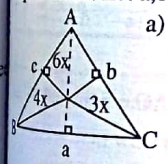
- **The altitude drawn from vertex C**
  Exercise left for you

Note • The altitude drawn to the longest side is **shortest altitude.**
• The altitude drawn to the smallest side is **longest altitude.**
Example: $\frac{12\sqrt{6}}{7}$ cm is shortest altitude. $\frac{12\sqrt{6}}{5}$ cm is longest altitude.

59. The length of three altitude of a triangle are 6x, 4x and 3x unit and the perimeter of the triangle is 18 unit.

Find a) the length of the sides of triangle
b) area of the triangle

Explanation: Let a,b and c are the length of the three sides of triangle



a) Perimeter ΔABC is P: a + b + c = 18

- Area of ΔABC is given by any one is holds true.

Area = $\frac{1}{2}(BC)(6x) = \frac{1}{2}(AC)(4x) = \frac{1}{2}(AB)(3x)$

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$\Rightarrow \frac{1}{2}(a)(6x) = \frac{1}{2}(b)(4x) = \frac{1}{2}(C)(3x)$

$\Rightarrow a(6x) = (b)(4x)$ and $(b)(4x) = (c)(3x)$

$\Rightarrow 6a = 4b$ and $4b = 3c$

$\Rightarrow a = \frac{4}{6}b = \frac{2}{3}b$ and $c = \frac{4}{3}b$

Since the perimeter P: $a + b + c = 18$

$\Rightarrow \frac{2}{3}b + b + \frac{4}{3}b = 18$ implies $3b = 18 \Rightarrow b = 6$ unit

and $a = \frac{2}{3}b = \frac{2}{3}(6) = 4$ unit

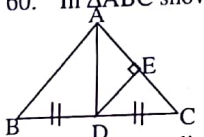
To find C: $C = \frac{4}{3}b = \frac{4}{3}(6) = 8$ unit

Therefore, the length of ΔABC are 4, 6 and 8 unit

b) To find area of ΔABC, use $A = \sqrt{S(S-a)(S-b)(S-C)}$

$\Rightarrow A = \sqrt{9(9-4)(9-6)(9-8)} = 3\sqrt{15}$ where $S = \frac{4+6+8}{2} = 9$

60. In ΔABC show below, $\overline{AD}$ be median and $\overline{DE} \perp \overline{AC}$. If the area of ΔABC is 54 sq. unit and $\overline{AC} = 12$ unit



What is the length of $\overline{DE}$ in (unit)

A. 4.5 B. 5.4 C. 6 D. 9

Solution: the median $\overline{AD}$ of a triangle divides it into two triangles of equal area.

$\Rightarrow$ area of $\Delta ABC = \frac{1}{2}$(area of ΔABC) $= \frac{1}{2}(54) = 27$sq. unit

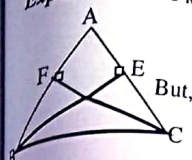
However, area of $\Delta ABC = \frac{1}{2}(AC)(DE)$

$\Rightarrow 27 = \frac{1}{2}(12)(DE)$ implies $\overline{DE}\ \frac{27}{6} = 4.5$ unit

Therefore, Answer: A

60 b) BE and CF are two altitude of ΔABC If AB = 6cm, AC = 5cm and CF = 4cm **Find the length of BE**

Explanation: we know that area of ΔABC is $A = \frac{1}{2}(AB)(CF) = \frac{1}{2}(6)(4) = 12\text{cm}^2$



But, $A = \frac{1}{2}(AC)(BE)$ implies $12 = \frac{1}{2}(5)\,BE$

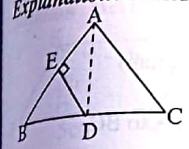
Therefore, BE $= \frac{24}{5}$ cm

60 c) In ΔABC, AB = 8cm, AC = 9cm and D is the mid – point of BC. If the area of ΔABC is 36cm^2,

Find a) the altitude from D to AB.

b) the altitude from D to AC.

Explanation: consider the triangle ABC we know that, area of triangle



ΔADB $= \frac{1}{2}$(ΔABC), since D is mid – point

$\Rightarrow a(\Delta ABC) = \frac{1}{2}(36) = 18\text{cm}^2$

We know that area of $\Delta ADB = \frac{1}{2}(AB)(DE)$

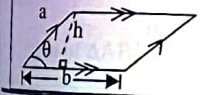
$\Rightarrow 18\text{cm}^2 = \frac{1}{2}(8)(DE)$ implies S, $DE = \frac{9}{2}$ cm

$\therefore DE = \frac{9}{2}$ cm ⟵ altitude from D to AB.

Area of parallelogram

- Any side of a parallelogram can be called **base (b)**. for each base there is a corresponding **altitude (h)**.
- The length of an altitude is called **the height** of the parallelogram.

• The area A of a parallelogram with base **'b'** and height **'h'** is given y



i) $A = bh \longrightarrow h = a\sin\theta$

ii) $A = ab\sin\theta$

Illustrative Example

1. ABCD is a parallelogram with two adjacent side length of $\overline{AB} = 4$cm, $\overline{AD} = 5$cm with $m(<B) = 60^0$.

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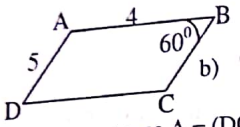
Find a) the area of parallelogram

b) the length of altitude from A to $\overline{BC}$

c) the length of altitude from A to DC

Explanation: (a) A = ab sinθ

$\Rightarrow$ A = (AB) (AD) sinθ, since $\overline{AD}$ = BC



$\Rightarrow$ A = (4) (5) sin60^0, $10\sqrt{3}$ sq. unit.

b) The altitude from A to DC is h

$\Rightarrow$ area A = (DC) (h) $\Rightarrow 10\sqrt{3}$ = (4) (h) implies h = $\frac{10\sqrt{3}}{4}cm$

Therefore, altitude from A to $\overline{DC}$ is $\frac{5\sqrt{3}}{2}cm$

c) The altitude from A to BC is H

$\Rightarrow$ Area: A = (H) (BC) implies $10\sqrt{3}$ = (H) (5cm)

$\Rightarrow$ H = $\frac{10\sqrt{3}}{5} = 2\sqrt{3}cm \longleftarrow$ altitude from A to BC

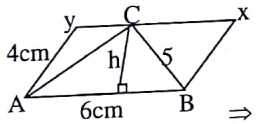
62. ABXY is a parallelogram of area 18cm^2, AB = 6cm, AY = 4cm and C is a point on YX extended such that BC = 5cm

Find a) area of ΔABC

b) The **altitude** from B to $\overline{AY}$

c) The **altitude** from A to $\overline{BC}$

Explanation: consider the parallelogram



Area of parallelogram

$\Rightarrow$ A = h(AB) = 18cm^2

$\Rightarrow$ h(AB) = 18cm^2

$\Rightarrow$ h(6) = 18cm^2 implies h = $\frac{18}{6}$ = 3cm

a) Area of ΔABC = $\frac{1}{2}$ (AB) (h) = $\frac{1}{2}$ (6) (3) = 9cm^2

b) **To find** distance from B to AY is H

Area of parallelogram A = (AY) let H

$\Rightarrow$ 18cm^2 = (4)H implies H = $\frac{9}{2}$ cm

$\therefore$ H = $\frac{9}{2}$ cm $\longleftarrow$ Distance from B to AY

c) **To find distance from A to BC** is let K

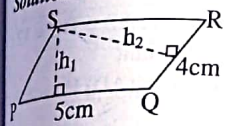
We know that area of ΔABC = $\frac{1}{2}$ (K) $\overline{BC}$

$\Rightarrow$ 9cm^2 = $\frac{1}{2}$ (K)(5) implies k = $\frac{18}{5}$ cm

$\therefore$ K = $\frac{18}{5}$ cm $\longleftarrow$ altitude from A to $\overline{BC}$

63. PQRS is a parallelogram of area 18sq. cm PQ = 5cm, QR = 4cm **Find** the length of corresponding altitude

Solution: **Here**



Area of PQRS: $h_1(PQ) = h_2(QR) = 18$

$\Rightarrow h_1$ (5)= 18cm^2 implies $h_1 = \frac{18}{5}$ cm

Similarly, $h_2(QR) = 18cm^2$

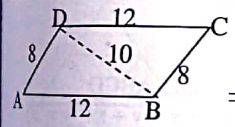
$\Rightarrow h_2$ (4) = 18 $\Rightarrow h_2 = \frac{18}{5} = \frac{9}{2}$ cm

Therefore, the two altitude are $\frac{18}{5}$ cm and $\frac{9}{2}$ cm

64. What is the area of parallelogram with sides 8cm and 12cm, the shorter diagonal is 10cm.

A. 96cm^2 B. $15\sqrt{7}$ cm^2 C. $30\sqrt{7}$ cm^2 D. 80cm^2

Explanation: consider the parallelogram.



Remember Heron's formula

Semi – perimeter of triangle ABD

$\Rightarrow$ S = $\frac{8+12+10}{12} = 15 \longleftarrow$ semi perimeter

Area of ΔABD = $\sqrt{15(15-12)(15-10)(15-8)} = 15\sqrt{7}$ cm^2

We know that area of ABCD = $2(15)(\sqrt{7}) = 30\sqrt{7}$ cm^2

$\therefore$ Answer: C

65. What is the area of parallelogram whose are 12cm and 16cm and intersect at an angle of 60^0?

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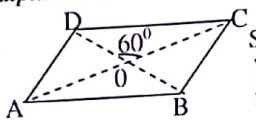
A. $96cm^2$ B. $48cm^2$ C. $96\sqrt{3}\ cm^2$ D. $48\sqrt{3}\ cm^2$

Explanation: Let DB = 12cm $\Rightarrow$ OD = OB = 6cm

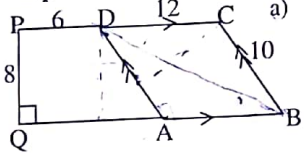


Let AC = 16cm $\Rightarrow$ OA = OC = 8cm
Since the diagonal of parallelogram bisect each other
We known that area of parallelogram is the sum of four small triangle.

$\Rightarrow$ Area of ABCD $= 4\left(\frac{1}{2}\right)(8)(6)\sin 60^0 = 48 = \sqrt{3}\ cm^2$ ∴ **Answer: D**

66. In the given figure, PD=r units, DC=12 unit, PQ=8 unit and BC=10 unit
    a) **Find** i) the area of ABCD
       ii) the length of altitude from B to AD
    b) Join AP, QD, QC, **Find the area of** triangle, ΔAPD and ΔQDE

Explanation: Here



a) i. area of ABCD=(h) (AB)←h=8, AB = DC = 12
∴ A(ABCD) = (8)(12) = 96Sq. unit
ii) we may have area of ABCD = (AD) H,← AD = BC
$\Rightarrow$ 96 = (10)H implies H = 9.6 unit
∴ H = 9.6 unit← altitude from B to AD

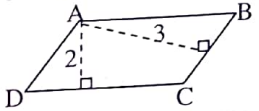
b) we know that the altitude of triangle APD is equal to PQ = 8 unit

therefore, area of $\Delta APD = \frac{1}{2}(PD)(PQ)\ \frac{1}{2}(6)(8) = 24$ sq. unit

area of $\Delta QDC = \frac{1}{2}(DC)(PQ) = \frac{1}{2}(12)(8) = 48$ sq. unit

67. The length of two altitude of parallelogram are 2cm, 3cm and the perimeter of parallelogram is 20cm.
    Find a) side length b) area of parallelogram

Solution: Let AB = DC = x and AD = BC = y



We know that perimeter ABCD
= AB + BC + CD + DA = x + y + x + y = 20
$\Rightarrow$ 2x + 2y = 20 implies x + y = 10

- Area of ABCD = (2) (DC) = 3BC implies 2x = 3y

  $\Rightarrow y = \frac{2}{3}x$ since x + y = 10 **provided** that

$\Rightarrow x + \left(\frac{2}{3}x\right) = 10$ implies $\frac{5x}{3} = 10 \Rightarrow x = 6cm$

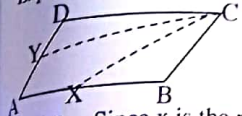
To find y, $y = \frac{2}{3}x = \frac{2}{3}(6) = 5cm$

Therefore, side length will be 6cm and 4cm.

b) area of ABCD = 2(DC) = 2(6) = $12cm^2$

68. ABCD is a parallelogram such that x and y are the mid – point of side AB and AD respectively.
    Find the area of Ax Cy interms of ABCD . . .

Explanation: consider the figure



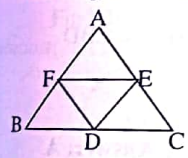
1. Draw AC ⟵ two distinct points determine a line
2. The diagonal AC divides the area of ABCD in to two equal area. That is a(Δ ABC) = a(ΔADC)
3. Since x is the mid – point of AB so that Cx divides the area of ΔABC in to two equal parts. That is $a(\Delta ACx) = a(\Delta BxC) = \frac{a(\Delta ABC)}{2}$

Similarly, area of $\Delta ACy = a(\Delta AxC)^2$

∴ area of ABCD = (area of ΔABC) + (area of ΔACD)
= 2(area of ΔAxC) + (area of ΔACy)
= 2(area of ΔAxCy)

Therefore, area of AxCy $= \frac{1}{2}$ area of ABCD.

69. D,E and F are the mid – points of the sides of ΔABC, then which of the following necessarily true.



A. DEAF is parallelogram
B. ΔBFD ≡ ΔCED
C. DEAF is a rhombus
D. Area of $\Delta AFE = \frac{1}{4}$ area of ΔABC

Explanation:

1. A line which joins the mid – point of two sides of a triangle is parallel to the third – side

∴ D and E is mid – point of $\overline{BC}$ and $\overline{AC}$

$\Rightarrow$ DE // AB and $\overline{DE}$ // $\overline{AF}$, since F is AB

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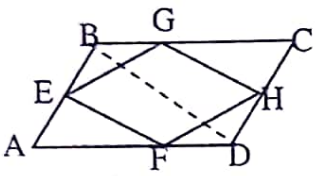
$\therefore$ D and F is mid – point of $\overline{BC}$ and $\overline{AB}$

$\Rightarrow$ FD // AE

If both pair of a quadrilateral is parallel then DEAF is parallelogram. **Answer: A**

70. In the figure, ABCD is a parallelogram E,F,G and H are mid – points of $\overline{AB}$, $\overline{AD}$ $\overline{BC}$, $\overline{DC}$ resp. which of the following is **Not true.**



A. $\Delta ABF \equiv \Delta CHB$

B. $\Delta AEF \sim \Delta CDB$

C. $\overline{EF} // \overline{GH}$

D. Area of $\Delta CGH = \frac{1}{8}$ a(ABCD)

Explanation:

A. $\overline{AB} \neq \overline{BC}$ implies $\Delta ABF \neq \Delta CBH$

B. $\Delta AEF \sim \Delta CDB$ because $\Delta AEF \sim \Delta ABD$ by AA and $\Delta CDB \equiv \Delta ABD$

C. EFHG is rhombus, so that $\overline{EF} // \overline{GH}$

D. G and H is mid – point of BC and DC

D. H and G is mid – point of $\overline{DC}$ and $\overline{BC}$

$\Rightarrow \Delta CGH \sim \Delta CBD$ by AA – similarity

$\therefore \frac{CG}{CB} = \frac{GH}{BD} = \frac{1}{2} \longleftarrow GH = \frac{1}{2}$ BD

$\therefore$ Area of $\left(\frac{\Delta CGH}{\Delta CBD}\right) = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$

$\Rightarrow$ Area of Δ CBD = 4 (area of ΔCGH)

But area of ABCD = area of (ΔCBD +ΔABD) $\leftarrow$ ABD $\equiv$ CBD

= 4 area of (ΔCGH +ΔCGH)

= 8 area of (ΔCGH)

$\therefore$ area of $\Delta CGH = \frac{1}{8}$ area of (ABCD) $\therefore$ **Answer: A**

The circle

Symmetrical properties of the circle

- A circle is symmetrical about any diameter. The locus of point equidistant from two fixed points is the perpendicular bisector of the line joining the two fixed points.

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$\Rightarrow x = \sqrt{20} = \sqrt{4x5}$

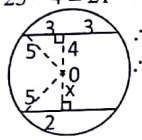
$\therefore x = 2\sqrt{5}$

73. In a circle of radius 5cm, there are two parallel chords of lengths 6cm, 4cm. find the distance between the chords.

Solution:

$x^2 + 2^2 = 5^2$

$\Rightarrow x^2 = 25 - 4 = 21$



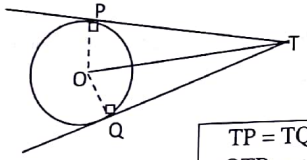
$\therefore x = \sqrt{21}$ and the other is 4

$\therefore$ distance between chords will be

$(\sqrt{21} + 4)$cm

Note: If two tangents are drawn to a circle from an external point then,

1. the tangents are equal in length
2. the line joining the centre to the external point bisects the angle between the tangent

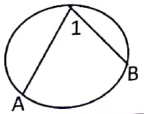


TP = TQ

$<OTP \equiv < OTQ$

Angles and Arcs determined by the line intersecting inside and on a circle

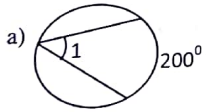
- An angle is inscribed in a circle and is an inscribed angle if its vertex lies on the circle and its side contains chord of the circle.
- The measure of an inscribed angle is equal to half the measure of its intercepted arc.

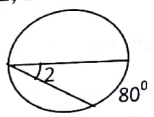


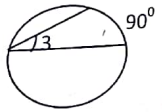
$m < 1 = \frac{1}{2} m(\widehat{AB})$

Illustrative example

74. Find the measure of <1, <2, and <3

a) 

b) 

c) 

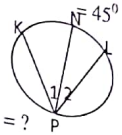
Solution:

a) $m < 1 = \frac{1}{2} \times 200^0 = 100^0$

b) $m < 2 = \frac{1}{2} \times 80^0 = 40^0$

c) $m < 3 = \frac{1}{2} \times 90^0$

75. $\widehat{PK}, \widehat{PN}$ and $\overline{PL}$ re chords of circle



a. If $\widehat{mkw} = 50$ then $m < 1 = ?$

Solution: $: m < 1 = \frac{1}{2} m \widehat{KN} = \frac{1}{2}(50^0) = 25^0$

b. If $m \widehat{NL} = 110^0$, then find, $m<2$

Solution: $m < 2 = \frac{1}{2} m \widehat{NL} = \frac{1}{2}(110^0) = 55^0$

c. If $m \overline{KN} = 70^0$ and $m \overline{NL} = 115^0$, then find, $m \widehat{KNL}$

Solution: $m \overline{KNL} = 185^0$

d. If $m<1 = 28^0$, then find, $m \widehat{KN}$

Solution: $m<1 = \frac{1}{2} m \widehat{KN}$

$\Rightarrow m \widehat{KN} = 2(28^0) = 56^0$

e. If $m < 2 = 76^0$, then find $m \widehat{NL}$

Solution: $m< 2 = \frac{1}{2} m \widehat{NL}$

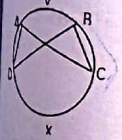
$\Rightarrow 76^0 = \frac{1}{2} m \widehat{NL}$

$\therefore \widehat{NL} = 152^0$

Note: 1) An angle inscribed in a semi circle is right angle

2) In a circle, inscribed angles subtended by the same arc are equal

$<A = \frac{1}{2} m \widehat{DXB} = < B \qquad <D = \frac{1}{2} m \widehat{AYB} = <C$



1) an angle inscribed in arc less than a semi-circle is obtuse

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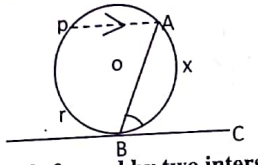
2) an angle inscribed in an arc greater than a semi-circle is Acute

Angle formed by tangent and chord

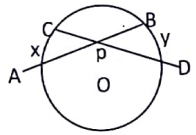
- An angle formed by a tangent and chord drawn from the point of tangency is measured by half the arc if intercepts



$$m(<ABC) = \frac{1}{2}\left(\widehat{A \times B}\right)$$

Angle formed by two intersecting chord

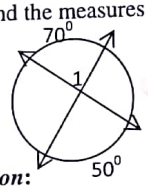
- The measure of an angle formed by two chord intersecting in side a circle is half the sum of the measures of the arc subtending the angle and its vertically opposite angle.

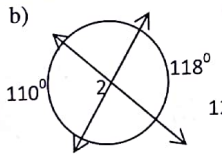


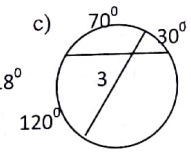
$$m(<BPD) = \frac{1}{2}m\left(\widehat{A \times B}\right) + \frac{1}{2}m\left(\widehat{BYD}\right)$$

Illustrative example

76. Find the measures of <1,<2 and <3

a) 

b) 

c) 

Solution:

$m<1 = \frac{1}{2}(50^0 + 70^0)$

$= 60^0$

$m<2 = \frac{1}{2}(110^0 + 118^0)$

$m<2 = \frac{1}{2} \times 228^0$

$m<2 = 114^0$

c) $m<3 = \frac{1}{2}(120^0 + 30^0$

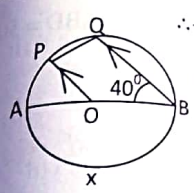
$= \frac{1}{2} \times 150^0$

$= 75^0$

77. In figure, $\overline{AB}$ is a diameter. 0 is the centre of the circle. If $\overline{OP} \parallel \overline{BQ}$ and $m(<ABQ) = 40^0$

Find $m(<OPQ)$

Solution:



$<AOP = 40^0$ (Explanation: Corresponding angle of parallel lines)

$\therefore < POB = 140^0$

And $< PQB = \frac{1}{2} m\left(\widehat{PxB}\right)$

$= \frac{1}{2}(40^0 + 180^0)$

$= 110^0$

Since OPQB is Quadrilateral, sum of vertex angle is 360^0

$\therefore 140^0 + 40^0 + 110^0 + < 0PQ = 360^0$

$\therefore <OPQ = 360^0 - 290^0 = 70^0$

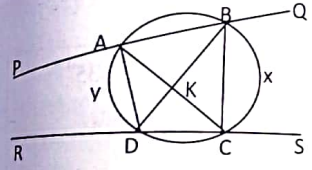
78. In figure PABQ, RDCS are straight line cutting the circle ABCD. Given short reason for each answer to the following

a) which angle equals to

Solution: <DAC= <DBC (subtend by the same arc) i) <DAC = _____

Solution: <BDC=<BAC (subtended by the same arc) ii) <BDC = _____

Solution: <BAD = <BAC + <CAD iii) <BAD = _____



iv) $<BKC = \frac{1}{2} m\left(\widehat{BxC}\right) + \left(\widehat{AyD}\right)$

vi)<PAD=100^0 ,<DAC = 30^0, then Find $m\left(\widehat{DC}\right)$ m (<DBC)

Solution: $<DAC = \frac{1}{2} m\left(\widehat{DC}\right) \Leftrightarrow m\left(\widehat{DC}\right) = 60^0$

$$m(<DBC) = \frac{1}{2} m\left(\widehat{DC}\right) = \frac{1}{2}(60^0) = 30^0$$

What follows if $\overline{BD}$ is a diameter

Solution: $<BCD = 90^0$

vii) If $<RDB = 140^0$ and $<QBD = 130^0$ then find m(<BKC) m(<AKB)

Solution:

$m(<BKC) = \frac{1}{2} m\left(\widehat{BxC} + \widehat{AyD}\right)$ if $<RDB = 140^0$, then $<BDC = 40^0$

$\Rightarrow <BDC = \frac{1}{2} m\left(\widehat{BxC}\right) = 40^0$

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$= m(\widehat{BxC}) = 80^0$

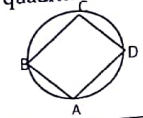
And $<QBD + <ABD = 180^0 \Rightarrow 130^0 + <ABD = 180^0 \therefore <ABD = 50^0$

And $<ABD + \frac{1}{2} m(\widehat{AyD}) = 50^0$

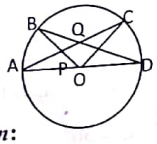
$\Rightarrow \widehat{AyD} = 100^0 \quad \therefore m(<BKC) = \frac{1}{2}(80^0 + 100^0) = 90^0$

And $m < AKB = 90^0$

Note. The vertices of quadrilateral ABCD lies on the circle O. Such a quadrilateral is called cyclic quadrilateral.



- Opposite angle of a cyclic quadrilateral are supplementary
- $m< A + m< C = 180^0$
- $m<B + m<D = 180^0$

viii) 

circle having center 0 such that $m(AOB) = 40^0$ and $m(<COD) = 30^0 \ldots$ .... (ESLCE 1985) then find

a) <AQB b) m(<BQC) c) m(APB)

Solution:

$m(<AQB) = \frac{1}{2}(\widehat{AB} + \widehat{CD}) = \frac{1}{2}(40+30) = 35^0$

$\therefore m(<BQC) = 180^0 - 35^0 = 145^0$

c) $m(\widehat{<APB})$

Extend $\overline{BO}$ to E

$m(<AOB) = m(<DOE) = 40^0$ (by vertically opposite angle)

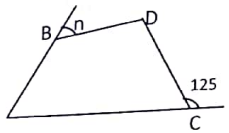
$\therefore m(\widehat{DE}) = 40^0$

$\therefore m(<APB) = \frac{1}{2} m(\widehat{AB} + \widehat{CDE}) = \frac{1}{2}(40^0 + (30 + 40))$

$= \frac{1}{2}(40 + 70) = 55^0$

Illustrative Example

79. a) find whether the points A,B,C,D are concyclic if i) $n = 55^0$ ii) $n = 45^0$



Solution: i) $n = 55^0$

$\Rightarrow <ABD = 125^0$

and $<ACD = 55^0$

A

$\therefore <ABD + <ACD = 125^0 + 55^0 = 180^0$

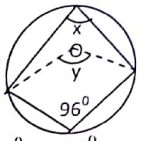
ii) if $n = 45^0$ $\therefore$ Yes, concyclic

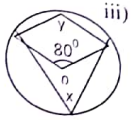
$\Rightarrow <ABD = 135^0$ and $<ACD = 55^0$

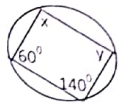
$\Rightarrow < ABD + <ACD = 135^0 + 55^0 = 190^0$

$\therefore$ Not concyclic

b) Find x,y, if O is center of circle

i) 

ii) 

iii) 

Solution:

i) $x + 96^0 = 180^0$

$\Rightarrow x = 180^0 - 96^0$

$= 84$

$\therefore y = 168$

ii) $x + y = 180^0$

$\therefore y = 180^0 - 40^0$

$\therefore y = 140^0$

$\therefore x = 40$

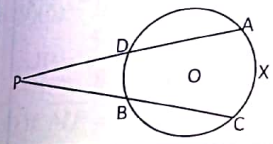
iii) $x + 140^0 = 180$

$\Rightarrow \frac{1}{2}(80) + y = 180^0 \Rightarrow x = 40^0$ and

$60 + y = 180^0$

$\Rightarrow y = 120^0$

- *The measure of an angle formed by two secants of a circle intersecting out side the circle is one half of the difference of the measure of the intercepted arcs.*

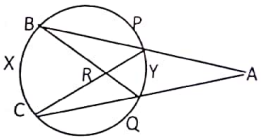


$m(p) = \frac{1}{2} m\left((\widehat{AxC}) - (\widehat{DB})\right)$

Illustrative example

80. In fig. $m(<BPC) = 48^0$, $m(<BRC) = 68^0$ and $m(<BCR) = 62^0$. Find angle of ΔABC

Solution:



Now, $<BPC = \frac{1}{2} m(BxC)$

$\Rightarrow 48^0 = \frac{1}{2} m(BxC)$

$\therefore BxC = 96$

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And $<BRC = \frac{1}{2} m\left((\widehat{BxC}) + (PyQ)\right)$

$\Rightarrow 68^0 = \frac{1}{2}(96^0 + PyQ)$

$\Rightarrow 136^0 - 96^0 = PyQ$

$\therefore PyQ = 40^0$

Hence $<A = \frac{1}{2} m(BxC - PyQ)$

$m(<A) = \frac{1}{2}(96 - 40)$

$\therefore m(<A) = 62^0$

And $m(<PCQ) = \frac{1}{2} m(\overline{PYQ}) = \frac{1}{2}(40^0) = 20^0$

$\therefore m(<BCA) = 62^0 + 20^0 = 82^0$

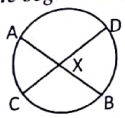
And $m(<ABC) = 180^0 - (82^0 + 28^0) = 70^0$

$\therefore$ The angle of ΔABC will be 28^0, 82^0 and 70^0

An unexpected relationship

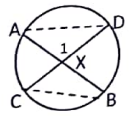
- *If two chords intersect inside a circle the product of the lengths of the segments of one chord equals the product of the lengths of the segments of the other*

$\overline{AX} \cdot \overline{XB} = \overline{CX} \cdot \overline{XD}$



Given $\overline{AB}$ *and* $\overline{CD}$ are chord of a circle. $\overline{AB}$ *and* $\overline{CD}$ intersect at point x inside the circle.

Proof:



1. $<A \equiv <C$ (subtended by the same arc)
2. $<1 \equiv <2$ (vertically opposite angle)
3. $<B \equiv <D$ (subtended by the same arc)

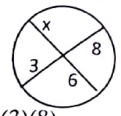
$\therefore \Delta AXD \sim \Delta CXB$

$\Rightarrow \frac{AX}{CX} = \frac{XD}{XB}$

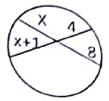
$\therefore \overline{AX} \cdot \overline{XB} = \overline{CX} \cdot \overline{XD}$

81. Find the length x.

a)



b)



Solution:

a) $6x = (3)(8)$

$\Rightarrow 6x = 24$

$\therefore x = \frac{24}{6} = 4$

b) $8x = 4(x+1)$

$\Rightarrow 8x = 4x+4$

$\Rightarrow 8x - 4x = 4$

$4x = 4$

$\therefore x = 1$

82. a) If IP = 4, $\overline{KP} = 6$, $\overline{MP} = 6$, then find $\overline{NP}$

Solution:

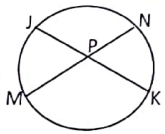
$\overline{JP} \cdot \overline{PK} = \overline{MP} \cdot \overline{PN}$

$\Rightarrow (4)\ (6) = (6)PN$

$\therefore PN = 4$

b) $\overline{MP} = 9$, $\overline{NP} = 6$, $\overline{KP} = 8$ then find JP _____

Solution:



MP.PN = JP.PK

$\Rightarrow (9)(6) = JP.(8)$

$\therefore JP = \frac{54}{8} = \frac{27}{4}$

c) If $\overline{MP} = \overline{NP}$, $\overline{JP} = 4, \overline{KP} = 9$, then find $\overline{MP}$

Solution:

Define: $\overline{MP} = \overline{NP} = x \Rightarrow MP \cdot NP = JP \cdot PK$

$(x)(x) = (4)(9)$

$\Rightarrow x^2 = 36$

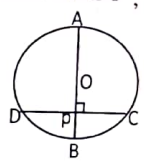
$\therefore x = 6$

i.e. MP = NP = 6

83. If AB is diameter, radius $\overline{AO}$ = 6cm and $\overline{PB}$ = 2cm find $\overline{DP}$, $\overline{OP}$

Solution:

$\overline{DP} \cdot \overline{PC} = \overline{AP} \cdot PB$



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But $\overline{DP} = \overline{PC}$ (diameter is ⊥ bisector of chord)

Let Define $\overline{DP} = \overline{PC}$ =x and $\overline{PO}$ = t so that

x.x = (AO + OP) PB

$\Rightarrow x^2 = (6 + t)2$

$\Rightarrow x^2 = 12 + 2t$ . . . .(1)

And $\left(\overline{DO}\right)^2 = \left(\overline{DP}\right)^2 + \left(\overline{OP}\right)^2$

$\Rightarrow 6^2 = x^2 + t^2$ (explanation: $\overline{DO} = \overline{OA} = CO = radius$)

From (1) we get

$\Rightarrow 36 = (12 + 2t) + t^2$

$\Rightarrow t^2 + 2t - 24 = 0$

(t-4)(t+6) = 0

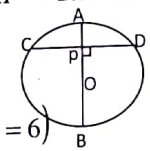
$\therefore t = 4 = \overline{OP}$

And DP = x = $\sqrt{12 + 2t}$

= $\sqrt{12+8}$

$\overline{DP} = 2\sqrt{5}cm$

84. If AB is diameter, Let $\overline{BO}$ = x, $\overline{AP}$ = 2cm and $\overline{PD}$ = 6, then find a) $\overline{BO}$ b) $\overline{PO}$



***Solution*:**

From the definition:

$\overline{CP} \cdot \overline{PD} = \overline{AP} \cdot \overline{PB}$ $\left(\overline{CP} = \overline{PD} = 6\right)$

$\Rightarrow \overline{AP} \cdot \overline{PB} = 6.6$

$\Rightarrow (2)\left(\overline{PO} + \overline{BO}\right) = 36$

Let Define $\overline{PO}$ = t and $\overline{BO}$ = x

$\Rightarrow 2(t + x) = 36$ $\Rightarrow t + x = 18$ . . . . (1)

And $PD^2 + PO^2 = OD^2$

$6^2 + t^2 = x^2$

$\Rightarrow x^2 - t^2 = 36 \Rightarrow (x - t)(x+t) = 36$ . . . . . (2)

From (1) we have t + x = 18

Hence put t+x = 18 in (2)

$\Rightarrow (x - t)(18) = 36$

$\Rightarrow x - t = 2$ . . . . . (3)

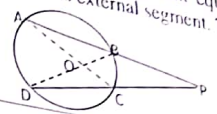
From (1) and (3)

$\begin{cases} x + t = 18 \\ x - t = 2 \end{cases}$

$2x = 20$ $\therefore x = 10 = \overline{BO}$ And $\overline{PO}$ = t = 8cm

Theorem. When two secants are drawn from an external point, the product of one secant and its external segment equals the product of the other secant and its external segment. That is

$\overline{AP} \cdot \overline{PB} = \overline{PD} \cdot \overline{PC}$



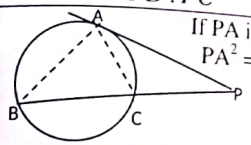
Given with secants $\overline{AP}$ and PC to prove: $\overline{AP} \cdot \overline{AB} = \overline{PD} \cdot \overline{PC}$

1. $m(<A) \cong \frac{1}{2} m(BC)$ and $m(<D) = \frac{1}{2} m(BC)$
2. <P = <P

$\therefore \Delta PAC \sim \Delta PDB$ . . . by AA

$\frac{PA}{PD} = \frac{PC}{PB} = \frac{AC}{DB}$

$\therefore \overline{AP} \cdot \overline{PB} = \overline{PD} \cdot \overline{PC}$



If PA is tangent to the circle then PA^2 = (PC)(PB)

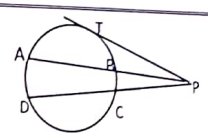
To prove PA^2 = PC.PB

1. <P = <P . . . . common angle
2. MABC = <CAP = $\frac{1}{2}$ m$\left(\overline{AC}\right)$... subtend by the same arc
3. $\therefore \Delta PBA \sim \Delta PAC$ . . . by AA similarity
4. $\frac{PB}{PA} = \frac{PA}{PC} \Leftrightarrow PA^2 = (PC)(PB)$

Illustrative example

85. $\overline{BP}$ = 12cm, $\overline{DC}$ = 10cm and $\overline{CP}$ = 14cm

Find the length of



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a. the chord $\overline{AB}$ b) the tangent PT

Solution:

a) We have: PA.PB = PC.PD

$\Rightarrow$ (AB+PB)PB = PC(PC+DC)

$\Rightarrow$ (AB + 12) 12 = 14(14+10)

$\Rightarrow$ 12AB+144 = 336

$\Rightarrow$ 12AB = 336 – 144 = 192

$\Rightarrow$ 12AB = 192

$\therefore$ AB = $\frac{192}{12}$ = 16cm

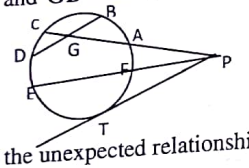
b) We have, PT^2 = (PA).PB

$\Rightarrow PT^2 = (12 + 16)12$

$\Rightarrow PT^2 = (28)(12)$

$\Rightarrow PT = \sqrt{(28)(12)}$ $\therefore PT = 4\sqrt{21}$

86. If PT tangent, secants PE, PC and chord BD, $\overline{CG}$ = 6, $\overline{GA}$ = 4, $\overline{PA}$ = 8, $\overline{PF}$ = 9 and $\overline{GB}$ = 3unit. Then find a) $\overline{DG}$ b) $\overline{EF}$ c) $\overline{PT}$



Solution:

Use the unexpected relationship of segment of a chord

a. Now: $\overline{DG}$. GB = CG.GA

$\Rightarrow \overline{DG}.3 = (6)(4) \Rightarrow \overline{DG} = \frac{24}{3} = 8$unit $\therefore$ $\overline{DG}$ = 8

b. PC.PA = PE.PF

(AC+PA).PA = (PE).PF (10+8).8 = PE.9 $\therefore$PE = $\frac{144}{9}$ = 16unit

$\Rightarrow$ PE = EF + FP $\Rightarrow$ 16 = EF + 9 $\Rightarrow \overline{EF} = 16 - 9 = 7$ $\therefore \overline{EF} = 7$

c. PT^2 = PE.PF

$\Rightarrow PT^2 = 16.9$ $\Rightarrow PT = \sqrt{(16)(9)} = 12$ unit $\therefore$ PT = 12 unit

87. The diagonal $\overline{AC}, \overline{BD}$ of parallelogram ABCD are of length 10cm and 8cm respectively. The circle BCD cuts $\overline{CA}$ at F.

Find the length of $\overline{AF}$.

Solution: Let the diagonal $\overline{AC}$ and $\overline{BD}$ intersecting at 0.

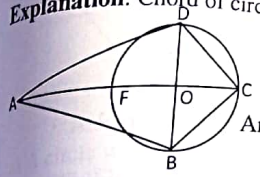
Given: $\overline{AC}$ = 10cm $\overline{BD}$ = 8cm

$\Rightarrow \overline{DO} = \overline{BO}$ = 4cm and OC = 5cm

Explanation: Diagonal of llgram bisect each other at 0

Hence $\overline{FO}$.OC = $\overline{DO}$. $\overline{BO}$

Explanation: Chord of circle $\overline{BD}$ and $\overline{FC}$ intersecting at O.



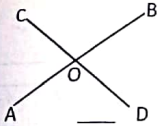
$\therefore$ FO.5 = (4)(4) $\therefore \overline{FO} = \frac{16}{5}$

And $\overline{AO} = \overline{AF} + \overline{OF} \Rightarrow 5 = \overline{AF} + \frac{16}{5}$

$\Rightarrow \overline{AF} = 5 - \frac{16}{5} = \frac{9}{5}$ $\therefore \overline{AF} = \frac{9}{5}$

88. $\overrightarrow{AOB}, \overrightarrow{COD}$ are two straight lines such that $\overline{AB}$ = 20cm, $\overline{CD}$ = 19cm, $\overline{AO}$ = 6cm, $\overline{CO}$ = 7cm prove that ACBD is a cyclic.

Solution:



Now, $\Rightarrow \overline{OB} = AB - \overline{AO} = 20 - 6 = 14cm$

And $\overline{OD} = \overline{CD} - \overline{OC} = 19 - 7 = 12cm$

If $\overline{AO}.\overline{OB}$ = CO.OD then ACBD is cyclic

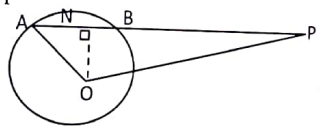
$\Rightarrow$ (6)(14) = (7)12 $\Rightarrow$ 84 = 84

$\therefore$ ACBD is cyclic quadrilateral.

89. From any point p outside circle, centre O, radius r a line is drawn cutting the circle at A and B.

Prove that (PA)(PB) = $(PO)^2 - r^2$

Solution:



Join $\overline{OA}$.OP, ON

Proof: $\overline{PA} \cdot \overline{PB}$ = (PN+AN).(PN-NB)

Explanation:

PA = PN + AN PB = PN – NB And AN = NB

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$\Rightarrow \overline{PA}\cdot\overline{PB} = (\overline{PN}+\overline{AN})(\overline{PN}-\overline{AN})$

$= PN^2 - AN^2$

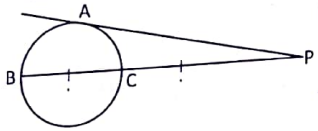
$= (PN^2 + ON^2) - (AN^2 + ON^2)$

$= (PO)^2 - AO^2$

$= (PO)^2 - r^2$ (Explanation: AO = r)

$\therefore \overline{PA}\cdot\overline{PB} = (PO)^2 - r^2$

90. Given that C is mid point of $\overline{BP}$ and $\overline{AP}$ is tangent to the circle. Find $\overline{AP}$ in terms of PC



Solution: $AP^2 = (\overline{PC})(\overline{PB})$

But $\overline{PB} = 2PC$

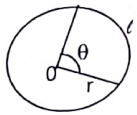
$\Rightarrow \overline{AP}^2 = (\overline{PC})(2\overline{PC}) = 2PC^2$

$\therefore AP = \sqrt{2}\,PC$

Arc lengths

- The length l of an arc of circle of radius r that subtends an angle of θ at the center is given by

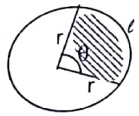
$$l = \frac{\theta}{360^0}\times 2\pi r = \frac{\pi r\theta}{180^0}$$



The area and perimeter of sector

- The area A of sector of radius r and central angleθ is given by

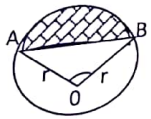
$$A = \frac{\theta}{360^0}\times \pi r^2 = \frac{\pi r^2\theta}{360^0}$$



- The perimeter P of the sector

$$P = 2r + \frac{\pi r\theta}{180^0}$$

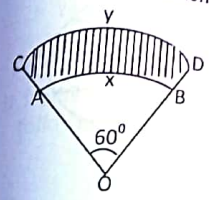
The area and perimeter of segment



- Area of segment $= \frac{\pi r^2\theta}{360^0} - \frac{1}{2}r^2\sin\theta$
- Perimeter of segment $= \left(2r\sin\left(\frac{\theta}{2}\right)\right) + \frac{\pi r\theta}{180^0}$

91. In the figure $\overset{\frown}{AxB}$ and $\overset{\frown}{CyD}$ are ares of concentric circles with center at O and with radii 2cm and 4cm respectively if m $(AOB) = 60^0$, then the area of the shaded part is:

A. πcm^2 B. $1.5\pi cm^2$ C. $3\pi cm^2$ D. $2\pi cm^2$



Solution: area of shaded

= Area of sector $(\overset{\frown}{OCD} - \overset{\frown}{OAB})$

$$= \frac{\pi(4^2)60^0}{360^0} - \frac{\pi(2)^2 60^0}{360^0}$$

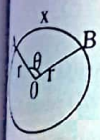
$$= \frac{16\pi - 4\pi}{6} = \frac{12\pi}{6}\,cm^2$$

$\therefore$ Area of shaded $= 2\pi cm^2$ $\therefore$ **Answer D**

91. a) A circle with center at a point '**0**' and radius 6cm then find the length of arc A×B if the central angle

a) $\theta = 60^0$ b) $\theta = 120^0$ c) $\theta = 150^0$

Solution: Arc length $AxB = \frac{\pi r\theta}{180^0}$, r = 6



a) Arc length $AxB = \frac{\pi(6)(60^0)}{180^0} = 2\pi$

b) Arc length $A\times B = \frac{\pi(6)(120^0)}{180^0} = 4\pi$

c) Exercise left for you

91. b) Find the radius of circle if

a) an arc length (ℓ) = 6π and area of sector 10π

b) an arc length (ℓ) = 4π and area of sector 12π

c) an arc length (ℓ) = 3π and area of sector 6π

a) $\frac{\text{length of arc}}{\text{circumference}} = \frac{\text{area of sector}}{\text{area of circle}}$

$\frac{6\pi}{2\pi} = \frac{10\pi}{\pi r^2}$ implies $\frac{3}{r} = \frac{10}{r^2} \Rightarrow r = \frac{10}{3}$ unit

b) $\frac{4\pi}{2\pi r} = \frac{12\pi}{\pi r^2}$ implies $\frac{2}{r} = \frac{12}{r^2} \Rightarrow r = \frac{12}{2} = 6$unit

c) Exercise left for you

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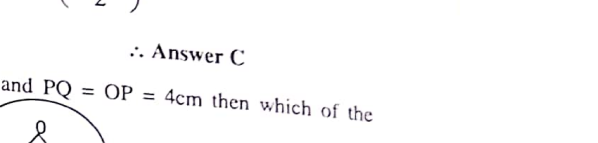
92. In the figure below, which of the following is true

A. $\frac{RX}{XY}=\frac{RT}{TS}$ C. $\frac{R(x)}{RT}=\frac{RS}{RY}$

B. $\frac{RX}{RT}=\frac{XY}{TS}$ D. $\frac{RX}{RY}=\frac{RT}{RS}$



Solution: $(\overline{RX})\overline{RY}=\overline{RT}.RS$

$\therefore \frac{RX}{RT}=\frac{RS}{RY}$ **∴Answer C**

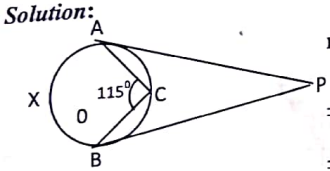
93. Consider the following graph

$m(A\hat{C}B)=115^0$. Then m(<APB) is equal to

A. 65^0 C. 115^0

B. 50^0 D. $57\frac{1}{2}^0$

Solution:



$m(<ACB)=\frac{1}{2}m\overset{\frown}{AxB}$

$\Rightarrow 115^0=\frac{1}{2}m(AxB)$

$\Rightarrow AxB=230^0$

$\therefore AxB=130$

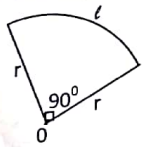
And $<P=\frac{1}{2}m\left(\overset{\frown}{AXB}-\overset{\frown}{ACB}\right)$

$=\frac{1}{2}(230^0-130^0)=50^0$ **∴Answer B**

94. A 300cm long piec of wire is bent into enclose quarter of a circular region center at 0 and radius r. then the radius of the region is equal to

A) $\frac{600}{\pi+2}cm$ C) $\frac{600}{\pi+4}$

B) $\frac{300}{\pi+2}$ D) $\frac{300}{\pi+4}$

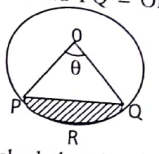


Perimeter $=2r+\frac{\pi r\theta}{180^0}\Leftrightarrow 2r+\frac{\pi r(90^0)}{180^0}=2r+\frac{1}{2}\pi r=300$

$\Rightarrow r\left(2+\frac{1}{2}\pi\right)=300\Leftrightarrow r\left(\frac{4+\pi}{2}\right)=300$

$\therefore r=\frac{600}{4+\pi}$ **∴ Answer C**

95. If in the circle O, is its center and PQ = OP = 4cm then which of the following **is not true**



A) The perimeter of the shaded region is $\frac{12+\pi}{3}cm$

B) Area of the sector is $\frac{8}{3}\pi cm^2$

C) Area of shaded region is $\frac{4}{3}(2\pi-3\sqrt{3})cm^2$

D) Area of ΔOPQ is $4\sqrt{3}\,cm^2$

Solution: Given $r=4,\ \theta=60^0$

A) perimeter of the shaded $=\overline{PQ}+\overset{\frown}{PRQ}$

$=4+\frac{\pi(4)60^0}{180^0}=\frac{12+4\pi}{3}$

B) Area of sector $=\frac{\pi r^2\theta}{360^0}=\frac{\pi(4^2)60^0}{360^0}=\frac{8}{3}\pi cm^2$

C) Area of shaded $=\frac{\pi r^2\theta}{360^0}-\frac{1}{2}r^2\sin\theta=\frac{\pi(4^2)60^0}{360^0}-\frac{1}{2}(4^2)\sin 60^0$

$=\frac{8}{3}\pi-4\sqrt{3}=\frac{4}{3}(2\pi-3\sqrt{3})cm^2$

D) Area of triangle $\Delta OPQ=\frac{1}{2}(4)(4)\sin 60^0=4\sqrt{3}\,cm^2$

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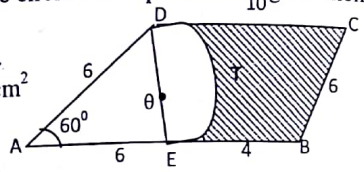
∴Answer A

96. In the figure, ABCD is parallelogram with $\overline{AD}$ = 6cm, $\overline{AB}$ = 10cm and BAD = 60^0. A is center of the circle which passes through D then which of the following **is not true**



A) length of arc $\overset{\frown}{DTE}$ is 2πcm.

B) Area of sector ADE is $6\pi cm^2$

C) The area of shaded part is $6\left(5\sqrt{3}-2\pi\right)cm^2$

D) Area of segment bounded by arc $\overset{\frown}{DTE}$ is $3\left(2\pi-3\sqrt{3}\right)cm^2$

Solution:

A) length of Arc $\overset{\frown}{DTE} = \frac{\pi r\theta}{180} = \pi(6)\left(\frac{60}{180}\right) = 2\pi$

B) Area of sector ADE = $\frac{\pi r^2\theta}{360^0} = \pi\frac{(36)60}{360} = 6\pi cm^2$

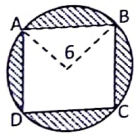
C) Area of shaded part
= Area of parallelogram – Area of sector
= (10)(6) sin 60^0 – $6\pi cm^2$
= $30\sqrt{3} - 6\pi cm^2$
= $6(5\sqrt{3} - \pi)cm^2$

D) Area of segment bounded by Arc $\overset{\frown}{DTE}$
= Area of sector – Area of triangle
= $6\pi cm^2 - \frac{1}{2}(6)(6)\sin 60^0 = 6\pi cm^2 - 9\sqrt{3}\ cm^2$
= $3(2\pi - 3\sqrt{3})\ cm^2$

∴**Answer C**

97. A square ABCD is inscribed in a circle of radius 6cm. find a) the area of the segment cut off by the chord.
b) the perimeter of segment

Solution:



a) Area of segment cut of by chord AB
= Area of sector – area of triangle
= $\frac{\pi r^2\theta}{360^0} - \frac{1}{2}r^2 \sin,\ (\theta = 90^0)$

$= \frac{1}{4}\pi(36) - \frac{1}{2}(36)\sin 90^0$
$= (9\pi - 18)cm^2$

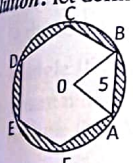
The total are of segment = $4(9\pi - 18)cm^2$
$= 36(\pi - 2)$

b) The perimeter of one segment = $\frac{\pi r\theta}{180^0} + 2r\sin\frac{\theta}{2}$
$= \frac{1}{2}\pi(6) + 2(6)\sin 45^0 = (3\pi + 6\sqrt{2})$cm

The perimeter of 4 segment = $4(3\pi + 6\sqrt{2})$cm
$= 12(\pi + 2\sqrt{2})$cm

98. A regular Hexagon ABCDEF is inscribed in a circle of side length $\overline{AB}$ = 4cm
Find a) The area of segment b) The perimeter of segment

Solution: let define $\overline{AB} = \overline{BC} = \overline{CD} = \overline{DE} = \overline{EF} = S$



$S = 2r\sin\left(\frac{180^0}{n}\right)$, η = 6, S = 4cm

$\Rightarrow 4 = 2r\sin\left(\frac{180^0}{6}\right) \Leftrightarrow 4 = 2r\sin 30^0$

$\Rightarrow r = 4$

Area of segment = (Area of sector– (Area pf triangle cot off by AB AOB) ΔAOB)

$= \frac{\pi r^2\theta}{360^0} + \frac{1}{2}r^2\sin 60^0,\ \theta = 60$

$= \frac{\pi(16)}{6} - \frac{1}{2}(16)\frac{\sqrt{3}}{2} = \frac{8}{3}\pi - 4\sqrt{3}$

Area of total segment = $6\left(\frac{8\pi}{3} - 4\sqrt{3}\right)$

Perimeter of segment = $\frac{\pi r\theta}{180^0} + 2r\sin\left(\frac{180^0}{n}\right)$

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$r = 4, \theta = 60^0, \eta = 6$

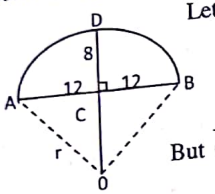
$= \frac{\pi(4)}{3} + 2(4)\sin 30^0 = \left(\frac{4\pi}{3} + 4\right) cm$

Total perimeter of segment $= 6\left(\frac{4\pi}{3} + 4\right) = 24\left(\frac{\pi}{3} + 1\right)$

$= 8(\pi + 3) cm$

99. In fig 0 is the center of circular arc $\overset{\frown}{ADB}$. Find the radius r of the circle

Solution: using def^n of chord of circle.



Let extend $\overline{DO}$ to F, F is point on circle.

$\Rightarrow \overline{AC} \cdot \overline{CB} = \overline{DC}.\overline{CF}$

$\Rightarrow 144 = (8)CF$

$\Rightarrow CF = 18$

But $\overline{CF} = \overline{CO}$ + DF, and DF = r $\Rightarrow$ DC + CO

$\Rightarrow 18 = r - 8 + r$

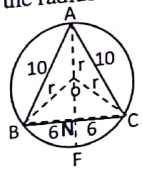
$\Rightarrow 26 = 2r$

$\therefore r = 13$unit

100. In fig. $\overline{AB} = \overline{AC} = 10$cm and $\overline{BC} = 12$cm

Find the radius r of the circle ABC.

Solution:



$AN^2 + BN^2 = AB^2$

Define AN = X

$\Rightarrow x^2 + 6^2 = 10^2$

$\Rightarrow x^2 = 100 - 36 = 64$

$\therefore x = AN = 8$

And Join $\overline{ON}$ to F on a circle.

$\therefore (BN)(NC) = (AN)(NF)$

$\Rightarrow (6)(6) = (8)NF$

$\Rightarrow NF = \frac{36}{8} = \frac{9}{2}$

So that the diameter $\overline{AF} = AN + NF$

$\Rightarrow AF = 8 + \frac{9}{2} = \frac{16+9}{2} \Rightarrow AF = \frac{25}{2}$

radius $r = \frac{AF}{2} = \frac{25}{4}$ $\therefore$ the radius, $r = \frac{25}{4}$ cm

92. The area of a regular hexagon is $96\sqrt{3}$ cm^2 what is its perimeter

A) 48cm B) 54cm C) 42cm D) 60cm

Solution:

We know Area $= \frac{1}{2} nr^2 \sin\left(\frac{360^0}{n}\right)$ area $= 96\sqrt{3}$, n = 6

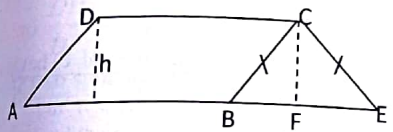
$\Rightarrow 96\sqrt{3}\ \frac{1}{2}(6)r^2 \sin 60^0 \Rightarrow 32\sqrt{3} = r^2 \frac{\sqrt{3}}{2} \Rightarrow r^2 = 64 \Rightarrow r = 8$

And $S = 2r \sin\left(\frac{180^0}{n}\right)$, n = 6 $\Rightarrow$ S = 2(8) sin 30^0 = $2(8)\left(\frac{1}{2}\right)$

$\therefore S = 8$ $\therefore$ Perimeter = 6 × 8 = 48

$\therefore$ **Answer A.**

93. In the following figure ABCD is parallelogram and ΔBCE is an isosceles triangle If $\overline{AB} = 8$ units, $\overline{CE} = 6$ units and the area of ABCD is 40 square units, which of the following is the area of ΔBCE in square units. Assume that A,B and E are on a line.



A) $\frac{5}{2}\sqrt{11}$ B. $5\sqrt{11}$ C) 15 D) 27.5

Solution: Area of ABCD = (h) $\overline{AB}$

$40 = h(8)$

$\therefore h = 5$

$\Rightarrow \overline{CF} = 5$

And $\overline{CF}^2 + \overline{FE}^2 = CE^2 \Rightarrow 25 + \overline{FE}^2 = 6^2\ \overline{FE}^2 = 36 - 25 = 11$

$\therefore FE = \sqrt{11} \Rightarrow \overline{BE} = 2\sqrt{11}$ $\therefore$ Area of $\Delta CBE = \frac{1}{2}\left(2\sqrt{11}\right)5)$

$= 5\sqrt{11}$

$\therefore$ **Answer B.**

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UNIT THIRTEEN

Measurement

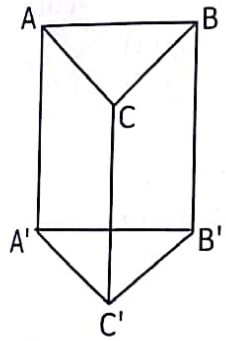
1. Surface areas and volume of prism

- **A solid figure**: is any portion of space bounded by one or more surfaces, plane or curved,
  These surfaces are called the face of the solid, and the intersections of adjacent faces are called edges.
- Prism, cylinder, sphere, pyramid,. . . etc are called solid figure

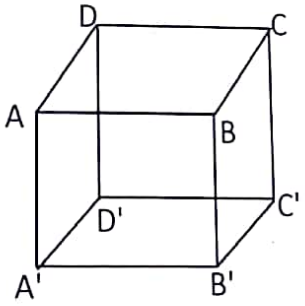
Prism

A solid figure bounded by plane faces of which two called the ends (bases) are congruent figures in parallel planes and the others called **side-faces are parallelogram**.

- The ends (bases) of a prism may be triangle, rectangular, hexagonal or any polygon of any number of sides. So that,
- If the base is triangle, then we call **triangular prism**
- If the base is rectangle, then we call **rectangular prism**
- If the base is hexagonal, then **hexagonal prism**
- The side-edge of every prism are all parallel and equal.
- A prism said to be right prism, if the side edge are perpendicular to the bases (ends) All other prisms are called oblique
- A regular prism is right prism whose base is a regular polygon.

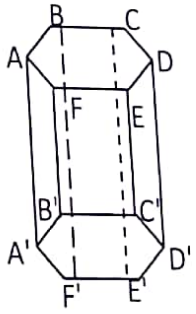


- Triangular prism
- 9 Edges • 6 vertices



Rectangular prism

- have 12 edges
- 8 vertices



Hexagonal prism

- have 18 edges.
- 12 vertices

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In the prism

1. $\overline{AC}, \overline{AB}, \overline{BC}$ are called edges of upper base of triangular prism.
2. $\overline{AB}, \overline{CB}$ *and* $\overline{AC}$ are called edges of lower base of triangular prism.
3. $\overline{AA}, \overline{CC}, \overline{BB}$ are called the lateral edges of prism.
4. the parallelogram region ACC'A; BCC'B' and ABB'A' are called lateral face of prism
5. The union of lateral face is called lateral surface
6. The union of lateral face and the two base are called **total surface**

- **Surface area and volume**

A) lateral surface area – **sum of the area of lateral face** = ph = (perimeter) (height)

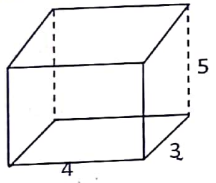
B) Total surface area sum of the area of lateral face and the two bases ph + 2(base area)

C) Volume= (base area) (height)

Illustrative example

1. If the dimension of aright prism (rectangular) prism are 3cm, 4cm and 5cm then find
   a) lateral surface area
   b) total surface area
   c) volume
   d) the length of its diagonal

Solution:

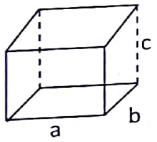


a) lateral surface area = (perimeter of base) height
LA = (3+4+3+4) (5) = $70cm^2$

b) Total surface area
= La + 2B
= $70cm^2 + 2(3\times4)$
= $70cm^2 + 24cm^2 = 94cm^2$

c) Volume = (base area) (height)
= (3×4) (5) = $60cm^3$

d) The diagonal of rectangular prism

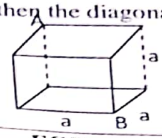


Let D be diagonal of rectangular prism.
Let a, b and c be the three edges of prism
$\therefore D^2 = a^2 + b^2 + c^2$
$\therefore D^2 = 3^2 + 4^2 + 5^2$
$\therefore D = \sqrt{9+16+25} = 5\sqrt{2}cm$

Note.

- If the three edges are equal (cube) then the diagonal of cube, $\overline{AB}$

$D^2 = a^2 + a^2 + a^2 = 3a^2$
$\therefore D = a\sqrt{3}$



2. If the diagonal of a cube is 6 units then find . . . EGSECE
   a) Lateral surface area of the cube
   b) Total surface area of the cube
   c) Volume of the cube of the cube

Solution: Let a be the side of cube

Let D be the diagonal of cube $\therefore D^2 = 3a^2 \Leftrightarrow D = a\sqrt{3}$

$\Rightarrow 6 = a\sqrt{3} \Rightarrow a = \frac{6}{\sqrt{3}} = 2\sqrt{3}$ unit

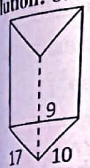
a) lateral surface area = $4a^2 = 4\left(2\sqrt{3}\right)^2 = 48$squ. Unit.

b) Total surface area = L.A + 2B = $48 + 2a^2 = 72$ squ. unit

c) Volume = $a^3 = \left(2\sqrt{3}\right)^2 = 24\sqrt{3}$ cubic unit

3. Aright prism stands on triangular base whose side are 17cm, 10cm, 9cm and the height is 10cm.
Find a) total surface area b) volume

Solution: base area of triangle using herons' formula



$A = \sqrt{S(s-a)(s-b)(s-c)}$ where $S = \frac{17+10+9}{2} = 18$

$= \sqrt{18(1)(8)(9)} = 36$ squ. unit

Lateral surface area = (perimeter of base) (height)
L.A = (17 + 10 + 9)10 = 360squ. unit

$\therefore$ Total surface area = L.A + 2(base area)
= (360 + 2(36)) squ. unit = 432 squ. unit

b) volume = (base area) height = (36) (10) cubic unit
= 360 cubic unit

The base of a prism a regular hexagon on side 8cm and height 6cm.
Find a) total surface area b) volume of prism

Solution:

Lateral surface area = (8+8+8+8+8+8)6 = $288cm^2$

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Base area $= \frac{1}{2}nr^2 \sin\left(\frac{360^0}{n}\right), n = 6 = \frac{1}{2}(6)(8^2)\sin 60^0$

$= (3)(64)\frac{\sqrt{3}}{2} = 96\sqrt{3}\ cm^2$

a) ∴Total surface area =

L.A + 2(B) $= 288+2(96\sqrt{3}) = (288+192\sqrt{3})cm^2$

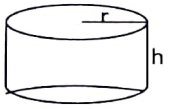
b) Volume = Bh= $(96\sqrt{3})6 = 576\sqrt{3}\ cm^3$

5. A lateral edge of aright prism is 8cm and the perimeter of its base is 12cm. Find lateral surface area.

Solution:

Lateral surface area = (perimeter) height $= (8)(12) = 96cm^2$

Right circular cylinders



1. Lateral surface Area = (circumference) (height)
   ∴ L.A = $(2\pi r)(h) = 2\pi rh$
2. Total surface area
   L.A + 2 (Base area) = $2\pi rh + 2\pi r^2$
3. Volume = (Base Area)(height) = $\pi r^2 h$

6. Aright circular cylinder has a 5cm radius and 3cm height
   Find a) the lateral surface area
   b) the total area c) the volume

Solution:

a) L.A = (circumference) (height) = $2\pi rh$
∴L.A = $2\pi(5)(3) = 30\pi cm^2$

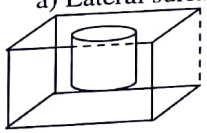
b) Total surface area = L.A + 2B = $30\pi + 2\pi r^2$
∴ T.S.A = $30\pi + 2x\pi \times 5^2 = 30\pi + 50\pi = 80\pi cm^2$

c) Volume = Bh = $\pi r^2 h$
$V = \pi(5^2)(3) = 75\pi cm^3$

7. Aright circular cylinder with radius 2cm is inscribed in a cube whose edges are 5cm long
   Find a) total surface area of resulting figure
   b) volume of resulting solid figure

Solution:

a) Lateral surface area of remaining cylinder and cube = ph + ch



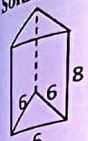
$= (5+5+5+5)5 + 2\pi rh$
$= 100 + 2\pi(2)(5) = 100 + 20\pi$

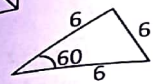
And area of resulting base :$5^2 - \pi r4^2 = 25 - \pi(4) = 25 - 4\pi$
∴Total surface area of resulting figure = $100 + 20\pi + 2(25 - 4\pi)$
$= 150 - 12\pi$

b) Volume of resulting solid is:
V_R = volume of cube – volume of cylinder
$= 5^3 - (\pi r^2)h = 125 - \pi(4)(5)$, radius = 2
$= (125 - 20\pi)cm^3$

8. The base of a prism area equilateral triangle of length of side 6cm and its altitude 8cm then find a) total surface area b) volume

Solution:





- lateral surface area = ph = (6+6+6) 8 = $124cm^2$
- Base area = $\frac{1}{2}(6)(6)\sin 60^0 = 9\sqrt{3}\ cm^2$

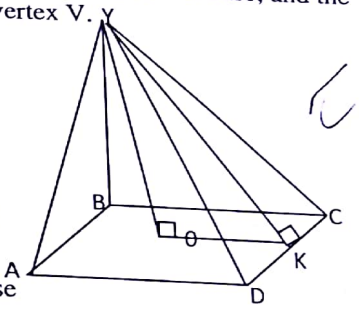
a) Total surface area = L.A + 2B
$= (124 + 2(9\sqrt{3}))cm^2$
$= (124 + 18\sqrt{3})cm^2$

b) Volume = Bh = $(9\sqrt{3})8 = 72\sqrt{3}\ cm^3$

Pyramids, Cones and spheres

Pyramid: is solid bounded by plane faces, of which one is called base, and the rest are triangles having common vertex V.

Regular pyramids



- The vertex: point V
- The base: ABCD
- A lateral edge: $\overline{VA}, \overline{VB}, \overline{VC}, \overline{VD}$
  $\overline{VA} = \overline{VB} = \overline{VC} = \overline{VD}$
- A lateral face: ΔVAB, ΔVAD, ΔVBC And ΔVDC
- The altitude: $\overline{VO}$ lies on center of base
- Aslant height: $\overline{VK}$
- Lateral surface area = $\frac{1}{2}p\ell$, where p is perimeter of the base and l is slant height.
- Total surface Area = $\frac{1}{2}p\ell$ + B.

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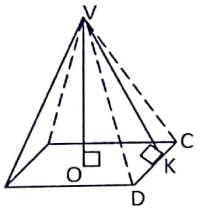
- Volume $= \frac{1}{3}$ Bh, where B-base area and h-is altitude.

Illustrative example

9. A regular pyramid has a square base whose side are 4cm each. The lateral edge are 5cm.
Find a) lateral surface area b) total surface area c) volume

Solution: Let l be the slant height and we have $\overline{VC} = 5$cm



$\overline{DC} = 4$

$\overline{KC} = 2$

$\therefore VK^2 + KC^2 = VC^2$

$\Rightarrow l^2 + 2^2 = 5^2$

$\Rightarrow l^2 = 25 - 4 = 21$

$\therefore l = \sqrt{21}$ cm

a) Lateral surface area $= \frac{1}{2} p\ell = \frac{1}{2}(4+4+4+4)\sqrt{21} = 8\sqrt{21}$ cm^2

b) Total surface area = L.A + B

$= 8\sqrt{21}$ cm^2 + 16cm^2 $= 8(\sqrt{21}+2)$cm^2

c) Volume $= \frac{1}{3}$ Bh

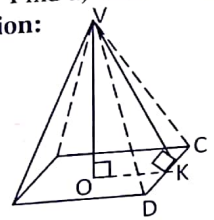
To find h use $Vo^2 + Ok^2 = VK^2 \Rightarrow Vo^2 + (2)^2 = \sqrt{21}^2$

$\Rightarrow Vo^2 = 21 - 4 = 17$

$\therefore \overline{Vo} = h = \sqrt{17}$ cm $\quad \therefore V = \frac{1}{3}(16)\sqrt{17}$ cm^3

10. A regular square pyramid has all its edge are 6cm
Find a) Total surface area b) volume

Solution:



$VK^2 + KC^2 = VC^2$

$VK^2 + 3^2 = 6^2 \Rightarrow VK^2 = 36 - 9 = 27$

$\therefore \overline{VK} = \ell = 3\sqrt{3}$

L.S.A $= \frac{1}{2} p\ell = (6+6+6+6)3\sqrt{3} = 72\sqrt{3}$ cm^2

a) Total surface area $= \frac{1}{2} p\ell$ + B

$= \frac{1}{2}(72\sqrt{3}) + 16 = (36\sqrt{3} + 6)$cm^2

b) Let $\overline{Vo} = h$

$\Rightarrow Vo^2 + OK^2 = VK^2$

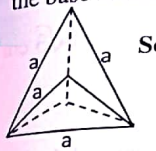
$\Rightarrow Vo^2 + 9 = (3\sqrt{3})^2$

$\Rightarrow Vo^2 = 27 - 9 = 18$

$\therefore \overline{Vo} = h = \sqrt{18} = 3\sqrt{2}$ cm

Volume $= \frac{1}{3}(36)\,3\sqrt{2}$ cm^3 $\therefore V = 36\sqrt{2}$ cm^3

11. The altitude of an equilateral triangular pyramid is 13cm. if one edge of the base is 8cm then find
a) its total surface area b) volume



Solution: In a regular polygon with n side you recall that

$S = 2r \sin\left(\frac{180^0}{n}\right)$

$8\text{cm} = 2r\left(\sin\left(\frac{180^0}{3}\right)\right.$

$\Rightarrow 8 = 2r \sin 60^0 \Rightarrow 4 = r\,\frac{\sqrt{3}}{2} \Rightarrow r = \frac{8}{\sqrt{3}}$

a) Let a be the length of an edge

$a^2 = r^2 + h^2 \quad \Rightarrow a^2 = \frac{64}{3} + 144$

And $\ell^2 + \left(\frac{a}{2}\right)^2 = a^2$ (Explanation: l slant height $\frac{a}{2}$ is the base

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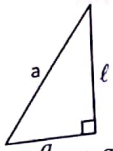
$\ell^2 = a^2 - \left(\frac{a}{2}\right)^2$ where l intersects

$= \frac{64}{3} + 144 - \left(\frac{64}{12} + \frac{144}{4}\right)$

$= \frac{256}{12} - \frac{64}{12} + 144 - 36 = \frac{192}{12} + 108$

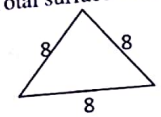


$\ell^2 = 124$

$\ell = \sqrt{124}$

$\therefore$ L.S.A $= \frac{1}{2} p\ell = \frac{1}{2}(3\times8)\sqrt{124} = 12\sqrt{124} = 24\sqrt{31}$

Total surface area $= \frac{1}{2} p\ell + B = 24\sqrt{3} + 64\left(\frac{\sqrt{3}}{4}\right)$



$= 24\sqrt{3} + 16\sqrt{3} = 40\sqrt{3}$ cm^2

Volume $= \frac{1}{3}Bh = \frac{1}{3}(16)\sqrt{3})(12) = 64\sqrt{3}$ cm^3

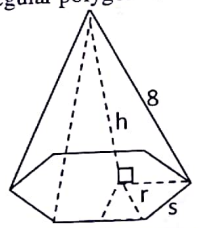
Area $= \frac{1}{2} 8^2 \sin 60 = \frac{64\sqrt{3}}{4}$

12. In a regular hexagonal pyramid, each side of the base is 6cm and the lateral edge is 8cm long. What is the volume of the pyramid? . . . EGSECE.

A) $54\sqrt{3}$ cm^3 B) $36\sqrt{21}$ cm^3 C. $108\sqrt{21}$ cm^3 D) $6\sqrt{21}$ cm^3

Solution:

In regular polygon with n side Recall that



$S = 2r\sin\left(\frac{180^0}{n}\right), n = 6$

$\Rightarrow 6 = 2r\sin 30^0$

$\Rightarrow 6 = 2(r)\frac{1}{2}$

$\therefore r = 6$

Hence, $h^2 + r^2 = 8^2$

$\Rightarrow h^2 + 6^2 = 64$

$\Rightarrow h^2 + 64 - 36 = 28 \quad \therefore h = 2\sqrt{7}$

Volume $= \frac{1}{3}Bh$

But $B = \frac{1}{2}\eta r^2 \sin\left(\frac{360^0}{\eta}\right) = \frac{1}{2}(6)(6^2)\sin\left(\frac{360^0}{6}\right)$

$= \frac{1}{2}(6)(36)\frac{\sqrt{3}}{2} = 54\sqrt{3}$

$\therefore V = \frac{1}{3}(54)\sqrt{3})(2\sqrt{7}) = 36\sqrt{21}cm^3$

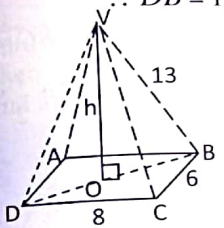
13. Aright pyramid with rectangular base of length 8m and width 6m, has a 13m long lateral edge. What is the volume of the pyramid in cubic meter . . . EGSECE

A) $64\sqrt{10}$ B) 160 C) 208 D) 192

Solution: $DB^2 = DC^2 + CB^2$

$DB^2 = 64 + 36 = 100$

$\therefore \overline{DB} = 10$



$\therefore OB = \frac{\overline{DB}}{2} = \frac{10}{2} = 10 = 5$

And $h^2 + OB^2 = 13^2$

$\Rightarrow h^2 + 25 = 169$

$\Rightarrow h^2 = 144$

$\therefore h = 12$m

Hence, volume of pyramid $= \frac{1}{3}Bh$

$V = \frac{1}{3}(6\times8)(12) \quad \therefore V = 192m^3 \quad \therefore$ **Answer D**

14. What is the lateral surface area of regular square pyramid of height 6m if its base has edge of length 16m . . . EGSECE.

A) $192m^2$ B) $320m^2$ C) $512m^2$ D. $576m^2$

Solution: To find lateral surface area, first, find, the slant height l.

So that

$\Rightarrow VO^2 + OK^2 = l^2$

$\Rightarrow 6^2 + 8^2 = l^2$

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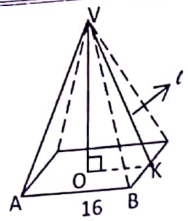


$\left[\text{Explanation}: OK = \frac{1}{2}\overline{AB} = \frac{1}{2}(16) = 8\right]$

$\therefore l = \sqrt{36+64} = \sqrt{100} \quad \therefore l = 10$

Hence, lateral surface area $= \frac{1}{2}p\ell$

$\Rightarrow$ L.S.A $= \frac{1}{2}(16+16+16+16)(10) \quad \therefore$ L.S.A $= 320\text{m}^2$

∴ Answer B

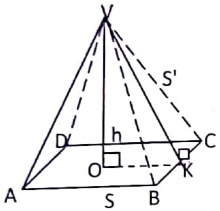
15. For a regular square pyramid whose lateral faces are equilateral triangle of side length S units.

Find a) lateral surface area of the pyramid

b) Altitude of the pyramid

c) Volume of the pyramid

Solution: first, we find slant height ℓ $(\overline{VK})$



Now, $\overline{VB} = \overline{VC} = \overline{VA} = \overline{VD} = \overline{BC} = \overline{AB} = \overline{DC}$

$\therefore \overline{VK}^2 + \overline{KC}^2 = \overline{VC}^2$

$\Rightarrow VK^2 + \left(\frac{S}{2}\right)^2 = S^2 \quad \left[\begin{array}{l}\text{Explanation}: BC = S \\ \therefore \overline{KC} = \frac{1}{2}\overline{BC} = \frac{1}{2}S\end{array}\right]$

Define: VK = ℓ

$\Rightarrow \ell^2 + \left(\frac{S}{2}\right)^2 = S^2$

$\therefore l = \sqrt{S^2 - \frac{S^2}{4}} \quad \therefore l = \sqrt{\frac{3S^2}{4}} = \frac{S\sqrt{3}}{2}$

a) Hence, L.S.A $= \frac{1}{2}p\ell = \frac{1}{2}(S+S+S+S)\frac{S}{2}\sqrt{3}$

$\therefore$ L.S.A $= S^2\sqrt{3}$ squ. unit

b) To find altitude: we use $V0^2 + OK^2 = VK^2$

$\Rightarrow h^2 + \left(\frac{S}{2}\right)^2 = \left(\frac{S}{2}\sqrt{3}\right)^2 \quad \left[\begin{array}{l}\text{Explanation}: BC = S \\ VK = \ell = \frac{S\sqrt{3}}{2}\end{array}\right]$

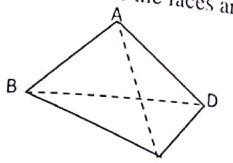
$\Rightarrow h^2 = \frac{3S^2}{4} - \frac{S^2}{4} = \frac{2S^2}{4}$

$\therefore h = \frac{S}{\sqrt{2}}$ unit $\quad \therefore$ Altitude $= \frac{S}{\sqrt{2}}$ unit

c) Volume $= \frac{1}{3}(B)h = \frac{1}{3}(S^2)\frac{S}{\sqrt{2}} = \frac{S^3\sqrt{2}}{6}$ cubic unit

A Tetrahedron is a pyramid on a triangular base, it has four triangular faces

- A regular tetrahedron if the faces are equilateral triangle

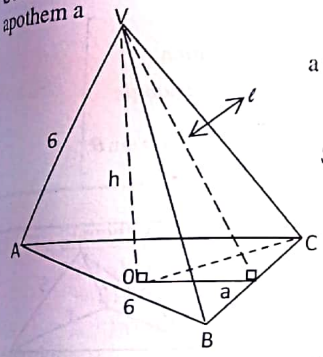


Illustrative example

16. The altitude of a regular triangular pyramid is h If one edge of the base is 6cm, then

Find a) its total surface area b) its volume

Solution: Recall that Regular polygon with n sides, the apothem a



$a = r\cos\left(\frac{180^0}{n}\right)$ and to find radius we use:

$S = 2r\sin\left(\frac{180^0}{n}\right)$

$\Rightarrow 6 = 2r\sin\left(\frac{180^0}{3}\right)$

$\Rightarrow 6 = 2r\sin 60^0$

$\Rightarrow 6 = 2r\frac{\sqrt{3}}{2} \quad \therefore r = \frac{6}{\sqrt{3}}$ i.e. $\overline{OC} = r = \frac{6}{\sqrt{3}}$

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Now, to find h, use $Vo^2 + OC^2 = VC^2$

$$\Rightarrow h^2 + \left(\frac{6}{\sqrt{2}}\right)^2 = 6^2 \quad \Rightarrow h^2 = 36 - \frac{36}{3} = 24$$

$$\therefore h = \sqrt{24} = 2\sqrt{6} \text{ cm}$$

And to find slant height l,

First, apothem $a = r\cos\left(\frac{180^0}{n}\right)$, $r = \frac{6}{\sqrt{3}}$, $n = 3$

$$\therefore a = \frac{6}{\sqrt{3}}\cos 60 = \frac{6}{\sqrt{3}}\left(\frac{1}{2}\right) \therefore a = \frac{3}{\sqrt{3}} \text{cm}$$

Hence, the slant height l,

$h^2 + a^2 = l^2$

$$\Rightarrow \left(\sqrt{24}\right)^2 + \left(\frac{3}{\sqrt{3}}\right)^2 = \ell^2$$

$$\Rightarrow l^2 = 24 + \frac{9}{3} = 24 + 3 = 27 \therefore l = \sqrt{27} = 3\sqrt{3} \text{ cm}$$

a) Hence, L.S.A $= \frac{1}{2}p\ell = \frac{1}{2}(6+6+6)3\sqrt{3} = 27\sqrt{3}$ cm^2

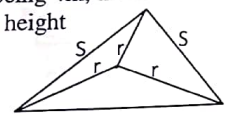
b) Volume $= \frac{1}{3}$Bh

$$V = \frac{1}{3}\left(\frac{1}{2}6\times 6\sin 60^0\right)\left(2\sqrt{6}\right)$$

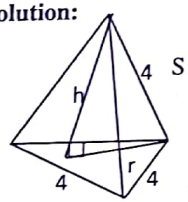
[*Explanation*: *Area of triangle* $= \frac{1}{2}ab\sin\theta$]

$$V = 18\sqrt{2} \text{ cm}^3$$

17. The length of each edge of regular tetrahedron being 4m, them
Find a) the altitude b) volume c) the slant height



Solution:



$S = 2r\sin\left(\frac{180^0}{n}\right)$, $\eta = 3$

$$\Rightarrow 4 = 2r\sin 60^0 \Rightarrow 4 = 2r\frac{\sqrt{3}}{2} \Rightarrow r = \frac{4}{\sqrt{3}}$$

a) The altitude, h is obtained by $h^2 + r^2 = 4^2$

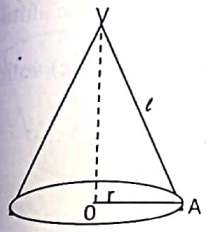
$$\Rightarrow h^2 + \left(\frac{4}{\sqrt{3}}\right)^2 = 4^2$$

$$\Rightarrow h^2 + 16 - \frac{16}{3} = \frac{32}{3} \therefore h = \sqrt{32} = \frac{4\sqrt{2}}{3} \text{ m}$$

b) Volume $= \frac{1}{3}$Bh $= \frac{1}{3}\left(\frac{1}{2}\times 4\times 4\sin 60^0\right)\frac{4\sqrt{2}}{3} = \frac{16}{9}\sqrt{6}$ cm^3

c) Exercise left for the reader

Cone Aright circular cone is solid generated by revolution of aright angled triangle about one of the sides containing the right angle as – axis.



- $\overline{VO}$ is called the altitude, VO = h
- $\overline{VA}$ is called a slant height $\overline{VA}$ = l
- The area of Base B = πr^2
- The lateral surface area $= \frac{1}{2}$(circumference)l
  $= \frac{1}{2}(2\pi r)\,l = \pi rl$
- Total surface area =
  = Base area + L.S.A
  = $\pi r^2 + \pi rl$
- Volume of the cone $= \frac{1}{3}$Bh $= \frac{1}{3}\pi r^2 h$.

Illustrative example

18. The altitude of a right circular cone is 4cm long. The radius is 3cm long.
Find a) the lateral area
b) total surface area
c) volume

Solution:

First find l, the length of the slant height

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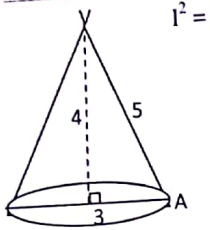


$l^2 = 3^2 + 4^2 = 9 + 16 = 25 \quad \therefore l = 5$

a) L.S.A $= \frac{1}{2}(2\pi r) \times l = \pi rl$

$\therefore$ L.S.A $= \pi(3)(5) = 15\pi cm^2$

b) T.S.A = L.S.A + B

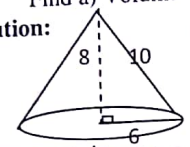
$= 15\pi + \pi \times 3^2 = 24\pi \text{ cm}^2$

c) $V = \frac{1}{3}\pi r^2 h. = \frac{1}{3}\pi(3^2)(4) \quad = 12\pi cm^3$

19. The slant height of aright circular cone is 10cm long. The radius of the base is 6cm long.

Find a) Volume b) Lateral surface area

Solution:



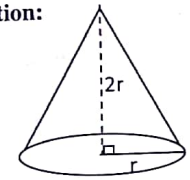
a) Volume $= \frac{1}{3}\pi r^2 h. = \frac{1}{3}\pi(6^2)(8) \quad = 96\pi cm^3$

b) L.S.A $= \pi rl = \pi(6)(10) = 60\pi cm^2$

20. In a certain cone, a diameter of the base has the same length as the altitude of the cone.

Find a) the lateral surface area b) the total surface area c) volume

Solution:



First find l,

Now, $(2r)^2 + r^2 = l^2$

$\Rightarrow l^2 = 5r^2 \qquad \therefore l = r\sqrt{5}$ unit

a) L.S.A $= \pi rl = \pi(r)\, r\sqrt{5}$

$\therefore$ L.S.A $= \sqrt{5}\,\pi r^2$ square unit

b) T.S.A $= B + \pi rl = \pi r^2 + \sqrt{5}\,\pi r^2 = \pi r^2(1+\sqrt{5})$

c) Volume $= \frac{1}{3}\pi r^2 h$

$= \frac{1}{3}\pi r^2 (2r) = \frac{2}{3}\pi r^3$ cubicic unit

A sphere

A sphere is a three dimensional figure made up of all points a given distance from center.

- Volume $= \frac{4}{3}\pi r^3$
- Surface area $= 4\pi r^2$

21. Find the radius of sphere whose surface is equal to the whole surface of a cylinder of height 16cm and radius 2cm.

Solution:

First, find the total surface area of cylinder

i) L.S.A $= 2\pi rh$

$\therefore$ L.S.A $= 2\pi(2)16 = 64\pi cm^2$

ii) T.S.A $= 2B + L.S.A$

$= 2(\pi r^2) + 64\pi \text{ cm}^2$

$= 2\pi(4) + 64\pi$

$= 72\pi cm^2$

Hence, surface area of sphere = surface area of cylinder

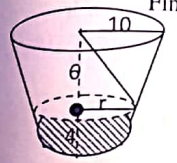
$\Rightarrow 4\pi r^2 = 72\pi cm^2 \quad \Rightarrow r^2 = \frac{72}{4} = 18$

$\therefore r = \sqrt{18} = 3\sqrt{2}cm$

22. A hemispherical bowl, internal diameter 20cm contains some water. If the greatest depth of water is 4cm.

Find a) radius of circle formed by water surface

b) volume of water



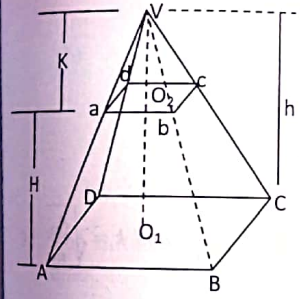
Solution: a) $r^2 + 6^2 = 10^2 \Rightarrow r^2 = 100 - 36 = 64$

$\therefore r = 8cm$

b) volume of water $= \frac{1}{2}\left(\frac{4}{3}\pi(8^3)\right) = \frac{1024}{3}\pi cm^3$

Frustums of pyramids and cones

Definition: A frustum (that is to say, a slice) of a pyramid or con is the part cut off between the base and a plane parallel to the base.



- The frustum of the pyramid (V, ABCD) is the part cut off between the base ABCD, and the parallel section abcd.
- ABCD, and abcd are called the base of the frustum.
- The base of the frustum is similar

$\therefore \frac{\overline{ad}}{\overline{AD}} = \frac{\overline{ab}}{\overline{AB}} = \frac{\overline{bc}}{\overline{BC}} = \frac{\overline{dc}}{\overline{DC}}$ and

$\frac{\overline{va}}{\overline{VA}} = \frac{\overline{vd}}{\overline{VD}} = \frac{k}{h}$ and

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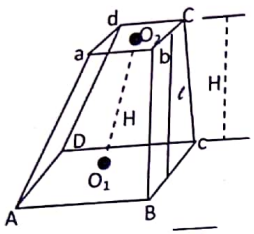
$$\frac{Area\ (abcd)}{area(ABCD} = \frac{k^2}{h^2} = \frac{A_2}{A_1} \leftarrow area\ of\ ABCD$$



- The slant surface of frustum of pyramid is made up of trapezium
- Lateral surface area of frustum = $\frac{1}{2}(p_1+p_2)l$ where p_1 is perimeter pf ABCD, p_2 is perimeter of the base of cross-section abcd, and l is the slant height of frustum
- Let Define: $\overline{VO_1}$ = h, and $\overline{VO_2}$ = k so that $\overline{VO_1} - \overline{VO_2}$ = h – k = H

Let area of the cross-section abcd is A_2 and area of base ABCD is A_1

Hence

Volume of frustum = volume of pyramid $[(V, ABCD) - (V, abcd)]$

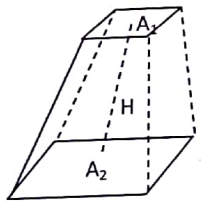
$$= \frac{1}{3}A_1h - \frac{1}{3}A_2k$$

From $\frac{A_2}{A_1} = \frac{k^2}{h^2}$, we get $A_2h^2 = A_1k^2$ gives $\frac{A_2}{k^2} = \frac{A_1}{h^2}$

Define: $\frac{A_2}{k^2} = \frac{A_1}{h^2}$ = m (where m is some constant)

$\Rightarrow A_2 = mk^2$ and $A_1 = mh^2$

$\therefore$ Volume of frustum $= \frac{1}{3}(mh^2)h - \frac{1}{3}(mk^2)k$



$$= \frac{1}{3}(h^3 - k^3)m$$

$$= \frac{1}{3}(h-k)(h^2+hk+k^2)m$$

$$= \frac{1}{3}(H)(mh^2+mhk+mk^2)$$

$$\left[Explanation\ mh^2 = A_1 \Rightarrow h = \sqrt{\frac{A_1}{m}}\ and\ k = \sqrt{\frac{A_2}{m}}\right.$$

$$= \frac{1}{3}H\left[A_1 + m\sqrt{\frac{A_1}{m}}\sqrt{\frac{A_2}{m}} + A_2\right]$$

$$\therefore \text{volume of frustum} = \frac{1}{3}H\left[A_1 + \sqrt{A_1A_2} + A_2\right] \text{ of pyramid}$$

Where A_1 is area of ABCD and H is altitude of frustum.

Illustrative example

23. A regular square pyramid has a base edge whose length is 6 unit and an altitude whose length is 10 unit.
Find a) the area of a horizontal cross section at a distance of 3 units above the base
b) The area of cross-section 3 unit from the vertex

Solution:

Given: Altitude of pyramid = 10 unit.
Base of pyramid = 6 unit
$\therefore$ The area of the base of pyramid = 36
Let A_2 – area of cross-section
Let A_1 – area of base of pyramid

a) the distance between the two base 3 unit
$\therefore$ The distance from vertex to cross-section is 10 – 3 = 7 = k

Now, from $\frac{A_2}{A_1} = \frac{k^2}{h^2}$, $A_1 = 36$, K = 7, h = 10

$$\Rightarrow \frac{A_2}{36} = \frac{49}{100} \Leftrightarrow A_2 = \frac{(49)(36)}{100} = \frac{441}{25}$$

$\therefore$ Area of cross-section = $\frac{441}{25}$ square unit

b) $A_1 = 36$, k = 3, h = 10

$$\Rightarrow \frac{A_2}{36} = \frac{9}{100}$$

$$\therefore A_2 = \frac{9\times36}{100} = \frac{81}{25} \text{ square unit}$$

24. The altitude of regular hexagon is 6 unit long and the side of the base $\sqrt{3}$ units.

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Find a) the area of horizontal cross-section at distance of 2 unit above the base

Solution:

For η side regular polygon

Recall the area, $A = \frac{1}{2}\eta r^2 \sin\left(\frac{360^0}{\eta}\right)$, $\eta = 6, r = \sqrt{3}$

$$\Rightarrow A = \frac{1}{2}(6)(\sqrt{3})^2 \sin 60^0 \quad \therefore A_1 = \frac{9\sqrt{3}}{2}$$

Now, we have, $h = 6$, $k = 6 - 2 = 4$

$$\Rightarrow \frac{A_2}{A_1} = \frac{k^2}{h^2} \Rightarrow \frac{A_2}{\frac{9\sqrt{3}}{2}} = \frac{16}{36}$$

$$\Rightarrow A_2 = \frac{16}{36} \times \frac{9\sqrt{3}}{2} = 2\sqrt{3} \text{ square unit}$$

b) From question # 24 the volume of frustum, if the plane cut off the pyramid at distance of

i) 2 unit above the base ii) 2 unit from the vertex

Solution: (i) Given: $A_1 = \frac{9\sqrt{3}}{2}$, $A_2 = 2\sqrt{3}$ and the altitude of frustum $H = 2$

Using the formula for volume of frustum

$$\Rightarrow VF = \frac{H}{3}\left(A_1 + \sqrt{A_1 A_2} + A_2\right)$$

$$= \frac{2}{3}\left[\frac{9\sqrt{3}}{2} + \sqrt{\left(\frac{9\sqrt{3}}{2}\right)(2\sqrt{3})} + 2\sqrt{3}\right] = \frac{2}{3}\left[\frac{9\sqrt{3}}{2} + 3\sqrt{3} + 2\sqrt{3}\right]$$

$\therefore$ Volume of frustum $= \frac{19\sqrt{3}}{3}$ cubic unit.

ii) Given, $A_1 = \frac{9\sqrt{3}}{2}$, $h = 6$, $k = 2$, $H = h - k = 6 - 2 = 4$

Now, First find A_2

$$\frac{A_2}{A_1} = \frac{k^2}{h^2}$$

$$\Rightarrow \frac{A_2}{\frac{9\sqrt{3}}{2}} = \frac{4}{36} \Rightarrow A_2 = \left(\frac{4}{36}\right)\frac{9\sqrt{3}}{2} = \frac{\sqrt{3}}{2}$$

Hence, using formula

$$V_f = \frac{H}{3}\left[A_1 + \sqrt{A_1 A_2} + A_2\right], \quad H = h - k = 6 - 2 = 4$$

$$\therefore V_f = \frac{4}{3}\left[\frac{9\sqrt{3}}{2} + \sqrt{\frac{9\sqrt{3}}{2}\left(\frac{\sqrt{3}}{2}\right)} + \frac{\sqrt{3}}{2}\right]$$

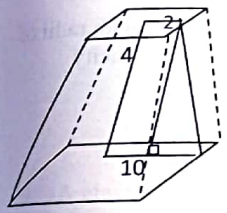
$$\therefore V_f = \frac{26}{3}\sqrt{3} \text{ cubic unit}$$

25. The bases of the frustum of a pyramid are squares on sides of 20cm and 4cm, if the altitude of frustum is 15cm, then

Find a) its slant height

b) its lateral surface area c) Volume of frustum

Solution: Given: $A_2 = 16$, $A_1 = 400$, $H = 15$



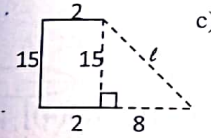
Now to find l, $l^2 = 8^2 + 15^2 = 64 + 225$

$$l = \sqrt{289} = 17$$

$\therefore$ slant height = 17cm

b) L.S.A $= \frac{1}{2}(p_1 + p_2)\, l$

$\therefore$ L.S.A $= \frac{1}{2}(4\times 20 + 4 \times 4)\, 17 = 816$ square unit



c) $V_f = \frac{H}{3}\left[A_1 + A_2 + \sqrt{A_1 A_2}\right]$ cubic unit

$$= \frac{15}{3}\left[400 + 16 + \sqrt{(400)16}\right]$$

$$= 5[496] = 2480 \text{ cubic unit}$$

26. Find the volume of frustum of pyramid, if the areas of bases of frustum of pyramid are $64cm^2$ and $36cm^2$, and its altitude is 3cm

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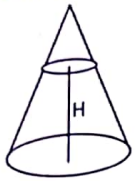
Solution: Given: $H = 3$, $A_2 = 36$, $A_1 = 64$

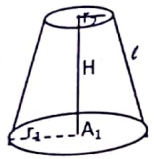
Use formula, $V_f = \frac{H}{3}\left[A_1 + A_2 + \sqrt{A_1 A_2}\right]$

$\Rightarrow V_f = \frac{3}{3}\left[64 + 36 + \sqrt{(64)(16)}\right]$

$\therefore \quad V_f = 148\text{cm}^3$

Frustum of cone





Lateral surface area of frustum of the cone given by

$\text{L.S.A} = \frac{1}{2}(c_1 + c_2)\ell$ where $c_1 = 2\pi r_1$

and $c_2 = 2\pi r_2$

$\therefore \quad \text{L.S.A} = \frac{1}{2}(2\pi r_1 + 2\pi r_2)\ell$ where

$= \pi\ell(r_1 + r_2)$

Volume of frustum $= \frac{H}{3}\left[A_1 + A_2 + \sqrt{A_1 A_2}\right]$

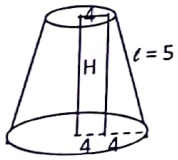
$= \frac{H}{3}\left[\pi r_1^2 + \pi r_2^2 + \pi r_1 r_2\right]$ $V_f = \frac{\pi H}{3}$

$\left[r_1^2 + r_2^2 + r_1 r_2\right]$

27. A water tank is in the shape of frustum of right circular cone with radii of bases 8m and 4m. If its slant height is 5m, then what is the capacity (volume) of tank in cubic meters . . . EGSECE

A) $\frac{320\pi}{3}$ B) 64π C) 112π D) 128π

Solution: Given: $r_1 = 8\text{m}$, $r_2 = 4\text{m}$, and $l = 5\text{m}$



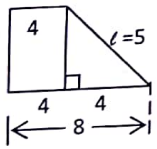
Now to find H

$H^2 + 4^2 = 5^2$

$H^2 = 25 - 16 = 9$

$\therefore H = 3$

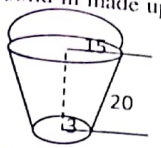
Volume of frustum $= \frac{H}{3}\pi[64 + 16 + (8)(4)]$



$\therefore V_f = \frac{3\pi}{3}[64 + 16 + (8)(4)]$

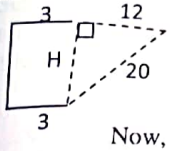
$\therefore V_f = 112\pi$ $\therefore$ Answer C

28. A solid in made up of a hemispherical top super imposed on a frustum of a right circular cone of base radii 3 units and 15 units and with slant height of 20 units as shown in the figure below. What is the volume of the solid in cubic units . . . EGSECE



A) 1938π B) 12310π C) 3738π D) 4110π

Solution:



$H^2 + 12^2 = 20^2$

$\Rightarrow H^2 = 400 - 144 = 256$

$\therefore H = \sqrt{256} = 16$

Now, volume of frustum $= \frac{H\pi}{3}\left[r_1^2 + r_2^2 + r_1 r_2\right]$

$\therefore V_f = \frac{16\pi}{3}\left[15^2 + 3^2 + 15.3\right]$

$\therefore V_f = 1488\pi$

And volume of hemisphere $= \frac{2}{3}\pi r^3$

Vol. sphere $= \frac{2}{3}\pi(15)^3 = 2250\pi$

Thus, volume of solid

= volume of frustum + volume of sphere

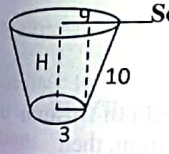
$= 1488\pi + 2250\pi$

$= 3738\pi$

Answer C

29. A glass in a shape of frustum of aright circular cone has slanted height 10cm, and bases of radii 3cm and 9cm. what is the volume of the glass in cm^3 . . . (EGSECE)

A) 240 B) 312π C) 192π D) 204π



Solution: $H^2 + 6^2 = 10^2$

$H^2 = 100 - 36 = 64$ $\therefore H = 8$

$\therefore V_f = \frac{\pi}{3}H\left[r_1^2 + r_2^2 + r_1 r_2\right]$

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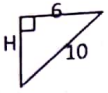


$$\therefore V_f = \frac{8}{3}\pi[9^2 + 3^2 + 9.3]$$

$$V_f = \frac{8}{3}\pi[81+9+27] = 312\pi \quad \therefore \textbf{Answer B}$$

30. Find the total surface area of Q# 29

Solution:

$$\text{L.S.A} = \frac{1}{2}(c_1 + c_2)\ell = \frac{1}{2}(2\pi r_1 + 2\pi r_2)\ell$$

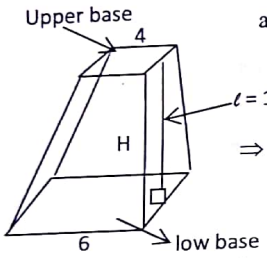
$$= \frac{1}{2}[2\pi(9) + 2\pi(3)]10 = 120\pi$$

$$\text{Base area} = \pi r_1^2 + \pi r_2^2 = 81\pi + 9\pi = 90\pi$$

$\therefore$ T.S.A = L.S.A + Base Area $= 120\pi + 90\pi = 210\pi \text{cm}^2$

31. The side of the upper and the lower bases of frustum regular square pyramid are 4cm and 6cm respectively if the slant height is 10m, then
Find a) lateral surface area
b) total surface area c) volume

Solution:



a) $\text{L.S.A} = \frac{1}{2}(p_1 + p_2)\,l$

(where p_1 and p_2 are perimeter of the lower and upper base)

$$\Rightarrow \text{L.S.A} = \frac{1}{2}(6\times4 + 4\times4)10$$

$$\therefore \text{L.S.A} = \frac{1}{2}(400) = 200\text{cm}^2$$

b) Total surface area
= L.S.A + A_1 + A_2, where
A_1 is lower base area
A_2 is upper base area

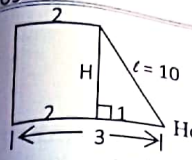
$A_1 = 6\times6 = 36$ and $A_2 = 4\times4 = 16$

$\Rightarrow$ T.S.A = 200 + 36 + 16

$\Rightarrow$ T.S.A = 252cm^2

c) To find volume of frustum, first, find the height of frustum
Let H be the height of frustum, then

$H^2 + 1^2 = 10^2$



$H^2 = 100 - 1 = 99$

$\therefore H = \sqrt{99}$

Hence, volume of frustum $= \frac{H}{3}\left[A_1 + A_2 + \sqrt{A_1 A_2}\right]$

$$\therefore V_f = \frac{\sqrt{99}}{3}\left[36 + 16 + \sqrt{(36)(16)}\right] \therefore V_f = 76\sqrt{11}\ cm^3\ \text{cm}^3$$

32. The areas of lower and the upper bases of frustum of a pyramid are 12cm^2 and 3cm^2 unit respectively. If the volume of frustum is 42cm^2, hence find the height of pyramid.

Solution: let H be the height of frustum and let h be the height of pyramid.
Now,

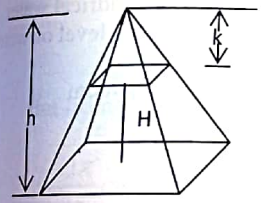
Volume of frustum $= \frac{H}{3}\left[A_1 + A_2 + \sqrt{A_1 A_2}\right]$

$$\Rightarrow V_f = \frac{H}{3}\left[12 + 3 + \sqrt{12(3)}\right]$$

$$\Rightarrow 42 = \frac{H}{3}[21] \qquad \Rightarrow 42 = 7H$$

$\therefore H = 6 \longleftarrow$ height of frustum

To find, the height of pyramid, h we use the relation



$$\Rightarrow \frac{A_2}{A_1} = \frac{k^2}{h^2}$$

$$\Rightarrow \frac{3}{12} = \frac{(h-6)^2}{h^2}$$

$\Rightarrow 4h - 24 = h$

$\Rightarrow 3h = 24$

But h − k = H

$\Rightarrow$ h − k = 6

$$\therefore h = \frac{24}{3} = 8 \longleftarrow \text{the height of pyramid}$$

33. A circular cone with altitude h and base radius r is cut at height of $\frac{1}{3}$ from the base to form frustum of cone. Then the volume of the frustum is equal to . . . ESLCE

Solution:

Let A_1 be area of the base

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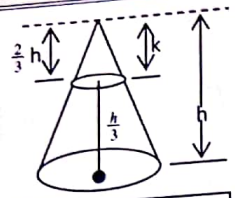


Let A_2 be area of cross-section

Now, we have the relation

$$\frac{A_2}{A_1}=\frac{k^2}{h^2}$$

Explanation:

$$k+\frac{h}{3}=h$$

$$k=h-\frac{h}{3}=\frac{2h}{3}$$

$$\Rightarrow\frac{A_2}{A_1}=\frac{\left(\frac{2}{3}h\right)^2}{h^2}=\frac{4}{9}$$

$$\Rightarrow\frac{A_2}{\pi r^2}=\frac{4}{9}\Rightarrow A_2=\frac{4}{9}\pi r^2$$

Hence vol. frustum $=\frac{H}{3}\left[A_1+A_2+\sqrt{A_1A_2}\right]$

$$\Rightarrow V_f=\frac{h}{9}\left[\pi r^2+\frac{4}{9}\pi r^2+\frac{2}{3}\pi r^2\right],H=\frac{h}{3}$$

$$\Rightarrow V_f=\frac{h}{9}\left[\frac{19}{9}\pi r^2\right]\quad\therefore V_f=\frac{19\pi r^2h}{81}$$

Solved problem

34. If a spherical stone with radius 60cm is submerged in a cylindrical water tank whose base radius is 2 meters, then by how much is the level of water raised?

A) 9cm B) 72cm C) 7.2cm D) 90cm

Solution: Given: R = 60cm $=\frac{3}{5}m$

At that level, h:

The volume of sphere = the volume of cylinder

$$\Rightarrow\frac{4}{3}\pi R^3=\pi r^2h$$

$$\Rightarrow\frac{4}{3}\pi\left(\frac{3}{5}\right)^3=\pi(4)h\qquad\left[\text{explanation}:\ 60cm=\frac{60}{100}m=\frac{3}{5}m\right]$$

$$\Rightarrow h=\frac{27}{(3)(125)}m=\frac{9}{125}=0.072m$$

$\therefore h=0.072\times100cm=7.2cm$

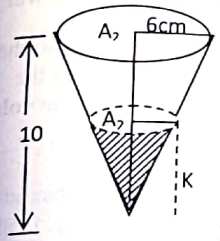
35. A right circular conical vessel of altitude 10cm and a vase radius 6cm is kept with its vertex down wards. If the volume of water in the vessel is $\frac{81}{25}\pi cm^3$. Then the depth of water in the vessel is:

A) 3cm B) 7.5cm C) 6cm D) $3\sqrt{3}$ cm

Solution: let A_1 be area of base

let A_2 be the area of cross-section.

Let K be the depth of water and h be the altitude of cone



So that $\frac{A_2}{A_1}=\frac{k^2}{h^2}$

$$=\frac{A_2}{36\pi}=\frac{k^2}{10^2}\Rightarrow A_2=\frac{K^2(36\pi)}{100}\ldots..$$

(1)

And volume of water

$$=\frac{1}{3}(\text{Base area})(\text{height of water})$$

$$\frac{81\pi}{25}cm^3=\frac{1}{3}(A_2)K\ldots\ldots\ldots\ldots(2)$$

But from (1), $A_2=\frac{K^2(36\pi)}{100}$ and from (2)

$$\therefore\frac{81\pi}{25}cm^3=\frac{1}{3}\frac{K^2(36\pi)}{100}\Rightarrow\frac{81\pi}{25}=\frac{3k^2}{25}\Rightarrow k^2=27\ \therefore K=3\sqrt{3}\text{ cm}$$

Hence the depth water is $k=3\sqrt{3}$ cm.

36. A frustum of height 6cm is formed from a right circular cone of height 12cm and base radius 4cm, then the volume of frustum is

A) $56\pi cm^3$ B) $72\pi cm^3$ C) $64\pi cm^3$ D) $8\pi cm^3$

Solution: Let A_2 be area of cross-section and A_1 be area of base

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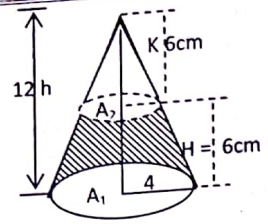


$$\frac{A_2}{A_1} = \frac{k^2}{h^2}$$

$$\Rightarrow \frac{A_2}{16\pi} = \frac{36}{144} \Rightarrow A_2 = \frac{16\pi \times 36}{144} = 4\pi$$

Using volume of frustum can obtained by formula,

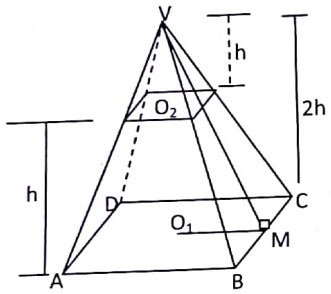
$$V_f = \frac{\pi H}{3}\left[r_1^2 + r_2^2 + r_1 r_2\right]$$

$$= \frac{\pi}{3}(6)[16 + 4 + 4(2)] = 2\pi\,[28) = 56\pi \qquad \therefore \textbf{Answer A}$$

37. A regular square pyramid is cut by a plane parallel to the base so that the plane bisected the altitude of the pyramid. If the slant height of the pyramid is 12 units and its lateral edge is 13 units long, then the volume of the frustum in cubic units is

A) $\frac{175\sqrt{119}}{6}$ B) 325 C) $\frac{175\sqrt{119}}{24}$ D) $\frac{2275}{5}$

Solution: Given: $\overline{VC} = \overline{VB} = \overline{VO} = \overline{VA} = 13 unit$



And $\overline{VM}$ slant height = 12 unit

From Pythagorean identity

$VM^2 + MC^2 = VC^2$

$\Rightarrow 12^2 + MC^2 = 13^2$

$\Rightarrow MC^2 = 169 - 144 = 25$

$MC = 5$

$\therefore \overline{BC} = 10 = \overline{AB}$

To find the altitude,

$\overline{VO_1}$, we have $\overline{OM} = 5$

$$\overline{VO_1^2} + \overline{OM}^2 = VM^2$$

$$\Rightarrow VO_1^2 + 5^2 = 12^2$$

$$VO_1^2 = 144 - 25 = 119$$

$\therefore$ The altitude $\overline{VO_1} = \sqrt{119}$

$\therefore$ The altitude $VO_2 = \frac{\sqrt{119}}{2}$

$\therefore$ The altitude of frustum $\overline{O_1O_2} = \frac{\sqrt{119}}{2} = H$

To find the Area of cross-section

$$\frac{A_2}{A_1} = \frac{h^2}{(2h)^2}$$

$$\Rightarrow \frac{A_2}{100} = \frac{1}{4} \Leftrightarrow A_2 = \frac{100}{4} = 25 \text{ square unit}$$

$\therefore$ Volume of frustum $= \frac{H}{3}\left[A_1 + A_2 + \sqrt{(25)(100)}\right]$

$$V_f = \frac{\sqrt{119}}{6}\left[100 + 25 + \sqrt{(25)(100)}\right] = \frac{\sqrt{119}}{6}[175] \quad \therefore \textbf{Answer A}$$

38. If the radius of sphere is doubled
What happens to the a) surface area b) volume

Solution: Surface area of sphere $= 4\pi r^2$

If the radius double $= 4\pi(2r)^2 = 16\pi r^2$

a) $\therefore$ It becomes four times as large as the original surface area

b) volume of sphere for the radius r, $v = \frac{4}{3}\pi r^3$ If the radius double, then

the volume $= \frac{4}{3}\pi(2r)^3 = \frac{32}{3}\pi r^3$

$\therefore$ It become $8V_0$

39. What is the volume of frustum of cone whose radius (R) of its larger base is twice that of the radius (R) of its smaller base and with height $x - y = 3y$ . . . EGSECE

A) $20\pi r^2$ B) $10\pi r^2 y$ C) $4\pi r^2 y$ D) $5\pi r^2 y$

Solution: Here

Volume of frustum of cone is given by

Volume of big con-volume of small cone.

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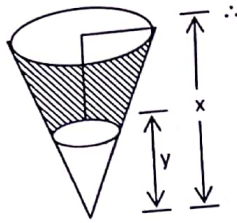


$$\therefore V_f = \frac{1}{3}\pi R^2 x - \frac{1}{3}\pi r^2 y$$

$$= \frac{1}{3}\pi(2r)^2(4y) - \frac{1}{3}\pi r^2 y$$

$$= \frac{16\pi r^2 y}{3} - \frac{\pi r^2 y}{3}$$

$$= \frac{15\pi r^2 y}{3} = 5\pi r^2 y \quad \therefore \textbf{Answer D}$$

40. The areas of the upper and the lower bases of frustum of a pyramid are 3 square units and 1.2 square respectively. If the volume of the frustum is 4.2 cubic units, then what is the height of the pyramid?
A) 12 units B) 9 units C) 6 units D) 15 units

Solution:

$$V_f = \frac{H}{3}\left[A_1 + A_2 + \sqrt{A_1 A_2}\right] \text{ where H is height of frustum}$$

$$\Rightarrow 42 = \frac{H}{3}\left[3 + 12 + \sqrt{(3)(12)}\right]$$

$$\Rightarrow 42 = \frac{H}{3}[15+6] \Rightarrow 42 = 7H \quad \therefore H = \frac{42}{7} = 6 \text{ unit}$$

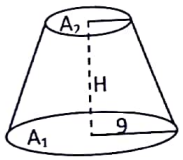
And

$$\frac{\text{Area of cross section}}{\text{Area of base}} = \frac{K^2}{h^2} \Rightarrow \frac{1}{4} = \left(\frac{k}{k+6}\right)^2 \Rightarrow \frac{k}{k+6} = \frac{1}{2}$$

$\Rightarrow 2k = k + 6 \Rightarrow k = 6 \therefore h = 6 + 6 = 12$

∴Answer A

41. The figure below represents a frustum formed from the base of aright circular cone with diameter of the base 18cm and height 12cm. if the ratio of the heights of frustum and the cone is 1:2, which of the following is the volume of the frustum?
A) $384\pi cm^3$ B) $283.5\pi cm^3$ C) $228\pi cm^3$ D) $336\pi cm^3$



Solution: Given
Ratio of height of frustum to height of the cone, let H be height of frustum, and let h be height of cone

So that $\frac{H}{h} = \frac{1}{2} \Leftrightarrow h = 2H$

Hence, $12 = 2H$

$\Rightarrow H = 6$

And height pf small cone $h - H = 12 - 6 = 6 = k$

Therefore, to find area of cross-section

$$\frac{\text{Area of cross} - \text{section}}{\text{Area of base}} = \frac{k^2}{h^2}$$

$$\Rightarrow \frac{A_2}{81\pi} = \frac{36}{144} \Rightarrow A_2 = \frac{36}{144}(81\pi) = \frac{81\pi}{4}$$

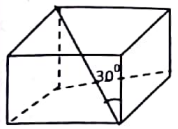
$$\therefore \text{Vol. of frustum} = \frac{H}{3}\left[A_1 + A_2 + \sqrt{A_1 A_2}\right]$$

$$= \frac{6}{3}\left[81\pi + \frac{81\pi}{4} + \sqrt{\frac{81\pi}{4}.81\pi}\right] = 2\left[81\pi + \frac{81\pi}{4} + \frac{81\pi}{2}\right]$$

$$= 283.5\pi cm^3$$

∴ Answer B

42. What is the volume of cube whose diagonal has length 4cm and makes 30^0 with it's adjacent edge as shown in the figure below . . . EGSEC



A) $8cm^3$ B) $24\sqrt{3}\ cm^3$
C) $12\sqrt{3}\ cm^3$ D) $\frac{64}{3\sqrt{3}} cm^3$

Solution: Let D be the diagonal of cube let **a** be the edge of cube

Hence $\cos 30 = \frac{a}{D}$

$$\Rightarrow \frac{\sqrt{3}}{2} = \frac{a}{4} \Rightarrow a = \frac{4\sqrt{3}}{2} = 2\sqrt{3}$$

$$\therefore \text{Volume of cube} = a^3 = \left(2\sqrt{3}\right)^3 = 24\sqrt{3} cm^3$$

∴Answer B

43. The slant height of a right circular cone is 8cm. of the angle between the slant height and the height at the vertex of the cone is 30^0, then what is the volume of the cone?

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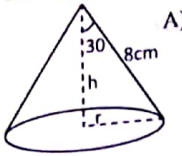


A) $\frac{64\sqrt{3}\pi}{3}cm^3$ B) $64\pi cm^3$ C) $\frac{128\pi}{3}cm^3$ D) $128\pi cm^3$

$\text{Cos}30^0 = \frac{h}{8} \Rightarrow h = 8\cos 30^0 = (8)\frac{\sqrt{3}}{2}$

$\Rightarrow h = 4\sqrt{3}$

And $\sin 30^0 = \frac{r}{8} \quad \Rightarrow \frac{1}{2} = \frac{r}{8} \Rightarrow r = 4$

$V = \frac{1}{3}Bh = \frac{1}{3}\pi r^2 h$

Hence volume $= \frac{1}{3}(\pi(16)4\sqrt{3})$

$\therefore V = \frac{64\sqrt{3}\pi}{3}cm^3$ **∴ Answer A**

44. Which of the following is true a bout the volume V of a cone determined by doubling both the base radius and height of a right circular cone whose volume was V_0

A) $V = 4V_0$ B) $V = 2V_0$ C) $V = 6V_0$ D) $V = 8V_0$

Solution: $V_0 = \frac{1}{3}\pi r^2 h$

But if $r \rightarrow 2r$ and $h \rightarrow 2h$

Then

$V = \frac{1}{3}\pi(2r)^2(2h) = \frac{1}{3}\pi(4r^2)(2h) = 8(\frac{1}{3}\pi r^2 h)$

$\therefore V = 8(V_0)$

∴ Answer D

45. From aright circular cone of height 6cm a frustum with bases of radii 2cm and 6cm is obtained. The volume of the frustum is equal to: . . . . EGSECE n2003/11

A) $52\pi cm^3$ B) $104\pi cm^3$ C) $\frac{104}{3}\pi cm^3$ D) $\frac{208}{3}\pi cm^3$

Solution: Vol. frustum $= \frac{H}{3}\left[A_1 + A_2 + \sqrt{A_1 \cdot A_2}\right]$

$H = A_1 = \pi r^2 = \pi(4) = 4\pi$

$A_2 = \pi(6^2) = 36\pi$

$\therefore V_f = \frac{4}{3}\left[4\pi + 36\pi + \sqrt{(36\pi)(4\pi)}\right]$

$= 2[52\pi]$

∴ Answer C

Explanation: height of cone,6

$\frac{k}{h} = \frac{2}{6}$

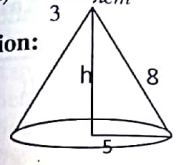
$\therefore k = 2$

$H + k = 6 \Rightarrow H = 4$

46. If a right circular cone ahs base radius 5cm and slanted height 8cm then the volume of this cone is equal to . . . EGSECE 2003/11

A) $\frac{65\sqrt{39}}{3}\pi cm^3$ B) $\frac{25\sqrt{39}}{3}\pi cm^3$ C) $\frac{200\pi}{3}\pi cm^3$ D) $200\pi cm^3$

Solution:



To find h, use $h^2 + 5^2 = 8^2$

$h^2 = 64 - 25 = 39$

$\therefore h = \sqrt{39}$

$\therefore$ Vol $= \frac{1}{3}Bh = \frac{1}{3}\pi(5^2)\sqrt{39} = \frac{25\sqrt{39}\pi}{3}$

∴ Answer B

47. If a metallic right circular cylinder of height 18cm and base radius 4cm is melted down to form a sphere, then which one of the following is correct about this sphere . . . EGSECE 2003/11

A) The surface area of the sphere is $144\pi cm^2$

B) The diameter of the sphere is $12\sqrt[3]{4\pi}$ cm

C) The radius of the sphere is $\frac{3}{\sqrt[3]{2}}\pi cm$

D) The volume of the sphere is $48\pi cm^3$

Solution:

Volume of cylinder $= \pi r^2 h$

$\Rightarrow V = (18)4^2\pi = (18)(16)\pi\ cm^3 \leftarrow$ volume of cylinder

But volume of sphere = volume of cylinder

$\Rightarrow \frac{4}{3}\pi r^3 = (18)(16)\pi \Rightarrow r^3 = (18)(16)\left(\frac{3}{4}\right) \Rightarrow r^3 = 216$

$\therefore$ Surface area of sphere $= 4\pi r^2$

$\therefore$ Surface area $= 4\pi(36) = 144\pi r^2$

$\therefore$ Volume of sphere $= \frac{4}{3}\pi r^3 = \frac{4}{3}\pi(6)^3 = 288\pi cm^3$

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∴Answer A

48. The base of a right triangular prism is right – angled triangle. If the height of this prism is 10cm and the longest side of the base is 5c, while the shortest one is 3cm, then what us the total surface area of this prism? . . . .
EGSECE 2003/11
A) $165cm^2$ B) $150cm^2$ C) $132cm^2$ D) $126cm^2$

Solution: Total surface area = Ph + 2B

$$= (3+4+5)10 + 2\left(\frac{1}{2}\times 3\times 4\right) = 120 + 12 = 132cm^2$$

∴Answer C

Supplementary Exercise

1. The base of a right prism is a triangle ABC right angled at C. If $\overline{AC}$ = 15cm, $\overline{CB}$ = 8cm, and the height of prism 12cm
Find a) lateral surface area b) volume
2. A right prism on triangular base whose sides are 17cm, 10cm and 9cm, and the height is 10cm.
Find a) Total surface area b) volume
3. The base of right prism is a trapezium whose parallel sides are 17cm and 13cm, the distance between them is 8cm. If the height of the prism is 100cm,
Find a) the volume
4. The diagonal of a cube is 8cm, then
Find a) Lateral surface area of cube b) volume of a cube
5. The altitude of right hexagonal prism is 8cm and the length of the edge of its bases are each 4cm. .
Find a) the volume
6. If the dimensions of right rectangular prism are 4cm, 5cm and 7cm, then
Find a) length of a diagonal b) volume
7. The base of a right pyramid of height 4cm, is square whose diagonal is 6cm
Find a) Lateral surface area
b) volume
8. The base of right pyramid is a square on side of 8cm, and the slant faces are equilateral triangle Find a) volume
9. A right pyramid of regular hexagon height 7cm and the base on side 5cm.
Find a) L.S.A b) volume

10. A right pyramids stands on a regular hexagon having a side of 5cm, and its slant face are inclined to the base at an angle of 60^0
Find the volume
11. The length of each edge of regular tetrahedron is 2m.
Find a) Total surface area b) volume
12. A regular square pyramid has all its edge 10cm long. Find volume
13. The altitude and radius of right circular cone area 10cm and 8cm respectively. Find its a) Total surface area b) volume
14. A frustum of height 6cm is formed from right circular cone of height 8cm and base radius 4cm.
Find a) a cone b) frustum
15. The lower base of a frustum of regular square pyramid is a square side of 2cm. and the upper base has side length of 4cm. if the slant height is 6cm, then
Find a) Lateral surface area b) volume
16. The base of frustum of a pyramid are triangle the side of the base measuring 13cm, 12cm and 5cm, and the sides of the top 6.5cm, 6cm, 2.5cm, of the height of frustum is 8. Find volume
17. The frustum of square pyramid is 6cm high and the area of one base is four times the area of the other. If the volume of frustum is $350cm^3$. Find the dimension of the bases.

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22. Use $PT^2 = PC.PD = PC(CD + CP)$

a) $36 = 4cm\ (CD + 4) \Rightarrow 36 = 4CD + 16$

$\therefore \overline{DC} = 5cm$

b) $PT^2 = \left(\overline{PB}\right) \cdot PA = PB\left(\overline{PB} + \overline{BA}\right) = PB(PB + 9)$

$36 = PB^2 + 9PB - 36 = 0$

$\Leftrightarrow (PB + 12)(PB - 3) = 0$

$\Rightarrow PB = 3cm \quad \therefore \overline{PB} = 3cm$

Answer key Unit 13

1. a) 480 square cm b) $720cm^3$
2. a) $432cm^2$ b) $360cm^3$
3. a) $12{,}000cm^3$
4. a) $\frac{256}{3}cm^2$ b) $\frac{64\sqrt{3}}{9}cm^3$
5. a) $192\sqrt{3}\ cm^3$
6. $a = \sqrt{4^2 + 5^2 + 7^2} = 3\sqrt{10}cm$ b) $140cm^3$
7. a) $3\sqrt{41}\ cm^2$ b) $24cm^2$
8. $\frac{256\sqrt{2}}{3}cm^3$
9. a) $\frac{15\sqrt{271}}{2}cm^2$ b) $\frac{175\sqrt{3}}{2}cm^3$
10. $162.4cm^3$
11. a) $4\sqrt{3}\ m^2$ b) $\frac{2}{3}\sqrt{2}m^3$
12. $\frac{500\sqrt{2}}{3}$
13. a) $\left(8\sqrt{164}\pi + 64\pi\right)$ b) $\frac{640\pi}{3}$
14. a) $\frac{128}{3}\pi cm^3$ b) $42\pi cm^3$
15. a) $72cm^2$ b) $\frac{28\sqrt{35}}{3}cm^3$
16. $140cm^3$
17. 10cm and 5cm

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